# IEC 62325-301

Edition 2.0 2018-03

# INTERNATIONAL STANDARD

# NORME INTERNATIONALE

colour inside

**Framework for energy market communications –
Part 301: Common information model (CIM) extensions for markets**

**Cadre pour les communications pour le marché de l'énergie –
Partie 301: Extensions du modèle d'information commun (CIM) pour les marchés**

IEC 62325-301:2018-03(en-fr)

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## Framework for energy market communications – Part 301: Common information model (CIM) extensions for markets

## Cadre pour les communications pour le marché de l'énergie – Partie 301: Extensions du modèle d'information commun (CIM) pour les marchés

INTERNATIONAL ELECTROTECHNICAL COMMISSION

COMMISSION ELECTROTECHNIQUE INTERNATIONALE

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CONTENTS

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# FRAMEWORK FOR ENERGY MARKET COMMUNICATIONS –

## Part 301: Common information model (CIM) extensions for markets

## FOREWORD

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International Standard IEC 62325-301 has been prepared by IEC technical committee 57: Power systems management and associated information exchange.

This second edition cancels and replaces the first edition published in 2014. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) Environmental weather support
b) Demand-side communication for North American markets
c) Transparency regulations and flow-based market coupling along with network codes support for European Markets
d) Registered resource updates.

The text of this standard is based on the following documents:

| FDIS | Report on voting |
|---|---|
| 57/1947/FDIS | 57/1961/RVD |

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

Throughout this document, references are made to the UML model. References to specific UML packages and classes are made by citing the name of package or class in single quotes, example, 'IEC 61970' or 'IdentifiedObject'.

A list of all parts of the IEC 62325 series, under the general title: *Framework for energy market communications,* can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC web site under "http://webstore.iec.ch" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

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# INTRODUCTION

This part of IEC 62325 is one of a series of standards for deregulated energy market data exchanges. This standard series was originally based upon the work of the European transmission system operators TF 14 EDI (Electronic data interchange) and the Electric Power Research Institute (EPRI) TR 1009455 and TR 1011431 (CIM extensions to support market operations, phase 1 and phase 2).

The principal objective of the IEC 62325 series is to produce standards which facilitate the integration of market application software developed independently by different vendors into a market management system, as well as integration of market management systems and market participant systems. This is accomplished by defining message exchanges to enable these applications or systems to access public data and exchange information independent of how such information is represented internally.

The common information model (CIM) specifies the basis for the semantics for this message exchange. The profile specifications, which are contained in other parts of IEC 62325, specify the content of the messages exchanged. Profile specifications in other parts of IEC 62325 support both European style markets based on the European regulations, Third Party Access and zonal market concepts, and North American style markets characterized by day ahead unit commitment by a market operator, intraday and real time balancing through central dispatch, and settlement based on locational marginal prices. These markets are used in the US and in other parts of the world and are not exclusively limited to North America.

The CIM is an abstract model that represents all the major objects in an electric utility enterprise typically needed to model the operational aspects of a utility. This model includes public classes and attributes for these objects, as well as the relationships between them.

The IEC 62325 series are part of a series of standards dealing with the CIM. IEC 61970 defines the EMS application programming interfaces and IEC 61968 defines System interfaces for distribution management. IEC 62325-301 corresponds to IEC 61970-301 and IEC 61968-11 which describe parts of the common information model relevant to modeling interfaces for EMS, DMS and MMS systems. While there are multiple IEC standard seriess dealing with different parts of the CIM, there is a single, unified information model comprising the CIM behind all these individual standard series.

# FRAMEWORK FOR ENERGY MARKET COMMUNICATIONS –

## Part 301: Common information model (CIM) extensions for markets

## 1 Scope

This part of IEC 62325 specifies the common information model (CIM) for energy market communications.

The CIM is an abstract model that represents all the major objects in an electric utility enterprise typically involved in utility operations and electricity market management. By providing a standard way of representing power system resources as object classes and attributes, along with their relationships, the CIM facilitates the integration of market management system (MMS) applications developed independently by different vendors, between entire MMS systems developed independently, or between an MMS system and other systems concerned with different aspects of market management, such as capacity allocation, day-ahead management, balancing, settlement, etc.

The CIM facilitates integration by defining a common language (i.e. semantics) based on the CIM to enable these applications or systems to access public data and exchange information independent of how such information is represented internally.

The object classes represented in the CIM are abstract in nature and may be used in a wide variety of applications. The use of the CIM goes far beyond its application in a market management system.

Due to the size of the complete CIM, the object classes contained in the CIM are grouped into a number of logical packages, each of which represents a certain part of the overall power system being modeled. Collections of these packages are progressed as separate international standards. This particular document specifies a set of packages which provide a logical view of the functional aspects of market management within an electricity market, and other functional aspects including environmental aspects that are closely related to electricity markets and that are shared between all applications. Other standards specify more specific parts of the model that are needed by only certain applications. Subclause 4.2 provides the current grouping of packages into standards documents.

This new edition of IEC 62325-301 contains support for demand-side communication within a wholesale market. The IEC 62325-301 additions include support for demand-side resource registration and enrollment of a market participating resource as well as support for deployment and performance evaluation of demand side resources. A new package has been included in this edition of IEC 62325-301 to support environmental (weather) data. This new package 'Environmental' provides support for weather conditions including forecasts, observations, measurements, phenomena, and alerts. Additional updates have been added within the 'MarketManagement' package to support the transparency regulations, flow based market coupling and new network codes to support the European Markets. These updates include new classes, attributes and associations within the IEC 62325 packages as well as updates to existing classes, attributes and associations to accurately represent the existing use cases.

## 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61968-11, *Application integration at electric utilities – System interfaces for distribution management – Part 11: Common information model (CIM) extensions for distribution*

IEC TS 61970-2:2004, *Energy management system application program interface (EMS-API) – Part 2: Glossary*

IEC 61970-301, *Energy management system application program interface (EMS-API) – Part 301: Common information model (CIM) base*

IEC 61970-302, *Energy management system application program interface (EMS-API) – Part 302: Common information model (CIM) for dynamics*

IEC 62325-450, *Framework for energy market communications – Part 450: Profile and context modelling rules*

Object management group: *UML 2.0 specification* – available at <http://www.omg.org>

## 3 Terms, definitions and abbreviated terms

For the purposes of this document, the terms and definitions given in IEC TS 61970-2 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

### 3.1 Terms and definitions

**3.1.1**
**application program interface**
**API**
set of public functions provided by an executable application component for use by other executable application components

[SOURCE: IEC 61970-2:2004, 3.4]

**3.1.2**
**extensible markup language**
**XML**
markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable

**3.1.3**
**XML schema description**
**XSD**
XML-based language used to describe and control XML document contents

**3.1.4**
**intra-day markets**
intra-day scheduling markets which refine the commitment and schedules produced by the day ahead markets

**3.1.5**
**load forecast**
load forecast system which predicts load for the specified time horizon

**3.1.6**
**locational marginal price**
**LMP**
marginal cost (currency/MWh) of serving the next increment of demand at a pricing node consistent with existing transmission constraints and the performance characteristics of the resources

**3.1.7**
**market management system**
**MMS**
computer system comprising a software platform that provides basic support services and a set of applications with the functionality needed for the effective management of the electricity market

Note 1 to entry: These software systems in an electricity market may include support for capacity allocation, scheduling energy, ancillary or other services, real-time operations and settlements.

**3.1.8**
**scheduler**
scheduling system for resources that produce, consume, or transport energy or ancillary services in an energy market

**3.1.9**
**third party access**
concept in market management systems which allows third parties to use the transmission system on an equal basis

**3.1.10**
**unified modeling language**
**UML**
modeling language in the field of object-oriented software engineering that is maintained by the Object Management Group

**3.1.11**
**RDF schema**
**RDFS**
W3C recommendation which provides basic elements for the description of ontologies

**3.1.12**
**ontology web language**
**OWL**
W3C recommendation which provides a knowledge representation language for authoring ontologies

### 3.2 Abbreviated terms

| | |
|---|---|
| **CRR** | congestion revenue rights |
| **DAM** | day ahead market which determines the commitment and schedules of resources |
| **DMS** | distribution management system |
| **EMS** | energy management system |
| **MDMS** | meter data management system |
| **RTM** | real time markets which finalize dispatch for system resources |

## 4 CIM specification

### 4.1 CIM modeling notation

The CIM is defined using object-oriented modeling techniques. Specifically, the CIM specification uses the unified modeling language (UML) notation, which defines the CIM as a group of packages.

Each package in the CIM contains one or more class diagrams showing graphically all the classes in that package and their relationships. Each class is then defined in text in terms of its attributes and relationships to other classes.

The UML notation is described in the Object Management Group (OMG) standard and several published textbooks.

### 4.2 CIM packages

The CIM is partitioned into a set of packages. A package is a general purpose means of grouping related model elements. The packages have been chosen to make the model easier to design, understand and review. The common information model consists of the complete set of packages. Entities may have associations that cross many package boundaries. Each application may use information represented in several packages.

The comprehensive CIM is partitioned into groups of packages for convenience in managing and maintaining them. IEC 62325-301 includes the following packages:

- ‘MarketCommon’
- ‘MarketManagement’
- ‘MarketOperations’
- ‘Environmental’

Packages used by IEC 61970-301 and packages used by IEC 61968-11 describe additional parts of the CIM that deal with other logical views of utility operations including network models, generation, load, outages, assets, location, activities, consumers, documentation, work management.

NOTE The package boundaries do not imply application boundaries. An application may use CIM entities from several packages.

The IEC 62325-301 is published based on an agreed upon stable version of IEC 61970 and IEC 61968.

Figure 1 shows package dependencies. The ‘IEC 62325’ package and its subpackages depend on the ‘IEC 61970’ and ‘IEC 61968’ packages. The ‘IEC 61970’ package is used by the IEC 61970 series of standards as a core model that represents power system resources and their relationships. The ‘IEC 61968’ package is used by the IEC 61968 series of standards describing logical views of utility operations, including assets, location, activities, customers, documentation, work management and metering models. The dashed line indicates a dependency relationship, with the arrowhead pointing from the dependent package to the package on which it has a dependency.

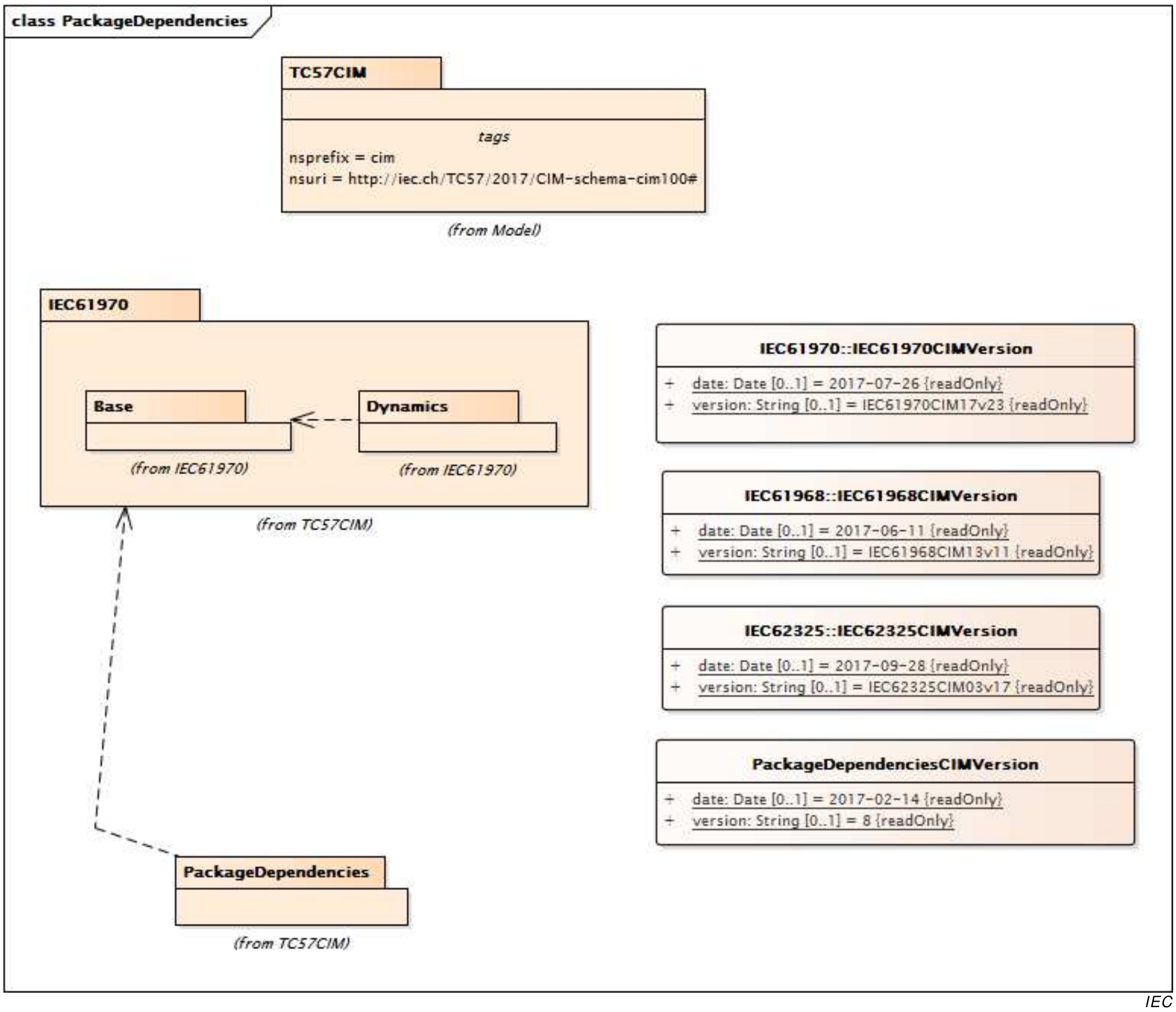


**Figure 1 – ‘TC 57CIM’ package dependency diagram**

Figure 2 shows the packages defined for IEC 62325-301 and their dependency relationships. The dashed line indicates a dependency relationship, with the arrowhead pointing from the dependent package to the package on which it has a dependency. The IEC 62325 packages include dependencies upon packages from the IEC 61970 series and the IEC 61968 series. These dependencies are shown in the diagram.

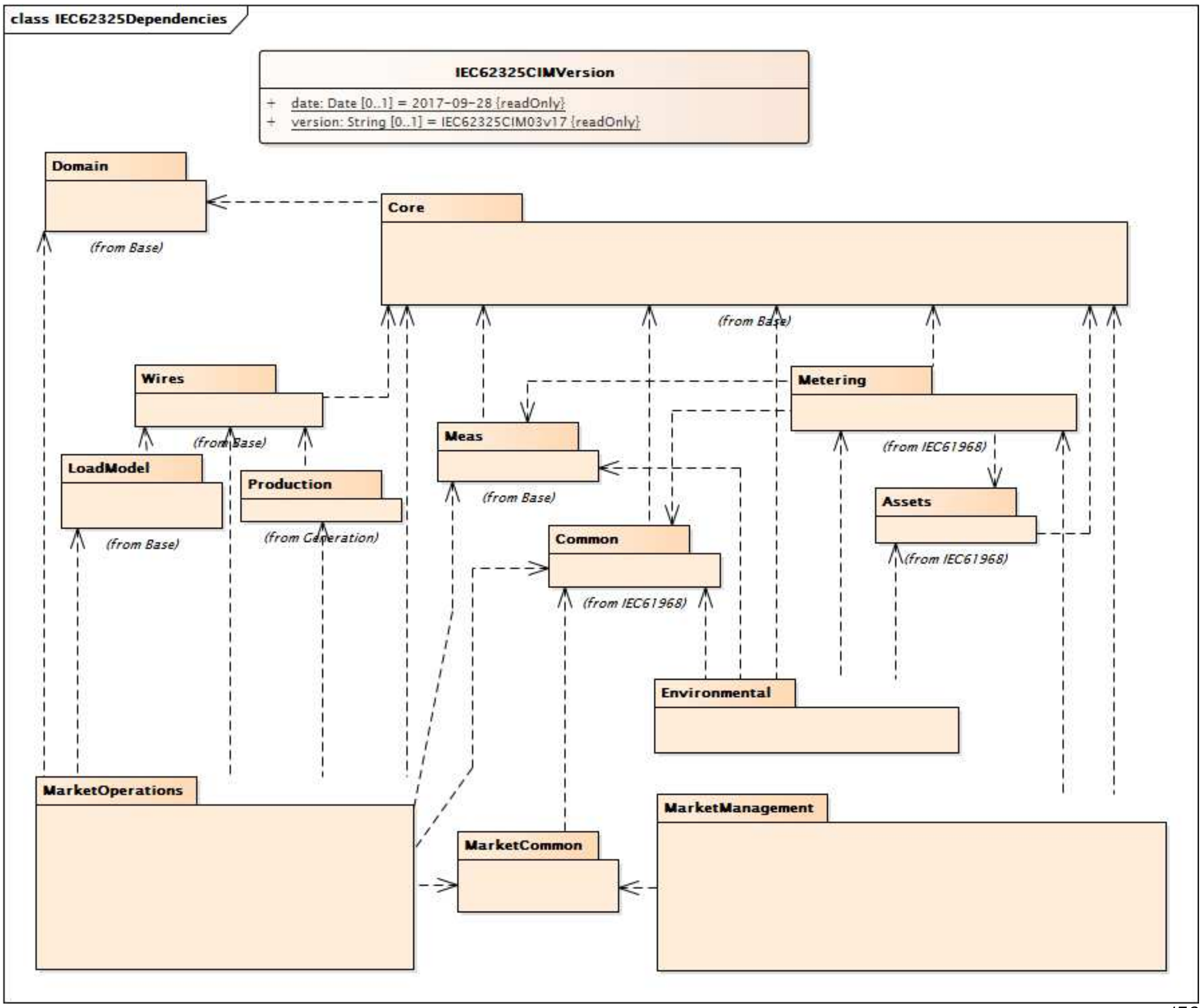


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**Figure 2 – 'IEC 62325' package dependency diagram**

Clause 6 contains the specification for each of the CIM packages.

NOTE The contents of the CIM defined in this specification were auto-generated from the CIM UML electronic model release IEC62325CIM03v17, which is available through the CIM Users Group.

## 4.3 CIM classes and relationships

### 4.3.1 Classes

The class diagram(s) for each CIM package show(s) all the classes in the package and their relationships. Where relationships exist with classes in other packages, those classes are also shown.

Classes and objects model what is in a power system that needs to be represented in a common way to market applications. A class is a description of an object found in the real world, such as a bid, registered resource, or registered load that needs to be represented as part of the overall electricity market in an MMS. Other types of objects include things such as schedules and measurements that MMS applications also need to process, analyze, and store. Such objects need a common representation to achieve the purposes of this standard for plug-compatibility and interoperability. A particular object in a power system with a unique identity is modeled as an instance of the class to which it belongs.

In this document when a header includes the term “root class”, it means the class does not have inheritance and does not have a stereotype.

The CIM is defined to facilitate data exchange. As defined in this document, CIM entities have no behavior. IEC 62325-450, *Framework for energy market communications – Part 450: Profile and context modeling rules* defines the modeling principles to generate a profile and the modeling framework principles to define the message payload.

Classes have attributes that describe the characteristics of the objects. Each class in the CIM contains the attributes that describe and identify a specific instance of the class. Only the attributes that are of public interest to MMS applications are included in the class descriptions.

Each attribute has a type, which identifies what kind of attribute it is. Typical attributes are of type ‘Integer’, ‘Float’, ‘Boolean’, ‘String’, ‘Datetime’, ‘Date’, ‘Duration’, and ‘Decimal’, which are called primitive types. However, many additional types are defined as part of the CIM specification. For example, ‘RegisteredGenerator’ has a “‘maxDependableCap’” attribute of type ‘ActivePower’. The definition of many shared types is contained in the ‘Domain’ package described in Clause 6. The UML stereotypes of ‘Primitive’, ‘enumeration’, ‘CIMDatatype’, and ‘Compound’ are added to classes used as types:

- The ‘CIMDatatype’ stereotype is used with a specific CIM semantic to describe a type that has several attributes {‘value’, ‘unit’, ‘multiplier’, ‘denominatorUnit’, ‘denominatorMultiplier’}, which implies custom mapping to serialization artifacts such as RDFS, OWL, and XSD. Classes with these stereotypes do not participate in generalization or association relationships and are simply used as types for attributes.
- The ‘enumeration’ stereotype is used to describe the type of an attribute with an enumerated list of choices.
- The ‘Compound’ stereotype is used to describe sets of related attributes that are commonly reused. ‘Compound’ classes may consist of attributes whose types are ‘Primitive’, ‘enumeration’, ‘CIMDatatype’ or other ‘compound’ classes as long as the ‘compound’ classes do not recurse.

All CIM attributes are implicitly optional in the sense that profiles using the CIM may eliminate any attributes.

Relationships between classes reveal how they are structured in terms of each other. CIM classes are related in a variety of ways, as described in 4.3.2, 4.3.3 and 4.3.4.

### 4.3.2 Generalization

A generalization is a relationship between a more general and a more specific class. The more specific class can contain only additional information. For example, ‘RegisteredGenerator’ is a specific subclass of ‘RegisteredResource’. Generalization provides for the specific class to inherit attributes and relationships from all the more general classes above it.

Figure 3 is an example of generalization. In this example taken from the ‘MarketCommon’ package, a ‘MarketParticipant’ is a more specific type of ‘Organisation’. ‘Organisation’ also inherits from ‘IdentifiedObject’.

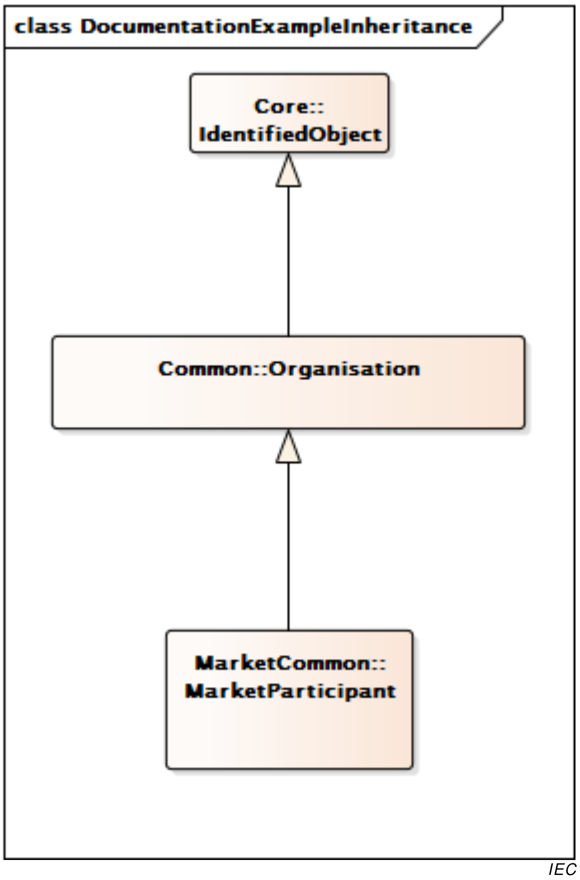


**Figure 3 – Example of generalization**

### 4.3.3 Simple association

An association is a conceptual connection between classes. Each association has two “association ends”. The “association ends” were called “roles” prior to the *UML 2.0 specification*. Each association end describes the role the target class (i.e., the class the association end goes *to*) has in relation to the source class (i.e., the class the association end goes *from*). Association ends are usually given the name of the target class with or without a qualifier term (verb, noun, adverb) phrase. Each association end also has multiplicity/cardinality, which is an indication of how many objects may participate in the given relationship. In the CIM, associations are not named, only association ends are named. For example, in the CIM there is an association between a ‘MarketParticipant’ and ‘MarketRole’ (see Figure 4 which is taken from the ‘MarketCommon’ package). Multiplicity is shown at both ends of the association. In this example, a ‘MarketParticipant’ object may reference 0 or more ‘MarketRole’ objects and a ‘MarketRole’ may be referenced by 0 or more ‘MarketParticipant’ objects.

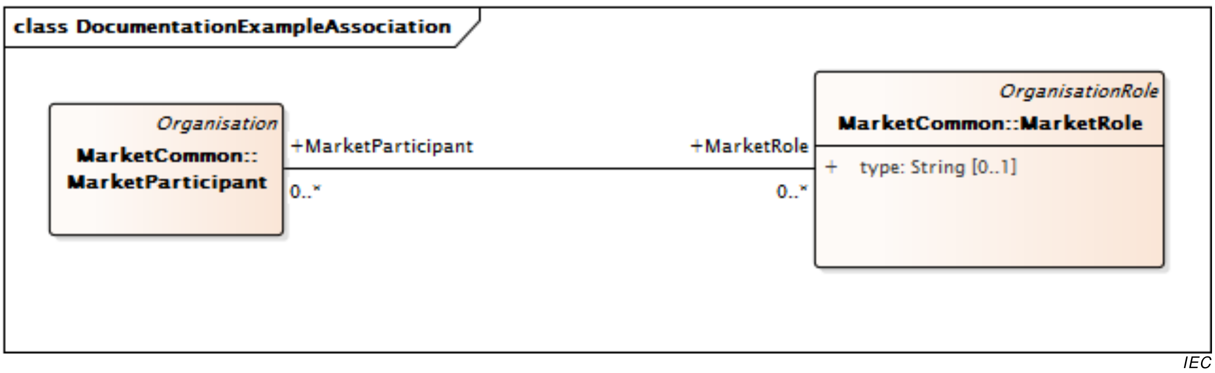


**Figure 4 – Example of simple association**

#### 4.3.4 Aggregation

Aggregation is a special case of association. Aggregation indicates that the relationship between the classes is some sort of whole-part relationship, where the whole class "consists of" or "contains" the part class, and the part class is "part of" the whole class. The part class does not inherit from the whole class as in generalization. Figure 5 illustrates an aggregation between the 'EnergyProfile' class and the 'EnergyTransaction' class, which is taken from the 'ExternalInputs' package. As shown, an 'EnergyProfile' can be a member of one 'EnergyTransaction' object, but an 'EnergyTransaction' object can contain any one or more of 'EnergyProfile' objects. In the context of using CIM as an information model, aggregation does not have a precise or formal interpretation beyond a simple association and is intended to visually assist in representing normal usage.

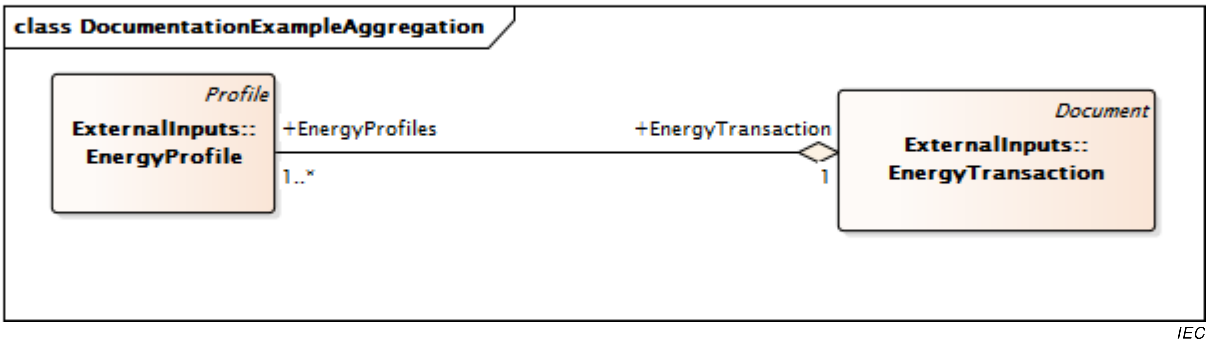


**Figure 5 – Example of aggregation**

### 4.4 CIM model concepts and examples

#### 4.4.1 'MarketCommon' package

The common market model describes the market participants and the role they are assuming in the market. A market participant could play several roles in a market, as shown in Figure 6.

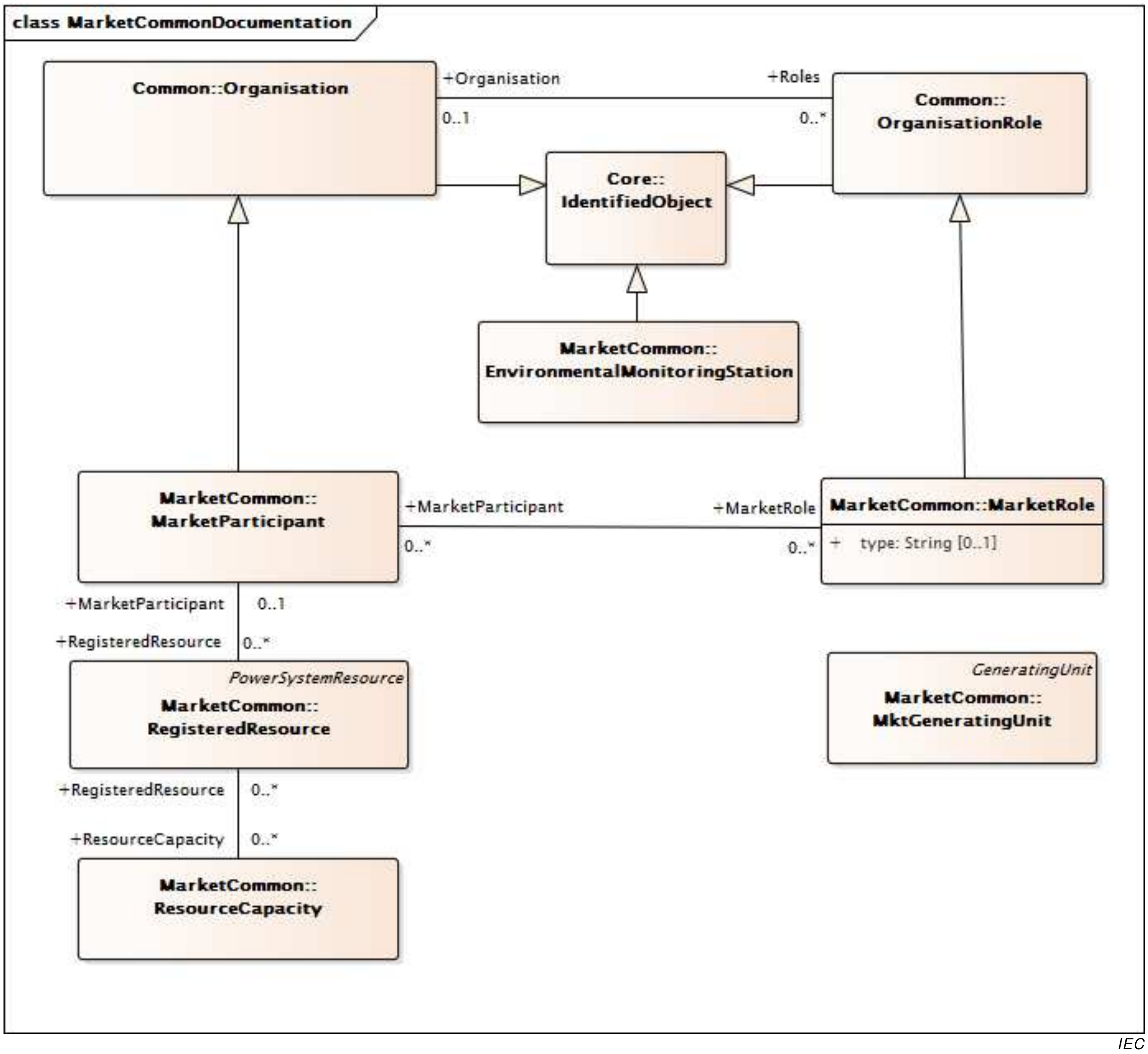


**Figure 6 – Common market model**

### 4.4.2 ‘MarketManagement’ package

The market management model describes a consistent set of classes to be used together with the common market model to generate a profile. The profile is to be used when the electricity market is mainly based on regulated Third Party Access, i.e. transmission system operators have to allow any electricity supplier non-discriminatory access to:

- the transmission network to supply customers
- the infrastructure that enables the exchange of energy by wholesale and retail transactions (bilateral or through a power exchange)

The ‘MarketManagement’ package refers to what is later called the European style market. Figure 7 shows an overview of the layered modeling framework that enables the specification of messages from the CIM down through the different layers. This layered modeling framework defines how the concepts of the 'MarketManagement' package originate from the CIM through the definition of a market profile that enables the generation of contextualized documents for information exchange.

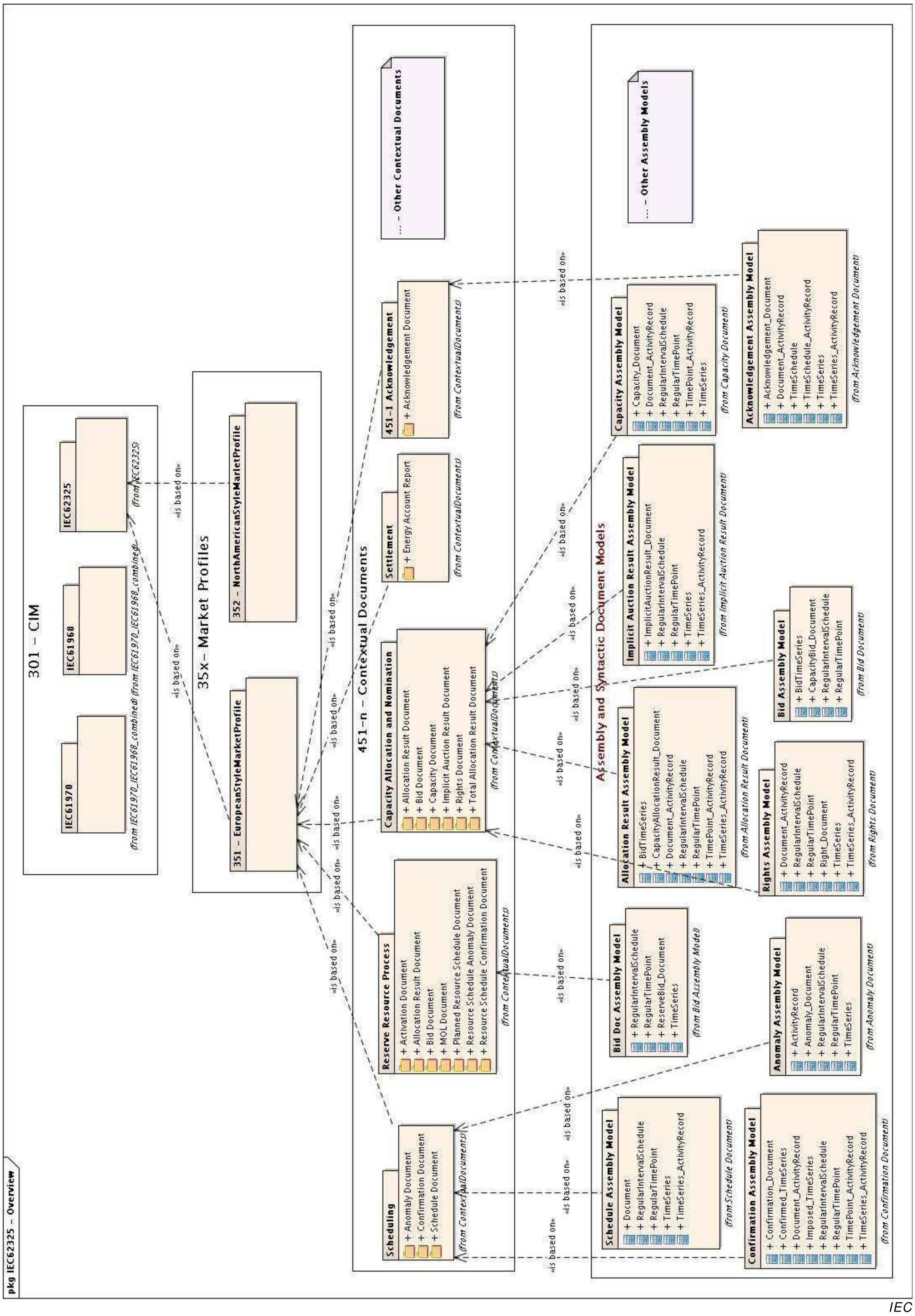

**Figure 7 – Market management model overview**

As shown in Figure 7, for each business process necessary to run an electricity market, a dedicated set of contextualized documents is provided. As indicated in Figure 7, the market profiles are specified in IEC 62325-351 and the contextual documents are described in IEC 62325-451-1, IEC 62325-451-2, etc.

Figure 8 shows the classes of the 'MarketManagement' package.

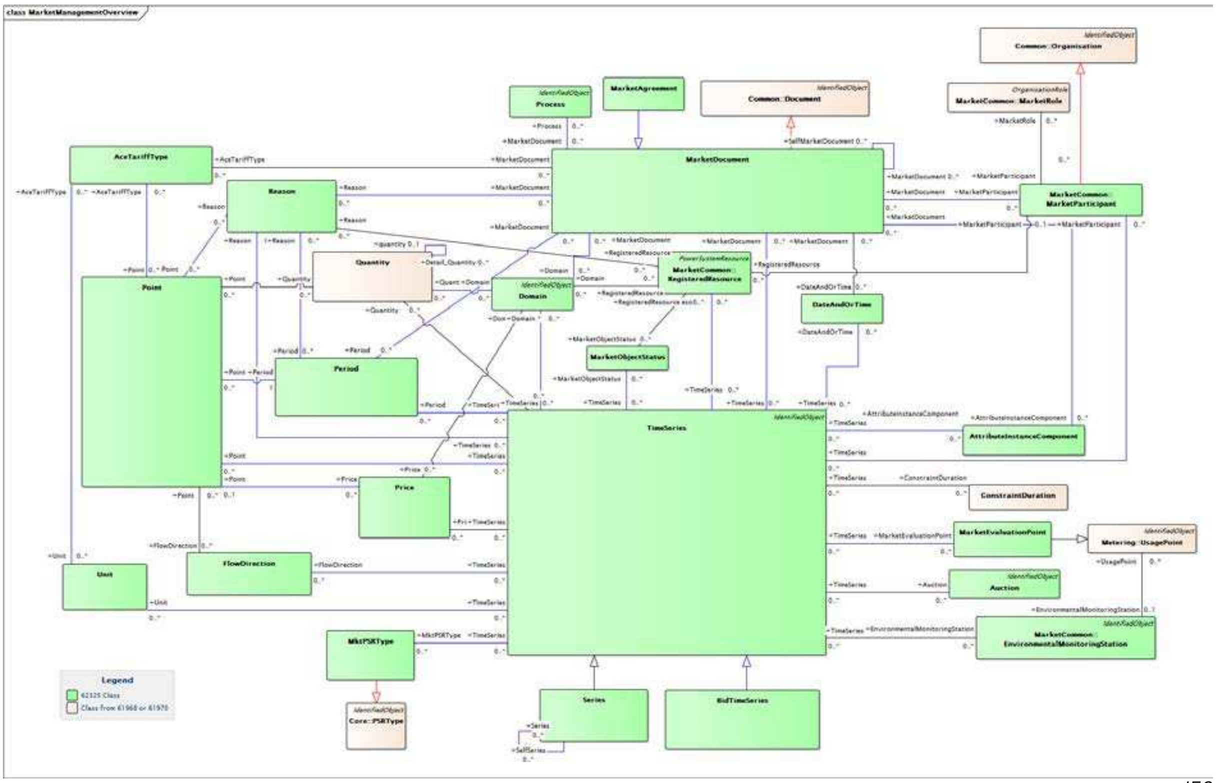


**Figure 8 – Market management model**

In this market management model a key role is given to the concept of 'MarketDocument', i.e. the transactions on the electricity market are based on contractual exchanges of information between market participants through a given set of documents depending upon the business process.

As shown in Figure 7, for each business process necessary to run an electricity market, a dedicated set of contextualized documents is provided. As an example, here is a non-exhaustive list of processes:

- the acknowledgment process: As the information exchange is based on contractual exchanges, it is required to have a functional acknowledgment for the exchanged document, and not only a technical acknowledgment;
- the scheduling process: This process describes the way of exchanging information related to schedules, i.e. generation schedules, load schedules, bilateral trade schedules, power exchanges trade schedules, etc.;
- the settlement process: This process describes how to exchange the information necessary to settle the electricity market, i.e. the comparison of the scheduled energy and the actual meters;
- the transmission capacity allocation and nomination process: This process describes the explicit and implicit auctions of transmission capacity for cross border trades. It includes also the secondary market of capacity rights, i.e. the resale of capacity rights;
- the reserve resource process: This process describes the way resources, i.e. the generation units and the dispatchable load, are scheduled, and capacity is auctioned and in particular the activation by the system operator of the tertiary reserve for balancing purpose;
- etc.

The ‘Process’ class (see Figure 8) enables the definition, for a given document, the process to which the information flow is directed. For example, the “schedule document” can be used in different processes such as “forecast”, “long term”, “day ahead”, “intra day” etc.

Within a ‘MarketDocument’, another important concept is. the ‘TimeSeries’ concept (see Figure 9) based on which market participants (i.e. ‘MarketParticipant’ class) playing a role (i.e. ‘MarketRole’ class) in a market business process (i.e. ‘Process’ class) can exchange schedules of volume/price related to an energy domain area (i.e. ‘Domain’ class) for a given business type (such as “internal trade”, “cross-border trade”, “primary reserve”, “secondary reserve”, “tertiary reserve”, etc.). These documents are the basis on which all exchanges are built in order to manage energy exchanges, bid, capacity allocation, energy reserve resources, and settlement.

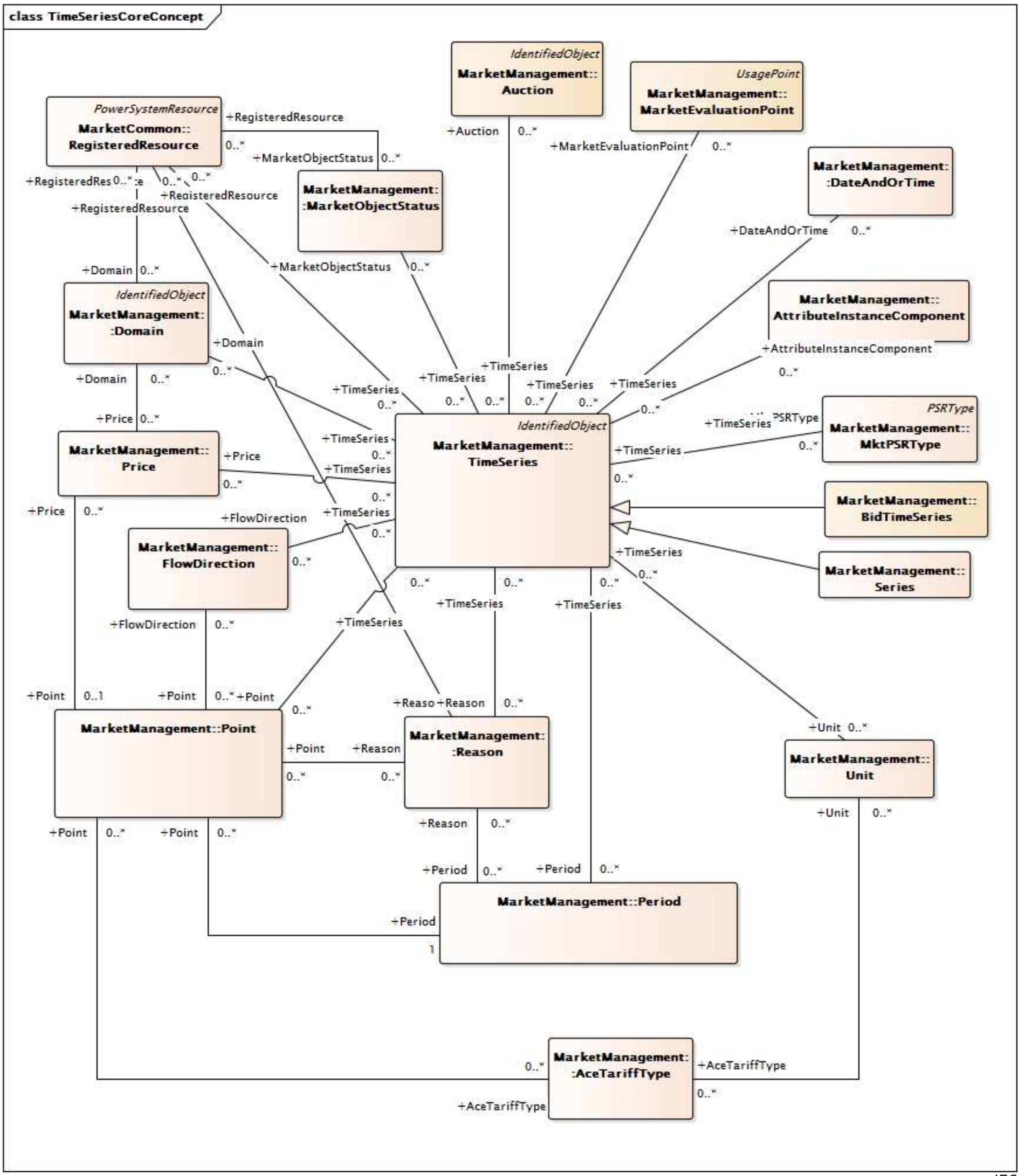

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**Figure 9 – ‘MarketManagement’ ‘TimeSeries’ core concept**

A specialization of the TimeSeries class is defined in order to provide an appropriate way to describe the configuration of a market object and the interrelation between these objects and market participants. This class is named Series.

To handle quantities and prices in a contractual way for the electricity market, attributes of type ‘Decimal’ have been introduced in the modeling process.

### 4.4.3 ‘MarketOperations’ package

#### 4.4.3.1 Market operations

The ‘MarketOperations’ package describes a consistent set of classes to be used together with the common market model (and other parts of the CIM described in IEC 61970-301 and IEC 61968-11) to generate a profile. The profile, in such a case, is to be used for North American style electricity market that is mainly characterized by day ahead unit commitment by a market operator, intraday and real time balancing through central dispatch, and settlement based on locational marginal prices (LMPs).

This kind of North American style electricity market also includes the auction of congestion revenue rights which are financial instruments that market participants purchase to hedge against congestion costs. Meter data management and billing and settlement are also included. The ‘MarketOperations’ package includes models to support these characteristics.

Subclauses 4.4.3.2 through 4.4.3.6 describe major concepts referenced in the ‘MarketOperations’ package.

#### 4.4.3.2 Market operator software systems

Figure 10 shows the software systems of a typical operator of North American style electricity markets that are referenced in the ‘MarketOperations’ package. The market operator may also be known as a regional transmission operator (RTO) or independent system operator (ISO). The systems are identified as follows:

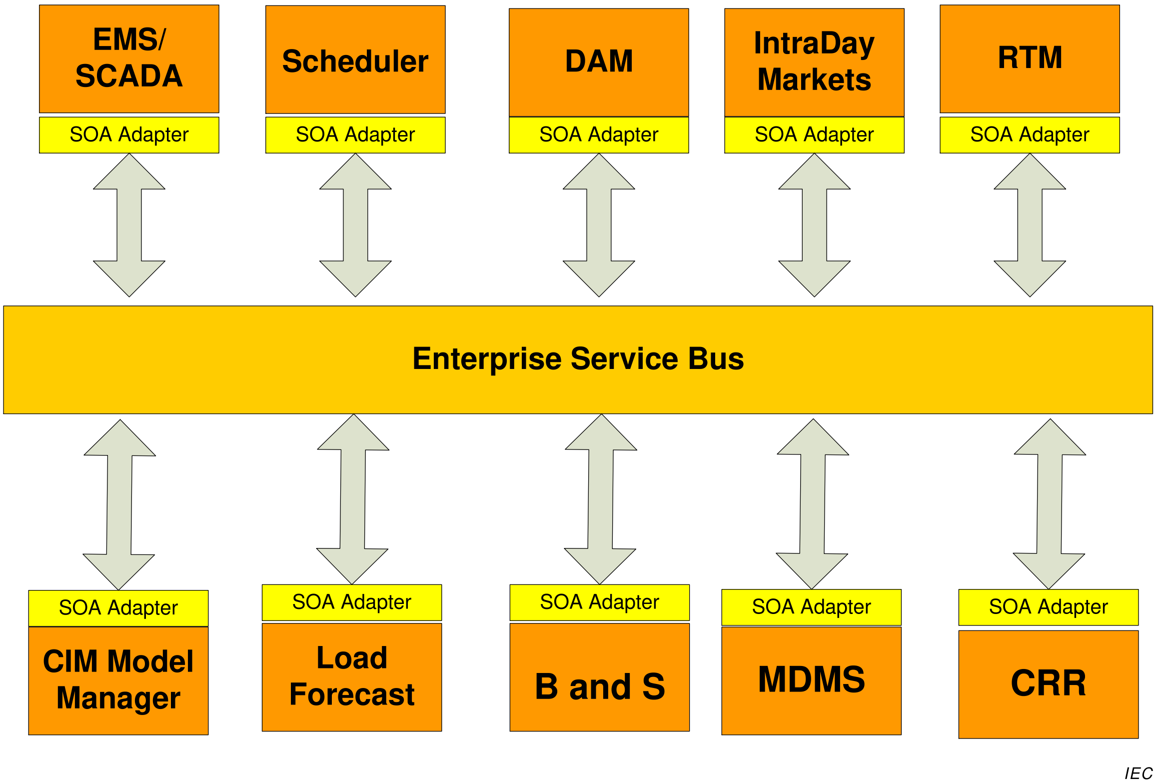


**Figure 10 – Market operator software systems for North American style electricity markets**

#### 4.4.3.3 Regional transmission operator interfaces

Figure 11 illustrates an overview of the UML interface model for a regional transmission operator (RTO) to support the interfaces illustrated in Figure 10. The major classes are briefly summarized: The classes 'AggregateNode' and 'Pnode' are used to model pricing nodes where locational marginal prices are averaged with a weighted average calculation. The class 'LocalReliabilityArea' is used to identify and model load pockets within the RTO. The class 'MktConnectivityNode' is used to extend the IEC 61970 'ConnectivityNode' class with market data. The classes 'TransmissionRightChain', 'ContractRight', and 'MSSAggregation' are used to model legacy contracts for use of the transmission system and to represent grandfathered transmission ownership rights. The classes 'Organisation' and 'MarketParticipant' identify the RTO. The classes 'SecurityConstraints' and 'SecurityConstraintSum' are used to model system constraints such as nomograms that approximate transient stability constraints. The classes 'HostControlArea', 'SubControlArea', and 'AdjacentCASet' are used to model data to support control area operations. Resource models and participating transmission owner data are included through associations with their associated classes. More details of the RTO model are included in Clause 6.

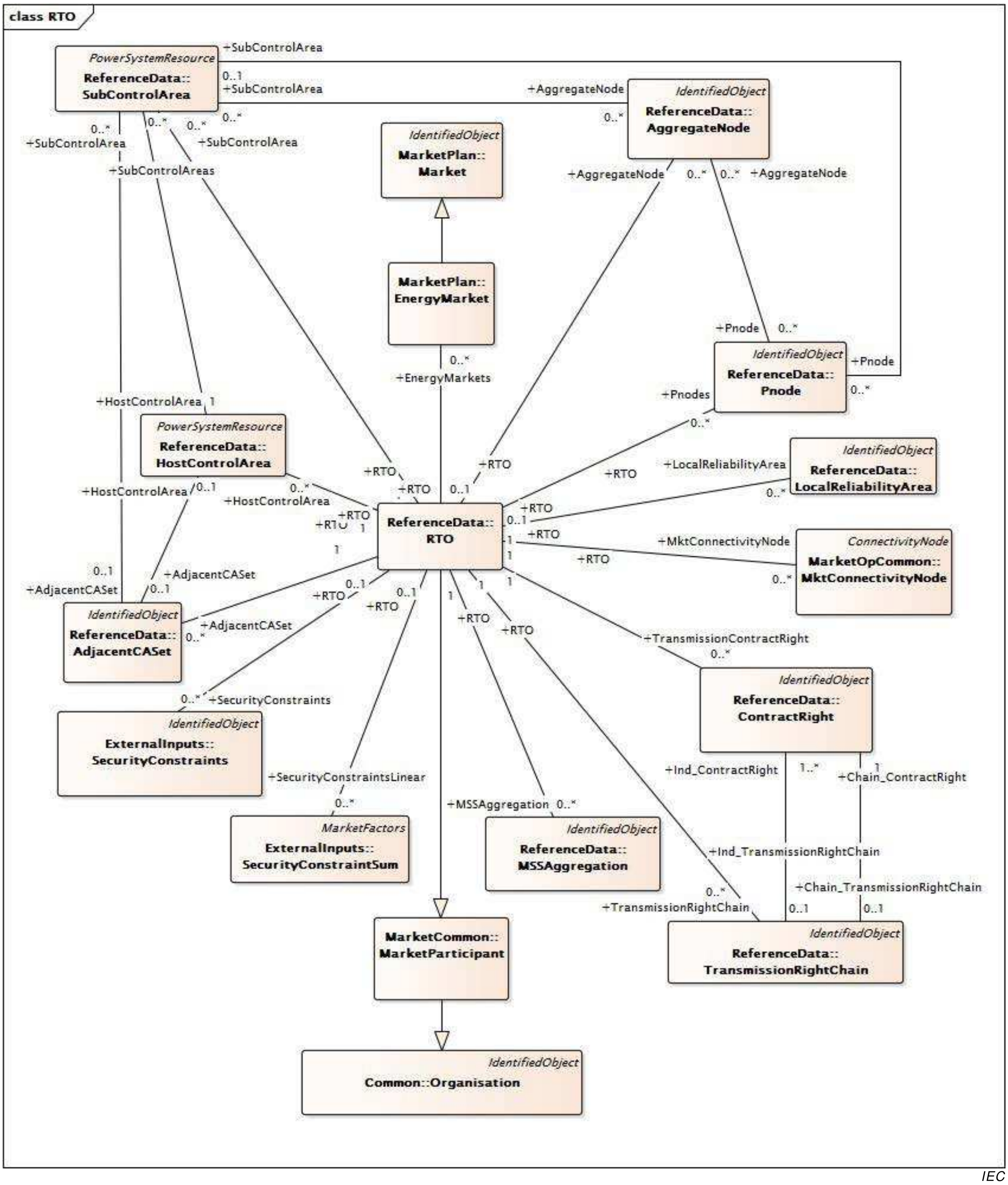


**Figure 11 – Regional transmission organization for North American style electricity market**

### 4.4.3.4 Resource model

Figure 12 illustrates a high level view of the UML model that supports the resource model data for North American style electricity market. The class ‘RegisteredResource’ is a parent class with the following subclasses: ‘RegisteredLoad’, ‘RegisteredInterTie’, ‘RegisteredGenerator’ and ‘RegisteredDistributedResource’. The ‘RegisteredResource’ class has associations with the classes ‘AggregateNode’, ‘Pnode’ and ‘MktConnectivityNode’ to support aggregated and physical resources. The associations to the ‘AggregateNode’ and to ‘AggregatedPnode’ (association is inherited from ‘Pnode’) support the aggregated resource definition on an aggregated node or aggregated pricing node basis. The associations to ‘IndividualPnode’ (inherited from ‘Pnode’) and ‘MktConnectivityNode’ support the physical resource definition on a pricing or connectivity node basis.

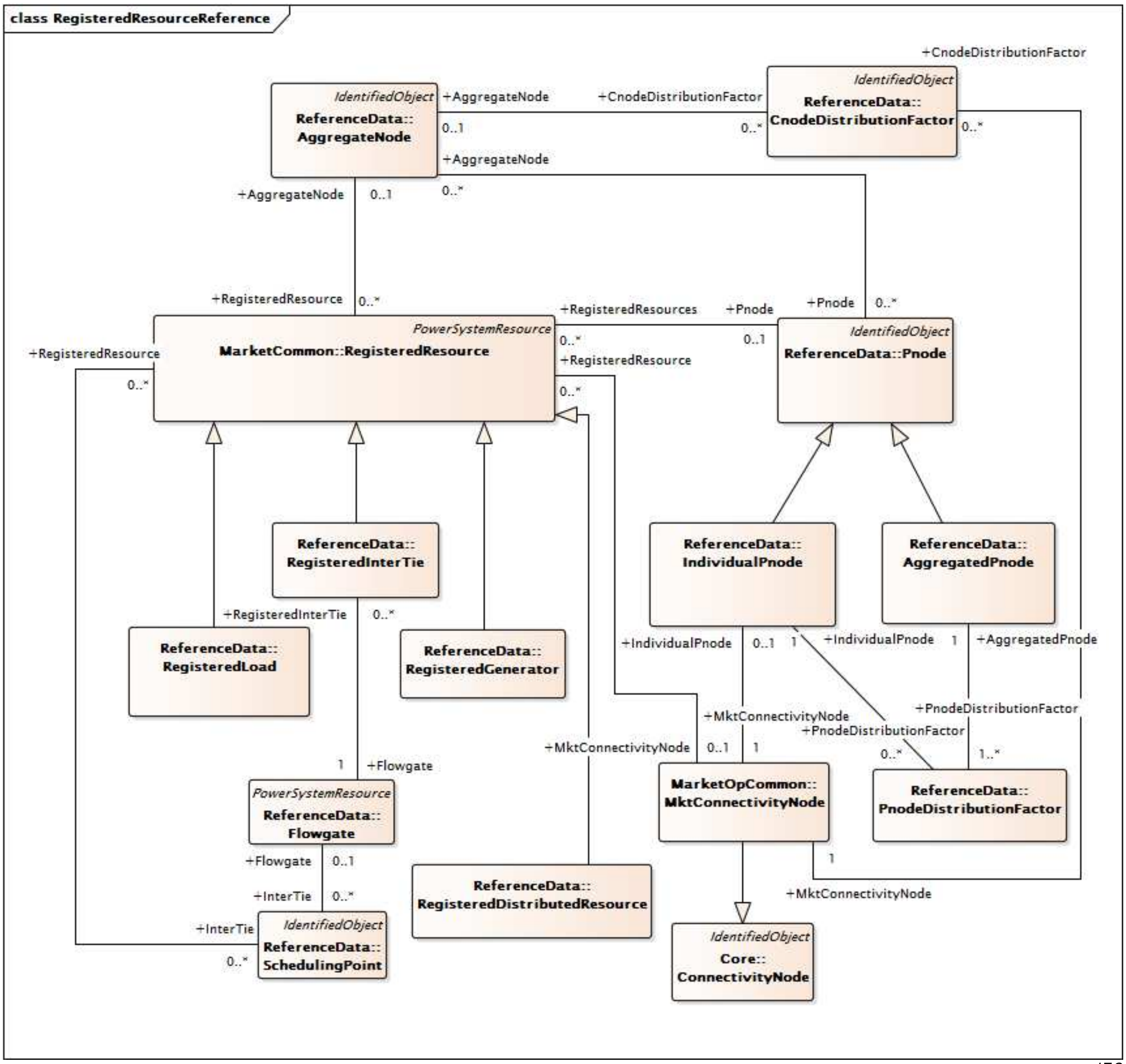


**Figure 12 – Registered resource reference definition for North American style electricity market**

### 4.4.3.5 Bid/Offer model

North American style electricity markets are based on offers to sell and bids to buy electricity products that are cleared by a market operator subject to network and resource constraints. Bids and offers include price quantity pairs and technical data related to the ability of the market participant to deliver the quoted products.

Figure 13 shows the overview of the classes and associations that are used to model offers and bids. The term bid is used to include offers to sell and bids to buy one or more electricity products. The ‘Bid’ class is a subclass of the ‘Document’ class from the ‘IEC 61968’ package. Bids are further classified as ‘ResourceBids’ or ‘TransactionBids’. ‘Resource’ bids are bids that are based on physical (or virtual) resources that are inside the footprint of the RTO and thus under the direct operational control of the RTO. ‘TransactionBids’ are bilateral agreements made between market participants that are reported to the RTO for inclusion as constraints in the market clearing. The RTO determines whether the bilateral agreements can be consummated while maintaining system reliability standards.

Bids are associated with market participants. Scheduling coordinators fulfil a specialized role of a market participant that submits bids on behalf of market participants. Each ‘Bid’ class must be associated to one ‘MarketParticipant’. This ‘MarketParticipant’ may be represented by a ‘SchedulingCoordinator’ as ‘SchedulingCoordinator’ inherits from ‘MarketParticipant’.

The bid class has a relationship to the ‘BidProduct’ class and to the ‘MarketProduct’ classes. These associations are used to model the submittal of bids for energy and ancillary services. A further association between the ‘Bid’ class and the ‘Market’ class indicates which market the bid is intended for (day ahead, real time, etc.).

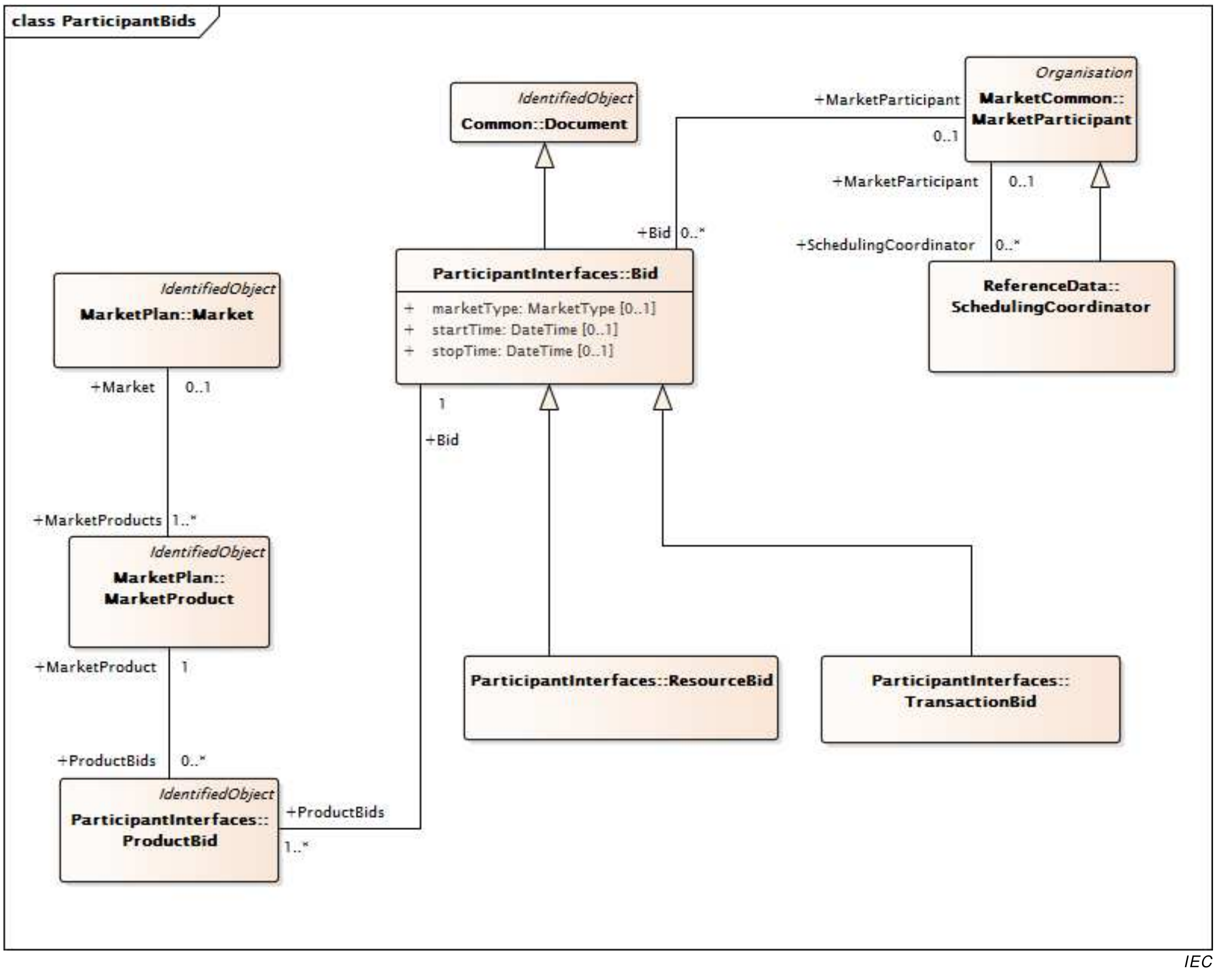


**Figure 13 – Bid definition for North American style electricity market**

Figure 14 shows further details of bids/offers.

The ‘Bid’ class is associated with a ‘ProductBid’ class, which in turn is associated with the ‘BidPriceSchedule’ class. The class ‘BidPriceSchedule’ defines bid schedules to allow a product bid to use specified bid price curves for different time intervals. This simplifies the modeling of bids that are made for multi-interval time horizons. The bid schedule is the set of consecutive intervals for which the bid is valid.

A bid may also be a self schedule, meaning that the market participant would like to operate the resource according to a certain (for example minimum) schedule. The market operator determines whether this resource can run with the submitted self schedule while system reliability criteria are met. These self schedules are settled at the LMPs determined during the market clearing. The class ‘Bid’ is associated with the class ‘ProductBid’, which is associated with the class ‘BidSelfSchedule’ to support the modeling of self schedules. The class ‘BidSelfSched’ is associated with the class ‘ContractRight’ which may be used to model the use of legacy contracts to support self schedules

This model also supports bids with part of the range of bid classified as a self schedule and part as regular bid.

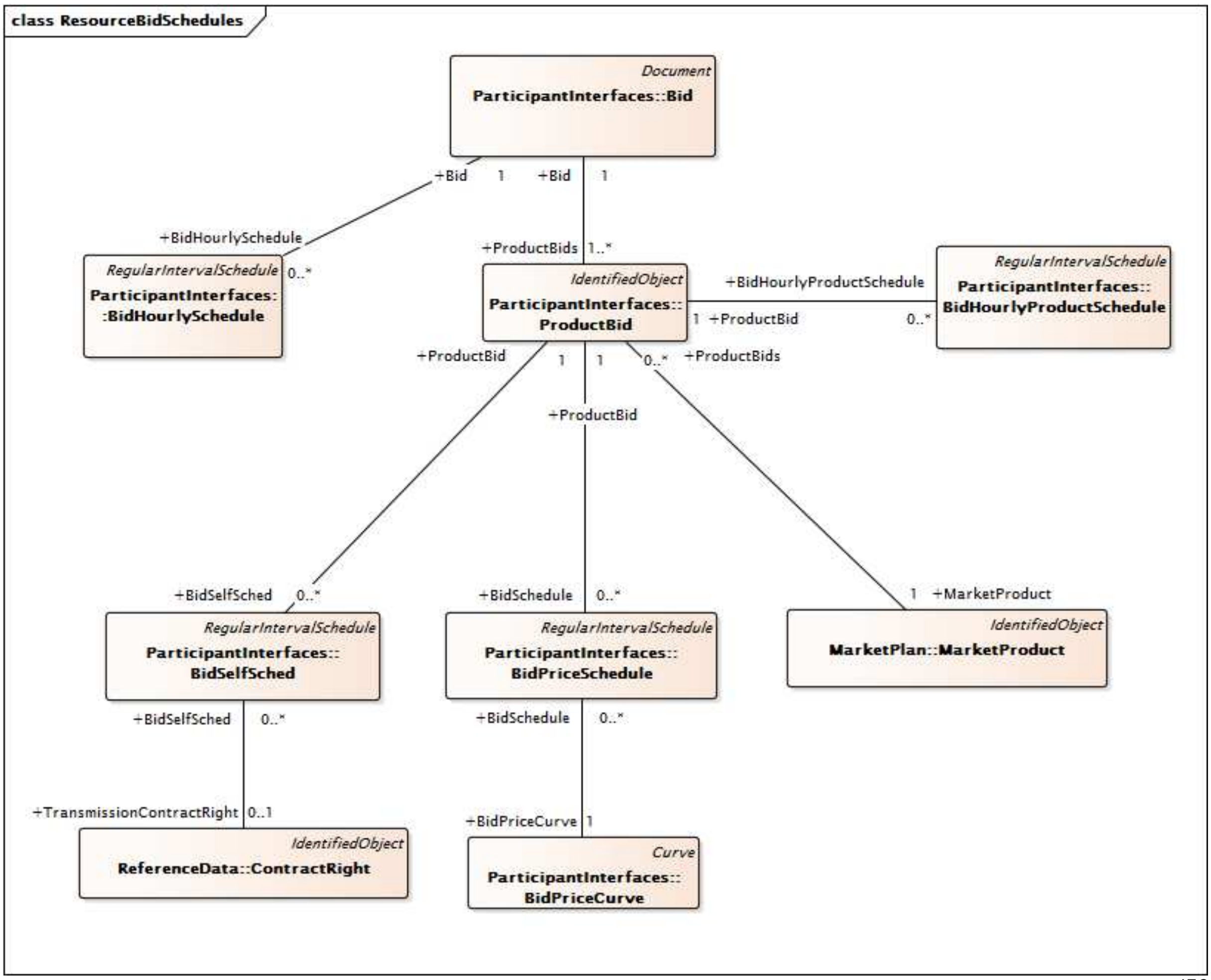


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**Figure 14 – Resource bid schedule definitions for North American style electricity market**

#### 4.4.3.6 Market clearing results

Figure 15 illustrates the major classes used to communicate the results of a market clearing run for a North American style electricity market. The class ‘Market’ is used to model the type of market (day ahead, real time, or intraday). The class ‘MarketFactors’ is used to model the market time horizon. The class ‘LossClearing’ is used to model electrical losses during the market horizon. The class ‘GeneralClearing’ is used to model the identity of the market interval. The class ‘PnodeClearing’ and its associations are used to model the cleared prices (locational marginal prices) of the run. The class ‘ResourceClearing’ is used to model market clearing results on a resource basis. The class ‘AncillaryServiceClearing’ is used to model the clearing results for ancillary services on a market region basis. The class ‘ResourceAwardClearing’ is used to model further details of the resource clearing. The class ‘ConstraintClearing’ is used to model the exchange of data on constraints which are binding constraints in the optimal market clearing solution.

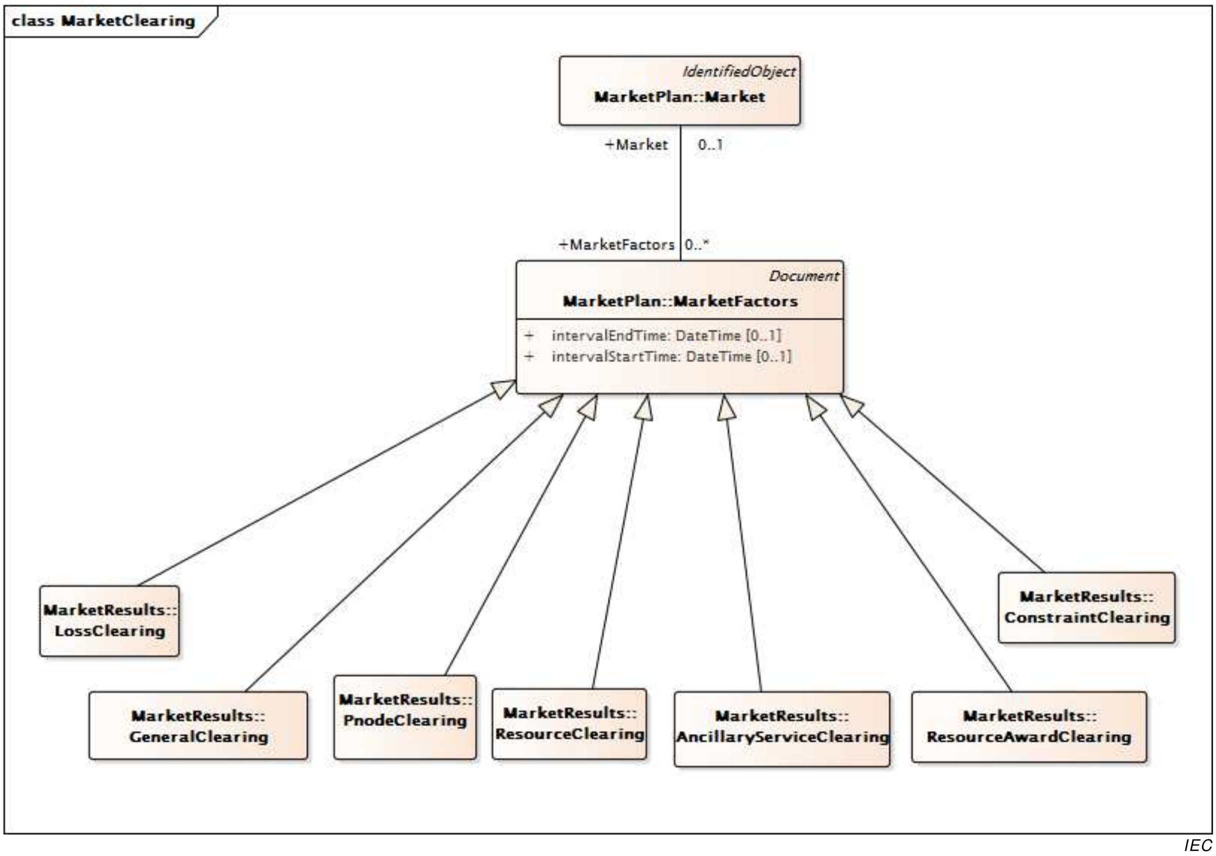


**Figure 15 – Market clearing for North American style electricity market**

### 4.4.3.7 Demand-Side Communication

Figure 16 illustrates the major classes used to communicate information between demand-side resources (demand response resources, storage resources, distribution generation resources, etc.) and the markets. The class 'RegisteredDistributedResource' is used to model the resource and is a new specialization of the 'RegisteredResource' class. The class 'ResourceCertification' is used to model different markets or services which the demand-side resource is qualified to provide; for example a demand response resource might be qualified to provide 10 MWh of Energy at 10 MW of power while a storage facility of 20 MWh might be qualified to provide 10 MW of up or down Regulation.

The 'MarketActualEvent' class has been extended to model demand-side events which have a targeted start and end time, along with different 'eventType' and 'eventStatus' indicators. The 'DistributedResourceActualEvent' class is a specialization of 'MarketActualEvent' that is used to capture system-wide demand response activity. Another important enhancement to the model is the extension of the 'Instructions' class to include a Dispatch Operating Delta should a relative amount of power be requested (as opposed to or in addition to an absolute power set-point, or Dispatch Operating Target, which was already in the model).

Performance of a demand-side resource is captured in the 'ResourcePerformanceEvaluation' class and may consist of one or more instances of a 'ResourcePerformanceTimeSeriesFactor' such as meter reads, baseline curves, correction curves and/or one or more instances of a 'ResourcePerformanceGlobalFactor' such as a temperature correction or load-offset. Performance over multiple events or dispatches is modeled in the 'ResourcePerformanceRating' class.

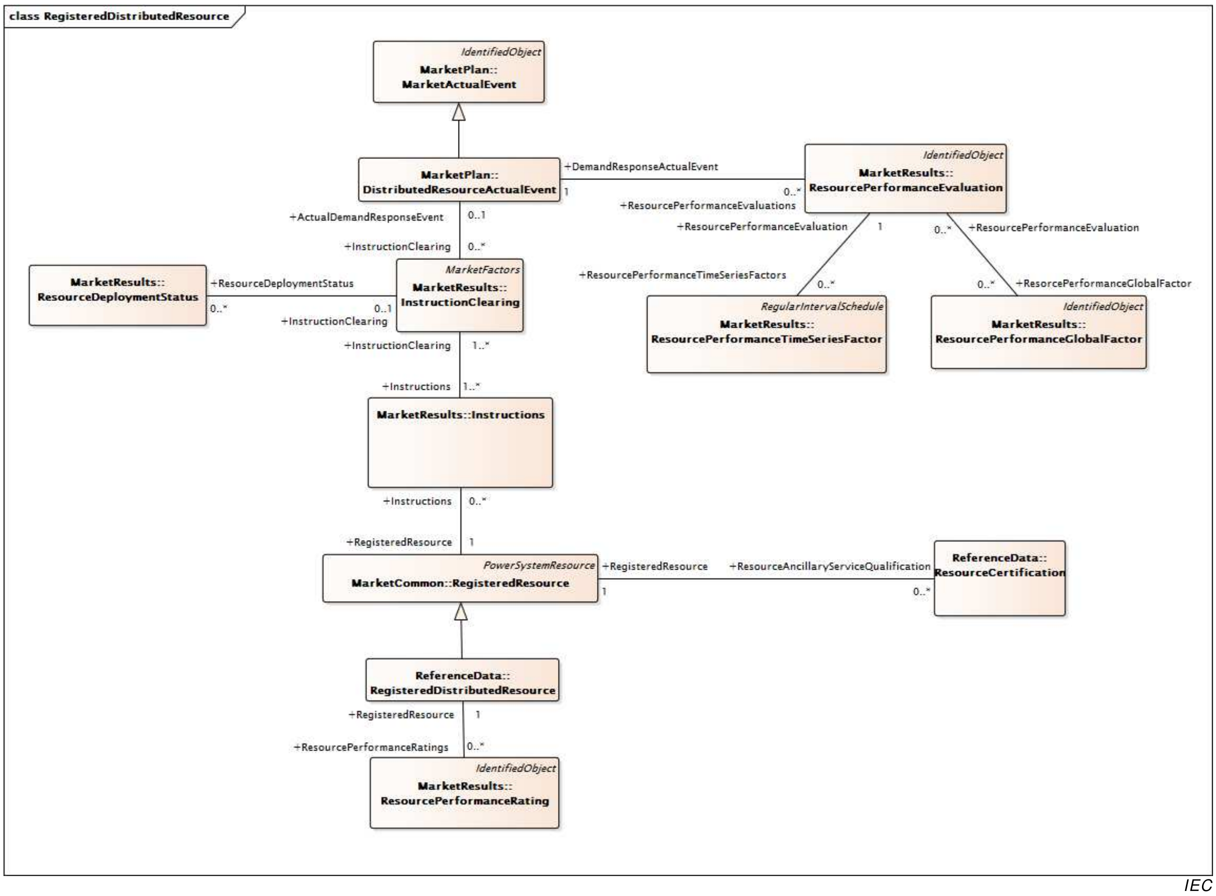


**Figure 16 – Demand Side Communication for North American style electricity market**

### 4.4.4 Environmental Data

Figure 17 provides an overview of the Environmental package, which contains a model of constructs typically used to describe environmental (weather) conditions including forecasts, observations, measurements, phenomena, alerts and events.

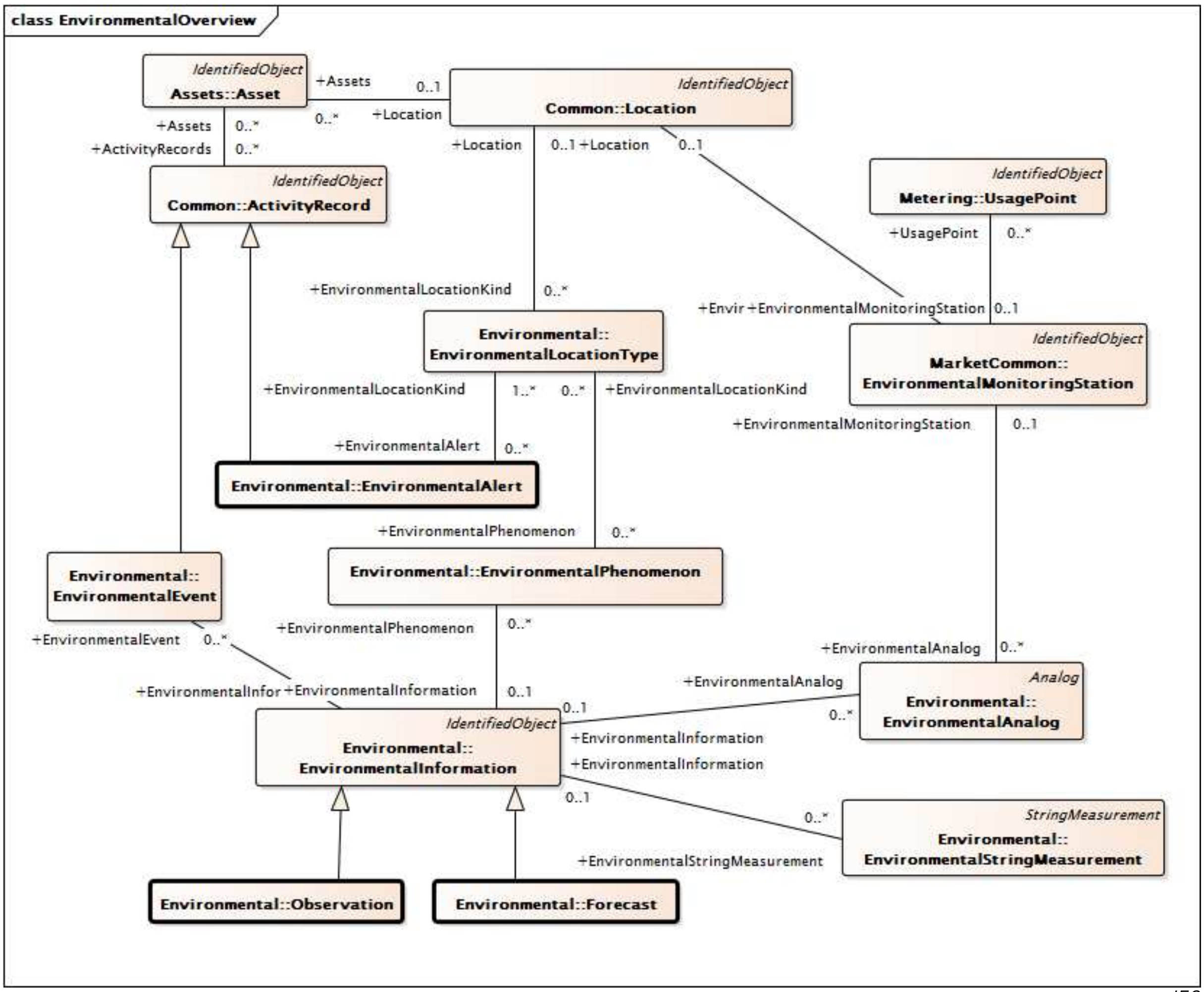


**Figure 17 – Main Environmental classes**

Environmental data is used by a diverse range of systems and applications within the power industry, from operations and asset management to customer billing and legal. It is vital to the behaviour of markets and activities as diverse as load and generation forecasting, storm restoration (crew pre-deployment, damage assessment, restoration), responding to environmental events (forest/grass fires, floods, tsunamis), emissions monitoring, geomagnetic induced current monitoring and accident claims investigation.

The classes of the Environmental package are intended to support the exchange of environmental data information inside a utility and to facilitate the association of environmental data with other utility enterprise information (location, asset, usage point) to which it might naturally relate. As illustrated in Figure 18, a Utility Environmental Data Service plays a key role in the use cases of environmental data exchange within the utility. The Environmental Data Service receives inputs from a variety of outside sources (in a variety of formats), stores and manages this data over time and provides it as required to a wide range of consuming applications at the utility.

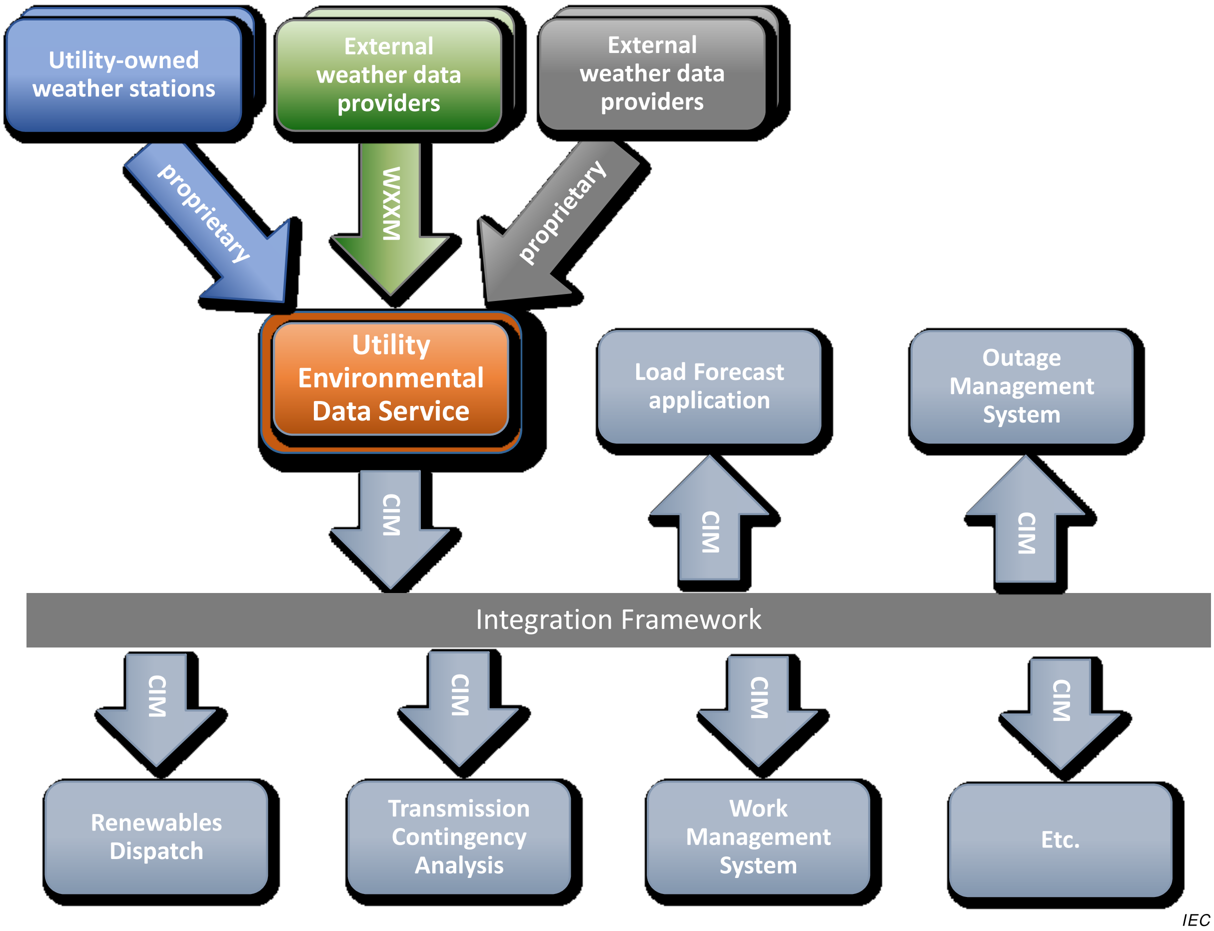


**Figure 18 – Utility Environmental Data Service**

As shown in Figure 19, the four high-level environmental data constructs are observation, forecast, alert and event. These constructs are represented by the 'Observation', 'Forecast', 'EnvironmentalAlert' and 'EnvironmentalEvent' classes respectively. 'Forecast' and 'Observation' are both child classes of 'EnvironmentalInformation' and the organization of data pertaining to each of them is identical.

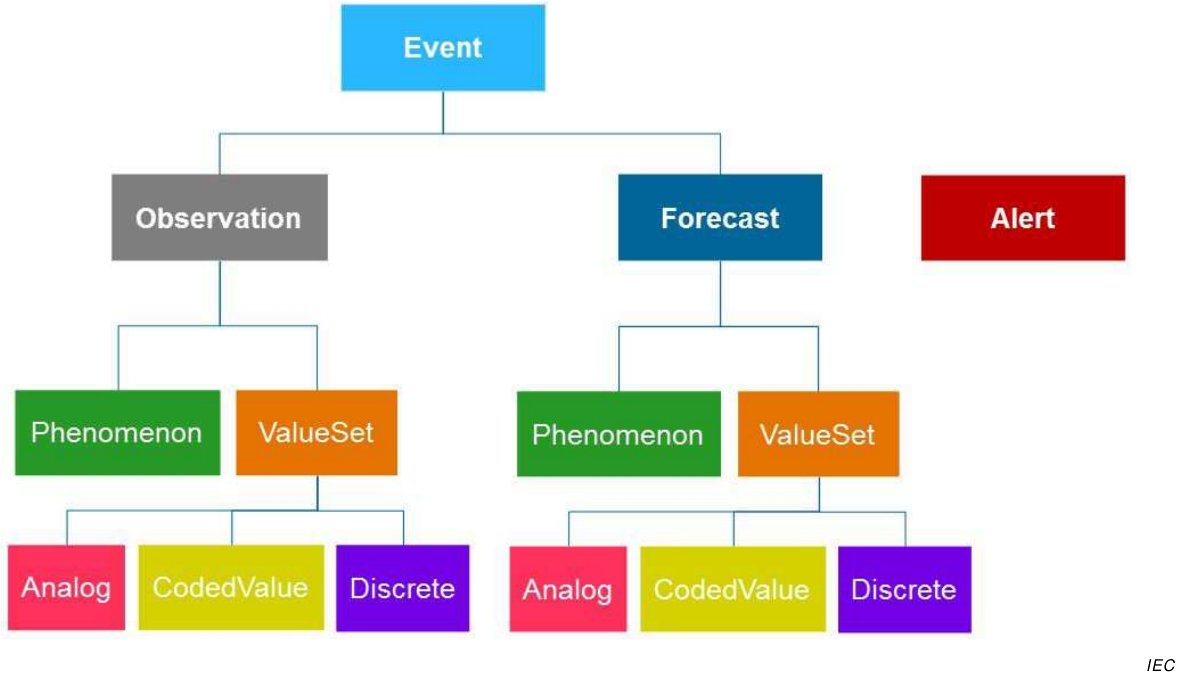


**Figure 19 – High Level Environmental Data Constructs**

As illustrated in Figure 20, each observation or forecast is created at a moment in time, can have a provider and contains any number of measurement values and phenomenon descriptions. Additionally, a forecast has a time for which it is valid. Locational information for forecasts and observations is provided at the measurement or phenomenon level.

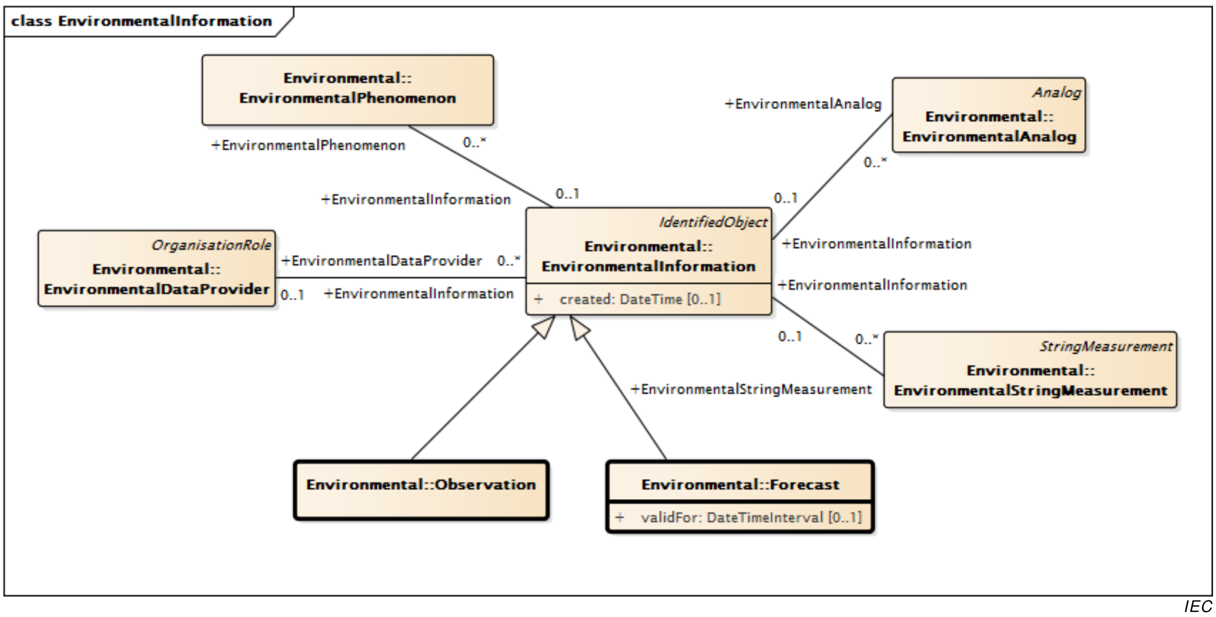


**Figure 20 – Forecasts and observations**

The measurement model for environment values is shown in Figure 21. Analogs associated with forecasts and observations are subclassed by the region in which they occur (atmospheric, geospheric, hydrospheric or space) and a kind attribute allows explicit definition of the value being measured (for example, atmospheric temperature, water surface temperature, humidity or irradiance). Analogs leverage the statistical calculation definitions provided by the ExtendedAnalog related classes (‘CalculationMethodHierarchy’, ‘CalculationMethodOrder’, ‘StatisticalCalculation’, and ‘PeriodicStatisticalCalculation’), which support clear definition of the meaning of an analog value. This allows, for example, an analog to be described as an ‘hourly peak of a 5 minute sampling of temperature’.

Environmental analog values use the usual IEC 61970 ‘AnalogValue’ class, while environment string values (used to express general weather conditions, like ‘heavy fog’) use a specialization of the IEC 61970 ‘StringMeasurementValue’ class.

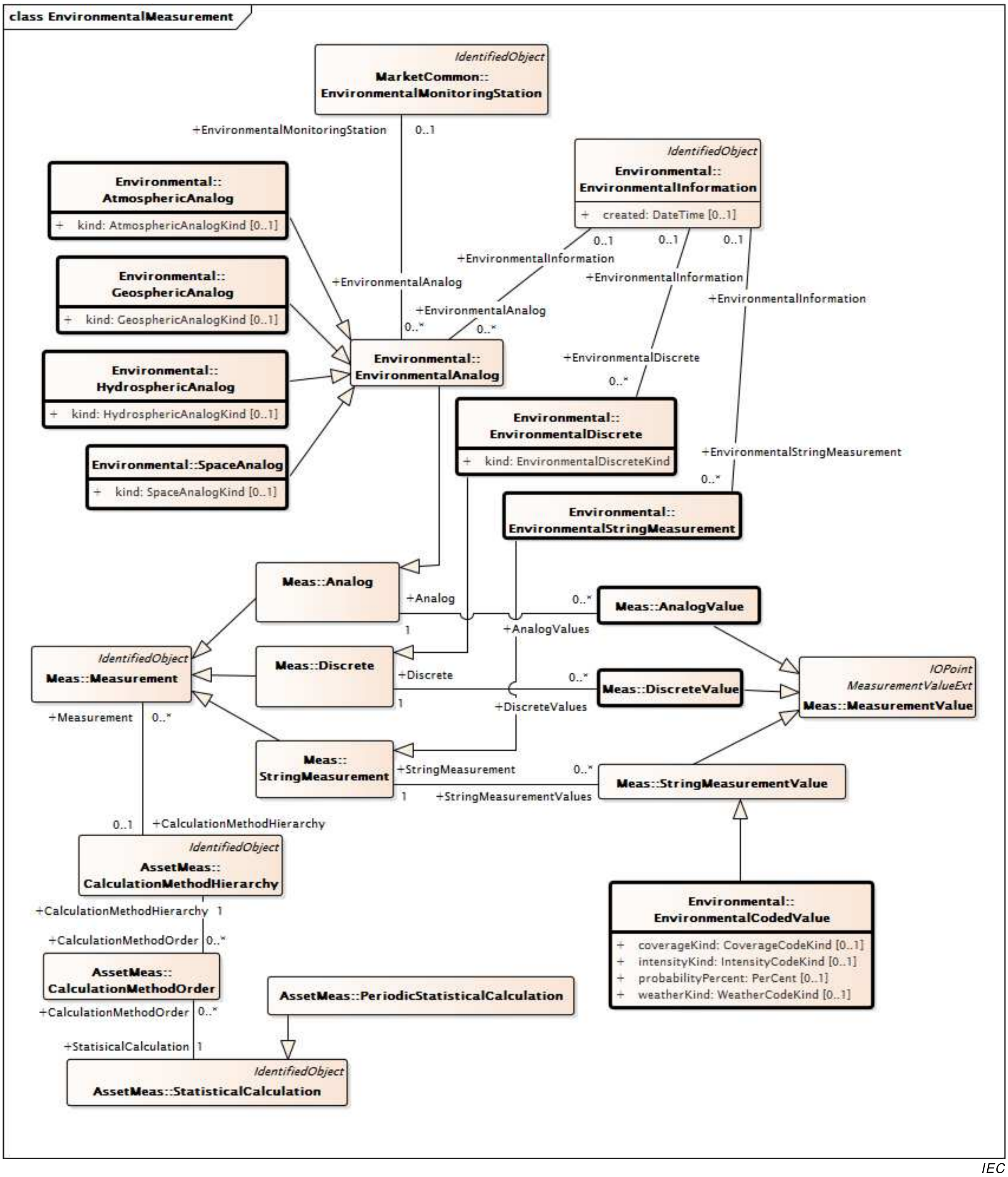


**Figure 21 – Environmental measurements**

Environmental monitoring stations, whose model is illustrated in Figure 22, can provide analog values for observations and have their capabilities defined by the ‘ReportingCapability’ class. The possibility that a monitoring station could be associated with a meter or customer is addressed by an association between ‘EnvironmentalMonitoringStation’ and ‘UsagePoint’ classes.

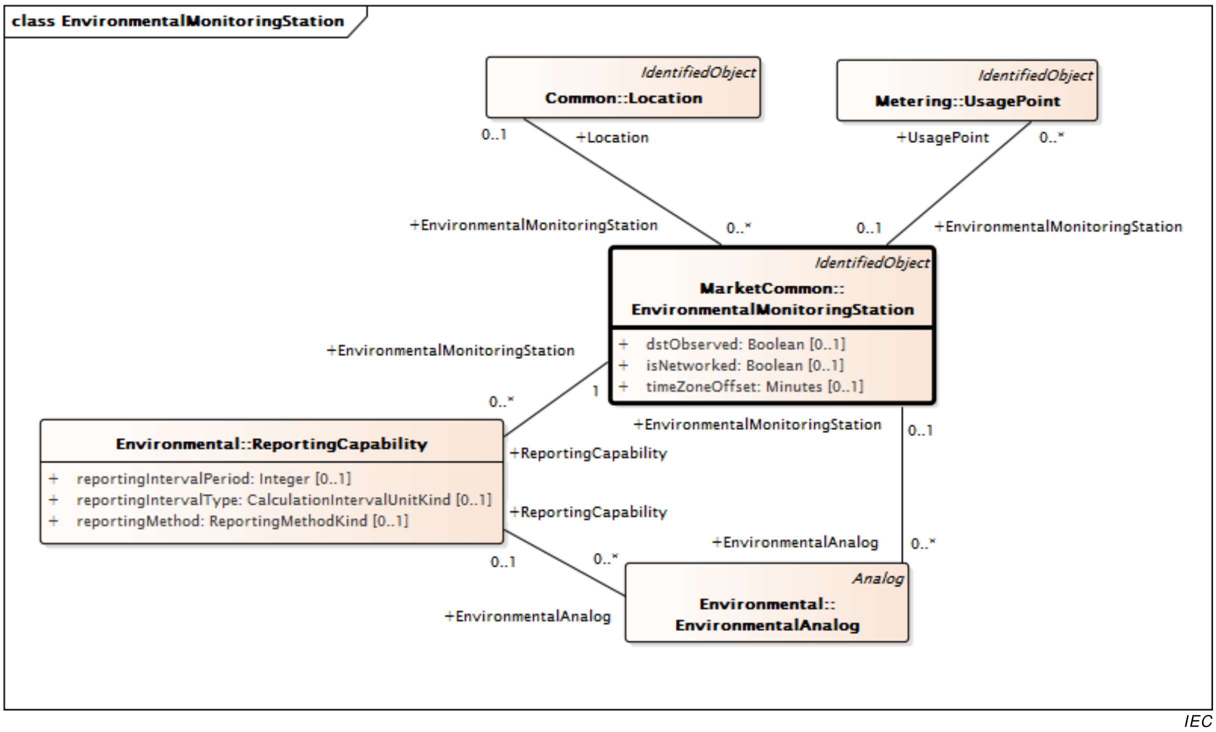


**Figure 22 – Environmental monitoring station**

Figure 23 overviews the modeling of environmental phenomena. The characteristics of environment phenomena (i.e., hurricanes, tornados, grass fires, floods, geo-magnetic space storms) at a moment in time are described by the ‘EnvironmentalPhenomenon’ class and its children, which are subclassed using the same regional groupings used for analogs (atmospheric, geospheric, hydrospheric, and space). The location of phenomena are often described by both a center and extent, and this is supported by means of the EnvironmentalLocationKind’ class.

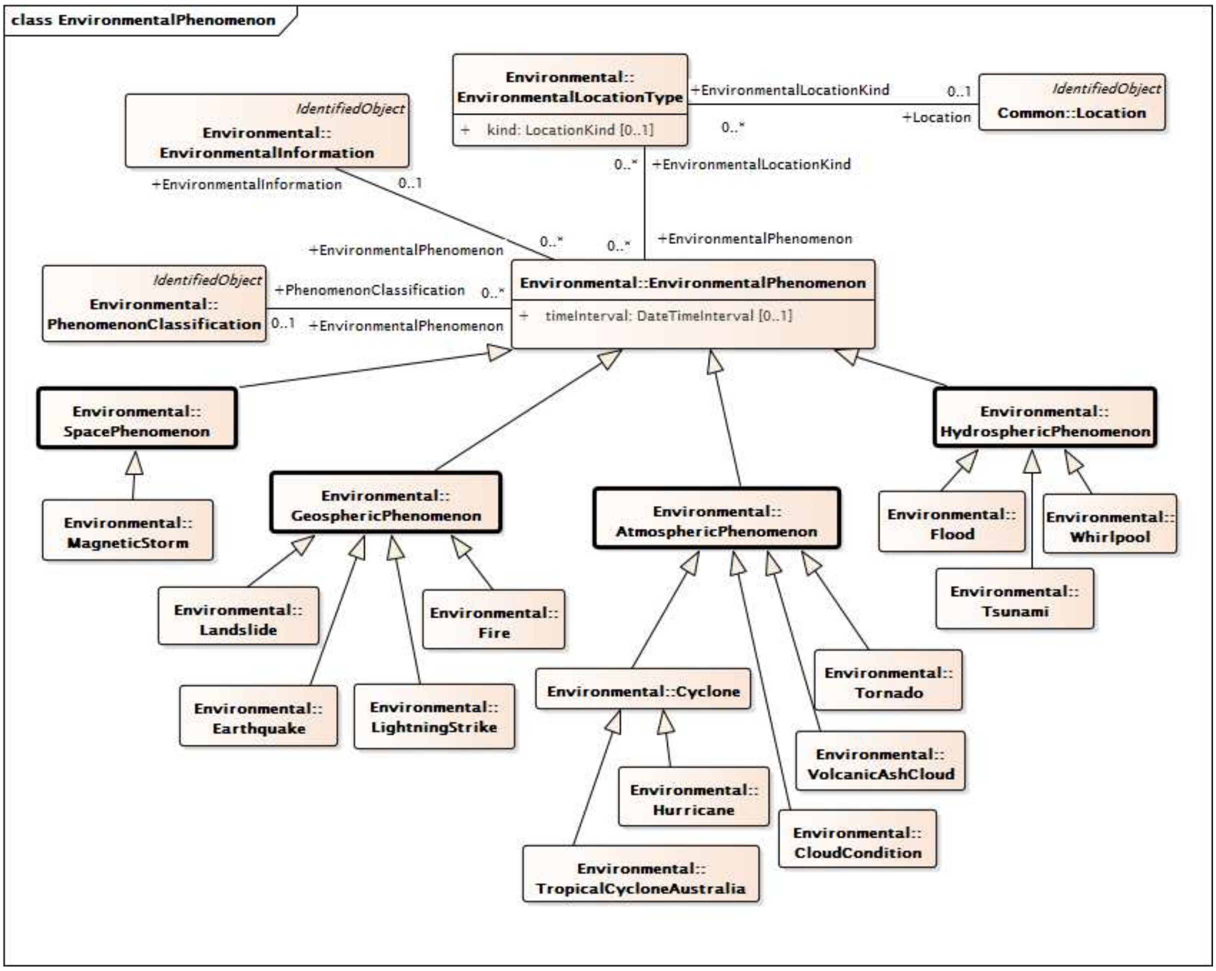


**Figure 23 – Environmental Phenomenon**

Additional definition of the severity of phenomena can be accomplished by using the ‘PhenomenonClassification’ class and associated classes, which allow an environmental data authority (for example a weather service or utility) to define threshold levels of measurement values associated with phenomenon classification conditions. These are shown in Figure 24.

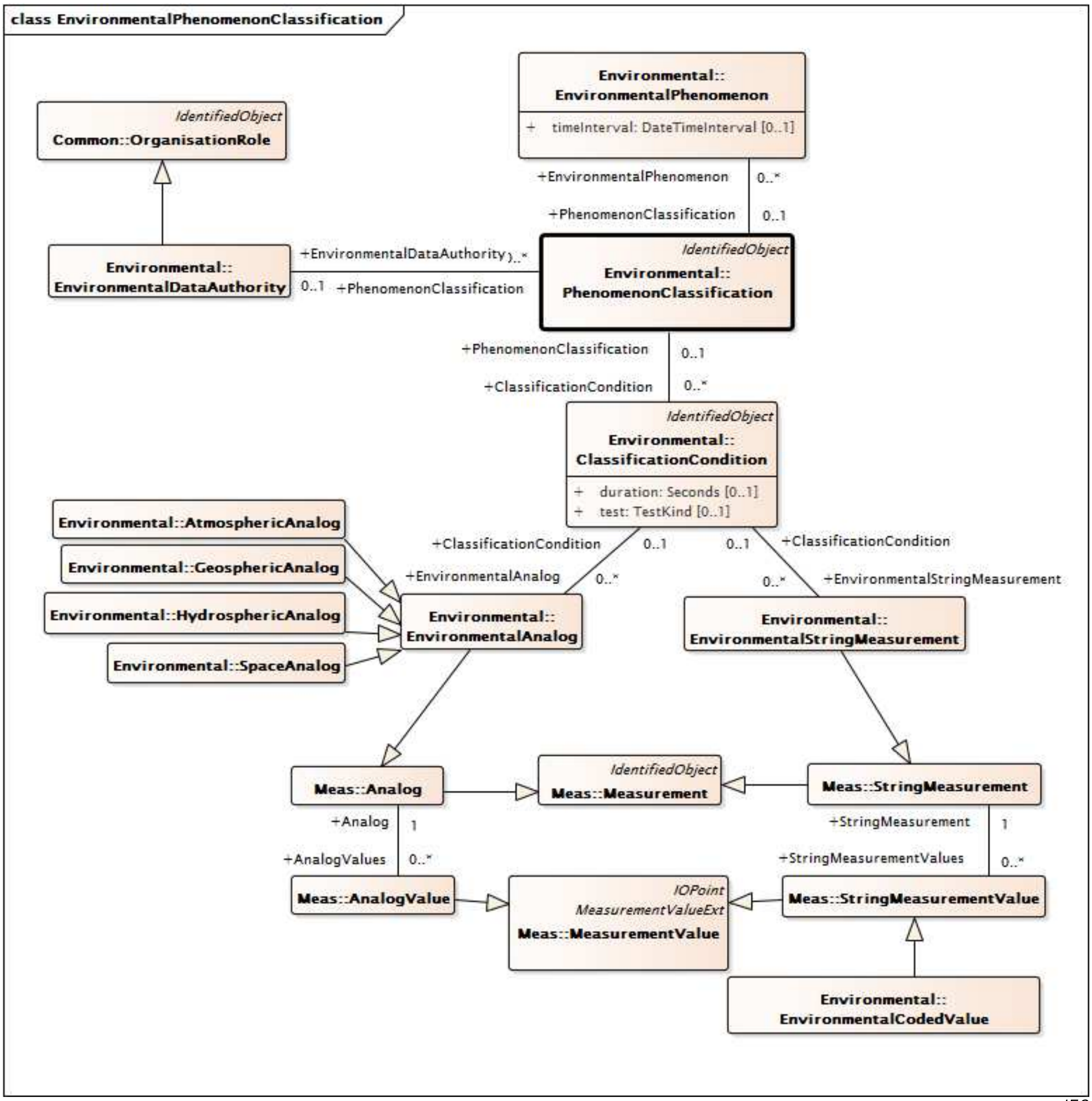


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**Figure 24 – Environmental Phenomenon classification**

Figure 25 illustrates the modeling of environmental events. Major events, like hurricane Katrina or the 2008 Sichuan earthquake, are represented by the EnvironmentalEvent class and can be described through time by means of multiple forecasts and/or observations (‘EnvironmentalInformation’ child class instances) each related to the environmental event and each including one or more phenomenon ‘snapshots’ (‘EnvironmentalPhenomenon’ child class instances).

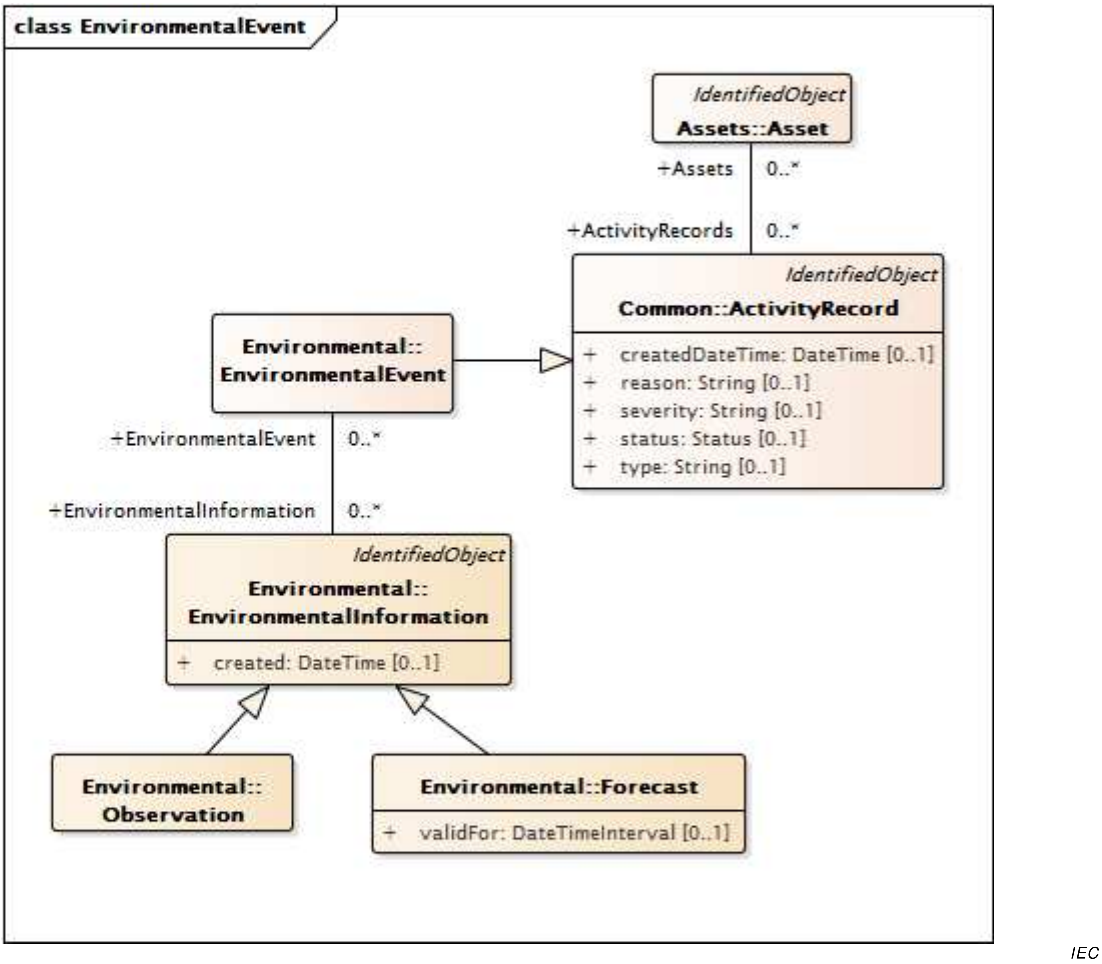


**Figure 25 – Environmental Event**

Environmental alerts are illustrated in Figure 26. Environmental alerts, like tornado watches and warnings, are described by the ‘EnvironmentalAlert’’ class, which supports the definition of alert provider, alerting authority, and alert location. Alerting authorities (like the US National Weather Service) periodically update the types and definitions of alerts they can issue and the ‘AlertTypeList’ class allows the specification of which version of alert definitions is being used. Some alerts have primary and secondary locations to which they apply and the ‘EnvironmentalLocationKind’ class allows this distinction to be made.

In addition to supporting the identification of major, named environmental events (like fires or tsunamis), the ‘EnvironmentalEvent’ class also supports the definition of smaller, local environmental events (like localized flooding) and allows them to be related to assets.

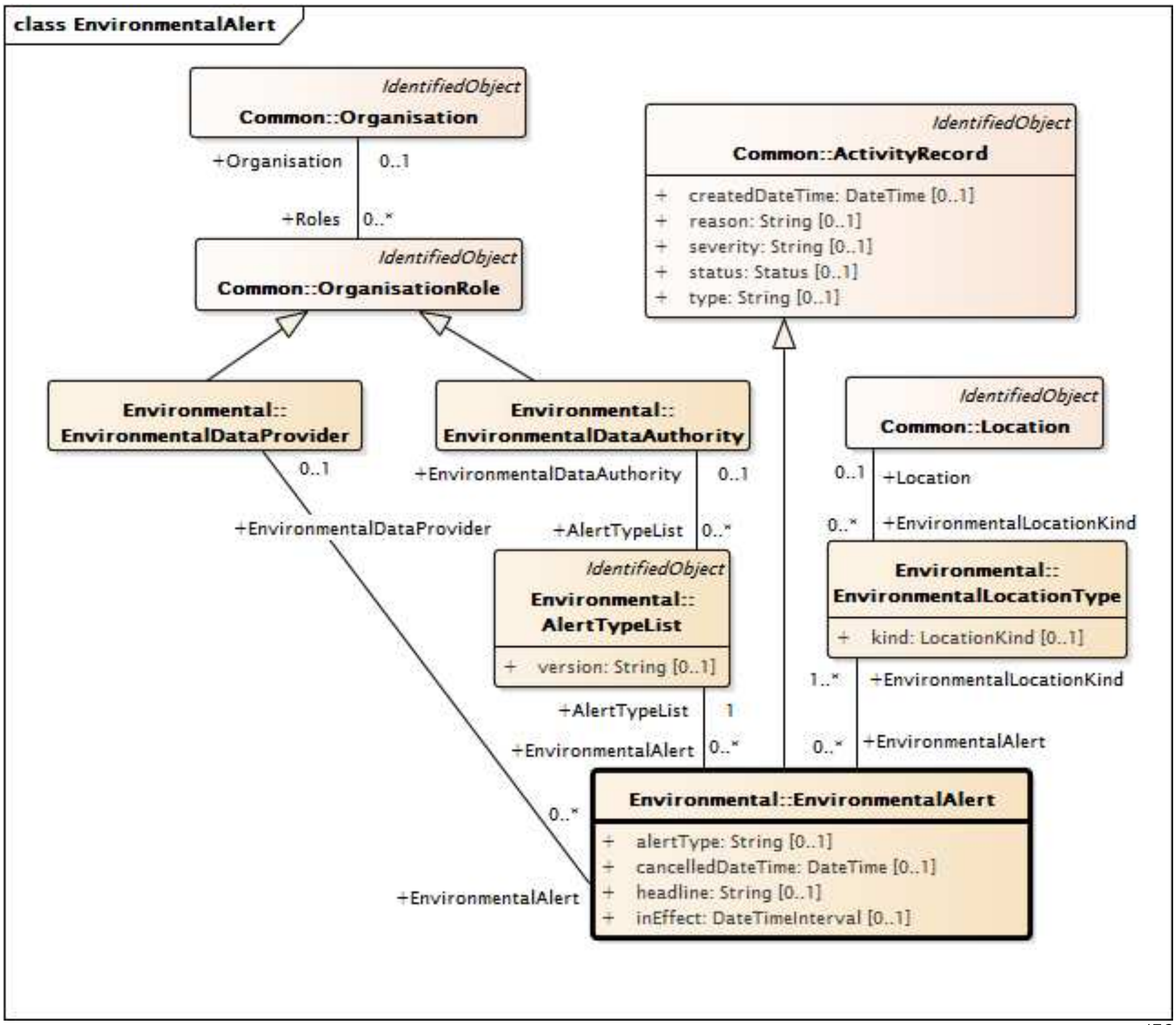


**Figure 26 – Environmental alert**

## 4.5 Modeling guidelines

### 4.5.1 Modeling for change

IEC 62325-301 defines the CIM along with IEC 61970-301, IEC 61970-302 and IEC 61968-11.

### 4.5.2 Process for amendments to the CIM

It may be desirable to amend the CIM to either revise the existing model or to extend the CIM to model additional elements of an electric utility power system. The recommended process for such amendments is as follows.

a) Prepare a use case(s) to describe the desired changes. This should include proposed changes to the appropriate class diagrams showing new/revised classes, attributes, and associations.
b) The use case(s) is then reviewed by the appropriate working group to decide if the requested changes should be treated as revisions to the current CIM standard, or if they should be treated as private amendments, not requiring a change to the standard itself.
c) Proposed amendments accepted by the working group will be added to a list of outstanding issues, and at the appropriate time, a new version of the CIM model will be prepared and an update made to the appropriate IEC CIM specification.

#### 4.5.3 Changes to the CIM UML model

From a modeling perspective, when the CIM is extended, the approach is to start with the existing CIM UML model. The extensions may be added in any of several ways that are available in UML, but in all cases the approach is to inspect the current model and determine the best way to build off of the existing class diagrams. The extensions may take the form of any of the following, starting from the simplest to the most complex:

- adding additional attributes to existing classes;
- adding additional associations between two existing classes;
- adding new classes that are specializations of existing classes;
- adding new classes via associations with existing classes.

The main objective is to reuse the existing CIM to the maximum extent possible. From a packaging point of view, extensions should be made to existing packages where possible. If the extensions comprise a new domain of application, then consideration should be given to creating a new package for the additions, but still creating the necessary associations to the existing package, keeping in mind that even though a new package is being created, the CIM is still a single model.

#### 4.5.4 Changes to the CIM standards documents

From a documentation perspective, when the CIM is extended, a decision shall be made whether the changes constitute updates to existing CIM standards documents, or whether a new part 3xx specification is required.

#### 4.5.5 CIM profiles

An implementation of the CIM need not include all classes, attributes, or associations in the standard CIM specification to be compliant with the CIM standard. Profiles may be defined to specify which elements shall be included (i.e., mandatory elements) in a particular use of the CIM, as well as which are optional. The CIM for Market profiles are defined in the Part IEC 62325-35X and -45X series and shall be based on the modeling rules defined in IEC 62325-450.

In IEC 62325-450, the definition of CIM profiles follows a layered modeling framework to go from the CIM information model down to the specification of messages based on CIM concepts through the definition of different regional contextual models and their subsequent contextualized documents for information exchange.

A full discussion of the rules for IEC 62325 series profiles is presented in IEC 62325-450. Please refer to IEC 62325-450 for a complete explanation and a full description of these rules.

### 4.6 Modeling tools

The entire CIM UML model exists as an Enterprise Architect project file and is viewable with that tool, including the class diagrams and descriptions of classes, attributes, types, and relationships. Viewing the CIM in this fashion provides a graphical navigation interface that permits all CIM specification data to be viewed via point-and-click from the class diagram in each package. A free viewer is available through Sparx Systems.

Ideally, the CIM information model is independent of any specific UML tool, though experience has shown that exchanges between different tools are often less than perfect. Until tool interoperability is proven effective, future changes to the CIM specification, resulting in new versions of this standard, will be incorporated first into the Enterprise Architect project description to ensure a single source for the CIM model data.

# 5 Detailed model

## 5.1 Overview

The common information model (CIM) represents a comprehensive logical view of market operations system information. This definition includes the public classes and attributes, as well as the relationships between them. Subclause 5.2 describes how Clause 6 is structured. Clause 6 is automatically generated from the CIM model.

## 5.2 Context

The CIM is partitioned into subpackages. Classes within the packages are listed alphabetically. Native class attributes are listed first, followed by inherited attributes in order of depth of inheritance, then by attribute name. Native associations are listed first for each class, followed by inherited associations in order of depth of inheritance, then alphabetically by class name, then alphabetically by association end name. The associations are described according to the role of each class participating in the association. The association ends are listed under the class at each end of an association.

For each package, the model information for each class is fully described in attribute and association end tables. Attribute and association end information for native and inherited attributes is listed as shown in Table 1 and Table 2. For any inherited attributes or association ends the “description” column will contain text indicating the attributes is inherited from a specific class. The description column for native attributes and association ends contains the actual description.

**Table 1 – Attribute documentation**

| name | type | description |
|---|---|---|
| native1 | Float | A floating point native attribute of the class is described here. |
| native2 | ActivePower | Documentation for another native attribute of type ActivePower. |
| Name | Float | Inherited from class IdentifiedObject |

As shown in Table 1, in an attribute table, in some cases, an attribute is a constant, in which case the phrase “(const)” is added in the name column of the attribute table. In such cases, the attribute normally has an initial value also which is preceded by an equal sign and appended to the attribute name.

**Table 2 – Association ends documentation**

| [mult from] | [mult to] name | type | description |
|---|---|---|---|
| [0..*] | [0..1] PSRType | PSRType | PSRType (custom classification) for this PowerSystemResource. |
| [0..1] | [0..*] Measurements | Measurement | The Measurements that are included in the naming hierarchy where the PSR is the containing object |
| [1..1] | [0..*] OperatingShare | OperatingShare | The linkage to any number of operating share objects. |
| [0..*] | [0..*] PsrLists | PsrList | |
| [1..1] | [0..1] OutageSchedule | OutageSchedule | A power system resource may have an outage schedule |
| [0..*] | [0..*] ReportingGroup | ReportingGroup | Reporting groups to which this PSR belongs. |
| [0..1] | [0..*] DiagramObjects | DiagramObject | inherited from: IdentifiedObject |
| [1..1] | [0..*] Names | Name | inherited from: IdentifiedObject |

As shown in Table 2, in an association ends table, the first column describes the multiplicity at the other end of the association (i.e., the multiplicity of the class for which association ends are being described). The multiplicity of the association end itself is included in brackets. The association end name is listed in plain text. The class at the other end of the association is listed in the “type” column. A multiplicity of zero indicates an optional association. A multiplicity of “*” indicates any number is allowed. For example, a multiplicity of [1..*] indicates a range from 1 to any larger number is allowed.

In the case that a class is an enumeration, the attributes table is replaced by the enums table as shown in Table 3, since enumeration literals have no type. There are no inherited enumeration literals for an enumeration class.

**Table 3 – Enums documentation**

| literal | description |
|---|---|
| native1 | This is the first native enumeration value. |
| native2 | This is the second native enumeration value. |
| native3 | There are typically no inherited attributes for enumerations. |

# 6 Top package IEC62325

## 6.1 General

The IEC 62325 subpackages of the CIM are developed, standardized and maintained by the IEC.

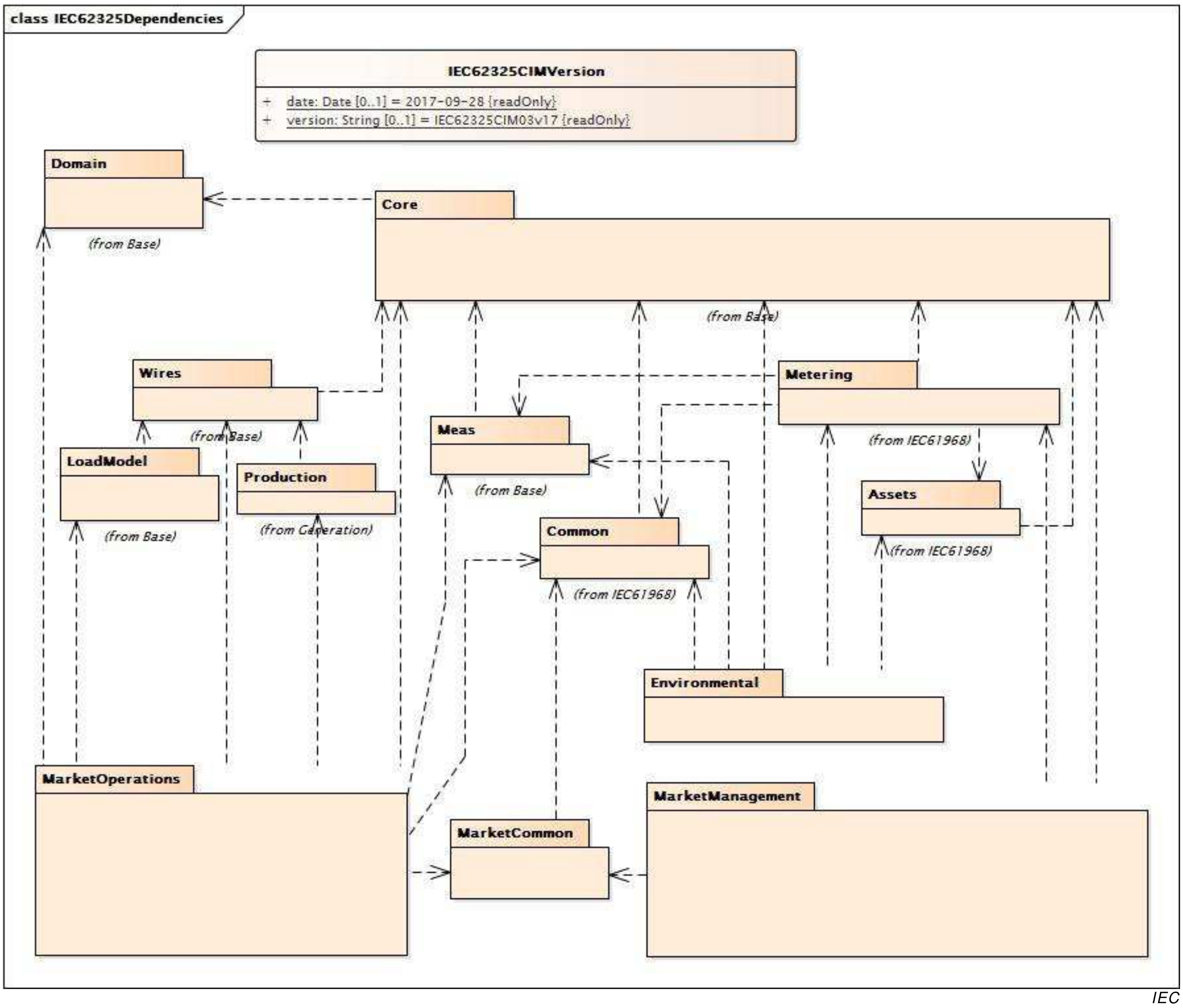


**Figure 27 – Class diagram IEC62325::IEC62325Dependencies**

Figure 27: Shows the version and top level IEC 62325 packages and their logical dependencies among packages based on inheritance only. Packages with references to an IEC standard are documented in other IEC standard specifications. Packages without references are documented in clause 6.

## 6.2 IEC62325CIMVersion root class

IEC 62325 version number assigned to this UML model.

Table 4 shows all attributes of IEC62325CIMVersion.

**Table 4 – Attributes of IEC62325::IEC62325CIMVersion**

| name | mult | type | description |
|---|---|---|---|
| date | 0..1 | Date | (const=2017-09-28) Form is YYYY-MM-DD for example for January 5, 2009 it is 2009-01-05. |
| version | 0..1 | String | (const=IEC62325CIM03v17) Form is IEC62325CIMXXvYY where XX is the major CIM package version and the YY is the minor version. For example IEC62325CIM10v03. |

## 6.3 Package MarketCommon

### 6.3.1 General

This package contains the common objects shared by MarketManagement, MarketOperations and Environmental packages.

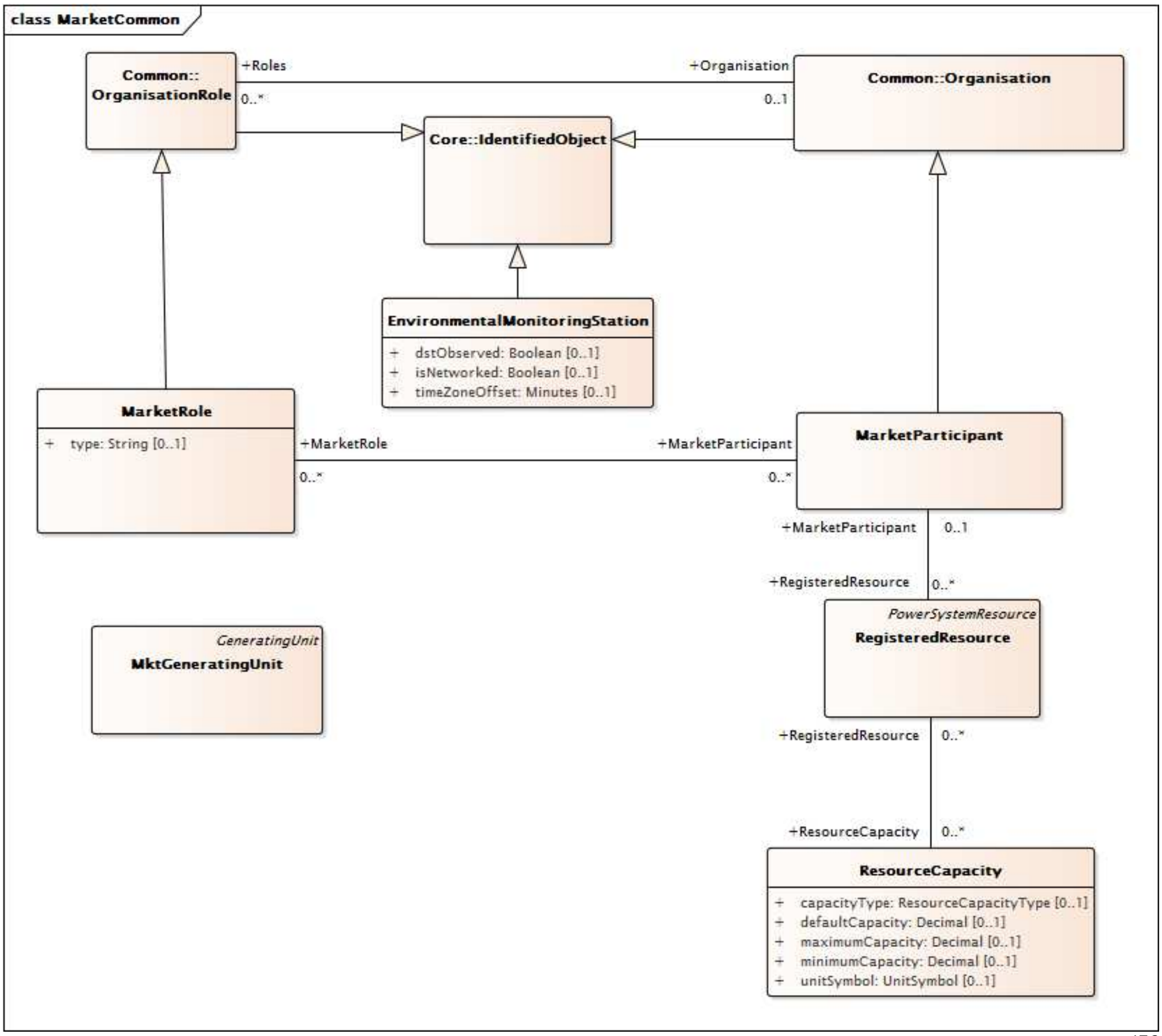


**Figure 28 – Class diagram MarketCommon::MarketCommon**

Figure 28: Shows common classes used by MarketManagement, MarketOperations and Environmental packages.

### 6.3.2 MarketParticipant

An identification of a party acting in a electricity market business process. This class is used to identify organizations that can participate in market management and/or market operations.

Table 5 shows all attributes of MarketParticipant.

**Table 5 – Attributes of MarketCommon::MarketParticipant**

| name | mult | type | description |
|---|---|---|---|
| streetAddress | 0..1 | StreetAddress | inherited from: Organisation |
| postalAddress | 0..1 | StreetAddress | inherited from: Organisation |
| phone1 | 0..1 | TelephoneNumber | inherited from: Organisation |
| phone2 | 0..1 | TelephoneNumber | inherited from: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Organisation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 6 shows all association ends of MarketParticipant with other classes.

**Table 6 – Association ends of MarketCommon::MarketParticipant with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketPerson | 0..* | MarketPerson | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | Bid | 0..* | Bid | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | MarketRole | 0..* | MarketRole | |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | inherited from: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | inherited from: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.3.3 MarketRole

The external intended behavior played by a party within the electricity market.

Table 7 shows all attributes of MarketRole.

**Table 7 – Attributes of MarketCommon::MarketRole**

| name | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | The kind of market roles that can be played by parties for given domains within the electricity market. Types are flexible using dataType of string for free-entry of role types. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 8 shows all association ends of MarketRole with other classes.

**Table 8 – Association ends of MarketCommon::MarketRole with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | inherited from: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.3.4 RegisteredResource

A resource that is registered through the market participant registration system. Examples include generating unit, load, and non-physical generator or load.

Table 9 shows all attributes of RegisteredResource.

**Table 9 – Attributes of MarketCommon::RegisteredResource**

| name | mult | type | description |
|---|---|---|---|
| ACAFlag | 0..1 | YesNo | Indication that this resource is associated with an Adjacent Control Area. |
| ASSPOptimizationFlag | 0..1 | YesNo | Indication that the resource participates in the optimization process by default. |
| commercialOpDate | 0..1 | DateTime | Resource Commercial Operation Date. |
| contingencyAvailFlag | 0..1 | YesNo | Contingent operating reserve availiability (Yes/No). Resource is availiable to participate with capacity in contingency dispatch. |
| dispatchable | 0..1 | Boolean | Dispatchable: indicates whether the resource is dispatchable. This implies that the resource intends to submit Energy bids/offers or Ancillary Services bids/offers, or self-provided schedules. |
| ECAFlag | 0..1 | YesNo | Indication that this resource is associated with an Embedded Control area. |
| flexibleOfferFlag | 0..1 | YesNo | Flexible offer flag (Y/N). |
| hourlyPredispatch | 0..1 | YesNo | Indicates need to dispatch before the start of the operating hour. Only relevant in Real-Time |

| name | mult | type | description |
|---|---|---|---|
| | | | Market. Applies to generation, intertie and participating load resource. Value (Y/N). |
| isAggregatedRes | 0..1 | YesNo | A flag to indicate if a resource is an aggregated resource. |
| lastModified | 0..1 | DateTime | Indication of the last time this item was modified/versioned. |
| LMPMFlag | 0..1 | YesNo | LMPM flag: indicates whether the resource is subject to the LMPM test (Yes/No). |
| marketParticipationFlag | 0..1 | YesNo | Market Participation flag: indicates whether the resource participate in the market. |
| maxBaseSelfSchedQty | 0..1 | Float | Maximum base self schedule quantity. |
| maxOnTime | 0..1 | Float | Maximum on time after start up. |
| minDispatchTime | 0..1 | Hours | Minimum number of consecutive hours a resource shall be dispatched if bid is accepted. |
| minOffTime | 0..1 | Float | Minimum off time after shut down. |
| minOnTime | 0..1 | Float | Minimum on time after start up. |
| mustOfferFlag | 0..1 | YesNo | Must offer flag: indicates whether the unit is subject to the must offer provisions (Y/N). |
| nonMarket | 0..1 | YesNo | Flag to indicate that the Resource is not participating in the Market Operations. |
| pointOfDeliveryFlag | 0..1 | YesNo | Indication that the registered resource is a Point of Delivery (YES) resource which implies there is a POD Loss Factor. |
| priceSetFlagDA | 0..1 | YesNo | Price setting flag: indicates whether a resource is capable of setting the Market Clearing Price (Y) for the DA market, and if not, indicates whether the resource shall submit bids for energy at $ 0 (S) or not (N).<br>Initially in the RegisteredGenerator class. It was moved to the RegisteredResource class for the participating load dispatch purpose. |
| priceSetFlagRT | 0..1 | YesNo | Price setting flag: indicates whether a resource is capable of setting the Market Clearing Price (Y) for the RT market, and if not, indicates whether the resource shall submit bids for energy at $ 0 (S) or not (N).<br>Initially in the RegisteredGenerator class. It was moved to the RegisteredResource class for the participating load dispatch purpose. |
| registrationStatus | 0..1 | ResourceRegistrationStatus | Registration Status of resource – Active, Mothballed, Planned, or Decommissioned. |
| resourceAdequacyFlag | 0..1 | YesNo | Indication that this resource participates in the resource adequacy function. |
| SMPMFlag | 0..1 | YesNo | SMPM flag: indicates whether the resource is subject to the SMPM test (Yes/No). |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 10 shows all association ends of RegisteredResource with other classes.

**Table 10 – Association ends of MarketCommon::RegisteredResource with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | |
| 0..* | AggregateNode | 0..1 | AggregateNode | An AggregateNode may be associated with up to many RegisteredResources. |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | RampRateCurve | 0..* | RampRateCurve | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | SubControlArea | 0..* | SubControlArea | |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | |
| 1..1 | FormerReference | 0..* | FormerReference | |
| 1..1 | Commitments | 0..* | Commitments | |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | RegisteredResources are qualified for resource ancillary service types (which include market product types as well as other types such as BlackStart) by the association to the class ResourceAncillaryServiceQualification. |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | InterTie | 0..* | SchedulingPoint | |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | |
| 0..1 | DopInstruction | 0..* | DopInstruction | |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 1..1 | DefaultBid | 0..1 | DefaultBid | |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | |
| 0..1 | ResourceAwardInstructi | 0..* | ResourceAwardInstructi | |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| | on | | on | |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | |
| 1..1 | Instructions | 0..* | Instructions | |
| 0..* | Pnode | 0..1 | Pnode | A registered resource injects power at one or more connectivity nodes related to a pnode |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | |
| 0..1 | DotInstruction | 0..* | DotInstruction | |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.3.5 MktGeneratingUnit

Subclass of IEC61970:Production:GeneratingUnit.

Table 11 shows all attributes of MktGeneratingUnit.

**Table 11 – Attributes of MarketCommon::MktGeneratingUnit**

| name | mult | type | description |
|---|---|---|---|
| allocSpinResP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| autoCntrlMarginP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| baseP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| controlDeadband | 0..1 | ActivePower | inherited from: GeneratingUnit |
| controlPulseHigh | 0..1 | Seconds | inherited from: GeneratingUnit |
| controlPulseLow | 0..1 | Seconds | inherited from: GeneratingUnit |
| controlResponseRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| efficiency | 0..1 | PerCent | inherited from: GeneratingUnit |
| genControlMode | 0..1 | GeneratorControlMode | inherited from: GeneratingUnit |
| genControlSource | 0..1 | GeneratorControlSource | inherited from: GeneratingUnit |
| governorMPL | 0..1 | PU | inherited from: GeneratingUnit |
| governorSCD | 0..1 | PerCent | inherited from: GeneratingUnit |
| highControlLimit | 0..1 | ActivePower | inherited from: GeneratingUnit |
| initialP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| longPF | 0..1 | Float | inherited from: GeneratingUnit |
| lowControlLimit | 0..1 | ActivePower | inherited from: GeneratingUnit |

| name | mult | type | description |
|---|---|---|---|
| lowerRampRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| maxEconomicP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | inherited from: GeneratingUnit |
| maxOperatingP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| minEconomicP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| minimumOffTime | 0..1 | Seconds | inherited from: GeneratingUnit |
| minOperatingP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| modelDetail | 0..1 | Classification | inherited from: GeneratingUnit |
| nominalP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| normalPF | 0..1 | Float | inherited from: GeneratingUnit |
| penaltyFactor | 0..1 | Float | inherited from: GeneratingUnit |
| raiseRampRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| ratedGrossMaxP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| ratedGrossMinP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| ratedNetMaxP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| shortPF | 0..1 | Float | inherited from: GeneratingUnit |
| startupCost | 0..1 | Money | inherited from: GeneratingUnit |
| startupTime | 0..1 | Seconds | inherited from: GeneratingUnit |
| tieLinePF | 0..1 | Float | inherited from: GeneratingUnit |
| totalEfficiency | 0..1 | PerCent | inherited from: GeneratingUnit |
| variableCost | 0..1 | Money | inherited from: GeneratingUnit |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 12 shows all association ends of MktGeneratingUnit with other classes.

**Table 12 – Association ends of MarketCommon::MktGeneratingUnit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | GeneratingUnitDynamicValues | 0..* | GeneratingUnitDynamicValues | |
| 0..1 | RotatingMachine | 0..* | RotatingMachine | inherited from: GeneratingUnit |
| 1..1 | GenUnitOpCostCurves | 0..* | GenUnitOpCostCurve | inherited from: GeneratingUnit |
| 1..1 | GenUnitOpSchedule | 0..1 | GenUnitOpSchedule | inherited from: GeneratingUnit |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | inherited from: GeneratingUnit |
| 1..1 | GrossToNetActivePowerCurves | 0..* | GrossToNetActivePowerCurve | inherited from: GeneratingUnit |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.3.6 EnvironmentalMonitoringStation

An environmental monitoring station, examples of which could be a weather station or a seismic monitoring station.

Table 13 shows all attributes of EnvironmentalMonitoringStation.

**Table 13 – Attributes of MarketCommon::EnvironmentalMonitoringStation**

| name | mult | type | description |
|---|---|---|---|
| dstObserved | 0..1 | Boolean | Whether this station is currently reporting using daylight saving time. Intended to aid a utility Weather Service in interpreting information coming from a station and has no direct relationship to the manner in which time is expressed in EnvironmentalValueSet. |
| isNetworked | 0..1 | Boolean | Indication that station is part of a network of stations used to monitor weather phenomena covering a large geographical area. |
| timeZoneOffset | 0..1 | Minutes | The time offset from UTC (a.k.a. GMT) configured in the station "clock", not (necessarily) the time zone in which the station is physically located.<br>This attribute exists to support management of utility monitoring stations and has no direct relationship to the manner in which time is expressed in EnvironmentalValueSet. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 14 shows all association ends of EnvironmentalMonitoringStation with other classes.

**Table 14 – Association ends of MarketCommon:: EnvironmentalMonitoringStation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Location | 0..1 | Location | Location of this monitoring station. |
| 0..1 | UsagePoint | 0..* | UsagePoint | |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Environmental analog measurement provided by this monitoring station. |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 1..1 | ReportingCapability | 0..* | ReportingCapability | One of the reporting capabilities of this monitoring station. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.3.7 ResourceCapacity root class

This class model the various capacities of a resource. A resource may have numbers of capacities related to operating, ancillary services, energy trade and so forth. Capacities may be defined for active power or reactive power.

Table 15 shows all attributes of ResourceCapacity.

**Table 15 – Attributes of MarketCommon::ResourceCapacity**

| name | mult | type | description |
|---|---|---|---|
| capacityType | 0..1 | ResourceCapacityType | capacity type<br>The types are but not limited to:<br>Regulation Up<br>Regulation Dn<br>Spinning Reserve<br>Non-Spinning Reserve<br>FOO capacity<br>MOO capacity |
| defaultCapacity | 0..1 | Decimal | default capacity |
| maximumCapacity | 0..1 | Decimal | maximum capacity |
| minimumCapacity | 0..1 | Decimal | minimum capacity |
| unitSymbol | 0..1 | UnitSymbol | Unit selection for the capacity values. |

Table 16 shows all association ends of ResourceCapacity with other classes.

**Table 16 – Association ends of MarketCommon::ResourceCapacity with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |

## 6.4 Package MarketManagement

### 6.4.1 General

This package contains all core CIM Market Extensions required for market management systems.

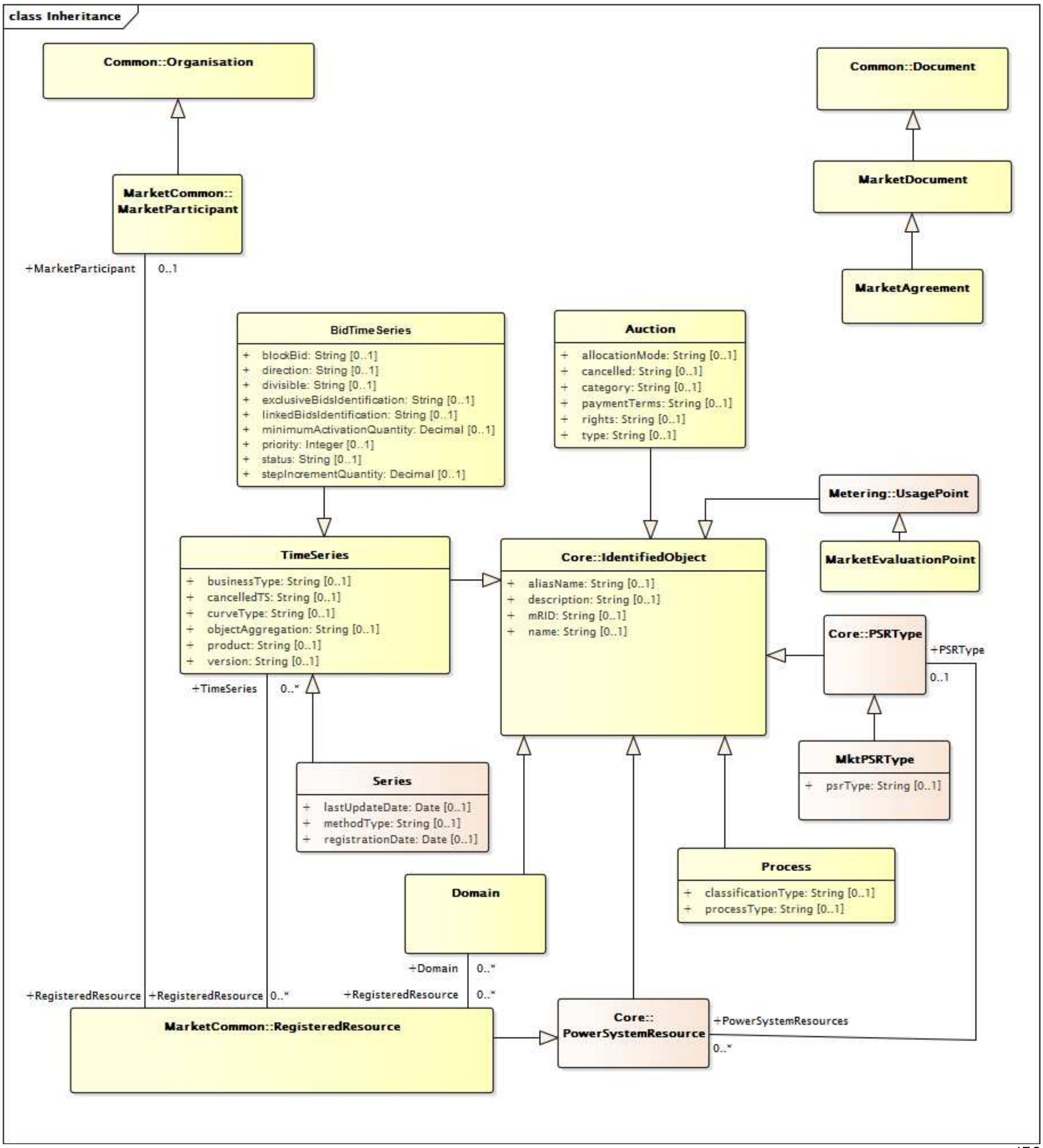

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**Figure 29 – Class diagram MarketManagement::Inheritance**

Figure 29: Provides the Market Management Inheritance structure.

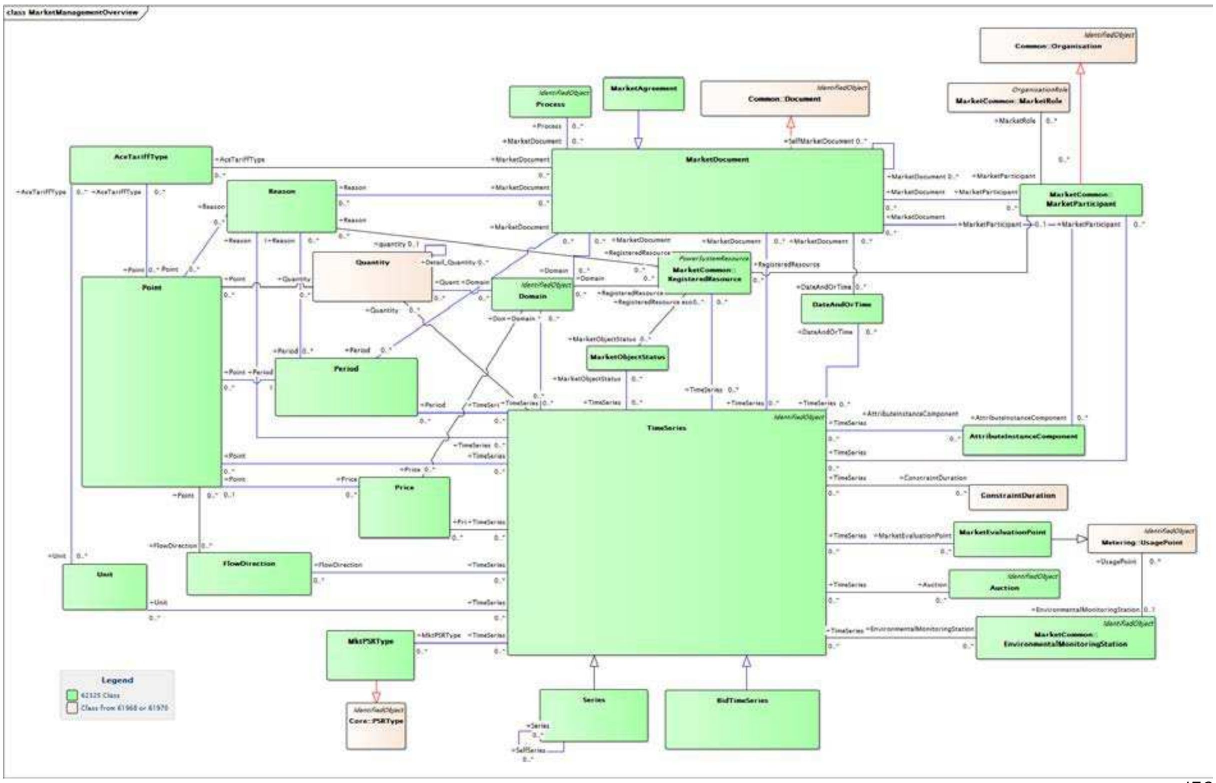

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**Figure 30 – Class diagram MarketManagement::MarketManagementOverview**

Figure 30: Describes all used classes for the definition of Market Management Profiles.

It is possible to see all dependencies with other common CIM concepts in Energy Management Systems and Distribution Management Systems.

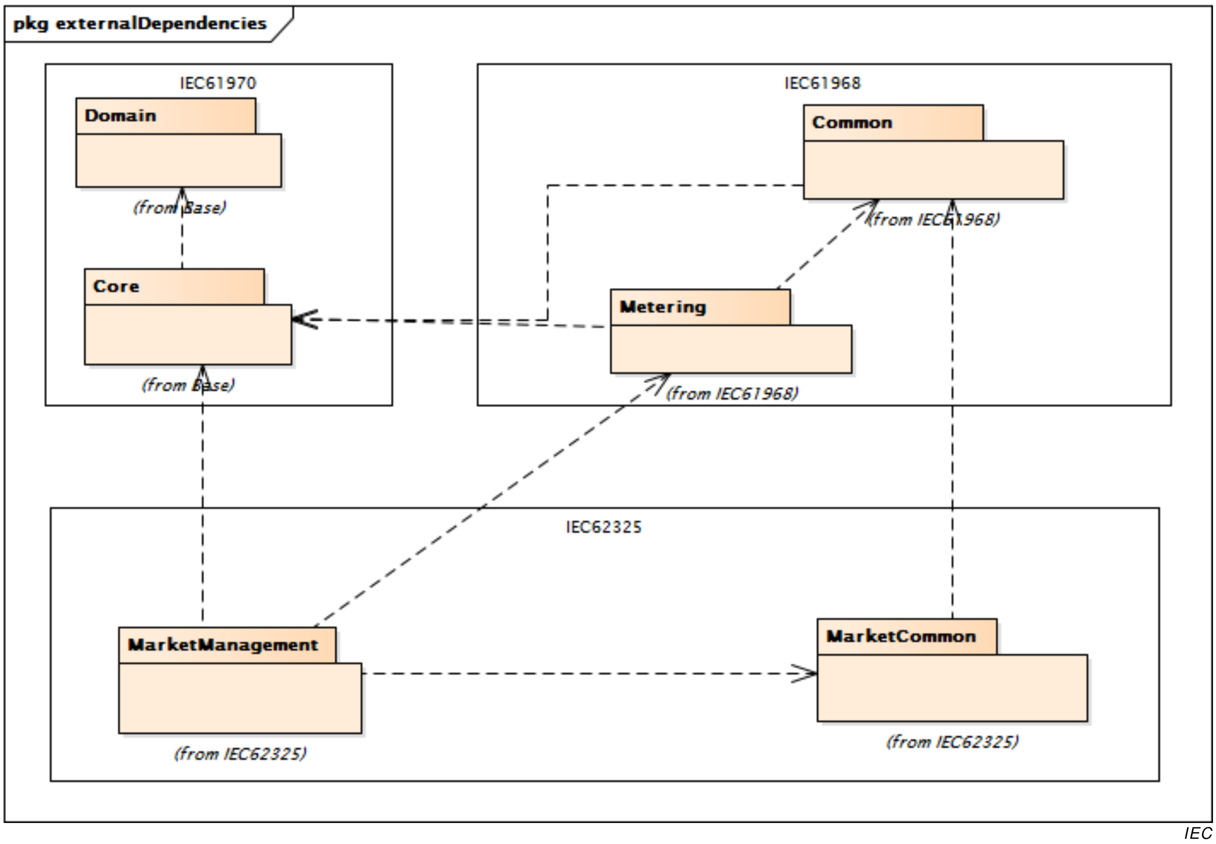



**Figure 31 – Package diagram MarketManagement::externalDependencies**

Figure 31: Provides the Market Management external dependencies.

### 6.4.2 AceTariffType root class

The Area Control Error tariff type that is applied or used.

Table 17 shows all attributes of AceTariffType.

**Table 17 – Attributes of MarketManagement::AceTariffType**

| name | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | The coded type of an ACE tariff. |

Table 18 shows all association ends of AceTariffType with other classes.

**Table 18 – Association ends of MarketManagement::AceTariffType with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Unit | 0..* | Unit | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Point | 0..* | Point | |

### 6.4.3 AttributeInstanceComponent root class

A class used to provide information about an attribute.

Table 19 shows all attributes of AttributeInstanceComponent.

**Table 19 – Attributes of MarketManagement::AttributeInstanceComponent**

| name | mult | type | description |
|---|---|---|---|
| attribute | 0..1 | String | The identification of the formal name of an attribute. |
| attributeValue | 0..1 | String | The instance value of the attribute. |
| position | 0..1 | Integer | A sequential value representing a relative sequence number. |

Table 20 shows all association ends of AttributeInstanceComponent with other classes.

**Table 20 – Association ends of MarketManagement:: AttributeInstanceComponent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | MarketDocument | 0..* | MarketDocument | |

### 6.4.4 Auction

A class providing the identification and type of an auction.

Table 21 shows all attributes of Auction.

**Table 21 – Attributes of MarketManagement::Auction**

| name | mult | type | description |
|---|---|---|---|
| allocationMode | 0..1 | String | Identification of the method of allocation in an auction. |
| cancelled | 0..1 | String | An indicator that signifies that the auction has been cancelled. |
| category | 0..1 | String | The product category of an auction. |
| paymentTerms | 0..1 | String | The terms which dictate the determination of the bid payment price. |
| rights | 0..1 | String | The rights of use the transmission capacity acquired in an auction. |
| type | 0..1 | String | The kind of the Auction (e.g. implicit, explicit ...). |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 22 shows all association ends of Auction with other classes.

**Table 22 – Association ends of MarketManagement::Auction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.5 BidTimeSeries

The formal specification of specific characteristics related to a bid.

Table 23 shows all attributes of BidTimeSeries.

**Table 23 – Attributes of MarketManagement::BidTimeSeries**

| name | mult | type | description |
|---|---|---|---|
| blockBid | 0..1 | String | Indication that the values in the period are considered as a whole. They cannot be changed or subdivided. |
| direction | 0..1 | String | The coded identification of the energy flow. |
| divisible | 0..1 | String | An indication whether or not each element of the bid may be partially accepted or not. |
| status | 0..1 | String | The information about the status of the bid, such as "shared", "restricted", ... |
| linkedBidsIdentification | 0..1 | String | Unique identification associated with all linked bids. |
| exclusiveBidsIdentificati on | 0..1 | String | Unique identification associated with all linked tenders.<br>The identification of a set of tenders that are linked together signifying that only one can be accepted.<br>This identification is defined by the tenderer and must be unique for a given auction. |
| minimumActivationQuant ity | 0..1 | Decimal | The minimum quantity of energy that can be activated at a given time interval. |
| stepIncrementQuantity | 0..1 | Decimal | The minimum increment that can be applied for an increase in an activation request. |
| priority | 0..1 | Integer | The numeric local priority given to a bid. Lower numeric values will have higher priority. |
| businessType | 0..1 | String | inherited from: TimeSeries |
| cancelledTS | 0..1 | String | inherited from: TimeSeries |
| curveType | 0..1 | String | inherited from: TimeSeries |
| objectAggregation | 0..1 | String | inherited from: TimeSeries |
| product | 0..1 | String | inherited from: TimeSeries |
| version | 0..1 | String | inherited from: TimeSeries |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 24 shows all association ends of BidTimeSeries with other classes.

**Table 24 – Association ends of MarketManagement::BidTimeSeries with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | inherited from: TimeSeries |
| 0..* | EnvironmentalMonitoring Station | 0..* | EnvironmentalMonitoring Station | inherited from: TimeSeries |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | inherited from: TimeSeries |
| 0..* | Domain | 0..* | Domain | inherited from: TimeSeries |
| 0..* | FlowDirection | 0..* | FlowDirection | inherited from: TimeSeries |
| 0..* | MarketDocument | 0..* | MarketDocument | inherited from: TimeSeries |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | inherited from: TimeSeries |
| 0..* | Period | 0..* | Period | inherited from: TimeSeries |
| 0..* | Reason | 0..* | Reason | inherited from: TimeSeries |
| 0..* | RegisteredResource | 0..* | RegisteredResource | inherited from: TimeSeries |
| 0..* | AttributeInstanceCompo nent | 0..* | AttributeInstanceCompo nent | inherited from: TimeSeries |
| 0..* | Auction | 0..* | Auction | inherited from: TimeSeries |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | inherited from: TimeSeries |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: TimeSeries |
| 0..* | MktPSRType | 0..* | MktPSRType | inherited from: TimeSeries |
| 0..* | Point | 0..* | Point | inherited from: TimeSeries |
| 0..* | Price | 0..* | Price | inherited from: TimeSeries |
| 0..* | Quantity | 0..* | Quantity | inherited from: TimeSeries |
| 0..* | Unit | 0..* | Unit | inherited from: TimeSeries |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.6 ConstraintDuration root class

Duration constraint to activate, to put in operation, to deactivate, ... a given event.

Table 25 shows all attributes of ConstraintDuration.

**Table 25 – Attributes of MarketManagement::ConstraintDuration**

| name | mult | type | description |
|---|---|---|---|
| duration | 0..1 | Duration | The duration of the constraint. |
| type | 0..1 | String | The type of the constraint. |

Table 26 shows all association ends of ConstraintDuration with other classes.

**Table 26 – Association ends of MarketManagement:: ConstraintDuration with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.7 DateAndOrTime root class

The date and/or the time.

Table 27 shows all attributes of DateAndOrTime.

**Table 27 – Attributes of MarketManagement::DateAndOrTime**

| name | mult | type | description |
|---|---|---|---|
| date | 0..1 | Date | Date as "yyyy-mm-dd", which conforms with ISO 8601 |
| time | 0..1 | Time | Time as "hh:mm:ss.sssZ", which conforms with ISO 8601. |

Table 28 shows all association ends of DateAndOrTime with other classes.

**Table 28 – Association ends of MarketManagement::DateAndOrTime with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.8 Domain

An area of activity defined within the energy market.

Table 29 shows all attributes of Domain.

**Table 29 – Attributes of MarketManagement::Domain**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 30 shows all association ends of Domain with other classes.

**Table 30 – Association ends of MarketManagement::Domain with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Price | 0..* | Price | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Quantity | 0..* | Quantity | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.9 FlowDirection root class

The coded identification of the direction of energy flow.

Table 31 shows all attributes of FlowDirection.

**Table 31 – Attributes of MarketManagement::FlowDirection**

| name | mult | type | description |
|---|---|---|---|
| direction | 0..1 | String | The coded identification of the direction of energy flow. |

Table 32 shows all association ends of FlowDirection with other classes.

**Table 32 – Association ends of MarketManagement::FlowDirection with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.10 MarketAgreement

An identification or eventually the contents of an agreement between two or more parties.

Table 33 shows all attributes of MarketAgreement.

**Table 33 – Attributes of MarketManagement::MarketAgreement**

| name | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 34 shows all association ends of MarketAgreement with other classes.

**Table 34 – Association ends of MarketManagement:: MarketAgreement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | inherited from: MarketDocument |
| 0..* | Domain | 0..* | Domain | inherited from: MarketDocument |
| 0..* | Period | 0..* | Period | inherited from: MarketDocument |
| 0..* | SelfMarketDocument | 0..* | MarketDocument | inherited from: MarketDocument |
| 0..* | MarketParticipant | 0..* | MarketParticipant | inherited from: MarketDocument |
| 0..* | AceTariffType | 0..* | AceTariffType | inherited from: MarketDocument |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | inherited from: MarketDocument |
| 0..* | Reason | 0..* | Reason | inherited from: MarketDocument |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: MarketDocument |
| 0..* | Process | 0..* | Process | inherited from: MarketDocument |
| 0..* | MarketDocument | 0..* | MarketDocument | inherited from: MarketDocument |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.11 MarketDocument

Electronic document containing the information necessary to satisfy a given business process set of requirements.

Table 35 shows all attributes of MarketDocument.

**Table 35 – Attributes of MarketManagement::MarketDocument**

| name | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 36 shows all association ends of MarketDocument with other classes.

**Table 36 – Association ends of MarketManagement::MarketDocument with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | Period | 0..* | Period | |
| 0..* | SelfMarketDocument | 0..* | MarketDocument | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..* | AceTariffType | 0..* | AceTariffType | |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Process | 0..* | Process | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.12 MarketEvaluationPoint

The identification of an entity where energy products are measured or computed.

Table 37 shows all attributes of MarketEvaluationPoint.

**Table 37 – Attributes of MarketManagement::MarketEvaluationPoint**

| name | mult | type | description |
|---|---|---|---|
| isSdp | 0..1 | Boolean | inherited from: UsagePoint |
| isVirtual | 0..1 | Boolean | inherited from: UsagePoint |
| phaseCode | 0..1 | PhaseCode | inherited from: UsagePoint |
| grounded | 0..1 | Boolean | inherited from: UsagePoint |
| servicePriority | 0..1 | String | inherited from: UsagePoint |
| serviceDeliveryRemark | 0..1 | String | inherited from: UsagePoint |
| estimatedLoad | 0..1 | CurrentFlow | inherited from: UsagePoint |
| checkBilling | 0..1 | Boolean | inherited from: UsagePoint |
| ratedCurrent | 0..1 | CurrentFlow | inherited from: UsagePoint |
| nominalServiceVoltage | 0..1 | Voltage | inherited from: UsagePoint |
| ratedPower | 0..1 | ActivePower | inherited from: UsagePoint |
| outageRegion | 0..1 | String | inherited from: UsagePoint |
| readCycle | 0..1 | String | inherited from: UsagePoint |
| readRoute | 0..1 | String | inherited from: UsagePoint |
| amiBillingReady | 0..1 | AmiBillingReadyKind | inherited from: UsagePoint |
| connectionState | 0..1 | UsagePointConnectedKind | inherited from: UsagePoint |
| minimalUsageExpected | 0..1 | Boolean | inherited from: UsagePoint |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 38 shows all association ends of MarketEvaluationPoint with other classes.

**Table 38 – Association ends of MarketManagement:: MarketEvaluationPoint with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Equipments | 0..* | Equipment | inherited from: UsagePoint |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: UsagePoint |
| 0..* | ServiceCategory | 0..1 | ServiceCategory | inherited from: UsagePoint |
| 0..1 | EndDevices | 0..* | EndDevice | inherited from: UsagePoint |
| 0..1 | ServiceMultipliers | 0..* | ServiceMultiplier | inherited from: UsagePoint |
| 0..* | Outage | 0..* | Outage | inherited from: UsagePoint |
| 0..* | Outage | 0..* | Outage | inherited from: UsagePoint |
| 0..* | CustomerAgreement | 0..1 | CustomerAgreement | inherited from: UsagePoint |
| 0..* | PricingStructures | 0..* | PricingStructure | inherited from: UsagePoint |
| 0..* | ServiceLocation | 0..1 | ServiceLocation | inherited from: UsagePoint |
| 0..* | EndDeviceControls | 0..* | EndDeviceControl | inherited from: UsagePoint |
| 0..1 | EndDeviceEvents | 0..* | EndDeviceEvent | inherited from: UsagePoint |
| 0..1 | MeterReadings | 0..* | MeterReading | inherited from: UsagePoint |
| 0..1 | MeterServiceWorkTasks | 0..* | MeterWorkTask | inherited from: UsagePoint |
| 0..* | MetrologyRequirements | 0..* | MetrologyRequirement | inherited from: UsagePoint |
| 0..* | ServiceSupplier | 0..1 | ServiceSupplier | inherited from: UsagePoint |
| 0..* | UsagePointGroups | 0..* | UsagePointGroup | inherited from: UsagePoint |
| 0..* | UsagePointLocation | 0..1 | UsagePointLocation | inherited from: UsagePoint |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.13 MarketObjectStatus root class

The condition or position of an object with regard to its standing.

Table 39 shows all attributes of MarketObjectStatus.

**Table 39 – Attributes of MarketManagement::MarketObjectStatus**

| name | mult | type | description |
|---|---|---|---|
| status | 0..1 | String | The coded condition or position of an object with regard to its standing. |

Table 40 shows all association ends of MarketObjectStatus with other classes.

**Table 40 – Association ends of MarketManagement:: MarketObjectStatus with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.14 MktPSRType

The type of a power system resource.

Table 41 shows all attributes of MktPSRType.

**Table 41 – Attributes of MarketManagement::MktPSRType**

| name | mult | type | description |
|---|---|---|---|
| psrType | 0..1 | String | The coded type of a power system resource. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 42 shows all association ends of MktPSRType with other classes.

**Table 42 – Association ends of MarketManagement::MktPSRType with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..1 | PowerSystemResources | 0..* | PowerSystemResource | inherited from: PSRType |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.15 Period root class

An identification of a time interval that may have a given resolution.

Table 43 shows all attributes of Period.

**Table 43 – Attributes of MarketManagement::Period**

| name | mult | type | description |
|---|---|---|---|
| resolution | 0..1 | Duration | The number of units of time that compose an individual step within a period. |
| timeInterval | 0..1 | DateTimeInterval | The start and end date and time for a given interval. |

Table 44 shows all association ends of Period with other classes.

**Table 44 – Association ends of MarketManagement::Period with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Reason | 0..* | Reason | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 1..1 | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.16 Point root class

An identification of a set of values beeing adressed within a specific interval of time.

Table 45 shows all attributes of Point.

**Table 45 – Attributes of MarketManagement::Point**

| name | mult | type | description |
|---|---|---|---|
| position | 0..1 | Integer | A sequential value representing the relative position within a given time interval. |
| quality | 0..1 | String | The quality of the information being provided. This quality may be estimated, not available, as provided, etc. |
| quantity | 0..1 | Decimal | Principal quantity identified for a point. |
| secondaryQuantity | 0..1 | Decimal | Secondary quantity identified for a point. |

Table 46 shows all association ends of Point with other classes.

**Table 46 – Association ends of MarketManagement::Point with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AceTariffType | 0..* | AceTariffType | |
| 0..* | Period | 1..1 | Period | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | FlowDirection | 0..* | FlowDirection | |
| 0..1 | Price | 0..* | Price | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | Quantity | 0..* | Quantity | |

### 6.4.17 Price root class

The cost corresponding to a specific measure and expressed in a currency.

Table 47 shows all attributes of Price.

**Table 47 – Attributes of MarketManagement::Price**

| name | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Decimal | A number of monetary units specified in a unit of currency. |
| category | 0..1 | String | The category of a price to be used in a price calculation. The price category is mutually agreed between System Operators. |
| direction | 0..1 | String | The direction indicates whether a System Operator pays the Market Parties or inverse. |

Table 48 shows all association ends of Price with other classes.

**Table 48 – Association ends of MarketManagement::Price with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Domain | 0..* | Domain | |
| 0..* | Point | 0..1 | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.18 Process

The formal specification of a set of business transactions having the same business goal.

Table 49 shows all attributes of Process.

**Table 49 – Attributes of MarketManagement::Process**

| name | mult | type | description |
|---|---|---|---|
| classificationType | 0..1 | String | The classification mechanism used to group a set of objects together within a business process. The grouping may be of a detailed or a summary nature. |
| processType | 0..1 | String | The kind of business process. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 50 shows all association ends of Process with other classes.

**Table 50 – Association ends of MarketManagement::Process with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.19 Quantity root class

Description of quantities needed in the data exchange.

The type of the quantity is described either by the role of the association or the type attribute.

The quality attribute provides the information about the quality of the quantity (measured, estimated, etc.).

Table 51 shows all attributes of Quantity.

**Table 51 – Attributes of MarketManagement::Quantity**

| name | mult | type | description |
|---|---|---|---|
| quantity | 0..1 | Decimal | The quantity value.<br>The association role provides the information about what is expressed. |
| type | 0..1 | String | The description of the type of the quantity. |
| quality | 0..1 | String | The quality of the information being provided. This quality may be estimated, not available, as provided, etc. |

Table 52 shows all association ends of Quantity with other classes.

**Table 52 – Association ends of MarketManagement::Quantity with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Domain | 0..* | Domain | |
| 0..* | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | quantity | 0..1 | Quantity | |
| 0..1 | Detail_Quantity | 0..* | Quantity | Additional information related to the associated quantity. |

### 6.4.20 Reason root class

The motivation of an act.

Table 53 shows all attributes of Reason.

**Table 53 – Attributes of MarketManagement::Reason**

| name | mult | type | description |
|---|---|---|---|
| code | 0..1 | String | The motivation of an act in coded form. |
| text | 0..1 | String | The textual explanation corresponding to the reason code. |

Table 54 shows all association ends of Reason with other classes.

**Table 54 – Association ends of MarketManagement::Reason with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Point | 0..* | Point | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | Period | 0..* | Period | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.21 Series

A set of similar physical or conceptual objects defined for the same period or point of time.

Table 55 shows all attributes of Series.

**Table 55 – Attributes of MarketManagement::Series**

| name | mult | type | description |
|---|---|---|---|
| lastUpdateDate | 0..1 | Date | The date of the last update related to this market object. |
| methodType | 0..1 | String | Type of method used in the business process related to this Series, e.g. metering method. |
| registrationDate | 0..1 | Date | For a market object, the date of registration of a contract, e.g. the date of change of supplier for a customer. |
| businessType | 0..1 | String | inherited from: TimeSeries |
| cancelledTS | 0..1 | String | inherited from: TimeSeries |
| curveType | 0..1 | String | inherited from: TimeSeries |
| objectAggregation | 0..1 | String | inherited from: TimeSeries |
| product | 0..1 | String | inherited from: TimeSeries |
| version | 0..1 | String | inherited from: TimeSeries |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 56 shows all association ends of Series with other classes.

**Table 56 – Association ends of MarketManagement::Series with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | SelfSeries | 0..* | Series | |
| 0..* | Series | 0..* | Series | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | inherited from: TimeSeries |
| 0..* | EnvironmentalMonitoring Station | 0..* | EnvironmentalMonitoring Station | inherited from: TimeSeries |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | inherited from: TimeSeries |
| 0..* | Domain | 0..* | Domain | inherited from: TimeSeries |
| 0..* | FlowDirection | 0..* | FlowDirection | inherited from: TimeSeries |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | inherited from: TimeSeries |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | inherited from: TimeSeries |
| 0..* | Period | 0..* | Period | inherited from: TimeSeries |
| 0..* | Reason | 0..* | Reason | inherited from: TimeSeries |
| 0..* | RegisteredResource | 0..* | RegisteredResource | inherited from: TimeSeries |
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | inherited from: TimeSeries |
| 0..* | Auction | 0..* | Auction | inherited from: TimeSeries |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | inherited from: TimeSeries |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: TimeSeries |
| 0..* | MktPSRType | 0..* | MktPSRType | inherited from: TimeSeries |
| 0..* | Point | 0..* | Point | inherited from: TimeSeries |
| 0..* | Price | 0..* | Price | inherited from: TimeSeries |
| 0..* | Quantity | 0..* | Quantity | inherited from: TimeSeries |
| 0..* | Unit | 0..* | Unit | inherited from: TimeSeries |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.22 TimeSeries

A set of regular time-ordered measurements or values of quantitative nature of an individual or collective phenomenon taken at successive, in most cases equidistant, periods / points of time.

Table 57 shows all attributes of TimeSeries.

**Table 57 – Attributes of MarketManagement::TimeSeries**

| name | mult | type | description |
|---|---|---|---|
| businessType | 0..1 | String | The identification of the nature of the time series. |
| cancelledTS | 0..1 | String | An indicator stating that the TimeSeries, identified by the mRID, is cancelled as well as all the values sent in a previous version of the TimeSeries in a previous document. |
| curveType | 0..1 | String | The coded representation of the type of curve being described. |
| objectAggregation | 0..1 | String | Identification of the object that is the common denominator used to aggregate a time series. |
| product | 0..1 | String | The type of the product such as Power, energy, reactive power, transport capacity that is the subject of the time series. |
| version | 0..1 | String | Version of the time series. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 58 shows all association ends of TimeSeries with other classes.

**Table 58 – Association ends of MarketManagement::TimeSeries with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..* | EnvironmentalMonitoring Station | 0..* | EnvironmentalMonitoring Station | |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | FlowDirection | 0..* | FlowDirection | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | |
| 0..* | Period | 0..* | Period | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | AttributeInstanceCompo nent | 0..* | AttributeInstanceCompo nent | |
| 0..* | Auction | 0..* | Auction | |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | |
| 0..* | MktPSRType | 0..* | MktPSRType | |
| 0..* | Point | 0..* | Point | |
| 0..* | Price | 0..* | Price | |
| 0..* | Quantity | 0..* | Quantity | |
| 0..* | Unit | 0..* | Unit | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.4.23 Unit root class

The identification of the unit name for the time series quantities.

Table 59 shows all attributes of Unit.

**Table 59 – Attributes of MarketManagement::Unit**

| name | mult | type | description |
|---|---|---|---|
| name | 0..1 | String | The coded representation of the unit. |

Table 60 shows all association ends of Unit with other classes.

**Table 60 – Association ends of MarketManagement::Unit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | AceTariffType | 0..* | AceTariffType | |

## 6.5 Package MarketOperations

### 6.5.1 General

This package contains all core CIM Market Extensions required for market operations systems.

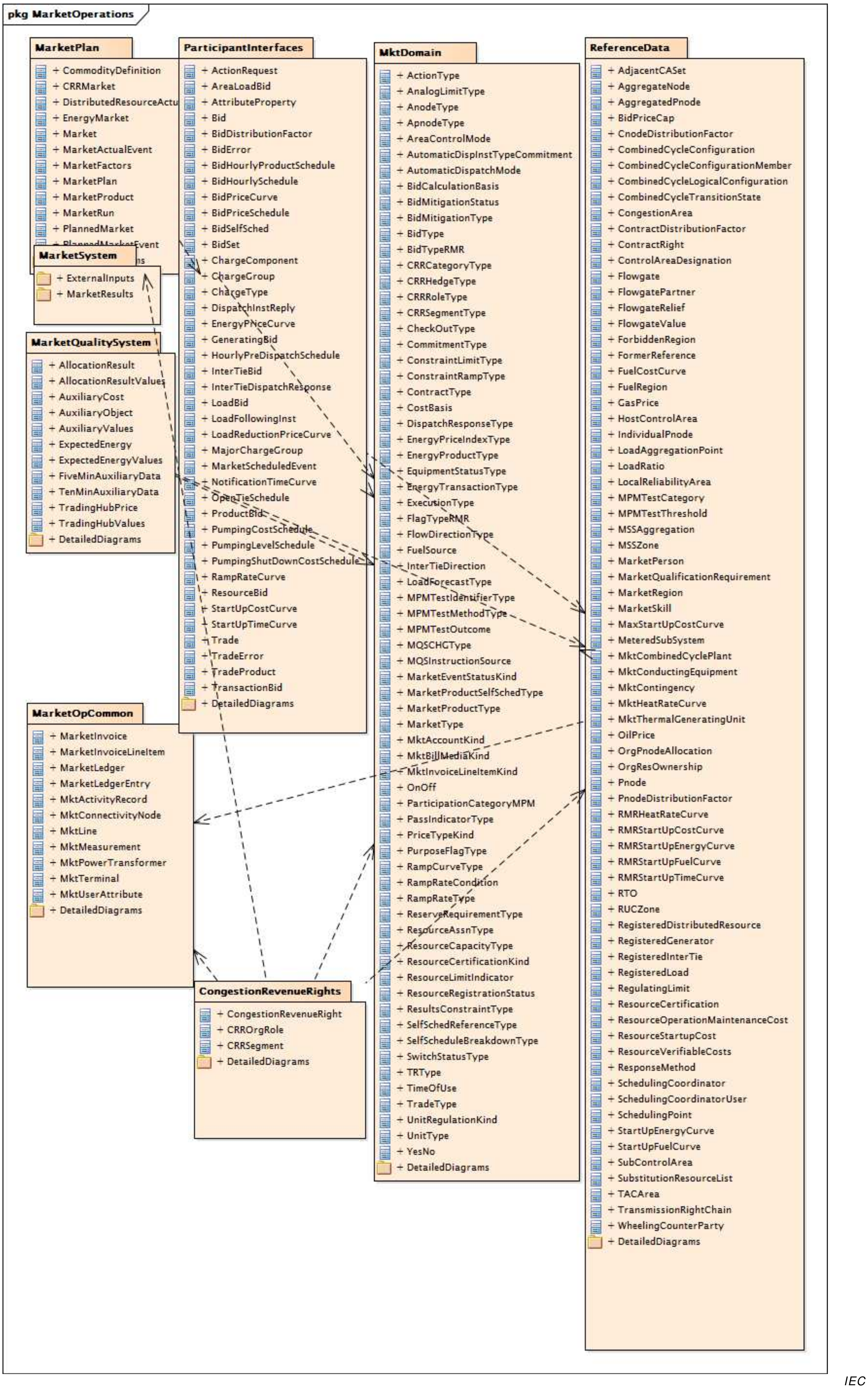



**Figure 32 – Package diagram MarketOperations::MarketOperations**

Figure 32: Shows the package containing all core CIM Market Extensions required for market operations systems.

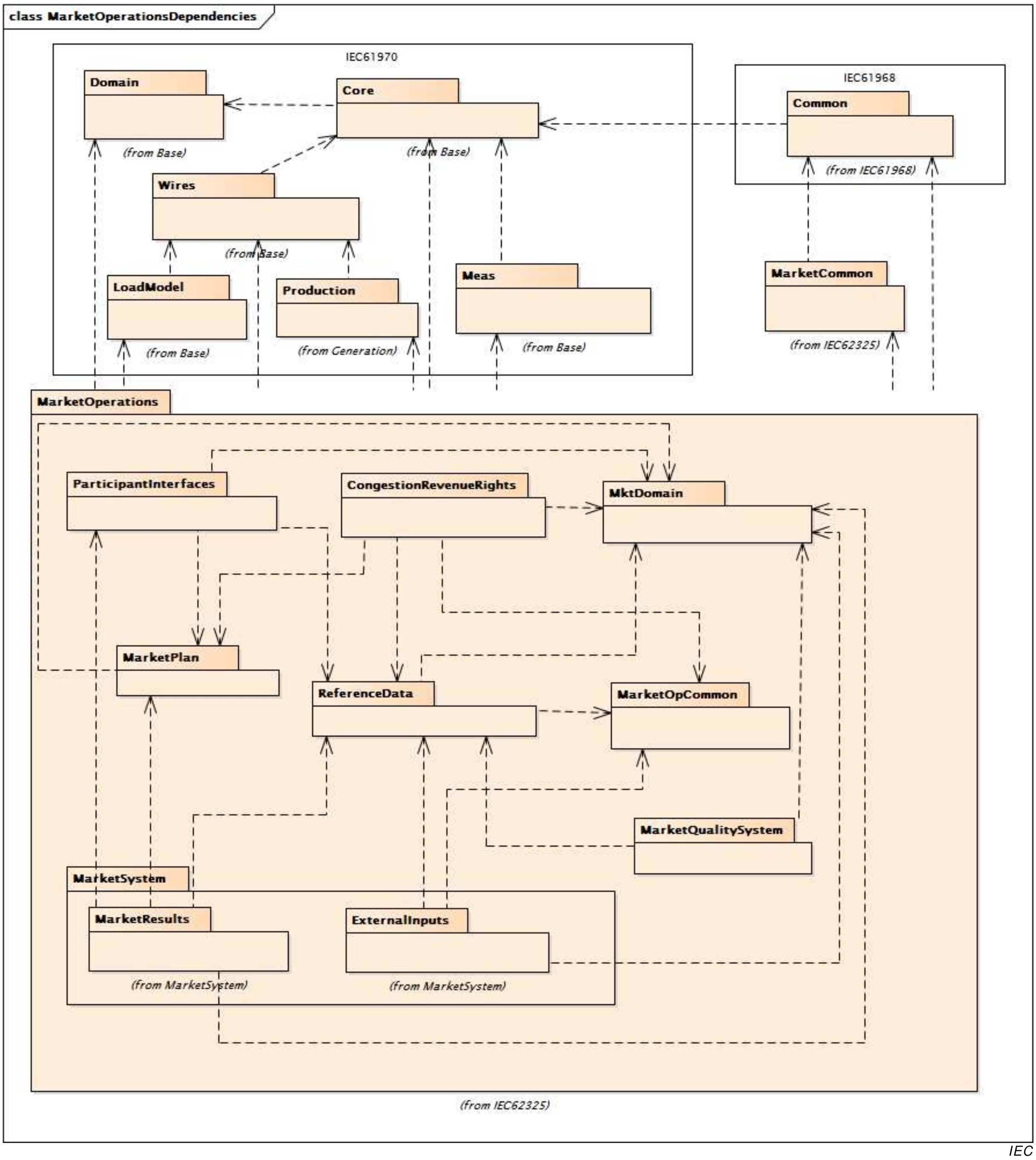



**Figure 33 – Class diagram MarketOperations::MarketOperationsDependencies**

Figure 33: Shows Market Operation internal package dependencies and external dependencies.

### 6.5.2 Package CongestionRevenueRights

#### 6.5.2.1 General

Congestion rent is a major, highly volatile charge currently faced by many participants in the LMP-based electrical energy markets. For this reason, the ISOs offer congestion revenue rights (CRR), also known as financial transmission rights or transmission congestion contracts. These are financial instruments that allow market participants to hedge against congestion charges when they schedule their generation, load and bilateral energy transactions.

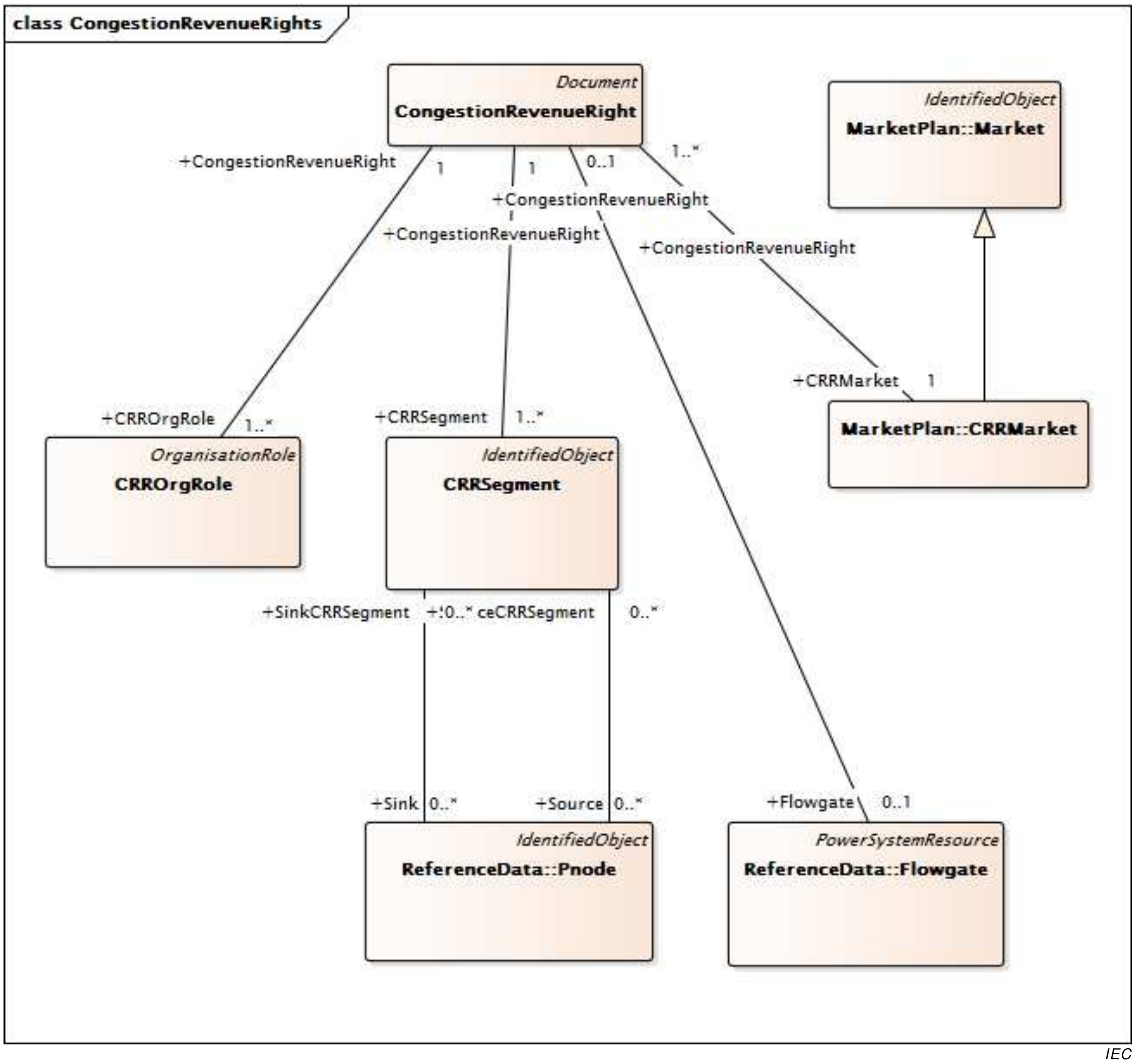



**Figure 34 – Class diagram CongestionRevenueRights::CongestionRevenueRights**

Figure 34: Shows a congestion revenue right (CRR), which is is a financial instrument which entitles the holder to a CRR payment when the Congestion is in the direction of the CRR source to the CRR sink. The classes shown are used to model the CRR data. The CRRSegment class contains information about CRR segments. The MktOrganization class contains information about the entity that holds the CRR. Associations from CRRSegment to Reference data show the source and sink Pnodes.

### 6.5.2.2 CongestionRevenueRight

Congestion Revenue Rights (CRR) class that is inherited from a Document class.

A CRR is a financial concept that is used to hedge congestion charges.

The CRR is usually settled based on the Locational Marginal Prices (LMPs) that are calculated in the day-ahead market. These LMPs are determined by the Day-ahead resource schedules/bids. CRRs will not hedge against marginal losses. If the congestion component of LMP at the sink is greater than at the source, then the CRR owner is entitled to receive a portion of congestion revenues. If the congestion component at the sink is less than at the source, then an obligation-type CRR owner will be charged, but an option-type CRR owner will not.

Table 61 shows all attributes of CongestionRevenueRight.

**Table 61 – Attributes of CongestionRevenueRights::CongestionRevenueRight**

| name | mult | type | description |
|---|---|---|---|
| cRRcategory | 0..1 | CRRCategoryType | CRR category represents 'PTP' for a point-to-point CRR, or 'NSR' for a Network Service Right. If CRR category is 'PTP', both Source ID and Sink ID fields are required. If CRR category is 'NSR' only one field, either Source ID or Sink ID, shall be not null and the other shall be null. However, the 'NSR' category will include at least three records. |
| cRRtype | 0..1 | CRRSegmentType | Type of the CRR, from the possible type definitions in the CRR System (e.g. 'LSE', 'ETC'). |
| hedgeType | 0..1 | CRRHedgeType | Hedge type Obligation or Option. An obligation type requires the holder to receive or pay the congestion rent. An option type gives the holder the option of receiving or paying the congestion rent. |
| timeOfUse | 0..1 | TimeOfUse | Time of Use flag of the CRR – Peak (ON), Offpeak (OFF) or all 24 hours (24HR). |
| tradeSliceID | 0..1 | String | Segment of the CRR described in the current record. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 62 shows all association ends of CongestionRevenueRight with other classes.

**Table 62 – Association ends of CongestionRevenueRights:: CongestionRevenueRight with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | Flowgate | 0..1 | Flowgate | |
| 1..* | CRRMarket | 1..1 | CRRMarket | |
| 1..1 | CRRSegment | 1..* | CRRSegment | |
| 1..1 | CRROrgRole | 1..* | CRROrgRole | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.2.3 CRROrgRole

Identifies a way in which an organisation may participate with a defined Congestion Revenue Right (CRR).

Table 63 shows all attributes of CRROrgRole.

**Table 63 – Attributes of CongestionRevenueRights::CRROrgRole**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | CRRRoleType | Kind of role the organisation is with regards to the congestion revenue rights. |
| status | 0..1 | Status | Status of congestion revenue rights organisation role. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 64 shows all association ends of CRROrgRole with other classes.

**Table 64 – Association ends of CongestionRevenueRights:: CRROrgRole with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | CongestionRevenueRigh t | 1..1 | CongestionRevenueRight | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | inherited from: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.2.4 CRRSegment

CRRSegment represents a segment of a CRR in a particular time frame.

The segment class contains amount, clearing price, start date and time, end date and time.

Table 65 shows all attributes of CRRSegment.

**Table 65 – Attributes of CongestionRevenueRights::CRRSegment**

| name | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Money | Dollar amount = quantity x clearingPrice |
| clearingPrice | 0..1 | Money | Clearing price of a CRR |
| endDateTime | 0..1 | DateTime | segment end date time |
| quantity | 0..1 | Float | The MW amount associated with the CRR |
| startDateTime | 0..1 | DateTime | segment start date time |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 66 shows all association ends of CRRSegment with other classes.

**Table 66 – Association ends of CongestionRevenueRights:: CRRSegment with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | CongestionRevenueRigh t | 1..1 | CongestionRevenueRigh t | |
| 0..* | Source | 0..* | Pnode | |
| 0..* | Sink | 0..* | Pnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.3 Package MarketOpCommon

#### 6.5.3.1 General

This package contains the common objects shared by MarketOperations packages.

#### 6.5.3.2 MarketInvoice root class

A roll up of invoice line items. The whole invoice has a due date and amount to be paid, with information such as customer, banks etc. being obtained through associations. The invoice roll up is based on individual line items that each contain amounts and descriptions for specific services or products.

Table 67 shows all attributes of MarketInvoice.

**Table 67 – Attributes of MarketOpCommon::MarketInvoice**

| name | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Money | Total amount due on this invoice based on line items and applicable adjustments. |
| billMediaKind | 0..1 | MktBillMediaKind | Kind of media by which the CustomerBillingInfo was delivered. |
| dueDate | 0..1 | Date | Calculated date upon which the Invoice amount is due. |
| kind | 0..1 | MktAccountKind | Kind of invoice (default is 'sales'). |
| mailedDate | 0..1 | Date | Date on which the customer billing statement/invoice was printed/mailed. |
| proForma | 0..1 | Boolean | True if payment is to be paid by a Customer to accept a particular ErpQuote (with associated Design) and have work initiated, at which time an associated ErpInvoice should automatically be generated. EprPayment.subjectStatus satisfies terms specificed in the ErpQuote. |
| referenceNumber | 0..1 | String | Number of an invoice to be reference by this invoice. |
| transactionDateTime | 0..1 | DateTime | Date and time when the invoice is issued. |
| transferType | 0..1 | String | Type of invoice transfer. |

Table 68 shows all association ends of MarketInvoice with other classes.

**Table 68 – Association ends of MarketOpCommon::MarketInvoice with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MajorChargeGroup | 1..* | MajorChargeGroup | |
| 1..1 | MarketInvoiceLineItems | 0..* | MarketInvoiceLineItem | |

#### 6.5.3.3 MarketInvoiceLineItem root class

An individual line item on an invoice.

Table 69 shows all attributes of MarketInvoiceLineItem.

**Table 69 – Attributes of MarketOpCommon::MarketInvoiceLineItem**

| name | mult | type | description |
|---|---|---|---|
| billPeriod | 0..1 | DateTimeInterval | Bill period for the line item. |
| glAccount | 0..1 | String | General Ledger account code, shall be a valid combination. |
| glDateTime | 0..1 | DateTime | Date and time line item will be posted to the General Ledger. |
| kind | 0..1 | MktInvoiceLineItemKind | Kind of line item. |
| lineAmount | 0..1 | Float | Amount due for this line item. |
| lineNumber | 0..1 | String | Line item number on invoice statement. |
| lineVersion | 0..1 | String | Version number of the bill run. |
| netAmount | 0..1 | Float | Net line item charge amount. |
| previousAmount | 0..1 | Float | Previous line item charge amount. |

Table 70 shows all association ends of MarketInvoiceLineItem with other classes.

**Table 70 – Association ends of MarketOpCommon:: MarketInvoiceLineItem with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketInvoice | 1..1 | MarketInvoice | |
| 0..* | Settlement | 0..* | Settlement | |
| 0..* | ContainerMarketInvoiceLineItem | 0..1 | MarketInvoiceLineItem | |
| 0..1 | ComponentMarketInvoiceLineItems | 0..* | MarketInvoiceLineItem | |

### 6.5.3.4 MarketLedger root class

In accounting transactions, a ledger is a book containing accounts to which debits and credits are posted from journals, where transactions are initially recorded. Journal entries are periodically posted to the ledger. Ledger actual represents actual amounts by account within ledger within company or within business area. Actual amounts may be generated in a source application and then loaded to a specific ledger within the enterprise general ledger or budget application.

Table 71 shows all association ends of MarketLedger with other classes.

**Table 71 – Association ends of MarketOpCommon::MarketLedger with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | MarketLedgerEntries | 0..* | MarketLedgerEntry | |

### 6.5.3.5 MarketLedgerEntry root class

Details of an individual entry in a ledger, which was posted from a journal on the posted date.

Table 72 shows all attributes of MarketLedgerEntry.

**Table 72 – Attributes of MarketOpCommon::MarketLedgerEntry**

| name | mult | type | description |
|---|---|---|---|
| accountID | 0..1 | String | Account identifier for this entry. |
| accountKind | 0..1 | MktAccountKind | Kind of account for this entry. |
| amount | 0..1 | Money | The amount of the debit or credit for this account. |
| postedDateTime | 0..1 | DateTime | Date and time this entry was posted to the ledger. |
| status | 0..1 | Status | Status of ledger entry. |
| transactionDateTime | 0..1 | DateTime | Date and time journal entry was recorded. |

Table 73 shows all association ends of MarketLedgerEntry with other classes.

**Table 73 – Association ends of MarketOpCommon:: MarketLedgerEntry with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Settlement | 0..* | Settlement | |
| 0..* | MarketLedger | 1..1 | MarketLedger | |

#### 6.5.3.6 MktActivityRecord

Subclass of IEC61968: Common:ActivityRecord.

Table 74 shows all attributes of MktActivityRecord.

**Table 74 – Attributes of MarketOpCommon::MktActivityRecord**

| name | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | inherited from: ActivityRecord |
| type | 0..1 | String | inherited from: ActivityRecord |
| severity | 0..1 | String | inherited from: ActivityRecord |
| reason | 0..1 | String | inherited from: ActivityRecord |
| status | 0..1 | Status | inherited from: ActivityRecord |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 75 shows all association ends of MktActivityRecord with other classes.

**Table 75 – Association ends of MarketOpCommon:: MktActivityRecord with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketFactors | 0..* | MarketFactors | |
| 0..* | Assets | 0..* | Asset | inherited from: ActivityRecord |
| 0..* | Author | 0..1 | Author | inherited from: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.3.7 MktConnectivityNode

Subclass of IEC61970:Topology:ConnectivityNode.

Table 76 shows all attributes of MktConnectivityNode.

**Table 76 – Attributes of MarketOpCommon::MktConnectivityNode**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 77 shows all association ends of MktConnectivityNode with other classes.

**Table 77 – Association ends of MarketOpCommon:: MktConnectivityNode with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 1..1 | NodeConstraintTerm | 0..* | NodeConstraintTerm | |
| 0..* | RTO | 1..1 | RTO | |
| 1..1 | SysLoadDistribuFactor | 0..1 | SysLoadDistributionFactor | |
| 1..1 | LossPenaltyFactor | 0..* | LossSensitivity | |
| 1..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 1..1 | IndividualPnode | 0..1 | IndividualPnode | |
| 0..* | TopologicalNode | 0..1 | TopologicalNode | inherited from: ConnectivityNode |
| 0..* | ConnectivityNodeContainer | 1..1 | ConnectivityNodeContainer | inherited from: ConnectivityNode |
| 0..1 | Terminals | 0..* | Terminal | inherited from: ConnectivityNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.3.8 MktLine

Subclass for IEC61970:Wires:Line.

Table 78 shows all attributes of MktLine.

**Table 78 – Attributes of MarketOpCommon::MktLine**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 79 shows all association ends of MktLine with other classes.

**Table 79 – Association ends of MarketOpCommon::MktLine with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..* | Flowgate | |
| 0..* | Region | 0..1 | SubGeographicalRegion | inherited from: Line |
| 0..1 | Equipments | 0..* | Equipment | inherited from: EquipmentContainer |
| 0..* | AdditionalGroupedEquipment | 0..* | Equipment | inherited from: EquipmentContainer |
| 1..1 | ConnectivityNodes | 0..* | ConnectivityNode | inherited from: ConnectivityNodeContainer |
| 0..1 | TopologicalNode | 0..* | TopologicalNode | inherited from: ConnectivityNodeContainer |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.3.9 MktMeasurement

Subclass of IEC61970:Meas:Measurement.

Table 80 shows all attributes of MktMeasurement.

**Table 80 – Attributes of MarketOpCommon::MktMeasurement**

| name | mult | type | description |
|---|---|---|---|
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 81 shows all association ends of MktMeasurement with other classes.

**Table 81 – Association ends of MarketOpCommon::MktMeasurement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..1 | Pnode | Allows Pnode an association to aggregated resources external DC ties or psuedo tie measurements. |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.3.10 MktPowerTransformer

Subclass of IEC61970:Wires:PowerTransformer.

Table 82 shows all attributes of MktPowerTransformer.

**Table 82 – Attributes of MarketOpCommon::MktPowerTransformer**

| name | mult | type | description |
|---|---|---|---|
| beforeShCircuitHighestOperatingCurrent | 0..1 | CurrentFlow | inherited from: PowerTransformer |
| beforeShCircuitHighestOperatingVoltage | 0..1 | Voltage | inherited from: PowerTransformer |
| beforeShortCircuitAnglePf | 0..1 | AngleDegrees | inherited from: PowerTransformer |
| highSideMinOperatingU | 0..1 | Voltage | inherited from: PowerTransformer |
| isPartOfGeneratorUnit | 0..1 | Boolean | inherited from: PowerTransformer |
| operationalValuesConsidered | 0..1 | Boolean | inherited from: PowerTransformer |
| vectorGroup | 0..1 | String | inherited from: PowerTransformer |
| aggregate | 0..1 | Boolean | inherited from: Equipment |

| name | mult | type | description |
|---|---|---|---|
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 83 shows all association ends of MktPowerTransformer with other classes.

**Table 83 – Association ends of MarketOpCommon:: MktPowerTransformer with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..* | Flowgate | |
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..1 | PowerTransformerEnd | 0..* | PowerTransformerEnd | inherited from: PowerTransformer |
| 0..1 | TransformerTanks | 0..* | TransformerTank | inherited from: PowerTransformer |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.3.11 MktTerminal

Subclass of IEC61970:Core:Terminal.

Table 84 shows all attributes of MktTerminal.

**Table 84 – Attributes of MarketOpCommon::MktTerminal**

| name | mult | type | description |
|---|---|---|---|
| phases | 0..1 | PhaseCode | inherited from: Terminal |
| connected | 0..1 | Boolean | inherited from: ACDCTerminal |
| sequenceNumber | 0..1 | Integer | inherited from: ACDCTerminal |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 85 shows all association ends of MktTerminal with other classes.

**Table 85 – Association ends of MarketOpCommon::MktTerminal with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..1 | Flowgate | |
| 1..1 | TerminalConstraintTerm | 0..* | TerminalConstraintTerm | |
| 0..* | ConductingEquipment | 1..1 | ConductingEquipment | inherited from: Terminal |
| 0..* | ConnectivityNode | 0..1 | ConnectivityNode | inherited from: Terminal |
| 0..* | TopologicalNode | 0..1 | TopologicalNode | inherited from: Terminal |
| 0..1 | RegulatingControl | 0..* | RegulatingControl | inherited from: Terminal |
| 1..* | NormalHeadFeeder | 0..1 | Feeder | inherited from: Terminal |
| 1..1 | HasFirstMutualCoupling | 0..* | MutualCoupling | inherited from: Terminal |
| 0..1 | TransformerEnd | 0..* | TransformerEnd | inherited from: Terminal |
| 1..1 | BranchGroupTerminal | 0..* | BranchGroupTerminal | inherited from: Terminal |
| 1..1 | SvPowerFlow | 0..* | SvPowerFlow | inherited from: Terminal |
| 1..1 | AuxiliaryEquipment | 0..* | AuxiliaryEquipment | inherited from: Terminal |
| 1..1 | HasSecondMutualCoupling | 0..* | MutualCoupling | inherited from: Terminal |
| 1..1 | TieFlow | 0..2 | TieFlow | inherited from: Terminal |
| 0..1 | EquipmentFaults | 0..* | EquipmentFault | inherited from: Terminal |
| 0..1 | ConverterDCSides | 0..* | ACDCConverter | inherited from: Terminal |
| 1..1 | RemoteInputSignal | 0..* | RemoteInputSignal | inherited from: Terminal |
| 1..* | BusNameMarker | 0..1 | BusNameMarker | inherited from: ACDCTerminal |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: ACDCTerminal |
| 0..1 | Measurements | 0..* | Measurement | inherited from: ACDCTerminal |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.3.12 MktUserAttribute

Subclass of IEC61968:Domain2:UserAttribute.

Table 86 shows all attributes of MktUserAttribute.

**Table 86 – Attributes of MarketOpCommon::MktUserAttribute**

| name | mult | type | description |
|---|---|---|---|
| sequenceNumber | 0..1 | Integer | inherited from: UserAttribute |
| name | 0..1 | String | inherited from: UserAttribute |
| value | 0..1 | StringQuantity | inherited from: UserAttribute |

Table 87 shows all association ends of MktUserAttribute with other classes.

**Table 87 – Association ends of MarketOpCommon::MktUserAttribute with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ChargeType | 0..* | ChargeType | |
| 0..* | MarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..* | PassThroughBill | 0..* | PassThroughBill | |
| 0..* | ChargeGroup | 0..* | ChargeGroup | |
| 1..1 | AttributeProperty | 0..* | AttributeProperty | |
| 0..* | BillDeterminant | 0..* | BillDeterminant | |
| 0..* | Transaction | 0..1 | Transaction | inherited from: UserAttribute |

### 6.5.4 Package MarketPlan

#### 6.5.4.1 General

Market plan definitions for planned markets, planned market events, actual market runs, actual market events.

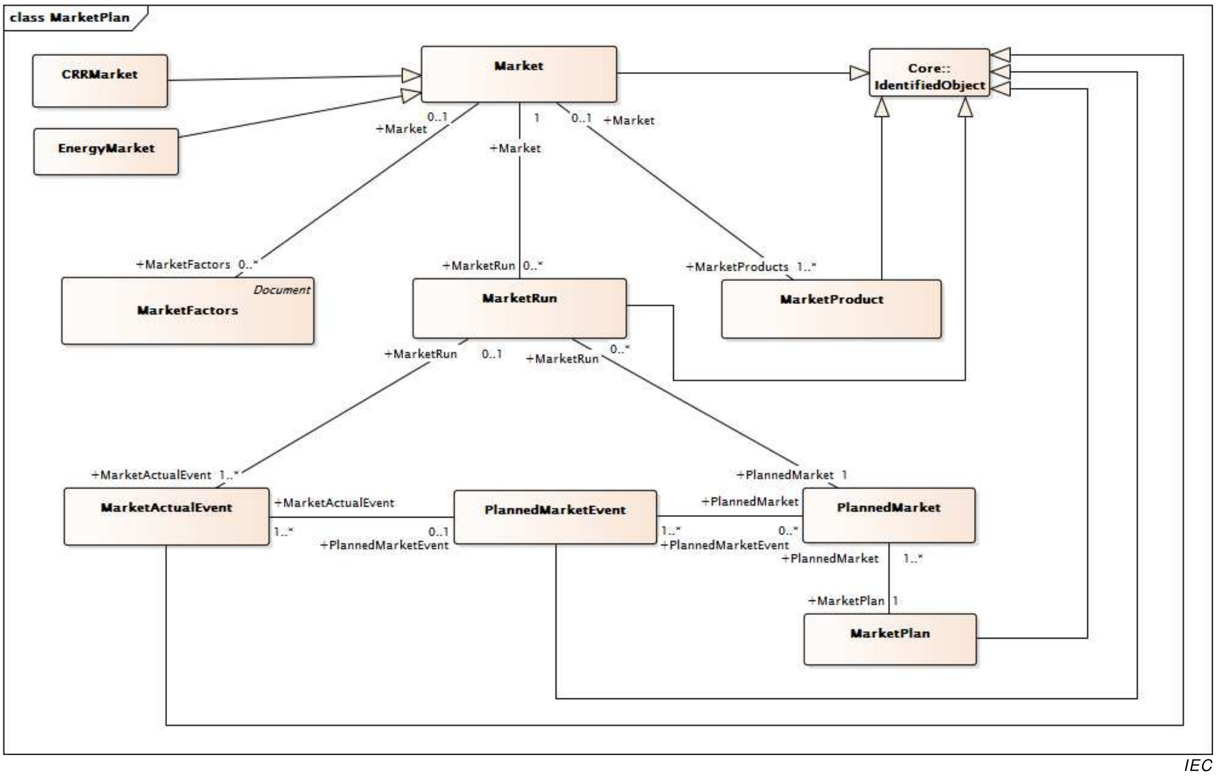



**Figure 35 – Class diagram MarketPlan::MarketPlan**

Figure 35: Shows the major classes that are used to identify an instance of a market clearing solution. The class Market identifies whether the instance is a Day Ahead, Hour-Ahead (Intraday) or Real Time market. The class MarketFactors describes a market interval as the start and end date/time of the market. The MarketProduct class indicates the market product such as energy or ancillary service. The class PlannedMarket describes the start and end times of the market. Relationships between classes are also shown.

#### 6.5.4.2 CommodityDefinition

Commodities in the context of IEC 62325 are MarketProducts (energy, regulation, reserve, etc) traded at a specific location, which in this case is a Pnode (either a specific pricing node or a pricing area or zone defined as a collection of pricing nodes). The CommodityDefinition is a container for these two parameters, plus the unit of measure and the currency in which the Commodity is traded. Each CommodityDefinition should be relatively static; defined once and rarely changed.

Table 88 shows all attributes of CommodityDefinition.

**Table 88 – Attributes of MarketPlan::CommodityDefinition**

| name | mult | type | description |
|---|---|---|---|
| commodityCurrency | 0..1 | Currency | The currency in which the Commodity is traded, using the standard conventions associated with the Currency enumeration. |
| commodityUnit | 0..1 | UnitSymbol | The unit of measure in which the Commodity is traded, using the standard conventions associated with the UnitSymbol enumeration. |
| commodityUnitMultiplier | 0..1 | UnitMultiplier | The unit multiplier, e.g. "k" to convert the unit "W-h" to "kW-h", using the standard conventions associated with the UnitMultiplier enumeration. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 89 shows all association ends of CommodityDefinition with other classes.

**Table 89 – Association ends of MarketPlan::CommodityDefinition with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 1..1 | Pnode | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..1 | CommodityPrice | 1..* | CommodityPrice | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.4.3 CRRMarket

Model that describes the Congestion Revenue Rights Auction Market.

Table 90 shows all attributes of CRRMarket.

**Table 90 – Attributes of MarketPlan::CRRMarket**

| name | mult | type | description |
|---|---|---|---|
| labelID | 0..1 | String | labelID – an ID for a set of apnodes/pnodes used in a CRR market |
| actualEnd | 0..1 | DateTime | inherited from: Market |
| actualStart | 0..1 | DateTime | inherited from: Market |
| dst | 0..1 | Boolean | inherited from: Market |
| end | 0..1 | DateTime | inherited from: Market |
| localTimeZone | 0..1 | String | inherited from: Market |
| start | 0..1 | DateTime | inherited from: Market |
| status | 0..1 | String | inherited from: Market |
| timeIntervalLength | 0..1 | Float | inherited from: Market |
| tradingDay | 0..1 | DateTime | inherited from: Market |
| tradingPeriod | 0..1 | String | inherited from: Market |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 91 shows all association ends of CRRMarket with other classes.

**Table 91 – Association ends of MarketPlan::CRRMarket with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | CongestionRevenueRight | 1..* | CongestionRevenueRight | |
| 0..1 | MarketFactors | 0..* | MarketFactors | inherited from: Market |
| 0..1 | MarketProducts | 1..* | MarketProduct | inherited from: Market |
| 1..1 | MarketRun | 0..* | MarketRun | inherited from: Market |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.4 DistributedResourceActualEvent

A demand response event is created when there is a need to call upon resources to respond to demand adjustment requests. These events are created by ISO/RTO system operations and managed by a demand response management system (DRMS). These events may or may not be coordinated with the Market Events and a defined Energy Market. The event will call for the deployment of a number of registered resources, or for deployment of resources within a zone (an organizational area within the power system that contains a number of resources).

Table 92 shows all attributes of DistributedResourceActualEvent.

**Table 92 – Attributes of MarketPlan::DistributedResourceActualEvent**

| name | mult | type | description |
|---|---|---|---|
| totalPowerAdjustment | 0..1 | ActivePower | Total active power adjustment (e.g. load reduction) requested for this demand response event. |
| eventComments | 1..1 | String | inherited from: MarketActualEvent |
| eventEndTime | 0..1 | DateTime | inherited from: MarketActualEvent |
| eventStartTime | 0..1 | DateTime | inherited from: MarketActualEvent |
| eventStatus | 1..1 | MarketEventStatusKind | inherited from: MarketActualEvent |
| eventType | 1..1 | String | inherited from: MarketActualEvent |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 93 shows all association ends of DistributedResourceActualEvent with other classes.

**Table 93 – Association ends of MarketPlan:: DistributedResourceActualEvent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | InstructionClearing | 0..* | InstructionClearing | ActualDemandResponseEvents may exist that are not part of a coordinated MarketActualEvent associated to a Market. These ActualDemandResponseEvents can have many InstructionClearing Instructions for specified RegisteredResources or DemandResponse AggregateNodes. |
| 1..1 | ResourcePerformanceEvaluations | 0..* | ResourcePerformanceEvaluation | |
| 0..1 | InstructionClearingDOT | 0..* | InstructionClearingDOT | |
| 1..* | MarketRun | 0..1 | MarketRun | inherited from: MarketActualEvent |
| 1..* | PlannedMarketEvent | 0..1 | PlannedMarketEvent | inherited from: MarketActualEvent |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.5 EnergyMarket

Energy and Ancillary Market (e.g. Energy, Spinning Reserve, Non-Spinning Reserve) with a description of the Market operation control parameters.

Table 94 shows all attributes of EnergyMarket.

**Table 94 – Attributes of MarketPlan::EnergyMarket**

| name | mult | type | description |
|---|---|---|---|
| actualEnd | 0..1 | DateTime | inherited from: Market |
| actualStart | 0..1 | DateTime | inherited from: Market |
| dst | 0..1 | Boolean | inherited from: Market |
| end | 0..1 | DateTime | inherited from: Market |
| localTimeZone | 0..1 | String | inherited from: Market |
| start | 0..1 | DateTime | inherited from: Market |
| status | 0..1 | String | inherited from: Market |
| timeIntervalLength | 0..1 | Float | inherited from: Market |
| tradingDay | 0..1 | DateTime | inherited from: Market |
| tradingPeriod | 0..1 | String | inherited from: Market |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 95 shows all association ends of EnergyMarket with other classes.

**Table 95 – Association ends of MarketPlan::EnergyMarket with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResources | 0..* | RegisteredResource | |
| 0..* | RTO | 0..1 | RTO | |
| 0..1 | Settlements | 0..* | Settlement | |
| 1..1 | MarketResults | 0..1 | MarketResults | |
| 1..1 | Bids | 0..* | Bid | |
| 0..1 | MarketFactors | 0..* | MarketFactors | inherited from: Market |
| 0..1 | MarketProducts | 1..* | MarketProduct | inherited from: Market |
| 1..1 | MarketRun | 0..* | MarketRun | inherited from: Market |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.6 Market

Market (e.g. Day Ahead Market, Real Time Market) with a description of the Market operation control parameters.

Table 96 shows all attributes of Market.

**Table 96 – Attributes of MarketPlan::Market**

| name | mult | type | description |
|---|---|---|---|
| actualEnd | 0..1 | DateTime | Market ending time – actual market end |
| actualStart | 0..1 | DateTime | Market starting time – actual market start |
| dst | 0..1 | Boolean | True if daylight savings time (DST) is in effect. |
| end | 0..1 | DateTime | Market end time. |
| localTimeZone | 0..1 | String | Local time zone. |
| start | 0..1 | DateTime | Market start time. |
| status | 0..1 | String | Market Status<br>'OPEN', 'CLOSED', 'CLEARED', 'BLOCKED' |
| timeIntervalLength | 0..1 | Float | Trading time interval length. |
| tradingDay | 0..1 | DateTime | Market trading date |
| tradingPeriod | 0..1 | String | Trading period that describes the market, possibilities could be for an Energy Market:<br>Day<br>Hour<br>For a CRR Market:<br>Year<br>Month<br>Season |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 97 shows all association ends of Market with other classes.

**Table 97 – Association ends of MarketPlan::Market with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MarketFactors | 0..* | MarketFactors | |
| 0..1 | MarketProducts | 1..* | MarketProduct | |
| 1..1 | MarketRun | 0..* | MarketRun | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.7 MarketActualEvent

This class represents the actual instance of an event.

Table 98 shows all attributes of MarketActualEvent.

**Table 98 – Attributes of MarketPlan::MarketActualEvent**

| name | mult | type | description |
|---|---|---|---|
| eventComments | 1..1 | String | Free format comments for the event, for any purpose needed. |
| eventEndTime | 0..1 | DateTime | End time of the event. |
| eventStartTime | 0..1 | DateTime | Start time of the event. |
| eventStatus | 1..1 | MarketEventStatusKind | Event status, e.g. active, canceled, expired, etc. |
| eventType | 1..1 | String | Actual event type. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 99 shows all association ends of MarketActualEvent with other classes.

**Table 99 – Association ends of MarketPlan::MarketActualEvent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | MarketRun | 0..1 | MarketRun | Market run triggered by this actual event. For example, the DA market run is triggered by the actual open bid submission event and terminated by the actual execution and completion of the DA market run captured by the runState of the MarketRun. |
| 1..* | PlannedMarketEvent | 0..1 | PlannedMarketEvent | Planned event executed by this actual event. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.8 MarketFactors

Aggregation of market information relative for a specific time interval.

Table 100 shows all attributes of MarketFactors.

**Table 100 – Attributes of MarketPlan::MarketFactors**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | The end of the time interval for which requirement is defined. |
| intervalStartTime | 0..1 | DateTime | The start of the time interval for which requirement is defined. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 101 shows all association ends of MarketFactors with other classes.

**Table 101 – Association ends of MarketPlan::MarketFactors with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | |
| 0..* | Market | 0..1 | Market | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.4.9 MarketPlan

This class identifies a set of planned markets.

Table 102 shows all attributes of MarketPlan.

**Table 102 – Attributes of MarketPlan::MarketPlan**

| name | mult | type | description |
|---|---|---|---|
| tradingDay | 0..1 | DateTime | Planned market trading day. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 103 shows all association ends of MarketPlan with other classes.

**Table 103 – Association ends of MarketPlan::MarketPlan with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | PlannedMarket | 1..* | PlannedMarket | A market plan has a number of markets (DA, HA, RT). |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.4.10 MarketProduct

A product traded by an RTO (e.g. energy, 10 minute spinning reserve). Ancillary service product examples include: Regulation, Regulation Up, Regulation Down, Spinning Reserve, Non-Spinning Reserve, etc.

Table 104 shows all attributes of MarketProduct.

**Table 104 – Attributes of MarketPlan::MarketProduct**

| name | mult | type | description |
|---|---|---|---|
| marketProductType | 0..1 | MarketProductType | Market product type examples:<br>EN (Energy)<br>RU (Regulation Up)<br>RD (Regulation Dn)<br>SR (Spinning Reserve)<br>NR (Non-Spinning Reserve)<br>RC (RUC) |
| rampInterval | 0..1 | Float | Ramping time interval for the specific market product type specified by marketProductType attribute. For example, if marketProductType = EN (from enumeration MarketProductType), then the rampInterval is the ramping time interval for Energy. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 105 shows all association ends of MarketProduct with other classes.

**Table 105 – Association ends of MarketPlan::MarketProduct with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | Market | 0..1 | Market | |
| 1..1 | ProductBids | 0..* | ProductBid | |
| 1..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | |
| 0..1 | BidError | 0..* | BidError | |
| 0..1 | BidPriceCap | 0..* | BidPriceCap | |
| 0..1 | MarketRegionResults | 0..1 | MarketRegionResults | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.11 MarketRun

This class represents an actual instance of a planned market. For example, a Day Ahead market opens with the Bid Submission, ends with the closing of the Bid Submission. The market run represent the whole process. MarketRuns can be defined for markets such as Day Ahead Market, Real Time Market, Hour Ahead Market, Week Ahead Market, etc.

Table 106 shows all attributes of MarketRun.

**Table 106 – Attributes of MarketPlan::MarketRun**

| name | mult | type | description |
|---|---|---|---|
| executionType | 0..1 | ExecutionType | The execution type; Day Ahead, Intra Day, Real Time Pre-Dispatch, Real Time Dispatch |
| marketApprovalTime | 0..1 | DateTime | Approved time for case. Identifies the time that the dispatcher approved a specific real time unit dispatch case |
| marketApprovedStatus | 0..1 | Boolean | Set to true when the plan is approved by authority and becomes the official plan for the day ahead market. Identifies the approved case for the market for the specified time interval. |
| marketEndTime | 0..1 | DateTime | The end time defined as the end of the market, market end time. |
| marketStartTime | 0..1 | DateTime | The start time defined as the beginning of the market, market start time. |
| marketType | 0..1 | MarketType | The market type, Day Ahead Market or Real Time Market. |
| reportedState | 0..1 | String | This is the state of market run activity as reported by market systems to the market definition services. |
| runState | 0..1 | String | This is the state controlled by market defintion service.<br>Possible values could be but not limited by: Open, Close. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 107 shows all association ends of MarketRun with other classes.

**Table 107 – Association ends of MarketPlan::MarketRun with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Market | 1..1 | Market | |
| 0..1 | MarketActualEvent | 1..* | MarketActualEvent | All actual events that trigger this market run. |
| 0..* | PlannedMarket | 1..1 | PlannedMarket | A planned market could have multiple market runs for the reason that a planned market could have a rerun. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.4.12 PlannedMarket root class

Represent a planned market. For example a planned DA/HA/RT market.

Table 108 shows all attributes of PlannedMarket.

**Table 108 – Attributes of MarketPlan::PlannedMarket**

| name | mult | type | description |
|---|---|---|---|
| marketEndTime | 0..1 | DateTime | Market end time. |
| marketStartTime | 0..1 | DateTime | Market start time. |
| marketType | 0..1 | MarketType | Market type. |

Table 109 shows all association ends of PlannedMarket with other classes.

**Table 109 – Association ends of MarketPlan::PlannedMarket with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | MarketRun | 0..* | MarketRun | A planned market could have multiple market runs for the reason that a planned market could have a rerun. |
| 0..* | PlannedMarketEvent | 1..* | PlannedMarketEvent | A planned market shall have a set of planned events |
| 1..* | MarketPlan | 1..1 | MarketPlan | A market plan has a number of markets (DA, HA, RT). |

### 6.5.4.13 PlannedMarketEvent

This class represents planned events. Used to model the various planned events in a market (closing time, clearing time, etc.)

Table 110 shows all attributes of PlannedMarketEvent.

**Table 110 – Attributes of MarketPlan::PlannedMarketEvent**

| name | mult | type | description |
|---|---|---|---|
| eventType | 0..1 | String | Planned event type. |
| plannedTime | 0..1 | Integer | This is relative time so that this attribute can be used by more than one planned market. For example the bid submission is 10am everyday. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 111 shows all association ends of PlannedMarketEvent with other classes.

**Table 111 – Association ends of MarketPlan::PlannedMarketEvent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MarketActualEvent | 1..* | MarketActualEvent | All actual events that execute this planned event. |
| 1..* | PlannedMarket | 0..* | PlannedMarket | A planned market shall have a set of planned events |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.5 Package MarketQualitySystem

#### 6.5.5.1 General

Post-market accounting, calculation and meter data corrections to reduce invoicing errors and disputes. Reduces manual validation, verification and correction of transactional data that could affect market settlements. Republishing of market results with affected data corrected.

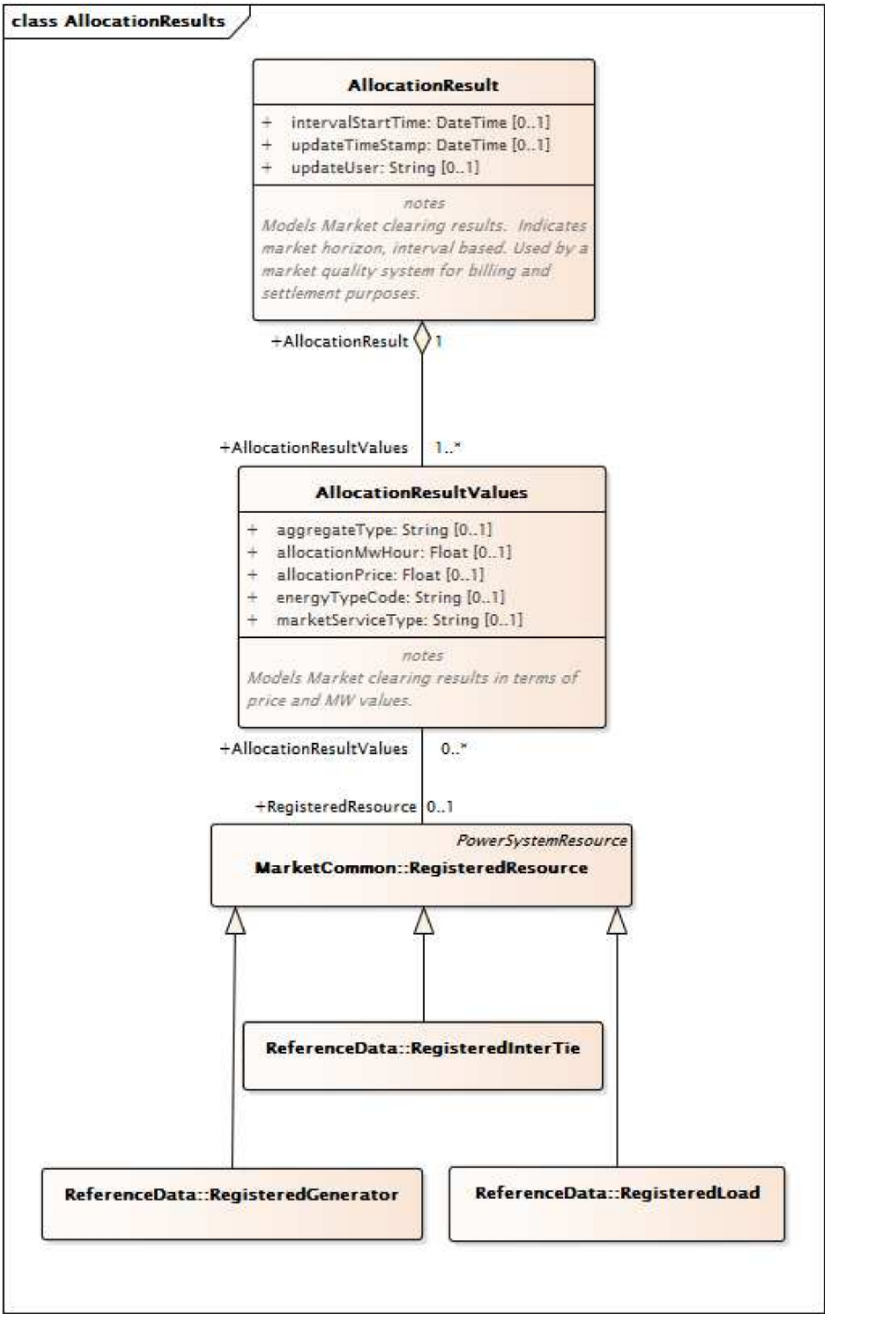



**Figure 36 – Class diagram MarketQualitySystem::AllocationResults**

Figure 36: Shows the major classes associated with the Market Quality Systems (MQS) which is typically used to perform analysis prior to Billing and Settlement. It describes the results that are sent to the MQS. Major classes include AllocationResult which define the market horizon. The allocated results are described in the classes AllocatinResultsValues and RegisteredResource class and subclasses for Generators, Interties, and Loads.

#### 6.5.5.2 AllocationResult root class

Models Market clearing results. Indicates market horizon, interval based. Used by a market quality system for billing and settlement purposes.

Table 112 shows all attributes of AllocationResult.

**Table 112 – Attributes of MarketQualitySystem::AllocationResult**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Table 113 shows all association ends of AllocationResult with other classes.

**Table 113 – Association ends of MarketQualitySystem:: AllocationResult with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | AllocationResultValues | 1..* | AllocationResultValues | |

#### 6.5.5.3 AllocationResultValues root class

Models Market clearing results in terms of price and MW values.

Table 114 shows all attributes of AllocationResultValues.

**Table 114 – Attributes of MarketQualitySystem::AllocationResultValues**

| name | mult | type | description |
|---|---|---|---|
| aggregateType | 0..1 | String | "1" -- "Detail",<br>"2" -- "Aggregate by Market service type", in which case, the "AllocationEnergyType" field will not be filled;<br>"3" -- "Aggregate by "AllocationEnergyType", in which case "MarketServiceType" will not be filled. |
| allocationMwHour | 0..1 | Float | |
| allocationPrice | 0..1 | Float | |
| energyTypeCode | 0..1 | String | |
| marketServiceType | 0..1 | String | Choices are:<br>ME – Market Energy Capacity;<br>SR – Spinning Reserve Capacity;<br>NR – Non-Spinning Reserve Capacity;<br>DAC – Day Ahead Capacity;<br>DEC – Derate Capacity |

Table 115 shows all association ends of AllocationResultValues with other classes.

**Table 115 – Association ends of MarketQualitySystem:: AllocationResultValues with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | AllocationResult | 1..1 | AllocationResult | |

### 6.5.5.4 AuxiliaryCost root class

Models Market clearing results for Auxiliary costs.

Table 116 shows all attributes of AuxiliaryCost.

**Table 116 – Attributes of MarketQualitySystem::AuxiliaryCost**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| marketType | 0..1 | MarketType | |
| updateTimeStamp | 0..1 | DateTime | |
| updateUser | 0..1 | String | |

Table 117 shows all association ends of AuxiliaryCost with other classes.

**Table 117 – Association ends of MarketQualitySystem::AuxiliaryCost with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryValues | 1..* | AuxiliaryValues | |

### 6.5.5.5 AuxiliaryObject root class

Models Auxiliary Values.

Table 118 shows all association ends of AuxiliaryObject with other classes.

**Table 118 – Association ends of MarketQualitySystem:: AuxiliaryObject with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |

### 6.5.5.6 AuxiliaryValues

Models Auxiliary Values.

Table 119 shows all attributes of AuxiliaryValues.

**Table 119 – Attributes of MarketQualitySystem::AuxiliaryValues**

| name | mult | type | description |
|---|---|---|---|
| minExpostCapacity | 0..1 | Float | |
| maxExpostCapacity | 0..1 | Float | |
| availUndispatchedQ | 0..1 | Float | |
| incrementalORAvail | 0..1 | Float | |
| startUpCost | 0..1 | Float | |
| startUpCostEligibilityFlag | 0..1 | YesNo | |
| noLoadCost | 0..1 | Float | |
| noLoadCostEligibilityFlag | 0..1 | YesNo | |

Table 120 shows all association ends of AuxiliaryValues with other classes.

**Table 120 – Association ends of MarketQualitySystem:: AuxiliaryValues with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | AuxillaryCost | 1..1 | AuxiliaryCost | |
| 1..* | TenMinAuxillaryData | 1..1 | TenMinAuxiliaryData | |
| 1..* | FiveMinAuxillaryData | 1..1 | FiveMinAuxiliaryData | |
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | inherited from: AuxiliaryObject |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | inherited from: AuxiliaryObject |

### 6.5.5.7 ExpectedEnergy root class

Model Expected Energy from Market Clearing, interval based.

Table 121 shows all attributes of ExpectedEnergy.

**Table 121 – Attributes of MarketQualitySystem::ExpectedEnergy**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Table 122 shows all association ends of ExpectedEnergy with other classes.

**Table 122 – Association ends of MarketQualitySystem:: ExpectedEnergy with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ExpectedEnergyValues | 1..* | ExpectedEnergyValues | |

#### 6.5.5.8 ExpectedEnergyValues root class

Model Expected Energy from Market Clearing.

Table 123 shows all attributes of ExpectedEnergyValues.

**Table 123 – Attributes of MarketQualitySystem::ExpectedEnergyValues**

| name | mult | type | description |
|---|---|---|---|
| energyTypeCode | 0..1 | String | |
| expectedMwh | 0..1 | Float | |

Table 124 shows all association ends of ExpectedEnergyValues with other classes.

**Table 124 – Association ends of MarketQualitySystem:: ExpectedEnergyValues with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ExpectedEnergy | 1..1 | ExpectedEnergy | |

#### 6.5.5.9 FiveMinAuxiliaryData root class

Models 5-Minutes Auxiliary Data.

Table 125 shows all attributes of FiveMinAuxiliaryData.

**Table 125 – Attributes of MarketQualitySystem::FiveMinAuxiliaryData**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Table 126 shows all association ends of FiveMinAuxiliaryData with other classes.

**Table 126 – Association ends of MarketQualitySystem:: FiveMinAuxiliaryData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryValues | 1..* | AuxiliaryValues | |

#### 6.5.5.10 TenMinAuxiliaryData root class

Models 10-Minutes Auxiliary Data.

Table 127 shows all attributes of TenMinAuxiliaryData.

**Table 127 – Attributes of MarketQualitySystem::TenMinAuxiliaryData**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Table 128 shows all association ends of TenMinAuxiliaryData with other classes.

**Table 128 – Association ends of MarketQualitySystem:: TenMinAuxiliaryData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryData | 1..* | AuxiliaryValues | |

#### 6.5.5.11 TradingHubPrice root class

Models prices at Trading Hubs, interval based.

Table 129 shows all attributes of TradingHubPrice.

**Table 129 – Attributes of MarketQualitySystem::TradingHubPrice**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| marketType | 0..1 | MarketType | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Table 130 shows all association ends of TradingHubPrice with other classes.

**Table 130 – Association ends of MarketQualitySystem:: TradingHubPrice with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TradingHubValues | 1..* | TradingHubValues | |

#### 6.5.5.12 TradingHubValues root class

Models prices at Trading Hubs.

Table 131 shows all attributes of TradingHubValues.

**Table 131 – Attributes of MarketQualitySystem::TradingHubValues**

| name | mult | type | description |
|---|---|---|---|
| price | 0..1 | Float | Utilizes the Market type. For DA, the price is hourly. For RTM the price is a 5 minute price. |

Table 132 shows all association ends of TradingHubValues with other classes.

**Table 132 – Association ends of MarketQualitySystem:: TradingHubValues with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 1..* | TradingHubPrice | 1..1 | TradingHubPrice | |

### 6.5.6 Package MarketSystem

#### 6.5.6.1 General

Market results and external inputs consumed by markets.

#### 6.5.6.2 Package ExternalInputs

##### 6.5.6.2.1 General

Inputs to the market system from external sources.

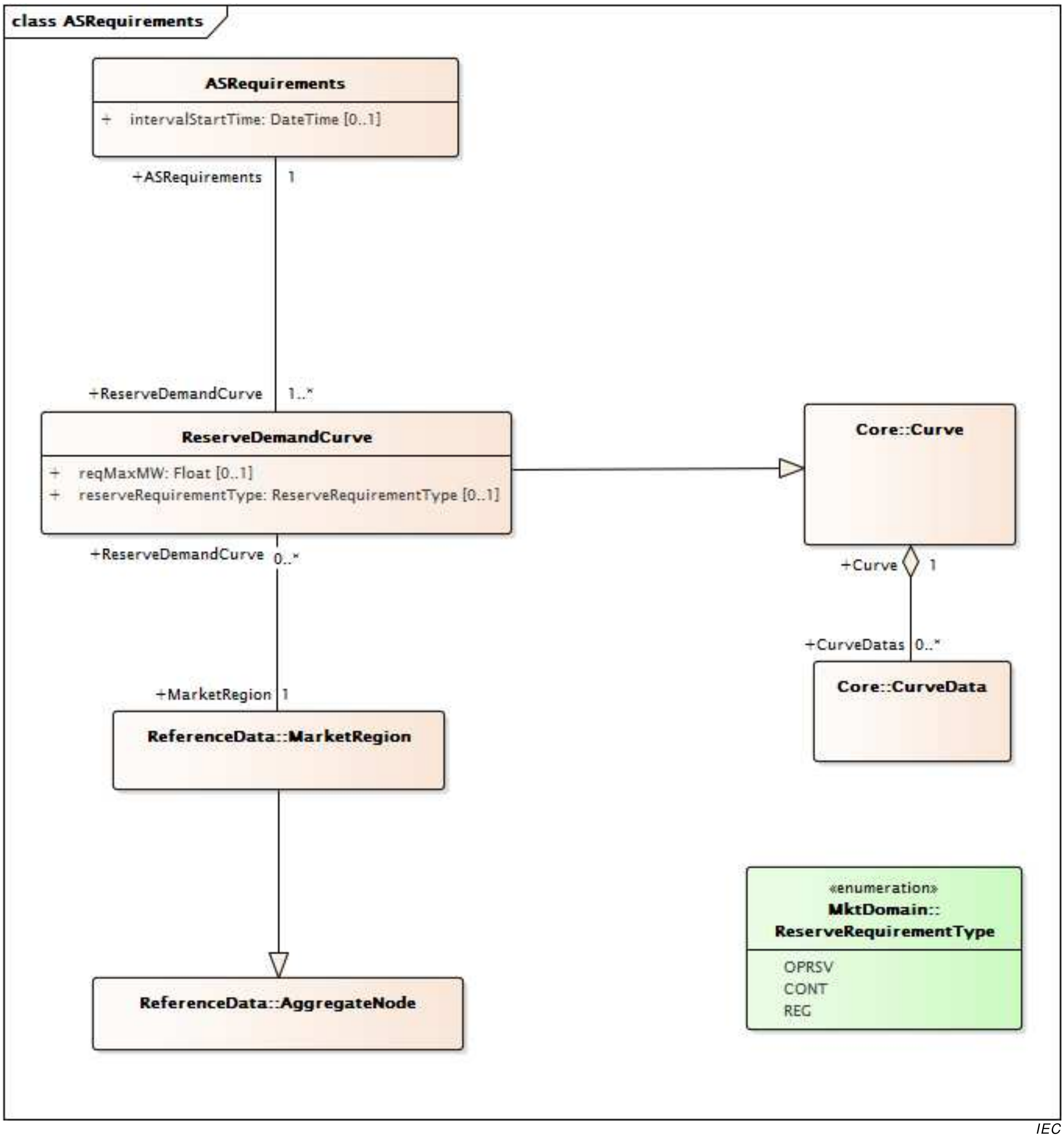



**Figure 37 – Class diagram ExternalInputs::ASRequirements**

Figure 37: Models Ancillary Service requirements. The class ASRequrements is used to describe the interval for which the Ancillary Service is required. The ASRequirement class is used to describe the quantities of required ancillary services. The association to MarketRegion models the regional requirements of ancillary services. The class MarketProduct is used to describe the particular class of ancillary service.

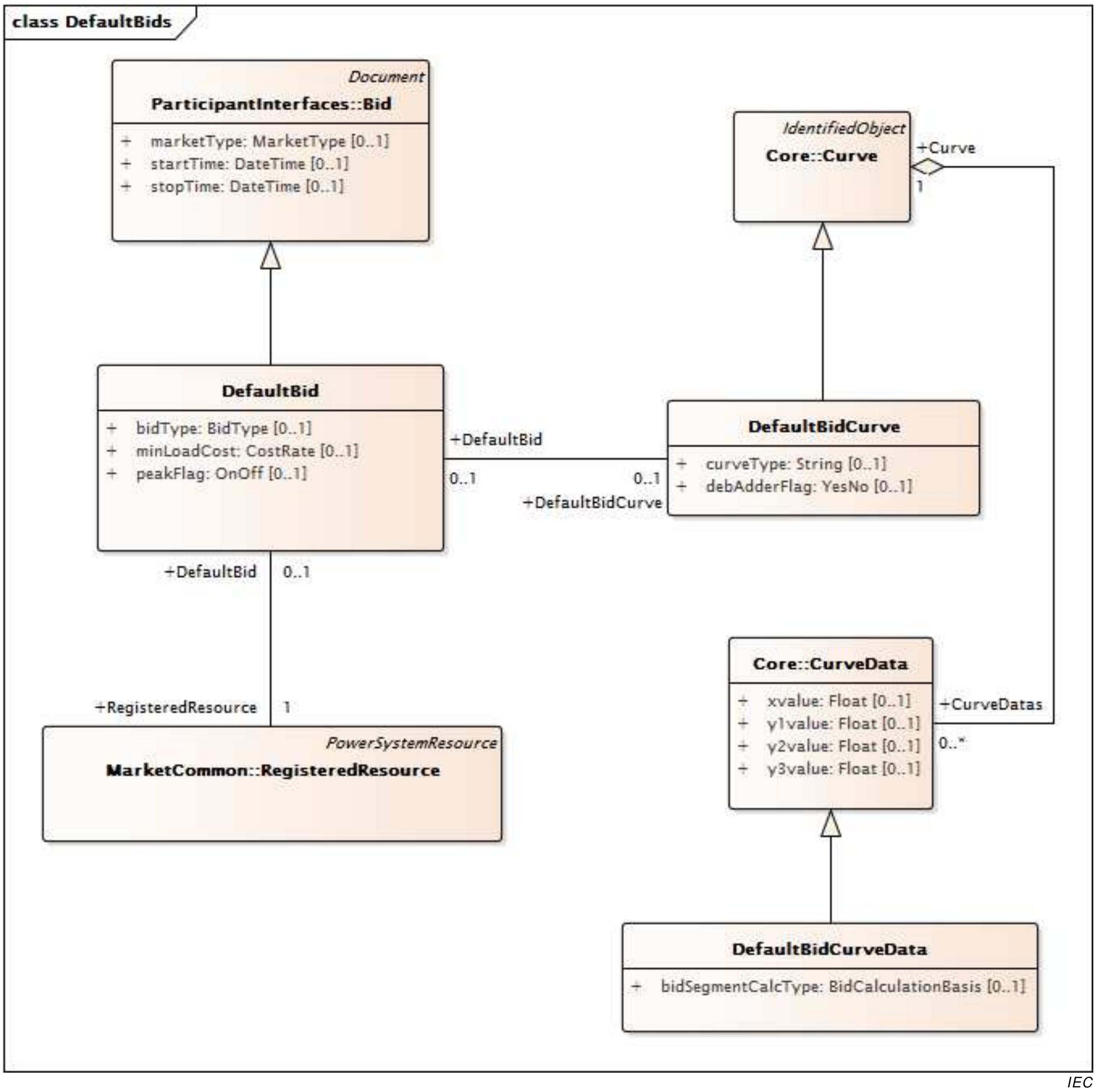



**Figure 38 – Class diagram ExternalInputs::DefaultBids**

Figure 38: Shows the classes used to model a default bid. Market Power Mitigation is a process that detects an attempt on the part of a market participant to exercise market power. Detection is followed by remediation. Remediation consist of subsitution of a default bid for the original bid. The classes include a Curve and Curve Data classes (from the IEC 61970 core package), as well as the generic class DefaultBid which may be used to further describe the market specific remediation methods. Three typical methods are used: LMP, cost based on gas index, or out-of-market negotiation.

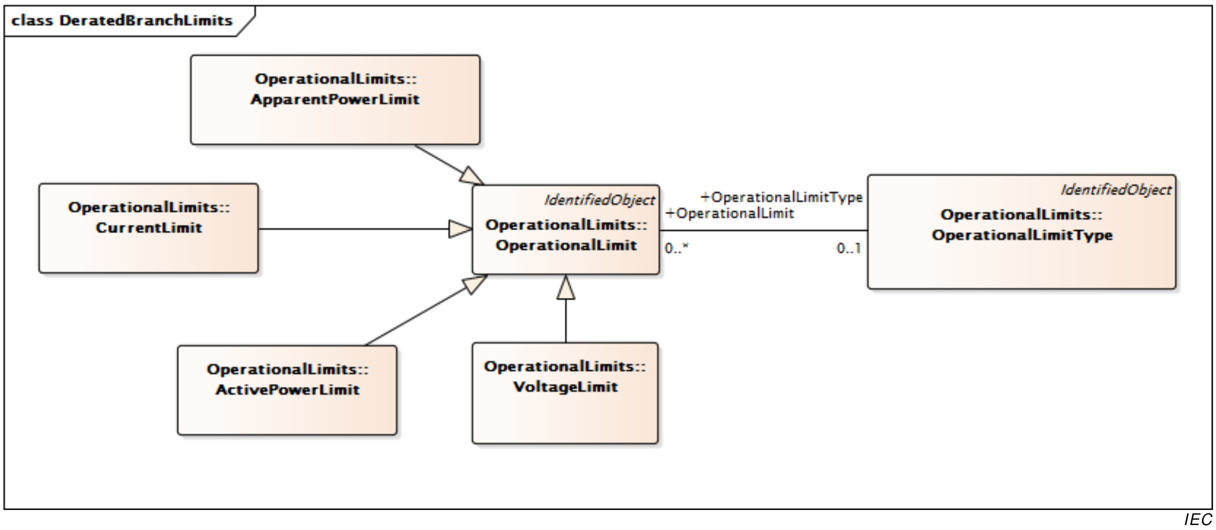
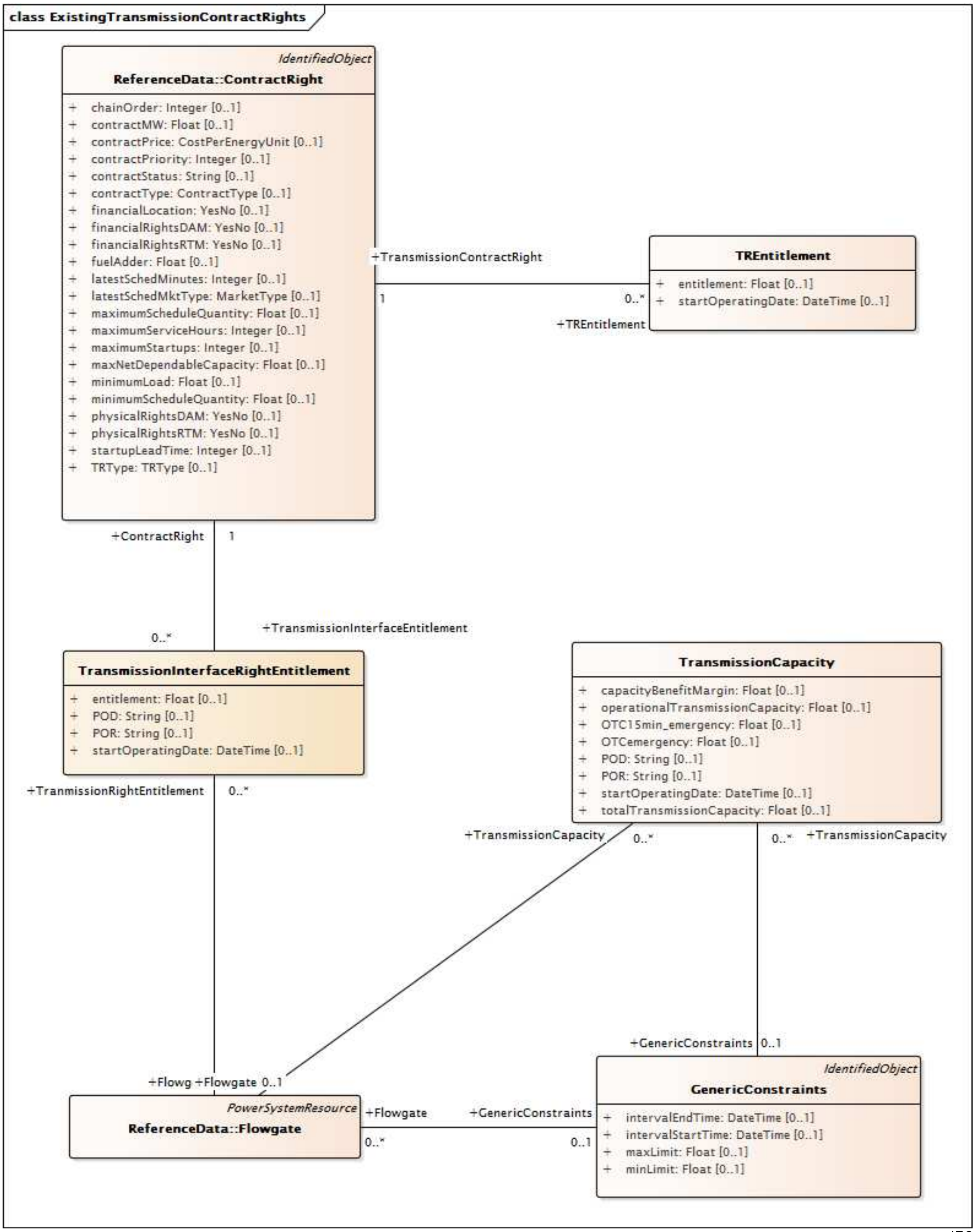



**Figure 39 – Class diagram ExternalInputs::DeratedBranchLimits**

Figure 39: Shows the major classes used to model operational limits. The classes are reused from the IEC 61970 OperationalLimits package and include Active Power, Voltage, and Current Limits.




IEC

**Figure 40 – Class diagram ExternalInputs::ExistingTransmissionContractRights**

Figure 40: Shows the major classes used to model these legacy rights. During the startup and evolution of LMP markets, the need arose to accommodate existing transmission rights in the market solution. The ContractRight class includes information describing the specific rights that are contracted in terms of active power and other contract details. The TransmissionInterfaceRightEntitlement class describs the physical aspects of the right in associated point of delivery and point of receipt of active power.

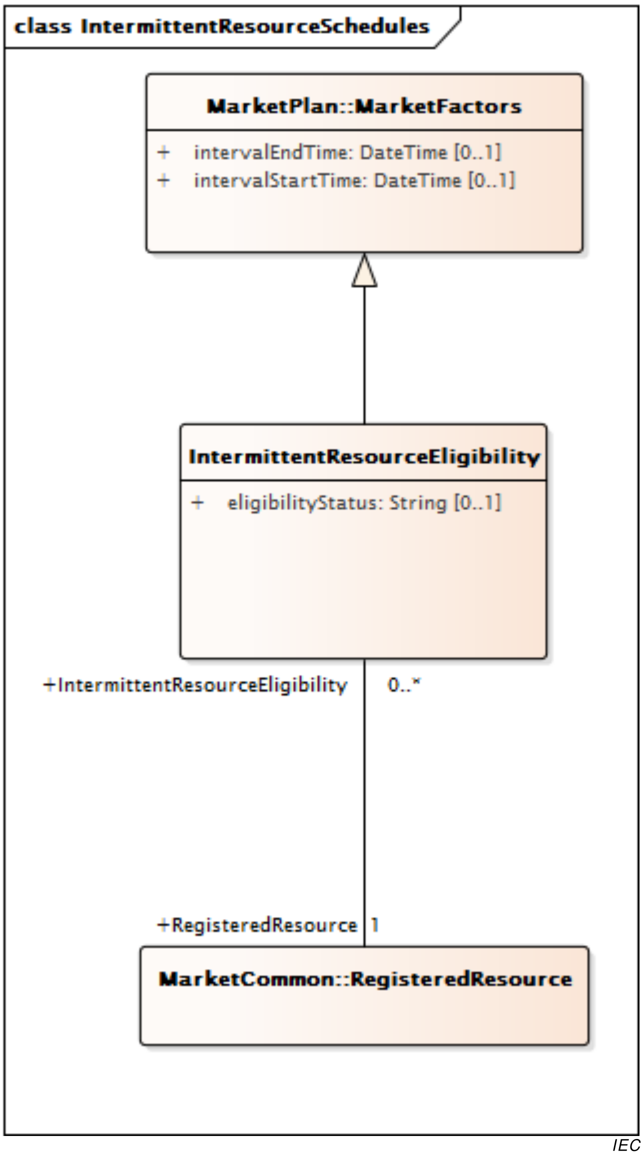



**Figure 41 – Class diagram ExternalInputs::IntermittentResourceSchedules**

Figure 41: Shows the major classes used to model intermittent (variable) resources. It is typically used to model renewables such as wind power. The IntermittentResourceEligbility class is used to indicate whether the resource is eligible to participate as a renewable resource governed by a generic renewable resource regime.

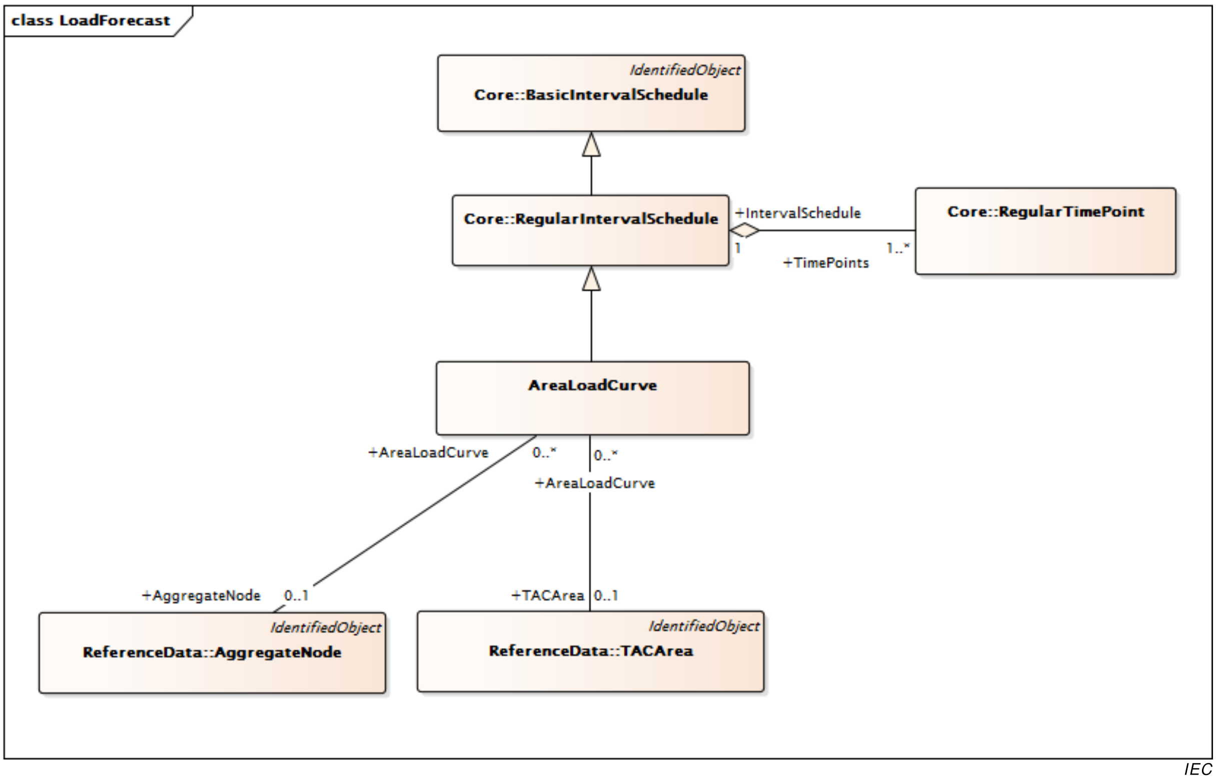
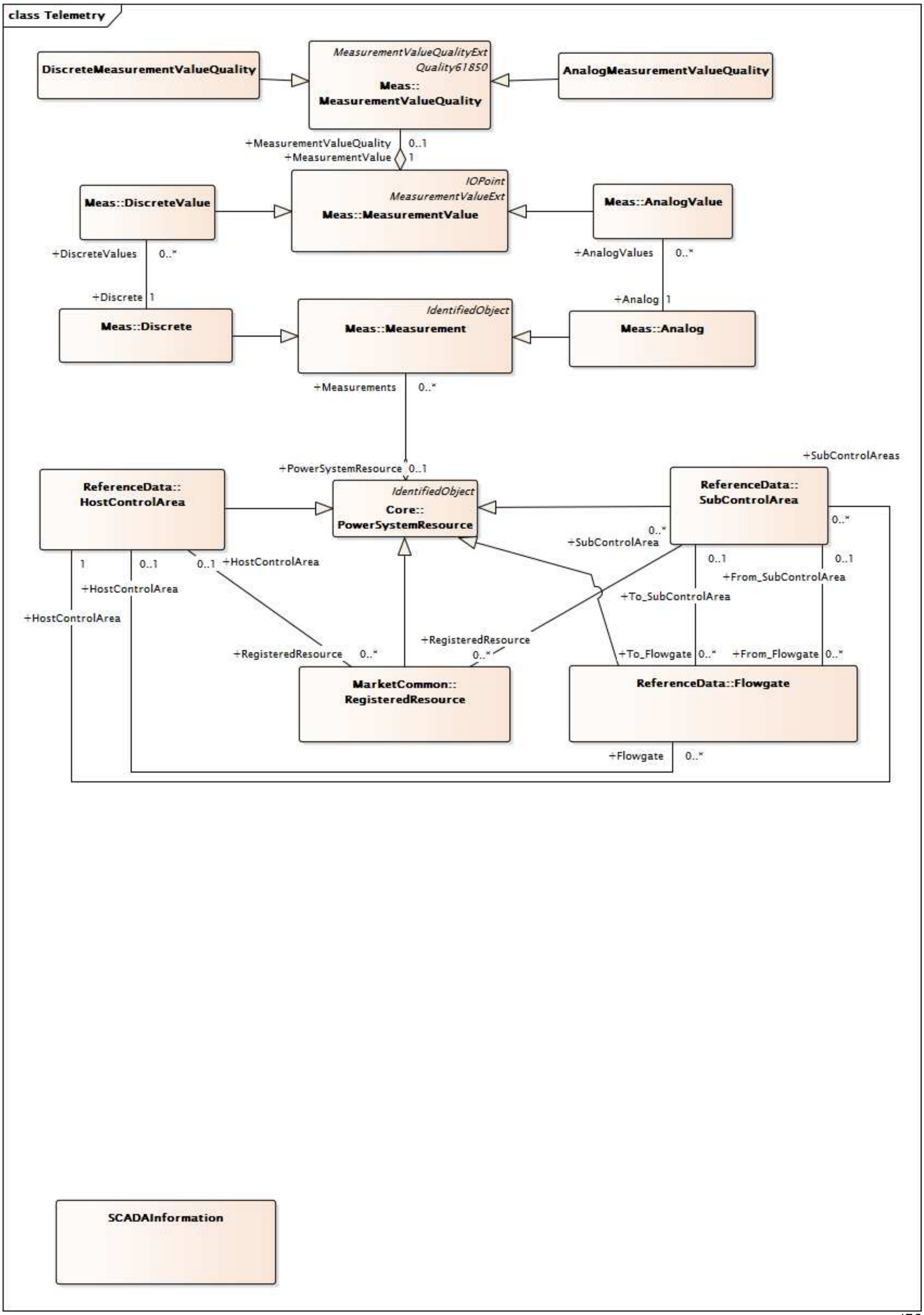



**Figure 42 – Class diagram ExternalInputs::LoadForecast**

Figure 42: Shows the classes used to model the load forecast over the market time horizon. The model reused classes from the IEC 61970 Core package and adds specific models for markets.




IEC

**Figure 43 – Class diagram ExternalInputs::Telemetry**

Figure 43: Shows the classes used to model telemetry that may be used (in the event of failure or non-availability of a state estimator solution) to initialize a real time market run. The

figure shows classes from the IEC 61970 package and additional market specific RegisteredResource and Flowgate classes and their relationships to the classes from the IEC 61970 packages.

#### 6.5.6.2.2 ASRequirements root class

Models Ancillary Service Requirements. Describes interval for which the requirement is applicable.

Table 133 shows all attributes of ASRequirements.

**Table 133 – Attributes of ExternalInputs::ASRequirements**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | The start of the time interval for which requirement is defined. |

Table 134 shows all association ends of ASRequirements with other classes.

**Table 134 – Association ends of ExternalInputs::ASRequirements with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ReserveDemandCurve | 1..* | ReserveDemandCurve | |

#### 6.5.6.2.3 AnalogMeasurementValueQuality

Measurement quality flags for Analog Values.

Table 135 shows all attributes of AnalogMeasurementValueQuality.

**Table 135 – Attributes of ExternalInputs::AnalogMeasurementValueQuality**

| name | mult | type | description |
|---|---|---|---|
| scadaQualityCode | 0..1 | String | The quality code for the given Analog Value. |
| badReference | 0..1 | Boolean | inherited from: Quality61850 |
| estimatorReplaced | 0..1 | Boolean | inherited from: Quality61850 |
| failure | 0..1 | Boolean | inherited from: Quality61850 |
| oldData | 0..1 | Boolean | inherited from: Quality61850 |
| operatorBlocked | 0..1 | Boolean | inherited from: Quality61850 |
| oscillatory | 0..1 | Boolean | inherited from: Quality61850 |
| outOfRange | 0..1 | Boolean | inherited from: Quality61850 |
| overFlow | 0..1 | Boolean | inherited from: Quality61850 |
| source | 0..1 | Source | inherited from: Quality61850 |
| suspect | 0..1 | Boolean | inherited from: Quality61850 |
| test | 0..1 | Boolean | inherited from: Quality61850 |
| validity | 0..1 | Validity | inherited from: Quality61850 |

Table 136 shows all association ends of AnalogMeasurementValueQuality with other classes.

**Table 136 – Association ends of ExternalInputs:: AnalogMeasurementValueQuality with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MeasurementValue | 1..1 | MeasurementValue | inherited from: MeasurementValueQuality |

#### 6.5.6.2.4 AreaLoadCurve

Area load curve definition.

Table 137 shows all attributes of AreaLoadCurve.

**Table 137 – Attributes of ExternalInputs::AreaLoadCurve**

| name | mult | type | description |
|---|---|---|---|
| forecastType | 0..1 | LoadForecastType | Load forecast area type. |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 138 shows all association ends of AreaLoadCurve with other classes.

**Table 138 – Association ends of ExternalInputs::AreaLoadCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AggregateNode | 0..1 | AggregateNode | |
| 0..* | TACArea | 0..1 | TACArea | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.5 BaseCaseConstraintLimit

Possibly time-varying max MW or MVA and optionally Min MW limit or MVA limit (Y1 and Y2, respectively) assigned to a contingency analysis base case. Use CurveSchedule XAxisUnits to specify MW or MVA. To be used only if the BaseCaseConstraintLimit differs from the DefaultConstraintLimit.

Table 139 shows all attributes of BaseCaseConstraintLimit.

**Table 139 – Attributes of ExternalInputs::BaseCaseConstraintLimit**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 140 shows all association ends of BaseCaseConstraintLimit with other classes.

**Table 140 – Association ends of ExternalInputs:: BaseCaseConstraintLimit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.6 BranchEndFlow root class

Dynamic flows and ratings associated with a branch end.

Table 141 shows all attributes of BranchEndFlow.

**Table 141 – Attributes of ExternalInputs::BranchEndFlow**

| name | mult | type | description |
|---|---|---|---|
| mwFlow | 0..1 | Float | The MW flow on the branch<br>Attribute Usage: Active power flow at the series device, transformer, phase shifter, or line end |
| mVARFlow | 0..1 | Float | The MVAR flow on the branch<br>Attribute Usage: Reactive power flow at the series device, transformer, phase shifter, or line end |
| normalRating | 0..1 | Float | The Normal Rating for the branch |
| longTermRating | 0..1 | Float | The Long Term Rating for the branch |
| shortTermRating | 0..1 | Float | The Short Term Rating for the branch |
| loadDumpRating | 0..1 | Float | The Load Dump Rating for the branch |

Table 142 shows all association ends of BranchEndFlow with other classes.

**Table 142 – Association ends of ExternalInputs::BranchEndFlow with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MktPowerTransformerEndBFlow | 0..* | MktPowerTransformer | |
| 0..1 | MktPowerTransformerEndAFlow | 0..* | MktPowerTransformer | |
| 0..1 | MktACLineSegmentEndAFlow | 0..* | MktACLineSegment | |
| 0..1 | MktACLineSegmentEndBFlow | 0..* | MktACLineSegment | |
| 0..1 | MktSeriesCompensatorEndBFlow | 0..* | MktSeriesCompensator | |
| 0..1 | MktSeriresCompensatorEndAFlow | 0..* | MktSeriesCompensator | |

#### 6.5.6.2.7 ConstraintTerm

A constraint term is one element of a linear constraint.

Table 143 shows all attributes of ConstraintTerm.

**Table 143 – Attributes of ExternalInputs::ConstraintTerm**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | |
| function | 0..1 | String | The function is an enumerated value that can be 'active', 'reactive', or 'VA' to indicate the type of flow. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 144 shows all association ends of ConstraintTerm with other classes.

**Table 144 – Association ends of ExternalInputs::ConstraintTerm with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.8 ContingencyConstraintLimit

Possibly time-varying max MW or MVA and optionally Min MW limit or MVA limit (Y1 and Y2, respectively) assigned to a constraint for a specific contingency. Use CurveSchedule XAxisUnits to specify MW or MVA.

Table 145 shows all attributes of ContingencyConstraintLimit.

**Table 145 – Attributes of ExternalInputs::ContingencyConstraintLimit**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 146 shows all association ends of ContingencyConstraintLimit with other classes.

**Table 146 – Association ends of ExternalInputs:: ContingencyConstraintLimit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktContingency | 1..1 | MktContingency | |
| 1..1 | MWLimitSchedules | 1..1 | MWLimitSchedule | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.9 ControlAreaSolutionData root class

State Estimator Solution Pool Interchange and Losses.

Table 147 shows all attributes of ControlAreaSolutionData.

**Table 147 – Attributes of ExternalInputs::ControlAreaSolutionData**

| name | mult | type | description |
|---|---|---|---|
| solvedLosses | 0..1 | Float | Pool Losses MW<br>Attribute Usage: The active power losses of the pool in MW |
| solvedInterchange | 0..1 | Float | Pool MW Interchange<br>Attribute Usage: The active power interchange of the pool |

Table 148 shows all association ends of ControlAreaSolutionData with other classes.

**Table 148 – Association ends of ExternalInputs:: ControlAreaSolutionData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktControlArea | 0..1 | MktControlArea | |

#### 6.5.6.2.10 DefaultBid

DefaultBid is a generic class to hold Default Energy Bid, Default Startup Bid, and Default Minimum Load Bid:

Default Energy Bid

A Default Energy Bid is a monotonically increasing staircase function consisting at maximum 10 economic bid segments, or 10 ($/MW, MW) pairs. There are three methods for determining the Default Energy Bid:

– Cost Based: derived from the Heat Rate or Average Cost multiplied by the Gas Price Index plus 10%.

– LMP Based: a weighted average of LMPs in the preceding 90 days.

– Negotiated: an amount negotiated with the designated Independent Entity.

Default Startup Bid

A Default Startup Bid (DSUB) shall be calculated for each RMR unit based on the Startup Cost stored in the Master File and the applicable GPI and EPI.

Default Minimum Load Bid

A Default Minimum Load Bid (DMLB) shall be calculated for each RMR unit based on the Minimum Load Cost stored in the Master File and the applicable GPI.

Table 149 shows all attributes of DefaultBid.

**Table 149 – Attributes of ExternalInputs::DefaultBid**

| name | mult | type | description |
|---|---|---|---|
| bidType | 0..1 | BidType | Default bid type such as Default Energy Bid, Default Minimum Load Bid, and Default Startup Bid |
| minLoadCost | 0..1 | CostRate | Minimum load cost in $/hr |
| peakFlag | 0..1 | OnOff | on-peak, off-peak, or all |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 150 shows all association ends of DefaultBid with other classes.

**Table 150 – Association ends of ExternalInputs::DefaultBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DefaultBidCurve | 0..1 | DefaultBidCurve | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.11 DefaultBidCurve

Default bid curve for default energy bid curve and default startup curves (cost and time).

Table 151 shows all attributes of DefaultBidCurve.

**Table 151 – Attributes of ExternalInputs::DefaultBidCurve**

| name | mult | type | description |
|---|---|---|---|
| curveType | 0..1 | String | To indicate a type used for a default energy bid curve, such as LMP, cost or consultative based. |
| debAdderFlag | 0..1 | YesNo | Default energy bid adder flag |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 152 shows all association ends of DefaultBidCurve with other classes.

**Table 152 – Association ends of ExternalInputs::DefaultBidCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | DefaultBid | 0..1 | DefaultBid | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.12 DefaultBidCurveData

Curve data for default bid curve and startup cost curve.

Table 153 shows all attributes of DefaultBidCurveData.

**Table 153 – Attributes of ExternalInputs::DefaultBidCurveData**

| name | mult | type | description |
|---|---|---|---|
| bidSegmentCalcType | 0..1 | BidCalculationBasis | Type of calculation basis used to define the default bid segment curve. |
| xvalue | 0..1 | Float | inherited from: CurveData |
| y1value | 0..1 | Float | inherited from: CurveData |
| y2value | 0..1 | Float | inherited from: CurveData |
| y3value | 0..1 | Float | inherited from: CurveData |

Table 154 shows all association ends of DefaultBidCurveData with other classes.

**Table 154 – Association ends of ExternalInputs:: DefaultBidCurveData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Curve | 1..1 | Curve | inherited from: CurveData |

#### 6.5.6.2.13 DefaultConstraintLimit

Possibly time-varying max MW or MVA and optionally Min MW limit or MVA limit (Y1 and Y2, respectively) applied as a default value if no specific constraint limits are specified for a contingency analysis. Use CurveSchedule XAxisUnits to specify MW or MVA.

Table 155 shows all attributes of DefaultConstraintLimit.

**Table 155 – Attributes of ExternalInputs::DefaultConstraintLimit**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 156 shows all association ends of DefaultConstraintLimit with other classes.

**Table 156 – Association ends of ExternalInputs:: DefaultConstraintLimit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.14 DiscreteMeasurementValueQuality

Measurement quality flags for Discrete Values.

Table 157 shows all attributes of DiscreteMeasurementValueQuality.

**Table 157 – Attributes of ExternalInputs::DiscreteMeasurementValueQuality**

| name | mult | type | description |
|---|---|---|---|
| manualReplaceIndicator | 0..1 | Boolean | Switch Manual Replace Indicator. Flag indicating that the switch is manual replace. |
| removeFromOperationIndicator | 0..1 | Boolean | Removed From Operation Indicator. Flag indicating that the switch is removed from operation. |
| badReference | 0..1 | Boolean | inherited from: Quality61850 |
| estimatorReplaced | 0..1 | Boolean | inherited from: Quality61850 |
| failure | 0..1 | Boolean | inherited from: Quality61850 |
| oldData | 0..1 | Boolean | inherited from: Quality61850 |
| operatorBlocked | 0..1 | Boolean | inherited from: Quality61850 |
| oscillatory | 0..1 | Boolean | inherited from: Quality61850 |
| outOfRange | 0..1 | Boolean | inherited from: Quality61850 |
| overFlow | 0..1 | Boolean | inherited from: Quality61850 |
| source | 0..1 | Source | inherited from: Quality61850 |
| suspect | 0..1 | Boolean | inherited from: Quality61850 |
| test | 0..1 | Boolean | inherited from: Quality61850 |
| validity | 0..1 | Validity | inherited from: Quality61850 |

Table 158 shows all association ends of DiscreteMeasurementValueQuality with other classes.

**Table 158 – Association ends of ExternalInputs:: DiscreteMeasurementValueQuality with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MeasurementValue | 1..1 | MeasurementValue | inherited from: MeasurementValueQuality |

#### 6.5.6.2.15 DistributionFactorSet root class

A containing class that groups all the distribution factors within a market.

This is calculated daily for DA factors and hourly for RT factors.

Table 159 shows all attributes of DistributionFactorSet.

**Table 159 – Attributes of ExternalInputs::DistributionFactorSet**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | The start of the time interval for which requirement is defined. |
| intervalEndTime | 0..1 | DateTime | The end of the time interval for which requirement is defined. |
| marketType | 0..1 | MarketType | Market type. |

Table 160 shows all association ends of DistributionFactorSet with other classes.

**Table 160 – Association ends of ExternalInputs:: DistributionFactorSet with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | GenDistributionFactor | 0..* | GenDistributionFactor | |
| 0..* | SysLoadDistribuFactor | 0..* | SysLoadDistributionFactor | |
| 0..* | LoadDistributionFactor | 0..* | LoadDistributionFactor | |

### 6.5.6.2.16 EnergyPriceIndex

An Energy Price Index for each Resource is valid for a period (e.g. daily) that is identified by a Valid Period Start Time and a Valid Period End Time. An Energy Price Index is in $/MWh.

Table 161 shows all attributes of EnergyPriceIndex.

**Table 161 – Attributes of ExternalInputs::EnergyPriceIndex**

| name | mult | type | description |
|---|---|---|---|
| energyPriceIndex | 0..1 | Float | Energy price index |
| energyPriceIndexType | 0..1 | EnergyPriceIndexType | EPI type such as wholesale or retail |
| lastModified | 0..1 | DateTime | Time updated |
| validPeriod | 0..1 | DateTimeInterval | Valid period for which the energy price index is valid. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 162 shows all association ends of EnergyPriceIndex with other classes.

**Table 162 – Association ends of ExternalInputs::EnergyPriceIndex with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | RegisteredGenerator | 1..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.17 EnergyProfile

Specifies the start time, stop time, level for an EnergyTransaction.

Table 163 shows all attributes of EnergyProfile.

**Table 163 – Attributes of ExternalInputs::EnergyProfile**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 164 shows all association ends of EnergyProfile with other classes.

**Table 164 – Association ends of ExternalInputs::EnergyProfile with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | TransactionBid | 1..1 | TransactionBid | |
| 1..* | EnergyTransaction | 1..1 | EnergyTransaction | An EnergyTransaction shall have at least one EnergyProfile. |
| 0..* | ProfileDatas | 0..* | ProfileData | inherited from: Profile |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.18 EnergyTransaction

Specifies the schedule for energy transfers between interchange areas that are necessary to satisfy the associated interchange transaction.

Table 165 shows all attributes of EnergyTransaction.

**Table 165 – Attributes of ExternalInputs::EnergyTransaction**

| name | mult | type | description |
|---|---|---|---|
| capacityBacked | 0..1 | Boolean | Interchange capacity flag. When the flag is set to true, it indicates a transaction is capacity backed. |
| congestChargeMax | 0..1 | Money | Maximum congestion charges in monetary units. |
| deliveryPointP | 0..1 | ActivePower | Delivery point active power. |
| energyMin | 0..1 | ActivePower | Transaction minimum active power if dispatchable. |
| firmInterchangeFlag | 0..1 | Boolean | Firm interchange flag indicates whether or not this energy transaction can be changed without potential financial consequences. |
| payCongestion | 0..1 | Boolean | Willing to Pay congestion flag |
| reason | 0..1 | String | Reason for energy transaction. |
| receiptPointP | 0..1 | ActivePower | Receipt point active power. |
| state | 0..1 | EnergyTransactionType | { Approve \| Deny \| Study } |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 166 shows all association ends of EnergyTransaction with other classes.

**Table 166 – Association ends of ExternalInputs::EnergyTransaction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnergyPriceCurves | 0..* | EnergyPriceCurve | |
| 1..1 | EnergyProfiles | 1..* | EnergyProfile | An EnergyTransaction shall have at least one EnergyProfile. |
| 0..* | Export_SubControlArea | 1..1 | SubControlArea | Energy is transferred between interchange areas |
| 0..* | Import_SubControlArea | 1..1 | SubControlArea | Energy is transferred between interchange areas |
| 0..1 | TransmissionReservatio n | 0..1 | TransmissionReservatio n | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.19 GenDistributionFactor root class

This class models the generation distribution factors. This class needs to be used along with the AggregatedPnode and the IndividualPnode to show the distribution of each individual party.

Table 167 shows all attributes of GenDistributionFactor.

**Table 167 – Attributes of ExternalInputs::GenDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Used to calculate generation "participation" of an individual pnond in an AggregatePnode. |

Table 168 shows all association ends of GenDistributionFactor with other classes.

**Table 168 – Association ends of ExternalInputs:: GenDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 1..* | AggregatedPnode | 0..1 | AggregatedPnode | |
| 0..1 | IndividualPnode | 0..1 | IndividualPnode | |

### 6.5.6.2.20 GeneratingUnitDynamicValues root class

Optimal Power Flow or State Estimator Unit Data for Operator Training Simulator. This is used for RealTime, Study and Maintenance Users.

Table 169 shows all attributes of GeneratingUnitDynamicValues.

**Table 169 – Attributes of ExternalInputs::GeneratingUnitDynamicValues**

| name | mult | type | description |
|---|---|---|---|
| lossFactor | 0..1 | Float | Loss Factor |
| maximumMW | 0..1 | Float | The maximum active power generation of the unit in MW |
| minimumMW | 0..1 | Float | The minimum active power generation of the unit in MW |
| mVAR | 0..1 | Float | Unit reactive power generation in MVAR |
| mw | 0..1 | Float | Unit active power generation in MW |
| sensitivity | 0..1 | Float | Unit sencivity factor. The distribution factors (DFAX) for the unit |

Table 170 shows all association ends of GeneratingUnitDynamicValues with other classes.

**Table 170 – Association ends of ExternalInputs:: GeneratingUnitDynamicValues with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktGeneratingUnit | 1..1 | MktGeneratingUnit | |
| 0..* | Flowgate | 0..1 | Flowgate | |

### 6.5.6.2.21 GenericConstraints

Generic constraints can represent secure areas, voltage profile, transient stability and voltage collapse limits.

The generic constraints can be one of the following forms:

a) Thermal MW limit constraints type
b) Group line flow constraint type

Table 171 shows all attributes of GenericConstraints.

**Table 171 – Attributes of ExternalInputs::GenericConstraints**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Interval End Time |
| intervalStartTime | 0..1 | DateTime | Interval Start Time |
| maxLimit | 0..1 | Float | Maximum Limit (MW) |
| minLimit | 0..1 | Float | Minimum Limit (MW) |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 172 shows all association ends of GenericConstraints with other classes.

**Table 172 – Association ends of ExternalInputs::GenericConstraints with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | Flowgate | 0..* | Flowgate | |
| 0..1 | TransmissionCapacity | 0..* | TransmissionCapacity | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.22 InterchangeETCData root class

Existing Transmission Contract data for an interchange schedule.

Table 173 shows all attributes of InterchangeETCData.

**Table 173 – Attributes of ExternalInputs::InterchangeETCData**

| name | mult | type | description |
|---|---|---|---|
| contractNumber | 0..1 | String | Existing transmission contract number |
| usageMW | 0..1 | Float | Existing transmission contract usage MW value |

Table 174 shows all association ends of InterchangeETCData with other classes.

**Table 174 – Association ends of ExternalInputs::InterchangeETCData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | InterchangeSchedule | 0..1 | InterchangeSchedule | |

#### 6.5.6.2.23 InterchangeSchedule

Interchange schedule class to hold information for interchange schedules such as import export type, energy type, and etc.

Table 175 shows all attributes of InterchangeSchedule.

**Table 175 – Attributes of ExternalInputs::InterchangeSchedule**

| name | mult | type | description |
|---|---|---|---|
| checkOutType | 0..1 | CheckOutType | To indicate a check out type such as adjusted capacity or dispatch capacity. |
| directionType | 0..1 | InterTieDirection | Import or export. |
| energyType | 0..1 | MarketProductType | Energy product type. |
| intervalLength | 0..1 | Integer | Interval length. |
| marketType | 0..1 | MarketType | Market type. |
| operatingDate | 0..1 | DateTime | Operating date, hour. |
| outOfMarketType | 0..1 | Boolean | To indicate an out-of-market (OOM) schedule. |
| scheduleType | 0..1 | EnergyProductType | Schedule type. |
| wcrID | 0..1 | String | Wheeling Counter-Resource ID (required when Schedule Type=Wheel). |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 176 shows all association ends of InterchangeSchedule with other classes.

**Table 176 – Association ends of ExternalInputs:: InterchangeSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | InterchangeETCData | 0..* | InterchangeETCData | |
| 0..* | InterTie | 0..1 | SchedulingPoint | |
| 0..* | RegisteredInterTie | 0..1 | RegisteredInterTie | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.24 IntermittentResourceEligibility

Indicates whether unit is eligible for treatment as a intermittent variable renewable resource.

Table 177 shows all attributes of IntermittentResourceEligibility.

**Table 177 – Attributes of ExternalInputs::IntermittentResourceEligibility**

| name | mult | type | description |
|---|---|---|---|
| eligibilityStatus | 0..1 | String | Indicates whether a resource is eligible for PIRP program for a given hour |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 178 shows all association ends of IntermittentResourceEligibility with other classes.

**Table 178 – Association ends of ExternalInputs:: IntermittentResourceEligibility with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.25 LoadDistributionFactor root class

This class models the load distribution factors. This class should be used in one of two ways:

Use it along with the AggregatedPnode and the IndividualPnode to show the distriubtion of each individual party

OR

Use it with Mkt_EnergyConsumer to represent the current MW/Mvar distribution within it's parnet load group.

Table 179 shows all attributes of LoadDistributionFactor.

**Table 179 – Attributes of ExternalInputs::LoadDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| pDistFactor | 0..1 | Float | Real power (MW) load distribution factor |
| qDistFactor | 0..1 | Float | Reactive power (MVAr) load distribution factor |

Table 180 shows all association ends of LoadDistributionFactor with other classes.

**Table 180 – Association ends of ExternalInputs:: LoadDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 0..1 | IndividualPnode | 0..1 | IndividualPnode | |
| 1..* | AggregatedPnode | 0..1 | AggregatedPnode | |

#### 6.5.6.2.26 LossSensitivity

Loss sensitivity applied to a ConnectivityNode for a given time interval.

Table 181 shows all attributes of LossSensitivity.

**Table 181 – Attributes of ExternalInputs::LossSensitivity**

| name | mult | type | description |
|---|---|---|---|
| lossFactor | 0..1 | Float | Loss penalty factor.<br>Defined as: 1 / ( 1 – Incremental Transmission Loss); with the Incremental Transmission Loss expressed as a plus or minus value. The typical range of penalty factors is (0,9 to 1,1). |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 182 shows all association ends of LossSensitivity with other classes.

**Table 182 – Association ends of ExternalInputs::LossSensitivity with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.27 MWLimitSchedule root class

Maximum MW and optionally Minimum MW (Y1 and Y2, respectively).

Table 183 shows all association ends of MWLimitSchedule with other classes.

**Table 183 – Association ends of ExternalInputs::MWLimitSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | SecurityConstraintLimit | 1..1 | ContingencyConstraintLimit | |

#### 6.5.6.2.28 MktACLineSegment

Subclass of IEC 61970:Wires:ACLineSegment.

Table 184 shows all attributes of MktACLineSegment.

**Table 184 – Attributes of ExternalInputs::MktACLineSegment**

| name | mult | type | description |
|---|---|---|---|
| b0ch | 0..1 | Susceptance | inherited from: ACLineSegment |
| bch | 0..1 | Susceptance | inherited from: ACLineSegment |
| g0ch | 0..1 | Conductance | inherited from: ACLineSegment |
| gch | 0..1 | Conductance | inherited from: ACLineSegment |
| r | 0..1 | Resistance | inherited from: ACLineSegment |
| r0 | 0..1 | Resistance | inherited from: ACLineSegment |
| shortCircuitEndTemperature | 0..1 | Temperature | inherited from: ACLineSegment |
| x | 0..1 | Reactance | inherited from: ACLineSegment |
| x0 | 0..1 | Reactance | inherited from: ACLineSegment |
| length | 0..1 | Length | inherited from: Conductor |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 185 shows all association ends of MktACLineSegment with other classes.

**Table 185 – Association ends of ExternalInputs::MktACLineSegment with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 1..1 | Clamp | 0..* | Clamp | inherited from: ACLineSegment |
| 0..* | PerLengthImpedance | 0..1 | PerLengthImpedance | inherited from: ACLineSegment |
| 1..1 | Cut | 0..* | Cut | inherited from: ACLineSegment |
| 0..1 | LineFaults | 0..* | LineFault | inherited from: ACLineSegment |
| 1..1 | ACLineSegmentPhases | 0..* | ACLineSegmentPhase | inherited from: ACLineSegment |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.29 MktAnalogLimit

Subclass of IEC 61970:Meas:AnalogLimit.

Table 186 shows all attributes of MktAnalogLimit.

**Table 186 – Attributes of ExternalInputs::MktAnalogLimit**

| name | mult | type | description |
|---|---|---|---|
| exceededLimit | 0..1 | Boolean | true if limit exceeded |
| limitType | 0..1 | AnalogLimitType | The type of limit the value represents<br>Branch Limit Types:<br>Short Term<br>Medium Term<br>Long Term<br>Voltage Limits:<br>High<br>Low |
| value | 0..1 | Float | inherited from: AnalogLimit |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 187 shows all association ends of MktAnalogLimit with other classes.

**Table 187 – Association ends of ExternalInputs::MktAnalogLimit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | LimitSet | 1..1 | AnalogLimitSet | inherited from: AnalogLimit |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.30 MktAnalogLimitSet

Subclass of IEC 61970:Meas:AnalogLimitSet.

Table 188 shows all attributes of MktAnalogLimitSet.

**Table 188 – Attributes of ExternalInputs::MktAnalogLimitSet**

| name | mult | type | description |
|---|---|---|---|
| ratingSet | 0..1 | Integer | Rating set numbers |
| isPercentageLimits | 0..1 | Boolean | inherited from: LimitSet |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 189 shows all association ends of MktAnalogLimitSet with other classes.

**Table 189 – Association ends of ExternalInputs::MktAnalogLimitSet with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | Limits | 0..* | AnalogLimit | inherited from: AnalogLimitSet |
| 0..* | Measurements | 0..* | Analog | inherited from: AnalogLimitSet |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.31 MktControlArea

Market subclass of IEC 61970:ControlArea.

Table 190 shows all attributes of MktControlArea.

**Table 190 – Attributes of ExternalInputs::MktControlArea**

| name | mult | type | description |
|---|---|---|---|
| netInterchange | 0..1 | ActivePower | inherited from: ControlArea |
| pTolerance | 0..1 | ActivePower | inherited from: ControlArea |
| type | 0..1 | ControlAreaTypeKind | inherited from: ControlArea |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 191 shows all association ends of MktControlArea with other classes.

**Table 191 – Association ends of ExternalInputs::MktControlArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ControlAreaSolutionData | 0..* | ControlAreaSolutionData | |
| 0..1 | EnergyArea | 0..1 | EnergyArea | inherited from: ControlArea |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | inherited from: ControlArea |
| 1..1 | TieFlow | 0..* | TieFlow | inherited from: ControlArea |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.32 MktSeriesCompensator

Subclass of IEC 61970:Wires:SeriesCompensator.

Table 192 shows all attributes of MktSeriesCompensator.

**Table 192 – Attributes of ExternalInputs::MktSeriesCompensator**

| name | mult | type | description |
|---|---|---|---|
| r | 0..1 | Resistance | inherited from: SeriesCompensator |
| r0 | 0..1 | Resistance | inherited from: SeriesCompensator |
| x | 0..1 | Reactance | inherited from: SeriesCompensator |
| x0 | 0..1 | Reactance | inherited from: SeriesCompensator |
| varistorPresent | 0..1 | Boolean | inherited from: SeriesCompensator |
| varistorRatedCurrent | 0..1 | CurrentFlow | inherited from: SeriesCompensator |
| varistorVoltageThreshold | 0..1 | Voltage | inherited from: SeriesCompensator |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 193 shows all association ends of MktSeriesCompensator with other classes.

**Table 193 – Association ends of ExternalInputs:: MktSeriesCompensator with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.33 MktShuntCompensator

Subclass of IEC 61970:Wires:ShuntCompensator.

Table 194 shows all attributes of MktShuntCompensator.

**Table 194 – Attributes of ExternalInputs::MktShuntCompensator**

| name | mult | type | description |
|---|---|---|---|
| aVRDelay | 0..1 | Seconds | inherited from: ShuntCompensator |
| grounded | 0..1 | Boolean | inherited from: ShuntCompensator |
| maximumSections | 0..1 | Integer | inherited from: ShuntCompensator |
| nomU | 0..1 | Voltage | inherited from: ShuntCompensator |
| normalSections | 0..1 | Integer | inherited from: ShuntCompensator |
| phaseConnection | 0..1 | PhaseShuntConnectionKind | inherited from: ShuntCompensator |
| switchOnCount | 0..1 | Integer | inherited from: ShuntCompensator |
| switchOnDate | 0..1 | DateTime | inherited from: ShuntCompensator |
| voltageSensitivity | 0..1 | VoltagePerReactivePower | inherited from: ShuntCompensator |
| sections | 0..1 | Float | inherited from: ShuntCompensator |
| controlEnabled | 0..1 | Boolean | inherited from: RegulatingCondEq |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 195 shows all association ends of MktShuntCompensator with other classes.

**Table 195 – Association ends of ExternalInputs:: MktShuntCompensator with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ShuntCompensatorDynamicData | 0..* | ShuntCompensatorDynamicData | |
| 1..1 | ShuntCompensatorPhase | 0..* | ShuntCompensatorPhase | inherited from: ShuntCompensator |
| 1..1 | SvShuntCompensatorSections | 0..1 | SvShuntCompensatorSections | inherited from: ShuntCompensator |
| 0..* | RegulatingControl | 0..1 | RegulatingControl | inherited from: RegulatingCondEq |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.34 MktSwitch

Subclass of IEC 61970:Wires:Switch.

Table 196 shows all attributes of MktSwitch.

**Table 196 – Attributes of ExternalInputs::MktSwitch**

| name | mult | type | description |
|---|---|---|---|
| normalOpen | 0..1 | Boolean | inherited from: Switch |
| ratedCurrent | 0..1 | CurrentFlow | inherited from: Switch |
| switchOnCount | 0..1 | Integer | inherited from: Switch |
| switchOnDate | 0..1 | DateTime | inherited from: Switch |
| retained | 0..1 | Boolean | inherited from: Switch |
| open | 0..1 | Boolean | inherited from: Switch |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |

| name | mult | type | description |
|---|---|---|---|
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 197 shows all association ends of MktSwitch with other classes.

**Table 197 – Association ends of ExternalInputs::MktSwitch with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | SwitchStatus | 0..* | SwitchStatus | |
| 1..1 | SwitchSchedules | 0..* | SwitchSchedule | inherited from: Switch |
| 1..1 | SwitchPhase | 0..* | SwitchPhase | inherited from: Switch |
| 0..* | CompositeSwitch | 0..1 | CompositeSwitch | inherited from: Switch |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.35 MktTapChanger

Subclass of IEC 61970:Wires:TapChanger.

Table 198 shows all attributes of MktTapChanger.

**Table 198 – Attributes of ExternalInputs::MktTapChanger**

| name | mult | type | description |
|---|---|---|---|
| controlEnabled | 0..1 | Boolean | inherited from: TapChanger |
| highStep | 0..1 | Integer | inherited from: TapChanger |
| initialDelay | 0..1 | Seconds | inherited from: TapChanger |
| lowStep | 0..1 | Integer | inherited from: TapChanger |
| ltcFlag | 0..1 | Boolean | inherited from: TapChanger |
| neutralStep | 0..1 | Integer | inherited from: TapChanger |
| neutralU | 0..1 | Voltage | inherited from: TapChanger |
| normalStep | 0..1 | Integer | inherited from: TapChanger |
| subsequentDelay | 0..1 | Seconds | inherited from: TapChanger |
| step | 0..1 | Float | inherited from: TapChanger |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 199 shows all association ends of MktTapChanger with other classes.

**Table 199 – Association ends of ExternalInputs::MktTapChanger with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TapChangerDynamicData | 0..* | TapChangerDynamicData | |
| 0..* | TapChangerControl | 0..1 | TapChangerControl | inherited from: TapChanger |
| 1..1 | TapSchedules | 0..* | TapSchedule | inherited from: TapChanger |
| 1..1 | SvTapStep | 0..1 | SvTapStep | inherited from: TapChanger |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.2.36 NodeConstraintTerm

To be used only to constrain a quantity that cannot be associated with a terminal. For example, a registered generating unit that is not electrically connected to the network.

Table 200 shows all attributes of NodeConstraintTerm.

**Table 200 – Attributes of ExternalInputs::NodeConstraintTerm**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | inherited from: ConstraintTerm |
| function | 0..1 | String | inherited from: ConstraintTerm |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 201 shows all association ends of NodeConstraintTerm with other classes.

**Table 201 – Association ends of ExternalInputs::NodeConstraintTerm with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | inherited from: ConstraintTerm |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.37 Profile

A profile is a simpler curve type.

Table 202 shows all attributes of Profile.

**Table 202 – Attributes of ExternalInputs::Profile**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 203 shows all association ends of Profile with other classes.

**Table 203 – Association ends of ExternalInputs::Profile with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ProfileDatas | 0..* | ProfileData | A profile has profile data associated with it. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.38 ProfileData root class

Data for profile.

Table 204 shows all attributes of ProfileData.

**Table 204 – Attributes of ExternalInputs::ProfileData**

| name | mult | type | description |
|---|---|---|---|
| bidPrice | 0..1 | Float | Bid price associated with contract |
| capacityLevel | 0..1 | ActivePower | Capacity level for the profile, in MW. |
| energyLevel | 0..1 | RealEnergy | Energy level for the profile, in MWH. |
| minimumLevel | 0..1 | Float | Minimum MW value of contract |
| sequenceNumber | 0..1 | Integer | Sequence to provide item numbering for the profile. { greater than or equal to 1 } |
| startDateTime | 0..1 | DateTime | Start date/time for this profile. |
| stopDateTime | 0..1 | DateTime | Stop date/time for this profile. |

Table 205 shows all association ends of ProfileData with other classes.

**Table 205 – Association ends of ExternalInputs::ProfileData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Profile | 0..* | Profile | A profile has profile data associated with it. |

#### 6.5.6.2.39 ReserveDemandCurve

Reserve demand curve. Models maximum quantities of reserve required per Market Region and models a reserve demand curve for the minimum quantities of reserve. The ReserveDemandCurve is a relationship between unit operating reserve price in $/MWhr (Y-axis) and unit reserves in MW (X-axis).

Table 206 shows all attributes of ReserveDemandCurve.

**Table 206 – Attributes of ExternalInputs::ReserveDemandCurve**

| name | mult | type | description |
|---|---|---|---|
| reqMaxMW | 0..1 | Float | Region requirement maximum limit |
| reserveRequirementType | 0..1 | ReserveRequirementType | Reserve requirement type that the max and curve apply to. For example, operating reserve, regulation and contingency. |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |

| name | mult | type | description |
|---|---|---|---|
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 207 shows all association ends of ReserveDemandCurve with other classes.

**Table 207 – Association ends of ExternalInputs:: ReserveDemandCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | ASRequirements | 1..1 | ASRequirements | |
| 0..* | MarketRegion | 1..1 | MarketRegion | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.40 SCADAInformation root class

Contains information about the update from SCADA.

Table 208 shows all attributes of SCADAInformation.

**Table 208 – Attributes of ExternalInputs::SCADAInformation**

| name | mult | type | description |
|---|---|---|---|
| timeStamp | 0..1 | DateTime | time of the update from SCADA |

#### 6.5.6.2.41 SecurityConstraintSum

Typically provided by RTO systems, constraints identified in both base case and critical contingency cases have to be transferred.

A constraint has N (>=1) constraint terms. A term is represented by an instance of TerminalConstraintTerm.

The constraint expression is:

minValue <= c1*x1 + c2*x2 + .... cn*xn + k <= maxValue

where:

- cn is ConstraintTerm.factor
- xn is the flow at the terminal

Flow into the associated equipment is positive for the purpose of ConnectivityNode NodeConstraintTerm.

k is SecurityConstraintsLinear.resourceMW.

The units of k are assumed to be same as the units of the flows, xn. The constants, cn, are dimensionless.

With these conventions, cn and k are all positive for a typical constraint such as "weighted sum of generation shall be less than limit". Furthermore, cn are all 1.0 for a case such as "interface flow shall be less than limit", assuming the terminals are chosen on the importing side of the interface.

Table 209 shows all attributes of SecurityConstraintSum.

**Table 209 – Attributes of ExternalInputs::SecurityConstraintSum**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 210 shows all association ends of SecurityConstraintSum with other classes.

**Table 210 – Association ends of ExternalInputs:: SecurityConstraintSum with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ConstraintTerms | 0..* | ConstraintTerm | |
| 0..* | RTO | 0..1 | RTO | |
| 1..1 | BaseCaseConstraintLimit | 0..1 | BaseCaseConstraintLimit | |
| 1..1 | ContingencyConstraintLimits | 0..* | ContingencyConstraintLimit | |
| 1..1 | DefaultConstraintLimit | 0..1 | DefaultConstraintLimit | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.42 SecurityConstraints

Typical for regional transmission operators (RTOs), these constraints include transmission as well as generation group constraints identified in both base case and critical contingency cases.

Table 211 shows all attributes of SecurityConstraints.

**Table 211 – Attributes of ExternalInputs::SecurityConstraints**

| name | mult | type | description |
|---|---|---|---|
| minMW | 0..1 | ActivePower | Minimum MW limit (only for transmission constraints). |
| maxMW | 0..1 | ActivePower | Maximum MW limit |
| actualMW | 0..1 | ActivePower | Actual branch or group of branches MW flow (only for transmission constraints) |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 212 shows all association ends of SecurityConstraints with other classes.

**Table 212 – Association ends of ExternalInputs::SecurityConstraints with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RTO | 0..1 | RTO | |
| 0..* | GeneratingBid | 0..1 | GeneratingBid | |
| 0..1 | Flowgate | 0..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.43 ServicePoint

The defined termination points of a transmission path. Service points are defined from the viewpoint of the transmission service. Each service point is contained within (or on the boundary of) an interchange area. A service point is source or destination of a transaction.

Table 213 shows all attributes of ServicePoint.

**Table 213 – Attributes of ExternalInputs::ServicePoint**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 214 shows all association ends of ServicePoint with other classes.

**Table 214 – Association ends of ExternalInputs::ServicePoint with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | PORTransmissionPath | 0..* | TransmissionPath | A transmission path has a "point-of-receipt" service point |
| 1..1 | PODTransmissionPath | 0..* | TransmissionPath | A transmission path has a "point-of-delivery" service point |
| 0..1 | SourceReservation | 0..* | TransmissionReservation | |
| 0..1 | SinkReservation | 0..* | TransmissionReservation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.44 ShuntCompensatorDynamicData root class

Optimal Power Flow or State Estimator Filter Bank Data for OTS. This is used for RealTime, Study and Maintenance Users.

Table 215 shows all attributes of ShuntCompensatorDynamicData.

**Table 215 – Attributes of ExternalInputs::ShuntCompensatorDynamicData**

| name | mult | type | description |
|---|---|---|---|
| mVARInjection | 0..1 | Float | The injection of reactive power of the filter bank in the NA solution or VCS reactive power production |
| connectionStatus | 0..1 | Integer | The current status for the Voltage Control Capacitor 1= Connected 0 = Disconnected |
| desiredVoltage | 0..1 | Float | The desired voltage for the Voltage Control Capacitor |
| voltageRegulationStatus | 0..1 | Boolean | Indicator if the voltage control this is regulating True = Yes, False = No |
| stepPosition | 0..1 | Integer | Voltage control capacitor step position |

Table 216 shows all association ends of ShuntCompensatorDynamicData with other classes.

**Table 216 – Association ends of ExternalInputs:: ShuntCompensatorDynamicData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktShuntCompensator | 1..1 | MktShuntCompensator | |

### 6.5.6.2.45 SwitchStatus root class

Optimal Power Flow or State Estimator Circuit Breaker Status.

Table 217 shows all attributes of SwitchStatus.

**Table 217 – Attributes of ExternalInputs::SwitchStatus**

| name | mult | type | description |
|---|---|---|---|
| switchStatus | 0..1 | SwitchStatusType | Circuit Breaker Status (closed or open) of the circuit breaker from the power flow. |

Table 218 shows all association ends of SwitchStatus with other classes.

**Table 218 – Association ends of ExternalInputs::SwitchStatus with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktSwitch | 1..1 | MktSwitch | |

### 6.5.6.2.46 SysLoadDistributionFactor root class

This class models the system distribution factors. This class needs to be used along with the HostControlArea and the ConnectivityNode to show the distribution of each individual party.

Table 219 shows all attributes of SysLoadDistributionFactor.

**Table 219 – Attributes of ExternalInputs::SysLoadDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Used to calculate load "participation" of a connectivity node in an host control area |

Table 220 shows all association ends of SysLoadDistributionFactor with other classes.

**Table 220 – Association ends of ExternalInputs:: SysLoadDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 0..* | HostControlArea | 1..1 | HostControlArea | |

#### 6.5.6.2.47 TREntitlement root class

A Transmission Right(TR) can be a chain of TR's or on individual.

When a transmission right is not a chain, this is formally the ETC/TOR Entitlement for each ETC/TOR contract with the inclusion of CVR(Converted Rights) as an ETC. This is the sum of all entitlements on all related transmission interfaces for the same TR.

When TR is a chain, its entitlement is the minimum of all entitlements for the individual TRs in the chain.

Table 221 shows all attributes of TREntitlement.

**Table 221 – Attributes of ExternalInputs::TREntitlement**

| name | mult | type | description |
|---|---|---|---|
| entitlement | 0..1 | Float | The entitlement |
| startOperatingDate | 0..1 | DateTime | Operating date and hour when the entitlement applies |

Table 222 shows all association ends of TREntitlement with other classes.

**Table 222 – Association ends of ExternalInputs::TREntitlement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TransmissionContractRight | 1..1 | ContractRight | |

#### 6.5.6.2.48 TapChangerDynamicData root class

Optimal Power Flow or State Estimator Phase Shifter Data. This is used for RealTime, Study and Maintenance Users. SE Solution Phase Shifter Measurements from the last run of SE.

Table 223 shows all attributes of TapChangerDynamicData.

**Table 223 – Attributes of ExternalInputs::TapChangerDynamicData**

| name | mult | type | description |
|---|---|---|---|
| tapPosition | 0..1 | Float | Tap position of the phase shifter, high-side tap position of the transformer, or low-side tap position of the transformer |
| desiredVoltage | 0..1 | Float | The desired voltage for the LTC |
| voltageRegulationStatus | 0..1 | Boolean | Indicator if the LTC transformer is regulating True = Yes, False = No |
| angleRegulationStatus | 0..1 | Boolean | True means the phase shifter is regulating. |
| desiredMW | 0..1 | Float | Phase Shifter Desired MW. The active power regulation setpoint of the phase shifter |
| solvedAngle | 0..1 | Float | Phase Shifter Angle. The solved phase angle shift of the phase shifter |
| minimumAngle | 0..1 | Float | The minimum phase angle shift of the phase shifter |
| maximumAngle | 0..1 | Float | The maximum phase angle shift of the phase shifter |

Table 224 shows all association ends of TapChangerDynamicData with other classes.

**Table 224 – Association ends of ExternalInputs:: TapChangerDynamicData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktTapChanger | 1..1 | MktTapChanger | |

#### 6.5.6.2.49 TerminalConstraintTerm

A constraint term associated with a specific terminal on a physical piece of equipment.

Table 225 shows all attributes of TerminalConstraintTerm.

**Table 225 – Attributes of ExternalInputs::TerminalConstraintTerm**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | inherited from: ConstraintTerm |
| function | 0..1 | String | inherited from: ConstraintTerm |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 226 shows all association ends of TerminalConstraintTerm with other classes.

**Table 226 – Association ends of ExternalInputs:: TerminalConstraintTerm with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktTerminal | 1..1 | MktTerminal | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | inherited from: ConstraintTerm |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.50 TransferInterface

A Transfer Interface is made up of branches such as transmission lines and transformers.

Table 227 shows all attributes of TransferInterface.

**Table 227 – Attributes of ExternalInputs::TransferInterface**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 228 shows all association ends of TransferInterface with other classes.

**Table 228 – Association ends of ExternalInputs::TransferInterface with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TransferInterfaceSolution | 1..1 | TransferInterfaceSolution | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.51 TransferInterfaceSolution root class

TNA Interface Definitions from OPF for VSA.

Table 229 shows all attributes of TransferInterfaceSolution.

**Table 229 – Attributes of ExternalInputs::TransferInterfaceSolution**

| name | mult | type | description |
|---|---|---|---|
| interfaceMargin | 0..1 | Float | The margin for the interface |
| transferLimit | 0..1 | Float | Transfer Interface + Limit<br>Attribute Usage: The absoloute of the maximum flow on the transfer interface. This is a positive MW value. |
| postTransferMW | 0..1 | Float | Post Transfer MW for step |

Table 230 shows all association ends of TransferInterfaceSolution with other classes.

**Table 230 – Association ends of ExternalInputs:: TransferInterfaceSolution with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TransferInterface | 1..1 | TransferInterface | |
| 0..1 | MktContingencyA | 0..1 | MktContingency | |
| 0..1 | MktContingencyB | 0..1 | MktContingency | |

#### 6.5.6.2.52 TransmissionCapacity root class

This class models the transmission (either a transmission interface or a POR/POD pair) capacity including Total Transfer Capacity (TTC), Operating Transfer Capacity (OTC), and Capacity Benefit Margin (CBM).

Table 231 shows all attributes of TransmissionCapacity.

**Table 231 – Attributes of ExternalInputs::TransmissionCapacity**

| name | mult | type | description |
|---|---|---|---|
| capacityBenefitMargin | 0..1 | Float | Capacity Benefit Margin (CBM) is used by Markets to calculate the transmission interface limits. This number could be manually or procedurally determined. The CBM is defined per transmission interface (branch group). |
| operationalTransmission Capacity | 0..1 | Float | The Operational Transmission Capacity (OTC) is the transmission capacity under the operating condition during a specific time period, incorporating the effects of derates and current settings of operation controls. The OTCs for all transmission interface (branch group) are always provided regardless of outage or switching conditions. |
| OTC15min_emergency | 0..1 | Float | The Operational Transmission Capacity (OTC) 15 minute Emergency Limit |
| OTCemergency | 0..1 | Float | The Operational Transmission Capacity (OTC) Emergency Limit. |
| POD | 0..1 | String | point of delivery |
| POR | 0..1 | String | point of receipt |
| startOperatingDate | 0..1 | DateTime | Operating date & hour when the entitlement applies |
| totalTransmissionCapaci ty | 0..1 | Float | Total Transmission Capacity |

Table 232 shows all association ends of TransmissionCapacity with other classes.

**Table 232 – Association ends of ExternalInputs:: TransmissionCapacity with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | GenericConstraints | 0..1 | GenericConstraints | |
| 0..* | Flowgate | 0..1 | Flowgate | |

#### 6.5.6.2.53 TransmissionInterfaceRightEntitlement root class

This is formally called the branch group ETC/TOR entitlement with the inclusion of CVR as ETC. This could be also used to represent the TR entitlement on a POR/POD.

Table 233 shows all attributes of TransmissionInterfaceRightEntitlement.

**Table 233 – Attributes of ExternalInputs::TransmissionInterfaceRightEntitlement**

| name | mult | type | description |
|---|---|---|---|
| entitlement | 0..1 | Float | the entitlement |
| POD | 0..1 | String | point of delivery |
| POR | 0..1 | String | point of receipt |
| startOperatingDate | 0..1 | DateTime | Operating date and hour when the entitlement applies |

Table 234 shows all association ends of TransmissionInterfaceRightEntitlement with other classes.

**Table 234 – Association ends of ExternalInputs:: TransmissionInterfaceRightEntitlement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ContractRight | 1..1 | ContractRight | |
| 0..* | Flowgate | 0..1 | Flowgate | |

#### 6.5.6.2.54 TransmissionPath

An electrical connection, link, or line consisting of one or more parallel transmission elements between two areas of the interconnected electric systems, or portions thereof. TransmissionCorridor and TransmissionRightOfWay refer to legal aspects. The TransmissionPath refers to the segments between a TransmissionProvider's ServicePoints.

Table 235 shows all attributes of TransmissionPath.

**Table 235 – Attributes of ExternalInputs::TransmissionPath**

| name | mult | type | description |
|---|---|---|---|
| availTransferCapability | 0..1 | ActivePower | The available transmission capability of a transmission path for the reference direction. |
| parallelPathFlag | 0..1 | Boolean | Flag which indicates if the transmission path is also a designated interconnection "parallel path". |
| totalTransferCapability | 0..1 | ActivePower | The total transmission capability of a transmission path in the reference direction. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 236 shows all association ends of TransmissionPath with other classes.

**Table 236 – Association ends of ExternalInputs::TransmissionPath with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TransmissionReservation | 0..* | TransmissionReservation | |
| 0..* | PointOfReceipt | 1..1 | ServicePoint | A transmission path has a "point-of-receipt" service point |
| 0..* | DeliveryPoint | 1..1 | ServicePoint | A transmission path has a "point-of-delivery" service point |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.2.55 TransmissionReservation root class

A transmission reservation is obtained from the OASIS system to reserve transmission for a specified time period, transmission path and transmission product.

Table 237 shows all association ends of TransmissionReservation with other classes.

**Table 237 – Association ends of ExternalInputs:: TransmissionReservation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | EnergyTransaction | 0..1 | EnergyTransaction | |
| 0..1 | TransactionBid | 0..1 | TransactionBid | |
| 0..* | Source | 0..1 | ServicePoint | |
| 0..* | Sink | 0..1 | ServicePoint | |
| 0..* | TransmissionPath | 1..1 | TransmissionPath | |

#### 6.5.6.2.56 UnitInitialConditions

Resource status at the end of a given clearing period.

Table 238 shows all attributes of UnitInitialConditions.

**Table 238 – Attributes of ExternalInputs::UnitInitialConditions**

| name | mult | type | description |
|---|---|---|---|
| cumEnergy | 0..1 | RealEnergy | Cumulative energy production over trading period. |
| cumStatusChanges | 0..1 | Integer | Cumulative number of status changes of the resource. |
| numberOfStartups | 0..1 | Integer | Number of start ups in the Operating Day until the end of previous hour. |
| onlineStatus | 0..1 | Boolean | 'true' if the GeneratingUnit is currently On-Line |
| resourceMW | 0..1 | ActivePower | Resource MW output at the end of previous clearing period. |
| resourceStatus | 0..1 | Integer | Resource status at the end of previous clearing period:<br>0 – off-line<br>1 – on-line production<br>2 – in shutdown process<br>3 – in startup process |
| statusDate | 0..1 | DateTime | Time and date for resourceStatus |
| timeInStatus | 0..1 | Float | Time in market trading intervals the resource is in the state as of the end of the previous clearing period. |
| timeInterval | 0..1 | DateTime | Time interval |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 239 shows all association ends of UnitInitialConditions with other classes.

**Table 239 – Association ends of ExternalInputs::UnitInitialConditions with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | GeneratingUnit | 0..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3 Package MarketResults

#### 6.5.6.3.1 General

Results from the execution of a market.

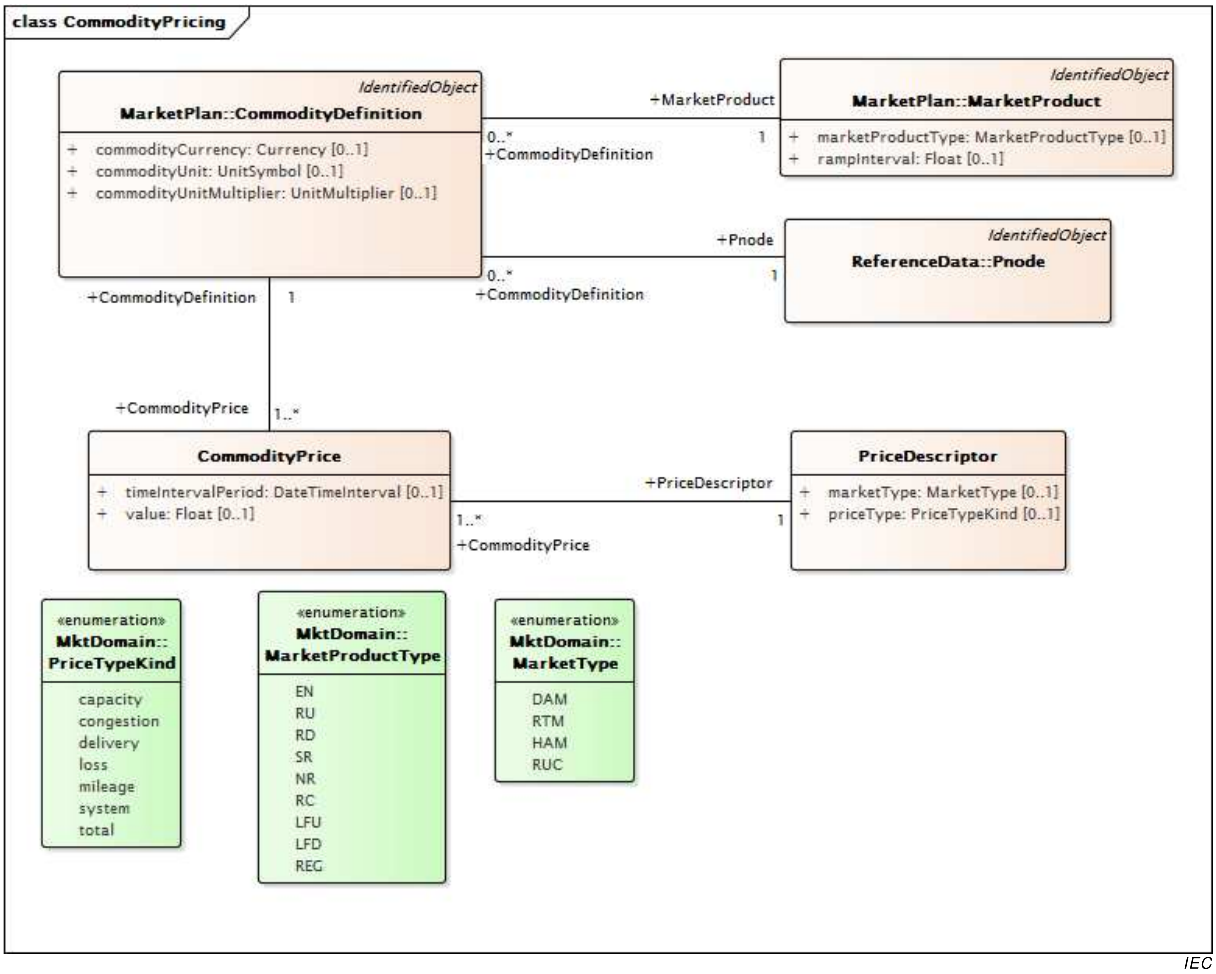



**Figure 44 – Class diagram MarketResults::CommodityPricing**

Figure 44: Illustrates how CommodityDefinitions are defined for a Market Product at a Pricing Node. CommodityPrices for specific time intervals can be defined for a CommodityDefinition. Each CommodityPrice has a PriceDescriptor which specifies the market type and pricing type.

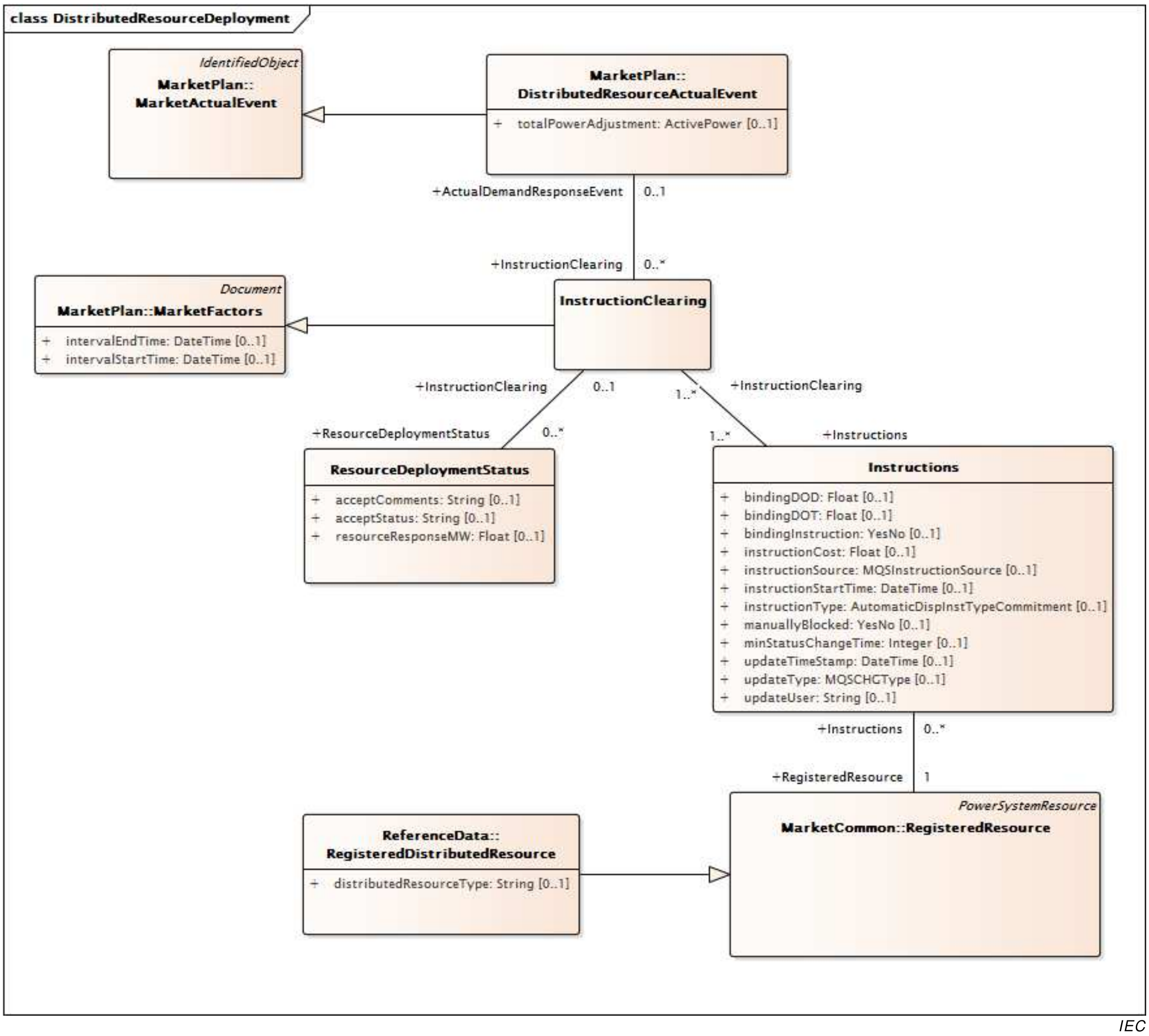



**Figure 45 – Class diagram MarketResults::DistributedResourceDeployment**

Figure 45: Illustrates the classes involved in a Registered Distributed Resource deployment. Event driven deployments are defined by the DistributedResourceActualEvent class which inherits from the MarketActualEvent class. Instructions which includes the binding dispatch operating target or binding dispatch operating delta along with the instruction start time and instruction type are sent to the RegisteredDistributedResources. Deployments that are resource specific, event driven or bulk are all support by these classes.

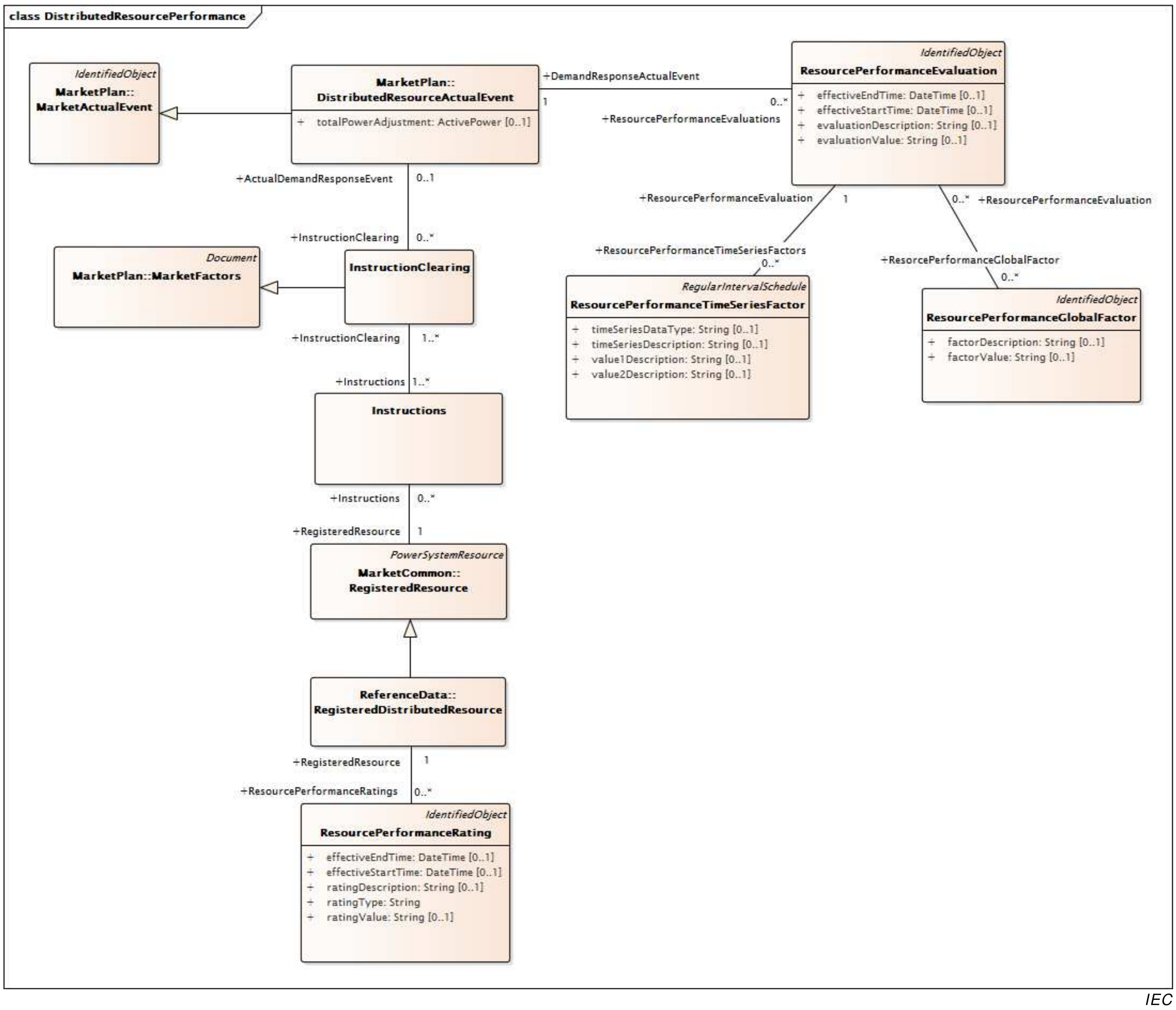



**Figure 46 – Class diagram MarketResults::DistributedResourcePerformance**

Figure 46: Illustrates performance evaluation support for RegisteredDistributedResources. After revenue-quality metering has been received, a ResourcePerformanceEvaluation instance may be created for each Resource per DR Event.

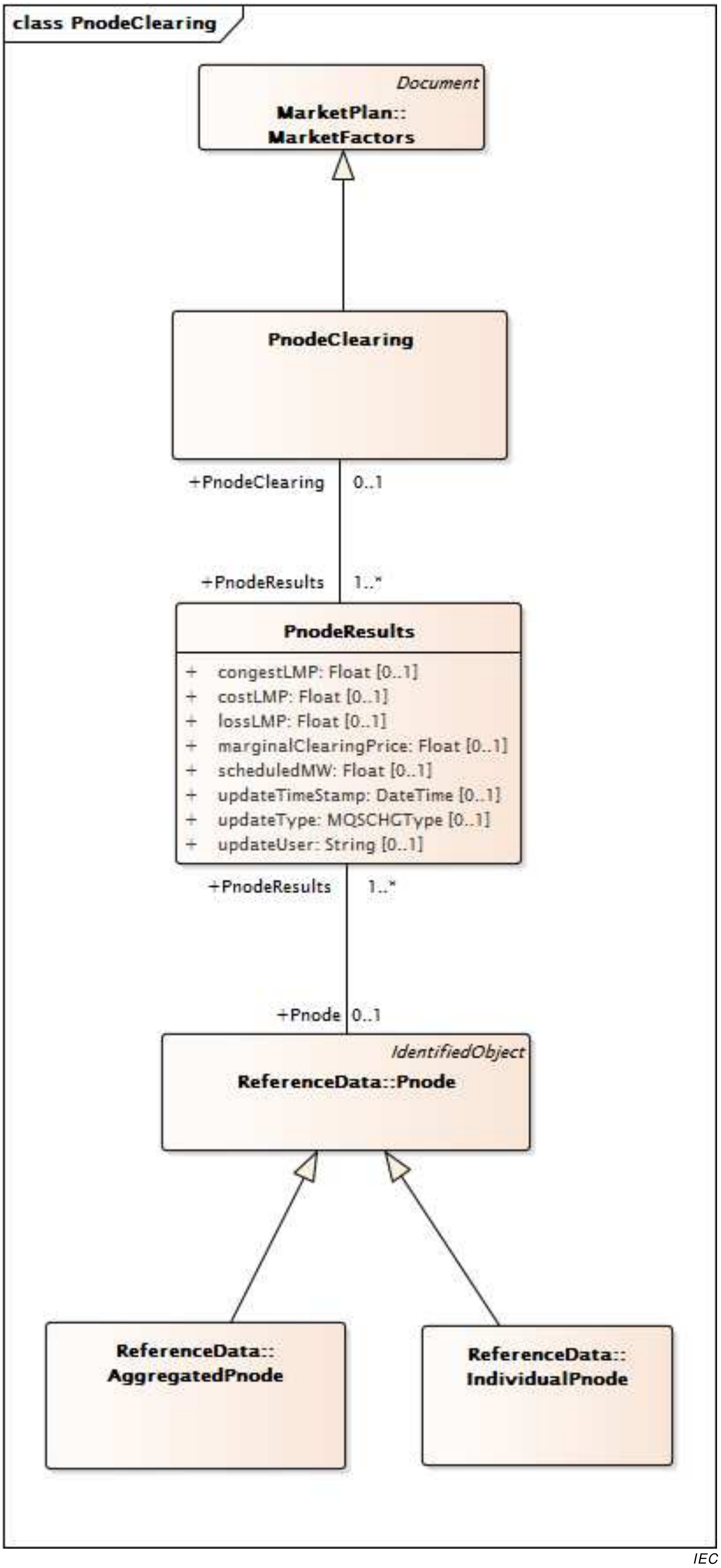



**Figure 47 – Class diagram MarketResults::PnodeClearing**

Figure 47: Shows the classes and relationships used to model the results of pnode clearing for each Pnode per interval. The intervals are defined by PnodeClearing which inherits from MarketFactors. PnodeResults contains the results for each Pnode.

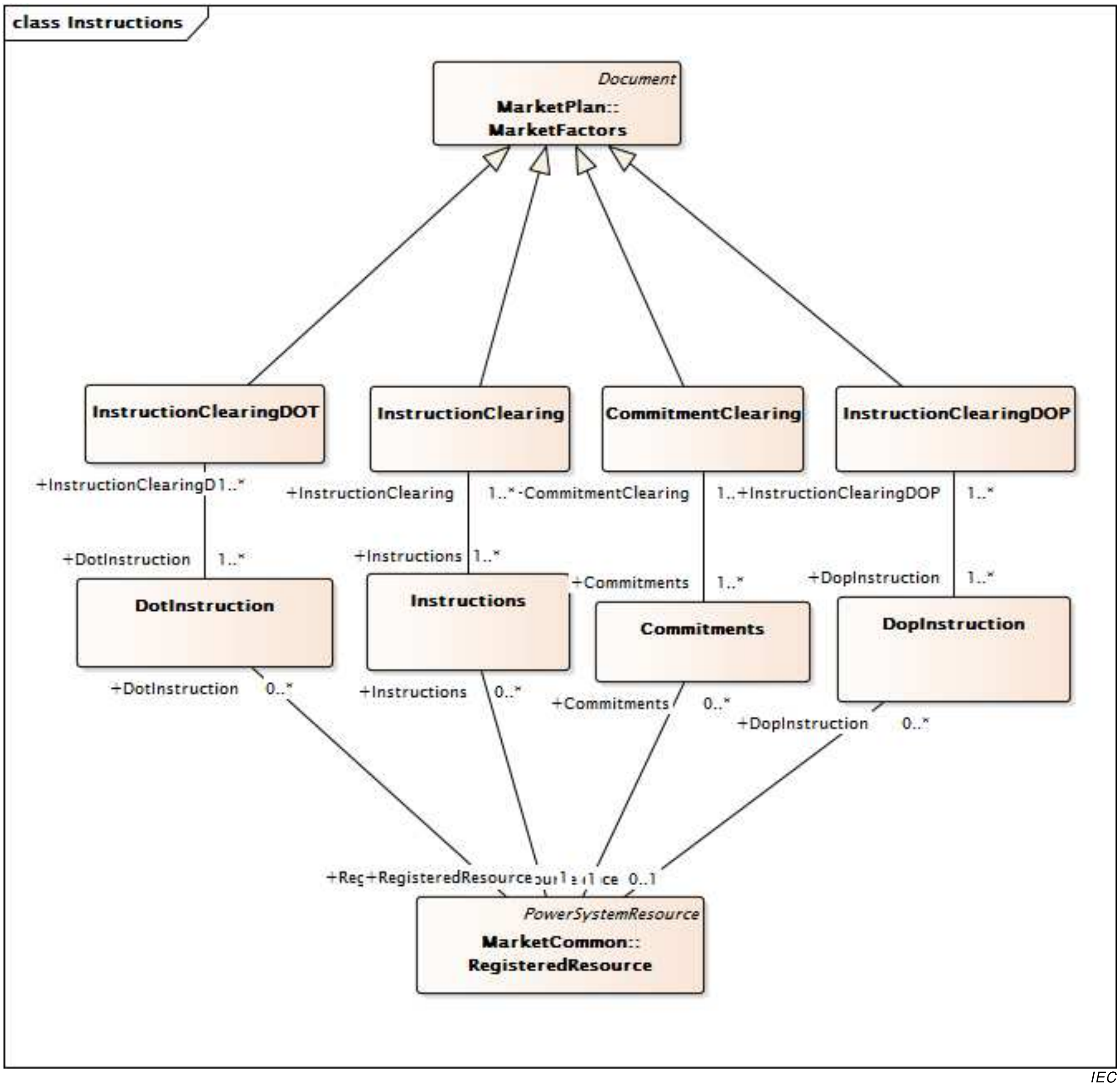



**Figure 48 – Class diagram MarketResults::Instructions**

Figure 48: Shows the classes used to model the instructions sent to market participants following a market clearing. The major classes and relationships include the RegisteredResource class that models the participating resource, the Commitment and CommitmentClearing classes which model results which calls for commitment of resource. The DotInstruction and DopInstruction allow the modeling of the results of a dynamic (or lookahead) dispatch that may be used in the real time market.

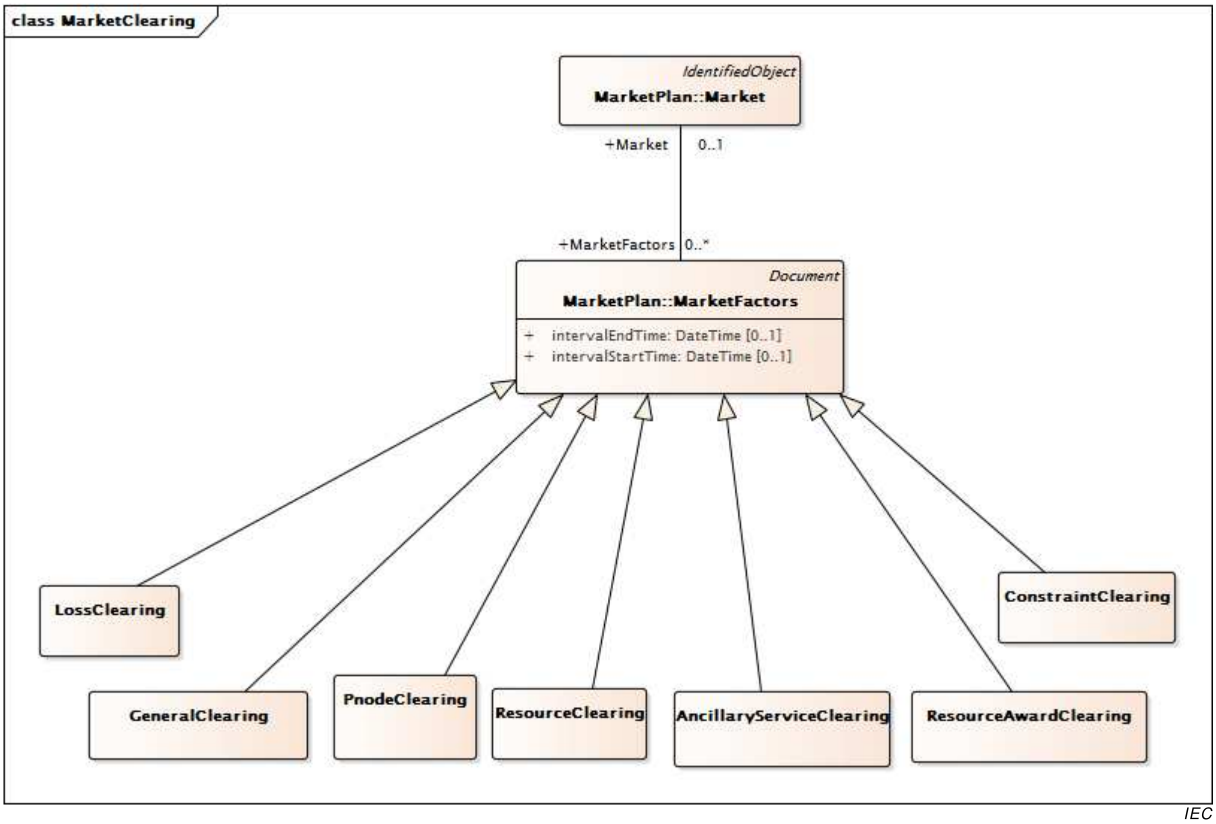



**Figure 49 – Class diagram MarketResults::MarketClearing**

Figure 49: Shows how each of the defined market clearing classes inherit from the MarketFactors class to allow definition of the market time intervals and the execution type within the market.

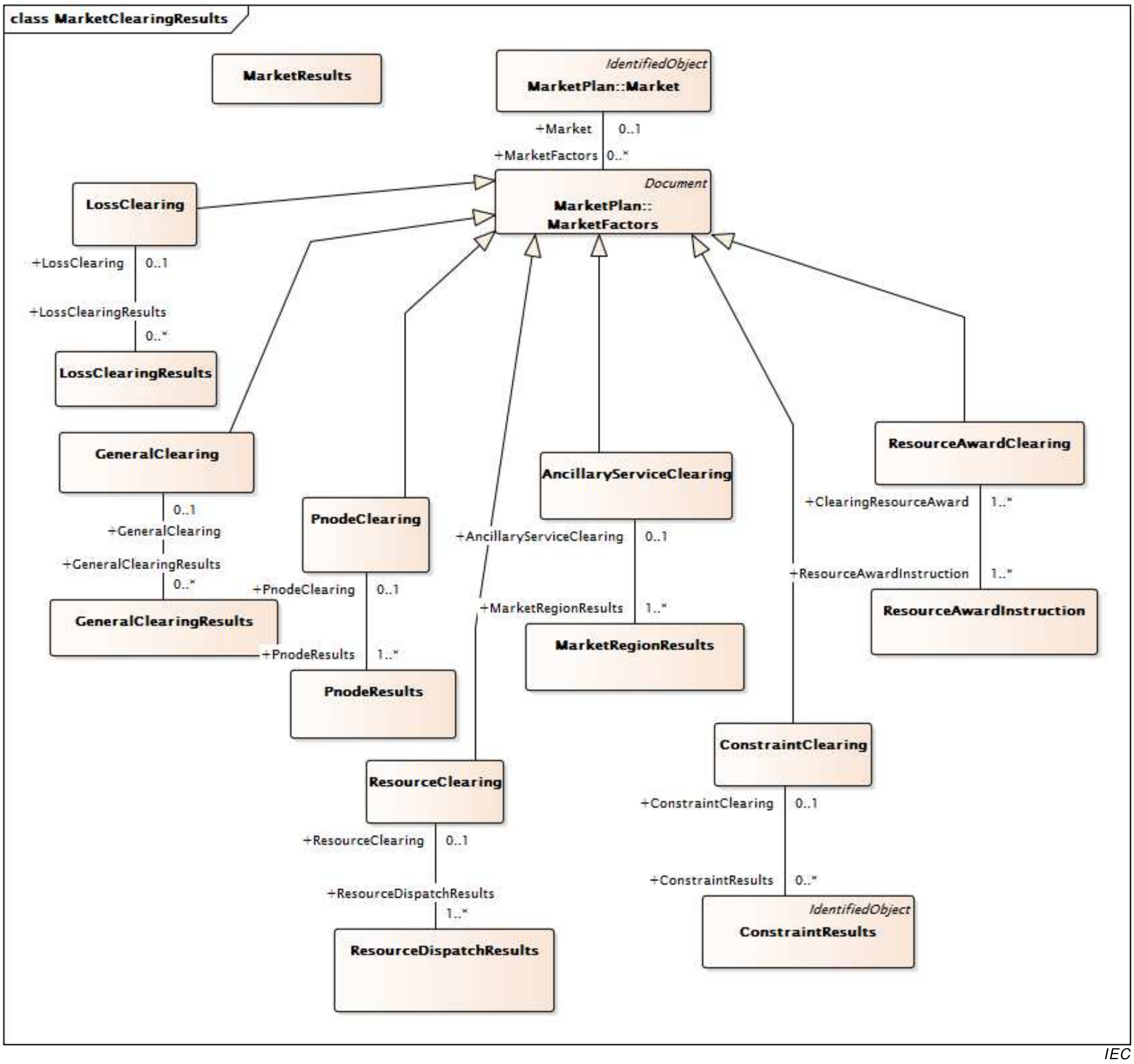



**Figure 50 – Class diagram MarketResults::MarketClearingResults**

Figure 50: Shows the classes and relationships used to model the results of the market clearing at an overview level. The Market Results class summarizes the market run. The LossClearingResults Class summarizes losses. The GeneralClearingResults is used to model the cleared load on a load zone basis. The PnodeClearing and PnodeClearingResults classes are used to model the locational marginal prices (LMPs) at the level of individual pricing nodes. The ResourceClearing and ResourceDispatchInstruction classes include the dispatch instructions and indicator of a normal case or contingency case dispatch. The AncillaryServiceClearing class includes information on ancillary services. The ConstraintResults class contains the constraints that are binding and their associated shadow costs. The ResourceAwardInstruction includes Active Power awards and LMP information.

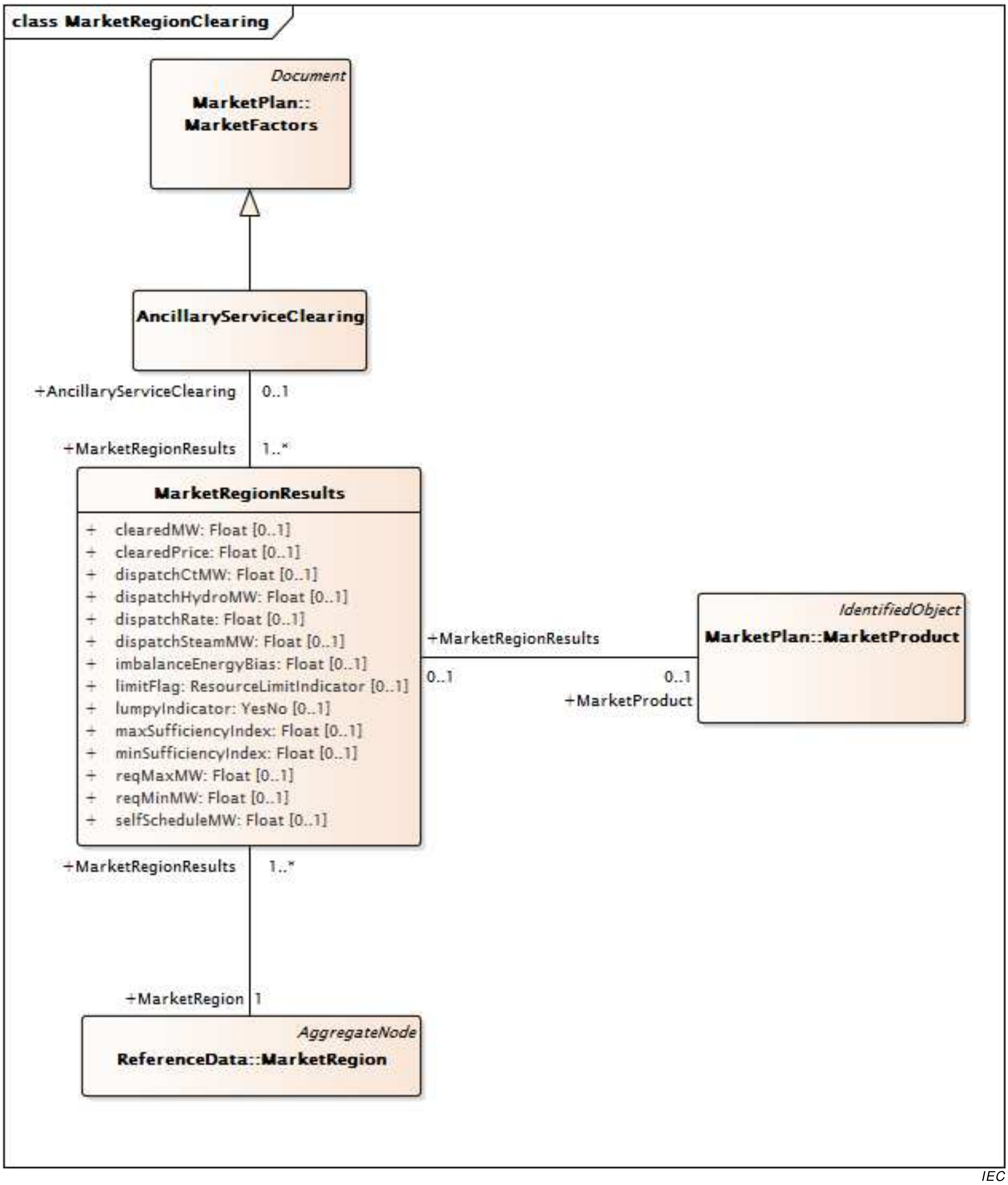



**Figure 51 – Class diagram MarketResults::MarketRegionClearing**

Figure 51: Shows the classes and relationships used to model the results of the market clearing of ancillary services at the level of a region of the market. The model includes results at the level of specific ancillary service products by means of the class MarketProduct and association to MarketRegionResults.

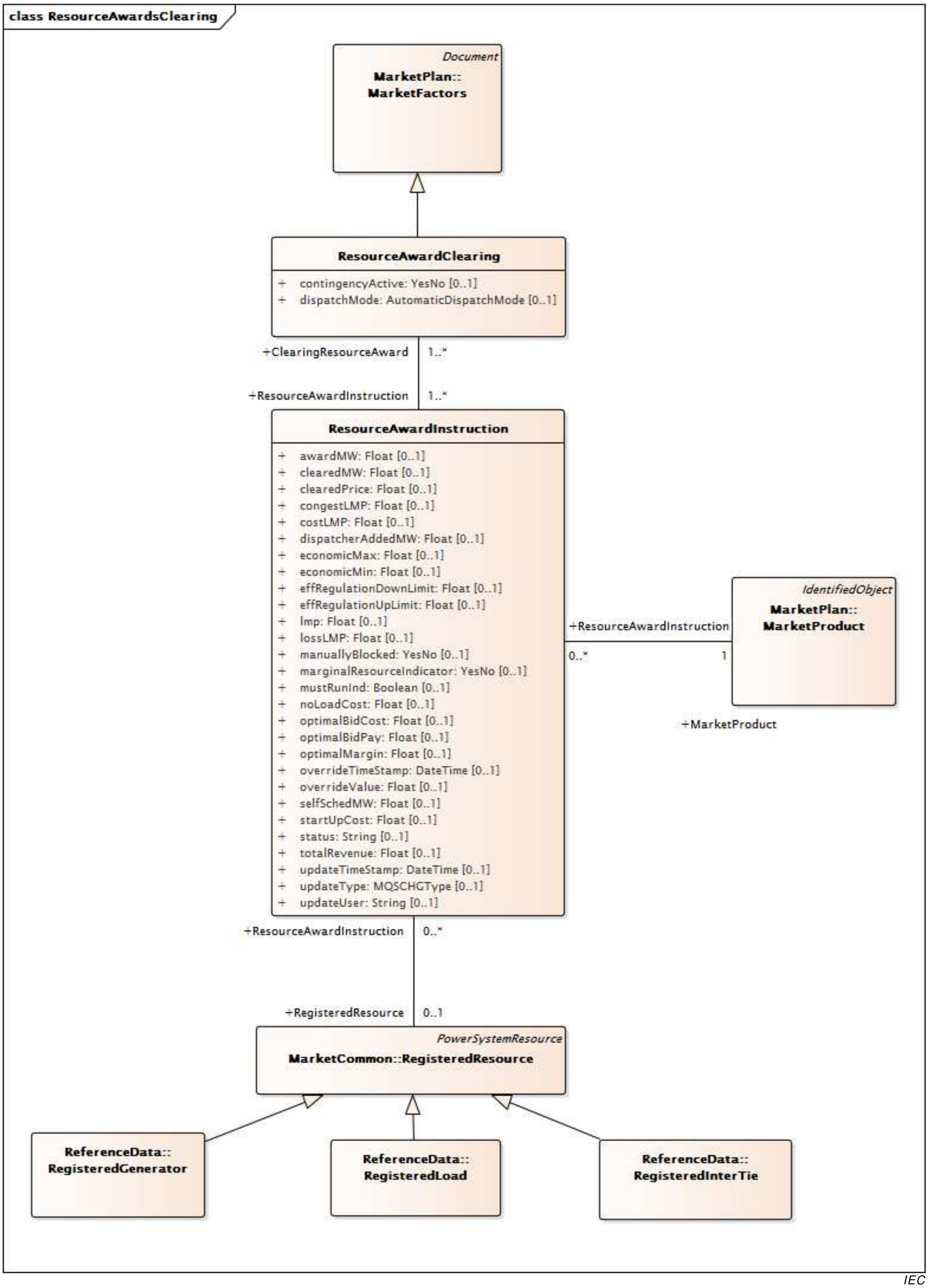



**Figure 52 – Class diagram MarketResults::ResourceAwardsClearing**

Figure 52: Shows the classes and relationships used to model the results of the market clearing with more emphasis on the ResourceAwardInstruction. The main class of this diagram is the ResourceAwardInstruction which models the details of the award to the resouce including the active power, the locational marginal price (LMP) and along with its

energy, congestion and cost components. the resource model includes RegisteredGenerator, RegisteredLoad and RegisteredInterTie as subclasses.

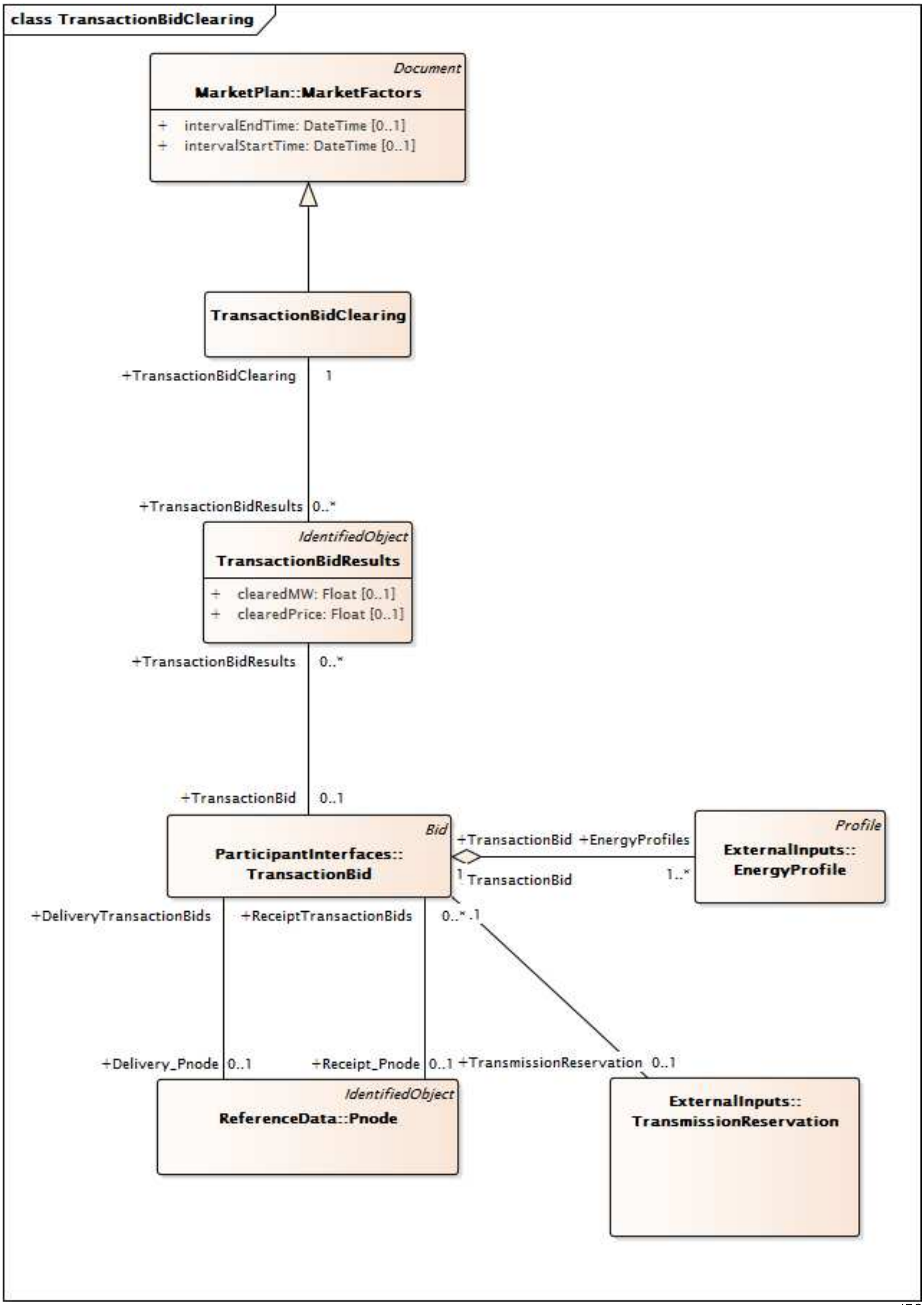


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**Figure 53 – Class diagram MarketResults::TransactionBidClearing**

Figure 53: Shows the classes and relationships used to model the results of the market clearing with more emphasis on Transaction Results. The model includes TransactionBid class and its relationship to Point of Delivery and Point of Receipt Pnodes.

#### 6.5.6.3.2 AncillaryServiceClearing

Model of results of market clearing with respect to Ancillary Service products.

Table 240 shows all attributes of AncillaryServiceClearing.

**Table 240 – Attributes of MarketResults::AncillaryServiceClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 241 shows all association ends of AncillaryServiceClearing with other classes.

**Table 241 – Association ends of MarketResults:: AncillaryServiceClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MarketRegionResults | 1..* | MarketRegionResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.3 BillDeterminant

Models various charges to support billing and settlement.

Table 242 shows all attributes of BillDeterminant.

**Table 242 – Attributes of MarketResults::BillDeterminant**

| name | mult | type | description |
|---|---|---|---|
| calculationLevel | 0..1 | String | Level in charge calculation order. |
| configVersion | 0..1 | String | The version of configuration of calculation logic in the settlement. |
| deleteStatus | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| exception | 0..1 | String | |
| factor | 0..1 | String | |
| frequency | 0..1 | String | |
| numberInterval | 0..1 | Integer | Number of intervals of bill determiant in trade day, e.g. 300 for five minute intervals. |
| offset | 0..1 | String | |
| precisionLevel | 0..1 | String | The level of precision in the current value. |
| primaryYN | 0..1 | String | |
| referenceFlag | 0..1 | String | |
| reportable | 0..1 | String | |
| roundOff | 0..1 | String | |
| source | 0..1 | String | |
| terminationDate | 0..1 | DateTime | |
| unitOfMeasure | 0..1 | String | The UOM for the current value of the Bill Determinant. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 243 shows all association ends of BillDeterminant with other classes.

**Table 243 – Association ends of MarketResults::BillDeterminant with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..1 | ChargeProfileData | 0..* | ChargeProfileData | |
| 0..* | ChargeComponents | 0..* | ChargeComponent | A BillDeterminant can have 0-n ChargeComponent and a ChargeComponent can associate to 0-n BillDeterminant. |
| 0..1 | ChargeProfile | 0..1 | ChargeProfile | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.4 ChargeProfile

A type of profile for financial charges.

Table 244 shows all attributes of ChargeProfile.

**Table 244 – Attributes of MarketResults::ChargeProfile**

| name | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | The type of profile. It could be amount, price, or quantity. |
| frequency | 0..1 | String | The calculation frequency, daily or monthly. |
| numberInterval | 0..1 | Integer | The number of intervals in the profile data. |
| unitOfMeasure | 0..1 | String | The unit of measure applied to the value attribute of the profile data. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 245 shows all association ends of ChargeProfile with other classes.

**Table 245 – Association ends of MarketResults::ChargeProfile with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | BillDeterminant | 0..1 | BillDeterminant | |
| 0..1 | ChargeProfileData | 0..* | ChargeProfileData | |
| 0..* | PassTroughBill | 0..1 | PassThroughBill | |
| 0..* | Bid | 0..1 | Bid | |
| 0..* | ProfileDatas | 0..* | ProfileData | inherited from: Profile |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.5 ChargeProfileData root class

Model of various charges associated with an energy profile to support billing and settlement.

Table 246 shows all attributes of ChargeProfileData.

**Table 246 – Attributes of MarketResults::ChargeProfileData**

| name | mult | type | description |
|---|---|---|---|
| sequence | 0..1 | Integer | The sequence number of the profile. |
| timeStamp | 0..1 | DateTime | The date and time of an interval. |
| value | 0..1 | Float | The value of an interval given a profile type (amount, price, or quantity), subject to the UOM. |

Table 247 shows all association ends of ChargeProfileData with other classes.

**Table 247 – Association ends of MarketResults::ChargeProfileData with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BillDeterminant | 0..1 | BillDeterminant | |
| 0..* | ChargeProfile | 0..1 | ChargeProfile | |

#### 6.5.6.3.6 CommitmentClearing

Models results of market clearing which call for commitment of units.

Table 248 shows all attributes of CommitmentClearing.

**Table 248 – Attributes of MarketResults::CommitmentClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 249 shows all association ends of CommitmentClearing with other classes.

**Table 249 – Association ends of MarketResults::
CommitmentClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | Commitments | 1..* | Commitments | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.7 Commitments root class

Provides the necessary information (on a resource basis) to capture the Startup/Shutdown commitment results. This information is relevant to all markets.

Table 250 shows all attributes of Commitments.

**Table 250 – Attributes of MarketResults::Commitments**

| name | mult | type | description |
|---|---|---|---|
| commitmentType | 0..1 | CommitmentType | the type of UC status (self commitment, ISO commitment, or SCUC commitment) |
| instructionCost | 0..1 | Float | Total cost associated with changing the status of the resource. |
| instructionType | 0..1 | AutomaticDispInstTypeCommitment | Indicator of either a Start-Up or a Shut-Down. |
| intervalEndTime | 0..1 | DateTime | End time for the commitment period. This will be on an interval boundary. |
| intervalStartTime | 0..1 | DateTime | Start time for the commitment period. This will be on an interval boundary. |
| minStatusChangeTime | 0..1 | Integer | SCUC commitment period start-up time. Calculated start up time based on the StartUpTimeCurve provided with the Bid.<br>This is a combination of StartUp time bid and Unit down time.<br>Units is minutes |
| noLoadCost | 0..1 | Float | Unit no load cost in case of energy commodity |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Table 251 shows all association ends of Commitments with other classes.

**Table 251 – Association ends of MarketResults::Commitments with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 1..* | CommitmentClearing | 1..* | CommitmentClearing | |

### 6.5.6.3.8 CommodityPrice root class

The CommodityPrice class is used to define the price of a commodity during a given time interval. The interval may be long, e.g. a year, or very short, e.g. 5 minutes. There will be many instances of the CommodityPrice class for each instance of the CommodityDefinition to which it is associated. Note that there may be more than once price associated with a given interval and these variances are described by the association (or associations) with the PriceDescriptor class.

Table 252 shows all attributes of CommodityPrice.

**Table 252 – Attributes of MarketResults::CommodityPrice**

| name | mult | type | description |
|---|---|---|---|
| timeIntervalPeriod | 0..1 | DateTimeInterval | The time interval over which the CommodityPrice is valid, using the standard conventions associated with the DateTimeInterval class. |
| value | 0..1 | Float | The price of the Commodity, expressed as a floating point value with the currency and unit of measure defined in the associated CommodityDefinition class. |

Table 253 shows all association ends of CommodityPrice with other classes.

**Table 253 – Association ends of MarketResults::CommodityPrice with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | CommodityDefinition | 1..1 | CommodityDefinition | |
| 1..* | PriceDescriptor | 1..1 | PriceDescriptor | |
| 0..* | PnodeClearing | 0..1 | PnodeClearing | |

### 6.5.6.3.9 ConstraintClearing

Groups all items associated with Binding Constraints and Constraint Violations per interval and market.

Table 254 shows all attributes of ConstraintClearing.

**Table 254 – Attributes of MarketResults::ConstraintClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 255 shows all association ends of ConstraintClearing with other classes.

**Table 255 – Association ends of MarketResults::ConstraintClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ConstraintResults | 0..* | ConstraintResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.10 ConstraintResults

Provides the Market results for the constraint processing for either the DAM or RTM. The data includes the constraint type (binding or violated), the solved value for the constraint, and the associated shadow price.

Table 256 shows all attributes of ConstraintResults.

**Table 256 – Attributes of MarketResults::ConstraintResults**

| name | mult | type | description |
|---|---|---|---|
| baseFlow | 0..1 | Float | Branch base Power Flow. |
| BGLimit | 0..1 | Float | This value is determined in DA and RTM. The SCUC optimization ensures that the MW flow on the Branch Group will not exceed this limit in the relevant direction. |
| BGTRResCap | 0..1 | Float | Branch Group TR Reservation Capacity – This value is determined in DA and RTM. It is the amount of spare transmission capacity that is left for the TR holder to use. |
| bindingLimit | 0..1 | Float | MW Limit. |
| clearedValue | 0..1 | Float | Cleared MW. |
| competitivePathConstraint | 0..1 | YesNo | Non-competitive path constraint Flag"(Y/N) indicating whether the shadow price on a non-competitive path was non-zero. |
| constraintType | 0..1 | ResultsConstraintType | Type of constraint. |
| limitFlag | 0..1 | ConstraintLimitType | Limit flag ('Maximum', 'Minimum'). |
| optimizationFlag | 0..1 | YesNo | Included in optimization Y/N. |
| overloadMW | 0..1 | Float | Transmission overload MW. |
| percentMW | 0..1 | Float | Actual MW flow as percent of limit. |
| shadowPrice | 0..1 | Float | Shadow Price ($/MW) for the commodity. Shadow price for the corresponding constraint. |
| updateTimeStamp | 0..1 | DateTime | Update time stamp. |
| updateType | 0..1 | MQSCHGType | MQS change type. |
| updateUser | 0..1 | String | Updated user. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 257 shows all association ends of ConstraintResults with other classes.

**Table 257 – Association ends of MarketResults::ConstraintResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktContingency | 1..1 | MktContingency | |
| 0..* | ConstraintClearing | 0..1 | ConstraintClearing | |
| 1..* | Flowgate | 1..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.11 DopInstruction root class

Provides the necessary information (on a resource basis) to capture the Dispatch Operating Point (DOP) results on a Dispatch interval. This information is only relevant to the RT interval market.

Table 258 shows all attributes of DopInstruction.

**Table 258 – Attributes of MarketResults::DopInstruction**

| name | mult | type | description |
|---|---|---|---|
| mwDOP | 0..1 | ActivePower | Dispatched Operating Point (MW) |
| timestampDOP | 0..1 | DateTime | DOP time stamp |
| plotPriority | 0..1 | Integer | A value used to establish priority of the DOP when plotting. This is only applicable when two DOPs exist for the same time, but with different MW values. E.g. when indicating a step in the curve. Its used to determine if the curve steps up or down. |
| runIndicatorDOP | 0..1 | YesNo | Indication of DOP validity.<br>Shows the DOP is calculated from the latest run (YES). A NO indicator shows that the DOP is copied from a previous execution.<br>Up to 2 intervals can be missed. |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Table 259 shows all association ends of DopInstruction with other classes.

**Table 259 – Association ends of MarketResults::DopInstruction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResouce | 0..1 | RegisteredResource | |
| 1..* | InstructionClearingDOP | 1..* | InstructionClearingDOP | |

#### 6.5.6.3.12 DotInstruction root class

Provides the necessary information (on a resource basis) to capture the Dispatch Operating Target (DOT) results on a Dispatch interval. This information is only relevant to the RT interval market.

Table 260 shows all attributes of DotInstruction.

**Table 260 – Attributes of MarketResults::DotInstruction**

| name | mult | type | description |
|---|---|---|---|
| actualRampRate | 0..1 | Float | Actual ramp rate. |
| compliantIndicator | 0..1 | YesNo | Flag indicating whether or not the resource was in compliance with the instruction (plus/minus 10%).<br>Directs if a unit is allowed to set the price (ex-post pricing). |
| DOT | 0..1 | Float | Dispatch operating target value. |
| economicMaxOverride | 0..1 | Float | Economic Max Limit override for unit, this value is null, if it is not, this value overrides the Energy column value.<br>Allows dispatcher to override the unit's energy value. |
| expectedEnergy | 0..1 | Float | Expected energy. |
| generatorPerformanceDegree | 0..1 | Float | The Degree of Generator Performance (DGP) used for the unit. Measure of how a generator responds to raise /lower signals. Calculated every five minutes. |
| hourAheadSchedEnergy | 0..1 | Float | HASP results. |
| hourlySchedule | 0..1 | Float | Hourly Schedule (DA Energy Schedule). |
| instructionTime | 0..1 | DateTime | The date/time for the instruction. |
| maximumEmergencyInd | 0..1 | Boolean | True if maximum emergency limit activated; false otherwise. If unit is requested to move up to its max emergency limit., this flag is set to true. |
| meterLoadFollowing | 0..1 | Float | Meter Sub System Load Following. |
| nonRampRestrictedMW | 0..1 | Float | Desired MW that is not ramp restricted. If no ramp rate limit existed for the unit, this is the MW value tha t the unit was requested to move to. |
| nonSpinReserve | 0..1 | Float | Non Spin Reserve used to procure energy. |
| previousDOTTimeStamp | 0..1 | DateTime | Timestamp when the previous DOT value was issued. |
| rampRateLimit | 0..1 | Float | The ramp rate limit for the unit in MWs per minute. Participant bidding data. |
| regulationStatus | 0..1 | YesNo | Regulation Status (Yes/No). |
| spinReserve | 0..1 | Float | Spin Reserve used to procure energy. |
| standardRampEnergy | 0..1 | Float | Standard ramping energy (MWH). |
| supplementalEnergy | 0..1 | Float | Supplemental Energy procure by Real Time Dispatch. |
| unitStatus | 0..1 | Integer | Output results from the case identifying the reason the unit was committed by the software. |

Table 261 shows all association ends of DotInstruction with other classes.

**Table 261 – Association ends of MarketResults::DotInstruction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | InstructionClearingDOT | 1..* | InstructionClearingDOT | |

#### 6.5.6.3.13 ExPostLoss

Model of ex-post calcultion of MW losses.

Table 262 shows all attributes of ExPostLoss.

**Table 262 – Attributes of MarketResults::ExPostLoss**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 263 shows all association ends of ExPostLoss with other classes.

**Table 263 – Association ends of MarketResults::ExPostLoss with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ExPostLossResults | 0..* | ExPostLossResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.14 ExPostLossResults root class

Model results of ex-post calculation of MW losses. Summarizes loss in two categories losses on the the extra high voltage transmission and total losses. Calculated for each subcontrol area.

Table 264 shows all attributes of ExPostLossResults.

**Table 264 – Attributes of MarketResults::ExPostLossResults**

| name | mult | type | description |
|---|---|---|---|
| totalLossMW | 0..1 | Float | Total MW losses in the company<br>Attribute Usage: Information purposes – Output of LPA engine. |
| ehvLossMW | 0..1 | Float | EHV MW losses in the company<br>Attribute Usage: Information purposes – Output of LPA engine. |

Table 265 shows all association ends of ExPostLossResults with other classes.

**Table 265 – Association ends of MarketResults::ExPostLossResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ExPostLoss | 1..1 | ExPostLoss | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |

#### 6.5.6.3.15 ExPostMarketRegion

Model of ex-post calculation of cleared MW on a regional basis.

Table 266 shows all attributes of ExPostMarketRegion.

**Table 266 – Attributes of MarketResults::ExPostMarketRegion**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 267 shows all association ends of ExPostMarketRegion with other classes.

**Table 267 – Association ends of MarketResults::ExPostMarketRegion with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ExPostMarketRegionResults | 0..1 | ExPostMarketRegionResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.16 ExPostMarketRegionResults root class

Model of expost calculation of cleared MW on a region basis. Includes cleared price.

Table 268 shows all attributes of ExPostMarketRegionResults.

**Table 268 – Attributes of MarketResults::ExPostMarketRegionResults**

| name | mult | type | description |
|---|---|---|---|
| exPostClearedPrice | 0..1 | Float | |

Table 269 shows all association ends of ExPostMarketRegionResults with other classes.

**Table 269 – Association ends of MarketResults:: ExPostMarketRegionResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ExPostMarketRegion | 1..1 | ExPostMarketRegion | |
| 0..* | MarketRegion | 1..1 | MarketRegion | |

#### 6.5.6.3.17 ExPostPricing

Model of ex-post pricing of nodes.

Table 270 shows all attributes of ExPostPricing.

**Table 270 – Attributes of MarketResults::ExPostPricing**

| name | mult | type | description |
|---|---|---|---|
| energyPrice | 0..1 | Float | market energy price |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 271 shows all association ends of ExPostPricing with other classes.

**Table 271 – Association ends of MarketResults::ExPostPricing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.18 ExPostPricingResults root class

Model of ex-post pricing of nodes. Includes LMP information, pnode based.

Table 272 shows all attributes of ExPostPricingResults.

**Table 272 – Attributes of MarketResults::ExPostPricingResults**

| name | mult | type | description |
|---|---|---|---|
| congestLMP | 0..1 | Float | Congestion component of Location Marginal Price (LMP) in monetary units per MW; congestion component of the hourly LMP at a specific pricing node<br>Attribute Usage: Result of the Security, Pricing, and Dispatch(SPD)/Simultaneous Feasibility Test(SFT) software and denotes the hourly congestion component of LMP for each pricing node. |
| lmp | 0..1 | Float | 5 min weighted average LMP; the Location Marginal Price of the Pnode for which price calculation is carried out.<br>Attribute Usage: 5 min weighted average LMP to be displayed on UI |
| lossLMP | 0..1 | Float | Loss component of Location Marginal Price (LMP) in monetary units per MW; loss component of the hourly LMP at a specific pricing node<br>Attribute Usage: Result of the Security, Pricing, and Dispatch(SPD)/Simultaneous Feasibility Test(SFT) software and denotes the hourly loss component of LMP for each pricing node. |

Table 273 shows all association ends of ExPostPricingResults with other classes.

**Table 273 – Association ends of MarketResults:: ExPostPricingResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ExPostPricing | 1..1 | ExPostPricing | |
| 0..* | Pnode | 1..1 | Pnode | |

### 6.5.6.3.19 ExPostResource

Model of ex-post pricing of resources.

Table 274 shows all attributes of ExPostResource.

**Table 274 – Attributes of MarketResults::ExPostResource**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 275 shows all association ends of ExPostResource with other classes.

**Table 275 – Association ends of MarketResults::ExPostResource with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ExPostResourceResults | 0..* | ExPostResourceResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.20 ExPostResourceResults root class

Model of ex-post pricing of resources contains components of LMPs: energy, congestion, loss. Resource based.

Table 276 shows all attributes of ExPostResourceResults.

**Table 276 – Attributes of MarketResults::ExPostResourceResults**

| name | mult | type | description |
|---|---|---|---|
| congestionLMP | 0..1 | Float | LMP component in USD (deprecated) |
| desiredMW | 0..1 | Float | Desired output of unit |
| dispatchRate | 0..1 | Float | Unit Dispatch rate from real time unit dispatch. |
| lmp | 0..1 | Float | LMP (Local Marginal Price) in USD at the equipment (deprecated) |
| lossLMP | 0..1 | Float | loss lmp (deprecated) |
| maxEconomicMW | 0..1 | Float | Economic Maximum MW |
| minEconomicMW | 0..1 | Float | Economic Minimum MW |
| resourceMW | 0..1 | Float | Current MW output of the equipment<br>Attribute Usage: Information purposes – Information purposes – Output of LPA engine. |
| status | 0..1 | EquipmentStatusType | Status of equipment |

Table 277 shows all association ends of ExPostResourceResults with other classes.

**Table 277 – Association ends of MarketResults:: ExPostResourceResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | ExPostResource | 1..1 | ExPostResource | |

#### 6.5.6.3.21 GeneralClearing

Model of clearing result of the market run at the market level. Identifies interval.

Table 278 shows all attributes of GeneralClearing.

**Table 278 – Attributes of MarketResults::GeneralClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 279 shows all association ends of GeneralClearing with other classes.

**Table 279 – Association ends of MarketResults::GeneralClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneralClearingResults | 0..* | GeneralClearingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.22 GeneralClearingResults root class

Provides the adjusted load forecast value on a load forecast zone basis.

Table 280 shows all attributes of GeneralClearingResults.

**Table 280 – Attributes of MarketResults::GeneralClearingResults**

| name | mult | type | description |
|---|---|---|---|
| loadForecast | 0..1 | ActivePower | Load Prediction/Forecast (MW), by Time Period (5', 10', 15') |
| totalLoad | 0..1 | Float | Amount of load in the control zone<br>Attribute Usage: hourly load value for the specific area |
| totalNetInterchange | 0..1 | Float | Amount of interchange for the control zone<br>Attribute Usage: hourly interchange value for the specific area |

Table 281 shows all association ends of GeneralClearingResults with other classes.

**Table 281 – Association ends of MarketResults:: GeneralClearingResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | GeneralClearing | 0..1 | GeneralClearing | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |

### 6.5.6.3.23 InstructionClearing

Model of market clearing, relating to commitment instructions. Identifies interval.

Table 282 shows all attributes of InstructionClearing.

**Table 282 – Attributes of MarketResults::InstructionClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 283 shows all association ends of InstructionClearing with other classes.

**Table 283 – Association ends of MarketResults::InstructionClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ResourceDeploymentStatus | 0..* | ResourceDeploymentStatus | |
| 0..* | ActualDemandResponseEvent | 0..1 | DistributedResourceActualEvent | ActualDemandResponseEvents may exist that are not part of a cordinated MarketActualEvent associated to a Market. These ActualDemandResponseEvents can have many InstructionClearing Instructions for specified RegisteredResources or Distributed Energy Resource type of AggregateNodes. |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | Instructions | 1..* | Instructions | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.24 InstructionClearingDOP

Model of market clearing, related to Dispatch Operating Point. Identifies interval.

Table 284 shows all attributes of InstructionClearingDOP.

**Table 284 – Attributes of MarketResults::InstructionClearingDOP**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 285 shows all association ends of InstructionClearingDOP with other classes.

**Table 285 – Association ends of MarketResults:: InstructionClearingDOP with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | DopInstruction | 1..* | DopInstruction | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.25 InstructionClearingDOT

Model of market clearing, related to Dispatch Operating Target (model of anticipatory dispatch). Identifies interval.

Table 286 shows all attributes of InstructionClearingDOT.

**Table 286 – Attributes of MarketResults::InstructionClearingDOT**

| name | mult | type | description |
|---|---|---|---|
| contingencyActive | 0..1 | YesNo | Indication that the system is currently operating in a contingency mode. |
| dispatchMode | 0..1 | AutomaticDispatchMode | |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 287 shows all association ends of InstructionClearingDOT with other classes.

**Table 287 – Association ends of MarketResults:: InstructionClearingDOT with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | DotInstruction | 1..* | DotInstruction | |
| 0..* | DemandResponseActualEvent | 0..1 | DistributedResourceActualEvent | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.26 Instructions root class

Provides the necessary information (on a resource basis) to capture the Startup/Shutdown instruction results. This information is relevant to the DA Market (RUC only) as well as the RT Market (HASP, Pre-dispatch, and Interval).

Table 288 shows all attributes of Instructions.

**Table 288 – Attributes of MarketResults::Instructions**

| name | mult | type | description |
|---|---|---|---|
| bindingDOD | 0..1 | Float | Binding dispatch operating delta provides a relative delta to be applied. Typically used in demand response instructions. The bindingDOD instructions are cumulative; in other words a second DOD instruction does not replace the previous DOD, instead the second DOD adds to the previous DODs. |
| bindingDOT | 0..1 | Float | |
| bindingInstruction | 0..1 | YesNo | |
| instructionCost | 0..1 | Float | Total cost associated with changing the status of the resource. |
| instructionSource | 0..1 | MQSInstructionSource | instruction source for market quality results (INS, ACT) |
| instructionStartTime | 0..1 | DateTime | Time the resource should be at Pmin (for start ups).<br>Time the resource is off line. |
| instructionType | 0..1 | AutomaticDispInstTypeCommitment | Indicator of either a Start-Up or a Shut-Down. |
| manuallyBlocked | 0..1 | YesNo | Manually Blocked Indicator (Yes/No). The instruction has been blocked by an Operator. |
| minStatusChangeTime | 0..1 | Integer | Minimum start up time required to bring the unit online (minutes). |

| name | mult | type | description |
|---|---|---|---|
| | | | SCUC commitment period start-up time. Calculated start up time based on the StartUpTimeCurve provided with the Bid.<br>This is a combination of StartUp time bid and Unit down time.<br>Units is minutes |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Table 289 shows all association ends of Instructions with other classes.

**Table 289 – Association ends of MarketResults::Instructions with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 1..* | InstructionClearing | 1..* | InstructionClearing | |
| 0..* | AggregateNode | 0..1 | AggregateNode | |

#### 6.5.6.3.27 LoadFollowingOperatorInput root class

Model of load following capabilities that are entered by operators on a temporary basis. Related to Registered Resources in Metered Subsystems.

Table 290 shows all attributes of LoadFollowingOperatorInput.

**Table 290 – Attributes of MarketResults::LoadFollowingOperatorInput**

| name | mult | type | description |
|---|---|---|---|
| dataEntryTimeStamp | 0..1 | DateTime | Time the data entry was performed |
| tempLoadFollowingUpManualCap | 0..1 | Float | temporarily manually entered LFU capacity. |
| tempLoadFollowingDownManualCap | 0..1 | Float | temporarily manually entered LFD capacity |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Table 291 shows all association ends of LoadFollowingOperatorInput with other classes.

**Table 291 – Association ends of MarketResults:: LoadFollowingOperatorInput with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |

#### 6.5.6.3.28 LossClearing

RT only and is published on 5 minute intervals for the previous RT time interval results.

Table 292 shows all attributes of LossClearing.

**Table 292 – Attributes of MarketResults::LossClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 293 shows all association ends of LossClearing with other classes.

**Table 293 – Association ends of MarketResults::LossClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.29 LossClearingResults root class

Provides the MW loss for RUC Zones, subcontrol areas, and the total loss.

Table 294 shows all attributes of LossClearingResults.

**Table 294 – Attributes of MarketResults::LossClearingResults**

| name | mult | type | description |
|---|---|---|---|
| lossMW | 0..1 | Float | |

Table 295 shows all association ends of LossClearingResults with other classes.

**Table 295 – Association ends of MarketResults::LossClearingResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | LossClearing | 0..1 | LossClearing | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | RUCZone | 0..1 | RUCZone | |

#### 6.5.6.3.30 MPMClearing

Model of results of Market Power tests, and possible mitigation. Interval based.

Table 296 shows all attributes of MPMClearing.

**Table 296 – Attributes of MarketResults::MPMClearing**

| name | mult | type | description |
|---|---|---|---|
| LMPMFinalFlag | 0..1 | YesNo | |
| SMPMFinalFlag | 0..1 | YesNo | |
| mitigationOccuredFlag | 0..1 | YesNo | |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 297 shows all association ends of MPMClearing with other classes.

**Table 297 – Association ends of MarketResults::MPMClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MPMTestResults | 0..* | MPMTestResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.31 MPMResourceStatus root class

Model of results of Market Power tests, gives status of resource for the associated interval.

Table 298 shows all attributes of MPMResourceStatus.

**Table 298 – Attributes of MarketResults::MPMResourceStatus**

| name | mult | type | description |
|---|---|---|---|
| resourceStatus | 0..1 | String | Interval Test Status<br>'N' – not applicable |

Table 299 shows all association ends of MPMResourceStatus with other classes.

**Table 299 – Association ends of MarketResults::MPMResourceStatus with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |

### 6.5.6.3.32 MPMTestResults root class

Provides the outcome and margin percent (as appropriate) result data for the MPM tests. There are relationships to Zone for Designated Congestion Area Tests, CurveSchedData for bid segment tests, to the SubControlArea for the system wide level tests, and Pnodes for the LMPM impact tests.

Table 300 shows all attributes of MPMTestResults.

**Table 300 – Attributes of MarketResults::MPMTestResults**

| name | mult | type | description |
|---|---|---|---|
| outcome | 0..1 | MPMTestOutcome | The results of the test. For the Price, Impact, and Conduct tests, typical values are NA, Pass, Fail, Disable, or Skip. |
| marginPercent | 0..1 | PerCent | Used to show the Margin % result of the Impact test |

Table 301 shows all association ends of MPMTestResults with other classes.

**Table 301 – Association ends of MarketResults::MPMTestResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MPMClearing | 0..1 | MPMClearing | |
| 1..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |

### 6.5.6.3.33 MarketRegionResults root class

Provides all Region Ancillary Service results for the DA and RT markets. The specific data is commodity type (Regulation Up, Regulation Down, Spinning Reserve, Non-spinning Reserve, or Total Up reserves) based for the cleared MW, cleared price, and total capacity required for the region.

Table 302 shows all attributes of MarketRegionResults.

**Table 302 – Attributes of MarketResults::MarketRegionResults**

| name | mult | type | description |
|---|---|---|---|
| clearedMW | 0..1 | Float | Cleared generation Value in MW. For AS, this value is clearedMW = AS Total. For AS, clearedMW – selfScheduleMW = AS Procured |
| clearedPrice | 0..1 | Float | Marginal Price ($/MW) for the commodity (Energy, Regulation Up, Regulation Down, Spinning Reserve, or Non-spinning reserve) based on the pricing run. |
| dispatchCtMW | 0..1 | Float | Dispatchable MW for Combustion units. |
| dispatchHydroMW | 0..1 | Float | Dispatchable MW for Hydro units. |
| dispatchRate | 0..1 | Float | Dispatch rate in MW/minutes. |
| dispatchSteamMW | 0..1 | Float | Dispatchable MW for Steam units. |
| imbalanceEnergyBias | 0..1 | Float | Imbalance Energy Bias (MW) by Time Period (5' only) |
| limitFlag | 0..1 | ResourceLimitIndicator | Locational AS Flags indicating whether the Upper or Lower Bound limit of the AS regional procurment is binding |
| lumpyIndicator | 0..1 | YesNo | The "Lumpy Flag"(Y/N) indicates whether the resource that sets the price is a lumpy generator by hour over the time horizon.<br>Only applicable for the Day Ahead Market |
| maxSufficiencyIndex | 0..1 | Float | Region requirement maximum limit |
| minSufficiencyIndex | 0..1 | Float | Region requirement minimum limit |
| reqMaxMW | 0..1 | Float | Region requirement maximum limit |

| name | mult | type | description |
|---|---|---|---|
| reqMinMW | 0..1 | Float | Region requirement minimum limit |
| selfScheduleMW | 0..1 | Float | Aof AS, selfScheduleMW = AS Self-Provided |

Table 303 shows all association ends of MarketRegionResults with other classes.

**Table 303 – Association ends of MarketResults:: MarketRegionResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MarketProduct | 0..1 | MarketProduct | |
| 1..* | AncillaryServiceClearing | 0..1 | AncillaryServiceClearing | |
| 1..* | MarketRegion | 1..1 | MarketRegion | |

#### 6.5.6.3.34 MarketResults root class

This class holds elements that are single values for the entire market time horizon. That is, for the Day Ahead market, there is 1 value for each element, not hourly based. Is a summary of the market run.

Table 304 shows all attributes of MarketResults.

**Table 304 – Attributes of MarketResults::MarketResults**

| name | mult | type | description |
|---|---|---|---|
| startUpCost | 0..1 | Float | Total Start-up Cost ($) over the time horizon |
| minimumLoadCost | 0..1 | Float | Total Minimum Load Cost ($) over the time horizon |
| contingentOperatingResAvail | 0..1 | YesNo | Global Contingent Operating Reserve Availability Indicator (Yes/No) |
| energyCost | 0..1 | Float | Total Energy Cost ($) over the time horizon |
| totalCost | 0..1 | Float | Total Cost (Energy + AS) cost ($) by over the time horizon |
| ancillarySvcCost | 0..1 | Float | Total AS Cost (i.e., payment) ($) over the time horizon |
| totalRucCost | 0..1 | Float | The total RUC capacity cost for this interval |

Table 305 shows all association ends of MarketResults with other classes.

**Table 305 – Association ends of MarketResults::MarketResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | EnergyMarket | 1..1 | EnergyMarket | |

#### 6.5.6.3.35 MarketStatement

A statement is a roll up of statement line items. Each statement along with its line items provide the details of specific charges at any given time. Used by Billing and Settlement.

Table 306 shows all attributes of MarketStatement.

**Table 306 – Attributes of MarketResults::MarketStatement**

| name | mult | type | description |
|---|---|---|---|
| tradeDate | 0..1 | DateTime | The date of which Settlement is run. |
| referenceNumber | 0..1 | String | The version number of previous statement (in the case of true up). |
| start | 0..1 | DateTime | The start of a bill period. |
| end | 0..1 | DateTime | The end of a bill period. |
| transactionDate | 0..1 | DateTime | The date of which this statement is issued. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 307 shows all association ends of MarketStatement with other classes.

**Table 307 – Association ends of MarketResults::MarketStatement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | MarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.36 MarketStatementLineItem

An individual line item on an ISO settlement statement.

Table 308 shows all attributes of MarketStatementLineItem.

**Table 308 – Attributes of MarketResults::MarketStatementLineItem**

| name | mult | type | description |
|---|---|---|---|
| currentAmount | 0..1 | Float | Current settlement amount. |
| currentISOAmount | 0..1 | Float | Current ISO settlement amount. |
| currentISOQuantity | 0..1 | Float | Current ISO settlement quantity. |
| currentPrice | 0..1 | Float | Current settlement price. |
| currentQuantity | 0..1 | Float | Current settlement quantity, subject to the UOM. |
| intervalDate | 0..1 | DateTime | The date of which the settlement is run. |
| intervalNumber | 0..1 | String | The number of intervals. |
| netAmount | 0..1 | Float | Net settlement amount. |
| netISOAmount | 0..1 | Float | Net ISO settlement amount. |
| netISOQuantity | 0..1 | Float | Net ISO settlement quantity. |
| netPrice | 0..1 | Float | Net settlement price. |
| netQuantity | 0..1 | Float | Net settlement quantity, subject to the UOM. |
| previousAmount | 0..1 | Float | Previous settlement amount. |
| previousISOAmount | 0..1 | Float | Previous ISO settlement amount. |
| previousISOQuantity | 0..1 | Float | Previous ISO settlement quantity. |
| previousPrice | 0..1 | Float | Previous settlement price. |
| previousQuantity | 0..1 | Float | Previous settlement quantity, subject to the UOM. |
| quantityUOM | 0..1 | String | The unit of measure for the quantity element of the line item. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 309 shows all association ends of MarketStatementLineItem with other classes.

**Table 309 – Association ends of MarketResults:: MarketStatementLineItem with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | MarketStatement | 1..1 | MarketStatement | |
| 0..* | ContainerMarketStatementLineItem | 0..1 | MarketStatementLineItem | |
| 0..1 | PassThroughBill | 0..1 | PassThroughBill | |
| 0..1 | ComponentMarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.37 MitigatedBid

Mitigated bid results posted for a given settlement period.

Table 310 shows all attributes of MitigatedBid.

**Table 310 – Attributes of MarketResults::MitigatedBid**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 311 shows all association ends of MitigatedBid with other classes.

**Table 311 – Association ends of MarketResults::MitigatedBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Bid | 0..1 | Bid | |
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.38 MitigatedBidClearing

Model of market power mitigation through reference or mitigated bids. Interval based.

Table 312 shows all attributes of MitigatedBidClearing.

**Table 312 – Attributes of MarketResults::MitigatedBidClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 313 shows all association ends of MitigatedBidClearing with other classes.

**Table 313 – Association ends of MarketResults::MitigatedBidClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 1..* | MitigatedBid | 0..* | MitigatedBid | |
| 1..* | RMRDetermination | 0..* | RMRDetermination | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.39 MitigatedBidSegment root class

Model of mitigated bid. Indicates segment of piece-wise linear bid, that has been mitigated.

Table 314 shows all attributes of MitigatedBidSegment.

**Table 314 – Attributes of MarketResults::MitigatedBidSegment**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| thresholdType | 0..1 | String | |
| segmentNumber | 0..1 | Integer | Mitigated Bid Segment Number |
| segmentMW | 0..1 | Float | Mitigated bid segment MW value |

Table 315 shows all association ends of MitigatedBidSegment with other classes.

**Table 315 – Association ends of MarketResults:: MitigatedBidSegment with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | |

#### 6.5.6.3.40 PassThroughBill

Pass Through Bill is used for:

1) Two sided charge transactions with or without ISO involvement
2) Specific direct charges or payments that are calculated outside or provided directly to settlements
3) Specific charge bill determinants that are externally supplied and used in charge calculations

Table 316 shows all attributes of PassThroughBill.

**Table 316 – Attributes of MarketResults::PassThroughBill**

| name | mult | type | description |
|---|---|---|---|
| adjustedAmount | 0..1 | Money | |
| amount | 0..1 | Money | The charge amount of the product/service. |
| billedTo | 0..1 | String | The company to which the PTB transaction is billed. |
| billEnd | 0..1 | DateTime | Bill period end date |
| billRunType | 0..1 | String | The settlement run type, for example: prelim, final, and rerun. |
| billStart | 0..1 | DateTime | Bill period start date |
| effectiveDate | 0..1 | DateTime | The effective date of the transaction |
| isDisputed | 0..1 | Boolean | Disputed transaction indicator |
| isProfiled | 0..1 | Boolean | A flag indicating whether there is a profile data associated with the PTB. |
| paidTo | 0..1 | String | The company to which the PTB transaction is paid. |
| previousEnd | 0..1 | DateTime | The previous bill period end date |
| previousStart | 0..1 | DateTime | The previous bill period start date |
| price | 0..1 | Money | The price of product/service. |
| productCode | 0..1 | String | The product identifier for determining the charge type of the transaction. |
| providedBy | 0..1 | String | The company by which the PTB transaction service is provided. |
| quantity | 0..1 | FloatQuantity | The product quantity. |
| serviceEnd | 0..1 | DateTime | The end date of service provided, if periodic. |
| serviceStart | 0..1 | DateTime | The start date of service provided, if periodic. |
| soldTo | 0..1 | String | The company to which the PTB transaction is sold. |
| taxAmount | 0..1 | Money | The tax on services taken. |
| timeZone | 0..1 | String | The time zone code |
| tradeDate | 0..1 | DateTime | The trade date |
| transactionDate | 0..1 | DateTime | The date the transaction occurs. |

| name | mult | type | description |
|---|---|---|---|
| transactionType | 0..1 | String | The type of transaction. For example, charge customer, bill customer, matching AR/AP, or bill determinant |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 317 shows all association ends of PassThroughBill with other classes.

**Table 317 – Association ends of MarketResults::PassThroughBill with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..1 | MarketStatementLineItem | 0..1 | MarketStatementLineItem | |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.41 PnodeClearing

Pricing node clearing results posted for a given settlement period.

Table 318 shows all attributes of PnodeClearing.

**Table 318 – Attributes of MarketResults::PnodeClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 319 shows all association ends of PnodeClearing with other classes.

**Table 319 – Association ends of MarketResults::PnodeClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | CommodityPrice | 0..* | CommodityPrice | |
| 0..1 | PnodeResults | 1..* | PnodeResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.42 PnodeResults root class

Provides the total price, the cost component, the loss component, and the congestion component for Pnodes for the forward and real time markets. There are several prices produced based on the run type (MPM, RUC, Pricing, or Scheduling/Dispatch).

Table 320 shows all attributes of PnodeResults.

**Table 320 – Attributes of MarketResults::PnodeResults**

| name | mult | type | description |
|---|---|---|---|
| marginalClearingPrice | 0..1 | Float | Locational Marginal Price (LMP) ($/MWh) |
| costLMP | 0..1 | Float | Cost component of Locational Marginal Pricing (LMP) in monetary units per MW. |
| lossLMP | 0..1 | Float | Loss component of Location Marginal Price (LMP) in monetary units per MW. |
| congestLMP | 0..1 | Float | Congestion component of Location Marginal Price (LMP) in monetary units per MW. |
| scheduledMW | 0..1 | Float | total MW schedule at the pnode |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Table 321 shows all association ends of PnodeResults with other classes.

**Table 321 – Association ends of MarketResults::PnodeResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | PnodeClearing | 0..1 | PnodeClearing | |
| 1..* | Pnode | 0..1 | Pnode | |

### 6.5.6.3.43 PriceDescriptor root class

The price of a Commodity during a given time interval may change over time. For example, a price may be forecasted a year ahead, a month ahead, a day ahead, an hour ahead, and in real time; this is defined using the MarketType. Additionally a price may have one or more components. For example, a locational marginal energy price may be the arithmetic sum of the system price, the congestion price, and the loss price. The priceType enumeration is used determine if the price is the complete price (priceType="total") or one of potentially many constituent components.

Table 322 shows all attributes of PriceDescriptor.

**Table 322 – Attributes of MarketResults::PriceDescriptor**

| name | mult | type | description |
|---|---|---|---|
| marketType | 0..1 | MarketType | The time frame for the price, using the standard conventions associated with the MarketType enumeration. |
| priceType | 0..1 | PriceTypeKind | The "kind" of price being described. In general, the priceType will either be "total" to signify that the price is the price paid to buy or sell the commodity, sometimes referred to as an "all-in" price, or one of potentially many components. |

Table 323 shows all association ends of PriceDescriptor with other classes.

**Table 323 – Association ends of MarketResults::PriceDescriptor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | CommodityPrice | 1..* | CommodityPrice | |

### 6.5.6.3.44 RMRDetermination root class

Indicates whether unit is a reliablity must run unit: required to be on to satisfy Grid Code Reliablitiy criteria, load demand, or voltage support.

Table 324 shows all association ends of RMRDetermination with other classes.

**Table 324 – Association ends of MarketResults::RMRDetermination with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |
| 0..* | Bid | 0..1 | Bid | |

### 6.5.6.3.45 RMROperatorInput

RMR Operator's entry of the RMR requirement per market interval.

Table 325 shows all attributes of RMROperatorInput.

**Table 325 – Attributes of MarketResults::RMROperatorInput**

| name | mult | type | description |
|---|---|---|---|
| manuallySchedRMRMw | 0..1 | Float | The lower of the original pre-dispatch or the AC run schedule (Also known as the RMR Reguirement) becomes the pre-dispatch value. |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 326 shows all association ends of RMROperatorInput with other classes.

**Table 326 – Association ends of MarketResults::RMROperatorInput with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.46 RUCAwardInstruction root class

This class models the information about the RUC awards.

Table 327 shows all attributes of RUCAwardInstruction.

**Table 327 – Attributes of MarketResults::RUCAwardInstruction**

| name | mult | type | description |
|---|---|---|---|
| clearedPrice | 0..1 | Float | Marginal Price ($/MW) for the commodity (Regulation Up, Regulation Down, Spinning Reserve, or Non-spinning reserve) for pricing run. |
| marketProductType | 0..1 | MarketProductType | major product type may include the following but not limited to:<br>Energy<br>Regulation Up<br>Regulation Dn<br>Spinning Reserve<br>Non-Spinning Reserve<br>Operating Reserve |
| RUCAward | 0..1 | Float | The RUC Award of a resource is the portion of the RUC Capacity that is not under RA or RMR contracts. The RUC Award of a resource is the portion of the RUC Capacity that is eligible for RUC Availability payment. |
| RUCCapacity | 0..1 | Float | The RUC Capacity of a resource is the difference between (i) the RUC Schedule and (ii) the higher of the DA Schedule and the Minimum Load. |
| RUCSchedule | 0..1 | Float | The RUC Schedule of a resource is its output level that balances the load forecast used in RUC. The RUC Schedule in RUC is similar to the DA Schedule in DAM. |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Table 328 shows all association ends of RUCAwardInstruction with other classes.

**Table 328 – Association ends of MarketResults:: RUCAwardInstruction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ClearingResourceAward | 1..* | ResourceAwardClearing | |

#### 6.5.6.3.47 ResourceAwardClearing

Models details of bid and offer market clearing. Class indicates whether a contingency is active and whether the automatic dispatching system is active for this interval of the market solution.

Table 329 shows all attributes of ResourceAwardClearing.

**Table 329 – Attributes of MarketResults::ResourceAwardClearing**

| name | mult | type | description |
|---|---|---|---|
| dispatchMode | 0..1 | AutomaticDispatchMode | |
| contingencyActive | 0..1 | YesNo | Indication that the system is currently operating in a contingency mode. |
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 330 shows all association ends of ResourceAwardClearing with other classes.

**Table 330 – Association ends of MarketResults:: ResourceAwardClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | RUCAwardInstruction | 1..* | RUCAwardInstruction | |
| 1..* | ResourceAwardInstruction | 1..* | ResourceAwardInstruction | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.48 ResourceAwardInstruction root class

Model of market results, instruction for resource. Contains details of award as attributes.

Table 331 shows all attributes of ResourceAwardInstruction.

**Table 331 – Attributes of MarketResults::ResourceAwardInstruction**

| name | mult | type | description |
|---|---|---|---|
| awardMW | 0..1 | Float | For DA Energy: Not Applicable;<br>For DA AS: DA AS market award;<br>For RT Energy: Not Applicable;<br>For RT AS: RT AS market award (excluding DA AS market or self-proviison awards) |
| clearedMW | 0..1 | Float | For DA Energy: Total Schedule = DA market schedule + DA self-schedule award;<br>For DA AS: DA Ancillary Service Awards = DA AS market award + DA AS self-provision award;<br>For RT Energy: Total Schedule = RT market schedule + RT self-schedule award;<br>For RT AS: RT Ancillary Service Awards = RT AS self-provision award + RT AS market award + DA AS market award + DA AS self-provision award; |
| clearedPrice | 0..1 | Float | Marginal Price ($/MW) for the commodity (Regulation Up, Regulation Down, Spinning Reserve, or Non-spinning reserve) for pricing run. |
| congestLMP | 0..1 | Float | Congestion component of Location Marginal Price (LMP) in monetary units per MW. |
| costLMP | 0..1 | Float | Cost component of Locational Marginal Pricing (LMP) in monetary units per MW. |
| dispatcherAddedMW | 0..1 | Float | The tier2 mw added by dispatcher action<br>Market results of the synchronized reserve market |
| economicMax | 0..1 | Float | Unit max output for dispatch; bid in economic maximum |
| economicMin | 0..1 | Float | Unit min output for dispatch; bid in economic minimum |
| effRegulationDownLimit | 0..1 | Float | Effective Regulation Down Limit (MW) |
| effRegulationUpLimit | 0..1 | Float | Effective Regulation Up Limit |
| lmp | 0..1 | Float | Locational marginal price value |
| lossLMP | 0..1 | Float | Loss component of Location Marginal Price (LMP) in monetary units per MW. |
| manuallyBlocked | 0..1 | YesNo | Indicates if an award was manually blocked (Y/N). Valid for Spinning and Non-spinning. |
| marginalResourceIndicator | 0..1 | YesNo | Indicator (Yes / No) that this resource set the price for this dispatch / schedule. |
| mustRunInd | 0..1 | Boolean | Identifes if the unit was set to must run by the market participant responsible for bidding in the unit |
| noLoadCost | 0..1 | Float | Unit no-load cost in case of energy commodity |
| optimalBidCost | 0..1 | Float | Optimal Bid cost |
| optimalBidPay | 0..1 | Float | Optimal Bid production payment based on LMP |
| optimalMargin | 0..1 | Float | Optimal Bid production margin |

| name | mult | type | description |
|---|---|---|---|
| overrideTimeStamp | 0..1 | DateTime | Time the manual data entry occured. |
| overrideValue | 0..1 | Float | Provides the ability for the grid operator to override items, such as spin capacity requirements, prior to running the algorithm. This value is market product based (spin, non-spin, reg up, reg down, or RUC). |
| selfSchedMW | 0..1 | Float | For DA Energy: DA total self-schedule award;<br>For DA AS: DA AS self-provision award;<br>For RT Energy: RT total self-schedule award;<br>For RT AS: RT AS self-provision award (excluding DA AS market or self-provision awards) |
| startUpCost | 0..1 | Float | Unit start up cost in case of energy commodity |
| status | 0..1 | String | In or out status of resource |
| totalRevenue | 0..1 | Float | Total bid revenue (startup_cost + no_load_cost + bid_pay) |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Table 332 shows all association ends of ResourceAwardInstruction with other classes.

**Table 332 – Association ends of MarketResults:: ResourceAwardInstruction with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..1 | SelfScheduleBreakdown | 0..* | SelfScheduleBreakdown | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..* | ClearingResourceAward | 1..* | ResourceAwardClearing | |

### 6.5.6.3.49 ResourceClearing

Model of market results, including cleaing result of resources. Associated with ResourceDispatchResults.

Table 333 shows all attributes of ResourceClearing.

**Table 333 – Attributes of MarketResults::ResourceClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 334 shows all association ends of ResourceClearing with other classes.

**Table 334 – Association ends of MarketResults::ResourceClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ResourceDispatchResults | 1..* | ResourceDispatchResults | |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.50 ResourceDeploymentStatus root class

Table 335 shows all attributes of ResourceDeploymentStatus.

**Table 335 – Attributes of MarketResults::ResourceDeploymentStatus**

| name | mult | type | description |
|---|---|---|---|
| acceptComments | 0..1 | String | Commenst to explain why the acceptance status. For example, to explain why a request is accepted only partially, instead of fully. |
| acceptStatus | 0..1 | String | Status of the resource for this deployment. Values include (full compliance, partial compliance, opt-out, etc.) |
| resourceResponseMW | 0..1 | Float | The MW amount the resource can contribute for this deployment. This is from the DR provider – as a response? Or actual? Does this belong in settlement? Discuss more. |

Table 336 shows all association ends of ResourceDeploymentStatus with other classes.

**Table 336 – Association ends of MarketResults:: ResourceDeploymentStatus with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | InstructionClearing | 0..1 | InstructionClearing | |

#### 6.5.6.3.51 ResourceDispatchResults root class

The ResourceDispatchResults class provides market results that can be provided to a SC. The specific data provided consists of several indicators such as contingency flags, blocked start up, and RMR dispatch. It also provides the projected overall and the regulating status of the resource.

Table 337 shows all attributes of ResourceDispatchResults.

**Table 337 – Attributes of MarketResults::ResourceDispatchResults**

| name | mult | type | description |
|---|---|---|---|
| blockedDispatch | 0..1 | String | Blocked Dispatch Indicator (Yes/No) |
| blockedPublishDOP | 0..1 | String | Block sending DOP to ADS (Y/N) |
| contingencyFlag | 0..1 | YesNo | Contingent Operating Reserve Indicator (Yes/No). Resource participating with AS capacity in contingency dispatch. |
| limitIndicator | 0..1 | String | indicate which limit is the constraints |
| lowerLimit | 0..1 | Float | resource energy ramping lower limit |
| maxRampRate | 0..1 | Float | maximum ramp rate |
| operatingLimitHigh | 0..1 | Float | The upper operating limit incorporating any derate used by the RTD for the Binding Interval. |
| operatingLimitLow | 0..1 | Float | The lower operating limit incorporating any derate used by the RTD for the Binding Interval. |
| penaltyDispatchIndicator | 0..1 | YesNo | Penalty Dispatch Indicator (Yes / No) indicating an un-economic adjustment. |
| regulatingLimitHigh | 0..1 | Float | The upper regulating limit incorporating any derate used by the RTD for the Binding Interval. |
| regulatingLimitLow | 0..1 | Float | The lower regulating limit incorporating any derate used by the RTD for the Binding Interval. |
| resourceStatus | 0..1 | String | Unit Commitment Status (On/Off/Starting) |
| totalSchedule | 0..1 | Float | Resource total upward schedule. total schedule = En + all AS per resource per interval |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |
| upperLimit | 0..1 | Float | resource energy ramping upper limit |

Table 338 shows all association ends of ResourceDispatchResults with other classes.

**Table 338 – Association ends of MarketResults:: ResourceDispatchResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ResourceClearing | 0..1 | ResourceClearing | |

#### 6.5.6.3.52 ResourceLoadFollowingInst root class

Model of market clearing results for resources that bid to follow load.

Table 339 shows all attributes of ResourceLoadFollowingInst.

**Table 339 – Attributes of MarketResults::ResourceLoadFollowingInst**

| name | mult | type | description |
|---|---|---|---|
| instructionID | 0..1 | String | Unique instruction id per instruction, assigned by the SC and provided to ADS. ADS passes through. |
| intervalStartTime | 0..1 | DateTime | The start of the time interval for which requirement is defined. |
| calcLoadFollowingMW | 0..1 | Float | weighted average for RTPD and RTCD and same for RTID |
| dispWindowLowLimt | 0..1 | Float | |
| dispWindowHighLimt | 0..1 | Float | |

Table 340 shows all association ends of ResourceLoadFollowingInst with other classes.

**Table 340 – Association ends of MarketResults:: ResourceLoadFollowingInst with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | ResourceClearing | 0..1 | ResourceClearing | |

#### 6.5.6.3.53 ResourcePerformanceEvaluation

Represents an the performance evaluation of a resource deployment. Every resource deployment may have many performance evaluations, using different evaluation metrics or algorithms, or produced by different evaluation authorities.

Table 341 shows all attributes of ResourcePerformanceEvaluation.

**Table 341 – Attributes of MarketResults::ResourcePerformanceEvaluation**

| name | mult | type | description |
|---|---|---|---|
| effectiveEndTime | 0..1 | DateTime | |
| effectiveStartTime | 0..1 | DateTime | |
| evaluationDescription | 0..1 | String | Description of the performance evaluation, e.g. the rating classification used (any is allowed), why the evaluation was performed, anything that describes the demand response performance evaluation. |
| evaluationValue | 0..1 | String | The value of the performance. as a String, any rating scheme is supported (e.g. "1","2","3" or "low", "medium", "high"). The rating scheme is described in the performanceValueDescription attribute. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 342 shows all association ends of ResourcePerformanceEvaluation with other classes.

**Table 342 – Association ends of MarketResults:: ResourcePerformanceEvaluation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ResourcePerformanceTimeSeriesFactors | 0..* | ResourcePerformanceTimeSeriesFactor | |
| 0..* | DemandResponseActualEvent | 1..1 | DistributedResourceActualEvent | |
| 0..* | ResorcePerformanceGlobalFactor | 0..* | ResourcePerformanceGlobalFactor | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.54 ResourcePerformanceGlobalFactor

Global factors are property/value pairs that are used to adjust resource performance values. Example include scale factors (e.g. scale a baseline up or down), adders (positive or negative), etc.

Table 343 shows all attributes of ResourcePerformanceGlobalFactor.

**Table 343 – Attributes of MarketResults::ResourcePerformanceGlobalFactor**

| name | mult | type | description |
|---|---|---|---|
| factorDescription | 0..1 | String | Description (name) of the property (factor). |
| factorValue | 0..1 | String | Value of the property (factor). |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 344 shows all association ends of ResourcePerformanceGlobalFactor with other classes.

**Table 344 – Association ends of MarketResults:: ResourcePerformanceGlobalFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ResourcePerformanceEvaluation | 0..* | ResourcePerformanceEvaluation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.55 ResourcePerformanceRating

Rating of a resource for its demand response performance. e.g. given a set on monthly resource demand response performance evaluations, the resource may be rated with excellent, average, or poor performance for the sample set.

Table 345 shows all attributes of ResourcePerformanceRating.

**Table 345 – Attributes of MarketResults::ResourcePerformanceRating**

| name | mult | type | description |
|---|---|---|---|
| effectiveEndTime | 0..1 | DateTime | starting date time that the rating is valid for |
| effectiveStartTime | 0..1 | DateTime | ending date time that the rating is valid for |
| ratingDescription | 0..1 | String | the resource's demand response rating description |
| ratingType | 1..1 | String | the type of performance rating, e.g. which market or product the rating is for |
| ratingValue | 0..1 | String | the resource's demand response rating |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 346 shows all association ends of ResourcePerformanceRating with other classes.

**Table 346 – Association ends of MarketResults:: ResourcePerformanceRating with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredDistributedResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.56 ResourcePerformanceTimeSeriesFactor

Represents the performance of a resource as time series data for a specified time period, time interval, and evaluation criteria.

Table 347 shows all attributes of ResourcePerformanceTimeSeriesFactor.

**Table 347 – Attributes of MarketResults::ResourcePerformanceTimeSeriesFactor**

| name | mult | type | description |
|---|---|---|---|
| timeSeriesDataType | 0..1 | String | Type of the time series data, e.g. baseline data, meter read data, computed performance data. |
| timeSeriesDescription | 0..1 | String | Optional description of the time series data, e.g. baseline data, meter read data, computed performance data. |
| value1Description | 0..1 | String | Description for the value1 contained within the TimeSeriesFactor. |
| value2Description | 0..1 | String | Description for the value2 contained within the TimeSeriesFactor. |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 348 shows all association ends of ResourcePerformanceTimeSeriesFactor with other classes.

**Table 348 – Association ends of MarketResults:: ResourcePerformanceTimeSeriesFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ResourcePerformanceEvaluation | 1..1 | ResourcePerformanceEvaluation | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.57 SelfScheduleBreakdown root class

Model of Self Schedules Results. Includes self schedule MW,and type of self schedule for each self schedule type included in total self schedule MW value found in ResourceAwardInstruction.

Table 349 shows all attributes of SelfScheduleBreakdown.

**Table 349 – Attributes of MarketResults::SelfScheduleBreakdown**

| name | mult | type | description |
|---|---|---|---|
| selfSchedMW | 0..1 | Float | Cleared value for the specific self schedule type listed. |
| selfSchedType | 0..1 | SelfScheduleBreakdownType | Self schedule breakdown type. |

Table 350 shows all association ends of SelfScheduleBreakdown with other classes.

**Table 350 – Association ends of MarketResults:: SelfScheduleBreakdown with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ResourceAwardInstruction | 1..1 | ResourceAwardInstruction | |

#### 6.5.6.3.58 Settlement

Specifies a settlement run.

Table 351 shows all attributes of Settlement.

**Table 351 – Attributes of MarketResults::Settlement**

| name | mult | type | description |
|---|---|---|---|
| tradeDate | 0..1 | DateTime | The trade date on which the settlement is run. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 352 shows all association ends of Settlement with other classes.

**Table 352 – Association ends of MarketResults::Settlement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketInvoiceLineItem | 0..* | MarketInvoiceLineItem | |
| 0..* | MarketLedgerEntry | 0..* | MarketLedgerEntry | |
| 0..* | EnergyMarket | 0..1 | EnergyMarket | |
| 0..* | MajorChargeGroup | 1..* | MajorChargeGroup | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.6.3.59 TransactionBidClearing

Contains the intervals relavent for the associated TransactionBidResults. For example, Day Ahead cleared results for the transaction bids for each interval of the market day.

Table 353 shows all attributes of TransactionBidClearing.

**Table 353 – Attributes of MarketResults::TransactionBidClearing**

| name | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | inherited from: MarketFactors |
| intervalStartTime | 0..1 | DateTime | inherited from: MarketFactors |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 354 shows all association ends of TransactionBidClearing with other classes.

**Table 354 – Association ends of MarketResults:: TransactionBidClearing with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TransactionBidResults | 0..* | TransactionBidResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | inherited from: MarketFactors |
| 0..* | Market | 0..1 | Market | inherited from: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.6.3.60 TransactionBidResults

Contains the cleared results for each TransactionBid submitted to and accepted by the market.

Table 355 shows all attributes of TransactionBidResults.

**Table 355 – Attributes of MarketResults::TransactionBidResults**

| name | mult | type | description |
|---|---|---|---|
| clearedMW | 0..1 | Float | The market transaction megawatt |
| clearedPrice | 0..1 | Float | The price of the market transaction |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 356 shows all association ends of TransactionBidResults with other classes.

**Table 356 – Association ends of MarketResults::TransactionBidResults with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TransactionBidClearing | 1..1 | TransactionBidClearing | |
| 0..* | TransactionBid | 0..1 | TransactionBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.7 Package MktDomain

#### 6.5.7.1 General

The MktDomain package is a data dictionary of quantities and units that define datatypes for attributes (properties) that may be used by any class in any other package within MarketOperations.

#### 6.5.7.2 ActionType enumeration

Action type associated with an ActionRequest against a ParticipantInterfaces::Trade.

Table 357 shows all literals of ActionType.

**Table 357 – Literals of MktDomain::ActionType**

| literal | value | description |
|---|---|---|
| CANCEL | | Cancel a trade. |

#### 6.5.7.3 AnalogLimitType enumeration

Limit type specified for AnalogLimits.

Table 358 shows all literals of AnalogLimitType.

**Table 358 – Literals of MktDomain::AnalogLimitType**

| literal | value | description |
|---|---|---|
| BranchMediumTerm | | Branch Medium Term Limit |
| BranchLongTerm | | Branch Long Term Limit |
| BranchShortTerm | | Branch Short Term Limit |
| VoltageHigh | | Voltage High Limit |
| VoltageLow | | Voltage Low Limit |

#### 6.5.7.4 AnodeType enumeration

Aggregated Nodes Types for example:

SYS – System Zone/Region;
RUC – RUC Zone;
LFZ – Load Forecast Zone;
REG – Market Energy/Ancillary Service Region;
AGR – Aggregate Generation Resource;
POD – Point of Delivery;
ALR – Aggregate Load Resource;
LTAC – Load TransmissionAccessCharge (TAC) Group;
ACA – Adjacent Control Area
ASR – Aggregated System Resource
ECA – Embedded Control Area

Table 359 shows all literals of AnodeType.

**Table 359 – Literals of MktDomain::AnodeType**

| literal | value | description |
|---|---|---|
| SYS | | System Zone/Region; |
| RUC | | RUC Zone |
| LFZ | | Load Forecast Zone |
| REG | | Market Energy/Ancillary Service Region; |
| AGR | | Aggregate Generation Resource; |
| POD | | Point of Delivery; |
| ALR | | Aggregate Load Resource; |
| LTAC | | Load TransmissionAccessCharge (TAC) Group; |
| ACA | | Adjacent Control Area |
| ASR | | Aggregated System Resource |
| ECA | | Embedded Control Area |
| DER | | Distributed Energy Resource. |

#### 6.5.7.5 ApnodeType enumeration

Aggregate Node Types for example:

AG – Aggregated Generation

CPZ – Custom Price Zone

DPZ – Default Price Zone

LAP – Load Aggregation Point

TH – Trading Hub

SYS – System Zone

CA – Control Area

GA – generic aggregation

EHV – 500 kV

GH – generic hub

ZN – zone

INT – Interface

BUS – Bus

Table 360 shows all literals of ApnodeType.

**Table 360 – Literals of MktDomain::ApnodeType**

| literal | value | description |
|---|---|---|
| AG | | Aggregated Generation |
| CPZ | | Custom Price Zone |
| DPZ | | Default Price Zone |
| TH | | Trading Hub |
| SYS | | System Zone |
| CA | | Control Area |
| DCA | | Designated Congestion Area |
| GA | | generic aggregation |
| GH | | generic hub |
| EHV | | 500 kV – Extra High Voltage aggregate price nodes |
| ZN | | Zone |
| INT | | Interface |
| BUS | | Bus |

### 6.5.7.6 AreaControlMode enumeration

Area's present control mode.

Table 361 shows all literals of AreaControlMode.

**Table 361 – Literals of MktDomain::AreaControlMode**

| literal | value | description |
|---|---|---|
| CF | | CF = Constant Frequency |
| CTL | | Constant Tie-Line |
| TLB | | Tie-Line Bias |
| OFF | | Off control |

#### 6.5.7.7 AutomaticDispInstTypeCommitment enumeration

Commitment instruction types.

Table 362 shows all literals of AutomaticDispInstTypeCommitment.

**Table 362 – Literals of MktDomain::AutomaticDispInstTypeCommitment**

| literal | value | description |
|---|---|---|
| START_UP | | Start up instruction type |
| SHUT_DOWN | | Shut down instruction type |
| DR_DEPLOY | | Distributed Resource Deployment |
| DR_RELEASE | | Distributed Resource Release |
| DR_ADJUSTMENT | | Distributed Resource adjustment for a distributed resource that is already deployed (committed). |

#### 6.5.7.8 AutomaticDispatchMode enumeration

Automatic Dispatch mode.

Table 363 shows all literals of AutomaticDispatchMode.

**Table 363 – Literals of MktDomain::AutomaticDispatchMode**

| literal | value | description |
|---|---|---|
| INTERVAL | | |
| CONTINGENCY | | Contingnency occurance, redispatch of contingency reserves |
| MANUAL | | Operator override |

#### 6.5.7.9 BidCalculationBasis enumeration

The basis used to calculate the bid price curve for an energy default bid.

Table 364 shows all literals of BidCalculationBasis.

**Table 364 – Literals of MktDomain::BidCalculationBasis**

| literal | value | description |
|---|---|---|
| LMP_BASED | | Based on prices paid at particular pricing location. |
| COST_BASED | | Based on unit generation characteristics and a cost of fuel. |
| NEGOTIATED | | An amount negotiated with the designated Independent Entity. |

#### 6.5.7.10 BidMitigationStatus enumeration

For example:

'S' – Mitigated by SMPM because of "misconduct"

'L; – Mitigated by LMPM because of "misconduct"

'R' – Modified by LMPM because of RMR rules

'M' – Mitigated because of "misconduct" both by SMPM and LMPM

'B' – Mitigated because of "misconduct" both by SMPM and modified by LMLM because of RMR rules

'O' – original

Table 365 shows all literals of BidMitigationStatus.

**Table 365 – Literals of MktDomain::BidMitigationStatus**

| literal | value | description |
|---|---|---|
| S | | |
| L | | |
| R | | |
| M | | |
| B | | |
| O | | |

#### 6.5.7.11 BidMitigationType enumeration

For example:

Initial

Final

Table 366 shows all literals of BidMitigationType.

**Table 366 – Literals of MktDomain::BidMitigationType**

| literal | value | description |
|---|---|---|
| I | | |
| F | | |

#### 6.5.7.12 BidType enumeration

For example:

DEFAULT_ENERGY_BID

DEFAULT_STARTUP_BID

DEFAULT_MINIMUM_LOAD_BID

Table 367 shows all literals of BidType.

**Table 367 – Literals of MktDomain::BidType**

| literal | value | description |
|---|---|---|
| DEFAULT_ENERGY_BID | | |
| DEFAULT_STARTUP_BID | | |
| DEFAULT_MINIMUM_LOAD_BID | | |

#### 6.5.7.13 BidTypeRMR enumeration

Bid self schedule type has two types as the required output of requirements and qualified pre-dispatch.

Table 368 shows all literals of BidTypeRMR.

**Table 368 – Literals of MktDomain::BidTypeRMR**

| literal | value | description |
|---|---|---|
| REQUIREMENTS | | Qualified pre-dispatch bid self schedule type. |
| QUALIFIED_PREDISPATCH | | Output of requirements bid self schedule type. |

#### 6.5.7.14 CRRCategoryType enumeration

Congestion Revenue Rights category types.

Table 369 shows all literals of CRRCategoryType.

**Table 369 – Literals of MktDomain::CRRCategoryType**

| literal | value | description |
|---|---|---|
| NSR | | Network Service |
| PTP | | Point to Point |

#### 6.5.7.15 CRRHedgeType enumeration

Congestion Revenue Right hedge type.

Table 370 shows all literals of CRRHedgeType.

**Table 370 – Literals of MktDomain::CRRHedgeType**

| literal | value | description |
|---|---|---|
| OBLIGATION | | |
| OPTION | | |

#### 6.5.7.16 CRRRoleType enumeration

Role types an organisation can play with respect to a congestion revenue right.

Table 371 shows all literals of CRRRoleType.

**Table 371 – Literals of MktDomain::CRRRoleType**

| literal | value | description |
|---|---|---|
| SELLER | | |
| BUYER | | |
| OWNER | | |

#### 6.5.7.17 CRRSegmentType enumeration

Type of the CRR, from the possible type definitions in the CRR System (e.g. 'LSE', 'ETC').

Table 372 shows all literals of CRRSegmentType.

**Table 372 – Literals of MktDomain::CRRSegmentType**

| literal | value | description |
|---|---|---|
| AUC | | |
| CAP | | |
| CF | | |
| CVR | | Converted rights. |
| ETC | | Existing Transmission Contract. |
| LSE | | Load Serving Entity. |
| MT | | Merchant transmission. |
| TOR | | Transmission Ownership Rights. |

#### 6.5.7.18 CheckOutType enumeration

To indicate a check out type such as adjusted capacity or dispatch capacity.

Table 373 shows all literals of CheckOutType.

**Table 373 – Literals of MktDomain::CheckOutType**

| literal | value | description |
|---|---|---|
| PRE_HOUR | | |
| PRE_SCHEDULE | | |
| AFTER_THE_FACT | | |

#### 6.5.7.19 CommitmentType enumeration

For example:

SELF – Self commitment
ISO – New commitment for this market period
UC – Existing commitment that was a hold over from a previous market

Table 374 shows all literals of CommitmentType.

**Table 374 – Literals of MktDomain::CommitmentType**

| literal | value | description |
|---|---|---|
| SELF | | |
| ISO | | |
| UC | | |

#### 6.5.7.20 ConstraintLimitType enumeration

Binding constraint results limit type, For example:

MAXIMUM

MINIMUM

Table 375 shows all literals of ConstraintLimitType.

**Table 375 – Literals of MktDomain::ConstraintLimitType**

| literal | value | description |
|---|---|---|
| MAXIMUM | | |
| MINIMUM | | |

#### 6.5.7.21 ConstraintRampType enumeration

Constraint Ramp type.

Table 376 shows all literals of ConstraintRampType.

**Table 376 – Literals of MktDomain::ConstraintRampType**

| literal | value | description |
|---|---|---|
| FAST | | |
| SLOW | | |

#### 6.5.7.22 ContractType enumeration

Transmission Contract Type, For example:

O – Other

TE – Transmission Export

TI – Transmission Import

ETC – Existing Transmission Contract

RMT – RMT Contract

TOR – Transmission Ownership Right

RMR – Reliability Must Run Contract

CVR – Converted contract

Table 377 shows all literals of ContractType.

**Table 377 – Literals of MktDomain::ContractType**

| literal | value | description |
|---|---|---|
| ETC | | ETC – Existing Transmission Contract |
| TOR | | TOR – Transmission Ownership Right |
| RMR | | RMR – Reliability Must Run Contract |
| RMT | | RMT – RMT Contract |
| O | | O – Other |
| TE | | TE – Transmission Export |
| TI | | TI – Transmission Import |
| CVR | | CVR – Converted contract. |

#### 6.5.7.23 CostBasis enumeration

For example:

Bid Cost
Proxy Cost
Registered Cost

Table 378 shows all literals of CostBasis.

**Table 378 – Literals of MktDomain::CostBasis**

| literal | value | description |
|---|---|---|
| BIDC | | |
| PRXC | | |
| REGC | | |

#### 6.5.7.24 DispatchResponseType enumeration

For example:

NON_RESPONSE
ACCEPT
DECLINE
PARTIAL

Table 379 shows all literals of DispatchResponseType.

**Table 379 – Literals of MktDomain::DispatchResponseType**

| literal | value | description |
|---|---|---|
| NON_RESPONSE | | |
| ACCEPT | | |
| DECLINE | | |
| PARTIAL | | |

#### 6.5.7.25 EnergyPriceIndexType enumeration

For example:

WHOLESALE
RETAIL
BOTH

Table 380 shows all literals of EnergyPriceIndexType.

**Table 380 – Literals of MktDomain::EnergyPriceIndexType**

| literal | value | description |
|---|---|---|
| WHOLESALE | | |
| RETAIL | | |
| BOTH | | |

#### 6.5.7.26 EnergyProductType enumeration

Energy product type.

Table 381 shows all literals of EnergyProductType.

**Table 381 – Literals of MktDomain::EnergyProductType**

| literal | value | description |
|---|---|---|
| FIRM | | Firm |
| NFRM | | Non Firm |
| DYN | | Dynamic |
| WHL | | Wheeling |

#### 6.5.7.27 EquipmentStatusType enumeration

Status of equipment.

Table 382 shows all literals of EquipmentStatusType.

**Table 382 – Literals of MktDomain::EquipmentStatusType**

| literal | value | description |
|---|---|---|
| In | | Equipment is in. |
| Out | | Equipment is out. |

#### 6.5.7.28 EnergyTransactionType enumeration

Defines the state of a transaction.

Table 383 shows all literals of EnergyTransactionType.

**Table 383 – Literals of MktDomain::EnergyTransactionType**

| literal | value | description |
|---|---|---|
| approve | | Approve |
| deny | | Deny |
| study | | Study |

#### 6.5.7.29 ExecutionType enumeration

Execution types of Market Runs.

Table 384 shows all literals of ExecutionType.

**Table 384 – Literals of MktDomain::ExecutionType**

| literal | value | description |
|---|---|---|
| DA | | Day Ahead |
| HASP | | Real TIme Hour Ahead Execution |
| RTPD | | Real Time Pre-dispatch |
| RTD | | Real Time Dispatch |

### 6.5.7.30 FlagTypeRMR enumeration

Indicates whether the unit is RMR and it's condition type, for example:

N' – not an RMR unit
'1' – RMR Condition 1 unit
'2' – RMR Condition 2 unit

Table 385 shows all literals of FlagTypeRMR.

**Table 385 – Literals of MktDomain::FlagTypeRMR**

| literal | value | description |
|---|---|---|
| N | | 'N' – not an RMR unit |
| 1 | | '1' – RMR Condition 1 unit |
| 2 | | '2' – RMR Condition 2 unit |

### 6.5.7.31 FlowDirectionType enumeration

Specifies the direction of energy flow in the flowgate.

Table 386 shows all literals of FlowDirectionType.

**Table 386 – Literals of MktDomain::FlowDirectionType**

| literal | value | description |
|---|---|---|
| Forward | | Forward direction. |
| Reverse | | Reverse direction. |

### 6.5.7.32 FuelSource enumeration

For example:

Bio Gas (Landfill, Sewage, Digester, etc.)
Biomass
Coal
DIST
Natural Gas
Geothermal
HRCV
None
Nuclear

Oil

Other

Solar

Waste to Energy

Water

Wind

Table 387 shows all literals of FuelSource.

**Table 387 – Literals of MktDomain::FuelSource**

| literal | value | description |
|---|---|---|
| NG | | Natural Gas |
| NNG | | Non-Natural Gas |
| BGAS | | Bio Gas (Landfill, Sewage, Digester, etc.) |
| BIOM | | Biomass |
| COAL | | Coal |
| DIST | | |
| GAS | | |
| GEOT | | GeoThermal |
| HRCV | | |
| NONE | | |
| NUCL | | Nuclear |
| OIL | | |
| OTHR | | Other |
| SOLR | | Solar |
| WAST | | Waste to Energy |
| WATR | | Water |
| WIND | | Wind |

#### 6.5.7.33 InterTieDirection enumeration

Direction of an intertie.

Table 388 shows all literals of InterTieDirection.

**Table 388 – Literals of MktDomain::InterTieDirection**

| literal | value | description |
|---|---|---|
| E | | Export. |
| I | | Import. |

#### 6.5.7.34 LoadForecastType enumeration

Load forecast zone types.

Table 389 shows all literals of LoadForecastType.

**Table 389 – Literals of MktDomain::LoadForecastType**

| literal | value | description |
|---|---|---|
| LFZ | | Load forecast zone. |
| LZMS | | Metered sub system zone. |

### 6.5.7.35 MPMTestIdentifierType enumeration

Market power mitigation test identifier type, for example:

1 – Global Price Test

2 – Global Conduct Test

3 – Global Impact Test

4 – Local Price Test

5 – Local Conduct Test

6 – Local Impact Test

Table 390 shows all literals of MPMTestIdentifierType.

**Table 390 – Literals of MktDomain::MPMTestIdentifierType**

| literal | value | description |
|---|---|---|
| 1 | | 1 – Global Price Test. |
| 2 | | 2 – Global Conduct Test. |
| 3 | | 3 – Global Impact Test. |
| 4 | | 4 – Local Price Test. |
| 5 | | 5 – Local Conduct Test. |
| 6 | | 6 – Local Impact Test. |

### 6.5.7.36 MPMTestMethodType enumeration

Market power mitigation test method type.

Tests with the normal (default) thresholds or tests with the alternate thresholds.

Table 391 shows all literals of MPMTestMethodType.

**Table 391 – Literals of MktDomain::MPMTestMethodType**

| literal | value | description |
|---|---|---|
| NORMAL | | Normal. |
| ALTERNATE | | Alternate. |

### 6.5.7.37 MPMTestOutcome enumeration

For example:

Passed

Failed

Disabled

Skipped

Table 392 shows all literals of MPMTestOutcome.

**Table 392 – Literals of MktDomain::MPMTestOutcome**

| literal | value | description |
|---|---|---|
| P | | Passed |
| F | | Failed |
| D | | Disabled |
| S | | Skipped |

### 6.5.7.38 MQSCHGType enumeration

For example:

ADD – add
CHG – change

Table 393 shows all literals of MQSCHGType.

**Table 393 – Literals of MktDomain::MQSCHGType**

| literal | value | description |
|---|---|---|
| ADD | | |
| CHG | | |

### 6.5.7.39 MQSInstructionSource enumeration

Valid values, for example:

INS – Instruction from RTM
ACT – Actual instruction after the fact

Table 394 shows all literals of MQSInstructionSource.

**Table 394 – Literals of MktDomain::MQSInstructionSource**

| literal | value | description |
|---|---|---|
| INS | | |
| ACT | | |

### 6.5.7.40 MarketEventStatusKind enumeration

Market event status types.

Table 395 shows all literals of MarketEventStatusKind.

**Table 395 – Literals of MktDomain::MarketEventStatusKind**

| literal | value | description |
|---|---|---|
| active | | The status of the event is currently in a active state.<br>Active (when sysdate is equal or greater than to planned start time) |
| cancelled | | The status of the event is currently in a cancelled state.<br>Cancelled (stopped before planned start time or planned end time) |
| completed | | The status of the event is currently in a completed state.<br>Complete (when sysdate is equal to the release time) |
| planned | | The status of the event is currently in a planned state.<br>Planned (sysdate is less than planned start time) |

### 6.5.7.41 MarketProductSelfSchedType enumeration

Market product self schedule bid types.

Table 396 shows all literals of MarketProductSelfSchedType.

**Table 396 – Literals of MktDomain::MarketProductSelfSchedType**

| literal | value | description |
|---|---|---|
| ETC | | Existing Transmission Contract. |
| TOR | | Transmission Ownership Right. |
| RMR | | Reliability Must Run. |
| RGMR | | Regulatory must run. |
| RMT | | Reliability must take. |
| PT | | Price taker. |
| LPT | | Low price taker. |
| SP | | Self provision. |
| RA | | Resource adequacy. |

### 6.5.7.42 MarketProductType enumeration

For example:

Energy, Reg Up, Reg Down, Spin Reserve, Nonspin Reserve, RUC, Load Folloing Up, and Load Following Down.

Table 397 shows all literals of MarketProductType.

**Table 397 – Literals of MktDomain::MarketProductType**

| literal | value | description |
|---|---|---|
| EN | | energy type |
| RU | | regulation up |
| RD | | regulation down |
| SR | | spinning reserve |
| NR | | non spinning reserve |
| RC | | Residual Unit Commitment |
| LFU | | Load following up |
| LFD | | Load following down |
| REG | | Regulation |

### 6.5.7.43 MarketType enumeration

Market type.

Table 398 shows all literals of MarketType.

**Table 398 – Literals of MktDomain::MarketType**

| literal | value | description |
|---|---|---|
| DAM | | Day ahead market. |
| RTM | | Real time market. |
| HAM | | Hour Ahead Market. |
| RUC | | Residual Unit Commitment. |

### 6.5.7.44 MktAccountKind enumeration

Kind of Market account.

Table 399 shows all literals of MktAccountKind.

**Table 399 – Literals of MktDomain::MktAccountKind**

| literal | value | description |
|---|---|---|
| normal | | |
| reversal | | |
| statistical | | |
| estimate | | |

### 6.5.7.45 MktBillMediaKind enumeration

Kind of bill media.

Table 400 shows all literals of MktBillMediaKind.

**Table 400 – Literals of MktDomain::MktBillMediaKind**

| literal | value | description |
|---|---|---|
| paper | | |
| electronic | | |
| other | | |

#### 6.5.7.46 MktInvoiceLineItemKind enumeration

Kind of invoice line item.

Table 401 shows all literals of MktInvoiceLineItemKind.

**Table 401 – Literals of MktDomain::MktInvoiceLineItemKind**

| literal | value | description |
|---|---|---|
| initial | | |
| recalculation | | |
| other | | |

#### 6.5.7.47 OnOff enumeration

ON

OFF

Table 402 shows all literals of OnOff.

**Table 402 – Literals of MktDomain::OnOff**

| literal | value | description |
|---|---|---|
| ON | | |
| OFF | | |

#### 6.5.7.48 ParticipationCategoryMPM enumeration

For example:

'Y' – Participates in both LMPM and SMPM

'N' – Not included in LMP price measures

'S' – Participates in SMPM price measures

'L' – Participates in LMPM price measures

Table 403 shows all literals of ParticipationCategoryMPM.

**Table 403 – Literals of MktDomain::ParticipationCategoryMPM**

| literal | value | description |
|---|---|---|
| Y | | |
| N | | |
| S | | |
| L | | |

#### 6.5.7.49 PassIndicatorType enumeration

Defines the individual passes that produce results per execution type/market type.

Table 404 shows all literals of PassIndicatorType.

**Table 404 – Literals of MktDomain::PassIndicatorType**

| literal | value | description |
|---|---|---|
| MPM-1 | | Market Power Mitigation Pass 1 |
| MPM-2 | | Market Power Mitigation Pass 2 |
| MPM-3 | | Market Power Mitigation Pass 3 |
| MPM-4 | | Market Power Mitigation Pass 4 |
| RUC | | Residual Unit Commitment |
| RTPD | | Real Time Pre Dispatch |
| RTED | | Real Time Economic Dispatch |
| HA-SCUC | | Hour Ahead Security Constrained Unit Commitment |
| DA | | Day Ahead |

#### 6.5.7.50 PriceTypeKind enumeration

Value of this enumeration for different prices include "total" for the complete/full/all-in price, "congestion" for the congestion cost associated with the total price, the "loss" for the loss price associated with the total price, "capacity" for prices related to installed or reserved capacity, "mileage" for use-based accounting, "system" for system-wide/copper-plate prices, and "delivery" for distribution-based prices.

Table 405 shows all literals of PriceTypeKind.

**Table 405 – Literals of MktDomain::PriceTypeKind**

| literal | value | description |
|---|---|---|
| capacity | | |
| congestion | | |
| delivery | | |
| loss | | |
| mileage | | |
| system | | |
| total | | |

#### 6.5.7.51 PurposeFlagType enumeration

MPM Purpose Flag, for example:

Nature of threshold data:

'M' – Mitigation threshold

'R' – Reporting threshold

Table 406 shows all literals of PurposeFlagType.

**Table 406 – Literals of MktDomain::PurposeFlagType**

| literal | value | description |
|---|---|---|
| M | | |
| R | | |

### 6.5.7.52 RampCurveType enumeration

For example:

0 – Fixed ramp rate independent of rate function unit MW output

1 – Static ramp rates as a function of unit MW output only

2 – Dynamic ramp rates as a function of unit MW output and ramping time

Table 407 shows all literals of RampCurveType.

**Table 407 – Literals of MktDomain::RampCurveType**

| literal | value | description |
|---|---|---|
| 0 | | Fixed ramp rate independent of rate function unit MW output |
| 1 | | Static ramp rates as a function of unit MW output only |
| 2 | | Dynamic ramp rates as a function of unit MW output and ramping time |

### 6.5.7.53 RampRateCondition enumeration

Ramp rate condition.

Table 408 shows all literals of RampRateCondition.

**Table 408 – Literals of MktDomain::RampRateCondition**

| literal | value | description |
|---|---|---|
| WORST | | |
| BEST | | |
| NORMAL | | |
| NA | | not applicable |

### 6.5.7.54 RampRateType enumeration

Ramp rate curve type.

Table 409 shows all literals of RampRateType.

**Table 409 – Literals of MktDomain::RampRateType**

| literal | value | description |
|---|---|---|
| OP | | Operational ramp rate. |
| REG | | Regulating ramp rate. |
| OP_RES | | Operating reserve ramp rate. |
| LD_DROP | | Load drop ramp rate. |
| LD_PICKUP | | Load pick up rate. |
| INTERTIE | | Intertie ramp rate. |

### 6.5.7.55 ReserveRequirementType enumeration

For example:

Operating Reserve, Regulation, Contingency

Table 410 shows all literals of ReserveRequirementType.

**Table 410 – Literals of MktDomain::ReserveRequirementType**

| literal | value | description |
|---|---|---|
| OPRSV | | Operating Reserve |
| CONT | | Contingency |
| REG | | Regulation |

### 6.5.7.56 ResourceAssnType enumeration

For example:

Asset Owner Sink designator for use by CRR
Asset Owner Source designator for use by CRR
Reliability Must Run
Scheduling Coordinator
Load Serving Entity

Table 411 shows all literals of ResourceAssnType.

**Table 411 – Literals of MktDomain::ResourceAssnType**

| literal | value | description |
|---|---|---|
| CSNK | | |
| CSRC | | |
| RMR | | |
| SC | | |
| LSE | | |

### 6.5.7.57 ResourceCapacityType enumeration

Resource capacity type.

Table 412 shows all literals of ResourceCapacityType.

**Table 412 – Literals of MktDomain::ResourceCapacityType**

| literal | value | description |
|---|---|---|
| RU | | Regulation Up. |
| RD | | Regulation Down. |
| SR | | Spinning reserve. |
| NR | | Non spinning reserve. |
| MO | | Must Offer. |
| FO | | Flexible Offer. |
| RA | | Resource Adequacy. |
| RMR | | Reliability Must Run. |

#### 6.5.7.58 ResourceCertificationKind enumeration

Types used for resource certification.

Table 413 shows all literals of ResourceCertificationKind.

**Table 413 – Literals of MktDomain::ResourceCertificationKind**

| literal | value | description |
|---|---|---|
| RegulationUp | | Regulation Up (RU, REGUP) |
| RegulationDown | | Regulation Down (RD, REGDN) |
| SpinningReserve | | Spinning Reserve (SR, RRSPIN) |
| NonSpinningReserve | | Non Spinning Reserve (NR, NONSPIN) |
| ReliabilityMustRun | | Reliability Must Run (RMR) |
| BLACKSTART | | Black start |
| DemandSideResponse | | Demand Side Reponse (DSR) |
| SynchronousCondenser | | Synchronous Condenser (SYNCCOND) |
| IntermittentResource | | Intermittent resource |
| ReliabilityUnitCommitment | | Reliability unit commitment (RUC) |
| Energy | | |
| Capacity | | |

#### 6.5.7.59 ResourceLimitIndicator enumeration

Locational AS Flags indicating whether the Upper or Lower Bound limit of the AS regional procurement is binding.

Table 414 shows all literals of ResourceLimitIndicator.

**Table 414 – Literals of MktDomain::ResourceLimitIndicator**

| literal | value | description |
|---|---|---|
| UPPER | | |
| LOWER | | |

#### 6.5.7.60 ResourceRegistrationStatus enumeration

Types of resource registration status, for example:

Active
Mothballed
Planned
Decommissioned

Table 415 shows all literals of ResourceRegistrationStatus.

**Table 415 – Literals of MktDomain::ResourceRegistrationStatus**

| literal | value | description |
|---|---|---|
| Active | | Resource registration is active |
| Planned | | Registration status is in the planning stage |
| Mothballed | | Resource registration has been suspended |
| Decommissioned | | Resource registration status is decommissioned |

#### 6.5.7.61 ResultsConstraintType enumeration

Market results binding constraint types.

Table 416 shows all literals of ResultsConstraintType.

**Table 416 – Literals of MktDomain::ResultsConstraintType**

| literal | value | description |
|---|---|---|
| FG_act | | Flowgate actual base case |
| Contingency | | Contingency. |
| Interface | | Interface. |
| Actual | | Actual. |

#### 6.5.7.62 SelfSchedReferenceType enumeration

Indication of which type of self schedule is being referenced.

Table 417 shows all literals of SelfSchedReferenceType.

**Table 417 – Literals of MktDomain::SelfSchedReferenceType**

| literal | value | description |
|---|---|---|
| ETC | | Existing transmission contract. |
| TOR | | Transmission ownership right. |

#### 6.5.7.63 SelfScheduleBreakdownType enumeration

Self schedule breakdown type.

Table 418 shows all literals of SelfScheduleBreakdownType.

**Table 418 – Literals of MktDomain::SelfScheduleBreakdownType**

| literal | value | description |
|---|---|---|
| ETC | | Existing transmission contract. |
| TOR | | Transmission ownership right. |
| LPT | | Low price taker. |

#### 6.5.7.64 SwitchStatusType enumeration

Circuit Breaker Status (closed or open) of the circuit breaker.

Table 419 shows all literals of SwitchStatusType.

**Table 419 – Literals of MktDomain::SwitchStatusType**

| literal | value | description |
|---|---|---|
| Closed | | Closed status. |
| Open | | Open status. |

#### 6.5.7.65 TRType enumeration

Transmission Contract Right type -for example:

individual or chain of contract rights

Table 420 shows all literals of TRType.

**Table 420 – Literals of MktDomain::TRType**

| literal | value | description |
|---|---|---|
| CHAIN | | TR chain |
| INDIVIDUAL | | Individual TR |

#### 6.5.7.66 TimeOfUse enumeration

Time of Use used by a CRR definition for specifying the time the CRR spans. ON – CRR spans the on peak hours of the day, OFF – CRR spans the off peak hours of the day, 24HR – CRR spans the entire day.

Table 421 shows all literals of TimeOfUse.

**Table 421 – Literals of MktDomain::TimeOfUse**

| literal | value | description |
|---|---|---|
| ON | | Time of use spans only the on peak hours of the day. |
| OFF | | Time of use spans only the off peak hours of the day. |
| 24HR | | Time of use spans the entire day, 24 hours. |

#### 6.5.7.67 TradeType enumeration

Trade type.

Table 422 shows all literals of TradeType.

**Table 422 – Literals of MktDomain::TradeType**

| literal | value | description |
|---|---|---|
| IST | | InterSC Trade. |
| AST | | Ancillary Services Trade. |
| UCT | | Unit Commitment Trade. |

#### 6.5.7.68 UnitRegulationKind enumeration

Unit regulation kind.

Table 423 shows all literals of UnitRegulationKind.

**Table 423 – Literals of MktDomain::UnitRegulationKind**

| literal | value | description |
|---|---|---|
| 0 | | Unit is not on regulation. |
| 1 | | Unit is on AGC and regulating. |
| 2 | | Unit is suppose to be on regulation but it is not under regulation now. |

#### 6.5.7.69 UnitType enumeration

Combined Cycle

Gas Turbine

Hydro Turbine

Other

Photovoltaic

Hydro Pump-Turbine

Reciprocating Engine

Steam Turbine

Synchronous Condenser

Wind Turbine

Table 424 shows all literals of UnitType.

**Table 424 – Literals of MktDomain::UnitType**

| literal | value | description |
|---|---|---|
| CCYC | | Combined Cycle |
| GTUR | | Gas Turbine |
| HYDR | | Hydro Turbine |
| OTHR | | Other |
| PHOT | | Photovoltaic |
| PTUR | | Hydro Pump-Turbine |
| RECP | | Reciprocating Engine |
| STUR | | Steam Turbine |
| SYNC | | Synchronous Condenser |
| WIND | | Wind Turbine |

#### 6.5.7.70 YesNo enumeration

Used as a flag set to Yes or No.

Table 425 shows all literals of YesNo.

**Table 425 – Literals of MktDomain::YesNo**

| literal | value | description |
|---|---|---|
| YES | | |
| NO | | |

### 6.5.8 Package ParticipantInterfaces

#### 6.5.8.1 General

Market participant interfaces for bids and trades.

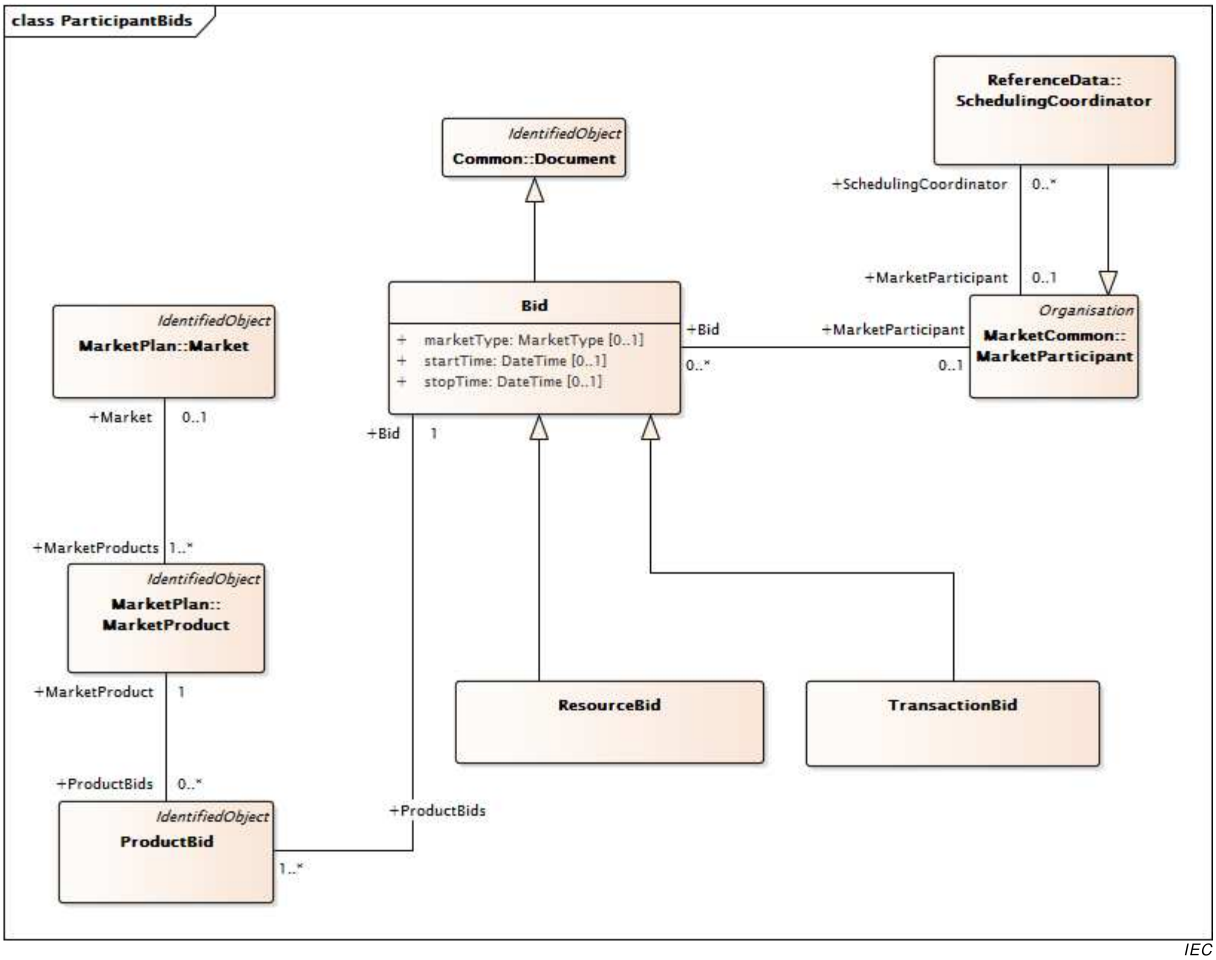



**Figure 54 – Class diagram ParticipantInterfaces::ParticipantBids**

Figure 54: Shows the overview of the class and associations that are used to model offers and bids. The term bid is used to include offers to sell and bids to buy one or more electrical products. The bid is a subclass of the document class from the IEC 61968 package. Bids are further classified as Resource Bids or Transaction Bids. Bids are associated with Scheduling Coordinators that submit them on behalf of market participants as shown by the association between the Bid class and the Scheduling Coordinator class.

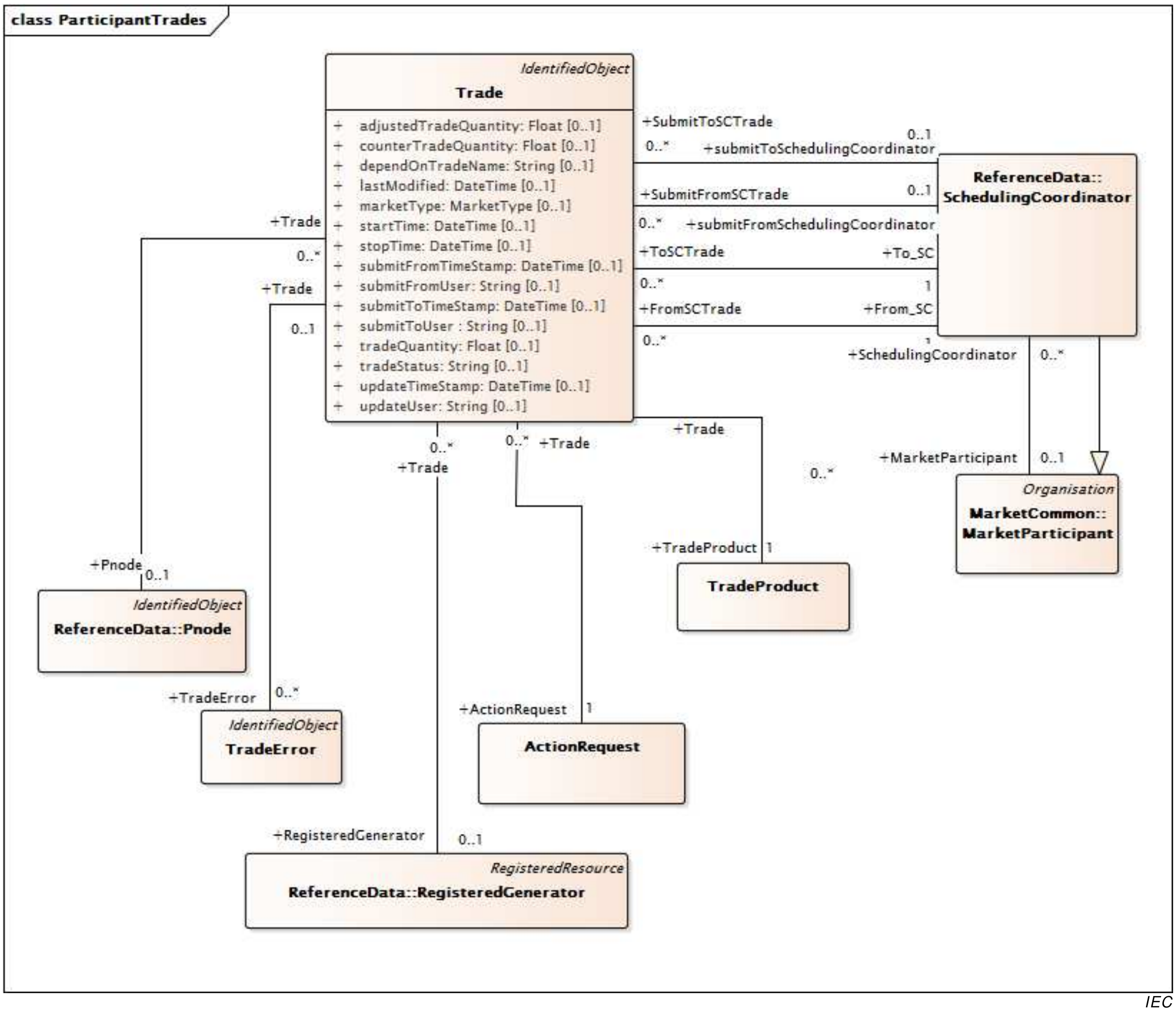



**Figure 55 – Class diagram ParticipantInterfaces::ParticipantTrades**

Figure 55: Shows the class and associations used to model inter scheduling coordinator financial trades. The TradeProduct class is used to indicate the electric product that is included in the trade. The SchedulingCoordinator class is used to indicate the scheduling coordinators involved in the trade.

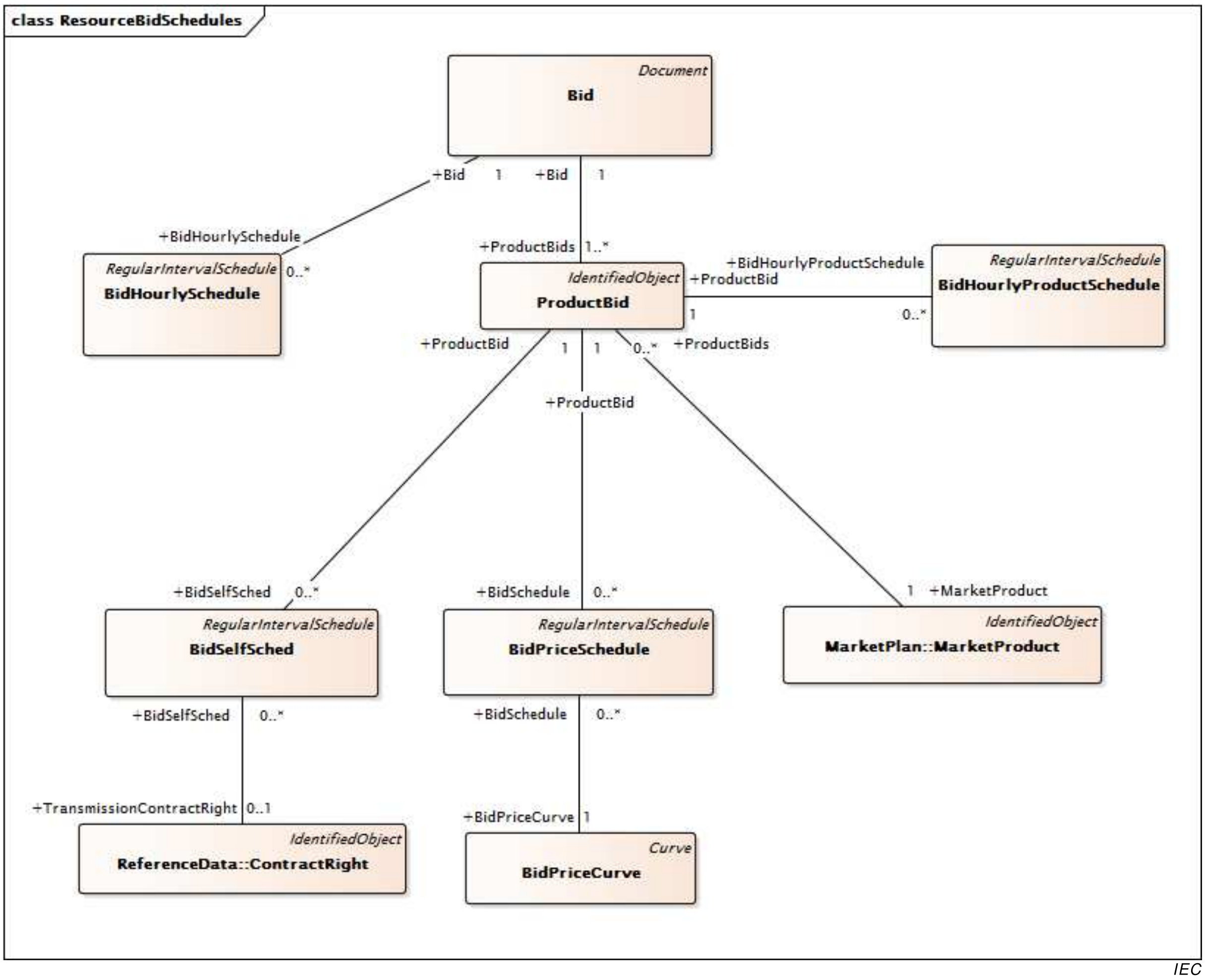



**Figure 56 – Class diagram ParticipantInterfaces::ResourceBidSchedules**

Figure 56: Shows the classes and associations used to model bid schedules. A scheduling coordinator may choose to submit a bid that is valid for several intervals of the market's time horizon. The classes of this model support this concept.

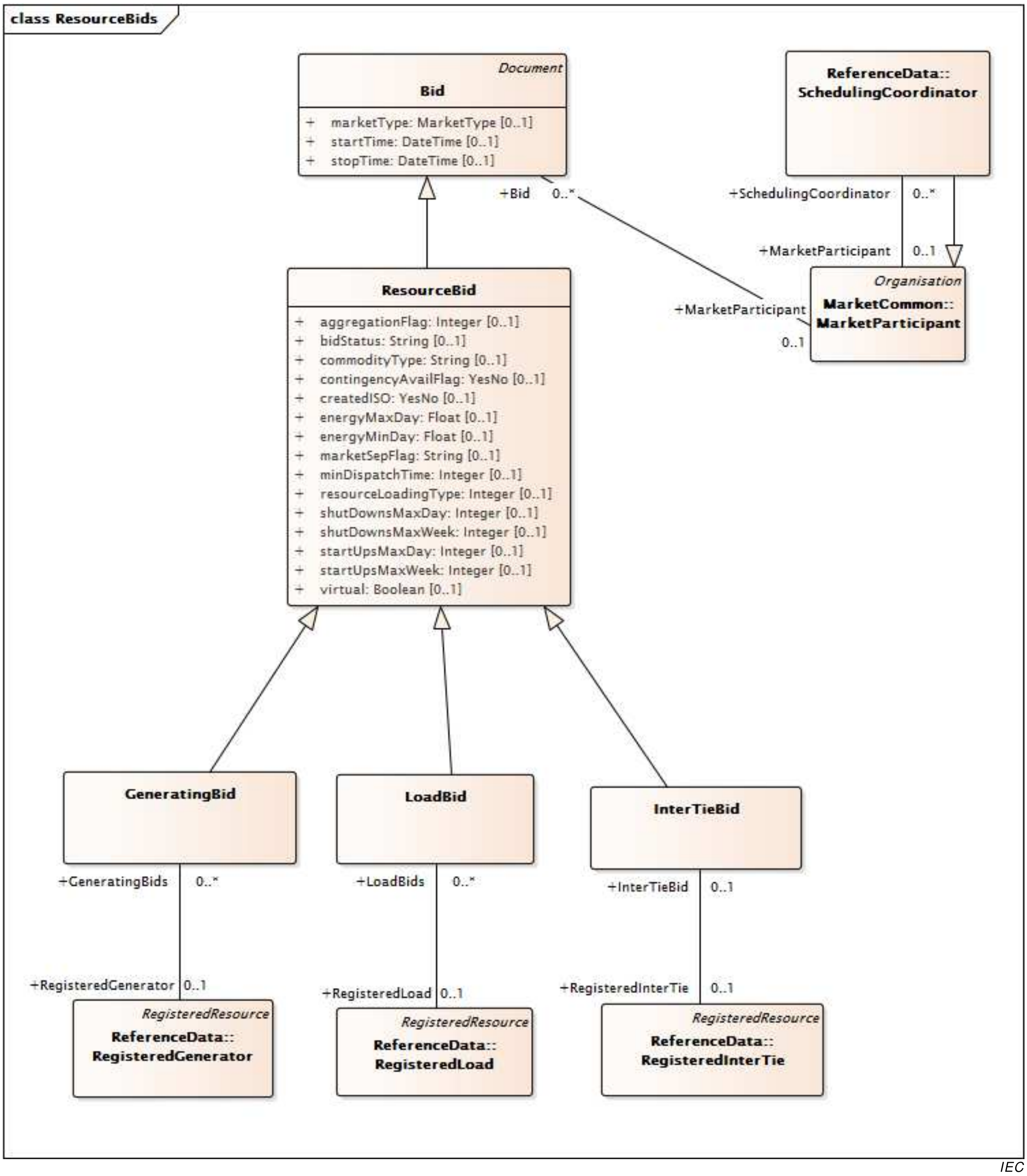



**Figure 57 – Class diagram ParticipantInterfaces::ResourceBids**

Figure 57: Shows the classes and associations used to model resource bids. The resource bid model includes the subclasses GeneratingBid, LoadBid, and IntertieBid to model the relationships to the ResourceBid and Bid classes.

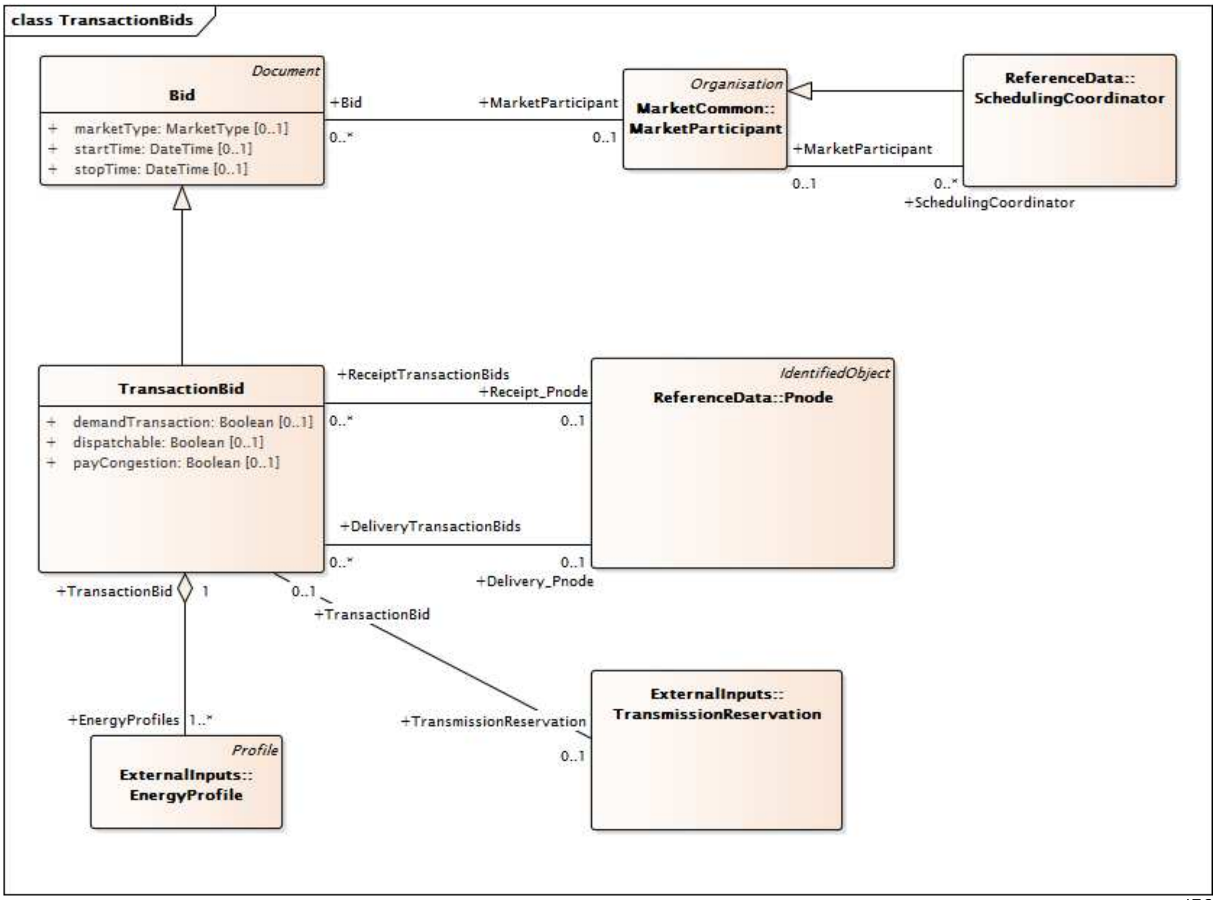



**Figure 58 – Class diagram ParticipantInterfaces::TransactionBids**

Figure 58: Shows the classes and associations used to model transaction bids. The TransactionBid model includes the subclasses GeneratingBid, LoadBid, and IntertieBid to model the relationships to the ResourceBid and Bid classes.

### 6.5.8.2 ActionRequest root class

Action request against an existing Trade.

Table 426 shows all attributes of ActionRequest.

**Table 426 – Attributes of ParticipantInterfaces::ActionRequest**

| name | mult | type | description |
|---|---|---|---|
| actionName | 0..1 | ActionType | Action name type for the action request. |

Table 427 shows all association ends of ActionRequest with other classes.

**Table 427 – Association ends of ParticipantInterfaces:: ActionRequest with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | Bid | 0..* | Bid | |
| 1..1 | Trade | 0..* | Trade | |

### 6.5.8.3 AreaLoadBid

AreaLoadBid is not submitted by a market participant into the Markets. Instead, it is simply an aggregation of all LoadBids contained wtihin a specific SubControlArea. This entity should inherit from Bid for representation of the timeframe (startTime, stopTime) and the market type.

Table 428 shows all attributes of AreaLoadBid.

**Table 428 – Attributes of ParticipantInterfaces::AreaLoadBid**

| name | mult | type | description |
|---|---|---|---|
| demandBidMW | 0..1 | Float | The Demand Bid Megawatt for the area case.<br>Attribute Usage: This is Scheduled demand MW in Day Ahead |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 429 shows all association ends of AreaLoadBid with other classes.

**Table 429 – Association ends of ParticipantInterfaces::AreaLoadBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | LoadBid | 0..* | LoadBid | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.4 AttributeProperty root class

Property for a particular attribute that contains name and value.

Table 430 shows all attributes of AttributeProperty.

**Table 430 – Attributes of ParticipantInterfaces::AttributeProperty**

| name | mult | type | description |
|---|---|---|---|
| sequence | 0..1 | String | |
| propertyName | 0..1 | String | |
| propertyValue | 0..1 | String | |

Table 431 shows all association ends of AttributeProperty with other classes.

**Table 431 – Association ends of ParticipantInterfaces::AttributeProperty with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 1..1 | MktUserAttribute | |

### 6.5.8.5 Bid

Represents both bids to purchase and offers to sell energy or ancillary services in an RTO-sponsored market.

Table 432 shows all attributes of Bid.

**Table 432 – Attributes of ParticipantInterfaces::Bid**

| name | mult | type | description |
|---|---|---|---|
| startTime | 0..1 | DateTime | Start time and date for which bid applies. |
| stopTime | 0..1 | DateTime | Stop time and date for which bid is applicable. |
| marketType | 0..1 | MarketType | The market type, DAM or RTM. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |

| name | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 433 shows all association ends of Bid with other classes.

**Table 433 – Association ends of ParticipantInterfaces::Bid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | |
| 0..* | ActionRequest | 1..1 | ActionRequest | |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | |
| 1..1 | ProductBids | 1..* | ProductBid | A bid comprises one or more product bids of market products |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.6 BidDistributionFactor root class

This class allows SC to input different time intervals for distribution factors.

Table 434 shows all attributes of BidDistributionFactor.

**Table 434 – Attributes of ParticipantInterfaces::BidDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| timeIntervalStart | 0..1 | DateTime | Start of the time interval in which bid is valid (yyyy-mm-dd hh24: mi: ss). |
| timeIntervalEnd | 0..1 | DateTime | End of the time interval n which bid is valid (yyyy-mm-dd hh24: mi: ss) |

Table 435 shows all association ends of BidDistributionFactor with other classes.

**Table 435 – Association ends of ParticipantInterfaces:: BidDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | PnodeDistributionFactor | 0..* | PnodeDistributionFactor | |
| 0..* | ProductBid | 1..1 | ProductBid | |

### 6.5.8.7 BidError

This class represent the error information for a bid that is detected during bid validation.

Table 436 shows all attributes of BidError.

**Table 436 – Attributes of ParticipantInterfaces::BidError**

| name | mult | type | description |
|---|---|---|---|
| errPriority | 0..1 | Integer | Priority number for the error message |
| errMessage | 0..1 | String | error message |
| ruleID | 0..1 | Integer | |
| startTime | 0..1 | DateTime | hour wihthin the bid for which the error applies |
| endTime | 0..1 | DateTime | hour wihthin the bid for which the error applies |
| logTimeStamp | 0..1 | DateTime | |
| componentType | 0..1 | String | |
| msgLevel | 0..1 | Integer | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 437 shows all association ends of BidError with other classes.

**Table 437 – Association ends of ParticipantInterfaces::BidError with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketProduct | 0..1 | MarketProduct | |
| 0..* | ResourceBid | 0..* | ResourceBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.8 BidHourlyProductSchedule

Containment for bid parameters that are dependent on a market product type.

Table 438 shows all attributes of BidHourlyProductSchedule.

**Table 438 – Attributes of ParticipantInterfaces::BidHourlyProductSchedule**

| name | mult | type | description |
|---|---|---|---|
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 439 shows all association ends of BidHourlyProductSchedule with other classes.

**Table 439 – Association ends of ParticipantInterfaces:: BidHourlyProductSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.9 BidHourlySchedule

Containment for bid hourly parameters that are not product dependent.

Table 440 shows all attributes of BidHourlySchedule.

**Table 440 – Attributes of ParticipantInterfaces::BidHourlySchedule**

| name | mult | type | description |
|---|---|---|---|
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 441 shows all association ends of BidHourlySchedule with other classes.

**Table 441 – Association ends of ParticipantInterfaces:: BidHourlySchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.10 BidPriceCurve

Relationship between unit operating price in $/hour (Y-axis) and unit output in MW (X-axis).

Table 442 shows all attributes of BidPriceCurve.

**Table 442 – Attributes of ParticipantInterfaces::BidPriceCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 443 shows all association ends of BidPriceCurve with other classes.

**Table 443 – Association ends of ParticipantInterfaces::BidPriceCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | BidSchedule | 0..* | BidPriceSchedule | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.11 BidPriceSchedule

Defines bid schedules to allow a product bid to use specified bid price curves for different time intervals.

Table 444 shows all attributes of BidPriceSchedule.

**Table 444 – Attributes of ParticipantInterfaces::BidPriceSchedule**

| name | mult | type | description |
|---|---|---|---|
| bidType | 0..1 | BidMitigationType | BID Type:<br>I – Initial Bid;<br>F – Final Bid |
| mitigationStatus | 0..1 | BidMitigationStatus | Mitigation Status:<br>'S' – Mitigated by SMPM because of "misconduct"<br>'L; – Mitigated by LMPM because of "misconduct"<br>'R' – Modified by LMPM because of RMR rules<br>'M' – Mitigated because of "misconduct" both by SMPM and LMPM<br>'B' – Mitigated because of "misconduct" both by SMPM and modified by LMLM because of RMR rules<br>'O' – original |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 445 shows all association ends of BidPriceSchedule with other classes.

**Table 445 – Association ends of ParticipantInterfaces:: BidPriceSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BidPriceCurve | 1..1 | BidPriceCurve | |
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.12 BidSelfSched

Defines self schedule values to be used for specified time intervals.

Table 446 shows all attributes of BidSelfSched.

**Table 446 – Attributes of ParticipantInterfaces::BidSelfSched**

| name | mult | type | description |
|---|---|---|---|
| balancingFlag | 0..1 | YesNo | This is a Y/N flag for a self-schedule of a resource per market per date and hour, using a specific TR ID. It indicates whether a self-schedule using a TR is balanced with another self-schedule using the same TR ID. |
| bidType | 0..1 | BidTypeRMR | bidType has two types as the required output of requirements and qualified pre-dispatch. |
| priorityFlag | 0..1 | YesNo | This is a Y/N flag for a self-schedule of a resource per market per date and hour, using a specific TR ID. It indicates whether a self-schedule using a TR has scheduling priority in DAM/RTM. |
| pumpSelfSchedMw | 0..1 | Float | Contains the PriceTaker, ExistingTransmissionContract, TransmissionOwnershipRights pumping self schedule quantity. If this value is not null, then the unit is in pumping mode. |
| referenceType | 0..1 | SelfSchedReferenceType | Indication of which type of self schedule is being referenced. |
| selfSchedMw | 0..1 | Float | Self scheduled value |
| selfSchedSptResource | 0..1 | String | Price Taker Export Self Sched Support Resource |
| selfSchedType | 0..1 | MarketProductSelfSchedType | This attribute is used to specify if a bid includes a self sched bid. If so what self sched type is it. The possible values are shown as follow but not limited to:<br>'ETC' – Existing transmission contract<br>'TOR' – Transmission ownership right<br>'RMR' – Reliability must run<br>'RGMR' – Regulatory must run<br>"RMT" – Relaiability must take<br>"PT" – Price taker<br>"LPT" – Low price taker<br>"SP" – Self provision<br>"RA" – Resource adequacy<br>This attribute is originally defined in the BidSelfSched class |
| updateType | 0..1 | MQSCHGType | |
| wheelingTransactionReference | 0..1 | String | A unique identifier of a wheeling transaction. A wheeling transaction is a balanced Energy exchange among Supply and Demand Resources. |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 447 shows all association ends of BidSelfSched with other classes.

**Table 447 – Association ends of ParticipantInterfaces::BidSelfSched with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.8.13 BidSet

As set of mutually exclusive bids for which a maximum of one may be scheduled.

Of these generating bids, only one generating bid can be scheduled at a time.

Table 448 shows all attributes of BidSet.

**Table 448 – Attributes of ParticipantInterfaces::BidSet**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 449 shows all association ends of BidSet with other classes.

**Table 449 – Association ends of ParticipantInterfaces::BidSet with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 1..* | GeneratingBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.14 ChargeComponent

A Charge Component is a list of configurable charge quality items to feed into settlement calculation and/or bill determinants.

Table 450 shows all attributes of ChargeComponent.

**Table 450 – Attributes of ParticipantInterfaces::ChargeComponent**

| name | mult | type | description |
|---|---|---|---|
| deleteStatus | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| message | 0..1 | String | |
| type | 0..1 | String | |
| sum | 0..1 | String | |
| roundOff | 0..1 | String | |
| equation | 0..1 | String | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 451 shows all association ends of ChargeComponent with other classes.

**Table 451 – Association ends of ParticipantInterfaces:: ChargeComponent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BillDeterminants | 0..* | BillDeterminant | A BillDeterminant can have 0-n ChargeComponent and a ChargeComponent can associate to 0-n BillDeterminant. |
| 0..* | ChargeTypes | 0..* | ChargeType | A ChargeType can have 0-n ChargeComponent and a ChargeComponent can associate to 0-n ChargeType |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.15 ChargeGroup

Charge Group is the grouping of Charge Types for settlement invoicing purpose. Examples such as Ancillary Services, Interests, etc.

Table 452 shows all attributes of ChargeGroup.

**Table 452 – Attributes of ParticipantInterfaces::ChargeGroup**

| name | mult | type | description |
|---|---|---|---|
| marketCode | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 453 shows all association ends of ChargeGroup with other classes.

**Table 453 – Association ends of ParticipantInterfaces::ChargeGroup with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | ChargeGroupParent | 0..1 | ChargeGroup | A ChargeGroup instance can have relationships with other ChargeGroup instances. |
| 0..* | ChargeType | 0..* | ChargeType | A ChargeGroup can have 0-n ChargeType. A ChargeType can associate to 0-n ChargeGroup. |
| 0..1 | ChargeGroupChild | 0..* | ChargeGroup | A ChargeGroup instance can have relationships with other ChargeGroup instances. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.16 ChargeType

Charge Type is the basic level configuration for settlement to process specific charges for invoicing purpose. Examples such as: Day Ahead Spinning Reserve Default Invoice Interest Charge, etc.

Table 454 shows all attributes of ChargeType.

**Table 454 – Attributes of ParticipantInterfaces::ChargeType**

| name | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| factor | 0..1 | String | |
| chargeOrder | 0..1 | String | |
| frequencyType | 0..1 | String | |
| chargeVersion | 0..1 | String | |
| totalInterval | 0..1 | String | |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 455 shows all association ends of ChargeType with other classes.

**Table 455 – Association ends of ParticipantInterfaces::ChargeType with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | ChargeComponents | 0..* | ChargeComponent | A ChargeType can have 0-n ChargeComponent and a ChargeComponent can associate to 0-n ChargeType |
| 0..* | ChargeGroup | 0..* | ChargeGroup | A ChargeGroup can have 0-n ChargeType. A ChargeType can associate to 0-n ChargeGroup. |
| 0..* | MajorChargeGroup | 0..* | MajorChargeGroup | A MajorChargeGroup can have 0-n ChargeType. A ChargeType can associate to 0-n MajorChargeGroup. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.8.17 DispatchInstReply

Response from registered resource acknowledging receipt of dispatch instructions.

Table 456 shows all attributes of DispatchInstReply.

**Table 456 – Attributes of ParticipantInterfaces::DispatchInstReply**

| name | mult | type | description |
|---|---|---|---|
| acceptMW | 0..1 | ActivePower | The accepted mw amount by the responder. aka response mw. |
| acceptStatus | 0..1 | DispatchResponseType | The accept status submitted by the responder. enumeration type needs to be defined |
| certificationName | 0..1 | String | The Subject DN is the X509 Certificate Subject DN. This is the essentially the certificate name presented by the client. In the case of ADS Certificates, this will be the user name. It may be from an API Client or the MP Client (GUI).<br>The Subject ID normally includes more than just the user name (Common Name), it can also contain information such as City, Company ID, etc. |
| clearedMW | 0..1 | ActivePower | MW amount associated with instruction. For 5 minute binding dispatches, this is the Goto MW or DOT |
| instructionTime | 0..1 | DateTime | The target date/time for the received instruction. |
| instructionType | 0..1 | String | instruction type:<br>commitment<br>out of sequence<br>dispatch |
| passIndicator | 0..1 | PassIndicatorType | The type of run for the market clearing. |
| receivedTime | 0..1 | DateTime | Timestamp indicating the time at which the instruction was received. |
| startTime | 0..1 | DateTime | start time |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 457 shows all association ends of DispatchInstReply with other classes.

**Table 457 – Association ends of ParticipantInterfaces:: DispatchInstReply with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.18 EnergyPriceCurve root class

Relationship between a price in $(or other monetary unit) /hour (Y-axis) and a MW value (X-axis).

Table 458 shows all association ends of EnergyPriceCurve with other classes.

**Table 458 – Association ends of ParticipantInterfaces:: EnergyPriceCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnergyTransactions | 0..* | EnergyTransaction | |

### 6.5.8.19 GeneratingBid

Offer to supply energy/ancillary services from a generating unit or resource.

Table 459 shows all attributes of GeneratingBid.

**Table 459 – Attributes of ParticipantInterfaces::GeneratingBid**

| name | mult | type | description |
|---|---|---|---|
| combinedCycleUnitOffer | 0..1 | String | Will indicate if the unit is part of a CC offer or not |
| downTimeMax | 0..1 | Float | Maximum down time. |
| installedCapacity | 0..1 | Float | Installed Capacity value |
| lowerRampRate | 0..1 | ActivePowerChangeRate | Maximum Dn ramp rate in MW/min |
| maxEmergencyMW | 0..1 | ActivePower | Power rating available for unit under emergency conditions greater than or equal to maximum economic limit. |
| maximumEconomicMW | 0..1 | Float | Maximum high economic MW limit, that should not exceed the maximum operating MW limit |
| minEmergencyMW | 0..1 | ActivePower | Minimum power rating for unit under emergency conditions, which is less than or equal to the economic minimum. |
| minimumEconomicMW | 0..1 | Float | Low economic MW limit that shall be greater than or equal to the minimum operating MW limit |
| noLoadCost | 0..1 | Float | Resource fixed no load cost. |
| notificationTime | 0..1 | Float | Time required for crew notification prior to start up of the unit. |
| operatingMode | 0..1 | String | Bid operating mode ('C' – cycling, 'F' – fixed, 'M' – must run, 'U' – unavailable) |

| name | mult | type | description |
|---|---|---|---|
| raiseRampRate | 0..1 | ActivePowerChangeRate | Maximum Up ramp rate in MW/min |
| rampCurveType | 0..1 | Integer | Ramp curve type:<br>0 – Fixed ramp rate independent of rate function unit MW output<br>1 – Static ramp rates as a function of unit MW output only<br>2 – Dynamic ramp rates as a function of unit MW output and ramping time |
| startupCost | 0..1 | Float | Startup cost/price |
| startUpRampRate | 0..1 | ActivePowerChangeRate | Resource startup ramp rate (MW/minute) |
| startUpType | 0..1 | Integer | Resource startup type:<br>1 – Fixed startup time and fixed startup cost<br>2 – Startup time as a function of down time and fixed startup cost<br>3 – Startup cost as a function of down time |
| upTimeMax | 0..1 | Float | Maximum up time. |
| aggregationFlag | 0..1 | Integer | inherited from: ResourceBid |
| bidStatus | 0..1 | String | inherited from: ResourceBid |
| commodityType | 0..1 | String | inherited from: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: ResourceBid |
| createdISO | 0..1 | YesNo | inherited from: ResourceBid |
| energyMaxDay | 0..1 | Float | inherited from: ResourceBid |
| energyMinDay | 0..1 | Float | inherited from: ResourceBid |
| marketSepFlag | 0..1 | String | inherited from: ResourceBid |
| minDispatchTime | 0..1 | Integer | inherited from: ResourceBid |
| resourceLoadingType | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |
| virtual | 0..1 | Boolean | inherited from: ResourceBid |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 460 shows all association ends of GeneratingBid with other classes.

**Table 460 – Association ends of ParticipantInterfaces::GeneratingBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraints | 0..* | SecurityConstraints | |
| 1..* | BidSet | 0..1 | BidSet | |
| 0..* | NotificationTimeCurve | 0..1 | NotificationTimeCurve | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..* | StartUpCostCurve | 0..1 | StartUpCostCurve | |
| 0..* | StartUpTimeCurve | 0..1 | StartUpTimeCurve | |
| 0..* | BidError | 0..* | BidError | inherited from: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.20 HourlyPreDispatchSchedule

An indicator specifying that a resource shall have an Hourly Pre-Dispatch. The resource could be a RegisteredGenerator or a RegisteredInterTie.

This schedule is assocated with the hourly parameters in a resource bid.

Table 461 shows all attributes of HourlyPreDispatchSchedule.

**Table 461 – Attributes of ParticipantInterfaces::HourlyPreDispatchSchedule**

| name | mult | type | description |
|---|---|---|---|
| value | 0..1 | Boolean | Flag defining that for this hour in the resource bid the resource shall have an hourly pre-dispatch. |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 462 shows all association ends of HourlyPreDispatchSchedule with other classes.

**Table 462 – Association ends of ParticipantInterfaces:: HourlyPreDispatchSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | inherited from: BidHourlySchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.21 InterTieBid

This class represents the inter tie bid.

Table 463 shows all attributes of InterTieBid.

**Table 463 – Attributes of ParticipantInterfaces::InterTieBid**

| name | mult | type | description |
|---|---|---|---|
| minHourlyBlock | 0..1 | Integer | The minimum hourly block for an Inter-Tie Resource supplied within the bid. |
| aggregationFlag | 0..1 | Integer | inherited from: ResourceBid |
| bidStatus | 0..1 | String | inherited from: ResourceBid |
| commodityType | 0..1 | String | inherited from: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: ResourceBid |
| createdISO | 0..1 | YesNo | inherited from: ResourceBid |
| energyMaxDay | 0..1 | Float | inherited from: ResourceBid |

| name | mult | type | description |
|---|---|---|---|
| energyMinDay | 0..1 | Float | inherited from: ResourceBid |
| marketSepFlag | 0..1 | String | inherited from: ResourceBid |
| minDispatchTime | 0..1 | Integer | inherited from: ResourceBid |
| resourceLoadingType | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |
| virtual | 0..1 | Boolean | inherited from: ResourceBid |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 464 shows all association ends of InterTieBid with other classes.

**Table 464 – Association ends of ParticipantInterfaces::InterTieBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredInterTie | 0..1 | RegisteredInterTie | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | BidError | 0..* | BidError | inherited from: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.22 InterTieDispatchResponse root class

Response from an intertie resource acknowledging receipt of dispatch instructions.

Table 465 shows all attributes of InterTieDispatchResponse.

**Table 465 – Attributes of ParticipantInterfaces::InterTieDispatchResponse**

| name | mult | type | description |
|---|---|---|---|
| acceptStatus | 0..1 | DispatchResponseType | The accept status submitted by the responder. Valid values are NON-RESPONSE, ACCEPT, DECLINE, PARTIAL. |
| acceptMW | 0..1 | Float | The accepted mw amount by the responder. aka response mw. |
| clearedMW | 0..1 | Float | MW amount associated with instruction. For 5 minute binding dispatches, this is the Goto MW or DOT |
| startTime | 0..1 | DateTime | Part of the Composite key that downstream app uses to match the instruction |
| passIndicator | 0..1 | PassIndicatorType | Part of the Composite key that downstream app uses to match the instruction |

Table 466 shows all association ends of InterTieDispatchResponse with other classes.

**Table 466 – Association ends of ParticipantInterfaces:: InterTieDispatchResponse with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredInterTie | 1..1 | RegisteredInterTie | |

### 6.5.8.23 LoadBid

Offer to supply energy/ancillary services from a load resource (participating load reduces consumption).

Table 467 shows all attributes of LoadBid.

**Table 467 – Attributes of ParticipantInterfaces::LoadBid**

| name | mult | type | description |
|---|---|---|---|
| dropRampRate | 0..1 | ActivePowerChangeRate | Maximum rate that load can be reduced (MW/minute) |
| loadRedInitiationCost | 0..1 | Money | load reduction initiation cost |
| loadRedInitiationTime | 0..1 | Float | load reduction initiation time |
| marketDate | 0..1 | DateTime | The date represents the NextMarketDate for which the load response bids apply to. |
| meteredValue | 0..1 | Boolean | Flag indicated that the load reduction is metered. (See above)<br>If priceSetting and meteredValue both equal 1, then the facility is eligible to set LMP in the real time market. |
| minLoad | 0..1 | ActivePower | Minimum MW load below which it may not be reduced. |
| minLoadReduction | 0..1 | ActivePower | Minimum MW for a load reduction (e.g. MW rating of a discrete pump. |
| minLoadReductionCost | 0..1 | Money | Cost in $ at the minimum reduced load |
| minLoadReductionInterval | 0..1 | Float | Shortest period load reduction shall be maintained before load can be restored to normal levels. |
| minTimeBetLoadRed | 0..1 | Float | Shortest time that load shall be left at normal levels before a new load reduction. |
| pickUpRampRate | 0..1 | ActivePowerChangeRate | Maximum rate load may be restored (MW/minute) |
| priceSetting | 0..1 | Boolean | Flag to indicate that the facility can set LMP Works in tandem with Metered Value. Greater chance of this being dynamic than the Metered Value, however, it is requested that Price Setting and Metered Value stay at the same source. Currently no customers have implemented the metering capability, but if this option is implemented, then Price Setting could become dynamic. However, Metered Value will remain static. |
| reqNoticeTime | 0..1 | Float | Time period that is required from an order to reduce a load to the time that it takes to get to the minimum load reduction. |
| shutdownCost | 0..1 | Money | The fixed cost associated with committing a load reduction. |
| aggregationFlag | 0..1 | Integer | inherited from: ResourceBid |
| bidStatus | 0..1 | String | inherited from: ResourceBid |
| commodityType | 0..1 | String | inherited from: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: ResourceBid |
| createdISO | 0..1 | YesNo | inherited from: ResourceBid |
| energyMaxDay | 0..1 | Float | inherited from: ResourceBid |
| energyMinDay | 0..1 | Float | inherited from: ResourceBid |
| marketSepFlag | 0..1 | String | inherited from: ResourceBid |
| minDispatchTime | 0..1 | Integer | inherited from: ResourceBid |
| resourceLoadingType | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | inherited from: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | inherited from: ResourceBid |

| name | mult | type | description |
|---|---|---|---|
| virtual | 0..1 | Boolean | inherited from: ResourceBid |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 468 shows all association ends of LoadBid with other classes.

**Table 468 – Association ends of ParticipantInterfaces::LoadBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | LoadReductionPriceCurve | 0..* | LoadReductionPriceCurve | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | AreaLoadBid | 0..1 | AreaLoadBid | |
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | |
| 0..* | BidError | 0..* | BidError | inherited from: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.8.24 LoadFollowingInst root class

Metered SubSystem Load Following Instruction.

Table 469 shows all attributes of LoadFollowingInst.

**Table 469 – Attributes of ParticipantInterfaces::LoadFollowingInst**

| name | mult | type | description |
|---|---|---|---|
| mssInstructionID | 0..1 | String | Unique instruction id per instruction, assigned by the SC and provided to ADS. ADS passes through. |
| startTime | 0..1 | DateTime | Instruction Start Time |
| endTime | 0..1 | DateTime | Instruction End Time |
| loadFollowingMW | 0..1 | Float | Load Following MW Positive for follow-up and negative for follow-down |

Table 470 shows all association ends of LoadFollowingInst with other classes.

**Table 470 – Association ends of ParticipantInterfaces:: LoadFollowingInst with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |

#### 6.5.8.25 LoadReductionPriceCurve

This is the price sensitivity that bidder expresses for allowing market load interruption. Relationship between price (Y1-axis) vs. MW (X-axis).

Table 471 shows all attributes of LoadReductionPriceCurve.

**Table 471 – Attributes of ParticipantInterfaces::LoadReductionPriceCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |

| name | mult | type | description |
|---|---|---|---|
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 472 shows all association ends of LoadReductionPriceCurve with other classes.

**Table 472 – Association ends of ParticipantInterfaces:: LoadReductionPriceCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | LoadBid | 1..1 | LoadBid | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.26 MajorChargeGroup

A Major Charge Group is the same as Invoice Type which provides the highest level of grouping for charge types configuration. Examples: Market, FERC, RMR.

Table 473 shows all attributes of MajorChargeGroup.

**Table 473 – Attributes of ParticipantInterfaces::MajorChargeGroup**

| name | mult | type | description |
|---|---|---|---|
| runType | 0..1 | String | |
| runVersion | 0..1 | String | |
| frequencyType | 0..1 | String | |
| invoiceType | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| requireAutorun | 0..1 | String | |
| revisionNumber | 0..1 | String | Revision number for the major charge group |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 474 shows all association ends of MajorChargeGroup with other classes.

**Table 474 – Association ends of ParticipantInterfaces:: MajorChargeGroup with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | Settlement | 0..* | Settlement | |
| 0..* | ChargeType | 0..* | ChargeType | A MajorChargeGroup can have 0-n ChargeType. A ChargeType can associate to 0-n MajorChargeGroup. |
| 0..1 | MktScheduledEvent | 0..* | MarketScheduledEvent | |
| 1..* | MarketInvoice | 0..* | MarketInvoice | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.27 MarketScheduledEvent

Signifies an event to trigger one or more activities, such as reading a meter, recalculating a bill, requesting work, when generating units shall be scheduled for maintenance, when a transformer is scheduled to be refurbished, etc.

Table 475 shows all attributes of MarketScheduledEvent.

**Table 475 – Attributes of ParticipantInterfaces::MarketScheduledEvent**

| name | mult | type | description |
|---|---|---|---|
| category | 0..1 | String | Category of scheduled event. |
| duration | 0..1 | Seconds | Duration of the scheduled event, for example, the time to ramp between values. |
| status | 0..1 | Status | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 476 shows all association ends of MarketScheduledEvent with other classes.

**Table 476 – Association ends of ParticipantInterfaces:: MarketScheduledEvent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MajorChargeGroup | 0..1 | MajorChargeGroup | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.8.28 NotificationTimeCurve

Notification time curve as a function of down time. Relationship between crew notification time (Y1-axis) and unit startup time (Y2-axis) vs. unit elapsed down time (X-axis).

Table 477 shows all attributes of NotificationTimeCurve.

**Table 477 – Attributes of ParticipantInterfaces::NotificationTimeCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 478 shows all association ends of NotificationTimeCurve with other classes.

**Table 478 – Association ends of ParticipantInterfaces:: NotificationTimeCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 0..* | GeneratingBid | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.8.29 OpenTieSchedule

Result of bid validation against conditions that may exist on an interchange that becomes disconnected or is heavily discounted with respect the MW flow.

This schedule is assocated with the hourly parameters in a resource bid.

Table 479 shows all attributes of OpenTieSchedule.

**Table 479 – Attributes of ParticipantInterfaces::OpenTieSchedule**

| name | mult | type | description |
|---|---|---|---|
| value | 0..1 | Boolean | |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 480 shows all association ends of OpenTieSchedule with other classes.

**Table 480 – Association ends of ParticipantInterfaces:: OpenTieSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | inherited from: BidHourlySchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.30 ProductBid

Component of a bid that pertains to one market product.

Table 481 shows all attributes of ProductBid.

**Table 481 – Attributes of ParticipantInterfaces::ProductBid**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 482 shows all association ends of ProductBid with other classes.

**Table 482 – Association ends of ParticipantInterfaces::ProductBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..* | Bid | 1..1 | Bid | A bid comprises one or more product bids of market products |
| 1..1 | BidDistributionFactor | 0..* | BidDistributionFactor | |
| 1..1 | BidSchedule | 0..* | BidPriceSchedule | |
| 1..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..1 | BidHourlyProductSchedule | 0..* | BidHourlyProductSchedule | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.31 PumpingCostSchedule

The operating cost of a Pump Storage Hydro Unit operating as a hydro pump.

This schedule is assocated with the hourly parameters in a resource bid associated with a specific product within the bid.

Table 483 shows all attributes of PumpingCostSchedule.

**Table 483 – Attributes of ParticipantInterfaces::PumpingCostSchedule**

| name | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 484 shows all association ends of PumpingCostSchedule with other classes.

**Table 484 – Association ends of ParticipantInterfaces:: PumpingCostSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | inherited from: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.32 PumpingLevelSchedule

The fixed operating level of a Pump Storage Hydro Unit operating as a hydro pump. Associated with the energy market product type.

This schedule is assocated with the hourly parameters in a resource bid associated with a specific product within the bid.

Table 485 shows all attributes of PumpingLevelSchedule.

**Table 485 – Attributes of ParticipantInterfaces::PumpingLevelSchedule**

| name | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 486 shows all association ends of PumpingLevelSchedule with other classes.

**Table 486 – Association ends of ParticipantInterfaces:: PumpingLevelSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | inherited from: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.33 PumpingShutDownCostSchedule

The cost to shutdown a Pump Storage Hydro Unit (in pump mode) or a pump.

This schedule is assocated with the hourly parameters in a resource bid associated with a specific product within the bid.

Table 487 shows all attributes of PumpingShutDownCostSchedule.

**Table 487 – Attributes of ParticipantInterfaces::PumpingShutDownCostSchedule**

| name | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | inherited from: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | inherited from: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | inherited from: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value1Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | inherited from: BasicIntervalSchedule |
| value2Unit | 0..1 | UnitSymbol | inherited from: BasicIntervalSchedule |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 488 shows all association ends of PumpingShutDownCostSchedule with other classes.

**Table 488 – Association ends of ParticipantInterfaces:: PumpingShutDownCostSchedule with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | inherited from: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | inherited from: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.34 RampRateCurve

Ramp rate as a function of resource MW output.

Table 489 shows all attributes of RampRateCurve.

**Table 489 – Attributes of ParticipantInterfaces::RampRateCurve**

| name | mult | type | description |
|---|---|---|---|
| condition | 0..1 | RampRateCondition | condition for the ramp rate |
| constraintRampType | 0..1 | ConstraintRampType | The condition that identifies whether a Generating Resource should be constrained from Ancillary Service provision if its Schedule or Dispatch change across Trading Hours or Trading Intervals requires more than a specified fraction of the duration of the Trading Hour or Trading Interval.<br>Valid values are Fast/Slow |
| rampRateType | 0..1 | RampRateType | How ramp rate is applied (e.g. raise or lower, as when applied to a generation resource) |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 490 shows all association ends of RampRateCurve with other classes.

**Table 490 – Association ends of ParticipantInterfaces:: RampRateCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | GeneratingBid | 0..1 | GeneratingBid | |
| 0..* | InterTieBid | 0..1 | InterTieBid | |
| 0..* | LoadBid | 0..1 | LoadBid | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.35 ResourceBid

Energy bid for generation, load, or virtual type for the whole of the market-trading period (i.e., one day in day ahead market or one hour in the real time market).

Table 491 shows all attributes of ResourceBid.

**Table 491 – Attributes of ParticipantInterfaces::ResourceBid**

| name | mult | type | description |
|---|---|---|---|
| aggregationFlag | 0..1 | Integer | Aggregation flag<br>0: individual resource level<br>1: Aggregated node location<br>2: Aggregated price location) |
| bidStatus | 0..1 | String | |
| commodityType | 0..1 | String | Energy product (commodity) type:<br>'En' – Energy<br>'Ru' – Regulation Up<br>'Rd' – Regulation Dn<br>'Sr' – Spinning Reserve<br>'Nr' – Non-Spinning Reserve<br>'Or' – Operating Reserve |
| contingencyAvailFlag | 0..1 | YesNo | contingent operating reserve availiability (Yes/No). Resource is availiable to participate with capacity only in contingency dispatch. |
| createdISO | 0..1 | YesNo | A Yes indicates that this bid was created by the ISO. |
| energyMaxDay | 0..1 | Float | Maximum amount of energy per day which can be produced during the trading period in MWh |
| energyMinDay | 0..1 | Float | Minimum amount of energy per day which has to be produced during the trading period in MWh |
| marketSepFlag | 0..1 | String | Market Separation Flag<br>'Y' – Enforce market separation constraints for this bid<br>'N' – Do not enforce market separation constraints for this bid. |

| name | mult | type | description |
|---|---|---|---|
| minDispatchTime | 0..1 | Integer | minimum number of consecutive hours a resource shall be dispatched if bid is accepted |
| resourceLoadingType | 0..1 | Integer | Resource loading curve type<br>1 – step-wise continuous loading<br>2 – piece-wise linear continuous loading<br>3 – block loading |
| shutDownsMaxDay | 0..1 | Integer | Maximum number of shutdowns per day. |
| shutDownsMaxWeek | 0..1 | Integer | Maximum number of shutdowns per week. |
| startUpsMaxDay | 0..1 | Integer | Maximum number of startups per day. |
| startUpsMaxWeek | 0..1 | Integer | Maximum number of startups per week. |
| virtual | 0..1 | Boolean | True if bid is virtual. Bid is assumed to be non-virtual if attribute is absent |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 492 shows all association ends of ResourceBid with other classes.

**Table 492 – Association ends of ParticipantInterfaces::ResourceBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BidError | 0..* | BidError | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.36 StartUpCostCurve

Startup costs and time as a function of down time. Relationship between unit startup cost (Y1-axis) vs. unit elapsed down time (X-axis).

Table 493 shows all attributes of StartUpCostCurve.

**Table 493 – Attributes of ParticipantInterfaces::StartUpCostCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 494 shows all association ends of StartUpCostCurve with other classes.

**Table 494 – Association ends of ParticipantInterfaces:: StartUpCostCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBid | 0..* | GeneratingBid | |
| 0..* | RegisteredGenerators | 0..* | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.37 StartUpTimeCurve

Startup time curve as a function of down time, where time is specified in minutes. Relationship between unit startup time (Y1-axis) vs. unit elapsed down time (X-axis).

Table 495 shows all attributes of StartUpTimeCurve.

**Table 495 – Attributes of ParticipantInterfaces::StartUpTimeCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 496 shows all association ends of StartUpTimeCurve with other classes.

**Table 496 – Association ends of ParticipantInterfaces:: StartUpTimeCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBid | 0..* | GeneratingBid | |
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.38 Trade

Inter Scheduling Coordinator Trades to model financial trades which may impact settlement.

Table 497 shows all attributes of Trade.

**Table 497 – Attributes of ParticipantInterfaces::Trade**

| name | mult | type | description |
|---|---|---|---|
| adjustedTradeQuantity | 0..1 | Float | The validated and current market accepted trade amount of a physical energy trade. |
| counterTradeQuantity | 0..1 | Float | MW quantity submitted by counter SC for the same trade |
| dependOnTradeName | 0..1 | String | The Depend On IST Name points to the unique IST Name in the chain of physical energy trades. |
| lastModified | 0..1 | DateTime | Time and date the trade was last modified. |
| marketType | 0..1 | MarketType | |
| startTime | 0..1 | DateTime | Start time and date for which trade applies. |
| stopTime | 0..1 | DateTime | Stop time and date for which trade is applicable. |
| submitFromTimeStamp | 0..1 | DateTime | Timestamp of submittal of submit From Scheduling Coordinator Trade to Market Participant Bid Submittal |
| submitFromUser | 0..1 | String | Userid of the submit From Scheduling Coordinator trade |
| submitToTimeStamp | 0..1 | DateTime | Timestamp of submittal of submit To Scheduling Coordinator Trade to Market Participant Bid Submittal |
| submitToUser | 0..1 | String | Userid of the submit To Scheduling Coordinator trade |
| tradeQuantity | 0..1 | Float | tradeQuantity:<br>If tradeType = IST, The amount of an Energy Trade.<br>If tradeType = AST, The amount of an Ancillary Service Obligation Trade.<br>If tradeType = UCT, The amount of a Unit Commitment Obligation Trade. |
| tradeStatus | 0..1 | String | Resulting status of the trade following the rule engine processing. |
| updateTimeStamp | 0..1 | DateTime | |

| name | mult | type | description |
|---|---|---|---|
| updateUser | 0..1 | String | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 498 shows all association ends of Trade with other classes.

**Table 498 – Association ends of ParticipantInterfaces::Trade with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | TradeError | 0..* | TradeError | |
| 0..* | ActionRequest | 1..1 | ActionRequest | |
| 0..* | Pnode | 0..1 | Pnode | |
| 0..* | submitToSchedulingCoordinator | 0..1 | SchedulingCoordinator | |
| 0..* | To_SC | 1..1 | SchedulingCoordinator | |
| 0..* | From_SC | 1..1 | SchedulingCoordinator | |
| 0..* | submitFromSchedulingCoordinator | 0..1 | SchedulingCoordinator | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..* | TradeProduct | 1..1 | TradeProduct | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.39 TradeError

Trade error and warning messages associated with the rule engine processing of the submitted trade.

Table 499 shows all attributes of TradeError.

**Table 499 – Attributes of ParticipantInterfaces::TradeError**

| name | mult | type | description |
|---|---|---|---|
| errPriority | 0..1 | Integer | Priority number for the error message |
| errMessage | 0..1 | String | error message |
| ruleID | 0..1 | Integer | Rule identifier which triggered the error/warning message |
| startTime | 0..1 | DateTime | hour wihthin the trade for which the error applies |
| endTime | 0..1 | DateTime | hour wihthin the trade for which the error applies |
| logTimeStamp | 0..1 | DateTime | Timestamp of logged error/warning message |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 500 shows all association ends of TradeError with other classes.

**Table 500 – Association ends of ParticipantInterfaces::TradeError with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Trade | 0..1 | Trade | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.8.40 TradeProduct root class

| | |
|---|---|
| TradeType | TradeProduct |
| IST (InterSC Trade) | PHY (Physical Energy Trade) |
| IST | APN (Energy Trades at Aggregated Pricing Nodes) |
| IST | CPT (Converted Physical Energy Trade) |
| AST (Ancilliary Services Trade) | RUT (Regulation Up Trade) |
| AST | RDT (Regulation Down Trade) |
| AST | SRT (Spinning Reserve Trade) |
| AST | NRT (Non-Spinning Reserve Trade) |
| UCT (Unit Commitment Trade) | null |

Table 501 shows all attributes of TradeProduct.

**Table 501 – Attributes of ParticipantInterfaces::TradeProduct**

| name | mult | type | description |
|---|---|---|---|
| tradeType | 0..1 | TradeType | IST – InterSC Trade;<br>AST – Ancilliary Services Trade;<br>UCT – Unit Commitment Trade |
| tradeProductType | 0..1 | String | PHY (Physical Energy Trade);<br>APN (Energy Trades at Aggregated Pricing Nodes);<br>CPT (Converted Physical Energy Trade);<br>RUT (Regulation Up Trade);<br>RDT (Regulation Down Trade);<br>SRT (Spinning Reserve Trade);<br>NRT (Non-Spinning Reserve Trade) |

Table 502 shows all association ends of TradeProduct with other classes.

**Table 502 – Association ends of ParticipantInterfaces::TradeProduct with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | Trade | 0..* | Trade | |

#### 6.5.8.41 TransactionBid

Bilateral or scheduled transactions for energy and ancillary services considered by market clearing process.

Table 503 shows all attributes of TransactionBid.

**Table 503 – Attributes of ParticipantInterfaces::TransactionBid**

| name | mult | type | description |
|---|---|---|---|
| demandTransaction | 0..1 | Boolean | Set true if this is a demand transaction. |
| dispatchable | 0..1 | Boolean | Set true if this is a dispatchable transaction. |
| payCongestion | 0..1 | Boolean | Set true if this is a willing to pay transaction. This flag is used to determine whether a schedule is willing-to-pay-congestion or not. |
| startTime | 0..1 | DateTime | inherited from: Bid |
| stopTime | 0..1 | DateTime | inherited from: Bid |
| marketType | 0..1 | MarketType | inherited from: Bid |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 504 shows all association ends of TransactionBid with other classes.

**Table 504 – Association ends of ParticipantInterfaces:: TransactionBid with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | EnergyProfiles | 1..* | EnergyProfile | |
| 0..1 | TransmissionReservation | 0..1 | TransmissionReservation | |
| 0..1 | TransactionBidResults | 0..* | TransactionBidResults | |
| 0..* | Receipt_Pnode | 0..1 | Pnode | |
| 0..* | Delivery_Pnode | 0..1 | Pnode | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | inherited from: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | inherited from: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | inherited from: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | inherited from: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | inherited from: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | inherited from: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | inherited from: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | inherited from: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9 Package ReferenceData

#### 6.5.9.1 General

Market static reference data.

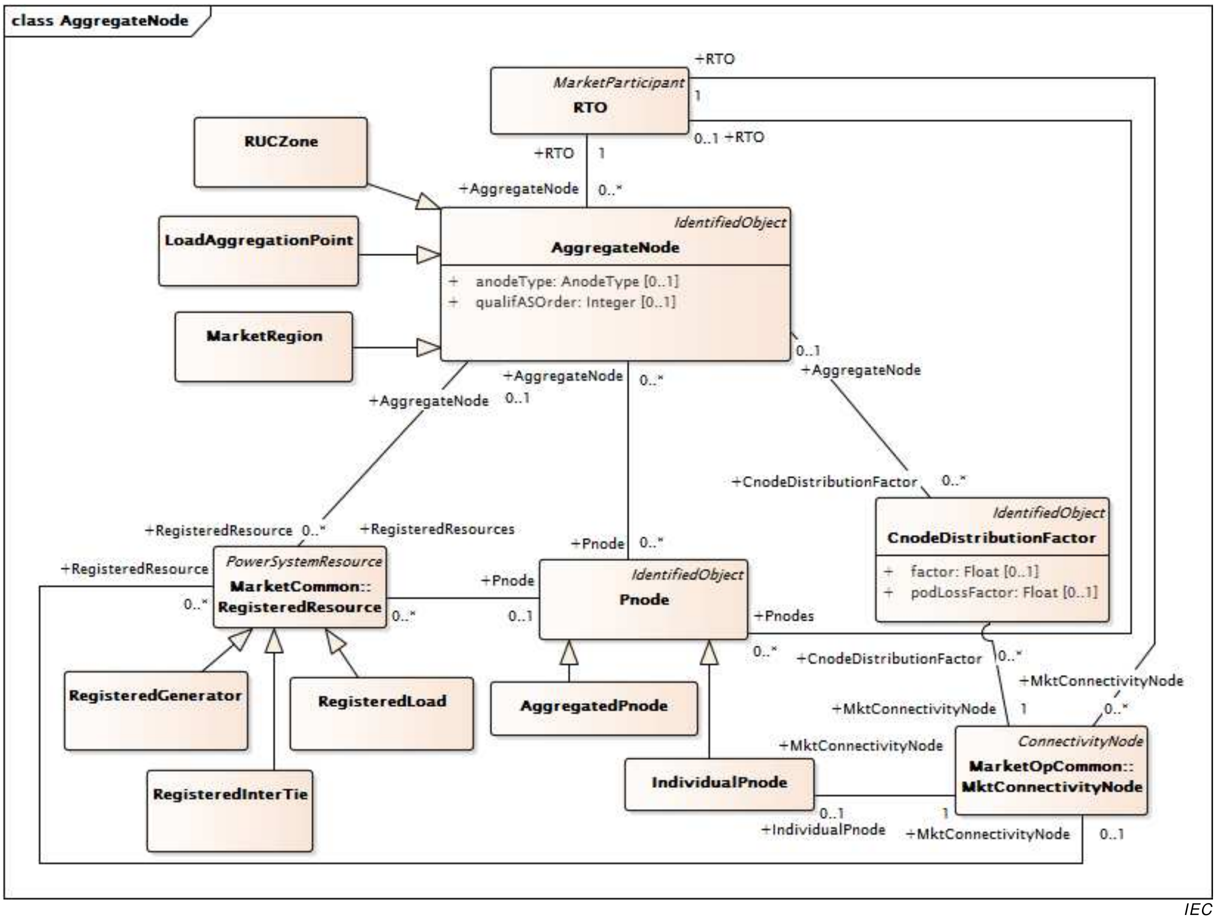



**Figure 59 – Class diagram ReferenceData::AggregateNode**

Figure 59: Shows classes and associations used to model aggregate nodes. The aggregated node allows a grouping of Pnode and Cnodes that is further defined by AnodeType. Reliability Unit Commitment zones are modeled by the RUCZone subclass. Load aggregation points are modeled by the LoadAggregationPoint subclass, and market regions by the MarketRegion subclass. Registered resources are associated with Aggregated nodes.

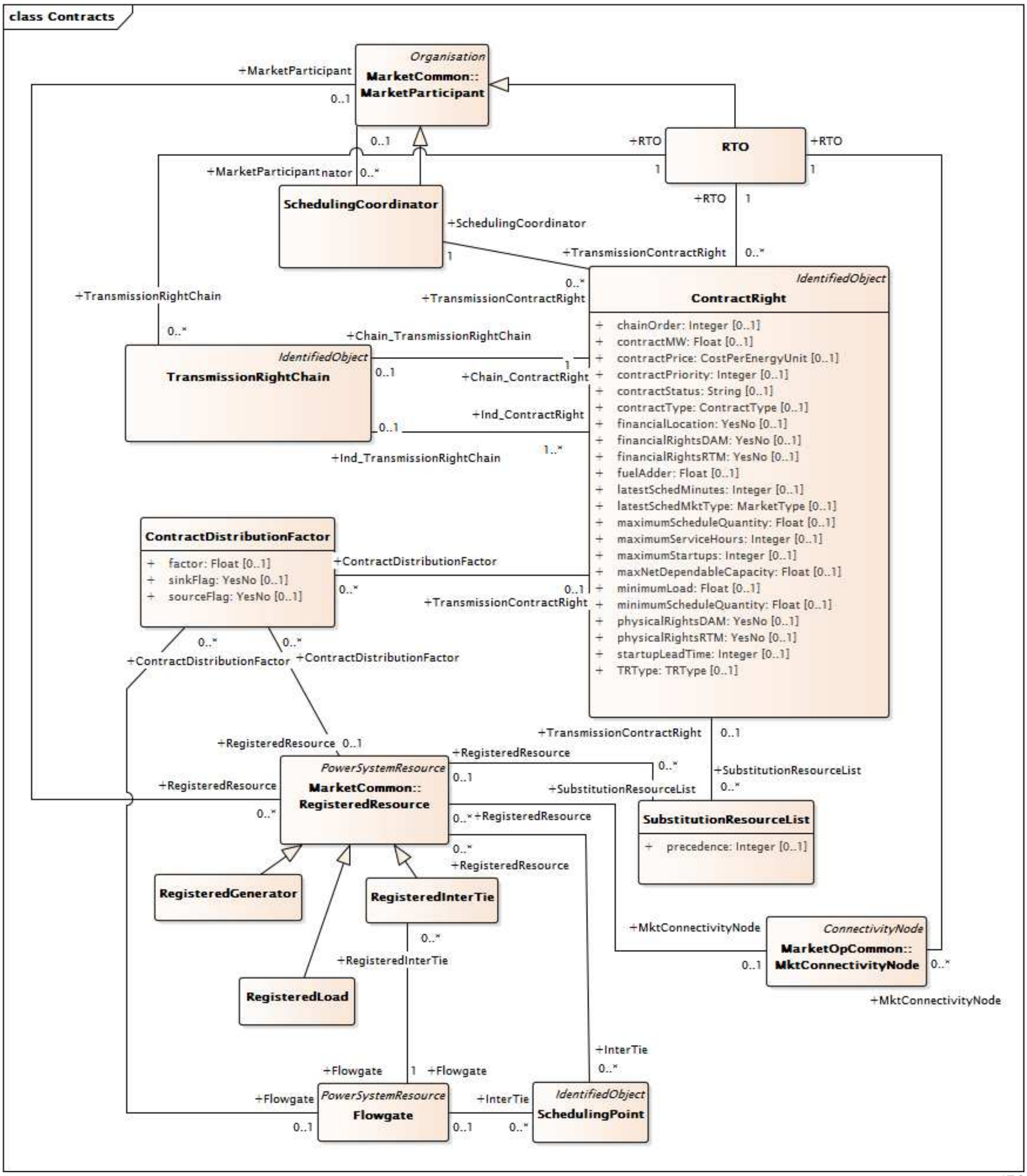



**Figure 60 – Class diagram ReferenceData::Contracts**

Figure 60: Shows the classes and associations used to model Contract rights. The class ContractRight is used to summarize contract information. Types of contracts include Existing Transmission Contracts (ETC), Transmission Ownership Rights (TOR), Reliability Must Run (RMR) and Reliability Must Take (RMT). ETCs and TOR represent grandfathered transmission use contracts that exist at the time of startup of an ISO. RMR and RMT affect generation. The figure shows associations with the TransmissionRightChain which may be used to model chaining of ETCs and with Scheduling Coordinators. Associations are also included to model distribution to sources and sinks (the Contract DistributionFactor class) and to registered resources (RegisteredResource class).

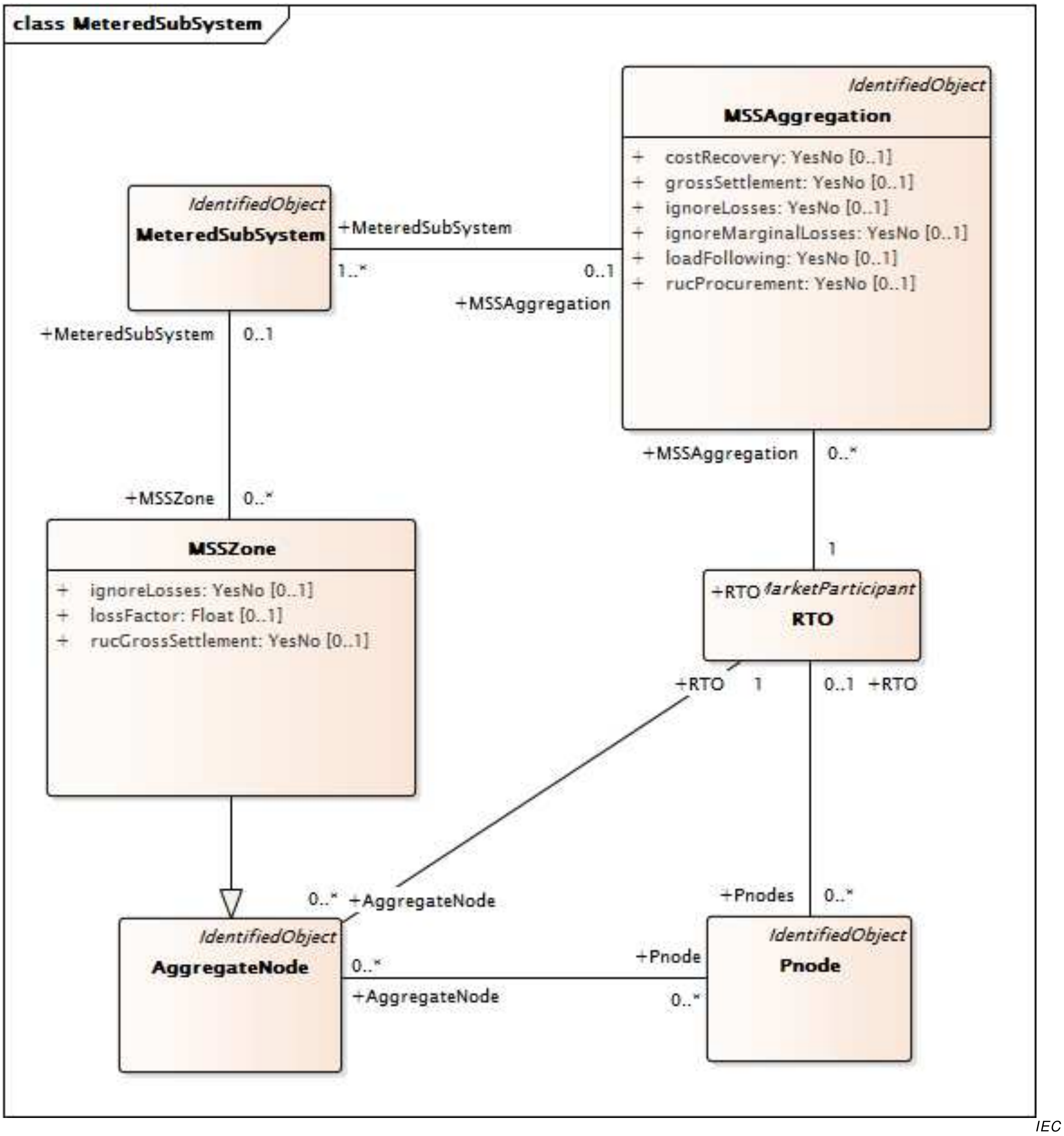



**Figure 61 – Class diagram ReferenceData::MeteredSubSystem**

Figure 61: Shows the classes and associations used to model metered subsystems. A metered subsystem (MSS) is a geographically contiguous system which was historically operated as a separate control area and which is now operated in accordance with a MSS agreement. Associations are shown with the class MSSZone which is used to model zones with an MSS. MSSZone is subtype of the Aggregated Node.

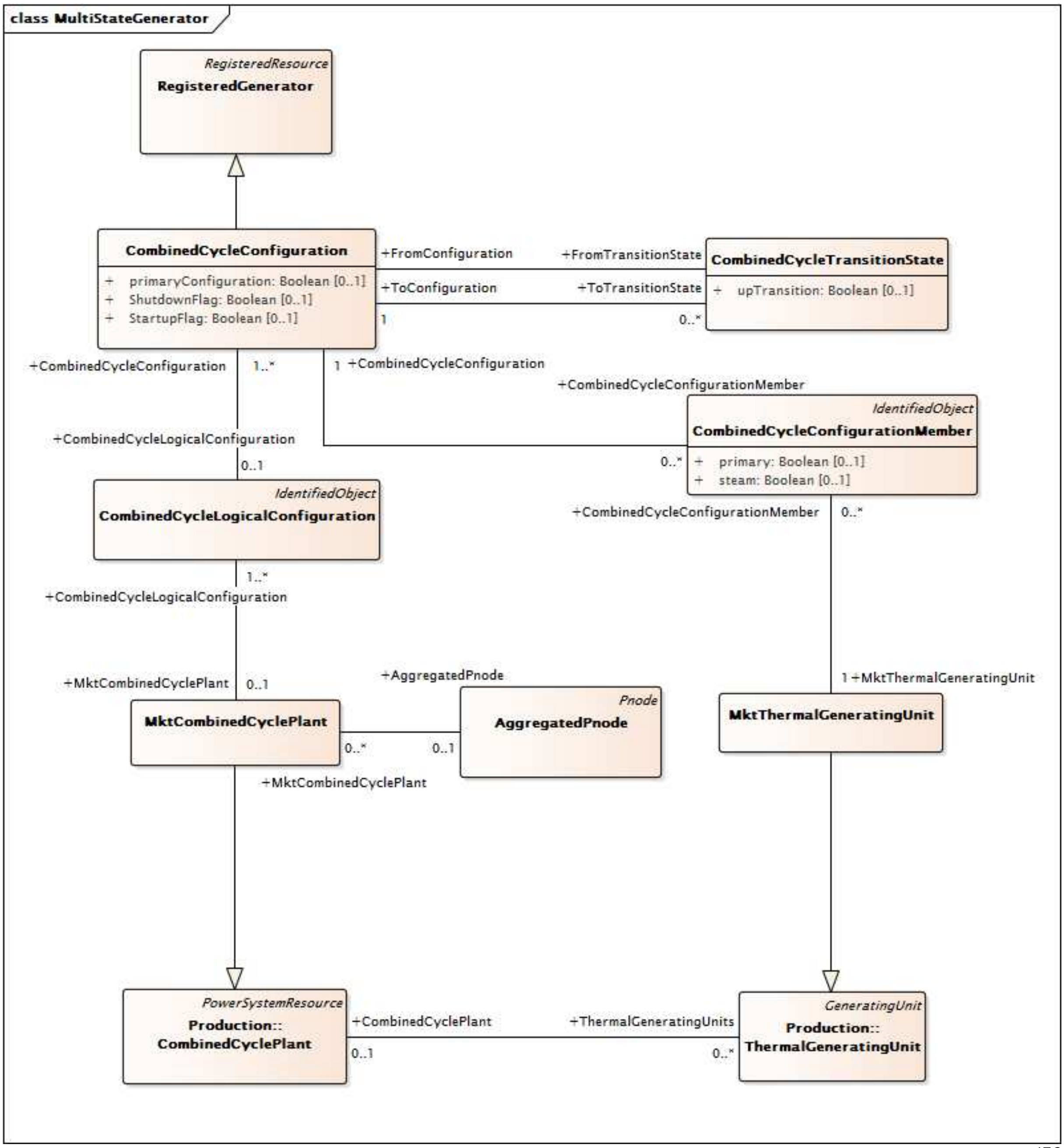



**Figure 62 – Class diagram ReferenceData::MultiStateGenerator**

Figure 62: Shows the classes and associations used to model MultiState Generating Units, such as combined cycle plants. The combined cycle plant is associated with the LogicalConfiguration class, the Configuration class and the ConfigurationMember class which are used to model the various modes of operation of the plant. The MktCombinedCyclePlant and the MktThermalGeneratingUnit classes are subclasses of the parent classes from IEC 61970 Production package.

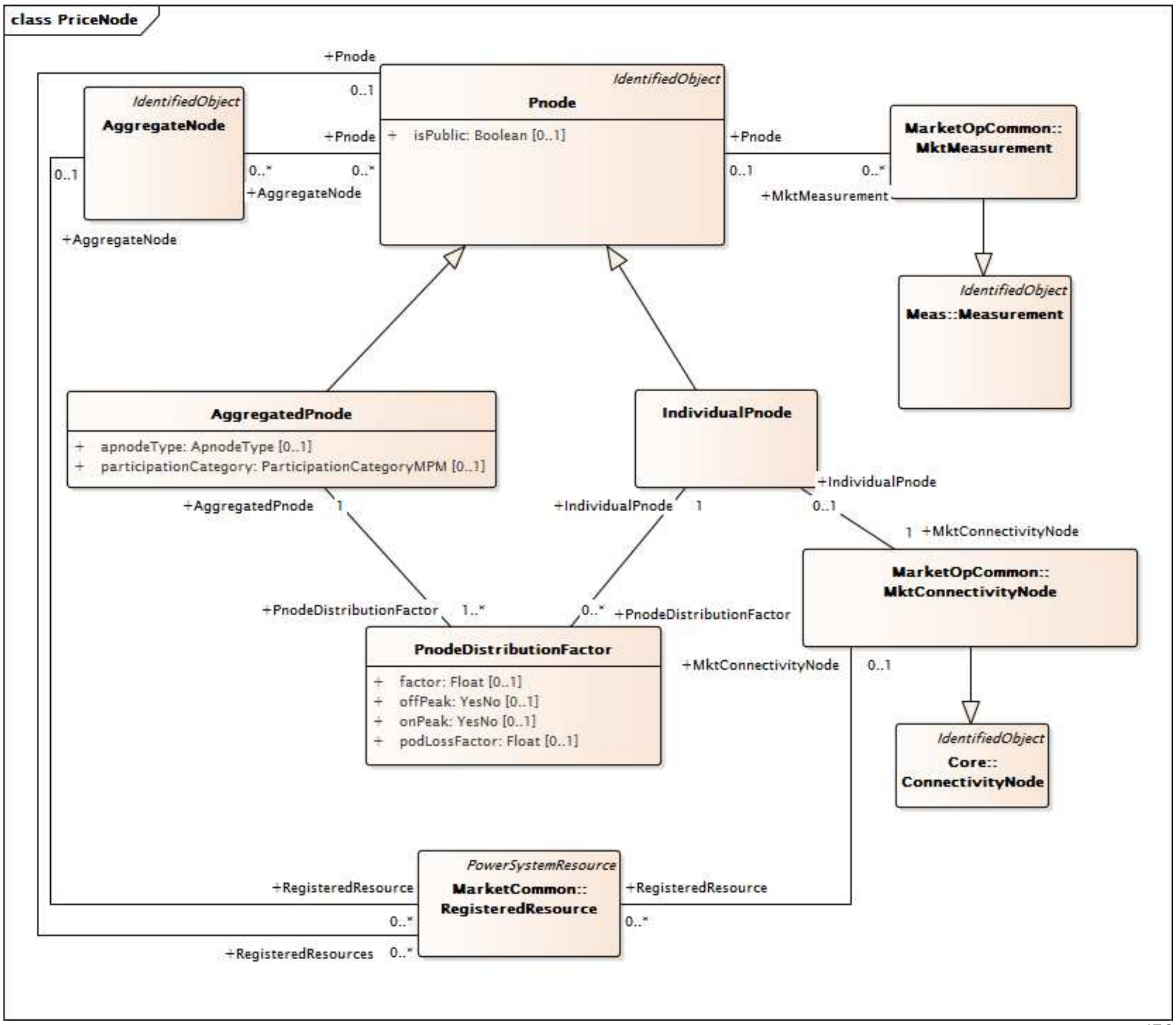



**Figure 63 – Class diagram ReferenceData::PriceNode**

Figure 63: Shows the classes and associations used to model pricing nodes. The pricing node is referenced by market participants, in bids/offers for energy, ancillary services, and CRRs. AggregatePnodes and IndividualPnodes are subclasses of the Pnode class. Pnodes are associated with registered resources and used in bids/offers. Pnodes are also associated with the MktMeasurement class which is used in Billing and Settlement.

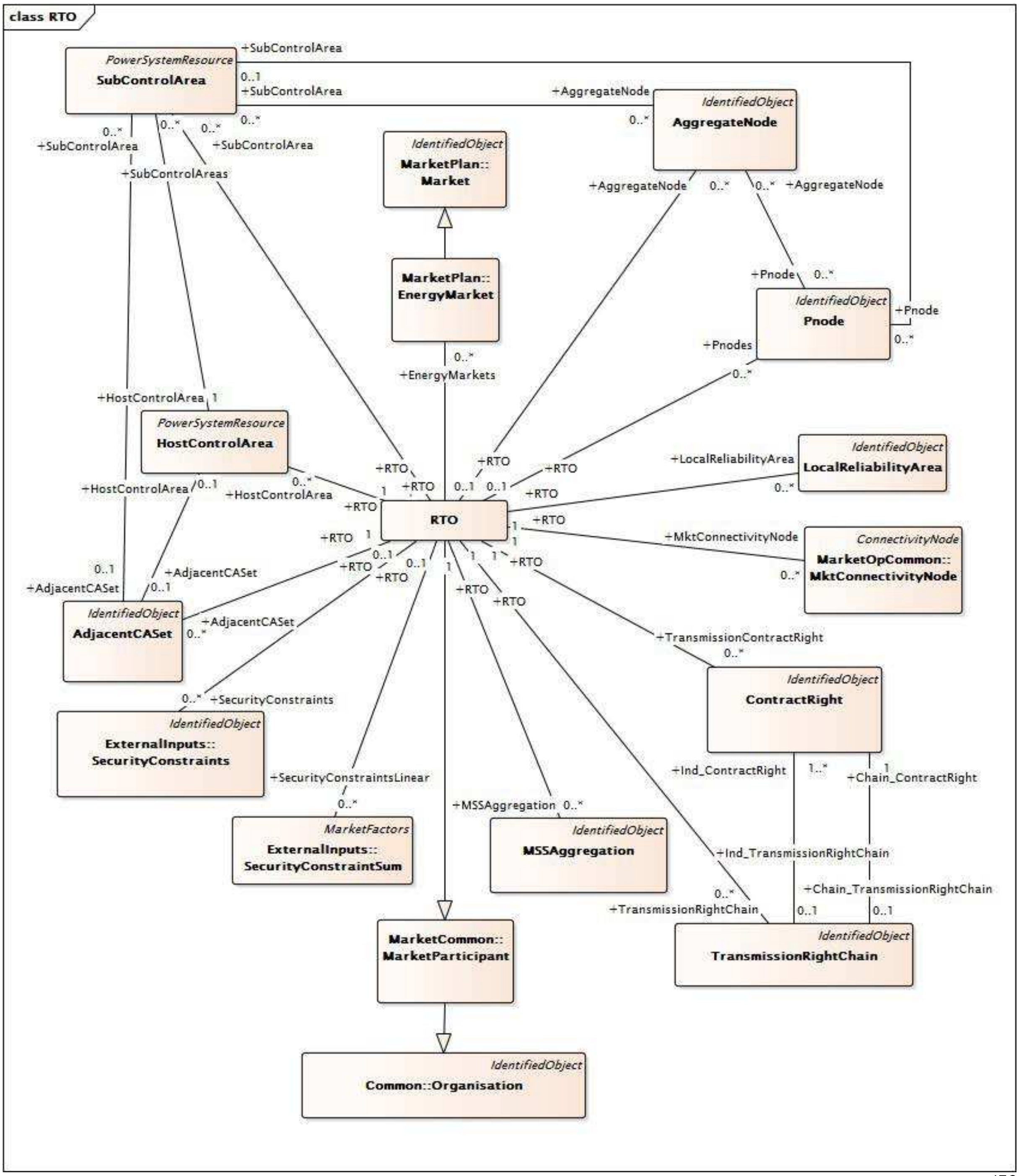



**Figure 64 – Class diagram ReferenceData::RTO**

Figure 64: Shows the classes and associations used to model a Regional Transmission Organization that operates an electricity market. The major classes and relationships are shown at an overview level. Supporting figures appear throughout the standard.

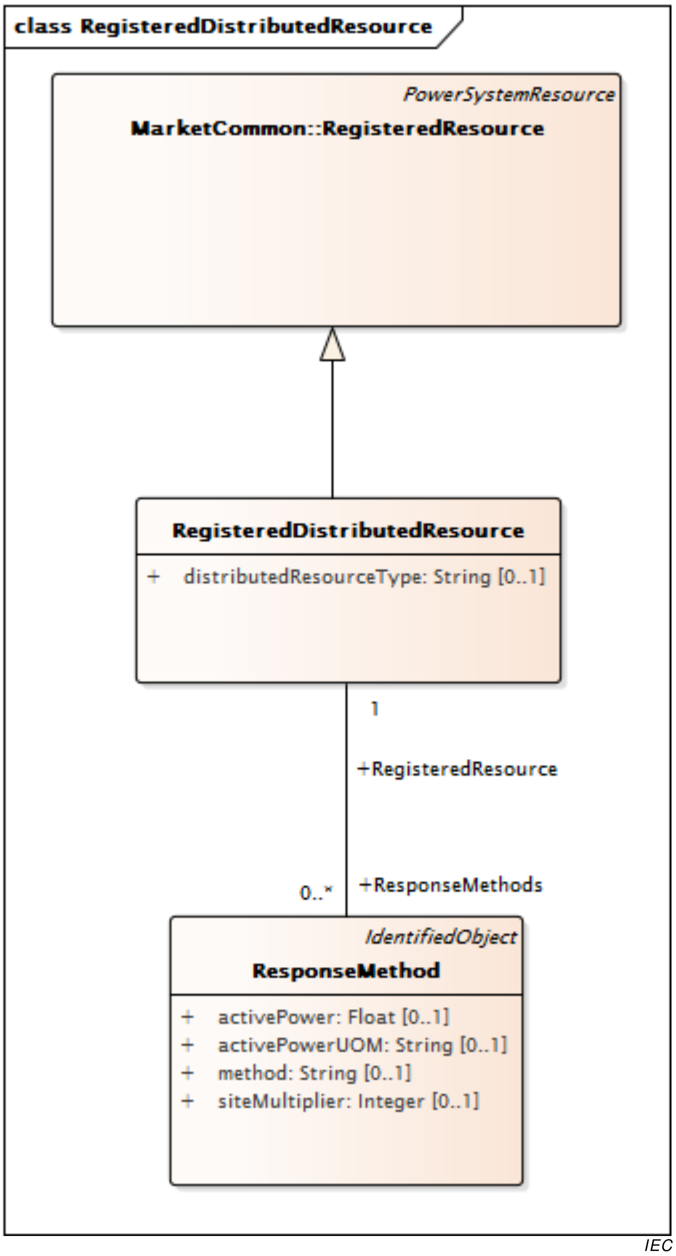



**Figure 65 – Class diagram ReferenceData::RegisteredDistributedResource**

Figure 65: Illustrates the classes and associations used to model registered distributed resources. RegisteredDistributedResources may represent demand response resources, storage resources, distribution generation resources, etc. RegisteredDistributedResource inherits from RegisteredResource and may be associated to ResponseMethods.

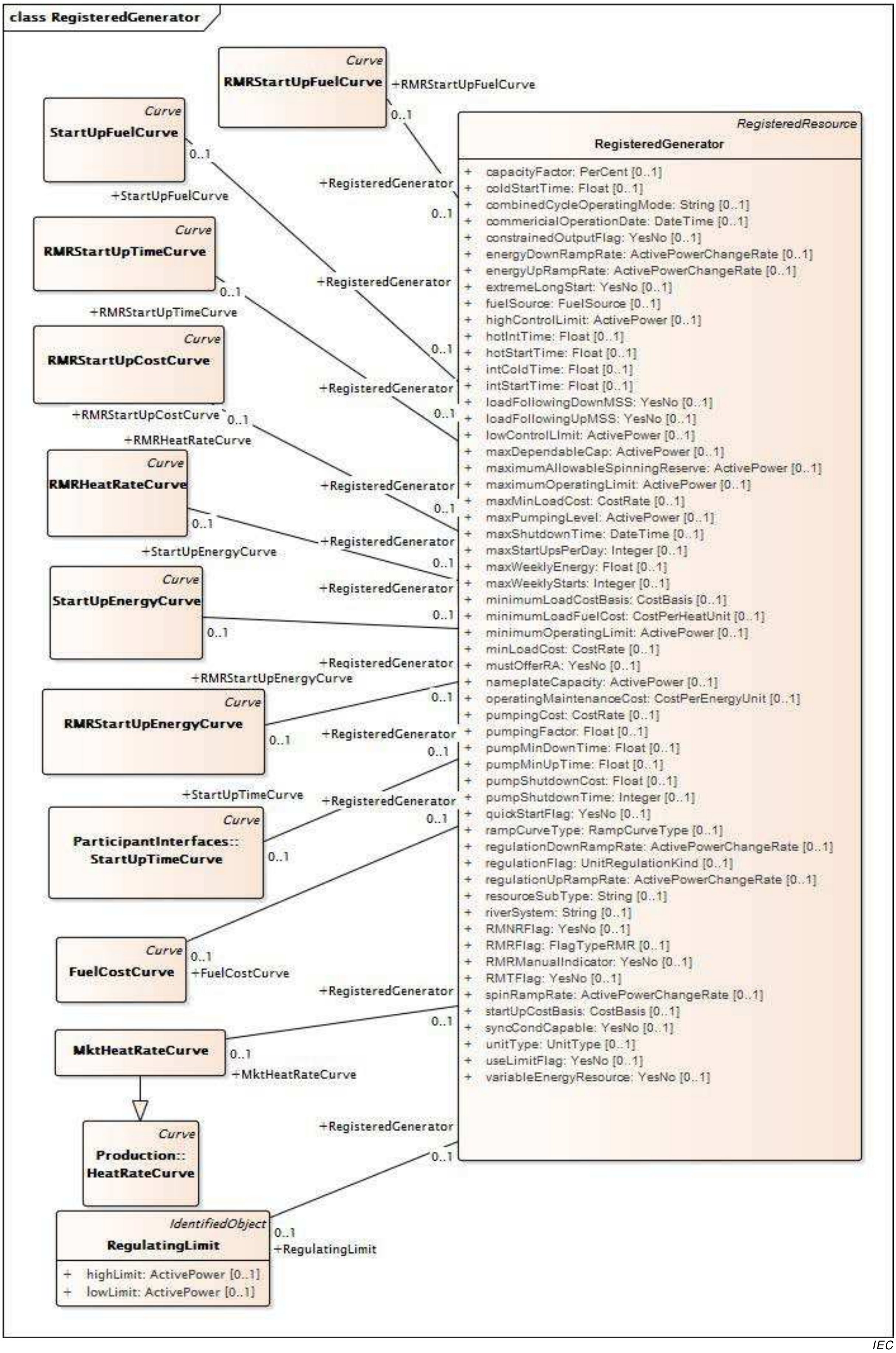



**Figure 66 – Class diagram ReferenceData::RegisteredGenerator**

Figure 66: Shows the classes and associations use to model registered generators. The RMRStartUpFuelCurve is used to model the cost of starting generating units classified as reliability must run units. Other classes are used to model startup cost, start up energy curve, heat rates, and start up time for RMR units. The class FuelCostCurve is used to model the relationship between startup fuel cost and unit active power output.

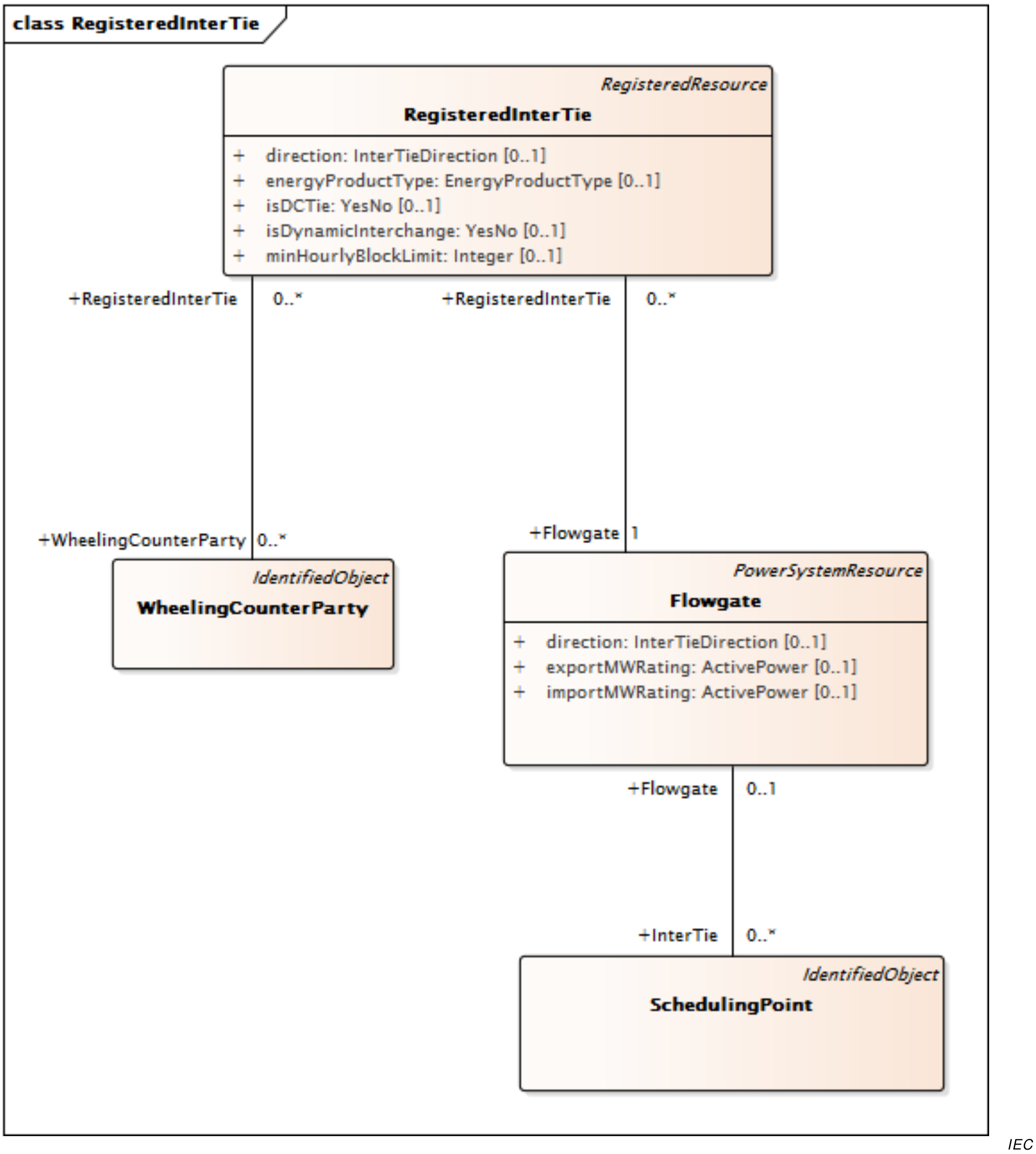



**Figure 67 – Class diagram ReferenceData::RegisteredInterTie**

Figure 67: Shows the classes and associations used to model registered interties. The SchedulingPoint class is associated with a flowgate which is in turn associated with a RegisteredInterTie. The RegisteredInterTie may be an AC or DC tie to a neighboring utility entity.

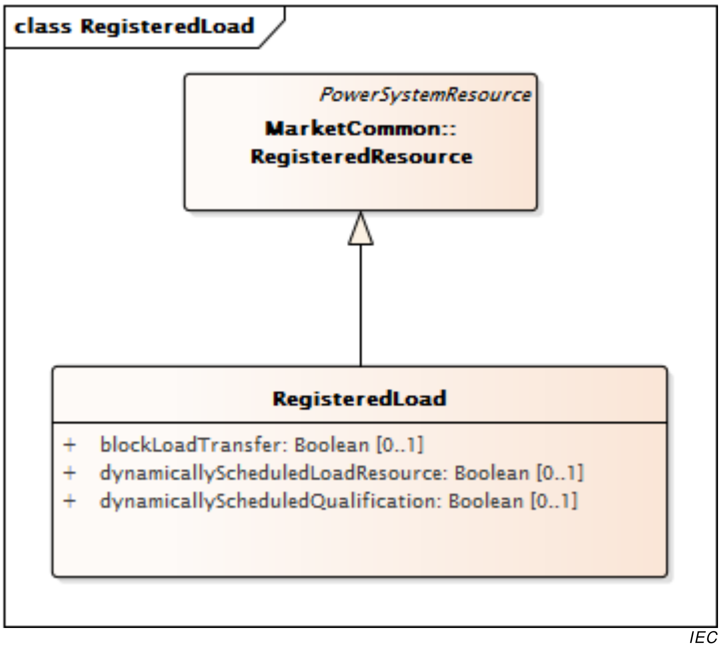



**Figure 68 – Class diagram ReferenceData::RegisteredLoad**

Figure 68: Shows the classes and associations use to model registered loads. RegisterLoads may participate in the market by offering to reduce baseline consumption of energy and one or more electrical products (energy, ancillary services) in return for a payment. The MktEnergyConsumer class models the location of the load in the network.

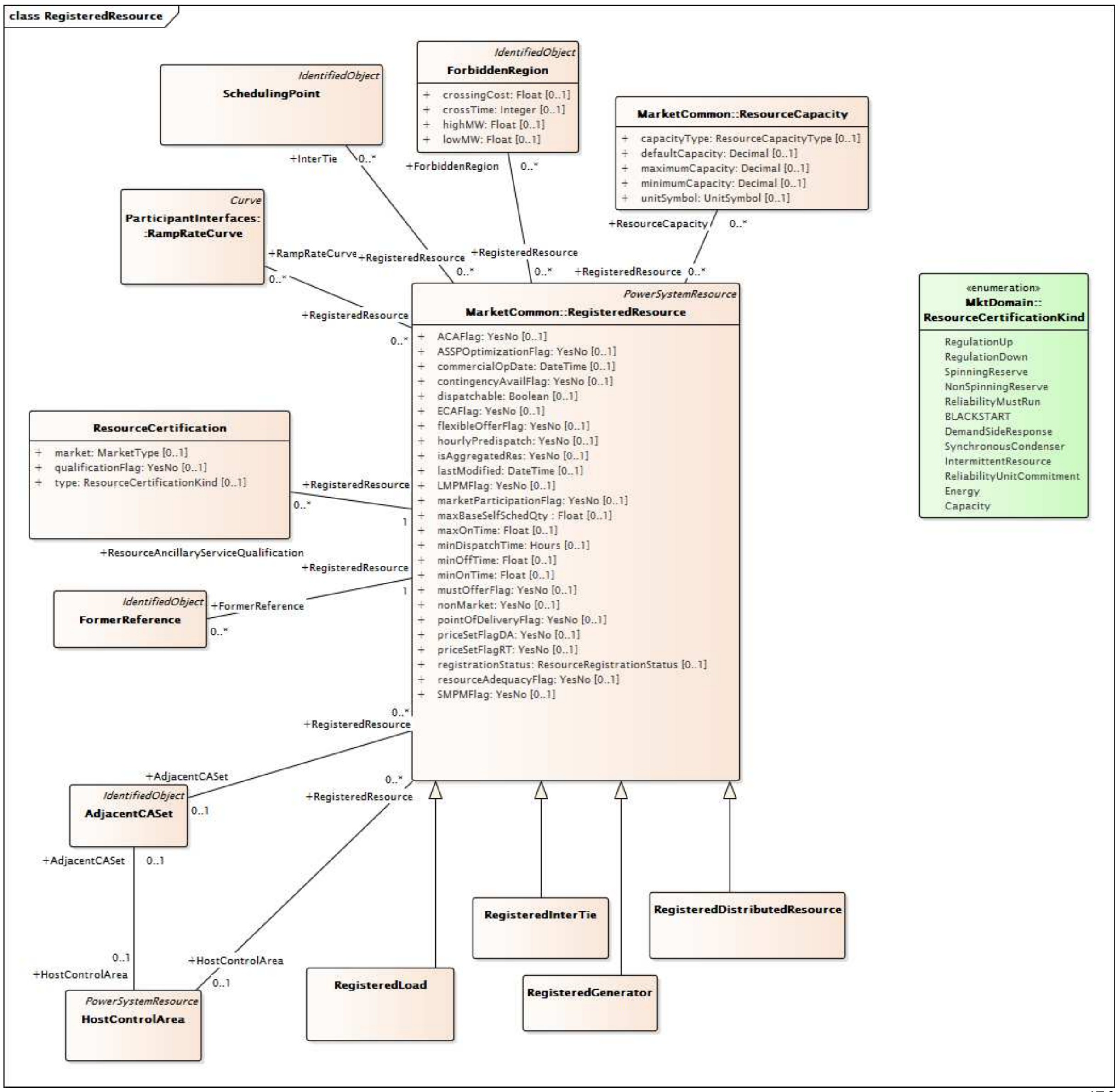



**Figure 69 – Class diagram ReferenceData::RegisteredResource**

Figure 69: Shows the classes and associations used to model registered resources. The RegisteredResource class is the parent class of the RegisteredGenerator, RegisteredInterTie, and RegisteredLoad subclasses. Major classes include the ResourceAncillaryServiceQualification class that is used to describe the ancillary services that the resource is qualified to offer. The ResourceCapability, the ForbiddenRegion and RampRateCurve classes are used to model technical capabilities of the resource. The SchedulingPoint class models the points on the boundary to other utility entities where the resource can deliver market products to the RTO.

### 6.5.9.2 AdjacentCASet

Groups Adjacent Control Areas.

Table 505 shows all attributes of AdjacentCASet.

**Table 505 – Attributes of ReferenceData::AdjacentCASet**

| name | mult | type | description |
|---|---|---|---|
| lossPercentage | 0..1 | Float | Loss percentage |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 506 shows all association ends of AdjacentCASet with other classes.

**Table 506 – Association ends of ReferenceData::AdjacentCASet with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | SubControlArea | 0..* | SubControlArea | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | HostControlArea | 0..1 | HostControlArea | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.3 AggregateNode

An aggregated node can define a typed grouping further defined by the AnodeType enumeratuion. Types range from System Zone/Regions to Market Energy Regions to Aggregated Loads and Aggregated Generators.

Table 507 shows all attributes of AggregateNode.

**Table 507 – Attributes of ReferenceData::AggregateNode**

| name | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | Type of aggregated node |
| qualifASOrder | 0..1 | Integer | Processing Order for AS self-provisions for this region. The priority of this attribute directs the awards of any resource that resides in overlapping regions. The regions are processed in priority manner. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 508 shows all association ends of AggregateNode with other classes.

**Table 508 – Association ends of ReferenceData::AggregateNode with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..* | Pnode | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..* | SubControlArea | 0..* | SubControlArea | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | A RegisteredResource can be associated to only one AggregateNode if not connected to a Pnode or MktConnectivityNode. |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | |
| 0..1 | Instruction | 0..* | Instructions | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.4 AggregatedPnode

An aggregated pricing node is a specialized type of pricing node used to model items such as System Zone, Default Price Zone, Custom Price Zone, Control Area, Aggregated Generation, Aggregated Particpating Load, Aggregated Non-Participating Load, Trading Hub, Designated Control Area(DCA) Zone.

Table 509 shows all attributes of AggregatedPnode.

**Table 509 – Attributes of ReferenceData::AggregatedPnode**

| name | mult | type | description |
|---|---|---|---|
| apnodeType | 0..1 | ApnodeType | Aggregate Price Node Types |
| participationCategory | 0..1 | ParticipationCategoryMPM | Designated Control Area participation in LMP price measurement<br>'Y' – Participates in both Local Market Power Mitigation (LMPM) and System Market Power Mitigation (SMPM)<br>'N' – Not included in LMP price measures<br>'S' – Participatesin SMPM price measures<br>'L' – Participatesin LMPM price measures |
| isPublic | 0..1 | Boolean | inherited from: Pnode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 510 shows all association ends of AggregatedPnode with other classes.

**Table 510 – Association ends of ReferenceData::AggregatedPnode with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GenDistributionFactor | 1..* | GenDistributionFactor | |
| 0..1 | LoadDistributionFactor | 1..* | LoadDistributionFactor | |
| 1..1 | MPMTestResults | 1..* | MPMTestResults | |
| 0..* | TACArea | 0..* | TACArea | |
| 1..1 | PnodeDistributionFactor | 1..* | PnodeDistributionFactor | |
| 1..1 | TradingHubValues | 0..* | TradingHubValues | |
| 0..1 | MktCombinedCyclePlant | 0..* | MktCombinedCyclePlant | |
| 0..* | MPMTestThreshold | 1..* | MPMTestThreshold | |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | inherited from: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | inherited from: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | inherited from: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | inherited from: Pnode |
| 0..1 | Trade | 0..* | Trade | inherited from: Pnode |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | inherited from: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | inherited from: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | inherited from: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | inherited from: Pnode |
| 0..* | RTO | 0..1 | RTO | inherited from: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.5 BidPriceCap root class

This class represent the bid price cap.

Table 511 shows all attributes of BidPriceCap.

**Table 511 – Attributes of ReferenceData::BidPriceCap**

| name | mult | type | description |
|---|---|---|---|
| marketType | 0..1 | MarketType | Market Type of the cap (DAM or RTM) |
| bidFloor | 0..1 | CostPerEnergyUnit | Bid Floor, ($/MWH) |
| bidCeiling | 0..1 | CostPerEnergyUnit | Bid Ceiling ($/MWH) |
| defaultPrice | 0..1 | CostPerEnergyUnit | Bid Default Price($/MWH) |
| bidFloorAS | 0..1 | CostPerEnergyUnit | Bid Floor ($/MWH) for generic AS versus a specific market product |
| bidCeilingAS | 0..1 | CostPerEnergyUnit | Bid Ceiling ($/MWH) for generic AS versus a specific market product |

Table 512 shows all association ends of BidPriceCap with other classes.

**Table 512 – Association ends of ReferenceData::BidPriceCap with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketProduct | 0..1 | MarketProduct | |

#### 6.5.9.6 CnodeDistributionFactor

Participation factors per Cnode. Used to calculate "participation" of Cnode in an AggregateNode. Each Cnode associated to an AggregateNode would be assigned a participation factor for its participation within the AggregateNode.

Table 513 shows all attributes of CnodeDistributionFactor.

**Table 513 – Attributes of ReferenceData::CnodeDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Used to calculate "participation" of Cnode in an AggregateNode |
| podLossFactor | 0..1 | Float | Point of delivery loss factor |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 514 shows all association ends of CnodeDistributionFactor with other classes.

**Table 514 – Association ends of ReferenceData:: CnodeDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | AggregateNode | 0..1 | AggregateNode | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.7 CombinedCycleConfiguration

Configuration options for combined cycle units.

For example, a Combined Cycle with (CT1, CT2, ST1) will have (CT1, ST1) and (CT2, ST1) configurations as part of(1CT + 1STlogicalconfiguration).

Table 515 shows all attributes of CombinedCycleConfiguration.

**Table 515 – Attributes of ReferenceData::CombinedCycleConfiguration**

| name | mult | type | description |
|---|---|---|---|
| primaryConfiguration | 0..1 | Boolean | Whether this CombinedCycleConfiguration is the primary configuration in the associated Logical configuration? |
| ShutdownFlag | 0..1 | Boolean | Whether Combined Cycle Plant can be shut-down in this Configuration? |
| StartupFlag | 0..1 | Boolean | Whether Combined Cycle Plant can be started in this Logical Configuration? |
| capacityFactor | 0..1 | PerCent | inherited from: RegisteredGenerator |
| coldStartTime | 0..1 | Float | inherited from: RegisteredGenerator |
| combinedCycleOperatingMode | 0..1 | String | inherited from: RegisteredGenerator |
| commericialOperationDate | 0..1 | DateTime | inherited from: RegisteredGenerator |
| constrainedOutputFlag | 0..1 | YesNo | inherited from: RegisteredGenerator |
| energyDownRampRate | 0..1 | ActivePowerChangeRate | inherited from: RegisteredGenerator |
| energyUpRampRate | 0..1 | ActivePowerChangeRate | inherited from: RegisteredGenerator |
| extremeLongStart | 0..1 | YesNo | inherited from: RegisteredGenerator |
| fuelSource | 0..1 | FuelSource | inherited from: RegisteredGenerator |
| highControlLimit | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| hotIntTime | 0..1 | Float | inherited from: RegisteredGenerator |
| hotStartTime | 0..1 | Float | inherited from: RegisteredGenerator |
| intColdTime | 0..1 | Float | inherited from: RegisteredGenerator |
| intStartTime | 0..1 | Float | inherited from: RegisteredGenerator |
| loadFollowingDownMSS | 0..1 | YesNo | inherited from: RegisteredGenerator |
| loadFollowingUpMSS | 0..1 | YesNo | inherited from: RegisteredGenerator |
| lowControlLImit | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| maxDependableCap | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| maximumOperatingLimit | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| maxMinLoadCost | 0..1 | CostRate | inherited from: RegisteredGenerator |
| maxPumpingLevel | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| maxShutdownTime | 0..1 | DateTime | inherited from: RegisteredGenerator |
| maxStartUpsPerDay | 0..1 | Integer | inherited from: RegisteredGenerator |
| maxWeeklyEnergy | 0..1 | Float | inherited from: RegisteredGenerator |
| maxWeeklyStarts | 0..1 | Integer | inherited from: RegisteredGenerator |
| minimumLoadCostBasis | 0..1 | CostBasis | inherited from: RegisteredGenerator |
| minimumLoadFuelCost | 0..1 | CostPerHeatUnit | inherited from: RegisteredGenerator |
| minimumOperatingLimit | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| minLoadCost | 0..1 | CostRate | inherited from: RegisteredGenerator |
| mustOfferRA | 0..1 | YesNo | inherited from: RegisteredGenerator |
| nameplateCapacity | 0..1 | ActivePower | inherited from: RegisteredGenerator |
| operatingMaintenanceCost | 0..1 | CostPerEnergyUnit | inherited from: RegisteredGenerator |
| pumpingCost | 0..1 | CostRate | inherited from: RegisteredGenerator |
| pumpingFactor | 0..1 | Float | inherited from: RegisteredGenerator |

| name | mult | type | description |
|---|---|---|---|
| pumpMinDownTime | 0..1 | Float | inherited from: RegisteredGenerator |
| pumpMinUpTime | 0..1 | Float | inherited from: RegisteredGenerator |
| pumpShutdownCost | 0..1 | Float | inherited from: RegisteredGenerator |
| pumpShutdownTime | 0..1 | Integer | inherited from: RegisteredGenerator |
| quickStartFlag | 0..1 | YesNo | inherited from: RegisteredGenerator |
| rampCurveType | 0..1 | RampCurveType | inherited from: RegisteredGenerator |
| regulationDownRampRat e | 0..1 | ActivePowerChangeRate | inherited from: RegisteredGenerator |
| regulationFlag | 0..1 | UnitRegulationKind | inherited from: RegisteredGenerator |
| regulationUpRampRate | 0..1 | ActivePowerChangeRate | inherited from: RegisteredGenerator |
| resourceSubType | 0..1 | String | inherited from: RegisteredGenerator |
| riverSystem | 0..1 | String | inherited from: RegisteredGenerator |
| RMNRFlag | 0..1 | YesNo | inherited from: RegisteredGenerator |
| RMRFlag | 0..1 | FlagTypeRMR | inherited from: RegisteredGenerator |
| RMRManualIndicator | 0..1 | YesNo | inherited from: RegisteredGenerator |
| RMTFlag | 0..1 | YesNo | inherited from: RegisteredGenerator |
| spinRampRate | 0..1 | ActivePowerChangeRate | inherited from: RegisteredGenerator |
| startUpCostBasis | 0..1 | CostBasis | inherited from: RegisteredGenerator |
| syncCondCapable | 0..1 | YesNo | inherited from: RegisteredGenerator |
| unitType | 0..1 | UnitType | inherited from: RegisteredGenerator |
| useLimitFlag | 0..1 | YesNo | inherited from: RegisteredGenerator |
| variableEnergyResource | 0..1 | YesNo | inherited from: RegisteredGenerator |
| ACAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | inherited from: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| dispatchable | 0..1 | Boolean | inherited from: RegisteredResource |
| ECAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | inherited from: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | inherited from: RegisteredResource |
| lastModified | 0..1 | DateTime | inherited from: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | inherited from: RegisteredResource |
| maxOnTime | 0..1 | Float | inherited from: RegisteredResource |
| minDispatchTime | 0..1 | Hours | inherited from: RegisteredResource |
| minOffTime | 0..1 | Float | inherited from: RegisteredResource |
| minOnTime | 0..1 | Float | inherited from: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| nonMarket | 0..1 | YesNo | inherited from: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | inherited from: RegisteredResource |

| name | mult | type | description |
|---|---|---|---|
| registrationStatus | 0..1 | ResourceRegistrationStatus | inherited from: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 516 shows all association ends of CombinedCycleConfiguration with other classes.

**Table 516 – Association ends of ReferenceData:: CombinedCycleConfiguration with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ToTransitionState | 0..* | CombinedCycleTransitionState | |
| 1..1 | FromTransitionState | 0..* | CombinedCycleTransitionState | |
| 1..1 | CombinedCycleConfigurationMember | 0..* | CombinedCycleConfigurationMember | |
| 1..* | CombinedCycleLogicalConfiguration | 0..1 | CombinedCycleLogicalConfiguration | |
| 0..1 | GeneratingBids | 0..* | GeneratingBid | inherited from: RegisteredGenerator |
| 0..1 | StartUpTimeCurve | 0..1 | StartUpTimeCurve | inherited from: RegisteredGenerator |
| 0..1 | Trade | 0..* | Trade | inherited from: RegisteredGenerator |
| 0..1 | MktHeatRateCurve | 0..1 | MktHeatRateCurve | inherited from: RegisteredGenerator |
| 0..1 | RMRHeatRateCurve | 0..1 | RMRHeatRateCurve | inherited from: RegisteredGenerator |
| 0..1 | RMRStartUpCostCurve | 0..1 | RMRStartUpCostCurve | inherited from: RegisteredGenerator |
| 0..1 | RMRStartUpEnergyCurve | 0..1 | RMRStartUpEnergyCurve | inherited from: RegisteredGenerator |
| 0..1 | RMRStartUpFuelCurve | 0..1 | RMRStartUpFuelCurve | inherited from: RegisteredGenerator |
| 0..1 | RMRStartUpTimeCurve | 0..1 | RMRStartUpTimeCurve | inherited from: RegisteredGenerator |
| 0..1 | RegulatingLimit | 0..1 | RegulatingLimit | inherited from: RegisteredGenerator |
| 0..1 | StartUpFuelCurve | 0..1 | StartUpFuelCurve | inherited from: RegisteredGenerator |
| 0..1 | StartUpEnergyCurve | 0..1 | StartUpEnergyCurve | inherited from: RegisteredGenerator |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | inherited from: RegisteredGenerator |
| 1..1 | EnergyPriceIndex | 1..1 | EnergyPriceIndex | inherited from: RegisteredGenerator |
| 0..1 | UnitInitialConditions | 0..* | UnitInitialConditions | inherited from: RegisteredGenerator |
| 0..* | StartUpCostCurves | 0..* | StartUpCostCurve | inherited from: RegisteredGenerator |
| 0..1 | FuelCostCurve | 0..1 | FuelCostCurve | inherited from: RegisteredGenerator |
| 0..* | FuelRegion | 0..1 | FuelRegion | inherited from: RegisteredGenerator |
| 0..* | LocalReliabilityArea | 0..1 | LocalReliabilityArea | inherited from: RegisteredGenerator |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | inherited from: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | inherited from: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | inherited from: RegisteredResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | inherited from: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | inherited from: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | inherited from: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | inherited from: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | inherited from: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | inherited from: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | inherited from: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | inherited from: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | inherited from: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | inherited from: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | inherited from: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | inherited from: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | inherited from: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | inherited from: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | inherited from: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | inherited from: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | inherited from: RegisteredResource |
| 0..* | Reason | 0..* | Reason | inherited from: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | inherited from: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | inherited from: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | inherited from: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | inherited from: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | inherited from: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | inherited from: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | inherited from: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | inherited from: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | inherited from: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | inherited from: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | inherited from: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | inherited from: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | inherited from: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | inherited from: RegisteredResource |
| 0..* | Domain | 0..* | Domain | inherited from: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | inherited from: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | inherited from: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.8 CombinedCycleConfigurationMember

Configuration Member of CCP Configuration.

Table 517 shows all attributes of CombinedCycleConfigurationMember.

**Table 517 – Attributes of ReferenceData::CombinedCycleConfigurationMember**

| name | mult | type | description |
|---|---|---|---|
| primary | 0..1 | Boolean | primary configuration. |
| steam | 0..1 | Boolean | Steam plant. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 518 shows all association ends of CombinedCycleConfigurationMember with other classes.

**Table 518 – Association ends of ReferenceData:: CombinedCycleConfigurationMember with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktThermalGeneratingUnit | 1..1 | MktThermalGeneratingUnit | |
| 0..* | CombinedCycleConfiguration | 1..1 | CombinedCycleConfiguration | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.9 CombinedCycleLogicalConfiguration

Logical Configuration of a Combined Cycle plant.

Operating Combined Cycle Plant (CCP) configurations are represented as Logical CCP Resources. Logical representation shall be used for Market applications to optimize and control Market Operations. Logical representation is also necessary for controlling the number of CCP configurations and to temper performance issues that may otherwise occur.

For example,(2CT configuration),(1CT + 1ST configuration) are examples of logical configuration, without specifying the specific CT and ST participating in the configuration.

Table 519 shows all attributes of CombinedCycleLogicalConfiguration.

**Table 519 – Attributes of ReferenceData::CombinedCycleLogicalConfiguration**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 520 shows all association ends of CombinedCycleLogicalConfiguration with other classes.

**Table 520 – Association ends of ReferenceData:: CombinedCycleLogicalConfiguration with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | CombinedCycleConfiguration | 1..* | CombinedCycleConfiguration | |
| 1..* | MktCombinedCyclePlant | 0..1 | MktCombinedCyclePlant | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.10 CombinedCycleTransitionState root class

Defines the available from and to Transition States for the Combine Cycle Configurations.

Table 521 shows all attributes of CombinedCycleTransitionState.

**Table 521 – Attributes of ReferenceData::CombinedCycleTransitionState**

| name | mult | type | description |
|---|---|---|---|
| upTransition | 0..1 | Boolean | Flag indicating whether this is an UP transition.<br>If not, it is a DOWN transition. |

Table 522 shows all association ends of CombinedCycleTransitionState with other classes.

**Table 522 – Association ends of ReferenceData:: CombinedCycleTransitionState with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ToConfiguration | 1..1 | CombinedCycleConfiguration | |
| 0..* | FromConfiguration | 1..1 | CombinedCycleConfiguration | |

### 6.5.9.11 CongestionArea

Designated Congestion Area Definition (DCA).

Table 523 shows all attributes of CongestionArea.

**Table 523 – Attributes of ReferenceData::CongestionArea**

| name | mult | type | description |
|---|---|---|---|
| apnodeType | 0..1 | ApnodeType | inherited from: AggregatedPnode |
| participationCategory | 0..1 | ParticipationCategoryMPM | inherited from: AggregatedPnode |
| isPublic | 0..1 | Boolean | inherited from: Pnode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 524 shows all association ends of CongestionArea with other classes.

**Table 524 – Association ends of ReferenceData::CongestionArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | IndividualPnode | 0..* | IndividualPnode | |
| 0..1 | GenDistributionFactor | 1..* | GenDistributionFactor | inherited from: AggregatedPnode |
| 0..1 | LoadDistributionFactor | 1..* | LoadDistributionFactor | inherited from: AggregatedPnode |
| 1..1 | MPMTestResults | 1..* | MPMTestResults | inherited from: AggregatedPnode |
| 0..* | TACArea | 0..* | TACArea | inherited from: AggregatedPnode |
| 1..1 | PnodeDistributionFactor | 1..* | PnodeDistributionFactor | inherited from: AggregatedPnode |
| 1..1 | TradingHubValues | 0..* | TradingHubValues | inherited from: AggregatedPnode |
| 0..1 | MktCombinedCyclePlant | 0..* | MktCombinedCyclePlant | inherited from: AggregatedPnode |
| 0..* | MPMTestThreshold | 1..* | MPMTestThreshold | inherited from: AggregatedPnode |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | inherited from: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | inherited from: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | inherited from: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | inherited from: Pnode |
| 0..1 | Trade | 0..* | Trade | inherited from: Pnode |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | inherited from: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | inherited from: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | inherited from: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | inherited from: Pnode |
| 0..* | RTO | 0..1 | RTO | inherited from: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.12 ContractDistributionFactor root class

Distribution among resources at the sink point or source point.

Table 525 shows all attributes of ContractDistributionFactor.

**Table 525 – Attributes of ReferenceData::ContractDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | MW value that this resource provides to the overall contract. |
| sourceFlag | 0..1 | YesNo | This value will be set to YES if the referenced Cnode is defined as the source point in the contract. |
| sinkFlag | 0..1 | YesNo | This value will be set to YES if the referenced Cnode is defined as the sink point in the contract. |

Table 526 shows all association ends of ContractDistributionFactor with other classes.

**Table 526 – Association ends of ReferenceData:: ContractDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |
| 0..* | Flowgate | 0..1 | Flowgate | |

#### 6.5.9.13 ContractRight

Provides definition of Transmission Ownership Right and Existing Transmission Contract identifiers for use by SCUC. RMR contract hosting: Startup lead time, Contract Service Limits, Max Service Hours, Max MWhs, Max Start-ups, Ramp Rate, Max Net Dependable Capacity, Min Capacity and Unit Substitution for DAM/RTM to retrieve.

Table 527 shows all attributes of ContractRight.

**Table 527 – Attributes of ReferenceData::ContractRight**

| name | mult | type | description |
|---|---|---|---|
| chainOrder | 0..1 | Integer | When used in conjunction with a Transmission Right contract chain, this is the precedence for the contracts. |
| contractMW | 0..1 | Float | MW value of the contract |
| contractPrice | 0..1 | CostPerEnergyUnit | Financial value of the contract |
| contractPriority | 0..1 | Integer | Priority for the contract. This should be unique amoung all contracts for a specific resource. This value is the directive for the SCUC algorithm on the order to satisfy/cut contracts. |
| contractStatus | 0..1 | String | Contract status |
| contractType | 0..1 | ContractType | type of the contract. Possible values are but not limited by:<br>ETC, TOR or RMR and RMT self schedules |
| financialLocation | 0..1 | YesNo | Indicator if the location associated with this contract is financial (e.g. pricing nodes) or physical (e.g. connectivity nodes). |
| financialRightsDAM | 0..1 | YesNo | Flag to indicate this contract provides financial rights in the DA Market |
| financialRightsRTM | 0..1 | YesNo | Flag to indicate this contract provides financial rights in the RT Market |
| fuelAdder | 0..1 | Float | Estimated Fuel Adder |
| latestSchedMinutes | 0..1 | Integer | This indicates the latest schedule minutes (e.g. t – xx) that this resource can be notified to respond. This attribute is only used if the market type is not supplied. |
| latestSchedMktType | 0..1 | MarketType | This indicates the latest schedule market type a contract can be applied to. This is used in conjunction with the latestSchedMinutes attribute to determine the latest time this contract can be called in. The possible values for this attribute are: DAM, RTM or it can be omitted. If omitted, the latestSchedMinutes attribute defines the value. |
| maximumScheduleQuantity | 0..1 | Float | Maximum schedule MW quantity |
| maximumServiceHours | 0..1 | Integer | Maximum service hours |
| maximumStartups | 0..1 | Integer | Maximum startups |
| maxNetDependableCapacity | 0..1 | Float | Maximum Net Dependable Capacity |
| minimumLoad | 0..1 | Float | Minimum Load |
| minimumScheduleQuantity | 0..1 | Float | Minimum schedule quanity |
| physicalRightsDAM | 0..1 | YesNo | Flag to indicate this contract provides physical rights in the DA Market |
| physicalRightsRTM | 0..1 | YesNo | Flag to indicate this contract provides physical rights in the RT Market |
| startupLeadTime | 0..1 | Integer | Start up lead time |
| TRType | 0..1 | TRType | Transmission Right type – is this an individual contract right or a chain contract right. Types = CHAIN or INDIVIDUAL |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 528 shows all association ends of ContractRight with other classes.

**Table 528 – Association ends of ReferenceData::ContractRight with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | TREntitlement | 0..* | TREntitlement | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | |
| 1..1 | TransmissionInterfaceEntitlement | 0..* | TransmissionInterfaceRightEntitlement | |
| 1..* | Ind_TransmissionRightChain | 0..1 | TransmissionRightChain | |
| 1..1 | Chain_TransmissionRightChain | 0..1 | TransmissionRightChain | |
| 0..* | SchedulingCoordinator | 1..1 | SchedulingCoordinator | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.14 ControlAreaDesignation root class

Indicates Control Area associated with self-schedule.

Table 529 shows all attributes of ControlAreaDesignation.

**Table 529 – Attributes of ReferenceData::ControlAreaDesignation**

| name | mult | type | description |
|---|---|---|---|
| attained | 0..1 | YesNo | Attained. |
| native | 0..1 | YesNo | Native. |

Table 530 shows all association ends of ControlAreaDesignation with other classes.

**Table 530 – Association ends of ReferenceData::
ControlAreaDesignation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | SubControlArea | 0..* | SubControlArea | |

### 6.5.9.15 Flowgate

A flowgate, is single or group of transmission elements intended to model MW flow impact relating to transmission limitations and transmission service usage.

Table 531 shows all attributes of Flowgate.

**Table 531 – Attributes of ReferenceData::Flowgate**

| name | mult | type | description |
|---|---|---|---|
| direction | 0..1 | InterTieDirection | The direction of the flowgate, export or import |
| exportMWRating | 0..1 | ActivePower | Export MW rating |
| importMWRating | 0..1 | ActivePower | Import MW rating |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 532 shows all association ends of Flowgate with other classes.

**Table 532 – Association ends of ReferenceData::Flowgate with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MktLine | 0..* | MktLine | |
| 0..* | MktPowerTransformer | 0..* | MktPowerTransformer | |
| 0..1 | SecurityConstraints | 0..1 | SecurityConstraints | |
| 0..1 | TransmissionCapacity | 0..* | TransmissionCapacity | |
| 0..1 | TranmissionRightEntitlement | 0..* | TransmissionInterfaceRightEntitlement | |
| 1..1 | ConstraintResults | 1..* | ConstraintResults | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 1..1 | FlowgateRelief | 0..* | FlowgateRelief | |
| 0..1 | InterTie | 0..* | SchedulingPoint | |
| 1..1 | RegisteredInterTie | 0..* | RegisteredInterTie | |
| 0..* | To_SubControlArea | 0..1 | SubControlArea | |
| 0..* | From_SubControlArea | 0..1 | SubControlArea | |
| 1..1 | FlowgateValue | 0..* | FlowgateValue | |
| 0..1 | CongestionRevenueRight | 0..1 | CongestionRevenueRight | |
| 0..1 | MktTerminal | 0..* | MktTerminal | |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingUnitDynamic Values | 0..* | GeneratingUnitDynamic Values | |
| 0..* | GenericConstraints | 0..1 | GenericConstraints | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.16 FlowgatePartner

Flowgate defined partner.

Table 533 shows all attributes of FlowgatePartner.

**Table 533 – Attributes of ReferenceData::FlowgatePartner**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 534 shows all association ends of FlowgatePartner with other classes.

**Table 534 – Association ends of ReferenceData::FlowgatePartner with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | FlowgateValue | 0..1 | FlowgateValue | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.17 FlowgateRelief root class

IDC (Interchange Distribution Calulator) sends data for a TLR (Transmission Loading Relief).

Table 535 shows all attributes of FlowgateRelief.

**Table 535 – Attributes of ReferenceData::FlowgateRelief**

| name | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | Date/Time when record becomes effective<br>Used to determine when a record becomes effective. |
| terminateDate | 0..1 | DateTime | Date/Time when record is no longer effective<br>Used to determine when a record is no longer effective |
| idcTargetMktFlow | 0..1 | Integer | Energy Flow level that should be maintained according to the TLR rules as specified by the IDC.<br>For Realtime Markets use in dispatch to control constraints under TLR and calculate unconstrained market flows |

Table 536 shows all association ends of FlowgateRelief with other classes.

**Table 536 – Association ends of ReferenceData::FlowgateRelief with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 1..1 | Flowgate | |

### 6.5.9.18 FlowgateValue root class

Day Ahead,  Network Native Load, Economic Dispatch, values used for calculation of Network Native Load (NNL) Determinator process.

Table 537 shows all attributes of FlowgateValue.

**Table 537 – Attributes of ReferenceData::FlowgateValue**

| name | mult | type | description |
|---|---|---|---|
| economicDispatchLimit | 0..1 | Integer | Limit for Economic Dispatch priority 6 energy flow on the specified flowgate for the specified time period. |
| effectiveDate | 0..1 | DateTime | Date/Time when record becomes effective<br>Used to determine when a record becomes effective |
| firmNetworkLimit | 0..1 | Integer | Limit for firm flow on the specified flowgate for the specified time period. The amount of energy flow over a specifed flowgate due to generation in the market which can be classified as Firm Network priority. |
| flowDirectionFlag | 0..1 | FlowDirectionType | Specifies the direction of energy flow in the flowgate |
| mktFlow | 0..1 | Integer | The amount of energy flow over a specifed flowgate due to generation in the market. |
| netFirmNetworkLimit | 0..1 | Integer | Net Energy flow in flowgate for the associated FlowgatePartner |

Table 538 shows all association ends of FlowgateValue with other classes.

**Table 538 – Association ends of ReferenceData::FlowgateValue with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 1..1 | Flowgate | |
| 0..1 | FlowgatePartner | 0..1 | FlowgatePartner | |

### 6.5.9.19 ForbiddenRegion

Forbbiden region is operating ranges where the units are unable to maintain steady operation without causing equipment damage. The four attributes that define a forbidden region are the low MW, the High MW, the crossing time, and the crossing cost.

Table 539 shows all attributes of ForbiddenRegion.

**Table 539 – Attributes of ReferenceData::ForbiddenRegion**

| name | mult | type | description |
|---|---|---|---|
| crossingCost | 0..1 | Float | Cost associated with crossing the forbidden region |
| crossTime | 0..1 | Integer | Time to cross the forbidden region in minutes. |
| highMW | 0..1 | Float | High end of the region definition |
| lowMW | 0..1 | Float | Low end of the region definition. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 540 shows all association ends of ForbiddenRegion with other classes.

**Table 540 – Association ends of ReferenceData::ForbiddenRegion with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.20 FormerReference

Used to indicate former references to the same piece of equipment. The ID, name, and effectivity dates are utilized.

Table 541 shows all attributes of FormerReference.

**Table 541 – Attributes of ReferenceData::FormerReference**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 542 shows all association ends of FormerReference with other classes.

**Table 542 – Association ends of ReferenceData::FormerReference with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.21 FuelCostCurve

Relationship between unit fuel cost in $/kWh(Y-axis) and unit output in MW (X-axis).

Table 543 shows all attributes of FuelCostCurve.

**Table 543 – Attributes of ReferenceData::FuelCostCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 544 shows all association ends of FuelCostCurve with other classes.

**Table 544 – Association ends of ReferenceData::FuelCostCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.22 FuelRegion

Indication of region for fuel inventory purposes.

Table 545 shows all attributes of FuelRegion.

**Table 545 – Attributes of ReferenceData::FuelRegion**

| name | mult | type | description |
|---|---|---|---|
| fuelRegionType | 0..1 | String | The type of fuel region |
| lastModified | 0..1 | DateTime | Time of last update |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 546 shows all association ends of FuelRegion with other classes.

**Table 546 – Association ends of ReferenceData::FuelRegion with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..* | RegisteredGenerator | |
| 1..1 | GasPrice | 1..1 | GasPrice | |
| 1..1 | OilPrice | 1..1 | OilPrice | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.23 GasPrice root class

Price of gas in monetary units.

Table 547 shows all attributes of GasPrice.

**Table 547 – Attributes of ReferenceData::GasPrice**

| name | mult | type | description |
|---|---|---|---|
| gasPriceIndex | 0..1 | Float | The average natural gas price at a defined fuel region. |

Table 548 shows all association ends of GasPrice with other classes.

**Table 548 – Association ends of ReferenceData::GasPrice with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | FuelRegion | 1..1 | FuelRegion | |

### 6.5.9.24 HostControlArea

A HostControlArea has a set of tie points and a set of generator controls (i.e., AGC). It also has a total load, including transmission and distribution losses.

Table 549 shows all attributes of HostControlArea.

**Table 549 – Attributes of ReferenceData::HostControlArea**

| name | mult | type | description |
|---|---|---|---|
| areaControlMode | 0..1 | AreaControlMode | The area's present control mode: (CF = constant frequency) or (CTL = constant tie-line) or (TLB = tie-line bias) or (OFF = off control) |
| freqSetPoint | 0..1 | Frequency | The present power system frequency set point for automatic generation control |
| frequencyBiasFactor | 0..1 | Float | The control area's frequency bias factor, in MW/0.1 Hz, for automatic generation control (AGC) |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 550 shows all association ends of HostControlArea with other classes.

**Table 550 – Association ends of ReferenceData::HostControlArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 1..1 | SysLoadDistribuFactor | 0..* | SysLoadDistributionFactor | |
| 0..1 | TransferInterface | 0..* | TransferInterface | |
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..1 | Flowgate | 0..* | Flowgate | |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | SubControlAreas | 0..* | SubControlArea | The interchange area may operate as a control area |
| 0..* | RTO | 1..1 | RTO | |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.25 IndividualPnode

Individual pricing node based on Pnode.

Table 551 shows all attributes of IndividualPnode.

**Table 551 – Attributes of ReferenceData::IndividualPnode**

| name | mult | type | description |
|---|---|---|---|
| isPublic | 0..1 | Boolean | inherited from: Pnode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 552 shows all association ends of IndividualPnode with other classes.

**Table 552 – Association ends of ReferenceData::IndividualPnode with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..1 | GenDistributionFactor | 0..1 | GenDistributionFactor | |
| 0..1 | LoadDistributionFactor | 0..1 | LoadDistributionFactor | |
| 1..1 | PnodeDistributionFactor | 0..* | PnodeDistributionFactor | |
| 0..* | CongestionArea | 0..* | CongestionArea | |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | inherited from: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | inherited from: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | inherited from: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | inherited from: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | inherited from: Pnode |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | Trade | 0..* | Trade | inherited from: Pnode |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | inherited from: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | inherited from: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | inherited from: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | inherited from: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | inherited from: Pnode |
| 0..* | RTO | 0..1 | RTO | inherited from: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.26 LoadAggregationPoint

A specialized class of type AggregatedNode type. Defines Load Aggregation Points.

Table 553 shows all attributes of LoadAggregationPoint.

**Table 553 – Attributes of ReferenceData::LoadAggregationPoint**

| name | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | inherited from: AggregateNode |
| qualifASOrder | 0..1 | Integer | inherited from: AggregateNode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 554 shows all association ends of LoadAggregationPoint with other classes.

**Table 554 – Association ends of ReferenceData:: LoadAggregationPoint with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..* | Pnode | inherited from: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | inherited from: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | inherited from: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | inherited from: AggregateNode |
| 0..* | RTO | 1..1 | RTO | inherited from: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.27 LoadRatio root class

Representing the ratio of the load share for the associated SC.

Table 555 shows all attributes of LoadRatio.

**Table 555 – Attributes of ReferenceData::LoadRatio**

| name | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | Interval Start Time |
| intervalEndTime | 0..1 | DateTime | Interval End Time |
| share | 0..1 | PerCent | Share in percentage of total Market load for the selected time interval. |

Table 556 shows all association ends of LoadRatio with other classes.

**Table 556 – Association ends of ReferenceData::LoadRatio with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | SchedulingCoordinator | 0..1 | SchedulingCoordinator | |

### 6.5.9.28 LocalReliabilityArea

Allows definition of reliability areas (e.g.. load pockets) within the ISO/RTO.

Table 557 shows all attributes of LocalReliabilityArea.

**Table 557 – Attributes of ReferenceData::LocalReliabilityArea**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 558 shows all association ends of LocalReliabilityArea with other classes.

**Table 558 – Association ends of ReferenceData::LocalReliabilityArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..* | RegisteredGenerator | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.29 MPMTestCategory

Provides a reference to the Market Power Mitigation test identifiers and methods for the results of the DA or RT markets. Specific data is the test identifier (Price, Conduct, or Impact) and the test method (System MPM, Local MPM, Alternate System MPM, or Alternate Local MPM).

Table 559 shows all attributes of MPMTestCategory.

**Table 559 – Attributes of ReferenceData::MPMTestCategory**

| name | mult | type | description |
|---|---|---|---|
| testIdentifier | 0..1 | MPMTestIdentifierType | 1 – Global Price Test<br>2 – Global Conduct Test<br>3 – Global Impact Test<br>4 – Local Price Test<br>5 – Local Conduct Test<br>6 – Local Impact Test |
| testMethod | 0..1 | MPMTestMethodType | The method of performing the market power monitoring. Examples are Normal (default) thresholds or Alternate thresholds. |
| purposeFlag | 0..1 | PurposeFlagType | Nature of threshold data:<br>'M' – Mitigation threshold<br>'R' – Reporting threshold |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 560 shows all association ends of MPMTestCategory with other classes.

**Table 560 – Association ends of ReferenceData::MPMTestCategory with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 1..1 | MPMTestResults | 0..* | MPMTestResults | |
| 1..1 | MPMTestThreshold | 0..* | MPMTestThreshold | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.30 MPMTestThreshold root class

Market Power Mitigation (MPM) test thresholds for resource as well as designated congestion areas (DCAs).

Table 561 shows all attributes of MPMTestThreshold.

**Table 561 – Attributes of ReferenceData::MPMTestThreshold**

| name | mult | type | description |
|---|---|---|---|
| price | 0..1 | CostPerEnergyUnit | Price Threshold in $/MW |
| percent | 0..1 | PerCent | Price Threshold in % |
| marketType | 0..1 | MarketType | Market Type (DAM, RTM) |

Table 562 shows all association ends of MPMTestThreshold with other classes.

**Table 562 – Association ends of ReferenceData::MPMTestThreshold with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 1..* | AggregatedPnode | 0..* | AggregatedPnode | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |

### 6.5.9.31 MSSAggregation

Metered Sub-System aggregation of MSS Zones.

Table 563 shows all attributes of MSSAggregation.

**Table 563 – Attributes of ReferenceData::MSSAggregation**

| name | mult | type | description |
|---|---|---|---|
| costRecovery | 0..1 | YesNo | Charge for Emission Costs, Start Up Costs, or Minimum Load Costs. |
| grossSettlement | 0..1 | YesNo | MSS Load Following may select Net vs. Gross settlement. Net Settlement requires the net Demand settled at the MSS LAP and Net Supply needs to settle at the equivalent to the weighted average price of the MSS generation. Gross load will be settled at the System LAP and the Gross supply will be settled at the LMP. MSS Aggregation that elects gross settlement shall have to identify if its resources are Load Following or not. |
| ignoreLosses | 0..1 | YesNo | Provides an indication if losses are to be ignored for this zone. Also refered to as Exclude Marginal Losses. |
| ignoreMarginalLosses | 0..1 | YesNo | Provides an indication if marginal losses are to be ignored for this zone. |
| loadFollowing | 0..1 | YesNo | Indication that this particular MSSA participates in the Load Following function. |
| rucProcurement | 0..1 | YesNo | Indicates that RUC will be procured by the ISO or self provided. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 564 shows all association ends of MSSAggregation with other classes.

**Table 564 – Association ends of ReferenceData::MSSAggregation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MeteredSubSystem | 1..* | MeteredSubSystem | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.32 MSSZone

Model to define a zone within a Metered Sub System.

Table 565 shows all attributes of MSSZone.

**Table 565 – Attributes of ReferenceData::MSSZone**

| name | mult | type | description |
|---|---|---|---|
| ignoreLosses | 0..1 | YesNo | Provides an indication if losses are to be ignored for this metered subsystem zone. |
| lossFactor | 0..1 | Float | This is the default loss factor for the Metered Sub-System (MSS) zone. The actual losses are calculated during the RT market. |
| rucGrossSettlement | 0..1 | YesNo | Metered Sub-System (MSS) Load Following may select Net vs. Gross settlement. Net Settlement requires the net Demand settled at the Metered Sub-Sustem (MSS) Load Aggregation Point (LAP) and Net Supply needs to settle at the equivalent to the weighted average price of the MSS generation. Gross load will be settled at the System LAP and the Gross supply will be settled at the LMP. MSS Aggregation that elects gross settlement shall have to identify if its resources are Load Following or not. |
| anodeType | 0..1 | AnodeType | inherited from: AggregateNode |
| qualifASOrder | 0..1 | Integer | inherited from: AggregateNode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 566 shows all association ends of MSSZone with other classes.

**Table 566 – Association ends of ReferenceData::MSSZone with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MeteredSubSystem | 0..1 | MeteredSubSystem | |
| 0..* | Pnode | 0..* | Pnode | inherited from: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | inherited from: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | inherited from: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | inherited from: AggregateNode |
| 0..* | RTO | 1..1 | RTO | inherited from: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.33 MarketPerson

General purpose information for name and other information to contact people.

Table 567 shows all attributes of MarketPerson.

**Table 567 – Attributes of ReferenceData::MarketPerson**

| name | mult | type | description |
|---|---|---|---|
| category | 0..1 | String | Category of this person relative to utility operations, classified according to the utility's corporate standards and practices. Examples include employee, contractor, agent, not affiliated, etc.<br>This field is not used to indicate whether this person is a customer of the utility. Often an employee or contractor is also a customer. Customer information is gained with relationship to Organisation and CustomerData. In similar fashion, this field does not indicate the various roles this person may fill as part of utility operations. |
| electronicAddressAlternate | 0..1 | ElectronicAddress | Alternate Electronic address. |
| electronicAddressPrimary | 0..1 | ElectronicAddress | Primary Electronic address. |
| firstName | 0..1 | String | Person's first name. |
| governmentID | 0..1 | String | Unique identifier for person relative to its governing authority, for example a federal tax identifier (such as a Social Security number in the United States). |
| landlinePhone | 0..1 | TelephoneNumber | Landline phone number. |
| lastName | 0..1 | String | Person's last (family, sir) name. |
| mName | 0..1 | String | Middle name(s) or initial(s). |
| mobilePhone | 0..1 | TelephoneNumber | Mobile phone number. |
| prefix | 0..1 | String | A prefix or title for the person's name, such as Miss, Mister, Doctor, etc. |
| specialNeed | 0..1 | String | Special service needs for the person (contact) are described; examples include life support, etc. |
| status | 0..1 | Status | |
| suffix | 0..1 | String | A suffix for the person's name, such as II, III, etc. |
| userID | 0..1 | String | The user name for the person; required to log in. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 568 shows all association ends of MarketPerson with other classes.

**Table 568 – Association ends of ReferenceData::MarketPerson with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MarketSkills | 0..* | MarketSkill | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.34 MarketQualificationRequirement

Certain skills are required and shall be certified in order for a person (typically a member of a crew) to be qualified to work on types of equipment.

Table 569 shows all attributes of MarketQualificationRequirement.

**Table 569 – Attributes of ReferenceData::MarketQualificationRequirement**

| name | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | Effective date of the privilege, terminate date of the privilege, or effective date of the application for the organization |
| expirationDate | 0..1 | DateTime | This is the terminate date of the application for the organization<br>The specific organization can no longer access the application as of the terminate date |
| qualificationID | 0..1 | String | Qualification identifier. |
| status | 0..1 | Integer | The status of the privilege. Shows the status of the user´s qualification.<br>The current statuses are: 1=New, 2=Active, 3=Refused, 4=Terminated, 5=Withdrawn<br>and it is subject to update. |
| statusType | 0..1 | String | This is the name of the status of the qualification and is used to display the status of the user's or organization's status. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 570 shows all association ends of MarketQualificationRequirement with other classes.

**Table 570 – Association ends of ReferenceData:: MarketQualificationRequirement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketSkills | 0..* | MarketSkill | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.35 MarketRegion

A specialized class of AggregatedNode type. Defines the MarketRegions. Regions could be system Market Regions, Energy Regions or Ancillary Service Regions.

Table 571 shows all attributes of MarketRegion.

**Table 571 – Attributes of ReferenceData::MarketRegion**

| name | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | inherited from: AggregateNode |
| qualifASOrder | 0..1 | Integer | inherited from: AggregateNode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 572 shows all association ends of MarketRegion with other classes.

**Table 572 – Association ends of ReferenceData::MarketRegion with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ReserveDemandCurve | 0..* | ReserveDemandCurve | |
| 1..1 | MarketRegionResults | 1..* | MarketRegionResults | |
| 1..1 | ExPostMarketRegionRes ults | 0..* | ExPostMarketRegionRes ults | |
| 0..* | Pnode | 0..* | Pnode | inherited from: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | inherited from: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | inherited from: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | inherited from: AggregateNode |
| 0..* | RTO | 1..1 | RTO | inherited from: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.36 MarketSkill

Proficiency level of a craft, which is required to operate or maintain a particular type of asset and/or perform certain types of work.

Table 573 shows all attributes of MarketSkill.

**Table 573 – Attributes of ReferenceData::MarketSkill**

| name | mult | type | description |
|---|---|---|---|
| certificationPeriod | 0..1 | DateTimeInterval | Interval between the certification and its expiry. |
| effectiveDateTime | 0..1 | DateTime | Date and time the skill became effective. |
| level | 0..1 | String | Level of skill for a Craft. |
| type | 0..1 | String | inherited from: Document |
| authorName | 0..1 | String | inherited from: Document |
| createdDateTime | 0..1 | DateTime | inherited from: Document |
| lastModifiedDateTime | 0..1 | DateTime | inherited from: Document |
| revisionNumber | 0..1 | String | inherited from: Document |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Document |
| subject | 0..1 | String | inherited from: Document |
| title | 0..1 | String | inherited from: Document |
| docStatus | 0..1 | Status | inherited from: Document |
| status | 0..1 | Status | inherited from: Document |
| comment | 0..1 | String | inherited from: Document |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 574 shows all association ends of MarketSkill with other classes.

**Table 574 – Association ends of ReferenceData::MarketSkill with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketQualificationRequirements | 0..* | MarketQualificationRequirement | |
| 0..* | MarketPerson | 0..1 | MarketPerson | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: Document |
| 0..* | Editor | 0..1 | Editor | inherited from: Document |
| 0..* | Approver | 0..1 | Approver | inherited from: Document |
| 0..* | Author | 0..1 | Author | inherited from: Document |
| 0..* | Issuer | 0..1 | Issuer | inherited from: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.37 MaxStartUpCostCurve

The maximum Startup costs and time as a function of down time. Relationship between unit startup cost (Y1-axis) vs. unit elapsed down time (X-axis). This is used to validate the information provided in the Bid.

Table 575 shows all attributes of MaxStartUpCostCurve.

**Table 575 – Attributes of ReferenceData::MaxStartUpCostCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 576 shows all association ends of MaxStartUpCostCurve with other classes.

**Table 576 – Association ends of ReferenceData:: MaxStartUpCostCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.38 MeteredSubSystem

A metered subsystem.

Table 577 shows all attributes of MeteredSubSystem.

**Table 577 – Attributes of ReferenceData::MeteredSubSystem**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 578 shows all association ends of MeteredSubSystem with other classes.

**Table 578 – Association ends of ReferenceData::MeteredSubSystem with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | MSSZone | 0..* | MSSZone | |
| 1..* | MSSAggregation | 0..1 | MSSAggregation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.39 MktCombinedCyclePlant

Subclass of Production: CombinedCyclePlant from IEC 61970 package.

A set of combustion turbines and steam turbines where the exhaust heat from the combustion turbines is recovered to make steam for the steam turbines, resulting in greater overall plant efficiency.

Table 579 shows all attributes of MktCombinedCyclePlant.

**Table 579 – Attributes of ReferenceData::MktCombinedCyclePlant**

| name | mult | type | description |
|---|---|---|---|
| combCyclePlantRating | 0..1 | ActivePower | inherited from: CombinedCyclePlant |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 580 shows all association ends of MktCombinedCyclePlant with other classes.

**Table 580 – Association ends of ReferenceData:: MktCombinedCyclePlant with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AggregatedPnode | 0..1 | AggregatedPnode | |
| 0..1 | CombinedCycleLogicalConfiguration | 1..* | CombinedCycleLogicalConfiguration | |
| 0..1 | ThermalGeneratingUnits | 0..* | ThermalGeneratingUnit | inherited from: CombinedCyclePlant |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.40 MktConductingEquipment

Subclass of IEC 61970:Core:ConductingEquipment.

Table 581 shows all attributes of MktConductingEquipment.

**Table 581 – Attributes of ReferenceData::MktConductingEquipment**

| name | mult | type | description |
|---|---|---|---|
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 582 shows all association ends of MktConductingEquipment with other classes.

**Table 582 – Association ends of ReferenceData:: MktConductingEquipment with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BaseVoltage | 0..1 | BaseVoltage | inherited from: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | inherited from: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | inherited from: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | inherited from: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.41 MktContingency

Subclass of IEC 61970:Contingency.

Table 583 shows all attributes of MktContingency.

**Table 583 – Attributes of ReferenceData::MktContingency**

| name | mult | type | description |
|---|---|---|---|
| loadRolloverFlag | 0..1 | Boolean | load change flag<br>Flag that indicates whether load rollover and load pickup should be processed for this contingency |
| ltcControlFlag | 0..1 | Boolean | ltc enable flag<br>Flag that indicates if LTCs regulate voltage during the solution of the contingency |
| participationFactorSet | 0..1 | String | Participation Factor flag<br>An indication which set of generator participation factors should be used to re-allocate generation in this contingency |
| screeningFlag | 0..1 | Boolean | sceening flag for outage<br>Flag that indicated whether screening is bypassed for the contingency |
| mustStudy | 0..1 | Boolean | inherited from: Contingency |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |

| name | mult | type | description |
|---|---|---|---|
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 584 shows all association ends of MktContingency with other classes.

**Table 584 – Association ends of ReferenceData::MktContingency with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ContingencyConstraintLimit | 0..* | ContingencyConstraintLimit | |
| 0..1 | TransferInterfaceSolutionA | 0..1 | TransferInterfaceSolution | |
| 0..1 | TransferInterfaceSolutionB | 0..1 | TransferInterfaceSolution | |
| 1..1 | ConstraintResults | 0..* | ConstraintResults | |
| 1..1 | ContingencyElement | 0..* | ContingencyElement | inherited from: Contingency |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.42 MktHeatRateCurve

Subclass of IEC 61970: Generation: Production:HeatRateCurve.

Table 585 shows all attributes of MktHeatRateCurve.

**Table 585 – Attributes of ReferenceData::MktHeatRateCurve**

| name | mult | type | description |
|---|---|---|---|
| isNetGrossP | 0..1 | Boolean | inherited from: HeatRateCurve |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 586 shows all association ends of MktHeatRateCurve with other classes.

**Table 586 – Association ends of ReferenceData::MktHeatRateCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 0..1 | ThermalGeneratingUnit | 1..1 | ThermalGeneratingUnit | inherited from: HeatRateCurve |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.43 MktThermalGeneratingUnit

Subclass of ThermalGeneratingUnit from Production Package in IEC 61970.

Table 587 shows all attributes of MktThermalGeneratingUnit.

**Table 587 – Attributes of ReferenceData::MktThermalGeneratingUnit**

| name | mult | type | description |
|---|---|---|---|
| oMCost | 0..1 | CostPerHeatUnit | inherited from: ThermalGeneratingUnit |
| allocSpinResP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| autoCntrlMarginP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| baseP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| controlDeadband | 0..1 | ActivePower | inherited from: GeneratingUnit |
| controlPulseHigh | 0..1 | Seconds | inherited from: GeneratingUnit |
| controlPulseLow | 0..1 | Seconds | inherited from: GeneratingUnit |
| controlResponseRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| efficiency | 0..1 | PerCent | inherited from: GeneratingUnit |
| genControlMode | 0..1 | GeneratorControlMode | inherited from: GeneratingUnit |
| genControlSource | 0..1 | GeneratorControlSource | inherited from: GeneratingUnit |
| governorMPL | 0..1 | PU | inherited from: GeneratingUnit |
| governorSCD | 0..1 | PerCent | inherited from: GeneratingUnit |
| highControlLimit | 0..1 | ActivePower | inherited from: GeneratingUnit |
| initialP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| longPF | 0..1 | Float | inherited from: GeneratingUnit |
| lowControlLimit | 0..1 | ActivePower | inherited from: GeneratingUnit |
| lowerRampRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| maxEconomicP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| maximumAllowableSpinn ingReserve | 0..1 | ActivePower | inherited from: GeneratingUnit |
| maxOperatingP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| minEconomicP | 0..1 | ActivePower | inherited from: GeneratingUnit |

| name | mult | type | description |
|---|---|---|---|
| minimumOffTime | 0..1 | Seconds | inherited from: GeneratingUnit |
| minOperatingP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| modelDetail | 0..1 | Classification | inherited from: GeneratingUnit |
| nominalP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| normalPF | 0..1 | Float | inherited from: GeneratingUnit |
| penaltyFactor | 0..1 | Float | inherited from: GeneratingUnit |
| raiseRampRate | 0..1 | ActivePowerChangeRate | inherited from: GeneratingUnit |
| ratedGrossMaxP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| ratedGrossMinP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| ratedNetMaxP | 0..1 | ActivePower | inherited from: GeneratingUnit |
| shortPF | 0..1 | Float | inherited from: GeneratingUnit |
| startupCost | 0..1 | Money | inherited from: GeneratingUnit |
| startupTime | 0..1 | Seconds | inherited from: GeneratingUnit |
| tieLinePF | 0..1 | Float | inherited from: GeneratingUnit |
| totalEfficiency | 0..1 | PerCent | inherited from: GeneratingUnit |
| variableCost | 0..1 | Money | inherited from: GeneratingUnit |
| aggregate | 0..1 | Boolean | inherited from: Equipment |
| inService | 0..1 | Boolean | inherited from: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | inherited from: Equipment |
| normallyInService | 0..1 | Boolean | inherited from: Equipment |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 588 shows all association ends of MktThermalGeneratingUnit with other classes.

**Table 588 – Association ends of ReferenceData:: MktThermalGeneratingUnit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | CombinedCycleConfigurationMember | 0..* | CombinedCycleConfigurationMember | |
| 0..1 | CAESPlant | 0..1 | CAESPlant | inherited from: ThermalGeneratingUnit |
| 0..* | CogenerationPlant | 0..1 | CogenerationPlant | inherited from: ThermalGeneratingUnit |
| 0..* | CombinedCyclePlant | 0..1 | CombinedCyclePlant | inherited from: ThermalGeneratingUnit |
| 1..1 | EmmissionAccounts | 0..* | EmissionAccount | inherited from: ThermalGeneratingUnit |
| 1..1 | EmissionCurves | 0..* | EmissionCurve | inherited from: ThermalGeneratingUnit |
| 1..1 | FossilFuels | 0..* | FossilFuel | inherited from: ThermalGeneratingUnit |
| 1..1 | FuelAllocationSchedules | 0..* | FuelAllocationSchedule | inherited from: ThermalGeneratingUnit |
| 1..1 | HeatInputCurve | 0..1 | HeatInputCurve | inherited from: ThermalGeneratingUnit |
| 1..1 | HeatRateCurve | 0..1 | HeatRateCurve | inherited from: ThermalGeneratingUnit |
| 1..1 | IncrementalHeatRateCurve | 0..1 | IncrementalHeatRateCurve | inherited from: ThermalGeneratingUnit |
| 1..1 | ShutdownCurve | 0..1 | ShutdownCurve | inherited from: ThermalGeneratingUnit |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | StartupModel | 0..1 | StartupModel | inherited from: ThermalGeneratingUnit |
| 0..1 | RotatingMachine | 0..* | RotatingMachine | inherited from: GeneratingUnit |
| 1..1 | GenUnitOpCostCurves | 0..* | GenUnitOpCostCurve | inherited from: GeneratingUnit |
| 1..1 | GenUnitOpSchedule | 0..1 | GenUnitOpSchedule | inherited from: GeneratingUnit |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | inherited from: GeneratingUnit |
| 1..1 | GrossToNetActivePowerCurves | 0..* | GrossToNetActivePowerCurve | inherited from: GeneratingUnit |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | inherited from: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | inherited from: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | inherited from: Equipment |
| 0..1 | Faults | 0..* | Fault | inherited from: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..* | EquipmentContainer | inherited from: Equipment |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.44 OilPrice root class

Price of oil in monetary units.

Table 589 shows all attributes of OilPrice.

**Table 589 – Attributes of ReferenceData::OilPrice**

| name | mult | type | description |
|---|---|---|---|
| oilPriceIndex | 0..1 | Float | The average oil price at a defined fuel region. |

Table 590 shows all association ends of OilPrice with other classes.

**Table 590 – Association ends of ReferenceData::OilPrice with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | FuelRegion | 1..1 | FuelRegion | |

#### 6.5.9.45 OrgPnodeAllocation

This class models the allocation between asset owners and pricing nodes.

Table 591 shows all attributes of OrgPnodeAllocation.

**Table 591 – Attributes of ReferenceData::OrgPnodeAllocation**

| name | mult | type | description |
|---|---|---|---|
| maxMWAllocation | 0..1 | ActivePower | Maximum MW for the Source/Sink for the Allocation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 592 shows all association ends of OrgPnodeAllocation with other classes.

**Table 592 – Association ends of ReferenceData::OrgPnodeAllocation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 1..1 | Pnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.46 OrgResOwnership

This class model the ownership percent and type of ownership between resource and organisation.

Table 593 shows all attributes of OrgResOwnership.

**Table 593 – Attributes of ReferenceData::OrgResOwnership**

| name | mult | type | description |
|---|---|---|---|
| asscType | 0..1 | ResourceAssnType | association type for the association between Organisation and Resource: |
| masterSchedulingCoordinatorFlag | 0..1 | YesNo | Flag to indicate that the SC representing the Resource is the Master SC. |
| ownershipPercent | 0..1 | PerCent | ownership percentage for each resource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 594 shows all association ends of OrgResOwnership with other classes.

**Table 594 – Association ends of ReferenceData::OrgResOwnership with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.47 Pnode

A pricing node is directly associated with a connectivity node. It is a pricing location for which market participants submit their bids, offers, buy/sell CRRs, and settle.

Table 595 shows all attributes of Pnode.

**Table 595 – Attributes of ReferenceData::Pnode**

| name | mult | type | description |
|---|---|---|---|
| isPublic | 0..1 | Boolean | If true, this Pnode is public (prices are published for DA/RT and FTR markets), otherwise it is private (location is not usable by market for bidding/FTRs/transactions). |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 596 shows all association ends of Pnode with other classes.

**Table 596 – Association ends of ReferenceData::Pnode with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResources | 0..* | RegisteredResource | A registered resource injects power at one or more connectivity nodes related to a pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | Allows Measurements to be associated to Pnodes. |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | |
| 0..1 | PnodeResults | 1..* | PnodeResults | |
| 0..1 | Trade | 0..* | Trade | |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 0..* | AggregateNode | 0..* | AggregateNode | |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | |
| 0..* | RTO | 0..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.48 PnodeDistributionFactor root class

This class allows SC to input different distribution factors for pricing node.

Table 597 shows all attributes of PnodeDistributionFactor.

**Table 597 – Attributes of ReferenceData::PnodeDistributionFactor**

| name | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Used to calculate "participation" of Pnode in an AggregatePnode. For example, for regulation region this factor is 1 and total sum of all factors for a specific regulation region does not have to be 1. For pricing zone the total sum of all factors has to be 1. |
| offPeak | 0..1 | YesNo | Indication that this distribution factor is to apply during off peak. |
| onPeak | 0..1 | YesNo | Indication that this factor is to apply during Peak periods. |
| podLossFactor | 0..1 | Float | Point of delivery loss factor |

Table 598 shows all association ends of PnodeDistributionFactor with other classes.

**Table 598 – Association ends of ReferenceData:: PnodeDistributionFactor with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | BidDistributionFactor | 0..1 | BidDistributionFactor | |
| 1..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 0..* | IndividualPnode | 1..1 | IndividualPnode | |

### 6.5.9.49 RMRHeatRateCurve

Model to support processing of reliability must run units.

Table 599 shows all attributes of RMRHeatRateCurve.

**Table 599 – Attributes of ReferenceData::RMRHeatRateCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 600 shows all association ends of RMRHeatRateCurve with other classes.

**Table 600 – Association ends of ReferenceData::RMRHeatRateCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.50 RMRStartUpCostCurve

Model to support processing of reliability must run units.

Table 601 shows all attributes of RMRStartUpCostCurve.

**Table 601 – Attributes of ReferenceData::RMRStartUpCostCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |

| name | mult | type | description |
|---|---|---|---|
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 602 shows all association ends of RMRStartUpCostCurve with other classes.

**Table 602 – Association ends of ReferenceData:: RMRStartUpCostCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.51 RMRStartUpEnergyCurve

Model to support processing of reliability must run units.

Table 603 shows all attributes of RMRStartUpEnergyCurve.

**Table 603 – Attributes of ReferenceData::RMRStartUpEnergyCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 604 shows all association ends of RMRStartUpEnergyCurve with other classes.

**Table 604 – Association ends of ReferenceData:: RMRStartUpEnergyCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.52 RMRStartUpFuelCurve

Model to support processing of reliability must run units.

Table 605 shows all attributes of RMRStartUpFuelCurve.

**Table 605 – Attributes of ReferenceData::RMRStartUpFuelCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 606 shows all association ends of RMRStartUpFuelCurve with other classes.

**Table 606 – Association ends of ReferenceData:: RMRStartUpFuelCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.53 RMRStartUpTimeCurve

Model to support processing of reliability must run units.

Table 607 shows all attributes of RMRStartUpTimeCurve.

**Table 607 – Attributes of ReferenceData::RMRStartUpTimeCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 608 shows all association ends of RMRStartUpTimeCurve with other classes.

**Table 608 – Association ends of ReferenceData:: RMRStartUpTimeCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.54 RTO

Regional transmission operator.

Table 609 shows all attributes of RTO.

**Table 609 – Attributes of ReferenceData::RTO**

| name | mult | type | description |
|---|---|---|---|
| streetAddress | 0..1 | StreetAddress | inherited from: Organisation |
| postalAddress | 0..1 | StreetAddress | inherited from: Organisation |
| phone1 | 0..1 | TelephoneNumber | inherited from: Organisation |
| phone2 | 0..1 | TelephoneNumber | inherited from: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Organisation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 610 shows all association ends of RTO with other classes.

**Table 610 – Association ends of ReferenceData::RTO with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | MktConnectivityNode | 0..* | MktConnectivityNode | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 1..1 | AdjacentCASet | 0..* | AdjacentCASet | |
| 1..1 | AggregateNode | 0..* | AggregateNode | |
| 1..1 | TransmissionContractRight | 0..* | ContractRight | |
| 1..1 | FuelRegion | 0..* | FuelRegion | |
| 1..1 | HostControlArea | 0..* | HostControlArea | |
| 1..1 | LocalReliabilityArea | 0..* | LocalReliabilityArea | |
| 0..1 | Pnodes | 0..* | Pnode | |
| 1..1 | TransmissionRightChain | 0..* | TransmissionRightChain | |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | SubControlArea | 0..* | SubControlArea | |
| 0..1 | EnergyMarkets | 0..* | EnergyMarket | |
| 0..1 | SecurityConstraintsLinear | 0..* | SecurityConstraintSum | |
| 0..1 | SecurityConstraints | 0..* | SecurityConstraints | |
| 1..1 | MSSAggregation | 0..* | MSSAggregation | |
| 0..* | MarketPerson | 0..* | MarketPerson | inherited from: MarketParticipant |
| 0..* | MarketDocument | 0..* | MarketDocument | inherited from: MarketParticipant |
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | inherited from: MarketParticipant |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: MarketParticipant |
| 0..1 | Bid | 0..* | Bid | inherited from: MarketParticipant |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: MarketParticipant |
| 0..* | MarketRole | 0..* | MarketRole | inherited from: MarketParticipant |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | inherited from: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | inherited from: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.55 RUCZone

A specialized class of type AggregatedNode type. Defines RUC Zones. A forecast region represents a collection of Nodes for which the Market operator has developed sufficient historical demand and relevant weather data to perform a demand forecast for such area. The Market Operator may further adjust this forecast to ensure that the Reliability Unit Commitment produces adequate local capacity procurement.

Table 611 shows all attributes of RUCZone.

**Table 611 – Attributes of ReferenceData::RUCZone**

| name | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | inherited from: AggregateNode |
| qualifASOrder | 0..1 | Integer | inherited from: AggregateNode |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 612 shows all association ends of RUCZone with other classes.

**Table 612 – Association ends of ReferenceData::RUCZone with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..* | Pnode | 0..* | Pnode | inherited from: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | inherited from: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | inherited from: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | inherited from: AggregateNode |
| 0..* | RTO | 1..1 | RTO | inherited from: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.56 RegisteredDistributedResource

A registered resource that represents a distributed energy resource, such as a micro-generator, fuel cell, photo-voltaic energy source, etc.

Table 613 shows all attributes of RegisteredDistributedResource.

**Table 613 – Attributes of ReferenceData::RegisteredDistributedResource**

| name | mult | type | description |
|---|---|---|---|
| distributedResourceType | 0..1 | String | The type of resource. Examples include: fuel cell, flywheel, photovoltaic, micro-turbine, CHP (combined heat power), V2G (vehicle to grid), DES (distributed energy storage), and others. |
| ACAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | inherited from: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| dispatchable | 0..1 | Boolean | inherited from: RegisteredResource |
| ECAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | inherited from: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | inherited from: RegisteredResource |
| lastModified | 0..1 | DateTime | inherited from: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | inherited from: RegisteredResource |
| maxOnTime | 0..1 | Float | inherited from: RegisteredResource |
| minDispatchTime | 0..1 | Hours | inherited from: RegisteredResource |
| minOffTime | 0..1 | Float | inherited from: RegisteredResource |
| minOnTime | 0..1 | Float | inherited from: RegisteredResource |

| name | mult | type | description |
|---|---|---|---|
| mustOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| nonMarket | 0..1 | YesNo | inherited from: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | inherited from: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationSt atus | inherited from: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 614 shows all association ends of RegisteredDistributedResource with other classes.

**Table 614 – Association ends of ReferenceData:: RegisteredDistributedResource with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ResourcePerformanceR atings | 0..* | ResourcePerformanceR ating | |
| 1..1 | ResponseMethods | 0..* | ResponseMethod | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | inherited from: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | inherited from: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | inherited from: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | inherited from: RegisteredResource |
| 0..1 | ContractDistributionFact or | 0..* | ContractDistributionFact or | inherited from: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | inherited from: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | inherited from: RegisteredResource |
| 0..1 | ResourceDispatchResult s | 0..* | ResourceDispatchResult s | inherited from: RegisteredResource |
| 0..1 | SubstitutionResourceLis t | 0..* | SubstitutionResourceLis t | inherited from: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | inherited from: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: RegisteredResource |
| 1..1 | IntermittentResourceElig ibility | 0..* | IntermittentResourceElig ibility | inherited from: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | inherited from: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | inherited from: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | inherited from: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | inherited from: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | inherited from: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | inherited from: RegisteredResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | inherited from: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | inherited from: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | inherited from: RegisteredResource |
| 0..* | Reason | 0..* | Reason | inherited from: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | inherited from: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | inherited from: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | inherited from: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | inherited from: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | inherited from: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | inherited from: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | inherited from: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | inherited from: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | inherited from: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | inherited from: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | inherited from: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | inherited from: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | inherited from: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | inherited from: RegisteredResource |
| 0..* | Domain | 0..* | Domain | inherited from: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | inherited from: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | inherited from: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.57 RegisteredGenerator

Model of a generator that is registered to participate in the market.

Table 615 shows all attributes of RegisteredGenerator.

**Table 615 – Attributes of ReferenceData::RegisteredGenerator**

| name | mult | type | description |
|---|---|---|---|
| capacityFactor | 0..1 | PerCent | The ratio of actual energy produced by resource divided by the maximum potential energy if the resource is fully utilized. As an example, wind farms. |
| coldStartTime | 0..1 | Float | Cold start time. |
| combinedCycleOperatingMode | 0..1 | String | Combined Cycle operating mode. |
| commericialOperationDate | 0..1 | DateTime | |
| constrainedOutputFlag | 0..1 | YesNo | Constrained Output Generator (COG) Indicator (Yes/No), per Generating Resource |
| energyDownRampRate | 0..1 | ActivePowerChangeRate | Response rate in MW per minute for ramping energy down. |
| energyUpRampRate | 0..1 | ActivePowerChangeRate | Response rate in MW per minute for ramping energy up. |
| extremeLongStart | 0..1 | YesNo | Some long-start up time units may need to receive start up instruction before DA market results are available. Long-Start resources may be either physical resources within the control with start-up times greater than 18 hours or the long-start contractual inter-tie commitment that shall be completed by 6 am one-day ahead. Therefore, there is a need for a process to determine the commitment of such resources before the DA market. |
| fuelSource | 0..1 | FuelSource | Values: Natural Gas Based Resource, Non Natural Gas Based Resource<br>"NG" – Natural-Gas-Based Resource – a Resource that is powered by Natural Gas<br>"NNG" – Non-Natural-Gas-Based Resource – a Resouce that is powered by some other fuel than Natural Gas |
| highControlLimit | 0..1 | ActivePower | High limit for secondary (AGC) control |
| hotIntTime | 0..1 | Float | Hot-to-intermediate time (Seasonal) |
| hotStartTime | 0..1 | Float | Hot start time. |
| intColdTime | 0..1 | Float | Intermediate-to-cold time (Seasonal) |
| intStartTime | 0..1 | Float | Intermediate start time. |
| loadFollowingDownMSS | 0..1 | YesNo | Certifies resources for use in MSS Load Following Down |
| loadFollowingUpMSS | 0..1 | YesNo | Certifies resources for use in MSS Load Following Up |
| lowControlLImit | 0..1 | ActivePower | Low limit for secondary (AGC) control |
| maxDependableCap | 0..1 | ActivePower | Maximum Dependable Capacity (MNDC). Maximun Net Dependable Capacity is used in association with an RMR contract. |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | Maximum allowable spinning reserve. Spinning reserve will never be considered greater than this value regardless of the current operating point. |
| maximumOperatingLimit | 0..1 | ActivePower | This is the maximum operating MW limit the dispatcher can enter for this unit |
| maxMinLoadCost | 0..1 | CostRate | The registered maximum Minimum Load Cost of a Generating Resource registered with a Cost Basis of "Bid Cost". |
| maxPumpingLevel | 0..1 | ActivePower | max pumping level of a hydro pump unit |
| maxShutdownTime | 0..1 | DateTime | Maximum time this device can be shut down. |
| maxStartUpsPerDay | 0..1 | Integer | maximum start ups per day |
| maxWeeklyEnergy | 0..1 | Float | Maximum weekly Energy (Seasonal) |
| maxWeeklyStarts | 0..1 | Integer | Maximum weekly starts (seasonal parameter) |

| name | mult | type | description |
|---|---|---|---|
| minimumLoadCostBasis | 0..1 | CostBasis | The cost basis for minimum load. |
| minimumLoadFuelCost | 0..1 | CostPerHeatUnit | The cost for the fuel required to get a Generating Resource to operate at the minimum load level |
| minimumOperatingLimit | 0..1 | ActivePower | This is the minimum operating MW limit the dispatcher can enter for this unit. |
| minLoadCost | 0..1 | CostRate | minimum load cost. Value is (currency/hr) |
| mustOfferRA | 0..1 | YesNo | Flag to indicate that this unit is a resource adequacy resource and must offer. |
| nameplateCapacity | 0..1 | ActivePower | MW value stated on the nameplate of the Generator -- the value it potentially could provide. |
| operatingMaintenanceCost | 0..1 | CostPerEnergyUnit | The portion of the Operating Cost of a Generating Resource that is not related to fuel cost. |
| pumpingCost | 0..1 | CostRate | |
| pumpingFactor | 0..1 | Float | Pumping factor for pump storage units, conversion factor between generating and pumping. |
| pumpMinDownTime | 0..1 | Float | The minimum down time for the pump in a pump storage unit. |
| pumpMinUpTime | 0..1 | Float | The minimum up time aspect for the pump in a pump storage unit |
| pumpShutdownCost | 0..1 | Float | The cost to shutdown a pump during the pump aspect of a pump storage unit. |
| pumpShutdownTime | 0..1 | Integer | The shutdown time (minutes) of the pump aspect of a pump storage unit. |
| quickStartFlag | 0..1 | YesNo | Quick start flag (Yes/No). Identifies the registered generator as a quick start unit. A quick start unit is a unit that has the ability to be available for load within a 30 minute period. |
| rampCurveType | 0..1 | RampCurveType | Ramp curve type. Identifies the type of curve which may be a fixed, static or dynamic. |
| regulationDownRampRate | 0..1 | ActivePowerChangeRate | Regulation down response rate in MW per minute |
| regulationFlag | 0..1 | UnitRegulationKind | Specifies if the unit is regulating or not regulating or expected to be regulating but is not. |
| regulationUpRampRate | 0..1 | ActivePowerChangeRate | Regulation up response rate in MW per minute. |
| resourceSubType | 0..1 | String | Unit sub type used by Settlements or scheduling application. Application use of the unit sub type may define the necessary types as applicable. |
| riverSystem | 0..1 | String | River System the Resource is tied to. |
| RMNRFlag | 0..1 | YesNo | Reliability must not run (RMNR) flag: indicated whether the RMR unit is set as an RMNR in the current market |
| RMRFlag | 0..1 | FlagTypeRMR | Reliability must run (RMR) flag: indicates whether the unit is RMR; Indicates whether the unit is RMR:<br>N' – not an RMR unit<br>'1' – RMR Condition 1 unit<br>'2' – RMR Condition 2 unit |
| RMRManualIndicator | 0..1 | YesNo | Indicates the RMR Manual pre-determination status [Y/N] |
| RMTFlag | 0..1 | YesNo | Reliability must take (RMT) flag (Yes/No): indicates whether the unit is RMT |
| spinRampRate | 0..1 | ActivePowerChangeRate | Response rate in MW per minute for spinning reserve. |
| startUpCostBasis | 0..1 | CostBasis | The cost basis for start up. |
| syncCondCapable | 0..1 | YesNo | Is the Resource Synchronous Condenser capable Resource? |
| unitType | 0..1 | UnitType | Generating unit type: Combined Cycle, Gas |

| name | mult | type | description |
|---|---|---|---|
| | | | Turbine, Hydro Turbine, Other, Photovoltaic, Hydro Pump-Turbine, Reciprocating Engine, Steam Turbine, Synchronous Condenser, Wind Turbine |
| useLimitFlag | 0..1 | YesNo | Use limit flag: indicates if the use-limited resource is fully scheduled (or has some slack for real-time dispatch) (Y/N) |
| variableEnergyResource | 0..1 | YesNo | Provides an indication that this resource is intending to participate in an intermittent resource program. |
| ACAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | inherited from: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| dispatchable | 0..1 | Boolean | inherited from: RegisteredResource |
| ECAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | inherited from: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | inherited from: RegisteredResource |
| lastModified | 0..1 | DateTime | inherited from: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | inherited from: RegisteredResource |
| maxOnTime | 0..1 | Float | inherited from: RegisteredResource |
| minDispatchTime | 0..1 | Hours | inherited from: RegisteredResource |
| minOffTime | 0..1 | Float | inherited from: RegisteredResource |
| minOnTime | 0..1 | Float | inherited from: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| nonMarket | 0..1 | YesNo | inherited from: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | inherited from: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | inherited from: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 616 shows all association ends of RegisteredGenerator with other classes.

**Table 616 – Association ends of ReferenceData:: RegisteredGenerator with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 0..* | GeneratingBid | |
| 0..1 | StartUpTimeCurve | 0..1 | StartUpTimeCurve | |
| 0..1 | Trade | 0..* | Trade | |
| 0..1 | MktHeatRateCurve | 0..1 | MktHeatRateCurve | |
| 0..1 | RMRHeatRateCurve | 0..1 | RMRHeatRateCurve | |
| 0..1 | RMRStartUpCostCurve | 0..1 | RMRStartUpCostCurve | |
| 0..1 | RMRStartUpEnergyCurve | 0..1 | RMRStartUpEnergyCurve | |
| 0..1 | RMRStartUpFuelCurve | 0..1 | RMRStartUpFuelCurve | |
| 0..1 | RMRStartUpTimeCurve | 0..1 | RMRStartUpTimeCurve | |
| 0..1 | RegulatingLimit | 0..1 | RegulatingLimit | |
| 0..1 | StartUpFuelCurve | 0..1 | StartUpFuelCurve | |
| 0..1 | StartUpEnergyCurve | 0..1 | StartUpEnergyCurve | |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | |
| 1..1 | EnergyPriceIndex | 1..1 | EnergyPriceIndex | |
| 0..1 | UnitInitialConditions | 0..* | UnitInitialConditions | |
| 0..* | StartUpCostCurves | 0..* | StartUpCostCurve | |
| 0..1 | FuelCostCurve | 0..1 | FuelCostCurve | |
| 0..* | FuelRegion | 0..1 | FuelRegion | |
| 0..* | LocalReliabilityArea | 0..1 | LocalReliabilityArea | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | inherited from: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | inherited from: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | inherited from: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | inherited from: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | inherited from: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | inherited from: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | inherited from: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | inherited from: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | inherited from: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | inherited from: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | inherited from: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | inherited from: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | inherited from: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | inherited from: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | inherited from: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | inherited from: RegisteredResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | inherited from: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | inherited from: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | inherited from: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | inherited from: RegisteredResource |
| 0..* | Reason | 0..* | Reason | inherited from: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | inherited from: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | inherited from: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | inherited from: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | inherited from: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | inherited from: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | inherited from: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | inherited from: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | inherited from: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | inherited from: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | inherited from: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | inherited from: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | inherited from: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | inherited from: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | inherited from: RegisteredResource |
| 0..* | Domain | 0..* | Domain | inherited from: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | inherited from: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | inherited from: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.58 RegisteredInterTie

This class represents the inter tie resource.

Table 617 shows all attributes of RegisteredInterTie.

**Table 617 – Attributes of ReferenceData::RegisteredInterTie**

| name | mult | type | description |
|---|---|---|---|
| direction | 0..1 | InterTieDirection | Indicates the direction (export/import) of an InterTie resource. |
| energyProductType | 0..1 | EnergyProductType | Under each major product type, the commodity type can be applied to further specify the type. |
| isDCTie | 0..1 | YesNo | Flag to indicated whether this Inter-tie is a DC Tie. |
| isDynamicInterchange | 0..1 | YesNo | Specifies whether the inter-tie resource is registered for the dynamic interchange. |
| minHourlyBlockLimit | 0..1 | Integer | The registered upper bound of minimum hourly block for an Inter-Tie Resource. |
| ACAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | inherited from: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| dispatchable | 0..1 | Boolean | inherited from: RegisteredResource |
| ECAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | inherited from: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | inherited from: RegisteredResource |
| lastModified | 0..1 | DateTime | inherited from: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | inherited from: RegisteredResource |
| maxOnTime | 0..1 | Float | inherited from: RegisteredResource |
| minDispatchTime | 0..1 | Hours | inherited from: RegisteredResource |
| minOffTime | 0..1 | Float | inherited from: RegisteredResource |
| minOnTime | 0..1 | Float | inherited from: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| nonMarket | 0..1 | YesNo | inherited from: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | inherited from: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | inherited from: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 618 shows all association ends of RegisteredInterTie with other classes.

**Table 618 – Association ends of ReferenceData::RegisteredInterTie with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | InterTieDispatchResponse | 0..* | InterTieDispatchResponse | |
| 0..1 | InterchangeSchedule | 0..* | InterchangeSchedule | |
| 0..1 | InterTieBid | 0..1 | InterTieBid | |
| 0..* | Flowgate | 1..1 | Flowgate | |
| 0..* | WheelingCounterParty | 0..* | WheelingCounterParty | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | inherited from: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | inherited from: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | inherited from: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | inherited from: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | inherited from: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | inherited from: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | inherited from: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | inherited from: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | inherited from: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | inherited from: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | inherited from: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | inherited from: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | inherited from: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | inherited from: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | inherited from: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | inherited from: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | inherited from: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | inherited from: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | inherited from: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | inherited from: RegisteredResource |
| 0..* | Reason | 0..* | Reason | inherited from: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | inherited from: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | inherited from: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | inherited from: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | inherited from: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | inherited from: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | inherited from: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | inherited from: RegisteredResource |
| 0..1 | LoadFollowingOperatorI | 0..* | LoadFollowingOperatorI | inherited from: RegisteredResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| | nput | | nput | |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | inherited from: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | inherited from: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | inherited from: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | inherited from: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | inherited from: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | inherited from: RegisteredResource |
| 0..* | Domain | 0..* | Domain | inherited from: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | inherited from: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | inherited from: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.59 RegisteredLoad

Model of a load that is registered to participate in the market.

RegisteredLoad is used to model any load that is served by the wholesale market directly. RegisteredLoads may be dispatchable or non-dispatchable and may or may not have bid curves. Examples of RegisteredLoads would include: distribution company load, energy retailer load, large bulk power system connected facility load.

Loads that are served indirectly, for example – through an energy retailer or a vertical utility, should be modeled as RegisteredDistributedResources. Examples of RegisteredDistributedResources would include: distribution level storage, distribution level generation and distribution level demand response.

Table 619 shows all attributes of RegisteredLoad.

**Table 619 – Attributes of ReferenceData::RegisteredLoad**

| name | mult | type | description |
|---|---|---|---|
| blockLoadTransfer | 0..1 | Boolean | Emergency operating procedure – Flag to indicate that the Resource is Block Load pseudo resource. |
| dynamicallyScheduledLoadResource | 0..1 | Boolean | Flag to indicate that a Load Resource is part of a DSR Load |
| dynamicallyScheduledQualification | 0..1 | Boolean | Qualification status (used for DSR qualification). |
| ACAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | inherited from: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| dispatchable | 0..1 | Boolean | inherited from: RegisteredResource |
| ECAFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | inherited from: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | inherited from: RegisteredResource |
| lastModified | 0..1 | DateTime | inherited from: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | inherited from: RegisteredResource |
| maxOnTime | 0..1 | Float | inherited from: RegisteredResource |
| minDispatchTime | 0..1 | Hours | inherited from: RegisteredResource |
| minOffTime | 0..1 | Float | inherited from: RegisteredResource |
| minOnTime | 0..1 | Float | inherited from: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| nonMarket | 0..1 | YesNo | inherited from: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | inherited from: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | inherited from: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | inherited from: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | inherited from: RegisteredResource |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 620 shows all association ends of RegisteredLoad with other classes.

**Table 620 – Association ends of ReferenceData::RegisteredLoad with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | LoadBids | 0..* | LoadBid | |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | inherited from: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | inherited from: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | inherited from: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | inherited from: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | inherited from: RegisteredResource |
| 0..1 | ContractDistributionFact or | 0..* | ContractDistributionFact or | inherited from: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | inherited from: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | inherited from: RegisteredResource |
| 0..1 | ResourceDispatchResult s | 0..* | ResourceDispatchResult s | inherited from: RegisteredResource |
| 0..1 | SubstitutionResourceLis t | 0..* | SubstitutionResourceLis t | inherited from: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | inherited from: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | inherited from: RegisteredResource |
| 1..1 | IntermittentResourceElig ibility | 0..* | IntermittentResourceElig ibility | inherited from: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | inherited from: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | inherited from: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | inherited from: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | inherited from: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | inherited from: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | inherited from: RegisteredResource |
| 0..1 | ResourceLoadFollowingI nst | 0..* | ResourceLoadFollowingI nst | inherited from: RegisteredResource |
| 1..1 | ResourceAncillaryServic eQualification | 0..* | ResourceCertification | inherited from: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | inherited from: RegisteredResource |
| 0..* | Reason | 0..* | Reason | inherited from: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | inherited from: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | inherited from: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | inherited from: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | inherited from: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | inherited from: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | inherited from: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | inherited from: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | inherited from: RegisteredResource |
| 0..1 | LoadFollowingOperatorI nput | 0..* | LoadFollowingOperatorI nput | inherited from: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | inherited from: RegisteredResource |
| 0..1 | ResourceAwardInstructi on | 0..* | ResourceAwardInstructi on | inherited from: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | inherited from: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | inherited from: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | inherited from: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | inherited from: RegisteredResource |
| 0..* | Domain | 0..* | Domain | inherited from: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | inherited from: RegisteredResource |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | DotInstruction | 0..* | DotInstruction | inherited from: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.60 RegulatingLimit

This class represents the physical characteristic of a generator regarding the regulating limit.

Table 621 shows all attributes of RegulatingLimit.

**Table 621 – Attributes of ReferenceData::RegulatingLimit**

| name | mult | type | description |
|---|---|---|---|
| highLimit | 0..1 | ActivePower | |
| lowLimit | 0..1 | ActivePower | |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 622 shows all association ends of RegulatingLimit with other classes.

**Table 622 – Association ends of ReferenceData::RegulatingLimit with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.61 ResourceCertification root class

Specifies certification for a resource to participate in a specific markets.

Table 623 shows all attributes of ResourceCertification.

**Table 623 – Attributes of ReferenceData::ResourceCertification**

| name | mult | type | description |
|---|---|---|---|
| market | 0..1 | MarketType | market type |
| qualificationFlag | 0..1 | YesNo | Status of the qualification ('Y' = Active, 'N' = Inactive) |
| type | 0..1 | ResourceCertificationKind | Type of service based on ResourceAncillaryServiceType enumeration |

Table 624 shows all association ends of ResourceCertification with other classes.

**Table 624 – Association ends of ReferenceData:: ResourceCertification with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | RegisteredResources are qualified for resource ancillary service types (which include market product types as well as other types such as BlackStart) by the association to the class ResourceAncillaryServiceQualification. |

### 6.5.9.62 ResourceOperationMaintenanceCost

To model the Operation and Maintenance (O and M) costs of a generation resource.

Table 625 shows all attributes of ResourceOperationMaintenanceCost.

**Table 625 – Attributes of ReferenceData::ResourceOperationMaintenanceCost**

| name | mult | type | description |
|---|---|---|---|
| gasPercentAboveLowSustainedLimit | 0..1 | PerCent | Percentage of Fuel Index Price (gas) for operating above Low Sustained Limit (LSL) |
| oilPercentAboveLowSustainedLimit | 0..1 | PerCent | Percentage of Fuel Oil Price (FOP) for operating above Low Sustained Limit (LSL) |
| omCostColdStartup | 0..1 | Float | Verifiable O&M Cost ($), Cold Startup |
| omCostHotStartup | 0..1 | Float | Verifiable O&M Cost ($), Hot Startup |
| omCostIntermediateStartup | 0..1 | Float | Verifiable O&M Cost ($), Intermediate Startup |
| omCostLowSustainedLimit | 0..1 | Float | Verifiable O&M Cost ($/MWh), LSL |
| solidfuelPercentAboveLowSustainedLimit | 0..1 | PerCent | Percentage of Solid Fuel for operating above Low Sustained Limit (LSL) |
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 626 shows all association ends of ResourceOperationMaintenanceCost with other classes.

**Table 626 – Association ends of ReferenceData:: ResourceOperationMaintenanceCost with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.63 ResourceStartupCost root class

To model the startup costs of a generation resource.

Table 627 shows all attributes of ResourceStartupCost.

**Table 627 – Attributes of ReferenceData::ResourceStartupCost**

| name | mult | type | description |
|---|---|---|---|
| fuelColdStartup | 0..1 | Float | Verifiable Cold Start Up Fuel (MMBtu per start) |
| fuelHotStartup | 0..1 | Float | Verifiable Hot Start Up Fuel (MMBtu per start) |
| fuelIntermediateStartup | 0..1 | Float | Verifiable Intermediate Start Up Fuel (MMBtu per start) |
| fuelLowSustainedLimit | 0..1 | Float | Minimum-Energy fuel, MMBtu/MWh |
| gasPercentColdStartup | 0..1 | PerCent | Percentage of Fuel Index Price (gas) for cold startup |
| gasPercentHotStartup | 0..1 | PerCent | Percentage of Fuel Index Price (gas) for hot startup |
| gasPercentIntermediateStartup | 0..1 | PerCent | Percentage of Fuel Index Price (gas) for intermediate startup |
| gasPercentLowSustainedLimit | 0..1 | PerCent | Percentage of FIP (gas) for operating at LSL |
| oilPercentColdStartup | 0..1 | PerCent | Percentage of Fuel Oil Price (FOP) for cold startup |
| oilPercentHotStartup | 0..1 | PerCent | Percentage of Fuel Oil Price (FOP) for hot startup |
| oilPercentIntermediateStartup | 0..1 | PerCent | Percentage of Fuel Oil Price (FOP) for intermediate startup |
| oilPercentLowSustainedLimit | 0..1 | PerCent | Percentage of FOP (oil) for operating at LSL |

| name | mult | type | description |
|---|---|---|---|
| solidfuelPercentColdStartup | 0..1 | PerCent | Percentage of Solid Fuel for cold startup |
| solidfuelPercentHotStartup | 0..1 | PerCent | Percentage of Solid Fuel for hot startup |
| solidfuelPercentIntermediateStartup | 0..1 | PerCent | Percentage of Solid Fuel for intermedite startup |
| solidfuelPercentLowSustainedLimit | 0..1 | PerCent | Percentage of Solid Fuel for operating at LSL |

Table 628 shows all association ends of ResourceStartupCost with other classes.

**Table 628 – Association ends of ReferenceData:: ResourceStartupCost with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ResourceVerifiableCosts | 1..1 | ResourceVerifiableCosts | |

#### 6.5.9.64 ResourceVerifiableCosts root class

This class is defined to describe the verifiable costs associated with a generation resource.

Table 629 shows all association ends of ResourceVerifiableCosts with other classes.

**Table 629 – Association ends of ReferenceData:: ResourceVerifiableCosts with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | MktHeatRateCurve | 1..1 | MktHeatRateCurve | |
| 0..1 | ResourceOperationMaintenanceCost | 1..1 | ResourceOperationMaintenanceCost | |
| 1..1 | ResourceStartupCost | 0..* | ResourceStartupCost | |

#### 6.5.9.65 ResponseMethod

Specifies a category of energy usage that the demand response applies for; e.g. energy from lighting, HVAC, other.

Table 630 shows all attributes of ResponseMethod.

**Table 630 – Attributes of ReferenceData::ResponseMethod**

| name | mult | type | description |
|---|---|---|---|
| activePower | 0..1 | Float | The active power value for the demand adjustment type. This supports requests to be made to a resource for some amount of active power provided by a particular response method, as specified by the method attribute (e.g. lighting, HVAC, wall mounted air conditioners, etc.). |
| activePowerUOM | 0..1 | String | The unit of measure of active power, e.g. kiloWatts (kW), megaWatts (mW), etc. |
| method | 0..1 | String | The response method (e.g. lighting, HVAC, wall mounted air conditioners, etc.). |
| siteMultiplier | 0..1 | Integer | This value provides for scaling of a response method's active power. For example, a response method of air conditioning could utilize a small amount of active power from each air conditioning unit (e.g. 0.1 kiloWatt), but the site multiplier could be used to produce a the total active power adjustment by multiplying the response method active power by this value (e.g. a building with 100 window air conditioning units, so 100 * 0.1 kW = 10 kW). |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 631 shows all association ends of ResponseMethod with other classes.

**Table 631 – Association ends of ReferenceData::ResponseMethod with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredDistributedResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.66 SchedulingCoordinator

Market participants could be represented by Scheduling Coordinators (SCs) that are registered with the RTO/ISO. One participant could register multiple SCs with the RTO/ISO. Many market participants can do business with the RTO/ISO using a single SC. One SC could schedule multiple generators. A load scheduling point could be used by multiple SCs. Each SC could schedule load at multiple scheduling points. An inter-tie scheduling point can be used by multiple SCs. Each SC can schedule interchange at multiple inter-tie scheduling points.

Table 632 shows all attributes of SchedulingCoordinator.

**Table 632 – Attributes of ReferenceData::SchedulingCoordinator**

| name | mult | type | description |
|---|---|---|---|
| creditFlag | 0..1 | YesNo | Flag to indicate creditworthiness (Y, N) |
| creditStartEffectiveDate | 0..1 | DateTime | Date that the scheduling coordinator becomes creditworthy. |
| lastModified | 0..1 | DateTime | Indication of the last time this scheduling coordinator information was modified. |
| qualificationStatus | 0..1 | String | Scheduling coordinator qualification status, Qualified, Not Qualified, or Disqualified. |
| scid | 0..1 | String | This is the short name or Scheduling Coordinator ID field. |
| streetAddress | 0..1 | StreetAddress | inherited from: Organisation |
| postalAddress | 0..1 | StreetAddress | inherited from: Organisation |
| phone1 | 0..1 | TelephoneNumber | inherited from: Organisation |
| phone2 | 0..1 | TelephoneNumber | inherited from: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | inherited from: Organisation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 633 shows all association ends of SchedulingCoordinator with other classes.

**Table 633 – Association ends of ReferenceData::SchedulingCoordinator with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..1 | SubmitToSCTrade | 0..* | Trade | |
| 1..1 | ToSCTrade | 0..* | Trade | |
| 1..1 | FromSCTrade | 0..* | Trade | |
| 0..1 | SubmitFromSCTrade | 0..* | Trade | |
| 1..1 | TransmissionContractRight | 0..* | ContractRight | |
| 0..1 | LoadRatio | 1..1 | LoadRatio | |
| 0..* | MarketPerson | 0..* | MarketPerson | inherited from: MarketParticipant |
| 0..* | MarketDocument | 0..* | MarketDocument | inherited from: MarketParticipant |
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | inherited from: MarketParticipant |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | inherited from: MarketParticipant |
| 0..1 | Bid | 0..* | Bid | inherited from: MarketParticipant |
| 0..* | TimeSeries | 0..* | TimeSeries | inherited from: MarketParticipant |
| 0..* | MarketRole | 0..* | MarketRole | inherited from: MarketParticipant |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | inherited from: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | inherited from: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.67 SchedulingCoordinatorUser root class

Describing users of a Scheduling Coordinator.

Table 634 shows all attributes of SchedulingCoordinatorUser.

**Table 634 – Attributes of ReferenceData::SchedulingCoordinatorUser**

| name | mult | type | description |
|---|---|---|---|
| loginID | 0..1 | String | Login ID |
| loginRole | 0..1 | String | Assigned roles (these are roles with either Read or Read/Write privileges on different Market Systems) |

#### 6.5.9.68 SchedulingPoint

Connection to other organizations at the boundary of the ISO/RTO.

Table 635 shows all attributes of SchedulingPoint.

**Table 635 – Attributes of ReferenceData::SchedulingPoint**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 636 shows all association ends of SchedulingPoint with other classes.

**Table 636 – Association ends of ReferenceData::SchedulingPoint with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | InterchangeSchedule | 0..* | InterchangeSchedule | |
| 0..* | Flowgate | 0..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.69 StartUpEnergyCurve

The energy consumption of a generating resource to complete a start-up from the StartUpEnergyCurve. Definition of the StartUpEnergyCurve includes, xvalue as the cooling time and y1value as the MW value.

Table 637 shows all attributes of StartUpEnergyCurve.

**Table 637 – Attributes of ReferenceData::StartUpEnergyCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 638 shows all association ends of StartUpEnergyCurve with other classes.

**Table 638 – Association ends of ReferenceData::StartUpEnergyCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.70 StartUpFuelCurve

The fuel consumption of a Generating Resource to complete a Start-Up.(x=cooling time) Form Startup Fuel Curve. xAxisData -> cooling time, y1AxisData -> MBtu.

Table 639 shows all attributes of StartUpFuelCurve.

**Table 639 – Attributes of ReferenceData::StartUpFuelCurve**

| name | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | inherited from: Curve |
| xMultiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| xUnit | 0..1 | UnitSymbol | inherited from: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y1Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y2Unit | 0..1 | UnitSymbol | inherited from: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | inherited from: Curve |
| y3Unit | 0..1 | UnitSymbol | inherited from: Curve |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 640 shows all association ends of StartUpFuelCurve with other classes.

**Table 640 – Association ends of ReferenceData::StartUpFuelCurve with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | inherited from: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.71 SubControlArea

An area defined for the purpose of tracking interchange with surrounding areas via tie points; may or may not serve as a control area.

Table 641 shows all attributes of SubControlArea.

**Table 641 – Attributes of ReferenceData::SubControlArea**

| name | mult | type | description |
|---|---|---|---|
| areaShortName | 0..1 | String | Market area short name, which is the regulation zone. It references AGC regulation zone name. |
| constantCoefficient | 0..1 | Float | Loss estimate constant coefficient |
| embeddedControlArea | 0..1 | YesNo | Used in conjunction with the InternalCA flag. If the InternalCA flag is YES, this flag does not apply. If the InternaCA flag is NO, this flag provides an indication of AdjacentCA (NO) or Embedded CA (YES). |
| internalCA | 0..1 | YesNo | A Yes/No indication that this control area is contained internal to the system. |

| name | mult | type | description |
|---|---|---|---|
| linearCoefficient | 0..1 | Float | Loss estimate linear coefficient |
| localCA | 0..1 | YesNo | Indication that this control area is the local control area. |
| maxSelfSchedMW | 0..1 | Float | Maximum amount of self schedule MWs allowed for an embedded control area. |
| minSelfSchedMW | 0..1 | Float | Minimum amount of self schedule MW allowed for an embedded control area. |
| quadraticCoefficient | 0..1 | Float | Loss estimate quadratic coefficient |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 642 shows all association ends of SubControlArea with other classes.

**Table 642 – Association ends of ReferenceData::SubControlArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 1..1 | Export_EnergyTransactions | 0..* | EnergyTransaction | Energy is transferred between interchange areas |
| 1..1 | Import_EnergyTransactions | 0..* | EnergyTransaction | Energy is transferred between interchange areas |
| 0..1 | GeneralClearingResults | 0..* | GeneralClearingResults | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | ExPostLossResults | 0..* | ExPostLossResults | |
| 0..1 | LossClearingResults | 1..* | LossClearingResults | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | AggregateNode | 0..* | AggregateNode | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | |
| 0..1 | To_Flowgate | 0..* | Flowgate | |
| 0..1 | From_Flowgate | 0..* | Flowgate | |
| 0..* | HostControlArea | 1..1 | HostControlArea | The interchange area may operate as a control area |
| 0..1 | Pnode | 0..* | Pnode | |
| 0..* | RTO | 1..1 | RTO | |
| 0..* | PSRType | 0..1 | PSRType | inherited from: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | inherited from: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | inherited from: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | inherited from: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | inherited from: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.5.9.72 SubstitutionResourceList root class

List of resources that can be substituted for within the bounds of a Contract definition. This class has a precedence and a resource.

Table 643 shows all attributes of SubstitutionResourceList.

**Table 643 – Attributes of ReferenceData::SubstitutionResourceList**

| name | mult | type | description |
|---|---|---|---|
| precedence | 0..1 | Integer | An indicator of the order a resource should be substituted. The lower the number the higher the precedence. |

Table 644 shows all association ends of SubstitutionResourceList with other classes.

**Table 644 – Association ends of ReferenceData:: SubstitutionResourceList with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |

### 6.5.9.73 TACArea

Transmission Access Charge Area. Charges assessed, on behalf of the Participating Transmission Owner, to parties who require access to the controlled grid.

Table 645 shows all attributes of TACArea.

**Table 645 – Attributes of ReferenceData::TACArea**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 646 shows all association ends of TACArea with other classes.

**Table 646 – Association ends of ReferenceData::TACArea with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | |
| 0..* | AggregatedPnode | 0..* | AggregatedPnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.74 TransmissionRightChain

Allows chaining of TransmissionContractRights. Many individual contract rights can be included in the definition of a TransmissionRightChain. A TransmissionRightChain is also defined as a TransmissionContractRight itself.

Table 647 shows all attributes of TransmissionRightChain.

**Table 647 – Attributes of ReferenceData::TransmissionRightChain**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 648 shows all association ends of TransmissionRightChain with other classes.

**Table 648 – Association ends of ReferenceData:: TransmissionRightChain with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | Ind_ContractRight | 1..* | ContractRight | |
| 0..1 | Chain_ContractRight | 1..1 | ContractRight | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

#### 6.5.9.75 WheelingCounterParty

Counter party in a wheeling transaction.

Table 649 shows all attributes of WheelingCounterParty.

**Table 649 – Attributes of ReferenceData::WheelingCounterParty**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 650 shows all association ends of WheelingCounterParty with other classes.

**Table 650 – Association ends of ReferenceData:: WheelingCounterParty with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | RegisteredInterTie | 0..* | RegisteredInterTie | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

## 6.6 Package Environmental

### 6.6.1 General

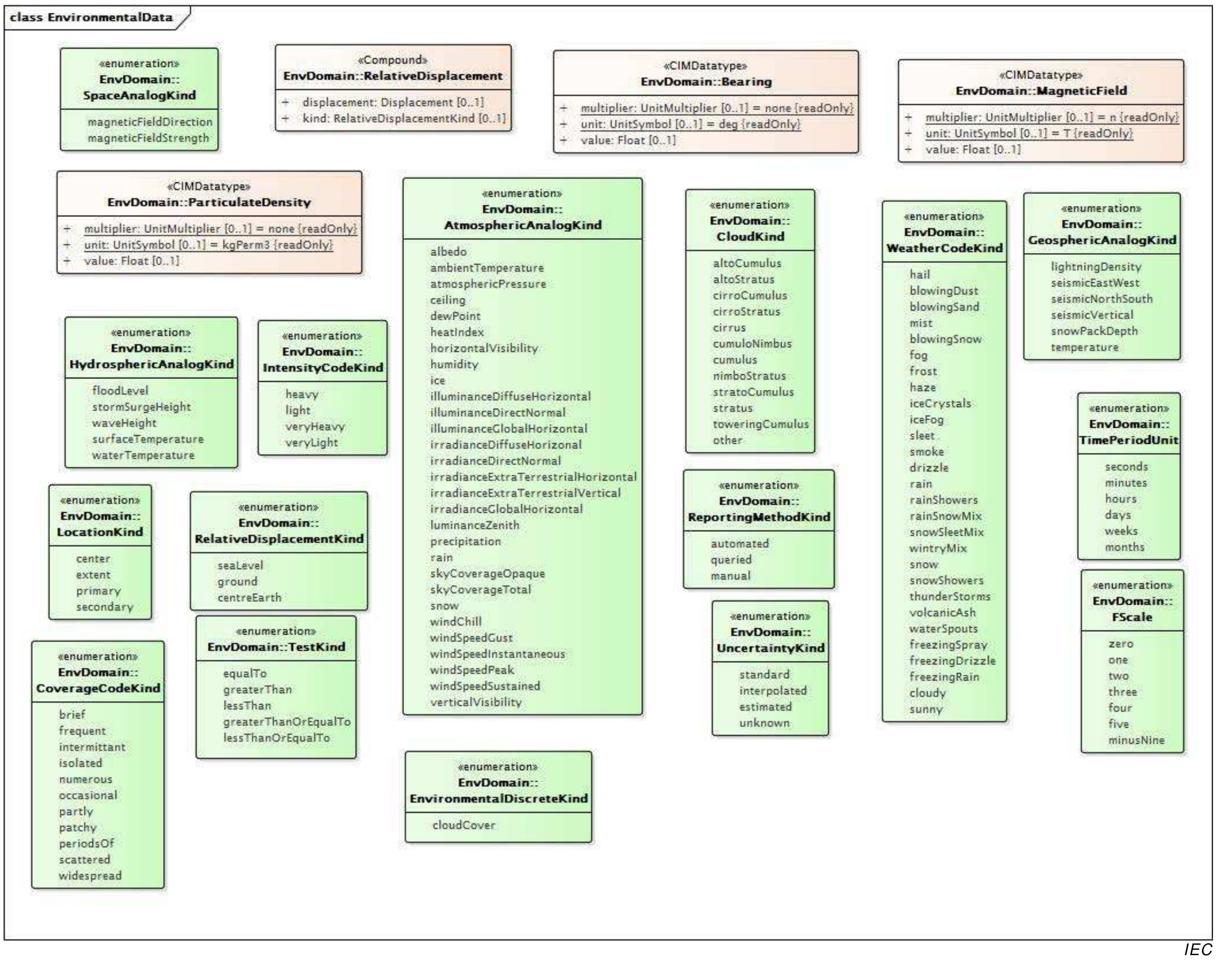
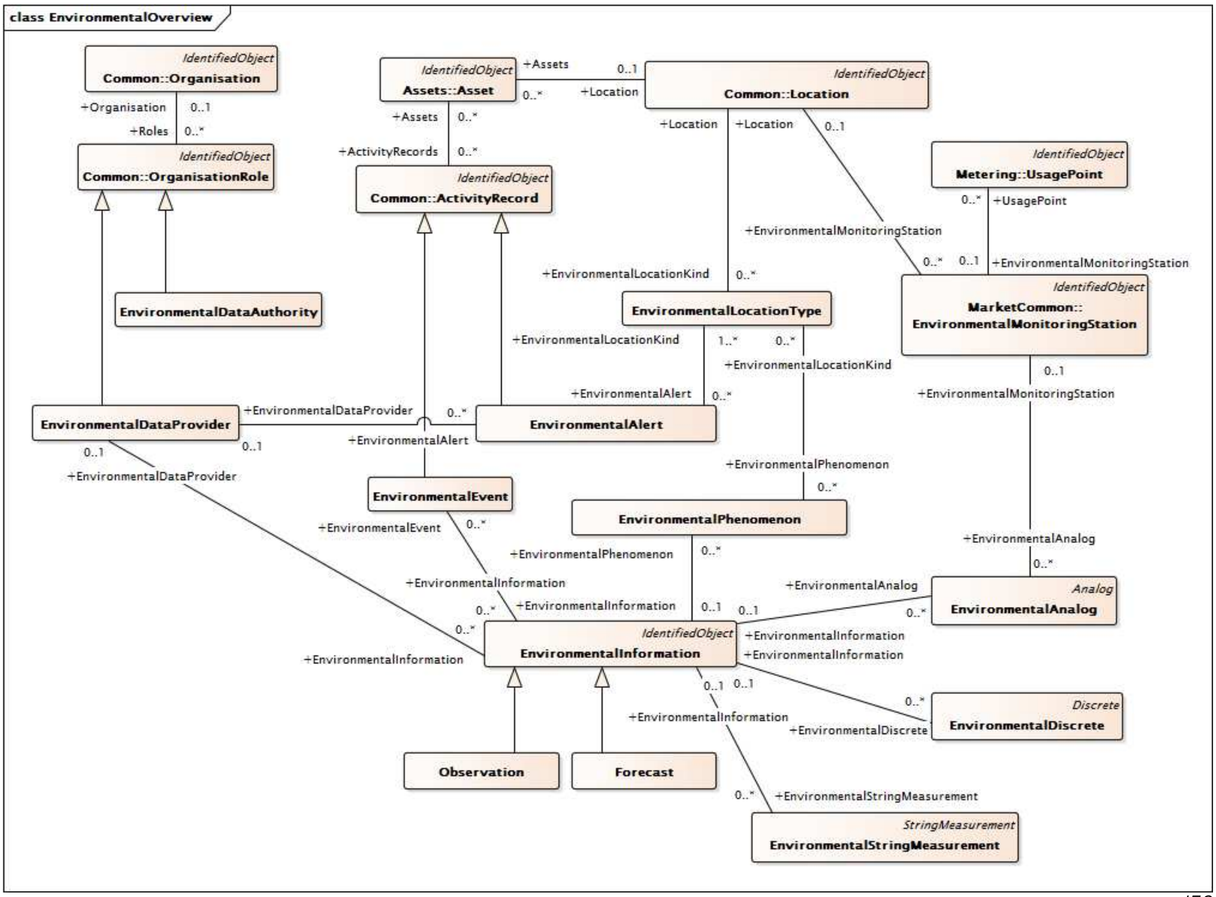



**Figure 70 – Class diagram Environmental::EnvironmentalData**

Figure 70: Illustrates the data types and enumerations which support the Environmental package classes in addition to the data types and enumerations which are contained within the IEC 61970, IEC 61968 and IEC 62325 packages.




IEC

**Figure 71 – Class diagram Environmental::EnvironmentalOverview**

Figure 71: Shows the Environmental package, which contains a model of constructs typically used to describe environmental (weather) conditions including forecasts, observations, measurements, phenomena, and alerts.

The foundational classes are Observation, Forecast and EnvironmentalAlert and EnvironmentalEvent. The remaining classes describe the content of information of the foundational classes.

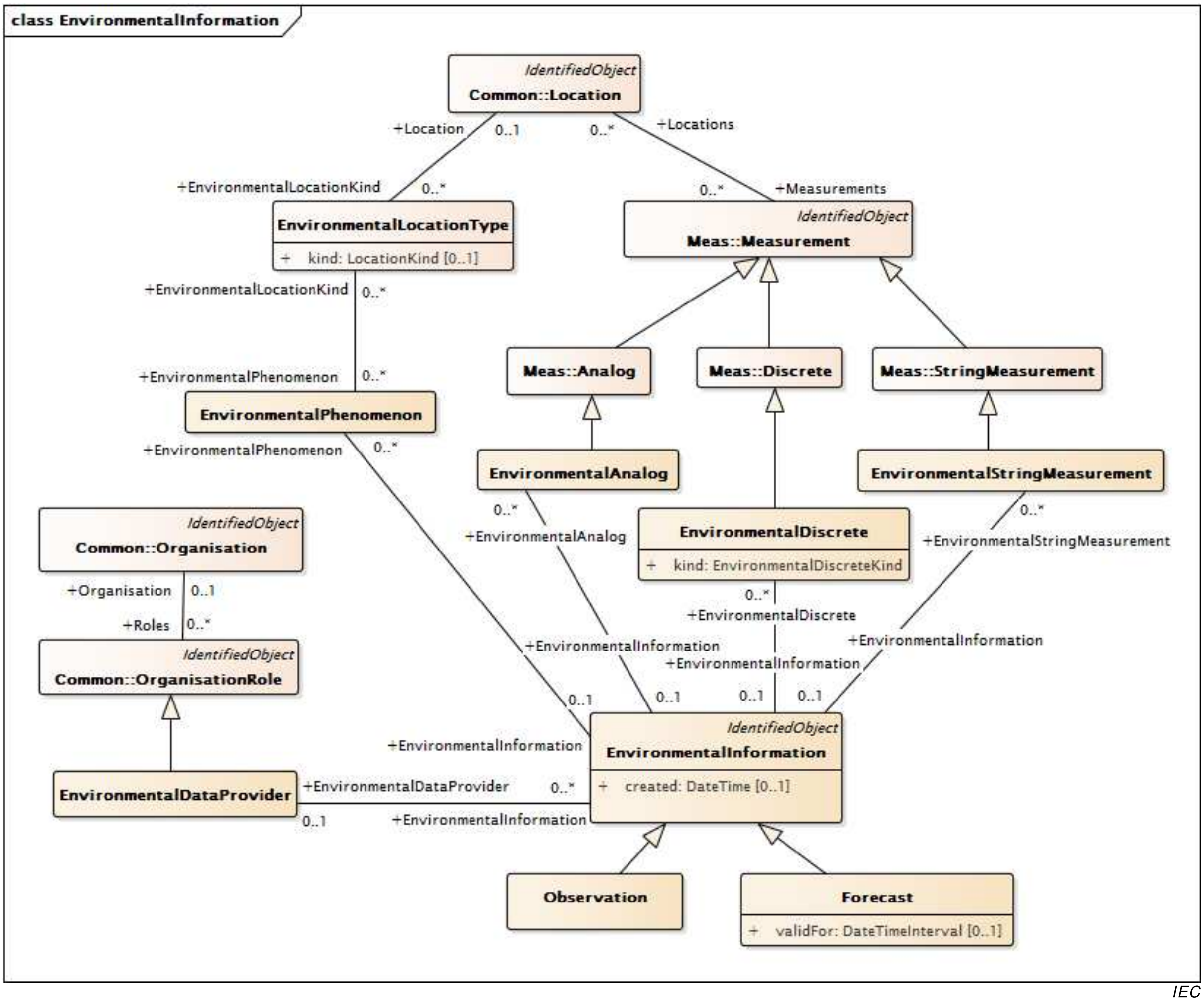
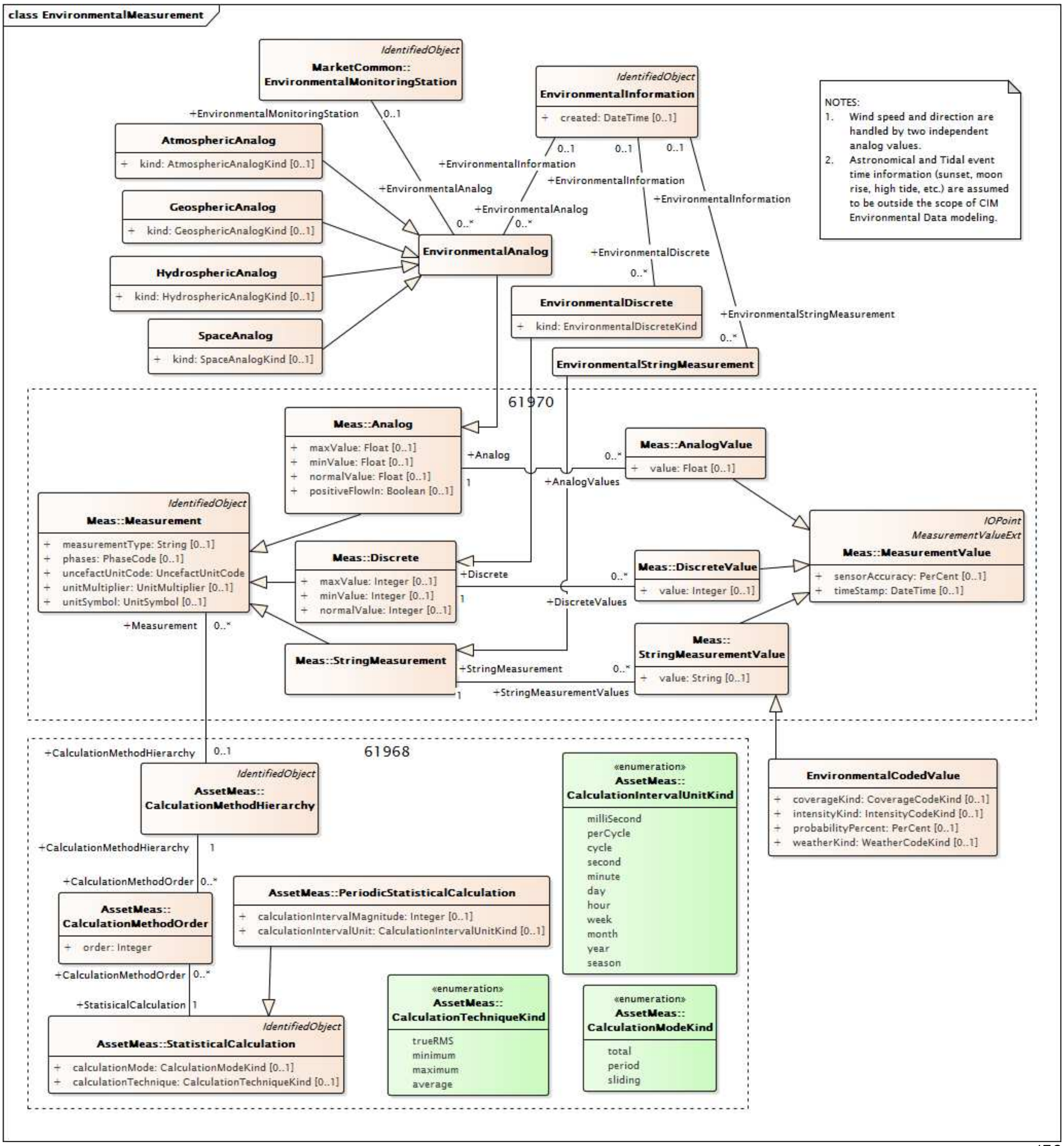



**Figure 72 – Class diagram Environmental::EnvironmentalInformation**

Figure 72: Shows Observation and Forecast, which are the two child classes of EnvironmentalInformation. Each has environmental data associated with it which is described in terms of measurements and phenomenon. Locational information for observations and forecasts is provided at the measurement or phenomenon level, not the observation or forecast level. Both observations and forecasts can have a data provider defined for them.




IEC

**Figure 73 – Class diagram Environmental::EnvironmentalMeasurement**

Figure 73: Shows observations and forecasts, which can include any number of analog (float), discrete (integer) or string values valid for various points or periods in time.

The nature of values is described by the Measurement child classes. Descriptions of periodic statistical calculations are made possible by the calculation method classes. Actual values which are a part of observations or forecasts are represented by MeasurementValue child classes.

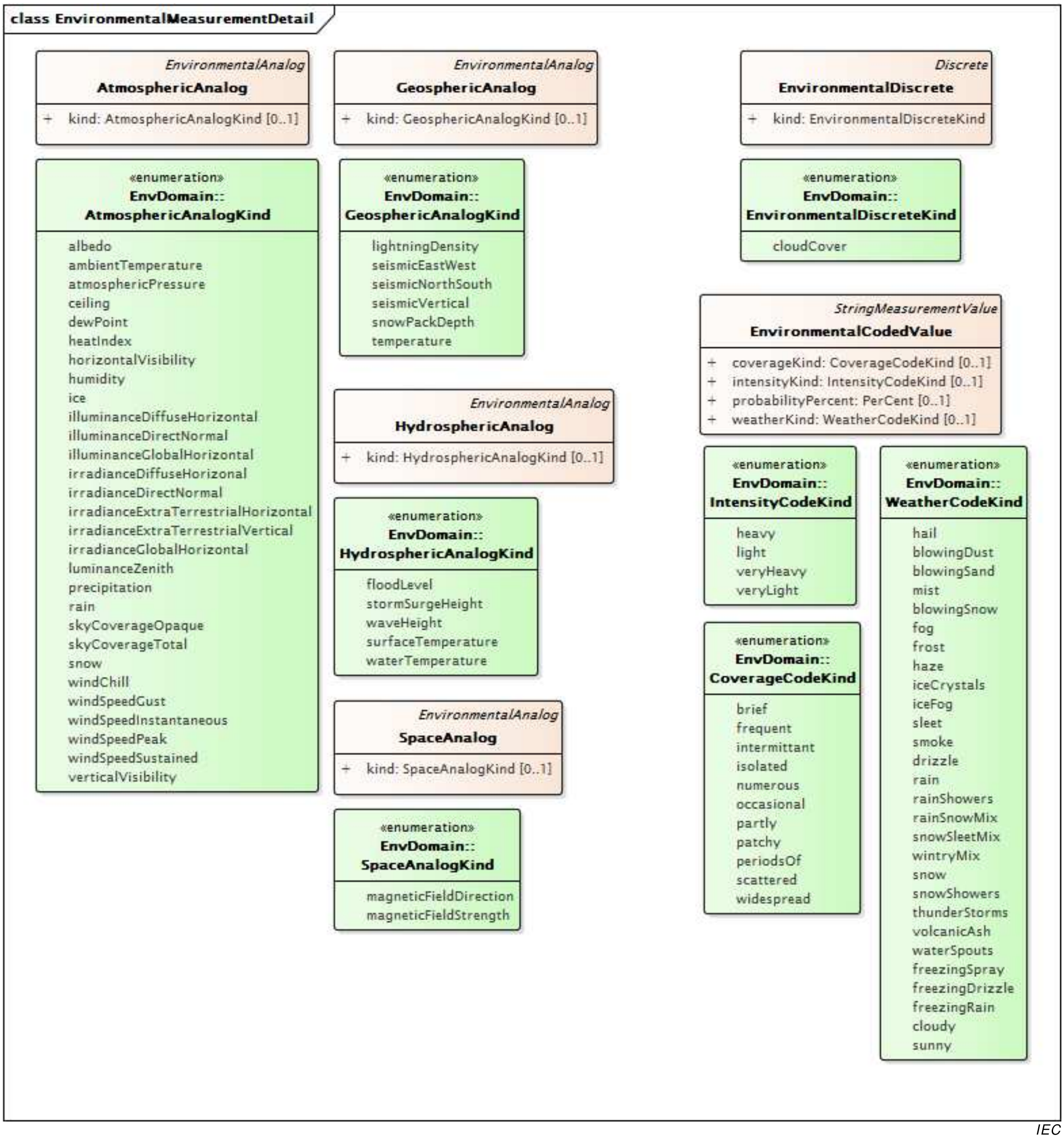



**Figure 74 – Class diagram Environmental::EnvironmentalMeasurementDetail**

Figure 74: Shows the child classes of EnvironmentalAnalog, each of which has an enumerated kind attribute whose values reflect what is being measured by the float value in that domain (atmospheric, geospheric, hydrospheric or space).

The EnvironmentalDiscrete class has an enumerated .kind attribute whose value reflects what is being measured by the integer value.

The EnvironmentalCodedValue class (a child of StringMeasurementValue) has additional string values which allow a clear description of weather conditions, like 'heavy isolated thunderstorms'.

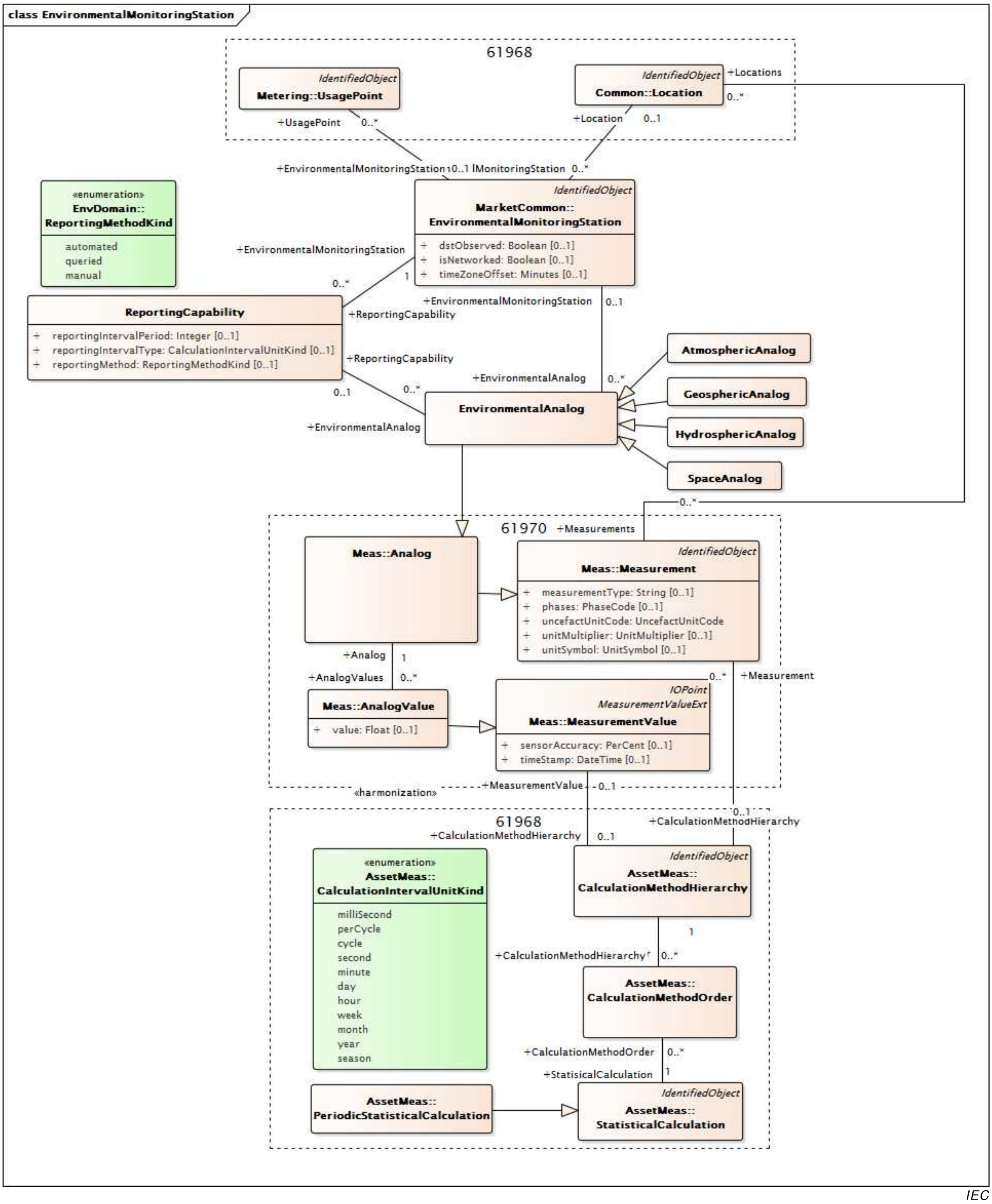



**Figure 75 – Class diagram Environmental::EnvironmentalMonitoringStation**

Figure 75: Shows environmental monitoring stations, which provide analog values for observations, which can be utility-owned or owned by an outside entity. Their capabilities can be defined by the ReportingCapability class. The ReportingCapability class utilizes EnvironmentalAnalog child classes to specifically describe the analogs the monitoring station can return. The possibility that a monitoring station could be associated with a meter or customer is addressed by an association between EnvironmentalMonitoringStation and UsagePoint.

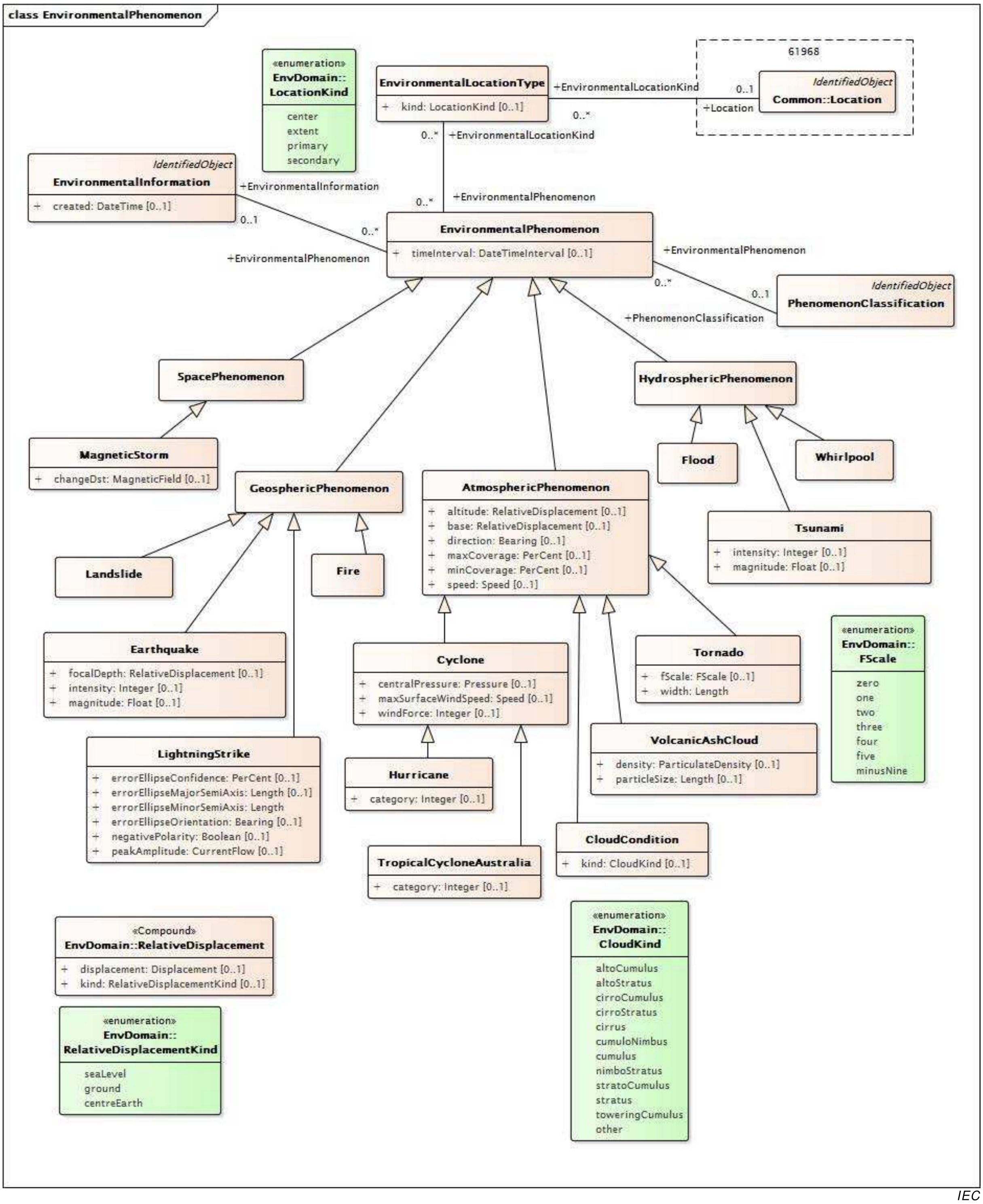



**Figure 76 – Class diagram Environmental::EnvironmentalPhenomenon**

Figure 76: Shows the EnvironmentalPhenomenon class and the specialized types of phenomenon classes. Environmental phenomena, like hurricanes, tornados, grass fires, floods, or geo-magnetic space storms are described by the Phenomenon class and its children, which are subclassed using the same regional groupings used for analogs (atmospheric, geospheric, hydrospheric, and space). Each EnvironmentalPhenomenon child class has attributes which reflect common information used to describe the nature or severity of that particular sort of phenomenon.

The location of phenomena are often described by both a center and extent, and this is supported by means of the EnvironmentalLocationKind class.

Major events, like hurricane Katrina or the 2008 Sichuan earthquake, can be described through time using multiple instances of a Phenomenon child class associated with an observation related to a 'named' event.

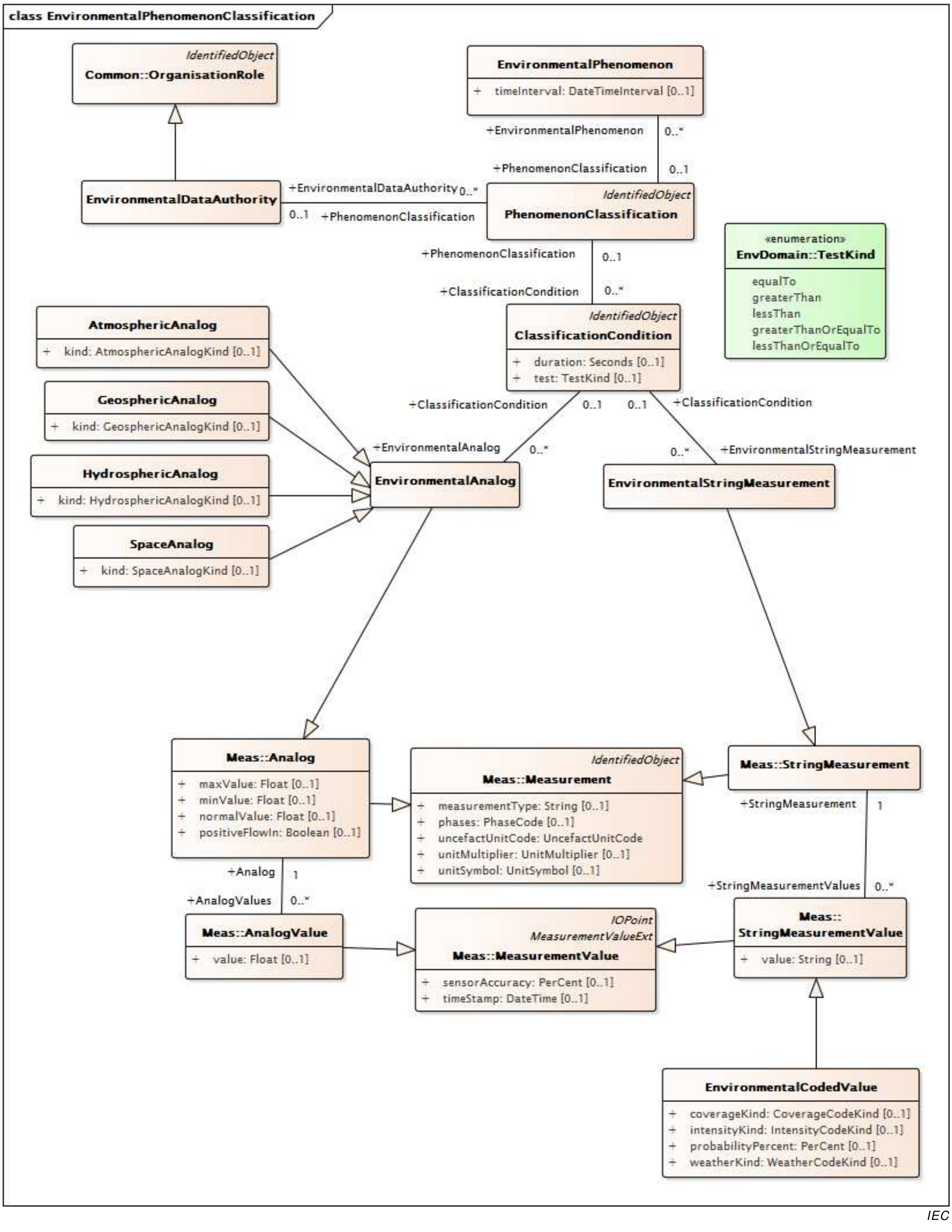



**Figure 77 – Class diagram Environmental::EnvironmentalPhenomenonClassification**

Figure 77: Shows the PhenomenonClassification class and associated classes which support the classification of phenomenon severity. This allows an environmental data authority (for example a weather service or utility) to define threshold levels of conditions for phenomenon classification. The threshold levels are described by means of the measurement and

measurement value child classes (which provide a value) in combination with a test (equal to, above, below, etc.) and a duration.

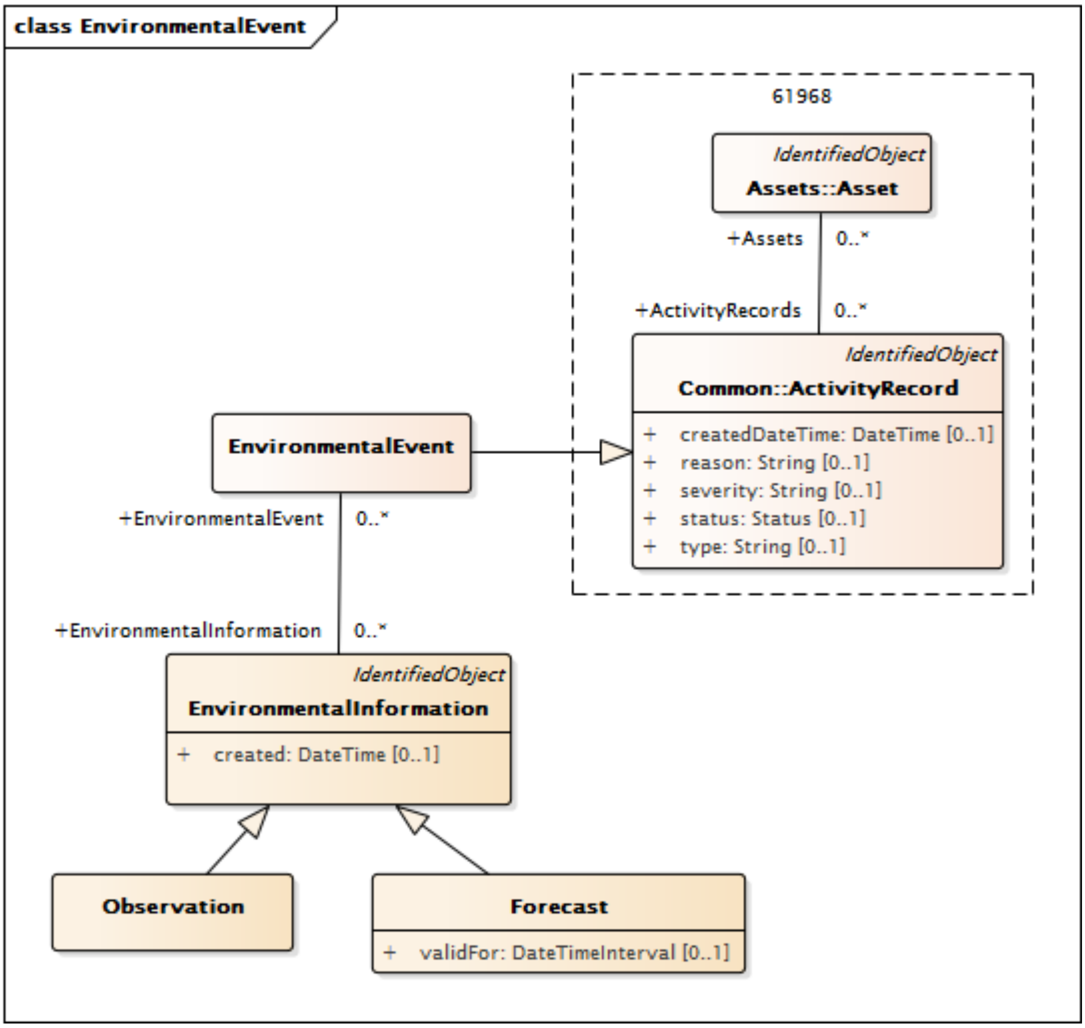



**Figure 78 – Class diagram Environmental::EnvironmentalEvent**

Figure 78: Shows the EnvironmentalEvent class as associated classes. Environmental events are occurrences to which one or more forecasts or observations may be tied and which may relate to or affect one or more assets. An event can be a 'named' occurrence like the 2011 Tohoku earthquake which could have multiple sets of associated observations or forecasts. Or an event could be simply a set of observations or forecasts that apply to an asset or set of assets.

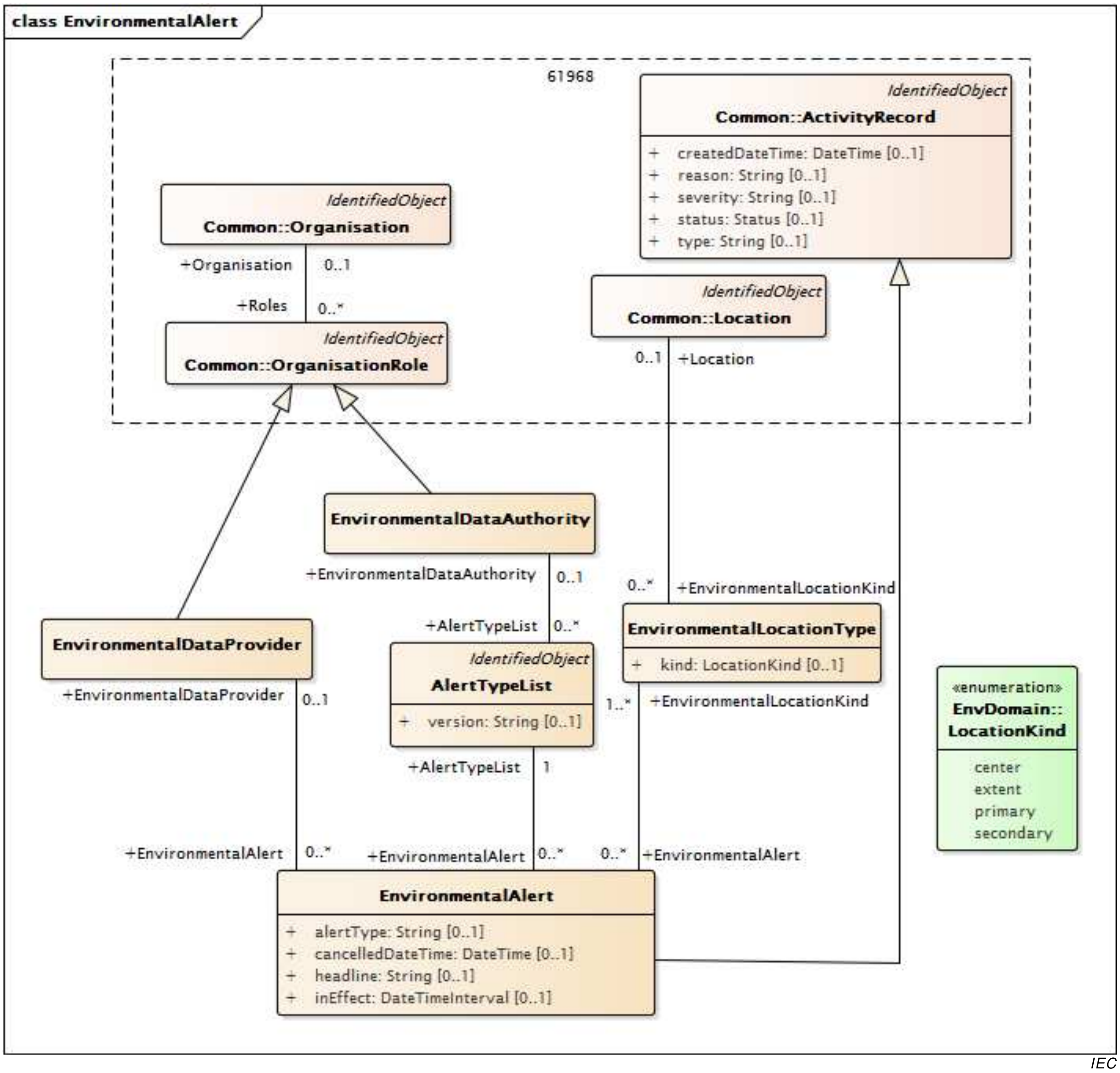



**Figure 79 – Class diagram Environmental::EnvironmentalAlert**

Figure 79: Shows EnvironmentalAlerts and associated classes. Environmental alerts, like tornado watches and warnings, are described by the EnvironmentalAlert class, which supports the definition of alert provider, alerting authority, and alert location. Alerting authorities (like the US National Weather Service) periodically update the types and definitions of alerts they can issue and the AlertTypeList class allows the specification of which version of alert definitions is being used. Some alerts have primary and secondary locations to which they apply and the EnvironmentalLocationKind class allows this distinction to be made.

### 6.6.2 AlertTypeList

A named list of alert types.

NOTE The name of the list is reflected in the .name attribute (inherited from IdentifiedObject).

Table 651 shows all attributes of AlertTypeList.

**Table 651 – Attributes of Environmental::AlertTypeList**

| name | mult | type | description |
|---|---|---|---|
| version | 0..1 | String | The version of the named list of alert types. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 652 shows all association ends of AlertTypeList with other classes.

**Table 652 – Association ends of Environmental::AlertTypeList with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataAuthority | 0..1 | EnvironmentalDataAuthority | The environmental data authority responsible for publishing this list of alert types. |
| 1..1 | EnvironmentalAlert | 0..* | EnvironmentalAlert | An alert whose type is drawn from this alert type list. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.3 AtmosphericAnalog

Analog (float) measuring an atmospheric condition.

Table 653 shows all attributes of AtmosphericAnalog.

**Table 653 – Attributes of Environmental::AtmosphericAnalog**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | AtmosphericAnalogKind | Kind of atmospheric analog. |
| maxValue | 0..1 | Float | inherited from: Analog |
| minValue | 0..1 | Float | inherited from: Analog |
| normalValue | 0..1 | Float | inherited from: Analog |
| positiveFlowIn | 0..1 | Boolean | inherited from: Analog |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 654 shows all association ends of AtmosphericAnalog with other classes.

**Table 654 – Association ends of Environmental::AtmosphericAnalog with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoring Station | 0..1 | EnvironmentalMonitoring Station | inherited from: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | inherited from: EnvironmentalAnalog |
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | inherited from: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | inherited from: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | inherited from: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.4 AtmosphericPhenomenon

An atmospheric phenomenon with a base and altitude providing the vertical coverage (combined with the Location to provide three dimensional space).

Table 655 shows all attributes of AtmosphericPhenomenon.

**Table 655 – Attributes of Environmental::AtmosphericPhenomenon**

| name | mult | type | description |
|---|---|---|---|
| altitude | 0..1 | RelativeDisplacement | The maximum altitude of the phenomenon. |
| base | 0..1 | RelativeDisplacement | The base altitude of the phenomenon. |
| direction | 0..1 | Bearing | The direction the phenomenon is moving. |
| maxCoverage | 0..1 | PerCent | The maximum percentage coverage |
| minCoverage | 0..1 | PerCent | The minimum percentage coverage |
| speed | 0..1 | Speed | The speed of the phenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 656 shows all association ends of AtmosphericPhenomenon with other classes.

**Table 656 – Association ends of Environmental:: AtmosphericPhenomenon with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.5 ClassificationCondition

A classification condition used to define preconditions that must be met by a phenomena classification.

Table 657 shows all attributes of ClassificationCondition.

**Table 657 – Attributes of Environmental::ClassificationCondition**

| name | mult | type | description |
|---|---|---|---|
| duration | 0..1 | Seconds | The duration of the of the condition in seconds |
| test | 0..1 | TestKind | The test applied to the value. |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 658 shows all association ends of ClassificationCondition with other classes.

**Table 658 – Association ends of Environmental:: ClassificationCondition with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Phenomenon classification to which this condition relates. |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | String measurement which contributes to the definition of this classification condition. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Analog which contributes to the definition of this classification condition. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.6 CloudCondition

A classified cloud phenomenon with a type.

Table 659 shows all attributes of CloudCondition.

**Table 659 – Attributes of Environmental::CloudCondition**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | CloudKind | The type of the cloud as defined by the CloudKind enumeration. |
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |

| name | mult | type | description |
|---|---|---|---|
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 660 shows all association ends of CloudCondition with other classes.

**Table 660 – Association ends of Environmental::CloudCondition with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.7 Cyclone

A cyclone (or tropical cyclone), a rapidly-rotating storm system characterized by a low-pressure center, strong winds, and a spiral arrangement of thunderstorms that produce heavy rain.

Table 661 shows all attributes of Cyclone.

**Table 661 – Attributes of Environmental::Cyclone**

| name | mult | type | description |
|---|---|---|---|
| centralPressure | 0..1 | Pressure | The central pressure of the cyclone during the time interval. |
| maxSurfaceWindSpeed | 0..1 | Speed | The maximum surface wind speed of the cyclone during the time interval. |
| windForce | 0..1 | Integer | Wind Force as classified on the Beaufort Scale (0-12) during the time interval. |
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 662 shows all association ends of Cyclone with other classes.

**Table 662 – Association ends of Environmental::Cyclone with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.8 Earthquake

An earthquake.

Table 663 shows all attributes of Earthquake.

**Table 663 – Attributes of Environmental::Earthquake**

| name | mult | type | description |
|---|---|---|---|
| focalDepth | 0..1 | RelativeDisplacement | The depth below the earth's surface of the earthquake's focal point. |
| intensity | 0..1 | Integer | The intensity of the earthquake as defined by the Modified Mercalli Intensity (MMI) scale. Possible values are 1-12, corresponding to I-XII. |
| magnitude | 0..1 | Float | The magnitude of the earthquake as defined on the Moment Magnitude (Mw) scale, which measures the size of earthquakes in terms of the energy released. Must be greater than zero. |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 664 shows all association ends of Earthquake with other classes.

**Table 664 – Association ends of Environmental::Earthquake with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.9 EnvironmentalAnalog

Analog (float) measurement of relevance in the environmental domain.

Table 665 shows all attributes of EnvironmentalAnalog.

**Table 665 – Attributes of Environmental::EnvironmentalAnalog**

| name | mult | type | description |
|---|---|---|---|
| maxValue | 0..1 | Float | inherited from: Analog |
| minValue | 0..1 | Float | inherited from: Analog |
| normalValue | 0..1 | Float | inherited from: Analog |
| positiveFlowIn | 0..1 | Boolean | inherited from: Analog |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 666 shows all association ends of EnvironmentalAnalog with other classes.

**Table 666 – Association ends of Environmental:: EnvironmentalAnalog with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoring Station | 0..1 | EnvironmentalMonitoring Station | Monitoring station which provides this environmental analog measurement. |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Classification condition which this analog helps define. |
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | Observation or forecast with which this environmental analog measurement is associated. |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | The reporting capability this environmental value set helps define. |
| 1..1 | AnalogValues | 0..* | AnalogValue | inherited from: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | inherited from: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.10 EnvironmentalAlert

An environmental alert issued by a provider or system.

Table 667 shows all attributes of EnvironmentalAlert.

**Table 667 – Attributes of Environmental::EnvironmentalAlert**

| name | mult | type | description |
|---|---|---|---|
| alertType | 0..1 | String | The type of the issued alert which is drawn from the specified alert type list. |
| cancelledDateTime | 0..1 | DateTime | Time and date alert cancelled. Used only if alert is cancelled before it expires. |
| headline | 0..1 | String | An abbreviated textual description of the alert issued.<br>Note: the full text of the alert appears in the .description attribute (inherited from IdentifiedObject). |
| inEffect | 0..1 | DateTimeInterval | The interval for which this weather alert is in effect. |
| createdDateTime | 0..1 | DateTime | inherited from: ActivityRecord |
| type | 0..1 | String | inherited from: ActivityRecord |
| severity | 0..1 | String | inherited from: ActivityRecord |
| reason | 0..1 | String | inherited from: ActivityRecord |
| status | 0..1 | Status | inherited from: ActivityRecord |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 668 shows all association ends of EnvironmentalAlert with other classes.

**Table 668 – Association ends of Environmental::EnvironmentalAlert with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | AlertTypeList | 1..1 | AlertTypeList | The list of alert types from which the type of this alert is drawn. |
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Environmental data provider for this alert. |
| 0..* | EnvironmentalLocationKind | 1..* | EnvironmentalLocationType | Type of location to which this environmental alert applies. |
| 0..* | Assets | 0..* | Asset | inherited from: ActivityRecord |
| 0..* | Author | 0..1 | Author | inherited from: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.11 EnvironmentalCodedValue

An environmental value described using a coded value. A triplicate of enumerated values representing intensity, coverage, type of weather is used. These may be concatenated into the string value.

Table 669 shows all attributes of EnvironmentalCodedValue.

**Table 669 – Attributes of Environmental::EnvironmentalCodedValue**

| name | mult | type | description |
|---|---|---|---|
| coverageKind | 0..1 | CoverageCodeKind | Code representing the coverage of the weather condition. |
| intensityKind | 0..1 | IntensityCodeKind | Code representing the intensity of the weather condition. |
| probabilityPercent | 0..1 | PerCent | Probability of weather condition occurring during the time interval expressed as a percentage. Applicable only when weather condition is related to a forecast (not an observation). |
| weatherKind | 0..1 | WeatherCodeKind | Code representing the type of weather condition. |
| value | 0..1 | String | inherited from: StringMeasurementValue |
| timeStamp | 0..1 | DateTime | inherited from: MeasurementValue |
| sensorAccuracy | 0..1 | PerCent | inherited from: MeasurementValue |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 670 shows all association ends of EnvironmentalCodedValue with other classes.

**Table 670 – Association ends of Environmental:: EnvironmentalCodedValue with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | StringMeasurement | 1..1 | StringMeasurement | inherited from: StringMeasurementValue |
| 1..1 | RemoteSource | 0..1 | RemoteSource | inherited from: MeasurementValue |
| 1..1 | MeasurementValueQuality | 0..1 | MeasurementValueQuality | inherited from: MeasurementValue |
| 0..* | MeasurementValueSource | 1..1 | MeasurementValueSource | inherited from: MeasurementValue |
| 0..1 | BilateralToIOPoint | 0..* | ProvidedBilateralPoint | inherited from: IOPoint |
| 0..* | IOPointSource | 0..1 | IOPointSource | inherited from: IOPoint |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.12 EnvironmentalDataAuthority

An entity defining classifications or categories of environmental information, like phenomena or alerts.

Table 671 shows all attributes of EnvironmentalDataAuthority.

**Table 671 – Attributes of Environmental::EnvironmentalDataAuthority**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 672 shows all association ends of EnvironmentalDataAuthority with other classes.

**Table 672 – Association ends of Environmental:: EnvironmentalDataAuthority with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | AlertTypeList | 0..* | AlertTypeList | A specific version of a list of alerts published by this environmental data authority. |
| 0..1 | PhenomenonClassification | 0..* | PhenomenonClassification | Phenomenon classification defined by this environmental data authority. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | inherited from: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.13 EnvironmentalDataProvider

Entity providing environmental data. Could be an observed weather data provider, an entity providing forecasts, an authority providing alerts, etc.

Table 673 shows all attributes of EnvironmentalDataProvider.

**Table 673 – Attributes of Environmental::EnvironmentalDataProvider**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 674 shows all association ends of EnvironmentalDataProvider with other classes.

**Table 674 – Association ends of Environmental:: EnvironmentalDataProvider with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | EnvironmentalAlert | 0..* | EnvironmentalAlert | Alert issued by this environmental data provider. |
| 0..1 | EnvironmentalInformation | 0..* | EnvironmentalInformation | Environmental information provided by this environmental data provider. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | inherited from: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | inherited from: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.14 EnvironmentalDiscrete

Discrete (integer) measurement of relevance in the environmental domain.

Table 675 shows all attributes of EnvironmentalDiscrete.

**Table 675 – Attributes of Environmental::EnvironmentalDiscrete**

| name | mult | type | description |
|---|---|---|---|
| kind | 1..1 | EnvironmentalDiscreteKind | Kind of environmental discrete (integer). |
| maxValue | 0..1 | Integer | inherited from: Discrete |
| minValue | 0..1 | Integer | inherited from: Discrete |
| normalValue | 0..1 | Integer | inherited from: Discrete |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 676 shows all association ends of EnvironmentalDiscrete with other classes.

**Table 676 – Association ends of Environmental:: EnvironmentalDiscrete with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Observation or forecast with which this environmental discrete (integer) is associated. |
| 1..1 | DiscreteValues | 0..* | DiscreteValue | inherited from: Discrete |
| 0..* | ValueAliasSet | 0..1 | ValueAliasSet | inherited from: Discrete |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.15 EnvironmentalEvent

An environmental event to which one or more forecasts or observations may be tied and which may relate to or affect one or more assets.

This class is intended to be used as a means of grouping forecasts and/or observations and could be used for a variety of purposes, including:

– to define a 'named' event like Hurricane Katrina and allow the historic (or forecast) values for phenomena and measurements (precipitation, temperature) across time to be associated with it

– to identify assets that were (or are forecast to be) affected by a phenomenon or set of measurements

Table 677 shows all attributes of EnvironmentalEvent.

**Table 677 – Attributes of Environmental::EnvironmentalEvent**

| name | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | inherited from: ActivityRecord |
| type | 0..1 | String | inherited from: ActivityRecord |
| severity | 0..1 | String | inherited from: ActivityRecord |
| reason | 0..1 | String | inherited from: ActivityRecord |
| status | 0..1 | Status | inherited from: ActivityRecord |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 678 shows all association ends of EnvironmentalEvent with other classes.

**Table 678 – Association ends of Environmental::EnvironmentalEvent with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..* | EnvironmentalInformation | Forecast or observation related to this environmental event. |
| 0..* | Assets | 0..* | Asset | inherited from: ActivityRecord |
| 0..* | Author | 0..1 | Author | inherited from: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.16 EnvironmentalInformation

Abstract class (with concrete child classes of Observation and Forecast) that groups phenomenon and/or environmental value sets.

Table 679 shows all attributes of EnvironmentalInformation.

**Table 679 – Attributes of Environmental::EnvironmentalInformation**

| name | mult | type | description |
|---|---|---|---|
| created | 0..1 | DateTime | The timestamp of when the forecast was created |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 680 shows all association ends of EnvironmentalInformation with other classes.

**Table 680 – Association ends of Environmental:: EnvironmentalInformation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Environmental data provider supplying this environmental information. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Environmental analog associated with this observation or forecast. |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | Environmental discrete (integer) associated with this observation or forecast. |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | Environmental event to which this forecast or observation relates. |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | Environmental string measurement associated with this forecast or observation. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.17 EnvironmentalLocationType root class

Type of environmental location. Used when an environmental alert or phenomenon has multiple locations associated with it.

Table 681 shows all attributes of EnvironmentalLocationType.

**Table 681 – Attributes of Environmental::EnvironmentalLocationType**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | LocationKind | The kind of location. Typical values might be center, extent, primary, secondary, etc. |

Table 682 shows all association ends of EnvironmentalLocationType with other classes.

**Table 682 – Association ends of Environmental:: EnvironmentalLocationType with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | Location | 0..1 | Location | Location of this instance of ths kind of environmental location. |
| 1..* | EnvironmentalAlert | 0..* | EnvironmentalAlert | Environmental alert applying to location of this type. |
| 0..* | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | Environmental phenomenon for which this location is of relevance. |

### 6.6.18 EnvironmentalPhenomenon root class

The actual or forecast characteristics of an environmental phenomenon at a specific point in time (or during a specific time interval) that may have both a center and area/line location.

Table 683 shows all attributes of EnvironmentalPhenomenon.

**Table 683 – Attributes of Environmental::EnvironmentalPhenomenon**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | The timestamp of the phenomenon as a single point or time interval. |

Table 684 shows all association ends of EnvironmentalPhenomenon with other classes.

**Table 684 – Association ends of Environmental:: EnvironmentalPhenomenon with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | The forecast or observation of which this phenomenon description is a part. |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Location of relevance to this environmental phenomenon. |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | The classification of this phenomenon. |

### 6.6.19 EnvironmentalStringMeasurement

String measurement of relevance in the environmental domain.

Table 685 shows all attributes of EnvironmentalStringMeasurement.

**Table 685 – Attributes of Environmental::EnvironmentalStringMeasurement**

| name | mult | type | description |
|---|---|---|---|
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 686 shows all association ends of EnvironmentalStringMeasurement with other classes.

**Table 686 – Association ends of Environmental:: EnvironmentalStringMeasurement with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Classification condition which this string measurement helps define. |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Observation or forecast with which this environmental string is associated. |
| 1..1 | StringMeasurementValues | 0..* | StringMeasurementValue | inherited from: StringMeasurement |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.20 Fire

A fire, often uncontrolled, covering an area of land which typically contains combustible vegetation. Associated location information is assumed to describe the total area burned as of a specified time.

Table 687 shows all attributes of Fire.

**Table 687 – Attributes of Environmental::Fire**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 688 shows all association ends of Fire with other classes.

**Table 688 – Association ends of Environmental::Fire with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.21 Flood

A flood, an overflowing of a large amount of water beyond its normal confines, esp. over what is normally dry land.

Table 689 shows all attributes of Flood.

**Table 689 – Attributes of Environmental::Flood**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 690 shows all association ends of Flood with other classes.

**Table 690 – Association ends of Environmental::Flood with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.22 Forecast

A forecast group of value sets and/or phenomena characteristics.

Table 691 shows all attributes of Forecast.

**Table 691 – Attributes of Environmental::Forecast**

| name | mult | type | description |
|---|---|---|---|
| validFor | 0..1 | DateTimeInterval | The interval for which the forecast is valid. For example, a forecast issued now for tomorrow might be valid for the next 2 hours. |
| created | 0..1 | DateTime | inherited from: EnvironmentalInformation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 692 shows all association ends of Forecast with other classes.

**Table 692 – Association ends of Environmental::Forecast with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | inherited from: EnvironmentalInformation |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | inherited from: EnvironmentalInformation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.23 GeosphericAnalog

Analog (float) measuring a geospheric condition.

Table 693 shows all attributes of GeosphericAnalog.

**Table 693 – Attributes of Environmental::GeosphericAnalog**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | GeosphericAnalogKind | Kind of geospheric analog. |
| maxValue | 0..1 | Float | inherited from: Analog |
| minValue | 0..1 | Float | inherited from: Analog |
| normalValue | 0..1 | Float | inherited from: Analog |
| positiveFlowIn | 0..1 | Boolean | inherited from: Analog |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 694 shows all association ends of GeosphericAnalog with other classes.

**Table 694 – Association ends of Environmental::GeosphericAnalog with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoring Station | 0..1 | EnvironmentalMonitoring Station | inherited from: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | inherited from: EnvironmentalAnalog |
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | inherited from: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | inherited from: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | inherited from: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.24 GeosphericPhenomenon

A geospheric phenomenon.

Table 695 shows all attributes of GeosphericPhenomenon.

**Table 695 – Attributes of Environmental::GeosphericPhenomenon**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 696 shows all association ends of GeosphericPhenomenon with other classes.

**Table 696 – Association ends of Environmental:: GeosphericPhenomenon with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.25 Hurricane

A hurricane, a subtype of cyclone occurring in the North Atlantic Ocean or North-eastern Pacific Ocean whose intensity is measured using the Saffir-Simpson Hurricane Scale.

Table 697 shows all attributes of Hurricane.

**Table 697 – Attributes of Environmental::Hurricane**

| name | mult | type | description |
|---|---|---|---|
| category | 0..1 | Integer | The hurricane's category during the time interval, using Saffir-Simpson Hurricane Wind Scale, a 1 to 5 rating based on a hurricane's sustained wind speed. |
| centralPressure | 0..1 | Pressure | inherited from: Cyclone |
| maxSurfaceWindSpeed | 0..1 | Speed | inherited from: Cyclone |
| windForce | 0..1 | Integer | inherited from: Cyclone |
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 698 shows all association ends of Hurricane with other classes.

**Table 698 – Association ends of Environmental::Hurricane with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.26 HydrosphericAnalog

Analog (float) measuring a hydrospheric condition.

Table 699 shows all attributes of HydrosphericAnalog.

**Table 699 – Attributes of Environmental::HydrosphericAnalog**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | HydrosphericAnalogKind | Kind of hydrospheric analog. |
| maxValue | 0..1 | Float | inherited from: Analog |
| minValue | 0..1 | Float | inherited from: Analog |
| normalValue | 0..1 | Float | inherited from: Analog |
| positiveFlowIn | 0..1 | Boolean | inherited from: Analog |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 700 shows all association ends of HydrosphericAnalog with other classes.

**Table 700 – Association ends of Environmental::HydrosphericAnalog with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoringStation | inherited from: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | inherited from: EnvironmentalAnalog |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | inherited from: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | inherited from: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | inherited from: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.27 HydrosphericPhenomenon

A hydrospheric phenomenon.

Table 701 shows all attributes of HydrosphericPhenomenon.

**Table 701 – Attributes of Environmental::HydrosphericPhenomenon**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 702 shows all association ends of HydrosphericPhenomenon with other classes.

**Table 702 – Association ends of Environmental:: HydrosphericPhenomenon with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKi nd | 0..* | EnvironmentalLocationT ype | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassificati on | 0..1 | PhenomenonClassificati on | inherited from: EnvironmentalPhenomenon |

### 6.6.28 Landslide

A landslide, a large mass of rocks and earth that suddenly and quickly moves down the side of a mountain or hill.

Table 703 shows all attributes of Landslide.

**Table 703 – Attributes of Environmental::Landslide**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 704 shows all association ends of Landslide with other classes.

**Table 704 – Association ends of Environmental::Landslide with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.29 LightningStrike

A cloud-to-ground lightning strike at a particular location.

Table 705 shows all attributes of LightningStrike.

**Table 705 – Attributes of Environmental::LightningStrike**

| name | mult | type | description |
|---|---|---|---|
| errorEllipseConfidence | 0..1 | PerCent | Likelihood that strike fell within errorEllipse. |
| errorEllipseMajorSemiAxis | 0..1 | Length | Length of major semi-axis (longest radius) of the error ellipse. |
| errorEllipseMinorSemiAxis | 1..1 | Length | Length of minor semi-axis (shortest radius) of the error ellipse. |
| errorEllipseOrientation | 0..1 | Bearing | The orientation of the major semi- axis in degrees from True North. |
| negativePolarity | 0..1 | Boolean | The polarity of the strike, with T meaning negative. About 90% of all lightning strokes are negative strokes, meaning that they were initiated by a large concentration of negative charge in the cloud-base; this tends to induce an area of positive charge on the ground. |
| peakAmplitude | 0..1 | CurrentFlow | Peak current of strike. |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 706 shows all association ends of LightningStrike with other classes.

**Table 706 – Association ends of Environmental::LightningStrike with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.30 MagneticStorm

A magnetic storm, a temporary disturbance of the earth's magnetic field, induced by radiation and streams of charged particles from the sun.

Table 707 shows all attributes of MagneticStorm.

**Table 707 – Attributes of Environmental::MagneticStorm**

| name | mult | type | description |
|---|---|---|---|
| changeDst | 0..1 | MagneticField | Change in the disturbance – storm time (Dst) index. The size of a geomagnetic storm is classified as:<br>– moderate ( -50 nT >minimum of Dst > -100 nT)<br>– intense (-100 nT > minimum Dst > -250 nT) or<br>– super-storm ( minimum of Dst < -250 nT). |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 708 shows all association ends of MagneticStorm with other classes.

**Table 708 – Association ends of Environmental::MagneticStorm with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.31 Observation

Observed (actual non-forecast) values sets and/or phenomena characteristics.

Table 709 shows all attributes of Observation.

**Table 709 – Attributes of Environmental::Observation**

| name | mult | type | description |
|---|---|---|---|
| created | 0..1 | DateTime | inherited from: EnvironmentalInformation |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 710 shows all association ends of Observation with other classes.

**Table 710 – Association ends of Environmental::Observation with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | inherited from: EnvironmentalInformation |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | inherited from: EnvironmentalInformation |
| 0..1 | EnvironmentalPhenome | 0..* | EnvironmentalPhenome | inherited from: EnvironmentalInformation |

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| | non | | non | |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | inherited from: EnvironmentalInformation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.32 PhenomenonClassification

A pre-defined phenomenon classification as defined by a particular authority.

Table 711 shows all attributes of PhenomenonClassification.

**Table 711 – Attributes of Environmental::PhenomenonClassification**

| name | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 712 shows all association ends of PhenomenonClassification with other classes.

**Table 712 – Association ends of Environmental:: PhenomenonClassification with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataAuthority | 0..1 | EnvironmentalDataAuthority | Authority defining this environmental phenomenon. |
| 0..1 | ClassificationCondition | 0..* | ClassificationCondition | Condition contributing to the classification of this phenomenon. |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.33 ReportingCapability root class

Definition of one set of reporting capabilities for this monitoring station. The associated EnvironmentalValueSets describe the maximum range of possible environmental values the station is capable of returning. This attribute is intended primarily to assist a utility in managing its stations.

Table 713 shows all attributes of ReportingCapability.

**Table 713 – Attributes of Environmental::ReportingCapability**

| name | mult | type | description |
|---|---|---|---|
| reportingIntervalPeriod | 0..1 | Integer | Number of units of time making up reporting period. |
| reportingIntervalType | 0..1 | CalculationIntervalUnitKind | Unit of time in which reporting period is expressed. |
| reportingMethod | 0..1 | ReportingMethodKind | Indicates how the weather station reports observations. |

Table 714 shows all association ends of ReportingCapability with other classes.

**Table 714 – Association ends of Environmental::ReportingCapability with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 1..1 | EnvironmentalMonitoringStation | The environmental monitoring station to which this set of reporting capabilities belong. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | One of the environmental value sets expressing one of the reporting capabilities. |

### 6.6.34 SpaceAnalog

Analog (float) measuring a space (extra-terrestrial) condition.

Table 715 shows all attributes of SpaceAnalog.

**Table 715 – Attributes of Environmental::SpaceAnalog**

| name | mult | type | description |
|---|---|---|---|
| kind | 0..1 | SpaceAnalogKind | Kind of space analog. |
| maxValue | 0..1 | Float | inherited from: Analog |
| minValue | 0..1 | Float | inherited from: Analog |
| normalValue | 0..1 | Float | inherited from: Analog |
| positiveFlowIn | 0..1 | Boolean | inherited from: Analog |
| measurementType | 0..1 | String | inherited from: Measurement |
| phases | 0..1 | PhaseCode | inherited from: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | inherited from: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | inherited from: Measurement |
| unitSymbol | 0..1 | UnitSymbol | inherited from: Measurement |
| aliasName | 0..1 | String | inherited from: IdentifiedObject |
| description | 0..1 | String | inherited from: IdentifiedObject |
| mRID | 0..1 | String | inherited from: IdentifiedObject |
| name | 0..1 | String | inherited from: IdentifiedObject |

Table 716 shows all association ends of SpaceAnalog with other classes.

**Table 716 – Association ends of Environmental::SpaceAnalog with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoring Station | 0..1 | EnvironmentalMonitoring Station | inherited from: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | inherited from: EnvironmentalAnalog |
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | inherited from: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | inherited from: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | inherited from: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | inherited from: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | inherited from: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | inherited from: IdentifiedObject |
| 1..1 | Names | 0..* | Name | inherited from: IdentifiedObject |
| 0..1 | PropertiesCIMDataObjec t | 0..1 | DataSetMember | inherited from: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | inherited from: IdentifiedObject |

### 6.6.35 SpacePhenomenon

An extra-terrestrial phenomenon.

Table 717 shows all attributes of SpacePhenomenon.

**Table 717 – Attributes of Environmental::SpacePhenomenon**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 718 shows all association ends of SpacePhenomenon with other classes.

**Table 718 – Association ends of Environmental::SpacePhenomenon with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKi nd | 0..* | EnvironmentalLocationT ype | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassificati on | 0..1 | PhenomenonClassificati on | inherited from: EnvironmentalPhenomenon |

### 6.6.36 Tornado

A tornado, a violent destructive whirling wind accompanied by a funnel-shaped cloud that progresses in a narrow path over the land.

Table 719 shows all attributes of Tornado.

**Table 719 – Attributes of Environmental::Tornado**

| name | mult | type | description |
|---|---|---|---|
| fScale | 0..1 | FScale | Fujita scale (referred to as EF-scale starting in 2007) for the tornado. |
| width | 1..1 | Length | Width of the tornado during the time interval. |
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 720 shows all association ends of Tornado with other classes.

**Table 720 – Association ends of Environmental::Tornado with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.37 TropicalCycloneAustralia

A tropical cyclone, a subtype of cyclone that forms to the east of 90°E in the Southern Hemisphere whose intensity is measured by the Australian tropical cyclone intensity scale.

Table 721 shows all attributes of TropicalCycloneAustralia.

**Table 721 – Attributes of Environmental::TropicalCycloneAustralia**

| name | mult | type | description |
|---|---|---|---|
| category | 0..1 | Integer | Strength of tropical cyclone during the time interval, based on Australian Bureau of Meteorology Category System where:<br>1 – tropical cyclone, with typical gusts over flat land 90-125 km/h<br>2 – tropical cyclone, with typical gusts over flat land 125-164 km/h<br>3 – severe tropical cyclone, with typical gusts over flat land 165-224 km/h<br>4 – severe tropical cyclone, with typical gusts over flat land 225-279 km/h<br>5 – severe tropical cyclone, with typical gusts over flat land greater than 280 km/h. |
| centralPressure | 0..1 | Pressure | inherited from: Cyclone |
| maxSurfaceWindSpeed | 0..1 | Speed | inherited from: Cyclone |
| windForce | 0..1 | Integer | inherited from: Cyclone |

| name | mult | type | description |
|---|---|---|---|
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 722 shows all association ends of TropicalCycloneAustralia with other classes.

**Table 722 – Association ends of Environmental:: TropicalCycloneAustralia with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.38 Tsunami

A tsunami (tidal wave), a long high sea wave caused by an earthquake, submarine landslide, or other disturbance.

Table 723 shows all attributes of Tsunami.

**Table 723 – Attributes of Environmental::Tsunami**

| name | mult | type | description |
|---|---|---|---|
| intensity | 0..1 | Integer | Tsunami intensity on the Papadopoulos-Imamura tsunami intensity scale. Possible values are 1-12, corresponding to I-XII. |
| magnitude | 0..1 | Float | Tsunami magnitude in the Tsunami Magnitude Scale (Mt). Is greater than zero. |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 724 shows all association ends of Tsunami with other classes.

**Table 724 – Association ends of Environmental::Tsunami with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.39 VolcanicAshCloud

An ash cloud formed as a result of a volcanic eruption.

Table 725 shows all attributes of VolcanicAshCloud.

**Table 725 – Attributes of Environmental::VolcanicAshCloud**

| name | mult | type | description |
|---|---|---|---|
| density | 0..1 | ParticulateDensity | Particulate density of the ash cloud during the time interval. |
| particleSize | 0..1 | Length | The diameter of the particles during the time interval. |
| altitude | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | inherited from: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | inherited from: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | inherited from: AtmosphericPhenomenon |
| speed | 0..1 | Speed | inherited from: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

Table 726 shows all association ends of VolcanicAshCloud with other classes.

**Table 726 – Association ends of Environmental::VolcanicAshCloud with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | inherited from: EnvironmentalPhenomenon |

### 6.6.40 Whirlpool

A whirlpool, a rapidly rotating mass of water in a river or sea into which objects may be drawn, typically caused by the meeting of conflicting currents.

Table 727 shows all attributes of Whirlpool.

**Table 727 – Attributes of Environmental::Whirlpool**

| name | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | inherited from: EnvironmentalPhenomenon |

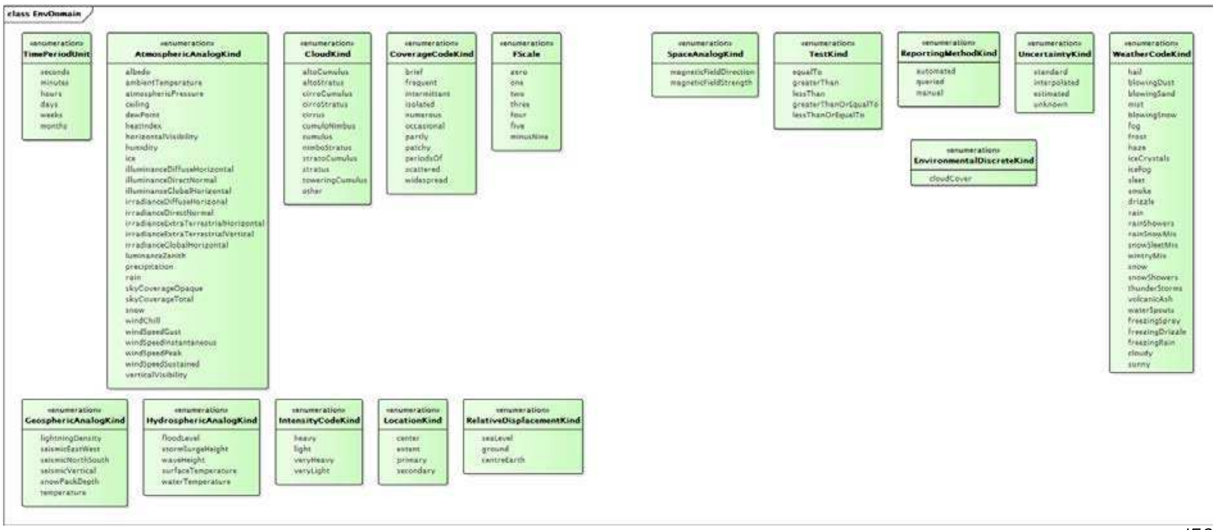
Table 728 shows all association ends of Whirlpool with other classes.

**Table 728 – Association ends of Environmental::Whirlpool with other classes**

| mult from | name | mult to | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformatio n | 0..1 | EnvironmentalInformatio n | inherited from: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKi nd | 0..* | EnvironmentalLocationT ype | inherited from: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassificati on | 0..1 | PhenomenonClassificati on | inherited from: EnvironmentalPhenomenon |

### 6.6.41 Package EnvDomain

#### 6.6.41.1 General



IEC

**Figure 80 – Class diagram EnvDomain::EnvDomain**

Figure 80: Illustrates the data types and enumerations which support the Environmental package classes in addition to the data types and enumerations which are contained within the IEC 61970, IEC 61968 and IEC 62325 packages.

#### 6.6.41.2 RelativeDisplacement compound

Vertical displacement relative to either sealevel, ground or the center of the earth.

Table 729 shows all attributes of RelativeDisplacement.

**Table 729 – Attributes of EnvDomain::RelativeDisplacement**

| name | mult | type | description |
|---|---|---|---|
| displacement | 0..1 | Displacement | |
| kind | 0..1 | RelativeDisplacementKind | |

#### 6.6.41.3 AtmosphericAnalogKind enumeration

Kinds of analogs (floats)  measuring an atmospheric condition.

Table 730 shows all literals of AtmosphericAnalogKind.

**Table 730 – Literals of EnvDomain::AtmosphericAnalogKind**

| literal | value | description |
|---|---|---|
| albedo | | |
| ambientTemperature | | The temperature measured by a thermometer exposed to the air in a place sheltered from direct solar radiation. Also known as "dry bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air. |
| atmosphericPressure | | |
| ceiling | | |
| dewPoint | | The temperature to which air must be cooled at constant pressure and constant water-vapor content in order for saturation to occur. In other words, it is the temperature at which water vapor starts to condense out of the air. |
| heatIndex | | The temperature of how hot it "feels like" for a given combination of warm air temperature and relative humidity. |
| horizontalVisibility | | |
| humidity | | |
| ice | | |
| illuminanceDiffuseHorizontal | | |
| illuminanceDirectNormal | | |
| illuminanceGlobalHorizontal | | |
| irradianceDiffuseHorizonal | | |
| irradianceDirectNormal | | |
| irradianceExtraTerrestrialHorizontal | | |
| irradianceExtraTerrestrialVertical | | |
| irradianceGlobalHorizontal | | |
| luminanceZenith | | |
| precipitation | | |
| rain | | |
| skyCoverageOpaque | | |
| skyCoverageTotal | | |
| snow | | Snow amount over a specified period of time. |
| windChill | | The temperature of how cold it "feels like" based on the rate of heat loss from exposed skin caused by the effects of wind and cold temperatures. |
| windSpeedGust | | Maximum instantaneous wind speed in the 10 minute period preceding a moment in time so long as more than 10 knots of difference has been exhibited between peaks and lulls during that 10 minute time period.<br>0 value means no gusts during preceding 10 minute period. |
| windSpeedInstantaneous | | Wind speed at a moment in time. |
| windSpeedPeak | | Peak instantaneous wind speed in the 60 minutes preceding a moment in time as long as peak speed greater than 25 knots.<br>0 value means speed did not exceed 25 knots during preceding 60 minutes. |
| windSpeedSustained | | Average instantaneous wind speed over the 2-minute time period preceding a moment in time. |
| verticalVisibility | | |

#### 6.6.41.4 CloudKind enumeration

Kind of cloud.

Table 731 shows all literals of CloudKind.

**Table 731 – Literals of EnvDomain::CloudKind**

| literal | value | description |
|---|---|---|
| altoCumulus | | |
| altoStratus | | |
| cirroCumulus | | |
| cirroStratus | | |
| cirrus | | |
| cumuloNimbus | | |
| cumulus | | |
| nimboStratus | | |
| stratoCumulus | | |
| stratus | | |
| toweringCumulus | | |
| other | | |

#### 6.6.41.5 CoverageCodeKind enumeration

Kinds of weather condition coverage.

Table 732 shows all literals of CoverageCodeKind.

**Table 732 – Literals of EnvDomain::CoverageCodeKind**

| literal | value | description |
|---|---|---|
| brief | | |
| frequent | | |
| intermittant | | |
| isolated | | |
| numerous | | |
| occasional | | |
| partly | | |
| patchy | | |
| periodsOf | | |
| scattered | | |
| widespread | | |

#### 6.6.41.6 EnvironmentalDiscreteKind enumeration

Discrete (integer) measuring an environmental condition.

Table 733 shows all literals of EnvironmentalDiscreteKind.

**Table 733 – Literals of EnvDomain::EnvironmentalDiscreteKind**

| literal | value | description |
|---|---|---|
| cloudCover | | Cloud cover in octa. |

### 6.6.41.7 FScale enumeration

Fujita scale (referred to as EF-scale starting in 2007) for tornado damage. A set of wind estimates (not measurements) based on damage. It uses three-second gusts estimated at the point of damage based on a judgment of 8 levels of damage to 28 indicators. These estimates vary with height and exposure.

The 3 second gust is not the same wind as in standard surface observations.

Enumerations based on NOAA conventions.

Table 734 shows all literals of FScale.

**Table 734 – Literals of EnvDomain::FScale**

| literal | value | description |
|---|---|---|
| zero | | 65-85 mph 3-second gust. |
| one | | 86-110 mph 3-second gust. |
| two | | 111-135 mph 3-second gust. |
| three | | 136-165 mph 3-second gust. |
| four | | 166-200 mph 3-second gust. |
| five | | Over 200 mph 3-second gust. |
| minusNine | | Unknown. |

### 6.6.41.8 GeosphericAnalogKind enumeration

Kinds of analogs (floats) measuring a geospheric condition.

Table 735 shows all literals of GeosphericAnalogKind.

**Table 735 – Literals of EnvDomain::GeosphericAnalogKind**

| literal | value | description |
|---|---|---|
| lightningDensity | | Flash rate in strikes/hour/km2. |
| seismicEastWest | | |
| seismicNorthSouth | | |
| seismicVertical | | |
| snowPackDepth | | |
| temperature | | |

### 6.6.41.9 HydrosphericAnalogKind enumeration

Kinds of analogs (floats) measuring a hydrospheric condition.

Table 736 shows all literals of HydrosphericAnalogKind.

**Table 736 – Literals of EnvDomain::HydrosphericAnalogKind**

| literal | value | description |
|---|---|---|
| floodLevel | | |
| stormSurgeHeight | | |
| waveHeight | | |
| surfaceTemperature | | |
| waterTemperature | | |

### 6.6.41.10 IntensityCodeKind enumeration

Kinds of weather condition intensity.

Table 737 shows all literals of IntensityCodeKind.

**Table 737 – Literals of EnvDomain::IntensityCodeKind**

| literal | value | description |
|---|---|---|
| heavy | | |
| light | | |
| veryHeavy | | |
| veryLight | | |

### 6.6.41.11 LocationKind enumeration

The nature of the location being defined for an environmental entity. Possible values are center, perimeter, primary, secondary.

Table 738 shows all literals of LocationKind.

**Table 738 – Literals of EnvDomain::LocationKind**

| literal | value | description |
|---|---|---|
| center | | The center of a phenomenon. Will typically be used with a Location with a single PositionPoint instance. |
| extent | | The area or line of a phenomenon, not the center. Will typically be used with a Location with multiple PositionPoint instances. |
| primary | | Primary area to which an environmental alert applies. |
| secondary | | Secondary area to which an environmental alert applies. |

### 6.6.41.12 RelativeDisplacementKind enumeration

The types of relative displacement

Table 739 shows all literals of RelativeDisplacementKind.

**Table 739 – Literals of EnvDomain::RelativeDisplacementKind**

| literal | value | description |
|---|---|---|
| seaLevel | | |
| ground | | |
| centreEarth | | |

#### 6.6.41.13 SpaceAnalogKind enumeration

Kinds of analogs (floats) measuring a space condition.

Table 740 shows all literals of SpaceAnalogKind.

**Table 740 – Literals of EnvDomain::SpaceAnalogKind**

| literal | value | description |
|---|---|---|
| magneticFieldDirection | | |
| magneticFieldStrength | | |

#### 6.6.41.14 TestKind enumeration

The test applied to determine if the condition is met.

Table 741 shows all literals of TestKind.

**Table 741 – Literals of EnvDomain::TestKind**

| literal | value | description |
|---|---|---|
| equalTo | | |
| greaterThan | | |
| lessThan | | |
| greaterThanOrEqualTo | | |
| lessThanOrEqualTo | | |

#### 6.6.41.15 TimePeriodUnit enumeration

Units in which reporting frequency is specified.

Table 742 shows all literals of TimePeriodUnit.

**Table 742 – Literals of EnvDomain::TimePeriodUnit**

| literal | value | description |
|---|---|---|
| seconds | | |
| minutes | | |
| hours | | |
| days | | |
| weeks | | |
| months | | |

#### 6.6.41.16 ReportingMethodKind enumeration

Method by which information is gathered from station.

Table 743 shows all literals of ReportingMethodKind.

**Table 743 – Literals of EnvDomain::ReportingMethodKind**

| literal | value | description |
|---|---|---|
| automated | | Station automatically reports. |
| queried | | Station must be queried to obtain observations. |
| manual | | Station must be physically visited to gather observations. |

#### 6.6.41.17 UncertaintyKind enumeration

The type of uncertainty for a reading.

Table 744 shows all literals of UncertaintyKind.

**Table 744 – Literals of EnvDomain::UncertaintyKind**

| literal | value | description |
|---|---|---|
| standard | | The value has standard uncertainty consistent with National Weather Service practises given the instrument and manner of observation. |
| interpolated | | The value is interpolated. |
| estimated | | The value is estimated. |
| unknown | | The process of value calculation or measurement is unknown. |

#### 6.6.41.18 WeatherCodeKind enumeration

Kinds of weather conditions.

Table 745 shows all literals of WeatherCodeKind.

**Table 745 – Literals of EnvDomain::WeatherCodeKind**

| literal | value | description |
|---|---|---|
| hail | | "A" weather code ("Hail"). |
| blowingDust | | "BD" weather code ("Blowing Dust"). |
| blowingSand | | "BN" weather code ("Blowing Sand"). |
| mist | | "BR" weather code ("Mist"), |
| blowingSnow | | "BS" weather code ("Blowing Snow"). |
| fog | | "F weather code ("Fog |
| frost | | "FR" weather code ("Frost"). |
| haze | | "H" weather code ("Haze"). |
| iceCrystals | | "IC" weather code ("Ice Crystals"). |
| iceFog | | "IF" weather code ("Ice Fog"). |
| sleet | | "IP" weather code ("Ice Pellets/Sleet"). |
| smoke | | "K" weather code ("Smoke"). |
| drizzle | | "L" weather code ("Drizzle"). |

| literal | value | description |
|---|---|---|
| rain | | "R" weather code ("Rain"). |
| rainShowers | | "RW" weather code ("Rain Showers"). |
| rainSnowMix | | "RS" weather code ("Rain/Snow Mix"). |
| snowSleetMix | | "SI" weather code ("Snow/Sleet Mix"). |
| wintryMix | | "WS" weather code ("Wintry Mix"). |
| snow | | "S" weather code ("Snow"). |
| snowShowers | | "SW" weather code ("Snow Showers"). |
| thunderStorms | | "T" weather code ("Thunder Storms"). |
| volcanicAsh | | "VA" weather code ("Volcanic Ash"). |
| waterSpouts | | "WP" weather code ("Water Spouts") |
| freezingSpray | | "ZF" weather code ("Freezing Spray"). |
| freezingDrizzle | | "ZL" weather code ("Freezing Drizzle"). |
| freezingRain | | "ZR" weather code ("Freezing Rain"). |
| cloudy | | |
| sunny | | |

#### 6.6.41.19 Bearing datatype

The bearing in degrees (with 360 degrees being True North). Measured in degrees clockwise from True North. 0 degrees indicates no direction being given.

Table 746 shows all attributes of Bearing.

**Table 746 – Attributes of EnvDomain::Bearing**

| name | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=none) |
| unit | 0..1 | UnitSymbol | (const=deg) |
| value | 0..1 | Float | |

#### 6.6.41.20 MagneticField datatype

Magnetic field in nanotesla.

Table 747 shows all attributes of MagneticField.

**Table 747 – Attributes of EnvDomain::MagneticField**

| name | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=n) |
| unit | 0..1 | UnitSymbol | (const=T) |
| value | 0..1 | Float | |

#### 6.6.41.21 ParticulateDensity datatype

Particulate density as kg/m$^3$.

Table 748 shows all attributes of ParticulateDensity.

**Table 748 – Attributes of EnvDomain::ParticulateDensity**

| name | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=none) |
| unit | 0..1 | UnitSymbol | (const=kgPerm3) |
| value | 0..1 | Float | |

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CIM Users Group – The community of CIM users. http://cimug.ucaiug.org

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SOMMAIRE

COMMISSION ÉLECTROTECHNIQUE INTERNATIONALE

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# CADRE POUR LES COMMUNICATIONS POUR LE MARCHÉ DE L'ÉNERGIE –

## Partie 301: Extensions du modèle d'information commun (CIM) pour les marchés

## AVANT-PROPOS

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La Norme internationale IEC 62325-301 a été établie par le comité d'études 57 de l'IEC: Gestion des systèmes de puissance et échanges d'informations associés.

Cette deuxième édition annule et remplace la première édition parue en 2014. Cette édition constitue une révision technique.

Cette édition inclut les modifications techniques majeures suivantes par rapport à l'édition précédente:

a) la prise en charge des conditions environnementales météorologiques;
b) la communication côté demande pour les marchés nord-américains;

c) les réglementations relatives à la transparence et le couplage de marchés fondé sur les flux, ainsi que les codes de réseaux pris en charge pour les marchés européens;
d) les mises à jour de la classe RegisteredResource.

Le texte de cette norme est issu des documents suivants:

| FDIS | Rapport de vote |
|---|---|
| 57/1947/FDIS | 57/1961/RVD |

Le rapport de vote indiqué dans le tableau ci-dessus donne toute information sur le vote ayant abouti à l'approbation de cette norme.

Cette publication a été rédigée selon les Directives ISO/IEC, Partie 2.

Tout au long du présent document, les références sont basées sur le modèle UML. Les références à des paquetages et classes UML spécifiques sont effectuées en citant le nom du paquetage ou de la classe entre guillemets simples (par exemple: 'IEC 61970' ou 'IdentifiedObject').

Une liste de toutes les parties de la série IEC 62325, publiées sous le titre général *Cadre pour les communications pour le marché de l'énergie,* peut être consultée sur le site web de l'IEC.

Le comité a décidé que le contenu de cette publication ne sera pas modifié avant la date de stabilité indiquée sur le site web de l'IEC sous "http://webstore.iec.ch" dans les données relatives à la publication recherchée. À cette date, la publication sera

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- remplacée par une édition révisée, ou
- amendée.

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# INTRODUCTION

La présente partie de l'IEC 62325 est une norme faisant partie d'une série de normes concernant les échanges d'informations relatives au marché déréglementé de l'énergie. Cette série de normes a été initialement fondée sur les travaux réalisés par les gestionnaires de réseaux de transport européens TF 14 EDI (Échange de données informatisé) et dans le cadre des publications de l'Institut de recherches sur la production d'électricité (EPRI – Electric Power Research Institute) TR 1009455 et TR 1011431 (Extensions du CIM pour la prise en charge des opérations du marché, phase 1 et phase 2).

Le principal objectif de la série IEC 62325 est de produire des normes destinées à faciliter l'intégration de logiciels d'application pour le marché, développés de façon indépendante par différents fournisseurs, dans un système de gestion de marché, ainsi que l'intégration des systèmes de gestion de marché et des systèmes participant au marché. Cela s'effectue par la définition d'échanges de messages pour permettre à ces applications ou systèmes d'accéder aux données publiques et d'échanger des informations, indépendamment de la façon dont ces informations sont représentées en interne.

Le modèle d'information commun (CIM) spécifie la base de la sémantique pour cet échange de messages. Les spécifications de profil, qui sont contenues dans d'autres parties de l'IEC 62325, précisent le contenu des messages échangés. Elles prennent en charge les marchés de style européen, fondés sur les règlements européens et sur les concepts d'accès de tiers au réseau et de répartition du marché en zones, et les marchés de style nord-américain, caractérisés par une gestion de la production à J-1 effectuée par un opérateur du marché, par un équilibrage intrajournalier et en temps réel par le biais d'un système de répartition central, et par règlement selon les prix marginaux locaux. Ces marchés sont utilisés aux États-Unis et dans d'autres régions du monde, et ne se limitent pas exclusivement à l'Amérique du Nord.

Le CIM est un modèle abstrait contenant tous les objets principaux d'une entreprise de distribution d'électricité habituellement nécessaires pour représenter les opérations d'une entreprise d'électricité. Ce modèle inclut les classes et les attributs publics de ces objets, ainsi que les relations entre eux.

La série IEC 62325 fait partie d'une série de normes traitant du CIM. L'IEC 61970 définit les interfaces de programmation d'application EMS et l'IEC 61968 définit les interfaces système pour la gestion de distribution. L'IEC 62325-301 correspond à l'IEC 61970-301 et à l'IEC 61968-11, qui décrivent des parties du modèle d'information commun applicables aux interfaces de modélisation des systèmes EMS, DMS et MMS. Alors qu'il existe plusieurs séries de normes de l'IEC qui traitent des différentes parties du CIM, un seul modèle d'information unifié comprenant le CIM est derrière toutes ces séries de normes individuelles.

# CADRE POUR LES COMMUNICATIONS POUR LE MARCHÉ DE L'ÉNERGIE –

# Partie 301: Extensions du modèle d'information commun (CIM) pour les marchés

## 1 Domaine d'application

La présente partie de la norme IEC 62325 spécifie le modèle d'information commun (CIM) relatif aux communications pour le marché de l'énergie.

Le CIM est un modèle abstrait de tous les objets principaux d'une entreprise de distribution d'électricité habituellement impliqués dans les opérations de l'entreprise et dans la gestion du marché de l'électricité. En fournissant une façon normalisée de représenter des ressources de systèmes électriques comme classes et attributs d'objets ainsi que leurs relations, le CIM facilite l'intégration des applications de système de gestion de marché (MMS) développées de façon indépendante par différents fournisseurs, entre des systèmes MMS complets développés de façon indépendante, ou entre un système MMS et d'autres systèmes concernés par différents aspects de la gestion de marché, tels que l'attribution de la capacité, la gestion à J-1, l'équilibrage, le règlement, etc.

Le CIM facilite l'intégration en définissant un langage commun (c'est-à-dire, une sémantique) fondé sur le modèle CIM pour permettre à ces applications ou systèmes d'accéder aux données publiques et d'échanger des informations indépendamment de la représentation interne de ces informations.

Les classes d'objets représentées dans le CIM sont de nature abstraite et peuvent être utilisées dans une large gamme d'applications. L'utilisation du CIM n'est pas limitée à son application dans un système de gestion de marché.

En raison de la taille du CIM complet, les classes d'objets qui le composent sont regroupées en un certain nombre de paquetages logiques, qui représentent chacun une certaine partie du système électrique modélisé. Les collections de ces paquetages sont fournies progressivement sous forme de Normes internationales distinctes. Le présent document particulier spécifie un ensemble de paquetages qui offrent une vue logique sur les aspects fonctionnels de la gestion d'un marché de l'électricité et sur d’autres aspects fonctionnels comprenant les aspects environnementaux qui sont étroitement liés aux marchés de l’électricité et qui sont partagés par toutes les applications. D'autres normes spécifient des aspects plus spécifiques du modèle qui ne sont nécessaires qu'à certaines applications. Le Paragraphe 4.2 définit le découpage actuel des paquetages dans les documents normatifs.

Cette nouvelle édition de l’IEC 62325-301 tient compte de la communication côté demande dans un marché de vente en gros. Les ajouts apportés à l’IEC 62325-301 comprennent la prise en charge de l’enregistrement des ressources et l’enregistrement d’une ressource participant à un marché côté demande ainsi que la prise en charge en vue du déploiement et de l’évaluation des performances des ressources côté demande. Un nouveau paquetage a été inclus dans cette édition de l’IEC 62325-301 afin de prendre en charge les données environnementales (météorologiques). Ce nouveau paquetage ‘Environmental’ fournit la prise en charge des conditions météorologiques, y compris les prévisions, les observations, les mesures, les phénomènes et les alertes. Des mises à jour supplémentaires ont été ajoutées au paquetage ‘MarketManagement’ afin de prendre en charge les réglementations relatives à la transparence, le couplage de marchés fondé sur les flux et les nouveaux codes de réseaux pris en charge pour les marchés européens. Ces mises à jour comprennent de nouvelles classes, de nouveaux attributs et de nouvelles associations dans les paquetages IEC 62325, ainsi que des mises à jour des classes, attributs et associations existants, afin de représenter avec exactitude les cas d'utilisation existants.

## 2 Références normatives

Les documents suivants cités dans le texte constituent, pour tout ou partie de leur contenu, des exigences du présent document. Pour les références datées, seule l'édition citée s'applique. Pour les références non datées, la dernière édition du document de référence s'applique (y compris les éventuels amendements).

IEC 61968-11, *Intégration d'applications pour les services électriques – Interfaces système pour la gestion de distribution – Partie 11: Extensions du modèle d'information commun (CIM) pour la distribution*

IEC TS 61970-2:2004, *Energy management system application program interface (EMS-API) – Part 2: Glossary* (disponible en anglais seulement)

IEC 61970-301, *Interface de programmation d'application pour système de gestion d'énergie (EMS-API) – Partie 301: Base de modèle d'information commun (CIM)*

IEC 61970-302, *Interface de programmation d'application pour système de gestion d'énergie (EMS-API) – Partie 302: CIM pour régimes dynamiques*

IEC 62325-450, *Cadre pour les communications pour le marché de l'énergie – Partie 450: Règles de modélisation de profils et de contextes*

Object management group: *UML 2.0 specification* – disponible à l'adresse <http://www.omg.org>

## 3 Termes, définitions et abréviations

Pour les besoins du présent document, les termes et définitions de l'IEC TS 61970-2 s'appliquent.

L'ISO et l'IEC tiennent à jour des bases de données terminologiques destinées à être utilisées en normalisation, consultables aux adresses suivantes:

- IEC Electropedia: disponible à l'adresse http://www.electropedia.org/
- ISO Online browsing platform: disponible à l'adresse http://www.iso.org/obp

### 3.1 Termes et définitions

**3.1.1**
**interface de programmation d'application**
**API**
ensemble des fonctions publiques offertes par un composant exécutable d'application et destinées à être utilisées par d'autres composants exécutables d'application

Note 1 à l'article: L'abréviation "API" est dérivée du terme anglais développé correspondant "application program interface".

[SOURCE: IEC 61970-2:2004, 3.4]

**3.1.2**
**langage de balisage extensible**
**XML**
langage de marquage qui définit un ensemble de règles pour le codage des documents dans un format lisible à la fois par l'homme et la machine

Note 1 à l'article: L'abréviation "XML" est dérivée du terme anglais développé correspondant "Extensible Markup Language".

**3.1.3**
**description de schéma XML**
**XSD**
langage XML utilisé pour décrire et contrôler le contenu des documents XML

Note 1 à l'article: L'abréviation "XSD" est dérivée du terme anglais développé correspondant "XML schema description".

**3.1.4**
**marchés intrajournaliers**
marchés de programmation intrajournalière qui affinent l'engagement et les programmes produits par les marchés à J-1

**3.1.5**
**prévision de charge**
système de prévision de charge qui prédit la charge pour l'horizon temporel spécifié

**3.1.6**
**prix marginal local**
**LMP**
coût marginal (devise/MWh) qui sert l'incrément suivant de demande au niveau d'un nœud de tarification en cohérence avec les contraintes de transport existantes et les caractéristiques de performances des ressources

Note 1 à l'article: L'abréviation "LMP" est dérivée du terme anglais développé correspondant "Locational Marginal Price".

**3.1.7**
**système de gestion de marché**
**MMS**
système informatique comprenant une plate-forme logicielle offrant les services de support de base et un ensemble d'applications offrant les fonctionnalités nécessaires à une gestion efficace du marché de l'électricité

Note 1 à l'article: L'abréviation "MMS" est dérivée du terme anglais développé correspondant "Market Management System".

Note 2 à l'article: Ces systèmes informatiques intégrés à un marché de l'électricité peuvent comprendre un service de support à l'attribution de la capacité, à la planification de l'énergie, aux services auxiliaires ou autres, aux opérations et aux règlements en temps réel.

**3.1.8**
**programmateur (scheduler)**
système de programmation pour les ressources qui produisent, consomment ou transportent de l'énergie ou des services auxiliaires dans un marché de l'énergie

**3.1.9**
**accès tiers**
concept dans les systèmes de gestion de marché qui permet aux tiers d'utiliser le système de transmission au même titre que les autres

**3.1.10**
**langage de modélisation unifié**
**UML**
langage de modélisation dans le domaine de l'ingénierie logicielle orientée objet qui est géré par l'Object Management Group (OMG)

Note 1 à l'article: L'abréviation "UML" est dérivée du terme anglais développé correspondant "Unified Modeling Language".

**3.1.11**
**schéma RDF**
**RDFS**
recommandation W3C qui fournit des éléments de base pour la description des ontologies

Note 1 à l'article: L'abréviation "RDFS" est dérivée du terme anglais développé correspondant "RDF schema".

**3.1.12**
**langage d'ontologies Web**
**OWL**
recommandation W3C qui fournit un langage de représentation de connaissances pour la création d'ontologies

Note 1 à l'article: L'abréviation "OWL" est dérivée du terme anglais développé correspondant "Ontology Web Language".

### 3.2 Abréviations

| | |
|---|---|
| **CRR** | congestion revenue rights (droits à revenu de congestion) |
| **DAM** | day ahead market (marché à J-1qui détermine l'engagement et les programmes des ressources) |
| **DMS** | distribution management system (système de gestion de la distribution) |
| **EMS** | energy management system (système de gestion d'énergie) |
| **MDMS** | meter data management system (système de gestion des données de compteur) |
| **RTM** | real time markets (marchés en temps réel qui finalisent la répartition pour les ressources système) |

## 4 Spécification du CIM

### 4.1 Notation de modélisation du CIM

Le CIM est défini à l'aide de techniques de modélisation orientées objet. Plus précisément, la spécification du CIM utilise la notation du langage de modélisation unifié (UML), qui définit le CIM comme un groupe de paquetages.

Chaque paquetage du CIM contient un ou plusieurs diagrammes de classe représentant sous forme graphique toutes les classes de ce paquetage et leurs relations. Chaque classe est ensuite définie de façon textuelle en mettant l'accent sur ses attributs et ses relations avec d'autres classes.

La notation UML est décrite dans la norme de l'Object Management Group (OMG) et dans plusieurs autres manuels publiés.

### 4.2 Paquetages CIM

Le CIM est fractionné en un ensemble de paquetages. Un paquetage est un moyen général permettant de grouper des éléments liés à un modèle. Les paquetages ont été choisis pour simplifier la conception, la compréhension et la révision du modèle. Le modèle d'information commun se compose de l'ensemble des paquetages. Des entités peuvent avoir des associations dépassant les limites de plusieurs paquetages. Chaque application peut utiliser des informations représentées dans plusieurs paquetages.

Le CIM complet est fractionné en groupes de paquetages, ce qui facilite la gestion et la maintenance. La norme IEC 62325-301 inclut les paquetages suivants:

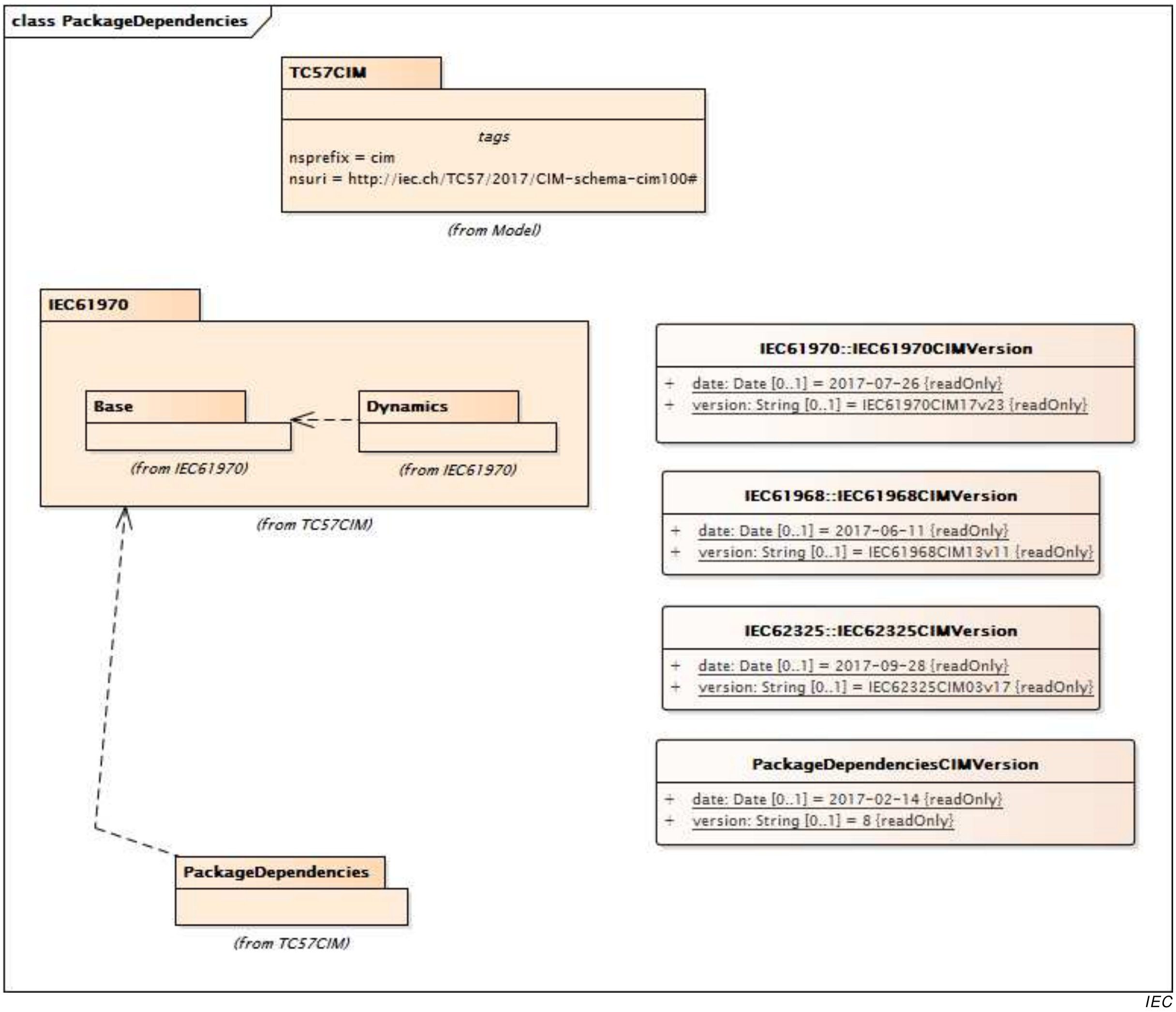
- 'MarketCommon'
- 'MarketManagement'
- 'MarketOperations'
- ‘Environmental’

Les paquetages utilisés par l'IEC 61970-301 et les paquetages utilisés par l'IEC 61968-11 décrivent des parties supplémentaires du CIM qui traitent d'autres vues logiques des opérations d'une entreprise de service public, comprenant les modèles de réseaux, la production, la charge, les indisponibilités, le patrimoine, les emplacements, les activités, les clients, la documentation et la gestion des travaux.

NOTE Les limites d'un paquetage n'impliquent pas de limites pour les applications. Une application peut utiliser des entités CIM de différents paquetages.

L'IEC 62325-301 est publiée sur la base des versions stables et convenues de l'IEC 61970 et de l'IEC 61968.

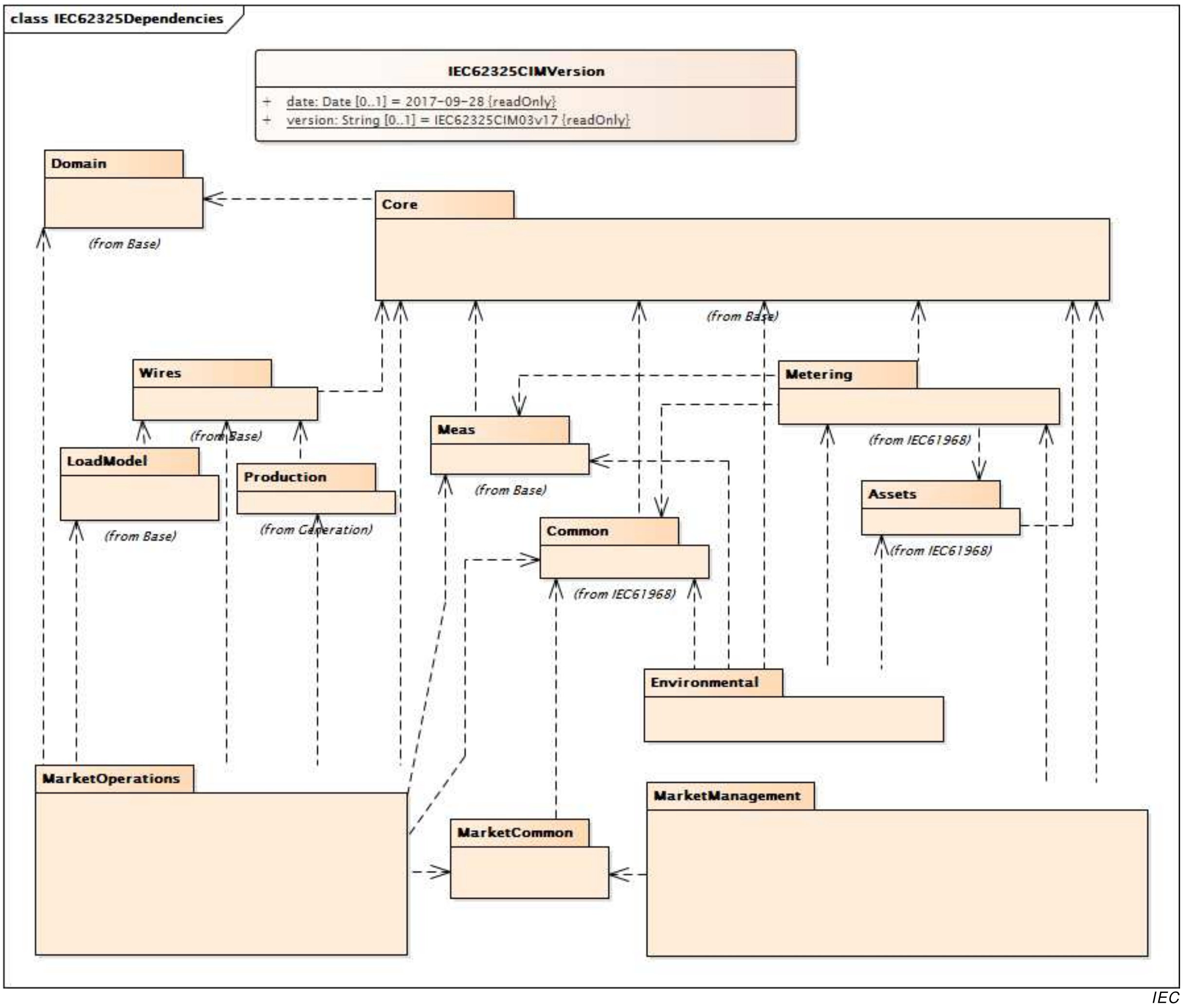
La Figure 1 représente les dépendances de paquetages. Le paquetage 'IEC 62325' et ses sous-paquetages dépendent des paquetages 'IEC 61970' et 'IEC 61968'. Le paquetage 'IEC 61970' est utilisé par la série de normes IEC 61970 comme un modèle central qui représente les ressources des systèmes électriques et leurs relations. Le paquetage 'IEC 61968' est utilisé par la série de normes IEC 61968 qui décrit les vues logiques des opérations d'une entreprise de service public, y compris vue du patrimoine, des emplacements, des activités, des clients, de la documentation, de la gestion des travaux et des modèles de comptage. Les lignes tiretées indiquent une relation de dépendance, la flèche allant du paquetage dépendant vers le paquetage dont il dépend.




| Anglais | Français |
|---|---|
| from | provenant de |

**Figure 1 – Diagramme des dépendances de paquetages 'TC 57CIM'**

La Figure 2 représente les paquetages définis pour l'IEC 62325-301 et leurs relations de dépendance. Les lignes tiretées indiquent une relation de dépendance, la flèche allant du paquetage dépendant vers le paquetage dont il dépend. Les paquetages de l'IEC 62325 incluent les dépendances à l'égard des paquetages décrits dans la série IEC 61970 et la série IEC 61968. Ces dépendances sont représentées dans le diagramme.




| Anglais | Français |
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| from | provenant de |

**Figure 2 – Diagramme des dépendances de paquetages ‘IEC 62325’**

L'Article 6 contient la spécification associée à chacun des paquetages du CIM.

NOTE Le contenu du CIM défini dans la présente spécification a été créé automatiquement à partir de la version IEC62325CIM03v17 du modèle électronique UML du CIM, qui est disponible auprès du groupe des utilisateurs du CIM (CIM Users Group).

## 4.3 Classes CIM et relations

### 4.3.1 Classes

Le ou les diagrammes de classe de chaque paquetage du CIM représentent toutes les classes du paquetage et leurs relations. Lorsque des relations existent avec des classes appartenant à d'autres paquetages, ces classes sont également représentées.

Les classes et les objets modélisent ce qui se trouve dans un système électrique et qui nécessite d’être représenté de la même façon pour les applications du marché. Une classe est une description d'un objet réel, tel qu'une offre, une ressource enregistrée ou une charge enregistrée, qui nécessite d’être représenté comme faisant partie de l'ensemble du marché de l'électricité dans un MMS. Parmi les autres types d'objets figurent des éléments tels que les programmes et les mesures que des applications MMS nécessitent également de traiter, d’analyser et d’enregistrer. De tels objets nécessitent une représentation commune pour atteindre les objectifs de compatibilité de connexion et d'interopérabilité de la présente norme. Un objet particulier d'un système électrique ayant une identité unique est modélisé comme une instance de la classe à laquelle il appartient.

Dans le présent document, lorsqu'un en-tête comprend l'expression “classe racine”, cela signifie que la classe n'a pas d'héritage et n'a pas de stéréotype.

Le CIM est défini pour faciliter l'échange de données. Comme défini dans le présent document, les entités du CIM n'ont pas de comportement. L'IEC 62325-450, *Cadre pour les communications pour le marché de l'énergie – Partie 450: Règles de modélisation de profils et de contextes* définit les principes de modélisation pour générer un profil et les principes du cadre de modélisation pour définir la charge utile de message.

Les classes comportent des attributs qui décrivent les caractéristiques des objets. Chaque classe du CIM contient les attributs décrivant et identifiant une instance spécifique de la classe. Seuls les attributs d'intérêt public pour les applications MMS sont inclus dans les descriptions de classes.

Chaque attribut possède un type qui identifie le type d'attribut dont il s'agit. Les attributs typiques sont de type 'Integer' (entier), 'Float' (flottant), 'Boolean' (booléen), 'String' (chaîne), 'Datetime' (date et heure), 'Date', 'Duration' (durée) et 'Decimal' (décimal), qui sont appelés types primitifs. Cependant, de nombreux types supplémentaires sont définis comme partie intégrante de la spécification CIM. Par exemple, 'RegisteredGenerator' a un attribut 'maxDependableCap' de type 'ActivePower'. La définition de nombreux types partagés est contenue dans le paquetage 'Domain' décrit à l'Article 6. Les stéréotypes 'Primitive', 'enumeration', 'CIMDatatype' et 'Compound' de l'UML sont ajoutés aux classes utilisées comme types:

- Le stéréotype 'CIMDatatype' est utilisé avec une sémantique spécifique du CIM pour décrire un type ayant plusieurs attributs {'value', 'unit', 'multiplier', 'denominatorUnit', 'denominatorMultiplier'}, qui implique une mise en correspondance personnalisée avec des artéfacts de sérialisation tels que RDFS, OWL et XSD. Les classes ayant ces stéréotypes ne participent pas à des relations de généralisation ou d'association et sont simplement utilisées comme types pour des attributs.
- Le stéréotype 'enumeration' est utilisé pour décrire le type d'un attribut avec une liste énumérée de choix.
- Le stéréotype 'Compound' est utilisé pour décrire des ensembles d'attributs liés qui sont couramment réutilisés. Les classes 'Compound' peuvent se composer d'attributs de type 'Primitive', 'enumeration', 'CIMDatatype' ou d'autres classes 'Compound', à condition que les classes 'Compound' ne soient pas récursives.

Tous les attributs du CIM sont implicitement facultatifs en ce sens que les profils utilisant le CIM peuvent les éliminer tous.

Les relations entre les classes révèlent la façon dont elles sont structurées les unes par rapport aux autres. Les classes CIM sont reliées de différentes façons, comme décrit en 4.3.2, 4.3.3 et 4.3.4.

### 4.3.2 Generalization (généralisation)

Une généralisation est une relation entre une classe générale et une classe plus spécifique. La classe la plus spécifique ne peut contenir que des informations supplémentaires. Par exemple, 'RegisteredGenerator' est une sous-classe spécifique de 'RegisteredResource'. La généralisation permet à une classe spécifique d'hériter d'attributs et de relations de toutes les classes plus générales situées au-dessus d'elle.

La Figure 3 est un exemple de généralisation. Dans cet exemple issu du paquetage 'MarketCommon', un 'MarketParticipant' est un type plus spécifique de 'Organisation'. 'Organisation' hérite également de 'IdentifiedObject'.

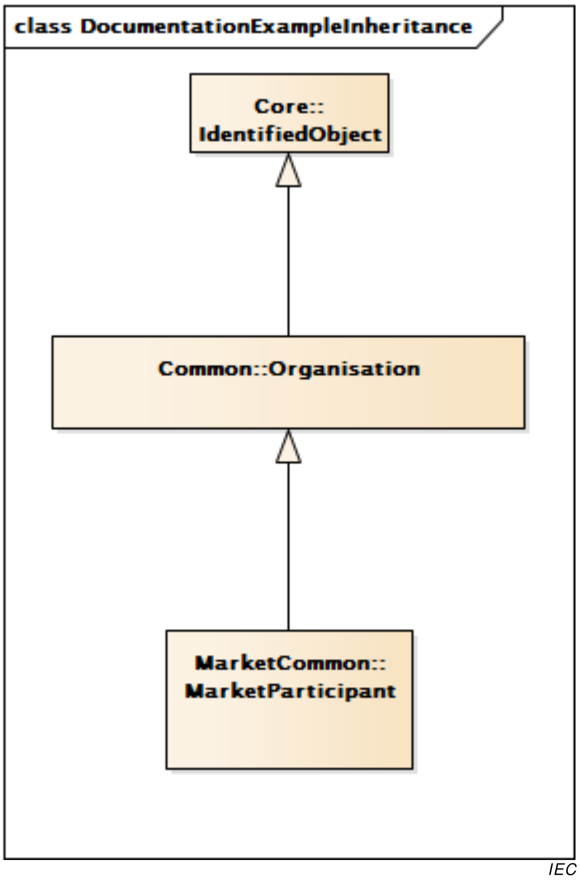



**Figure 3 – Exemple de généralisation**

### 4.3.3 Simple association (association simple)

Une association est une connexion conceptuelle entre des classes. Chaque association a deux “extrémités d'association”. Les “extrémités d'association” étaient appelées “rôles” avant la spécification UML 2.0. Chaque extrémité d'association décrit le rôle que la classe cible (c'est-à-dire, la classe *vers* laquelle l'extrémité d'association part) a en relation avec la classe source (c'est-à-dire, la classe *à partir de* laquelle l'extrémité d'association part). La coutume est d'attribuer aux extrémités d'association le nom de la classe cible avec ou sans qualificatif (verbe, nom ou adverbe). Chaque extrémité d'association possède aussi une cardinalité/multiplicité, qui est une indication du nombre d'objets qui peuvent prendre part à la relation décrite. Dans le CIM, les associations ne sont pas nommées, seules les extrémités d'association le sont. Par exemple, dans le CIM, il existe une association entre un 'MarketParticipant' et un 'MarketRole' (voir la Figure 4, qui est issue du paquetage 'MarketCommon'). La multiplicité est représentée aux deux extrémités de l'association. Dans cet exemple, un objet 'MarketParticipant' peut faire référence à 0 ou à plusieurs objets 'MarketRole', et un 'MarketRole' peut être référencé dans 0 ou plusieurs objets 'MarketParticipant'.

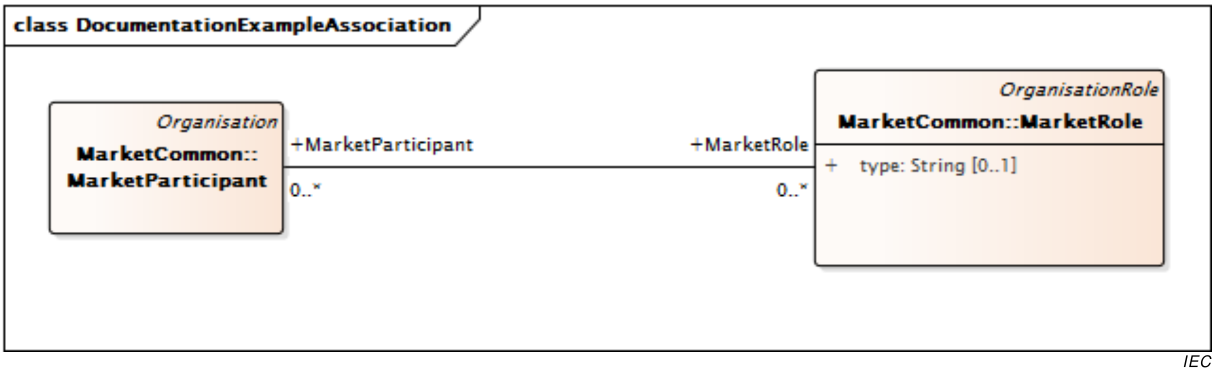



**Figure 4 – Exemple d'association simple**

### 4.3.4 Aggregation (agrégation)

L'agrégation est un cas spécial d'association. L'agrégation indique que la relation entre les classes est une sorte de relation classe entière/classe partielle, dans laquelle la classe entière "consists of" (se compose de) ou "contains" (contient) la classe partielle et cette classe partielle est une "part of" (partie de) la classe entière. La classe partielle n'hérite pas de la classe entière comme dans la généralisation. La Figure 5 représente une agrégation entre la classe 'EnergyProfile' et la classe 'EnergyTransaction', qui est issue du paquetage 'ExternalInputs'. Elle indique également qu'un 'EnergyProfile' peut faire partie d'un seul objet 'EnergyTransaction', mais qu'un objet 'EnergyTransaction' peut contenir n'importe quel nombre d'objets 'EnergyProfile'. Dans le cadre de l'utilisation du CIM comme modèle d'information, l'agrégation n'a pas d'interprétation précise ou formelle au-delà d'une association simple, et est destinée à représenter visuellement une utilisation normale.

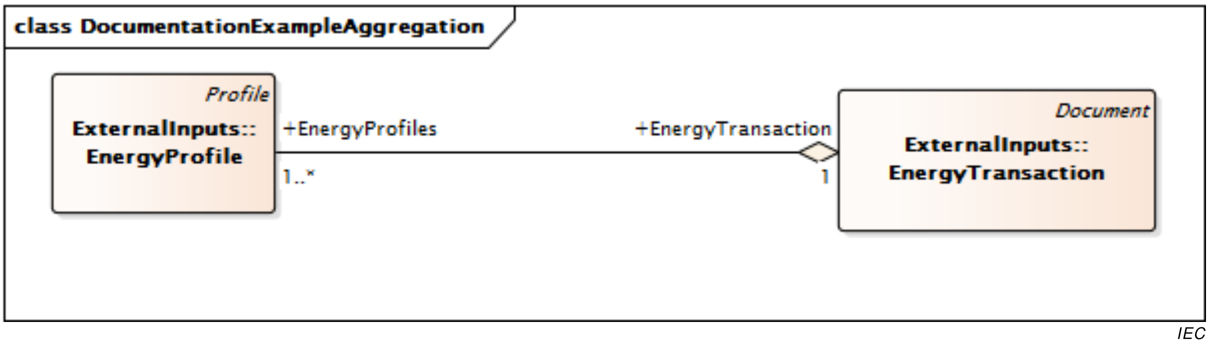



**Figure 5 – Exemple d'agrégation**

## 4.4 Concepts du modèle CIM et exemples

### 4.4.1 Paquetage 'MarketCommon'

Le modèle Common Market décrit les participants du marché ainsi que leurs rôles. Un participant du marché peut jouer plusieurs rôles dans un marché (voir Figure 6).

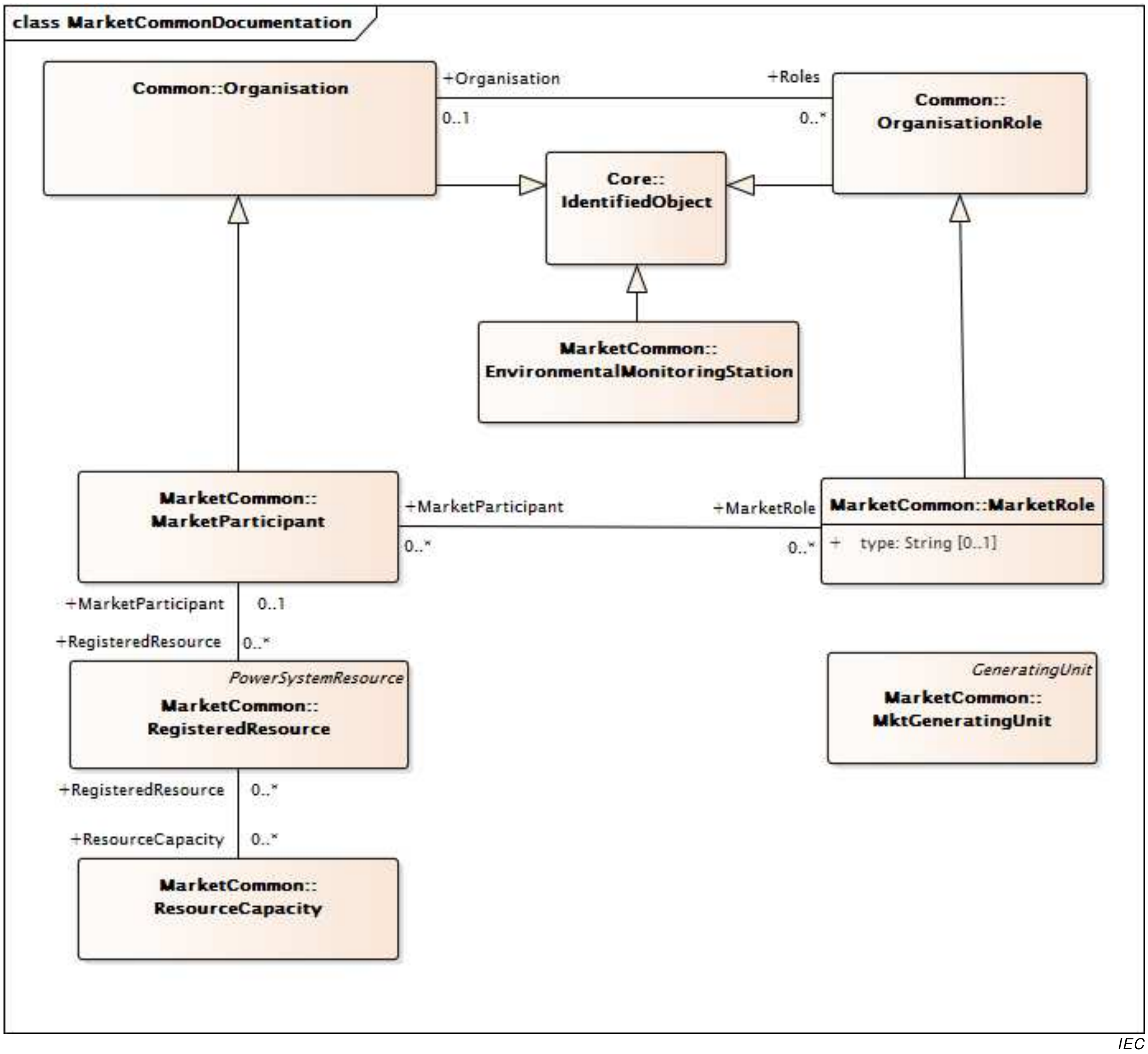


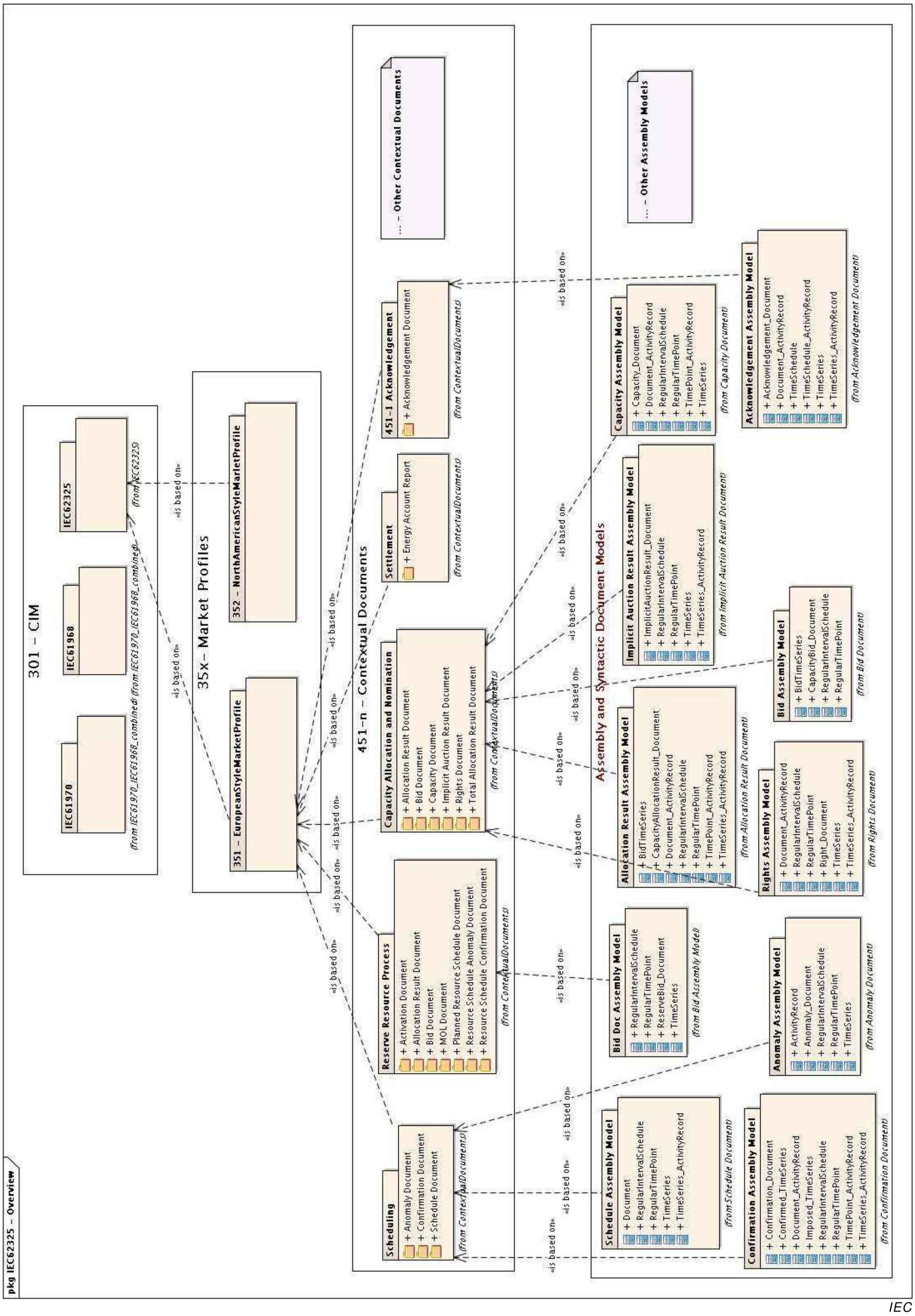

**Figure 6 – Modèle Common Market**

### 4.4.2 Paquetage 'MarketManagement'

Le modèle Market Management décrit un ensemble cohérent de classes à utiliser avec le modèle Common Market pour générer un profil. Le profil doit être utilisé lorsque le marché de l'électricité est principalement basé sur un accès réglementé de tiers au réseau, c'est-à-dire que les gestionnaires de réseaux de transport doivent donner aux fournisseurs d'électricité un accès non discriminatoire:

- au réseau de transport afin de leur permettre de fournir de l'énergie à leurs clients;
- à l'infrastructure qui permet l'échange d'énergie par les transactions de gros et de détail (bilatérales ou par le biais d'une installation d'échange d'énergie entre réseaux)

Le paquetage 'MarketManagement' fait référence à ce qui est ultérieurement appelé le marché de style européen (ESMP). La Figure 7 représente le cadre de modélisation en couches permettant la spécification des messages du CIM aux différentes couches. Ce cadre de modélisation définit la façon dont les concepts du paquetage 'MarketManagement' sont issus du CIM par le biais de la définition d'un profil de marché permettant la production de documents contextualisés pour l'échange d'informations.




**Légende**

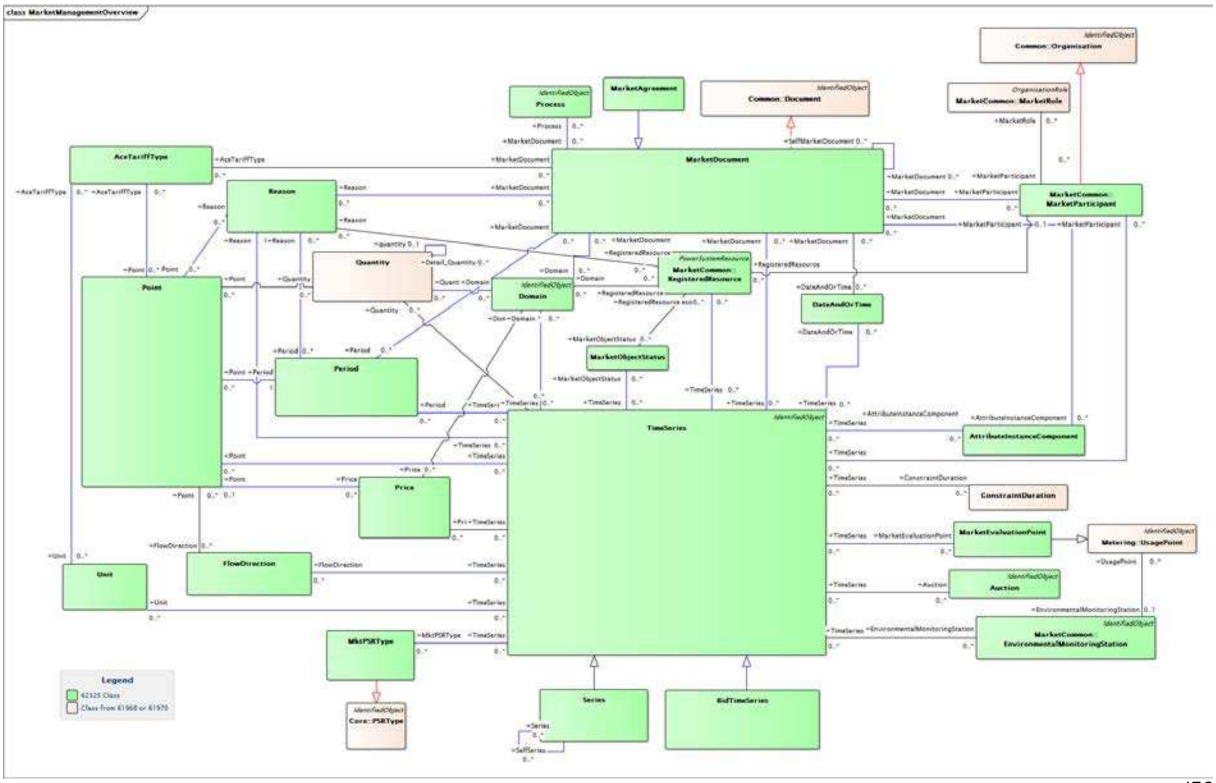
| Anglais | Français |
|---|---|
| From IEC 61970_IEC 61968_combined | À partir de IEC 61970_IEC 61968_combined |
| From IEC 62325 | À partir de IEC 62325 |
| Is based on | Est basé sur |

| Anglais | Français |
|---|---|
| Market Profiles | Profils de marché |
| Contextual Documents | Documents contextuels |
| Assembly and Syntactic Document Models | Modèles de documents d'assemblage et de syntaxe |
| Other Contextual Documents | Autres documents contextuels |
| Other Assembly Models | Autres modèles d'assemblage |
| Scheduling | Programmation |
| Reserve Resource Process | Processus de ressources en réserve |
| Capacity Allocation and Nomination | Attribution de la capacité et désignation |
| Settlement | Règlement |
| 451-1 Acknowledgement | 451-1 Acknowledgement (Acquittement) |
| From ContextualDocuments | À partir de ContextualDocuments |
| Schedule Assembly Model | Modèle d'assemblage de programmation |
| Bid Doc Assembly Model | Modèle d'assemblage de document d'offre |
| Allocation Result Assembly Model | Modèle d'assemblage de résultat de l'attribution |
| Implicit Auction Result Assembly Model | Modèle d'assemblage de résultat des ventes aux enchères implicites |
| Capacity Assembly Model | Modèle d'assemblage de capacité |
| From Schedule Document | À partir du document de programme |
| From Bid Assembly Model | À partir du modèle d'assemblage d'offre |
| From Allocation Result Document | À partir du document de résultat de l'attribution |
| From Implicit Auction Result Document | À partir du document de résultat des ventes aux enchères implicites |
| From Capacity Document | À partir du document de capacité |
| Confirmation Assembly Model | Modèle d'assemblage de confirmation |
| Anomaly Assembly Model | Modèle d'assemblage d'anomalie |
| Rights Assembly Model | Modèle d'assemblage de droits |
| Bid Assembly Model | Modèle d'assemblage d'offre |
| Acknowledgement Assembly Model | Modèle d'assemblage d’acquittement |
| From Confirmation Document | À partir du document de confirmation |
| From Anomaly Document | À partir du document d'anomalie |
| From Rights Document | À partir du document de droits |
| From Bid Document | À partir du document d'offre |
| From Acknowledgement document | À partir du document d’acquittement |

**Figure 7 – Présentation du modèle Market Management**

Comme indiqué dans la Figure 7, un jeu spécifique de documents contextualisés est fourni pour chaque processus métier nécessaire à l'exploitation d'un marché de l'électricité. Comme indiqué dans la Figure 7, les profils de marché sont spécifiés dans l'IEC 62325-351 et les documents contextualisés sont décrits dans les normes IEC 62325-451-1, IEC 62325-451-2, etc.

La Figure 8 représente les classes du paquetage 'MarketManagement'.




IEC

| Anglais | Français |
|---|---|
| Legend | Légende |
| 62325 Class | Classe 62325 |
| Class from 61968 or 61970 | Classe issue de 61968 ou 61970 |

**Figure 8 – Modèle Market Management**

Dans ce modèle Market Management, un rôle clé est donné au concept de 'MarketDocument', c'est-à-dire que les transactions du marché de l'électricité sont basées sur des échanges contractuels d'informations entre les participants du marché par le biais d’un jeu de documents donné en fonction du processus métier.

Comme indiqué dans la Figure 7, un jeu spécifique de documents contextualisés est fourni pour chaque processus métier nécessaire à l'exploitation d'un marché de l'électricité. À titre d'exemple, voici une liste non exhaustive des processus:

- processus d'acquittement: étant donné que l'échange d'informations est basé sur des échanges contractuels, il est exigé d'obtenir un acquittement fonctionnel du document échangé et pas seulement un acquittement technique;
- processus de programmation: ce processus décrit la façon d'échanger les informations relatives aux programmes (programmes de production, programmes de charge, programmes d'échanges commerciaux bilatéraux, programmes d'échange d'énergie entre réseaux, etc.);
- processus de règlement: ce processus décrit comment échanger les informations nécessaires au règlement du marché de l'électricité, c’est-à-dire, la comparaison entre l'énergie programmée et les compteurs relevés;
- processus d'attribution de la capacité de transport et de désignation: ce processus décrit les ventes aux enchères explicites et implicites de la capacité de transport dans le cadre du commerce transfrontalier. Il inclut également le marché secondaire des droits de capacité, c'est-à-dire la revente de droits de capacité;

- processus de ressources en réserve: ce processus décrit la programmation des ressources, c'est-à-dire les unités de production et la charge pouvant être répartie, et la vente aux enchères de la capacité et notamment l'activation par le gestionnaire de réseau de la réserve tertiaire à des fins d'équilibrage;
- etc.

La classe 'Process' (voir Figure 8) permet de définir, pour un document donné, le processus vers lequel le flux d'informations est dirigé. Par exemple, le Schedule Document peut être utilisé dans différents processus tels que "forecast", "long term", "day ahead", "intra day", etc.

Dans un 'MarketDocument', le concept 'TimeSeries' (voir Figure 9) basé sur les participants du marché (classe 'MarketParticipant') est également important, jouant un rôle (classe 'MarketRole') dans un processus métier du marché (classe 'Process') qui peuvent échanger des programmes de volume/prix relatifs à un domaine d'énergie (classe ‘Domain’) pour un type de métier donné (tel que le "commerce intérieur", le "commerce transfrontalier", la "réserve primaire", la "réserve secondaire", la "réserve tertiaire", etc.). Ces documents constituent la base sur laquelle sont fondés les échanges afin de gérer les échanges d'énergie, les d'offres, l'attribution de capacité, les ressources d'énergie en réserve et le règlement.

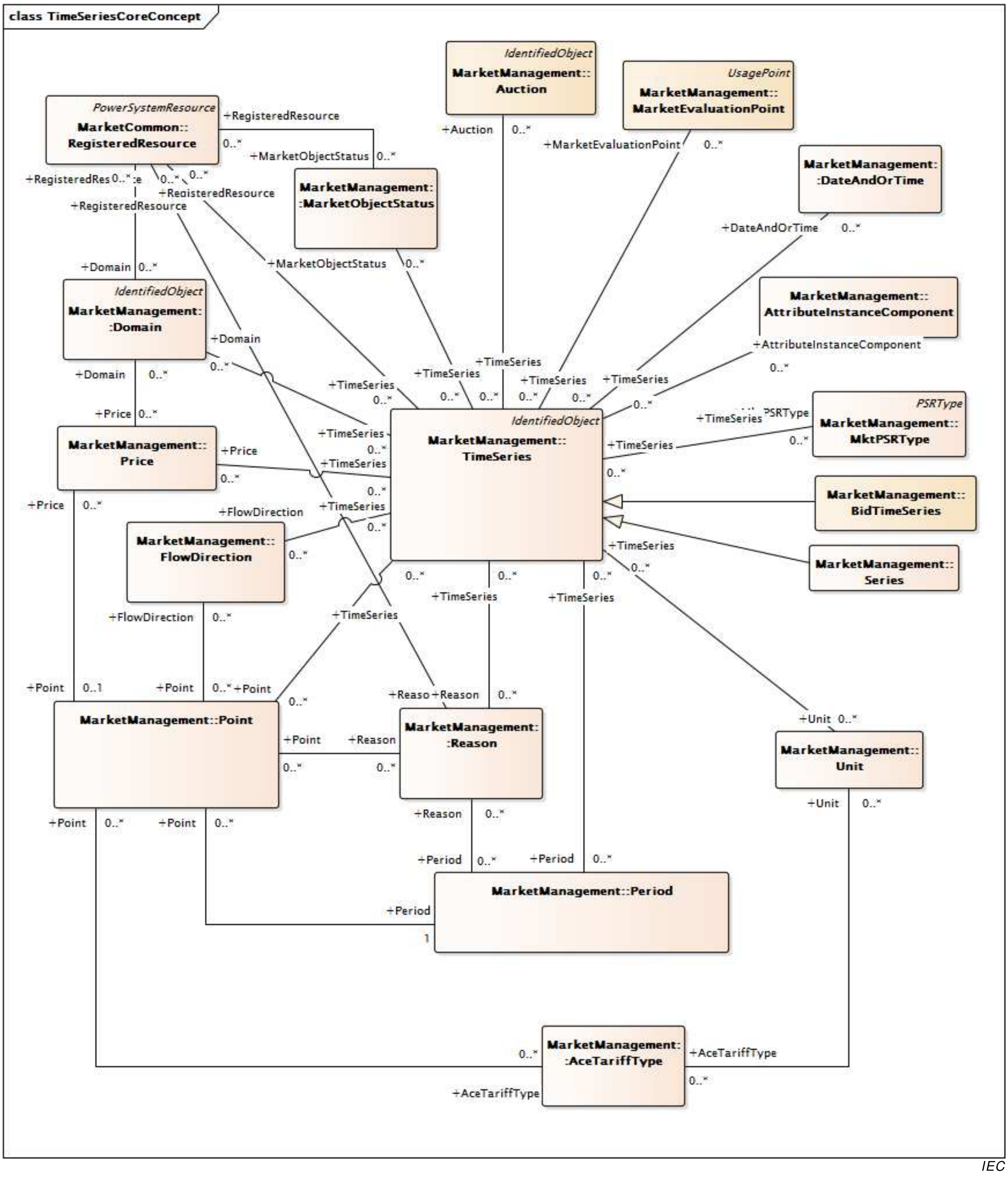


**Figure 9 – Concept de base 'TimeSeries' 'MarketManagement'**

Une spécialisation de la classe TimeSeries est définie afin de fournir une description appropriée de la configuration d'un objet de marché et de l'interrelation entre les objets du marché et les participants du marché. Cette classe est appelée Series.

Pour gérer les quantités et les prix de manière contractuelle dans le marché de l'électricité, les attributs de type 'Decimal' ont été intégrés au processus de modélisation.

### 4.4.3 Paquetage 'MarketOperations'

#### 4.4.3.1 Opérations du marché

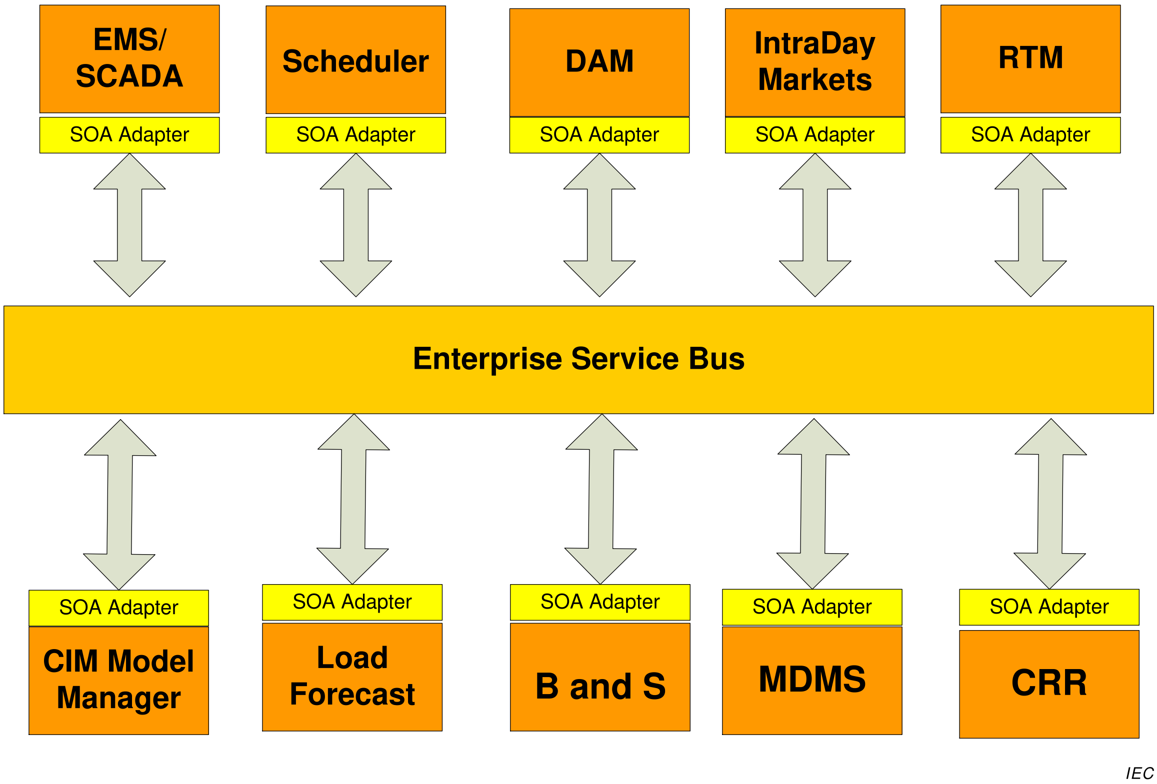
Le paquetage 'MarketOperations' décrit un ensemble cohérent de classes à utiliser avec le modèle Common Market (et d'autres parties du CIM décrites dans les normes IEC 61970-301 et IEC 61968-11) pour générer un profil. Dans ce cas, le profil doit être utilisé pour le marché de l'électricité de style nord-américain principalement caractérisé par une gestion de la production à J-1 effectuée par un opérateur du marché, par un équilibrage intrajournalier et en temps réel par le biais d’un système de répartition central, et par le règlement selon les prix marginaux locaux (LMP).

Le marché de l'électricité de style nord-américain inclut également la vente aux enchères des droits à revenu de congestion, qui sont des instruments financiers que les participants du marché achètent pour se couvrir des frais de congestion. Sont également inclus la gestion des données de compteur, la facturation et le règlement. Le paquetage 'MarketOperations' comprend des modèles prenant en charge ces caractéristiques.

Les Paragraphes 4.4.3.2 à 4.4.3.6 décrivent les principaux concepts référencés dans le paquetage 'MarketOperations'.

#### 4.4.3.2 Systèmes informatiques d'un opérateur du marché

La Figure 10 représente les systèmes informatiques d'un opérateur type des marchés de l'électricité de style nord-américain qui sont référencés dans le paquetage 'MarketOperations'. L'opérateur du marché peut également être un gestionnaire des réseaux de transport régionaux (RTO – Regional Transmission Operator) ou un gestionnaire de réseau indépendant (ISO – Independent System Operator). Ces systèmes informatiques sont identifiés comme suit:




**Légende**

| Anglais | Français |
|---|---|
| Scheduler | Programmateur |
| IntraDay Markets | Marchés intrajournaliers |
| SOA Adapter | Adaptateur SOA |
| Enterprise Service Bus | Bus de service d'entreprise |

**Figure 10 – Systèmes informatiques de l'opérateur d'un marché de l'électricité de style nord-américain**

#### 4.4.3.3 Interfaces du gestionnaire des réseaux de transport régionaux

La Figure 11 donne une vue d'ensemble du modèle d'interface UML d'un gestionnaire des réseaux de transport régionaux (RTO) capable de prendre en charge les interfaces de la Figure 10. Les principales classes sont brièvement résumées: les classes 'AggregateNode' et 'Pnode' sont utilisées afin de modéliser les nœuds de tarification pour lesquels les prix marginaux locaux sont calculés selon une moyenne pondérée. La classe 'LocalReliabilityArea' est utilisée pour identifier et modéliser les poches de charge dans le RTO. La classe 'MktConnectivityNode' est utilisée pour étendre la classe 'ConnectivityNode' de l'IEC 61970 avec les données du marché. Les classes 'TransmissionRightChain', 'ContractRight' et 'MSSAggregation' permettent de modéliser les contrats existants afin d'utiliser le réseau de transport et de représenter les droits de propriété de transport maintenus en vertu d'une clause d'antériorité. Les classes 'Organisation' et 'MarketParticipant' identifient le RTO. Les classes 'SecurityConstraints' et 'SecurityConstraintSum' sont utilisées pour représenter les contraintes du système, telles que les nomogrammes qui sont proches des contraintes de stabilité transitoire. Les classes 'HostControlArea', 'SubControlArea' et 'AdjacentCASet' sont utilisées pour modéliser les données et prendre en charge les opérations des zones de commande. Les modèles de ressource et les données du propriétaire de transport participant sont inclus dans leurs classes associées. Pour plus de détails sur le modèle RTO, voir l'Article 6.

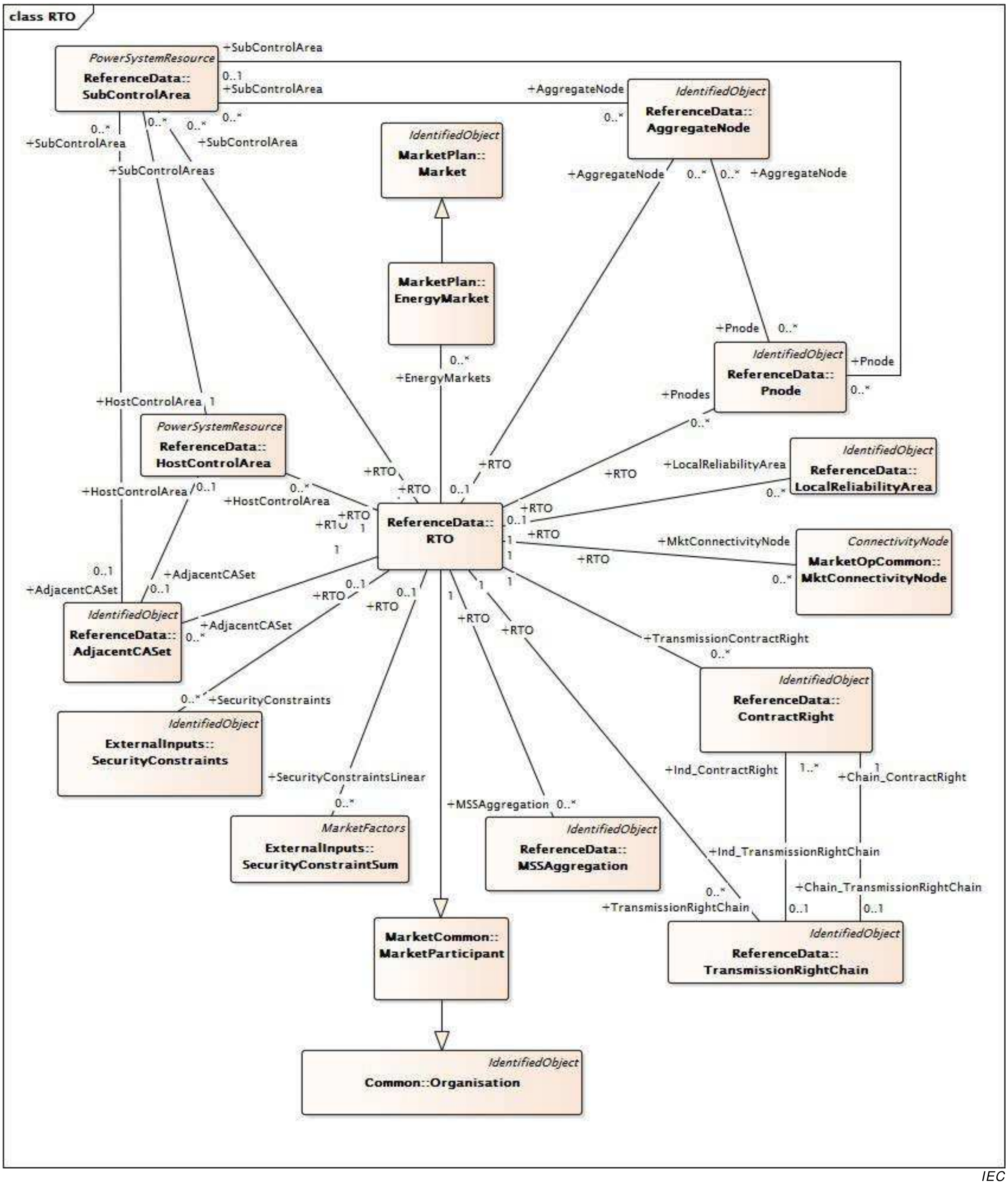



**Figure 11 – Organisation de transport régional pour le marché de l'électricité de style nord-américain**

### 4.4.3.4 Modèle Resource

La Figure 12 donne un aperçu de haut niveau du modèle UML qui prend en charge les données du modèle de ressource du marché de l'électricité de style nord-américain. La classe 'RegisteredResource' est une classe parent qui présente les sous-classes suivantes: 'RegisteredLoad', 'RegisteredInterTie', 'RegisteredGenerator' et ‘RegisteredDistributedResource’. Elle est associée aux classes 'AggregateNode', 'Pnode' et 'MktConnectivityNode' afin de prendre en charge les ressources agrégées et physiques. Les associations avec 'AggregateNode' et 'AggregatedPnode' (l'association est héritée de 'Pnode') prennent en charge la définition des ressources agrégées sur la base d'un nœud agrégé ou d'un nœud de tarification agrégé. Les associations avec 'IndividualPnode' (héritées de

'Pnode') et 'MktConnectivityNode' prennent en charge la définition des ressources physiques sur la base d'un nœud de tarification ou de connectivité.

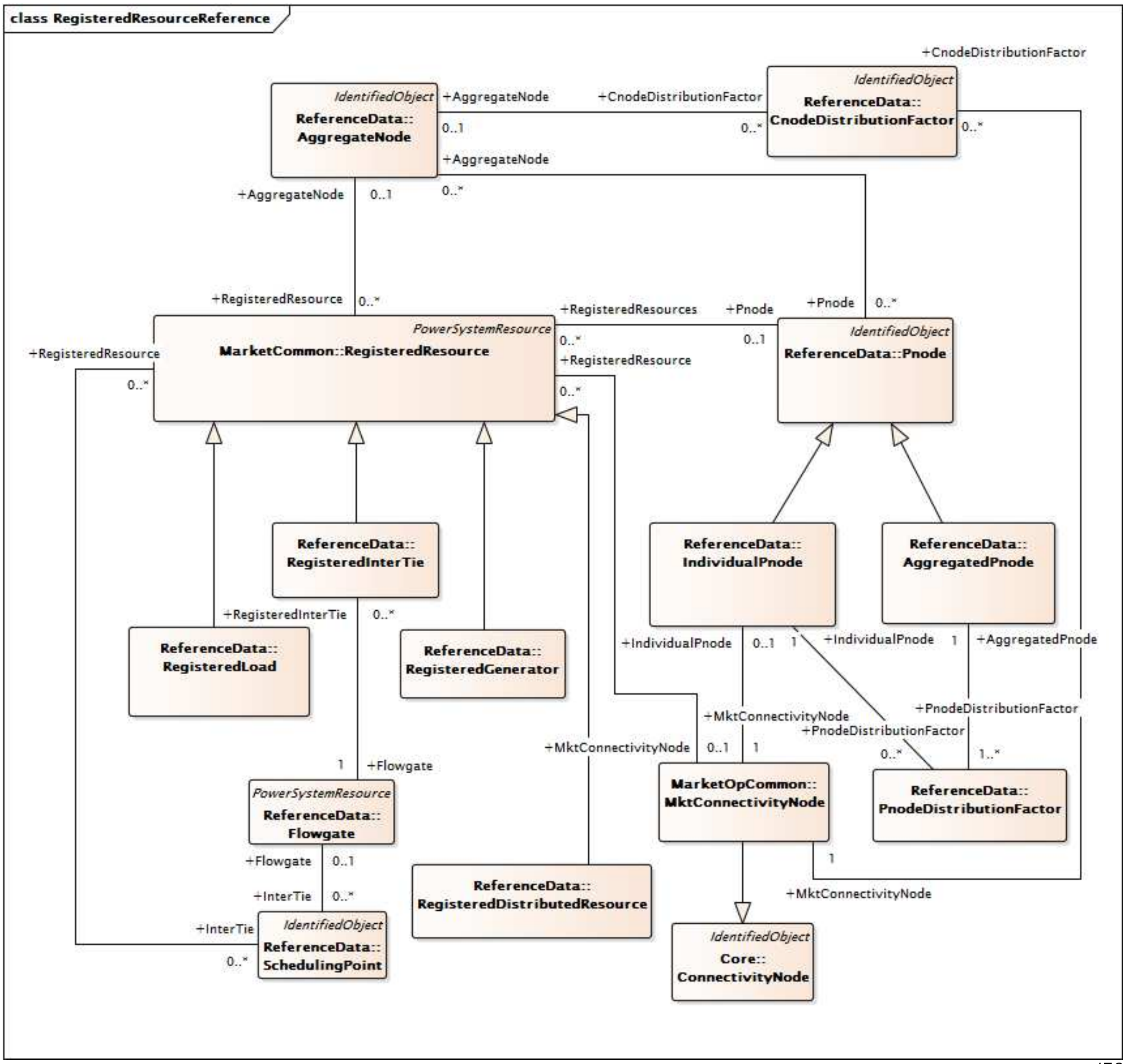



**Figure 12 – Définition de référence de RegisteredResource pour le marché de l'électricité de style nord-américain**

#### 4.4.3.5 Modèle Bid/Offer

Les marchés de l'électricité de style nord-américain sont basés sur des offres de vente et d'achat de produits d'électricité qui sont négociées par un opérateur du marché soumis aux contraintes du réseau et des ressources. Les enchères et les offres comprennent des paires de grandeurs de prix et des données techniques relatives à la capacité du participant du marché à fournir les produits mentionnés.

La Figure 13 donne une vue d'ensemble des classes et des associations qui sont utilisées pour modéliser les offres de vente et d'achat. Le terme "offre" ("bid") inclut les offres de vente et d'achat d'un ou plusieurs produits d'électricité. La classe 'Bid' est une sous-classe de la classe 'Document' du paquetage 'IEC 61968'. Les Bid sont elles-mêmes classées en 'ResourceBid' et 'TransactionBid'. Les 'ResourceBid' sont des offres basées sur des ressources physiques (ou virtuelles) qui sont situées dans la zone de couverture du RTO, c'est-à-dire directement sous le contrôle opérationnel du RTO. Les 'TransactionBid' sont des accords bilatéraux conclus entre les participants du marché, qui sont mentionnés au RTO comme étant des contraintes dans la mise en équilibre du marché. Le RTO décide si les accords bilatéraux peuvent être conclus dans le respect des normes de fiabilité du système.

Les offres sont associées aux participants du marché. Les coordinateurs de programmation remplissent un rôle spécifique de participant du marché qui envoie les offres au nom des participants du marché. Chaque classe 'Bid' doit être associée à une classe 'MarketParticipant'. Cette classe 'MarketParticipant' peut être représentée par un 'SchedulingCoordinator' étant donné que 'SchedulingCoordinator' hérite de 'MarketParticipant'.

La classe ‘Bid’ est reliée à la classe 'BidProduct' et aux classes 'MarketProduct'. Ces associations sont utilisées pour modéliser les offres relatives aux services énergétiques et auxiliaires. Une autre association entre la classe 'Bid' et la classe 'Market' indique le marché visé par l'offre (J-1, temps réel, etc.).

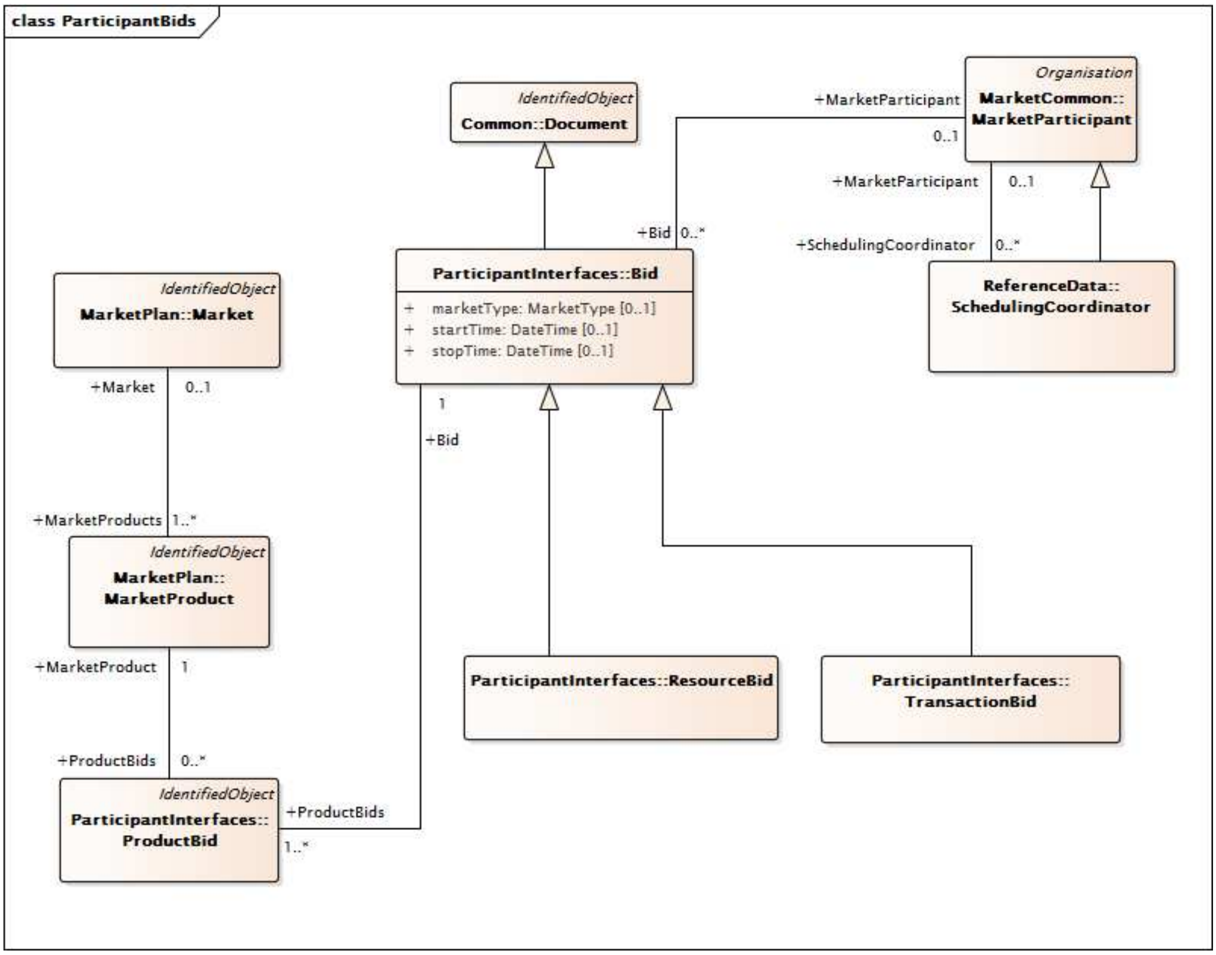



**Figure 13 – Définition de l'offre pour le marché de l'électricité de style nord-américain**

La Figure 14 donne des détails supplémentaires sur les offres d’achat/de vente.

La classe 'Bid' est associée à la classe 'ProductBid', qui est elle-même associée à la classe 'BidPriceSchedule'. La classe 'BidPriceSchedule' définit des programmes d'offre pour autoriser une offre de produit à utiliser des courbes de prix d'offre spécifiques sur plusieurs intervalles de temps. Cela simplifie la modélisation des offres qui sont effectuées sur des périodes couvrant plusieurs intervalles. Le programme d'offre est l'ensemble d'intervalles consécutifs pour lequel l'offre est valide.

Une offre peut également être un programme individuel indiquant que le participant du marché aimerait exploiter les ressources d'après un programme spécifique (par exemple, un programme minimum). L'opérateur du marché détermine si les ressources peuvent fonctionner avec le programme individuel envoyé tout en satisfaisant aux critères de fiabilité du système. Ces programmes individuels sont établis aux prix marginaux locaux (LMP) déterminés pendant la mise en équilibre du marché. La classe 'Bid' est associée à la classe 'ProductBid', qui est associée à la classe 'BidSelfSchedule' pour prendre en charge la modélisation des programmes individuels. La classe 'BidSelfSched' est associée à la classe 'ContractRight', qui peut être utilisée pour modéliser l'utilisation de contrats existants afin de prendre en charge les programmes individuels.

Ce modèle prend également en charge les offres dont une partie est classée comme programme individuel et l'autre partie comme offre ordinaire.

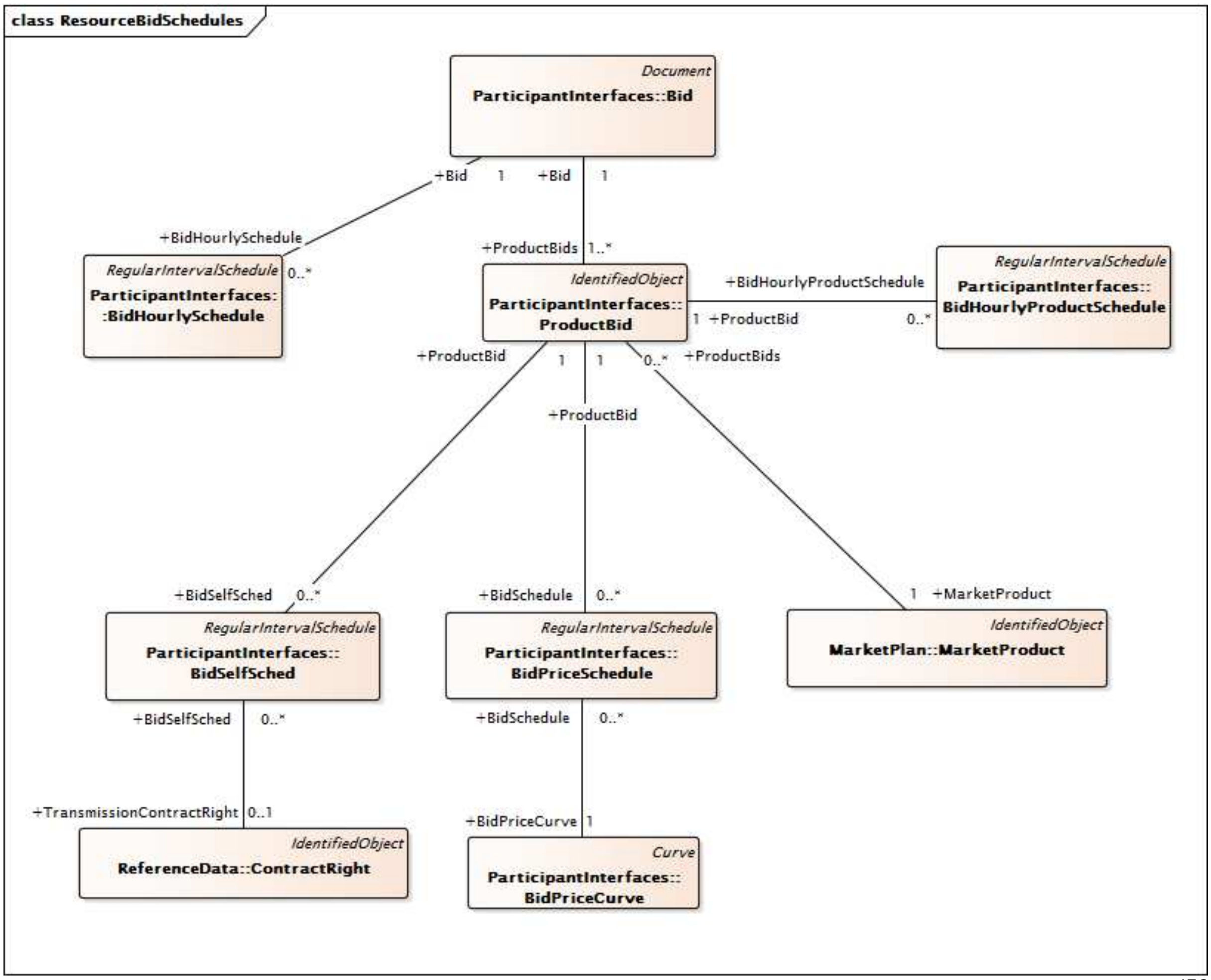



**Figure 14 – Définitions des programmes d'offre des ressources pour le marché de l'électricité de style nord-américain**

### 4.4.3.6 Résultats d'équilibre du marché

La Figure 15 représente les principales classes utilisées pour communiquer les résultats du processus d'ajustement d'un marché de l'électricité de style nord-américain. La classe 'Market' est utilisée pour modéliser le type de marché (J-1, temps réel ou intrajournalier). La classe 'MarketFactors' est utilisée pour modéliser l'horizon temporel du marché. La classe 'LossClearing' est utilisée pour modéliser les pertes électriques pendant l'horizon temporel du marché. La classe 'GeneralClearing' est utilisée pour modéliser l'identité de l'intervalle du marché. La classe 'PnodeClearing' et ses associations sont utilisées pour modéliser les prix d'équilibre (prix marginaux locaux) du processus. La classe 'ResourceClearing' est utilisée pour modéliser les résultats d'équilibre du marché en fonction des ressources. La classe 'AncillaryServiceClearing' est utilisée pour modéliser les résultats de l'équilibre des services auxiliaires en fonction des régions du marché. La classe 'ResourceAwardClearing' est utilisée pour modéliser d'autres informations sur l'équilibre des ressources. La classe 'ConstraintClearing' est utilisée pour modéliser l'échange de données selon des contraintes avec obligation dans la solution optimale d'équilibre du marché.

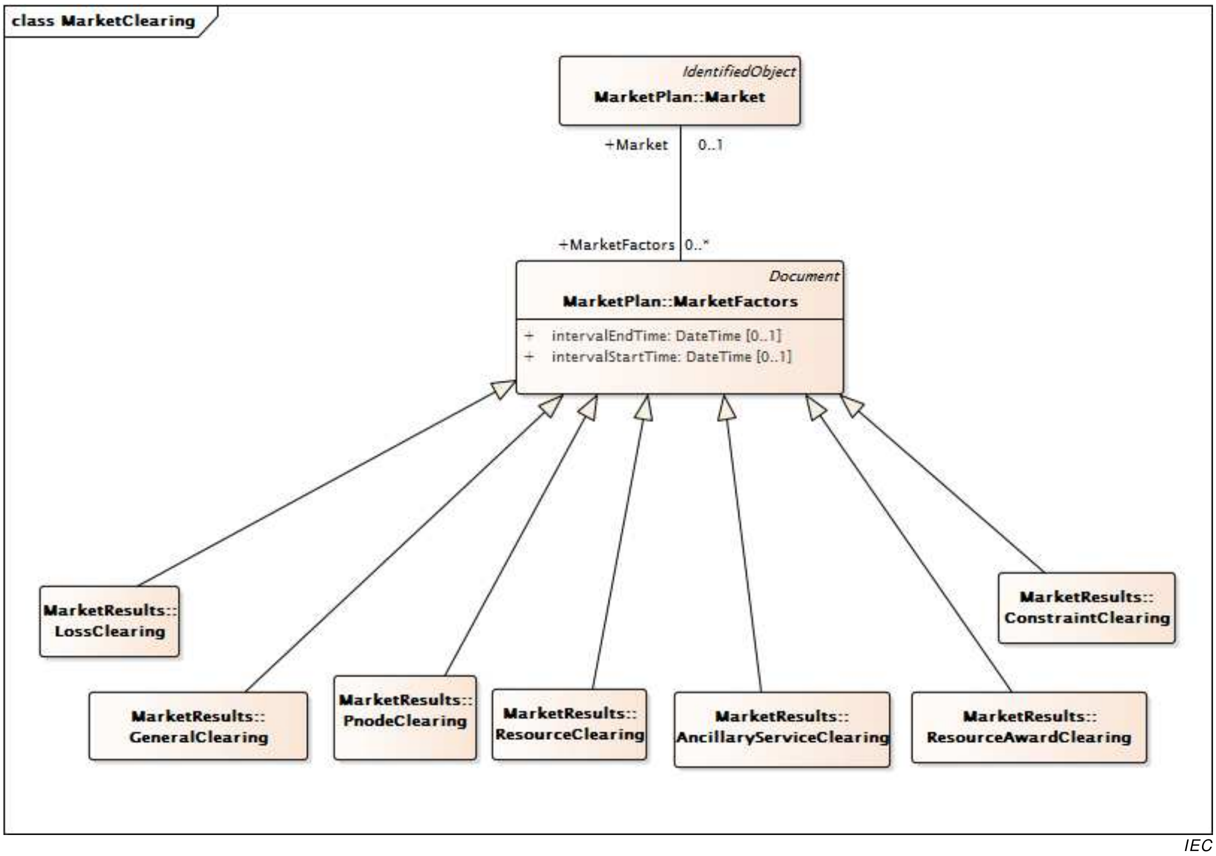



**Figure 15 – Équilibre du marché de l'électricité de style nord-américain**

### 4.4.3.7 Communication côté demande

La Figure 16 représente les principales classes utilisées pour communiquer des informations entre les ressources côté demande (ressources de réponse à la demande, ressources de stockage, ressources de production de distribution, etc.) et les marchés. La classe ‘RegisteredDistributedResource’ est utilisée pour modéliser la ressource et constitue une nouvelle spécialisation de la classe ‘RegisteredResource’. La classe ‘ResourceCertification’ est utilisée pour modéliser différents marchés ou services pour la fourniture desquels la ressource côté demande est qualifiée; par exemple, une ressource de réponse à la demande peut être qualifiée pour fournir 10 MWh d’énergie à une puissance de 10 MW tandis qu’une installation de stockage de 20 MWh peut être qualifiée pour fournir 10 MW de régulation (Regulation) à la hausse ou à la baisse.

La classe ‘MarketActualEvent’ a été étendue afin de modéliser les événements côté demande qui ont des heures de début et de fin ciblées, ainsi que des indicateurs ‘eventType’ et ‘eventStatus’ différents. La classe ‘DistributedResourceActualEvent’ est une spécialisation de ‘MarketActualEvent’ qui est utilisée pour la collecte des activités de réponse à la demande dans l’ensemble du système. L’extension de la classe ‘Instructions’ de façon à inclure un Dispatch Operating Delta (delta de répartition d’exploitation) constitue une autre amélioration importante du modèle lorsqu’une certaine quantité d’énergie est demandée (par opposition à ou en plus d’un point de consigne absolu d’énergie, ou d’une Dispatch Operating Target (cible de répartition d’exploitation), qui était déjà présent(e) dans le modèle).

Les performances d’une ressource côté demande sont enregistrées dans la classe ‘ResourcePerformanceEvaluation’ et peuvent être constituées d’une ou de plusieurs instances d’un ‘ResourcePerformanceTimeSeriesFactor’, telles que les relevés de compteurs, les courbes de base, les courbes de correction et/ou une ou plusieurs instances d'un 'ResourcePerformanceGlobalFactor', telles qu'une correction de température ou un décalage de charge. Les performances de plusieurs événements ou de plusieurs répartitions sont modélisées dans la classe 'ResourcePerformanceRating'.

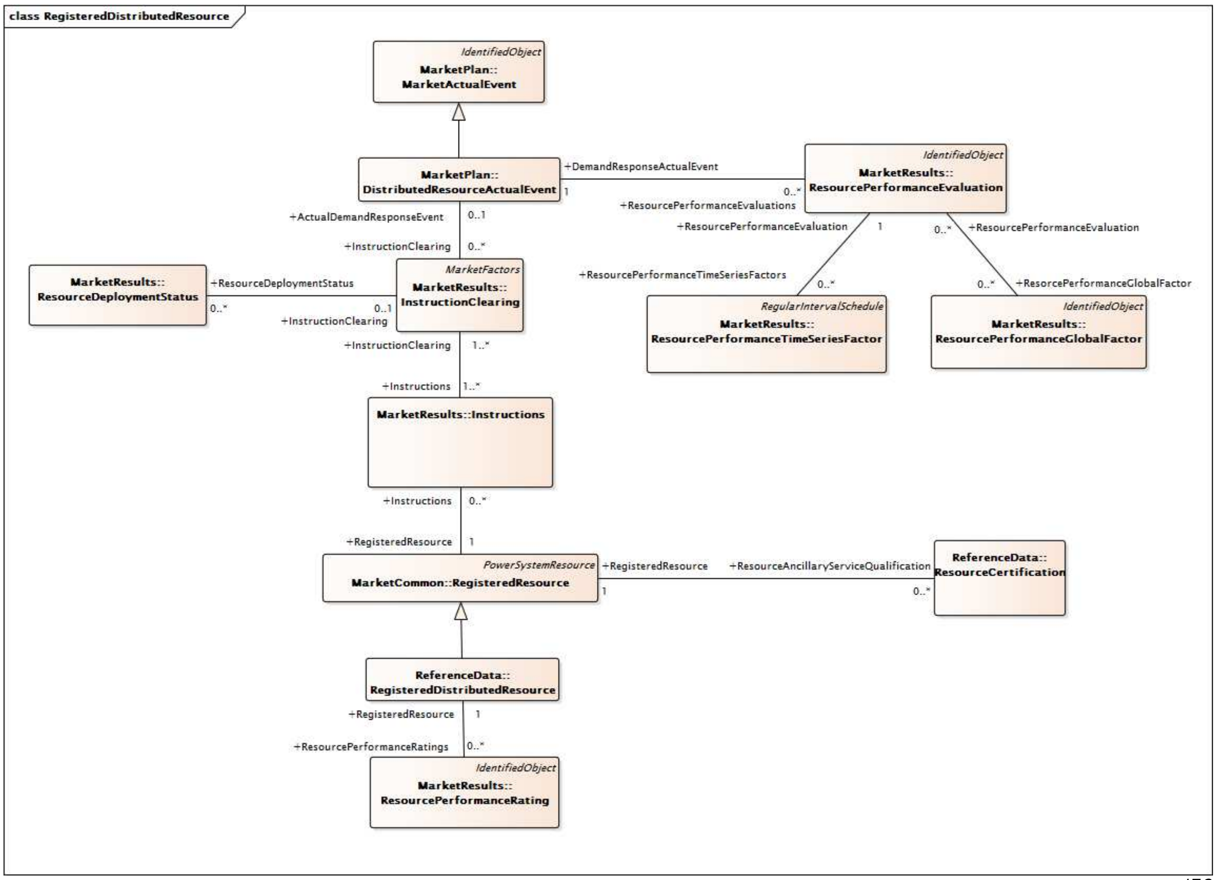



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**Figure 16 – Communication côté demande pour le marché de l'électricité de style nord-américain**

### 4.4.4 Données environnementales

La Figure 17 donne une vue d'ensemble du paquetage 'Environmental', qui comporte un modèle d'éléments généralement utilisés pour décrire les conditions environnementales (météorologiques), y compris les prévisions, les observations, les mesures, les phénomènes, les alertes et les événements.

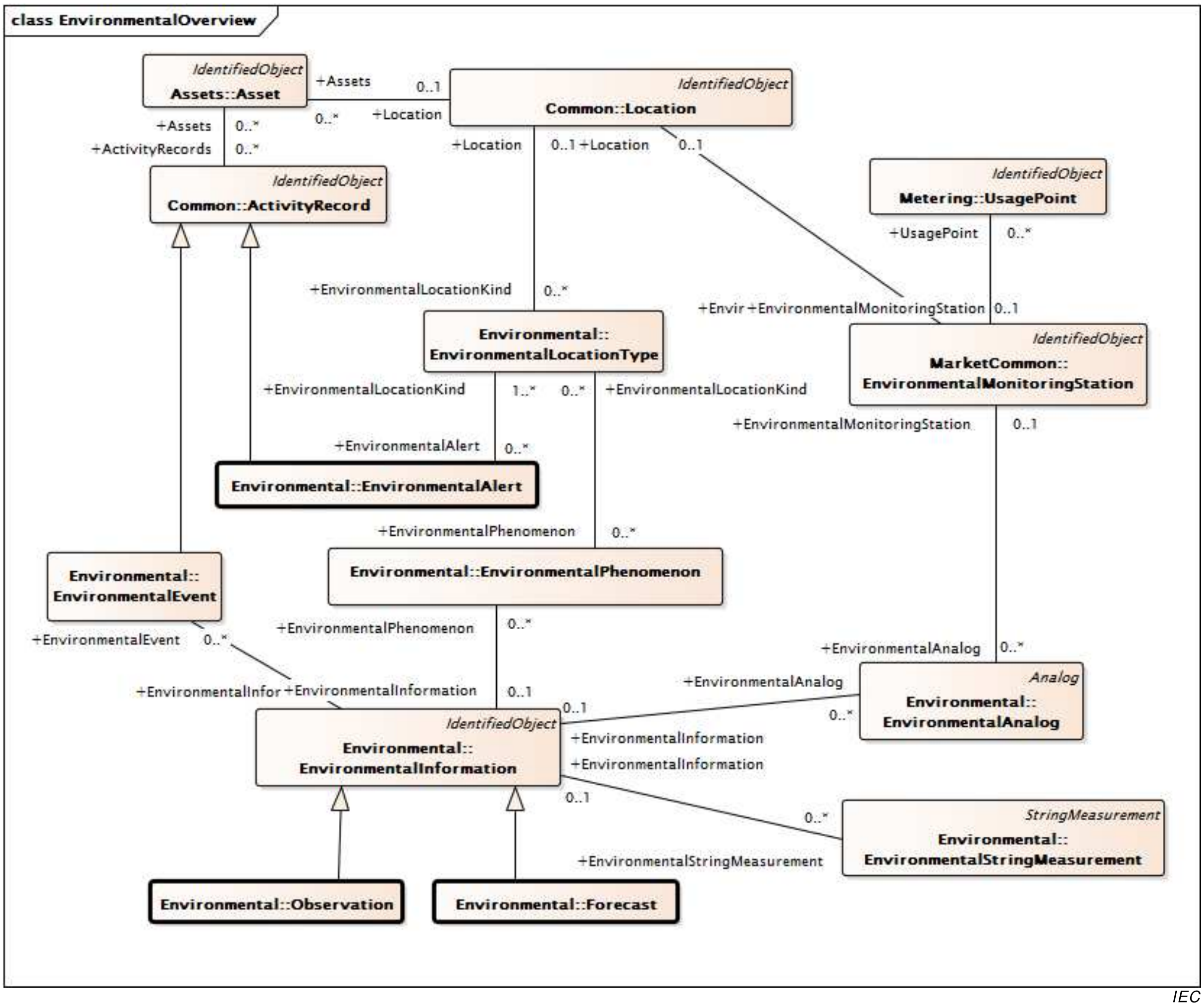
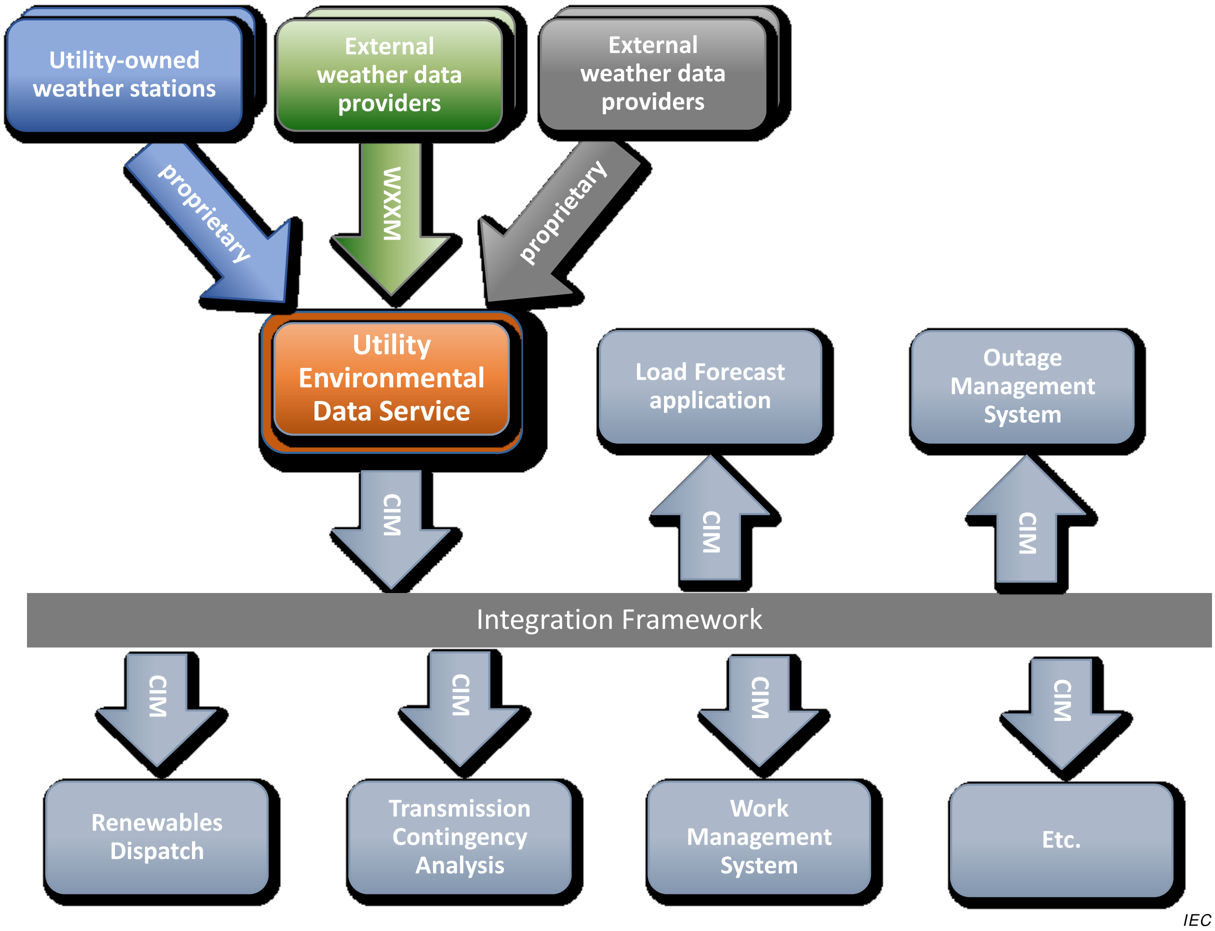



**Figure 17 – Principales classes environnementales**

Les données environnementales sont utilisées par une grande variété de systèmes et d'applications au sein de l'industrie électrique, de la gestion des opérations et du patrimoine aux aspects juridiques et relatifs à la facturation du client. Il est essentiel que le comportement des marchés et des activités soit aussi varié que la prévision de la charge et de la production, le rétablissement par suite d'une tempête (déploiement préalable du personnel, évaluation des dommages, rétablissement) et réponde aux événements environnementaux (feux de forêt/prairie, inondations, tsunamis), à la surveillance d'émissions, à la surveillance des courants induits géomagnétiquement et aux enquêtes menées dans le cadre de réclamations par suite d'accidents.

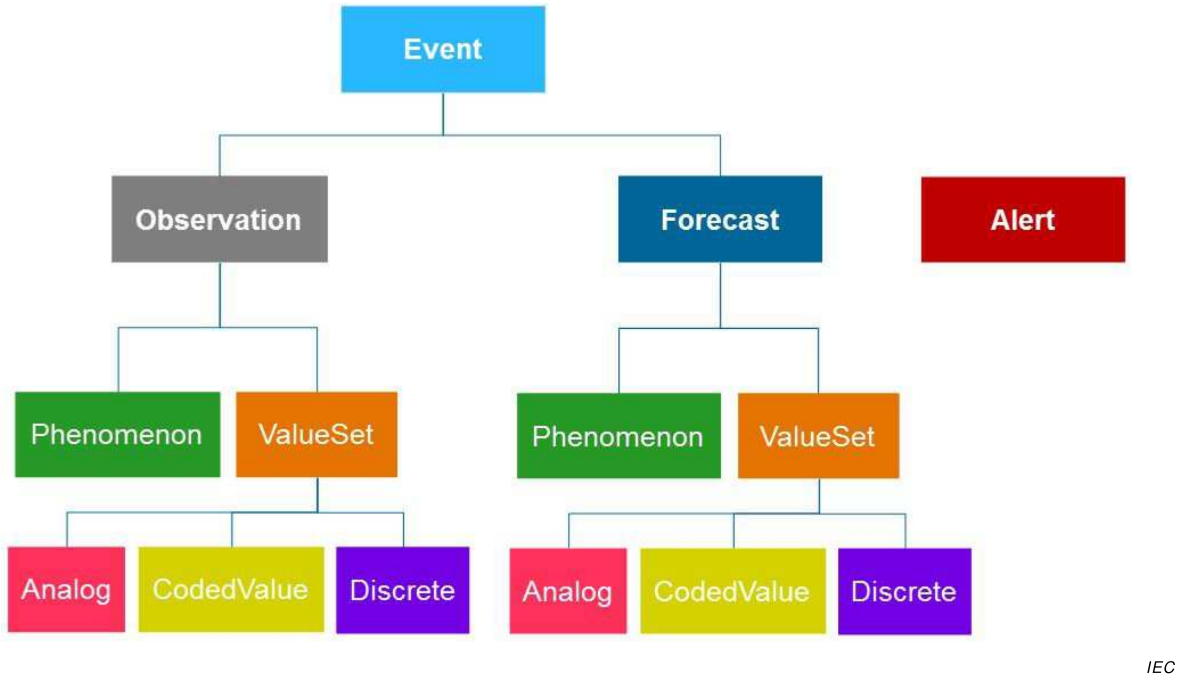
Les classes du paquetage Environmental sont destinées à prendre en charge l'échange d'informations sur les données environnementales au sein d'une entreprise de distribution d'électricité, ainsi qu'à simplifier l'association des données environnementales à d'autres informations d'entreprises de distribution d'électricité (emplacement, patrimoine, point d'usage) auxquelles ces données peuvent être naturellement liées. Comme représenté dans la Figure 18, un service de données environnementales d'entreprise de distribution d'électricité (Utility Environmental Data Service) joue un rôle majeur dans les cas d'utilisation de l'échange de données environnementales au sein de l'entreprise. Le service de données environnementales reçoit des données provenant de différentes sources externes (de différents formats), stocke et gère ces données dans le temps et les fournit, comme exigé, à une grande variété d'applications consommatrices au sein de l'entreprise.




| Anglais | Français |
|---|---|
| Utility-owned weather stations | Stations météorologiques appartenant à l'entreprise |
| External weather data providers | Fournisseurs externes de données météorologiques |
| Proprietary | Propriétaire |
| Utility environmental data service | Service de données environnementales d'entreprise de distribution d'électricité |
| Load forecast application | Application de prévision de charge |
| Outage management system | Système de gestion des indisponibilités |
| Integration framework | Cadre d'intégration |
| Renewables dispatch | Répartition des énergies renouvelables |
| Transmission contingency analysis | Analyse de contingence du transport |
| Work management system | Système de gestion des travaux |

**Figure 18 – Service de données environnementales de l'entreprise de distribution d'électricité**

Comme représenté dans la Figure 19, les quatre éléments de niveau supérieur des données environnementales sont l'observation, la prévision, l'alerte et l'événement. Ces éléments sont représentés par les classes 'Observation', 'Forecast', 'EnvironmentalAlert' et 'EnvironmentalEvent', respectivement. 'Forecast' et 'Observation' sont des classes enfants de 'EnvironmentalInformation' et l'organisation des données appartenant à ces classes est identique.




| Anglais | Français |
|---|---|
| Event | Événement |
| Forecast | Prévision |
| Alert | Alerte |
| Phenomenon | Phénomène |
| Analog | Grandeur analogique |
| Discrete | Grandeur discrète |

**Figure 19 – Éléments de niveau supérieur des données environnementales**

Comme représenté dans la Figure 20, chaque observation ou prévision est créée à un certain moment, peut avoir un fournisseur et comporte un nombre quelconque de valeurs de mesure et de descriptions de phénomènes. En outre, une prévision n'est valide qu'au cours d'une certaine période. Les informations de localisation relatives aux prévisions et aux observations sont fournies au niveau de la mesure ou du phénomène.

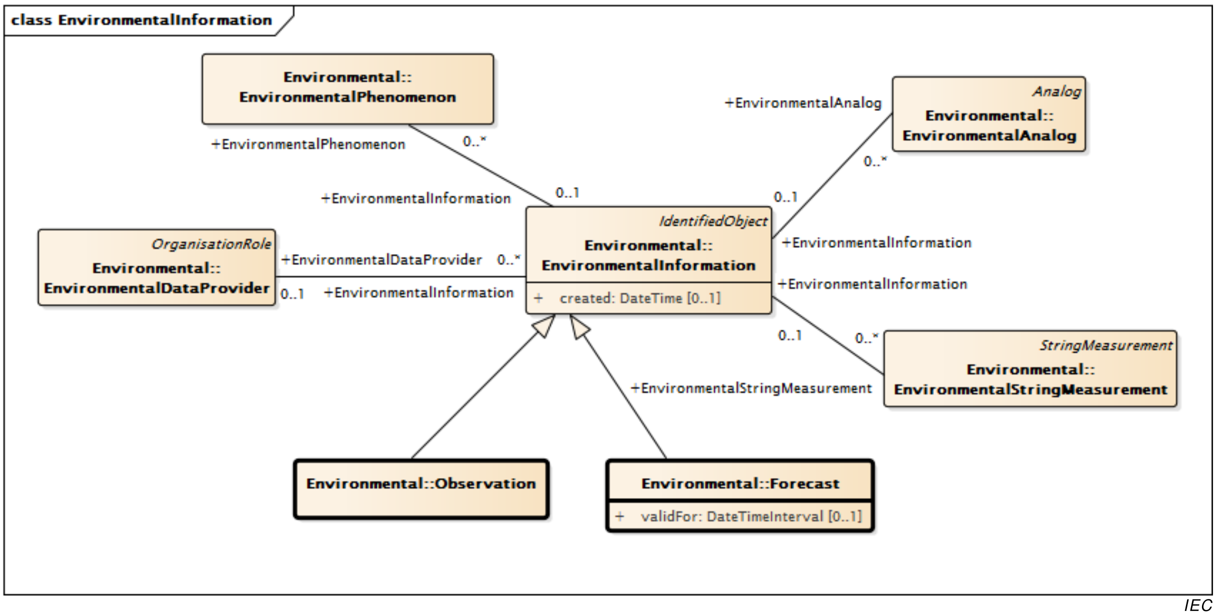



**Figure 20 – Prévisions et observations**

Le modèle de mesure pour les valeurs relatives à l'environnement est représenté dans la Figure 21. Les grandeurs analogiques associées aux prévisions et aux prévisions et aux observations sont réparties en sous-classes selon la région dans laquelle elles apparaissent (atmosphère, géosphère, hydrosphère ou espace) et un attribut type ('kind') permet la définition explicite de la grandeur mesurée (par exemple, la température atmosphérique, la température superficielle de l'eau, l'humidité ou l'éclairement énergétique). Les grandeurs analogiques s'appuient sur les définitions des calculs statistiques fournies par les classes ExtendedAnalog associées ('CalculationMethodHierarchy', 'CalculationMethodOrder', 'StatisticalCalculation', et 'PeriodicStatisticalCalculation'), qui prennent en charge la définition claire de la signification d'une grandeur analogique. Ceci permet, par exemple, de décrire une grandeur analogique comme une 'pointe horaire d'un échantillonnement de température de 5 min'. Les grandeurs environnementales analogiques utilisent la classe usuelle IEC 61970 'AnalogValue', tandis que les grandeurs environnementales de chaîne (utilisées pour exprimer des conditions météorologiques, comme 'brouillard dense') utilisent une spécialisation de la classe IEC 61970 'StringMeasurementValue'.

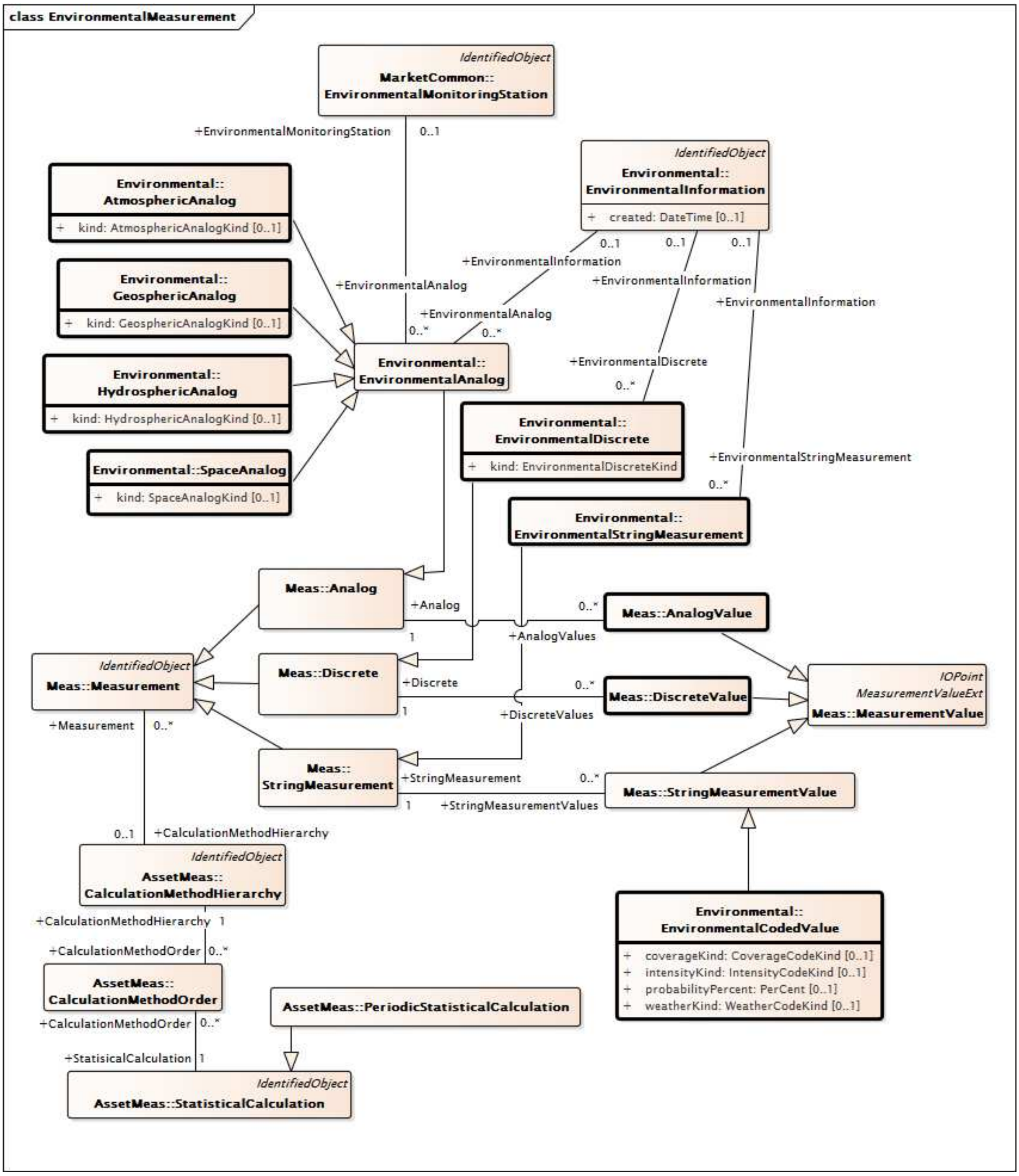



**Figure 21 – Measurements environnementaux**

Les stations de surveillance environnementale, dont un modèle est représenté dans la Figure 22, peuvent fournir des valeurs analogiques correspondant aux observations. Les capacités de ces stations sont définies par la classe 'ReportingCapability'. Une association entre les classes 'EnvironmentalMonitoringStation' et 'UsagePoint' empêche l'éventualité qu'une station de surveillance puisse être associée à un compteur ou à un client.

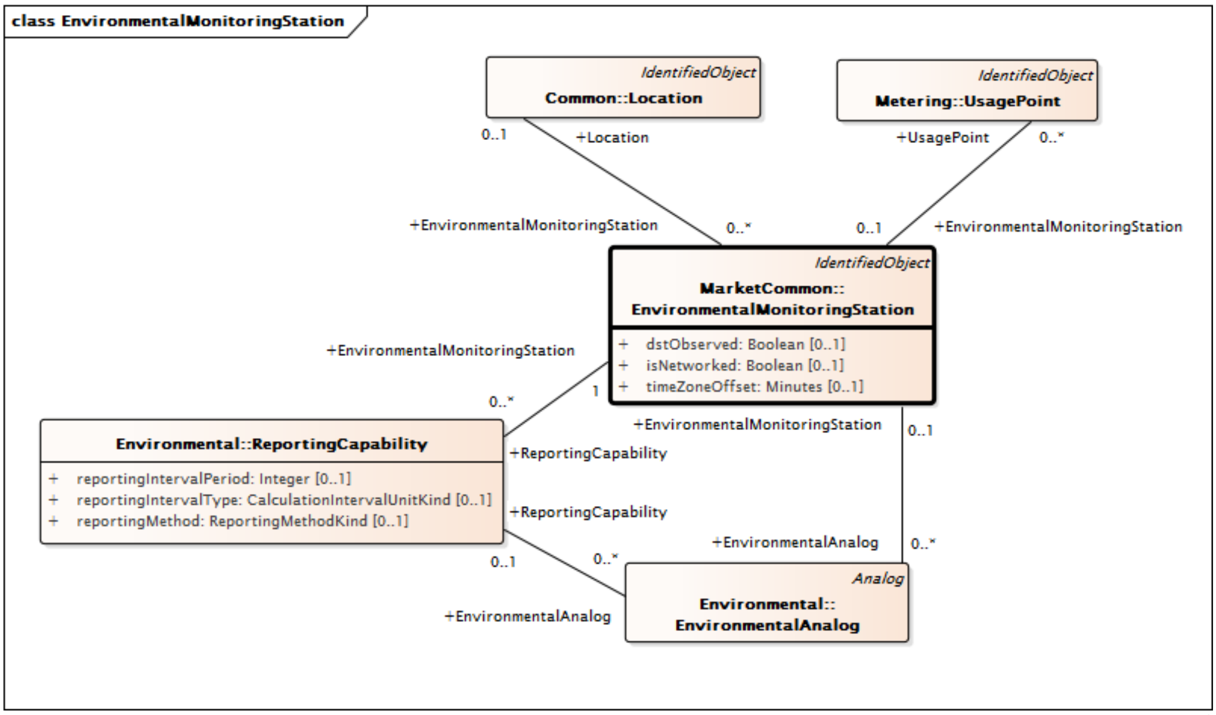



**Figure 22 – Station de surveillance environnementale**

La Figure 23 donne une vue d'ensemble de la modélisation des phénomènes environnementaux. Les caractéristiques des phénomènes environnementaux (c'est-à-dire, les ouragans, les tornades, les feux de prairie, les inondations, les tempêtes dans le champ géomagnétique) à un moment donné sont décrites par la classe 'EnvironmentalPhenomenon' et ses enfants, qui sont sous-classés à l'aide des groupements régionaux utilisés pour les valeurs analogiques (atmosphère, géosphère, hydrosphère et espace). Les emplacements des phénomènes sont souvent décrits par un centre et un encombrement. Ils sont pris en charge par la classe 'EnvironmentalLocationKind'.

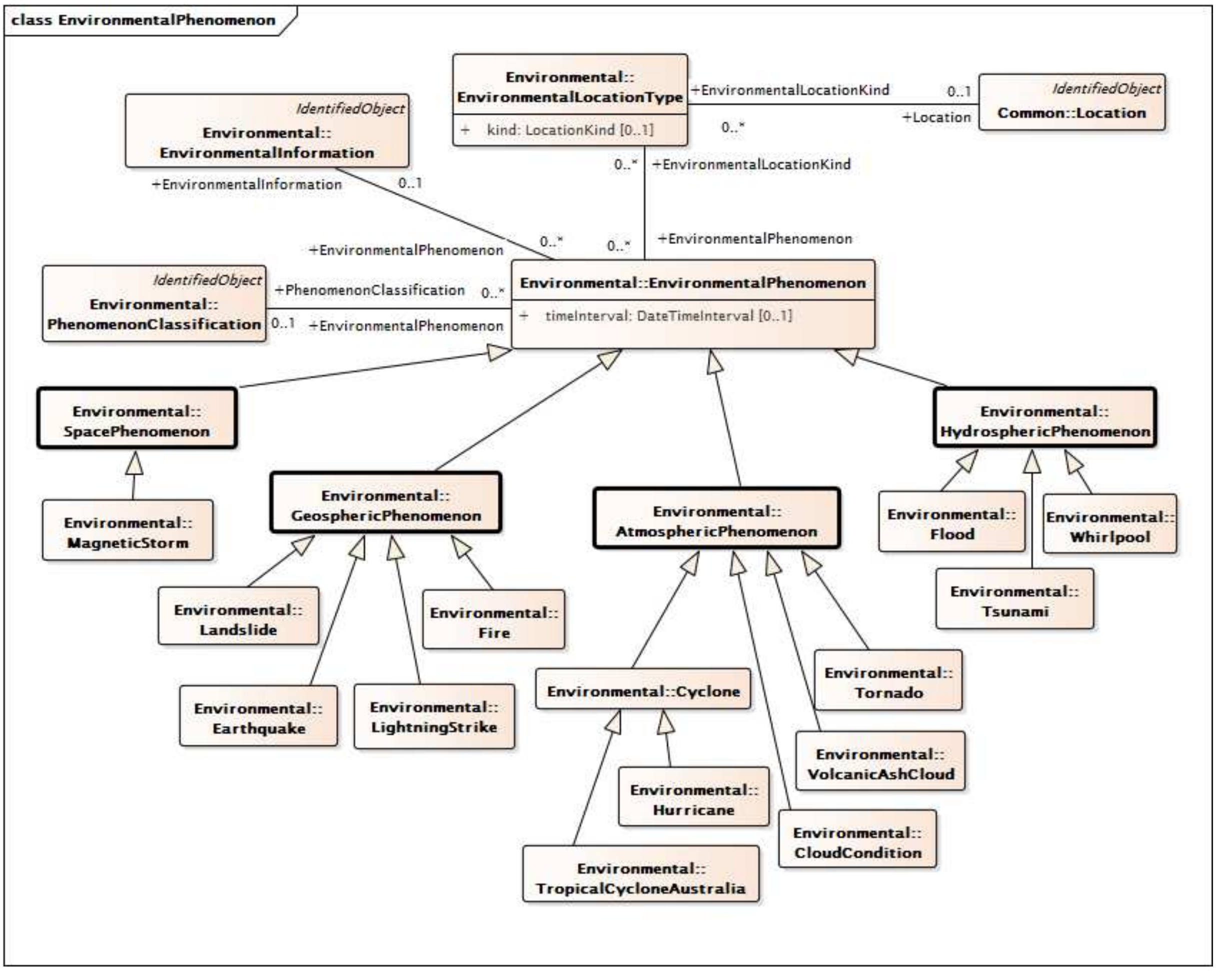



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**Figure 23 – Phénomène environnemental**

Une définition supplémentaire de la sévérité des phénomènes peut être réalisée à l'aide de la classe 'PhenomenonClassification' et des classes associées, ce qui permet à une autorité responsable des données environnementales (par exemple, un service météorologique ou une entreprise d'électricité) de définir les niveaux seuils des valeurs de mesure associées aux conditions de classification du phénomène. Ces niveaux sont représentés dans la Figure 24.

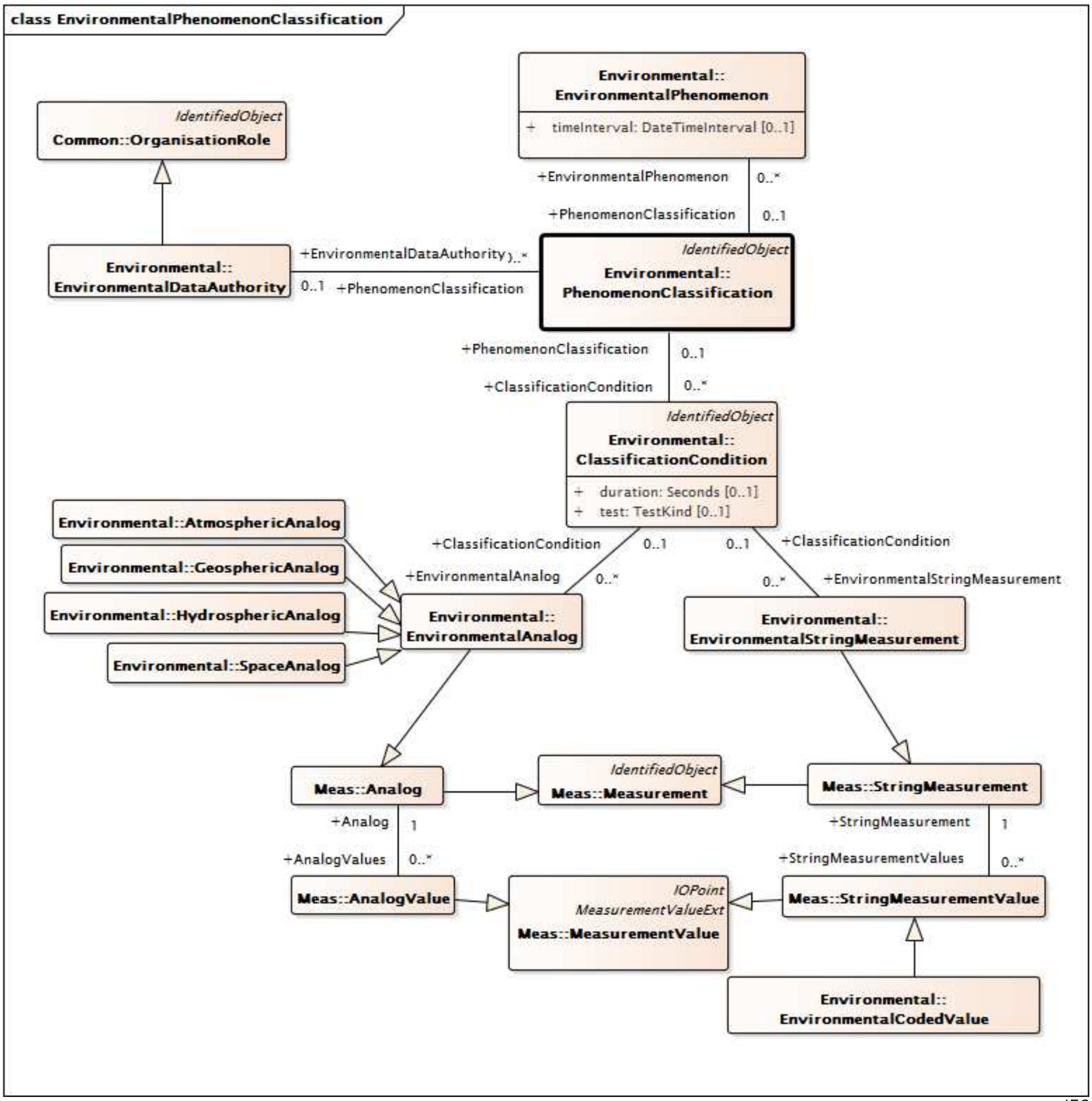


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**Figure 24 – Classification environnementale d'un phénomène**

La Figure 25 représente la modélisation des événements environnementaux. Les principaux événements, comme l'ouragan Katrina ou le séisme de 2008 au Sichuan, sont représentés par la classe EnvironmentalEvent et peuvent être décrits dans le temps au moyen de diverses prévisions et/ou observations (instance de la classe enfant 'EnvironmentalInformation'), chacune des ces prévisions/observations étant liée à l'événement environnemental et chacune comprenant un ou plusieurs phénomènes 'snapshots' (instantanés) (instances de la classe enfant 'EnvironmentalPhenomenon').

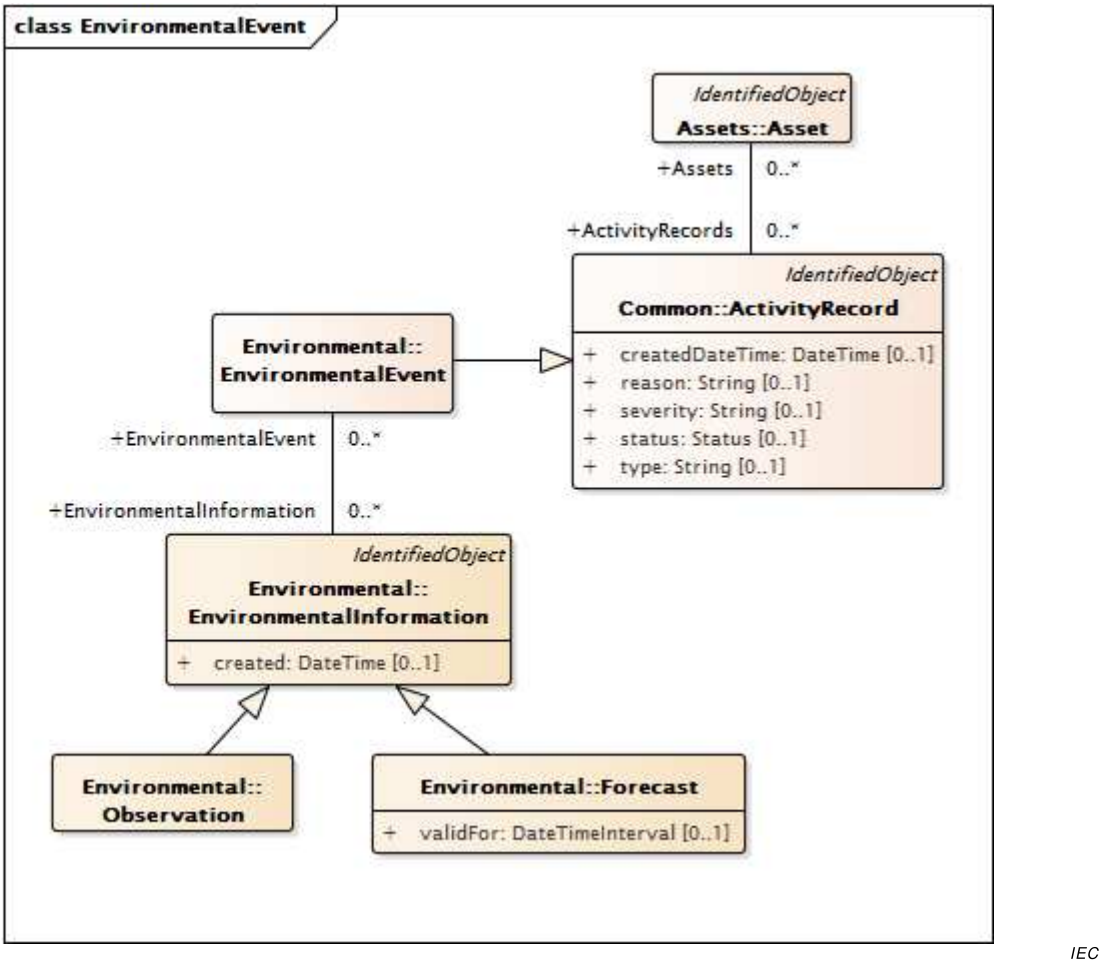



**Figure 25 – Événement environnemental**

Les alertes environnementales sont représentées dans la Figure 26. Les alertes environnementales, telles que les veilles et les avertissements de tornade, sont décrites par la classe ‘EnvironmentalAlert’, qui prend en charge la définition du fournisseur d’alerte, de l’autorité compétente en matière d’alerte, et de l’emplacement de l’alerte. Les autorités compétentes en matière d’alerte (comme le service météorologique national des États-Unis – National Weather Service) mettent régulièrement à jour les types et les définitions des alertes qu’elles peuvent émettre. La classe ‘AlertTypeList’ permet de spécifier la version utilisée des définitions d’alerte. Certaines alertes s’appliquent à des emplacements primaires et secondaires. La classe ‘EnvironmentalLocationKind’ permet d’effectuer la distinction correspondante.

En plus de prendre en charge l’identification des principaux événements, appelés événements environnementaux (tels que les incendies ou les tsunamis), la classe ‘EnvironmentalEvent’ prend également en charge la définition des événements environnementaux locaux de moindre ampleur (tels que les inondations localisées) et leur permet d’être mis en relation avec les constituants du patrimoine.

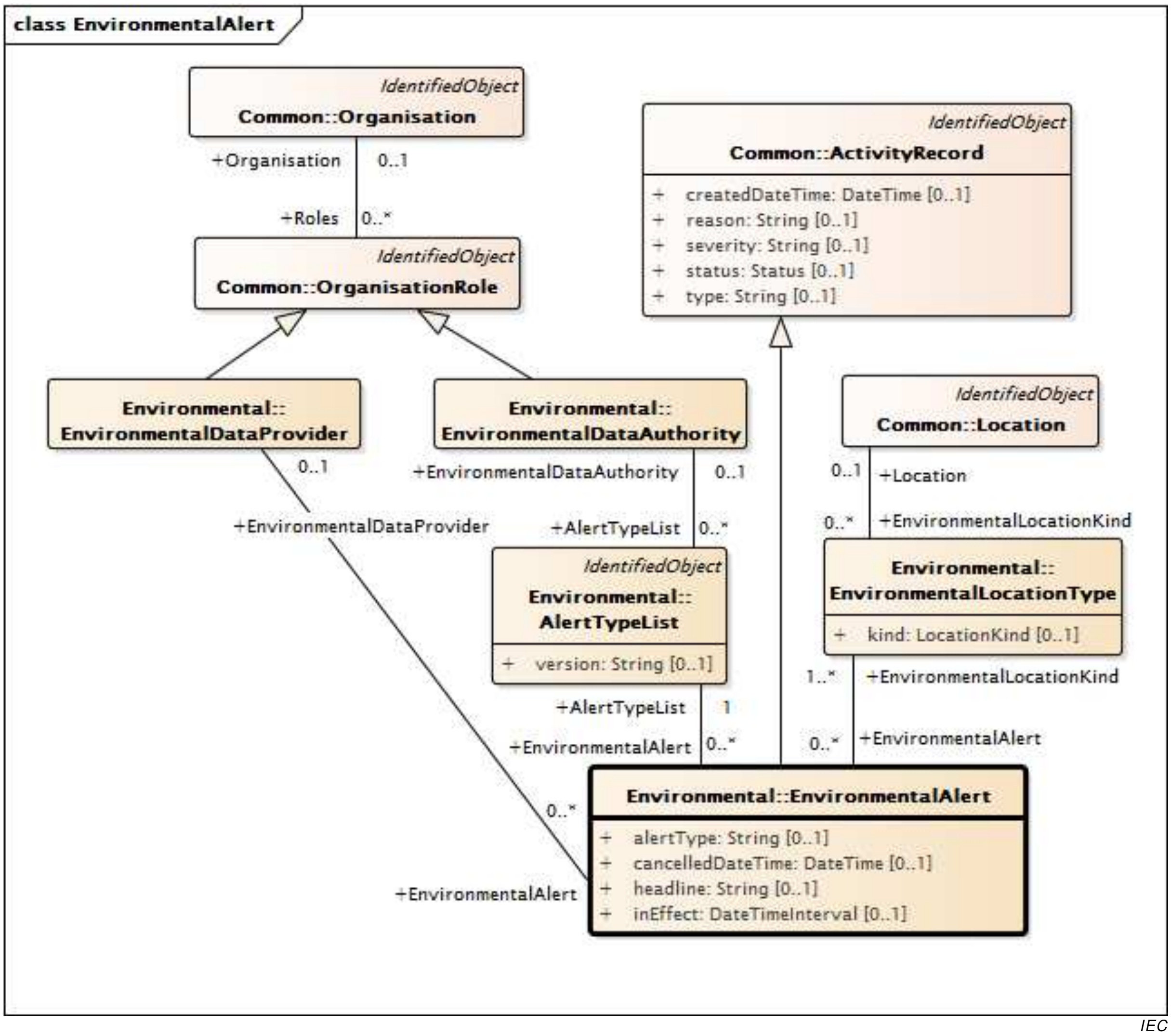



**Figure 26 – Alerte environnementale**

## 4.5 Lignes directrices relatives à la modélisation

### 4.5.1 Modélisation pour changements

L'IEC 62325-301 et l'IEC 61970-301, ainsi que l'IEC 61970-302 et l'IEC 61968-11, définissent le CIM.

### 4.5.2 Processus pour les amendements au CIM

Il peut être souhaitable d'amender le CIM pour réviser le modèle ou pour étendre le CIM afin de modéliser des éléments supplémentaires d'un système électrique. Le processus recommandé pour ces amendements est le suivant.

a) Préparer un ou plusieurs cas d'utilisation (Use Case) décrivant les changements souhaités. Il convient que cela inclue les changements proposés aux diagrammes de classe concernés avec représentation des classes, attributs et associations nouveaux ou révisés.

b) Le ou les cas d'utilisation sont ensuite examinés par le groupe de travail approprié, qui va décider s'il convient que les changements demandés soient traités en tant que révisions de la norme CIM en vigueur ou s'il convient qu'ils soient traités comme des amendements privés sans exiger de changement pour la norme.

c) Les propositions d'amendements acceptées par le groupe de travail seront ajoutées à la liste des sujets majeurs et, le moment venu, une nouvelle version du modèle CIM sera préparée et une mise à jour de la spécification IEC du CIM appropriée sera faite.

#### 4.5.3 Changements apportés au modèle UML du CIM

Du point de vue de la modélisation, lorsque le CIM est étendu, l'approche doit partir du modèle UML de CIM existant. Les extensions peuvent être ajoutées de plusieurs façons disponibles avec UML, mais dans tous les cas, la bonne approche doit examiner le modèle en vigueur et déterminer la meilleure extension à partir des diagrammes de classe existants. Les extensions peuvent prendre l'une des formes suivantes, de la plus simple à la plus complexe:

- ajouter des attributs supplémentaires à des classes préexistantes;
- ajouter des associations supplémentaires entre deux classes préexistantes;
- ajouter de nouvelles classes constituant des spécialisations de classes préexistantes;
- ajouter de nouvelles classes à l'aide d'associations avec des classes préexistantes.

L'objectif principal est de réutiliser au maximum le CIM. Du point de vue des paquetages, il convient que les extensions soient faites aux paquetages existants chaque fois que cela est possible. Si les extensions comprennent un nouveau domaine d'application, il convient d'envisager de créer un nouveau paquetage pour les additions, mais en créant toujours les associations nécessaires au paquetage préexistant, en gardant à l'esprit que, bien qu'un nouveau paquetage soit créé, le CIM continue d'être un seul modèle.

#### 4.5.4 Changements apportés aux documents normatifs du CIM

D'un point de vue documentaire, lorsque le CIM est étendu, une décision doit être prise quant à la question de savoir si les changements constituent des mises à jour de documents normatifs sur le CIM ou si une nouvelle spécification partie 3xx est exigée.

#### 4.5.5 Profils CIM

Il n'est pas nécessaire qu'une mise en œuvre du CIM implique la conformité de tous les classes, attributs et associations de la spécification CIM à la norme CIM. Des profils peuvent être définis pour spécifier les éléments devant être inclus (c'est-à-dire, les éléments obligatoires) pour une utilisation particulière du CIM, ainsi que ceux qui sont facultatifs. Le CIM pour les profils de marché est défini dans les Parties IEC 62325-35X et 45X de la série et doit respecter les règles de modélisation de l'IEC 62325-450.

Dans l'IEC 62325-450, la définition des profils CIM suit un cadre de modélisation en couches qui va du CIM jusqu'à la spécification des messages en fonction des concepts CIM, à travers la définition de plusieurs modèles contextuels régionaux et de leurs documents contextualisés ultérieurs pour l'échange d'informations.

Une analyse complète des règles associées aux profils de la série IEC 62325 est présentée dans l'IEC 62325-450. Se reporter à l'IEC 62325-450 pour obtenir une explication et une description complètes de ces règles.

### 4.6 Outils de modélisation

Le modèle UML du CIM complet existe sous la forme d'un fichier de projet Enterprise Architect. Il est possible de le visualiser avec l'outil Enterprise Architect, y compris les diagrammes de classe et les descriptions des classes, des attributs, des types et des relations. La visualisation du CIM de cette façon offre une interface de navigation graphique qui permet de visualiser toutes les données de spécification du CIM par simple pointer-cliquer sur le diagramme de classe dans chaque paquetage. Une visionneuse gratuite est disponible auprès de Sparx Systems.

Idéalement, le modèle d'information du CIM est indépendant de tout outil UML spécifique, bien que l'expérience ait montré que les échanges entre des outils différents sont souvent loin d'être parfaits. En attendant que l'interopérabilité des outils devienne réalité, les futurs changements apportés à la spécification du CIM, débouchant sur de nouvelles versions de la présente norme, seront incorporés en premier lieu dans la description du projet Enterprise Architect pour garantir une source unique pour les données de modèle CIM.

# 5 Modèle détaillé

## 5.1 Vue d'ensemble

Le modèle d'information commun (CIM) représente une vue logique complète des informations d'un système des opérations du marché. Cette définition inclut les classes et les attributs publics, ainsi que leurs relations. Le Paragraphe 5.2 décrit comment l'Article 6 est structuré. L'Article 6 est automatiquement généré à partir du modèle CIM.

## 5.2 Contexte

Le CIM est fractionné en sous-paquetages. Les classes au sein des paquetages sont énumérées alphabétiquement. Les attributs natifs d'une classe sont énumérés en premier, suivis des attributs hérités, dans l'ordre de profondeur d'héritage, puis du nom d'attribut. Les associations natives sont énumérées en premier pour chaque classe, suivies, dans l'ordre de profondeur d'héritage, des associations héritées, suivies, dans l'ordre alphabétique, du nom de classe, suivi, dans l'ordre alphabétique, du nom d'extrémité d'association. Les associations sont décrites selon le rôle de chaque classe participant à l'association. Les extrémités d'association sont répertoriées pour chaque classe à chaque extrémité d'une association.

Pour chaque paquetage, le modèle d'information de chaque classe est entièrement décrit dans des tableaux d'attributs et d'extrémités d'association. Les informations relatives aux attributs et extrémités d'association concernant les attributs natifs et hérités sont énumérées comme indiqué dans le Tableau 1 et le Tableau 2. Pour chaque attribut ou extrémité d'association hérités, la colonne "description" contiendra le texte indiquant que les attributs sont hérités d'une classe spécifique. La colonne "description" des attributs et extrémités d'association natifs contient la description réelle.

**Tableau 1 – Documentation d'attribut**

| nom | type | description |
|---|---|---|
| native1 | Float | Un attribut natif à virgule flottante de la classe est décrit ici. |
| native2 | ActivePower | Documentation pour un autre attribut natif du type ActivePower. |
| Name | Float | Hérité de la classe IdentifiedObject |

Comme indiqué dans le Tableau 1, dans un tableau attributs, un attribut est dans certains cas une constante; l'expression "(const)" est alors ajoutée dans la colonne "nom" du tableau d'attributs. Dans de tels cas, l'attribut a normalement une valeur initiale qui est précédée du signal "égal" et accolée au nom d'attribut.

**Tableau 2 – Documentation des extrémités d'association**

| [mult à partir de] | [mult vers] nom | type | description |
|---|---|---|---|
| [0..*] | [0..1] PSRType | PSRType | PSRType (classification personnalisée) pour cette PowerSystemResource. |
| [0..1] | [0..*] Measurements | Measurement | Measurements inclus dans la hiérarchie de dénomination dans laquelle le PSR est l'élément conteneur. |
| [1..1] | [0..*] OperatingShare | OperatingShare | Lien à n'importe quel nombre d'objets de quote-part d'opérateur. |
| [0..*] | [0..*] PsrLists | PsrList | |
| [1..1] | [0..1] OutageSchedule | OutageSchedule | Une ressource de système électrique peut avoir un programme d'indisponibilité. |
| [0..*] | [0..*] ReportingGroup | ReportingGroup | Groupes de production de rapports auxquels appartient cette PSR. |
| [0..1] | [0..*] DiagramObjects | DiagramObject | Hérité de: IdentifiedObject |
| [1..1] | [0..*] Names | Name | Hérité de: IdentifiedObject |

Comme indiqué dans le Tableau 2, dans un tableau d'extrémités d'association, la première colonne décrit la multiplicité à l'autre extrémité de l'association (c'est-à-dire, la multiplicité de la classe pour laquelle des extrémités d'association sont décrites). La multiplicité de l'extrémité d'association elle-même est incluse entre crochets. Le nom de l'extrémité d'association est répertorié en texte en clair. La classe à l'autre extrémité de l'association est énumérée dans la colonne "type". Une multiplicité zéro indique une association facultative. Une multiplicité "*" indique que n'importe quel nombre est admis. Par exemple, une multiplicité [1..*] indique qu'une plage allant de 1 à n'importe quel nombre supérieur est admise.

Lorsqu'une classe est une classe d'énumeration, le tableau d'attributs est remplacé par le tableau d'enums comme indiqué dans le Tableau 3, car les libellés enumeration n'ont pas de type. Il n'existe pas de libellés d'énumération hérités pour une classe d'énumération.

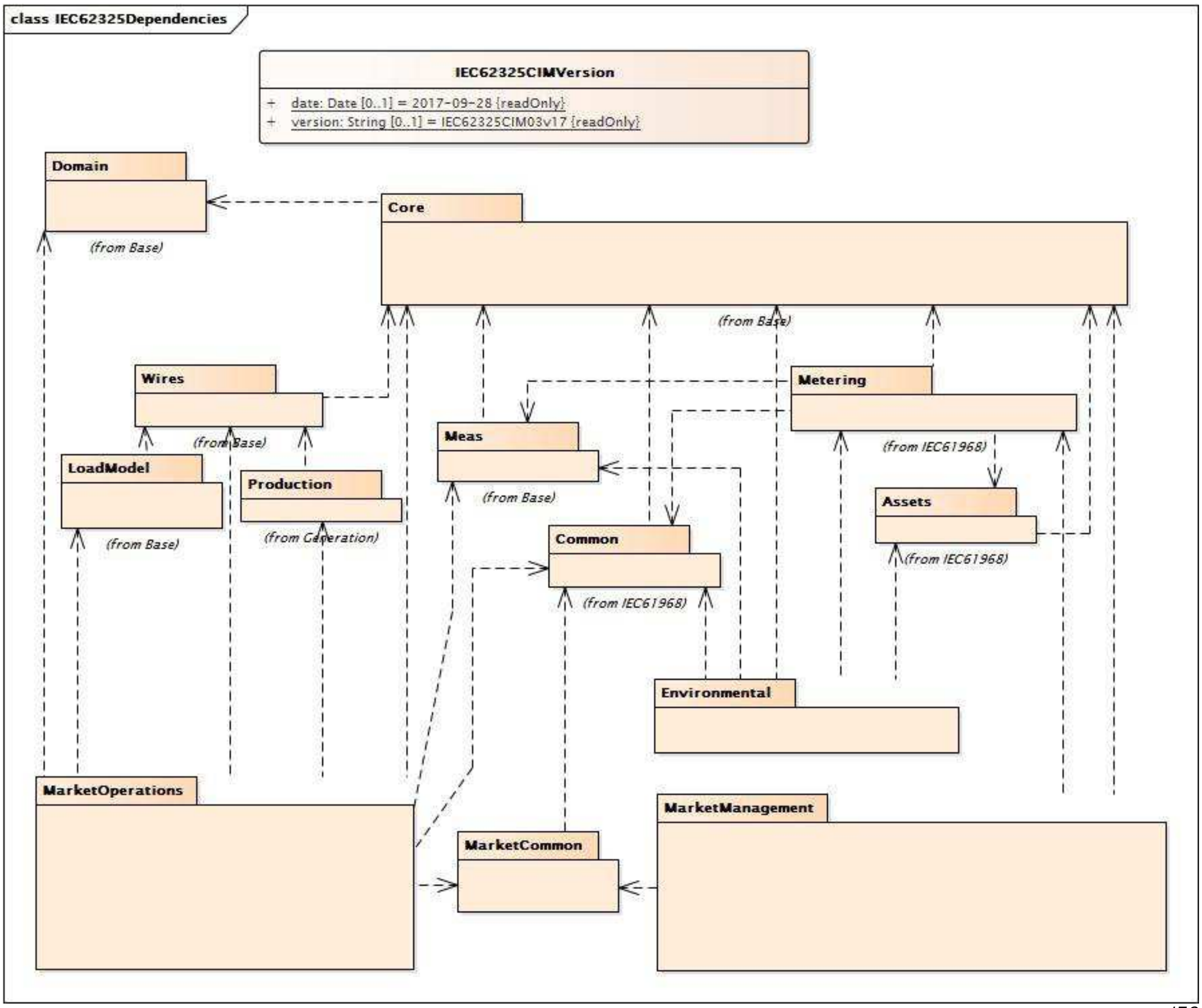
**Tableau 3 – Documentation des Enums**

| libellé | description |
|---|---|
| native1 | Il s'agit de la première valeur enumeration native. |
| native2 | Il s'agit de la deuxième valeur enumeration native. |
| native3 | Il n'y a typiquement pas d'attributs hérités pour les énumérations. |

# 6 Paquetage supérieur IEC62325

## 6.1 Généralités

Les sous-paquetages IEC 62325 du CIM sont développés, normalisés et maintenus par l'IEC.




IEC

| Anglais | Françaiqs |
|---|---|
| from | provenant de |

**Figure 27 – Diagramme de classe IEC62325::IEC62325Dependencies**

Figure 27: Représente les paquetages IEC 62325 du niveau de version et de niveau supérieur ainsi que leurs dépendances logiques avec les autres paquetages par héritage uniquement. Les paquetages qui font référence à une norme IEC sont documentés dans les autres spécifications de normes IEC. Les paquetages sans références sont documentés dans l'Article 6.

## 6.2 Classe racine IEC62325CIMVersion

Numéro de version de l'IEC 62325 attribué à ce modèle UML.

Le Tableau 4 présente tous les attributs de IEC62325CIMVersion.

**Tableau 4 – Attributs de IEC62325::IEC62325CIMVersion**

| nom | mult | type | description |
|---|---|---|---|
| date | 0..1 | Date | (const=2017-09-28) La forme est AAAA-MM-JJ; par exemple, le 5 janvier 2009 est 2009-01-05. |
| version | 0..1 | String | (const=IEC62325CIM03v17) La forme est IEC62325CIMXXvYY où XX est la version principale du paquetage du CIM et YY la version secondaire. Par exemple, IEC62325CIM10v03. |

## 6.3 Paquetage MarketCommon

### 6.3.1 Généralités

Ce paquetage contient les objets communs partagés par les paquetages MarketManagement, MarketOperations et Environmental.

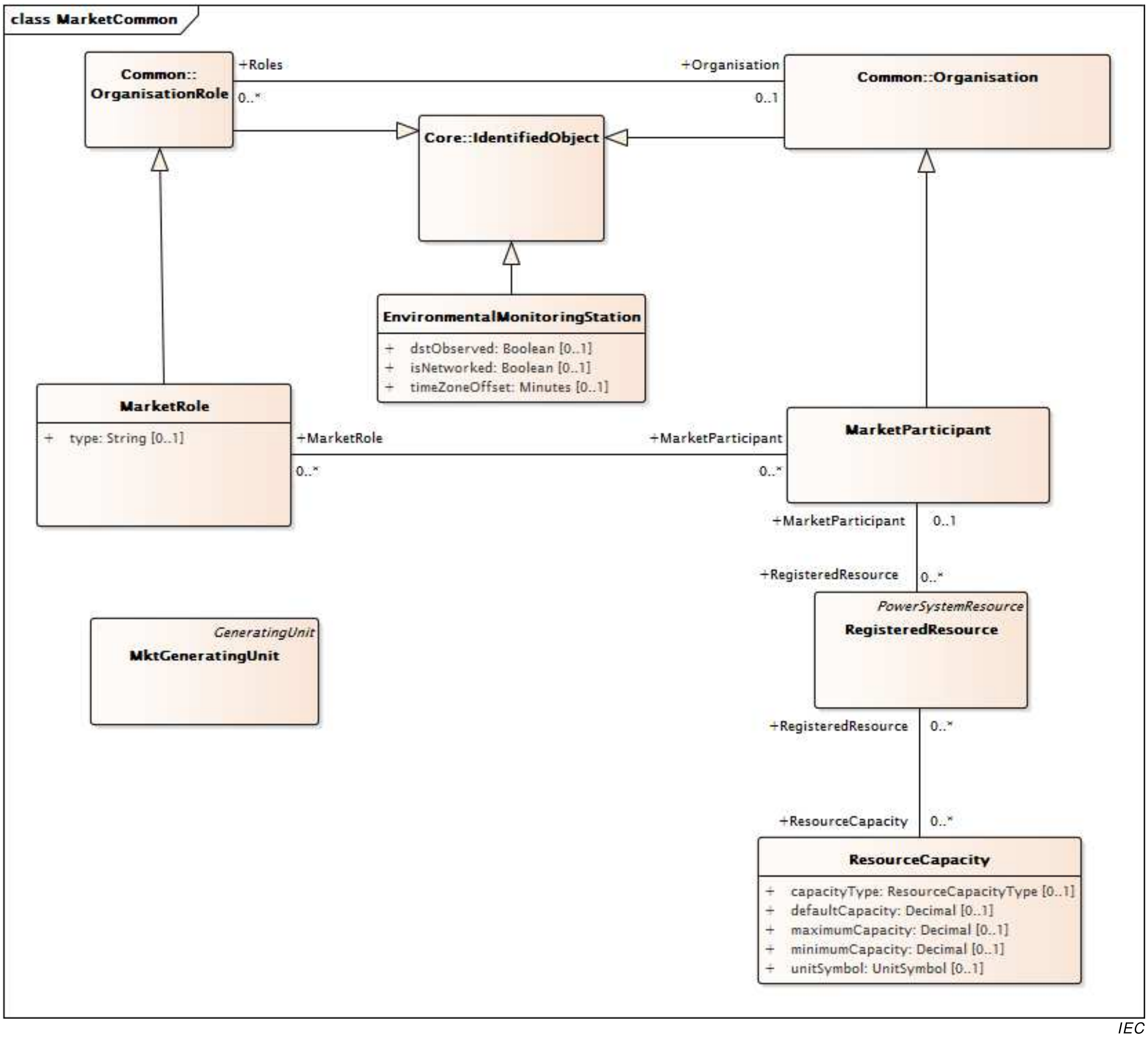



**Figure 28 – Diagramme de classe MarketCommon::MarketCommon**

Figure 28: représente les classes communes utilisées par les paquetages MarketManagement, MarketOperations et Environmental.

### 6.3.2 MarketParticipant

Identification d'une partie agissant dans un processus métier du marché de l'électricité. Cette classe est utilisée pour identifier les organisations qui peuvent participer à la gestion et/ou aux opérations du marché.

Le Tableau 5 présente tous les attributs de MarketParticipant.

**Tableau 5 – Attributs de MarketCommon::MarketParticipant**

| nom | mult | type | description |
|---|---|---|---|
| streetAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| postalAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| phone1 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| phone2 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Organisation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 6 présente toutes les extrémités d'association de MarketParticipant avec d'autres classes.

**Tableau 6 – Extrémités d'association de MarketCommon:: MarketParticipant avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketPerson | 0..* | MarketPerson | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | Bid | 0..* | Bid | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | MarketRole | 0..* | MarketRole | |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | Hérité de: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | Hérité de: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.3.3 MarketRole

Comportement externe joué par une partie du marché de l'électricité.

Le Tableau 7 présente tous les attributs de MarketRole.

**Tableau 7 – Attributs de MarketCommon::MarketRole**

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Type de rôle du marché que les parties peuvent jouer dans certains domaines du marché de l'électricité. Les types sont souples; ils utilisent le type de données String pour la libre entrée des types de rôles. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 8 présente toutes les extrémités d'association de MarketRole avec d'autres classes.

**Tableau 8 – Extrémités d'association de MarketCommon:: MarketRole avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | Hérité de: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.3.4 RegisteredResource

Ressource enregistrée via le système d'enregistrement des participants du marché. Par exemple: unité de production, charge, générateur ou charge non physique.

Le Tableau 9 présente tous les attributs de RegisteredResource.

**Tableau 9 – Attributs de MarketCommon::RegisteredResource**

| nom | mult | type | description |
|---|---|---|---|
| ACAFlag | 0..1 | YesNo | Indique que cette ressource est associée à une zone de réglage adjacente. |
| ASSPOptimizationFlag | 0..1 | YesNo | Indique que cette ressource participe au processus d'optimisation par défaut. |
| commercialOpDate | 0..1 | DateTime | Date d'exploitation commerciale de la ressource. |
| contingencyAvailFlag | 0..1 | YesNo | Disponibilité de la réserve d'exploitation contingente (Yes/No). La ressource est disponible pour participer avec la capacité dans la répartition de la contingence. |
| dispatchable | 0..1 | Boolean | Répartissable: indique si la ressource est répartissable. Ceci implique que la ressource est destinée à soumettre des offres d’achat/de vente d’énergie ou de services auxiliaires, ou des programmes autonomes. |

| nom | mult | type | description |
|---|---|---|---|
| ECAFlag | 0..1 | YesNo | Indique que cette ressource est associée à une zone de contrôle embarqué. |
| flexibleOfferFlag | 0..1 | YesNo | Indicateur d'offre flexible (Y/N). |
| hourlyPredispatch | 0..1 | YesNo | Indique une répartition nécessaire avant le début de l'heure d'exploitation. Uniquement pertinent sur le marché en temps réel. S'applique aux ressources de production, d'interconnexion et de charge participante. Valeur (Y/N). |
| isAggregatedRes | 0..1 | YesNo | Indicateur spécifiant si une ressource est une ressource agrégée. |
| lastModified | 0..1 | DateTime | Indique la date et l'heure à laquelle cet élément a été modifié/versionné pour la dernière fois. |
| LMPMFlag | 0..1 | YesNo | Indicateur LMPM: indique si la ressource est soumise à l'essai LMPM (Yes/No). |
| marketParticipationFlag | 0..1 | YesNo | Indicateur de participation au marché: indique si la ressource participe au marché. |
| maxBaseSelfSchedQty | 0..1 | Float | Quantité de base maximale à programmation automatique. |
| maxOnTime | 0..1 | Float | Temps d'exploitation maximal après démarrage. |
| minDispatchTime | 0..1 | Hours | Nombre minimal d'heures consécutives pendant lesquelles une ressource doit être répartie en cas d'acceptation de l'offre. |
| minOffTime | 0..1 | Float | Temps d'extinction minimal après arrêt. |
| minOnTime | 0..1 | Float | Temps d'exploitation minimal après démarrage. |
| mustOfferFlag | 0..1 | YesNo | Indicateur d'offre obligatoire: indique si l'unité est soumise aux dispositions d'offre obligatoire (Y/N). |
| nonMarket | 0..1 | YesNo | Indicateur spécifiant que la ressource ne participe pas aux opérations du marché. |
| pointOfDeliveryFlag | 0..1 | YesNo | Indique que la ressource enregistrée est une ressource Point de livraison (YES), ce qui implique qu'il existe un facteur de perte POD. |
| priceSetFlagDA | 0..1 | YesNo | Indicateur de définition du prix: indique si une ressource est capable de définir le prix d'équilibre sur le marché (Y) pour le marché DA et, si ce n'est pas le cas, indique si la ressource doit soumettre des offres pour l'énergie à 0 $ (S) ou non (N).<br>À l'origine, il était dans la classe RegisteredGenerator mais a été déplacé dans la classe RegisteredResource pour la répartition de la charge participante. |
| priceSetFlagRT | 0..1 | YesNo | Indicateur de définition du prix: indique si une ressource est capable de définir le prix d'équilibre sur le marché (Y) pour le marché RT et, si ce n'est pas le cas, indique si la ressource doit soumettre des offres pour l'énergie à 0 $ (S) ou non (N).<br>À l'origine, il était dans la classe RegisteredGenerator mais a été déplacé dans la classe RegisteredResource pour la répartition de la charge participante. |

| nom | mult | type | description |
|---|---|---|---|
| registrationStatus | 0..1 | ResourceRegistrationStatus | Statut d'enregistrement de la ressource: actif, suspendu, planifié ou hors service. |
| resourceAdequacyFlag | 0..1 | YesNo | Indique que cette ressource participe à la fonction d'adéquation de ressource. |
| SMPMFlag | 0..1 | YesNo | Indicateur SMPM: indique si la ressource est soumise à l'essai SMPM (Yes/No). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| nom | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 10 présente toutes les extrémités d'association de RegisteredResource avec d'autres classes.

**Tableau 10 – Extrémités d'association de MarketCommon:: RegisteredResource avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | |
| 0..* | AggregateNode | 0..1 | AggregateNode | Un AggregateNode peut être associé à plusieurs RegisteredResources. |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | RampRateCurve | 0..* | RampRateCurve | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | SubControlArea | 0..* | SubControlArea | |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | |
| 1..1 | FormerReference | 0..* | FormerReference | |
| 1..1 | Commitments | 0..* | Commitments | |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Les RegisteredResources sont qualifiées pour les types de services auxiliaires de ressource (qui incluent les types de produits du marché ainsi que d'autres types, tels que BlackStart) en étant associées à la classe ResourceAncillaryServiceQualification. |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | InterTie | 0..* | SchedulingPoint | |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | |
| 0..1 | DopInstruction | 0..* | DopInstruction | |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 1..1 | DefaultBid | 0..1 | DefaultBid | |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | |
| 1..1 | Instructions | 0..* | Instructions | |
| 0..* | Pnode | 0..1 | Pnode | Une ressource enregistrée injecte de la puissance au niveau d'un ou de plusieurs nœuds de connectivité lié à un pnode. |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | |
| 0..1 | DotInstruction | 0..* | DotInstruction | |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.3.5 MktGeneratingUnit

Sous-classe de IEC61970:Production:GeneratingUnit

Le Tableau 11 présente tous les attributs de MktGeneratingUnit.

**Tableau 11 – Attributs de MarketCommon::MktGeneratingUnit**

| nom | mult | type | description |
|---|---|---|---|
| allocSpinResP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| autoCntrlMarginP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| baseP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| controlDeadband | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| controlPulseHigh | 0..1 | Seconds | Hérité de: GeneratingUnit |
| controlPulseLow | 0..1 | Seconds | Hérité de: GeneratingUnit |
| controlResponseRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| efficiency | 0..1 | PerCent | Hérité de: GeneratingUnit |
| genControlMode | 0..1 | GeneratorControlMode | Hérité de: GeneratingUnit |
| genControlSource | 0..1 | GeneratorControlSource | Hérité de: GeneratingUnit |
| governorMPL | 0..1 | PU | Hérité de: GeneratingUnit |
| governorSCD | 0..1 | PerCent | Hérité de: GeneratingUnit |
| highControlLimit | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| initialP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| longPF | 0..1 | Float | Hérité de: GeneratingUnit |
| lowControlLimit | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| lowerRampRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| maxEconomicP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| maxOperatingP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| minEconomicP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| minimumOffTime | 0..1 | Seconds | Hérité de: GeneratingUnit |
| minOperatingP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| modelDetail | 0..1 | Classification | Hérité de: GeneratingUnit |
| nominalP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| normalPF | 0..1 | Float | Hérité de: GeneratingUnit |
| penaltyFactor | 0..1 | Float | Hérité de: GeneratingUnit |
| raiseRampRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| ratedGrossMaxP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| ratedGrossMinP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| ratedNetMaxP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| shortPF | 0..1 | Float | Hérité de: GeneratingUnit |
| startupCost | 0..1 | Money | Hérité de: GeneratingUnit |
| startupTime | 0..1 | Seconds | Hérité de: GeneratingUnit |
| tieLinePF | 0..1 | Float | Hérité de: GeneratingUnit |
| totalEfficiency | 0..1 | PerCent | Hérité de: GeneratingUnit |
| variableCost | 0..1 | Money | Hérité de: GeneratingUnit |

| nom | mult | type | description |
|---|---|---|---|
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 12 présente toutes les extrémités d'association de MktGeneratingUnit avec d'autres classes.

**Tableau 12 – Extrémités d'association de MarketCommon:: MktGeneratingUnit avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | GeneratingUnitDynamicValues | 0..* | GeneratingUnitDynamicValues | |
| 0..1 | RotatingMachine | 0..* | RotatingMachine | Hérité de: GeneratingUnit |
| 1..1 | GenUnitOpCostCurves | 0..* | GenUnitOpCostCurve | Hérité de: GeneratingUnit |
| 1..1 | GenUnitOpSchedule | 0..1 | GenUnitOpSchedule | Hérité de: GeneratingUnit |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | Hérité de: GeneratingUnit |
| 1..1 | GrossToNetActivePowerCurves | 0..* | GrossToNetActivePowerCurve | Hérité de: GeneratingUnit |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.3.6 EnvironmentalMonitoringStation

Station de surveillance environnementale; par exemple, une station météorologique ou une station de surveillance sismique.

Le Tableau 13 présente tous les attributs de EnvironmentalMonitoringStation.

**Tableau 13 – Attributs de MarketCommon::EnvironmentalMonitoringStation**

| nom | mult | type | description |
|---|---|---|---|
| dstObserved | 0..1 | Boolean | Indique si cette station produit actuellement un rapport en utilisant l'heure d'été. Destiné à venir en aide à un service météorologique d'une entreprise d'électricité lors de l'interprétation des informations provenant d'une station. N'a aucune relation directe avec la manière dont le temps est exprimé dans EnvironmentalValueSet. |
| isNetworked | 0..1 | Boolean | Indique que cette station fait partie d'un réseau de stations utilisé pour surveiller un phénomène météorologique couvrant une zone géographique importante. |
| timeZoneOffset | 0..1 | Minutes | Décalage temporel par rapport à UTC (aussi appelé GMT) configuré dans la station "clock", pas (nécessairement) la zone temporelle dans laquelle la station est physically située.<br>Cet attribut existe afin de prendre en charge la gestion des stations de surveillance d'une entreprise d'électricité et n'a aucune relation directe avec la manière dont le temps est exprimé dans EnvironmentalValueSet. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 14 présente toutes les extrémités d'association de EnvironmentalMonitoringStation avec d'autres classes.

**Tableau 14 – Extrémités d'association de MarketCommon:: EnvironmentalMonitoringStation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Location | 0..1 | Location | Emplacement de cette station de surveillance. |
| 0..1 | UsagePoint | 0..* | UsagePoint | |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Mesure environnementale analogique fournie par cette station de surveillance. |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 1..1 | ReportingCapability | 0..* | ReportingCapability | L'une des capacités de production de rapports de cette station de surveillance. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.3.7 Classe racine ResourceCapacity

Cette classe modélise les différentes capacités d'une ressource. Une ressource peut avoir un certain nombre de capacités liées à l'exploitation, aux services auxiliaires, au commerce d'énergie, etc. Les capacités peuvent être définies pour la puissance active ou la puissance réactive.

Le Tableau 15 présente tous les attributs de ResourceCapacity.

**Tableau 15 – Attributs de MarketCommon::ResourceCapacity**

| nom | mult | type | description |
|---|---|---|---|
| capacityType | 0..1 | ResourceCapacityType | Type de capacité<br>Exemples de types:<br>Régulation à la hausse<br>Régulation à la baisse<br>Réserve tournante<br>Réserve non tournante<br>Capacité FOO<br>Capacité MOO |
| defaultCapacity | 0..1 | Decimal | Capacité par défaut |
| maximumCapacity | 0..1 | Decimal | Capacité maximale |
| minimumCapacity | 0..1 | Decimal | Capacité minimale |
| unitSymbol | 0..1 | UnitSymbol | Sélection d'unité pour les valeurs de capacité. |

Le Tableau 16 présente toutes les extrémités d'association de ResourceCapacity avec d'autres classes.

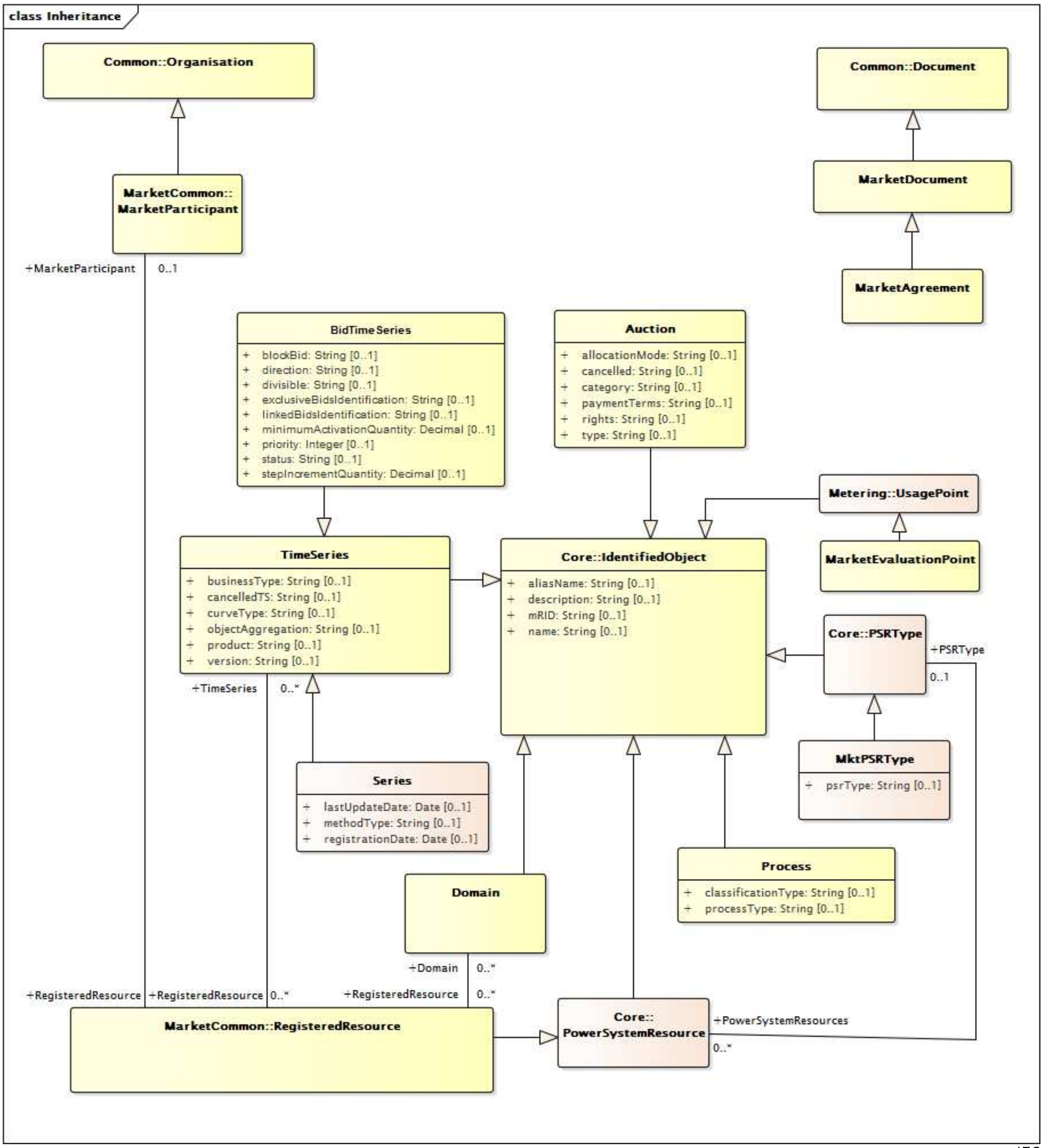
**Tableau 16 – Extrémités d'association de MarketCommon:: ResourceCapacity avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |

## 6.4 Paquetage MarketManagement

### 6.4.1 Généralités

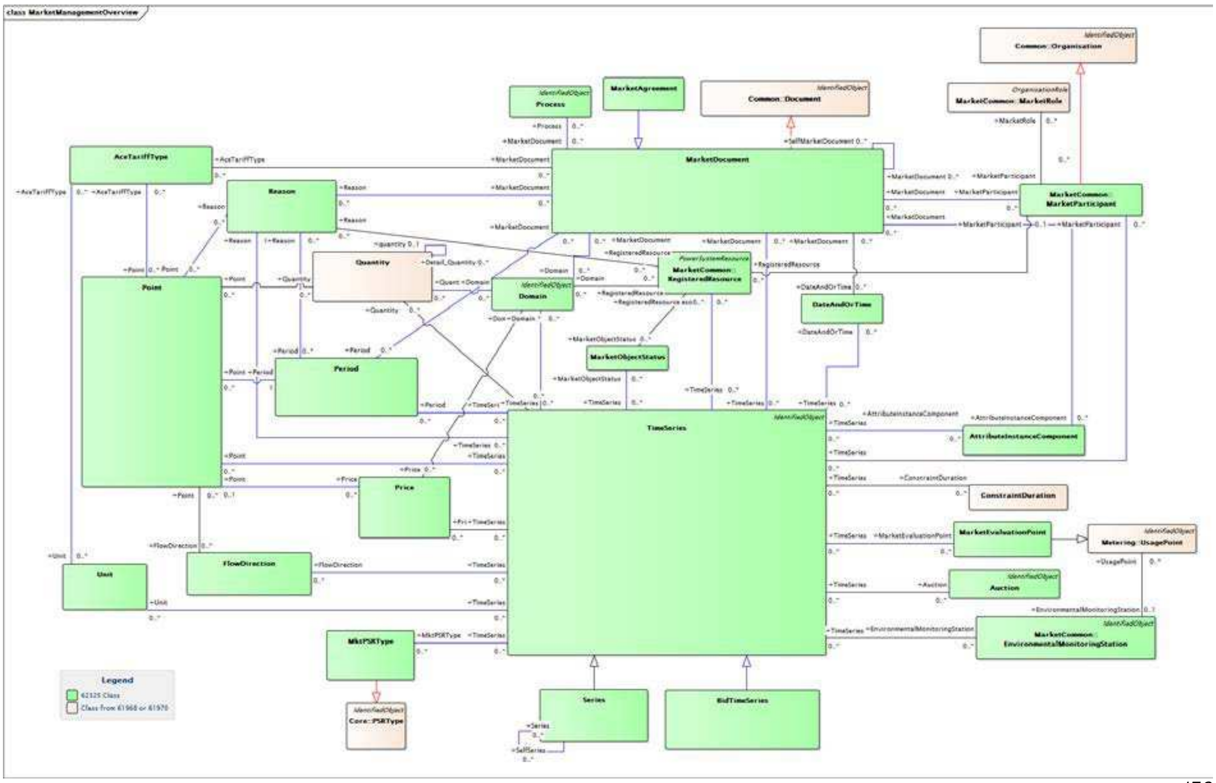
Ce paquetage contient toutes les principales extensions de marché du CIM exigées pour les systèmes de gestion de marché.



IEC

**Figure 29 – Diagramme de classe MarketManagement::Inheritance**

Figure 29: Fournit la structure des héritages de la gestion de marché.



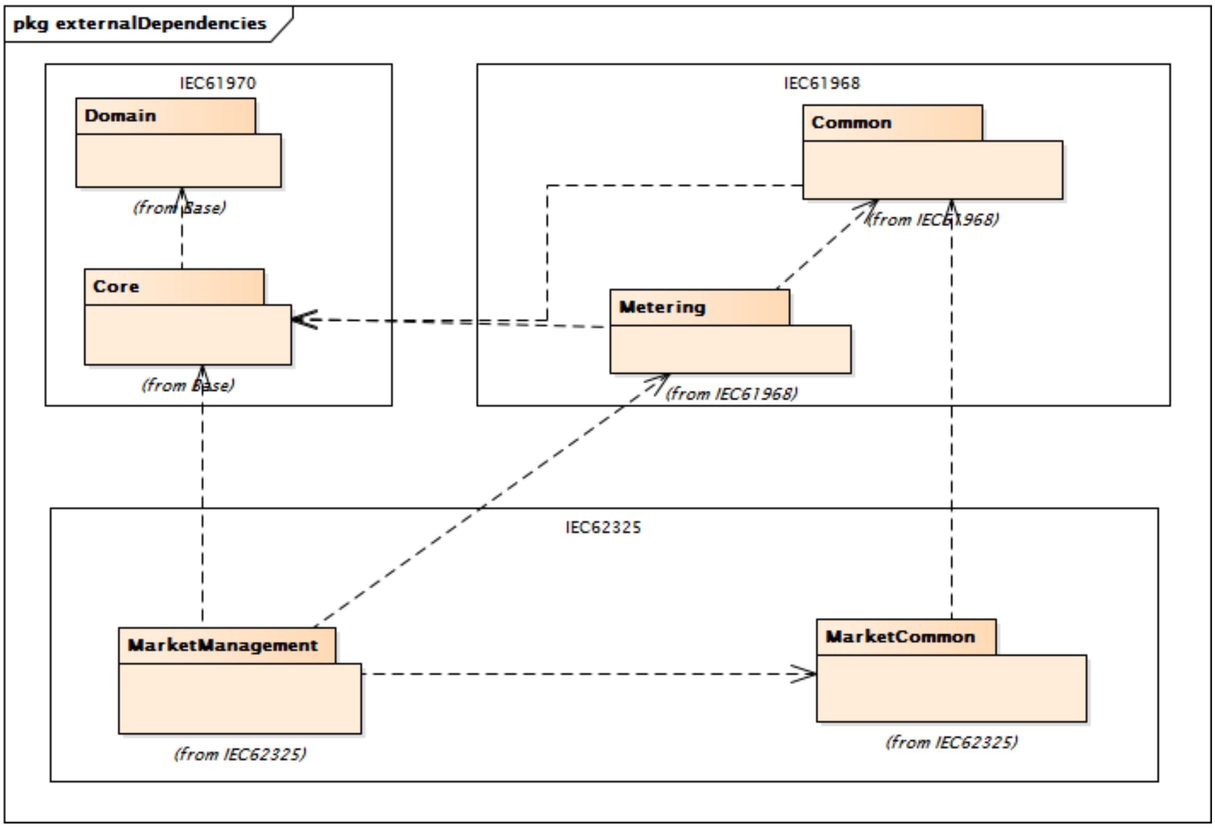

IEC

| Anglais | Français |
|---|---|
| Legend | Légende |
| 62325 class | Classe 62325 |
| Class from 61968 or 61970 | Classe issue de 61968 ou 61970 |

**Figure 30 – Diagramme de classe MarketManagement::MarketManagementOverview**

Figure 30: Décrit toutes les classes utilisées pour la définition des profils de gestion de marché.

Il est possible de voir toutes les dépendances avec d'autres concepts communs du CIM dans les systèmes de gestion d'énergie et les systèmes de gestion de la distribution.




| Anglais | Français |
|---|---|
| from Base | provenant de Base |
| from IEC61968 | issu de IEC61968 |
| from IEC62325 | issu de IEC62325 |

**Figure 31 – Diagramme de paquetages MarketManagement::externalDependencies**

Figure 31: Fournit les dépendances externes de la gestion du marché.

### 6.4.2 Classe racine AceTariffType

Type de tarif des erreurs de Zone de réglage (ACE – Area Control Error) qui est appliqué ou utilisé.

Le Tableau 17 présente tous les attributs de AceTariffType.

**Tableau 17 – Attributs de MarketManagement::AceTariffType**

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Type codé d'un tarif ACE. |

Le Tableau 18 présente toutes les extrémités d'association de AceTariffType avec d'autres classes.

**Tableau 18 – Extrémités d'association de MarketManagement:: AceTariffType avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Unit | 0..* | Unit | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Point | 0..* | Point | |

### 6.4.3 Classe racine AttributeInstanceComponent

Classe utilisée pour fournir des informations sur un attribut.

Le Tableau 19 présente tous les attributs de AttributeInstanceComponent.

**Tableau 19 – Attributs de MarketManagement::AttributeInstanceComponent**

| nom | mult | type | description |
|---|---|---|---|
| attribute | 0..1 | String | Identification du nom formel de l'attribut. |
| attributeValue | 0..1 | String | Valeur d'instance de l'attribut. |
| position | 0..1 | Integer | Valeur séquentielle représentant un numéro de séquence relatif. |

Le Tableau 20 présente toutes les extrémités d'association de AttributeInstanceComponent avec d'autres classes.

**Tableau 20 – Extrémités d'association de MarketManagement:: AttributeInstanceComponent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | MarketDocument | 0..* | MarketDocument | |

### 6.4.4 Auction

Classe fournissant l'identification et le type d'une vente aux enchères.

Le Tableau 21 présente tous les attributs de Auction.

**Tableau 21 – Attributs de MarketManagement::Auction**

| nom | mult | type | description |
|---|---|---|---|
| allocationMode | 0..1 | String | Identification de la méthode d'attribution dans une vente aux enchères. |
| cancelled | 0..1 | String | Indicateur spécifiant que la vente aux enchères a été annulée. |
| category | 0..1 | String | Catégorie de produit d'une vente aux enchères. |
| paymentTerms | 0..1 | String | Conditions régissant la détermination du prix d'offre. |
| rights | 0..1 | String | Droits d'utilisation de la capacité de transport acquise lors d'une vente aux enchères. |

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Type de vente aux enchères (par exemple: implicite, explicite...). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 22 présente toutes les extrémités d'association de Auction avec d'autres classes.

**Tableau 22 – Extrémités d’association de MarketManagement:: Auction avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.5 BidTimeSeries

Spécification formelle de caractéristiques spécifiques relatives à une offre.

Le Tableau 23 présente tous les attributs de BidTimeSeries.

**Tableau 23 – Attributs de MarketManagement::BidTimeSeries**

| nom | mult | type | description |
|---|---|---|---|
| blockBid | 0..1 | String | Indique que les valeurs de la période sont considérées comme un ensemble. Elles ne peuvent être ni modifiées ni subdivisées. |
| direction | 0..1 | String | Identification codée du flux de l'énergie. |
| divisible | 0..1 | String | Indique si chaque élément de l'offre peut être accepté en partie ou non. |
| status | 0..1 | String | Information concernant le statut de l’offre, tel que "shared" (partagé), "restricted" (restreint), ... |
| linkedBidsIdentification | 0..1 | String | Identification unique associée à toutes les offres liées. |
| exclusiveBidsIdentification | 0..1 | String | Identification unique associée à toutes les offres liées.<br>Identification d’un ensemble d’offres qui sont reliées ensemble signifiant que seule une offre peut être acceptée.<br>L’identification est définie par le soumissionnaire et doit être unique pour une vente aux enchères donnée. |
| minimumActivationQuantity | 0..1 | Decimal | Quantité minimale d'énergie qui peut être activée à un intervalle de temps donné. |

| nom | mult | type | description |
|---|---|---|---|
| stepIncrementQuantity | 0..1 | Decimal | Incrément minimal qui peut s'appliquer à une augmentation lors une demande d'activation. |
| priority | 0..1 | Integer | Priorité numérique locale accordée à une offre. Les valeurs numériques inférieures ont une priorité supérieure. |
| businessType | 0..1 | String | Hérité de: TimeSeries |
| cancelledTS | 0..1 | String | Hérité de: TimeSeries |
| curveType | 0..1 | String | Hérité de: TimeSeries |
| objectAggregation | 0..1 | String | Hérité de: TimeSeries |
| product | 0..1 | String | Hérité de: TimeSeries |
| version | 0..1 | String | Hérité de: TimeSeries |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 24 présente toutes les extrémités d'association de BidTimeSeries avec d'autres classes.

**Tableau 24 – Extrémités d’association de MarketManagement:: BidTimeSeries avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | Hérité de: TimeSeries |
| 0..* | EnvironmentalMonitoringStation | 0..* | EnvironmentalMonitoringStation | Hérité de: TimeSeries |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | Hérité de: TimeSeries |
| 0..* | Domain | 0..* | Domain | Hérité de: TimeSeries |
| 0..* | FlowDirection | 0..* | FlowDirection | Hérité de: TimeSeries |
| 0..* | MarketDocument | 0..* | MarketDocument | Hérité de: TimeSeries |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | Hérité de: TimeSeries |
| 0..* | Period | 0..* | Period | Hérité de: TimeSeries |
| 0..* | Reason | 0..* | Reason | Hérité de: TimeSeries |
| 0..* | RegisteredResource | 0..* | RegisteredResource | Hérité de: TimeSeries |
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | Hérité de: TimeSeries |
| 0..* | Auction | 0..* | Auction | Hérité de: TimeSeries |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | Hérité de: TimeSeries |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: TimeSeries |
| 0..* | MktPSRType | 0..* | MktPSRType | Hérité de: TimeSeries |
| 0..* | Point | 0..* | Point | Hérité de: TimeSeries |
| 0..* | Price | 0..* | Price | Hérité de: TimeSeries |
| 0..* | Quantity | 0..* | Quantity | Hérité de: TimeSeries |
| 0..* | Unit | 0..* | Unit | Hérité de: TimeSeries |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.6 Classe racine ConstraintDuration

Durée de contrainte à activer, à mettre en exploitation, à désactiver, … dans un événement donné.

Le Tableau 25 présente tous les attributs de ConstraintDuration.

**Tableau 25 – Attributs de MarketManagement::ConstraintDuration**

| nom | mult | type | description |
|---|---|---|---|
| duration | 0..1 | Duration | Durée de la contrainte. |
| type | 0..1 | String | Type de contrainte. |

Le Tableau 26 présente toutes les extrémités d’association de ConstraintDuration avec d’autres classes.

**Tableau 26 – Extrémités d’association de MarketManagement:: ConstraintDuration avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.7 Classe racine DateAndOrTime

La date et/ou l'heure.

Le Tableau 27 présente tous les attributs de DateAndOrTime.

**Tableau 27 – Attributs de MarketManagement::DateAndOrTime**

| nom | mult | type | description |
|---|---|---|---|
| date | 0..1 | Date | Date sous la forme "aaaa-mm-jj", qui est conforme à l’ISO 8601. |
| time | 0..1 | Time | Heure sous la forme "hh:mm:ss.sssZ", qui est conforme à l’ISO 8601. |

Le Tableau 28 présente toutes les extrémités d'association de DateAndOrTime avec d'autres classes.

**Tableau 28 – Extrémités d’association de MarketManagement:: DateAndOrTime avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.8 Domain

Domaine d'activité défini dans le marché de l'énergie.

Le Tableau 29 présente tous les attributs de Domain.

**Tableau 29 – Attributs de MarketManagement::Domain**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 30 présente toutes les extrémités d'association de Domain avec d'autres classes.

**Tableau 30 – Extrémités d’association de MarketManagement:: Domain avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Price | 0..* | Price | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Quantity | 0..* | Quantity | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.9 Classe racine FlowDirection

Identification codée du sens du flux de l'énergie.

Le Tableau 31 présente tous les attributs de FlowDirection.

**Tableau 31 – Attributs de MarketManagement::FlowDirection**

| nom | type | description |
|---|---|---|
| direction | String | Identification codée du sens du flux de l'énergie. |

Le Tableau 32 présente toutes les extrémités d'association de FlowDirection avec d'autres classes.

**Tableau 32 – Extrémités d'association de MarketManagement:: FlowDirection avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.10 MarketAgreement

Identification ou éventuellement le contenu d'un accord entre au moins deux entités.

Le Tableau 33 présente tous les attributs de MarketAgreement.

**Tableau 33 – Attributs de MarketManagement::MarketAgreement**

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 34 présente toutes les extrémités d'association de MarketAgreement avec d'autres classes.

**Tableau 34 – Extrémités d'association de MarketManagement:: MarketAgreement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | Hérité de: MarketDocument |
| 0..* | Domain | 0..* | Domain | Hérité de: MarketDocument |
| 0..* | Period | 0..* | Period | Hérité de: MarketDocument |
| 0..* | SelfMarketDocument | 0..* | MarketDocument | Hérité de: MarketDocument |
| 0..* | MarketParticipant | 0..* | MarketParticipant | Hérité de: MarketDocument |
| 0..* | AceTariffType | 0..* | AceTariffType | Hérité de: MarketDocument |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | Hérité de: MarketDocument |
| 0..* | Reason | 0..* | Reason | Hérité de: MarketDocument |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: MarketDocument |
| 0..* | Process | 0..* | Process | Hérité de: MarketDocument |
| 0..* | MarketDocument | 0..* | MarketDocument | Hérité de: MarketDocument |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.11 MarketDocument

Document électronique contenant les informations nécessaires pour satisfaire à un ensemble donné d'exigences de processus métier.

Le Tableau 35 présente tous les attributs de MarketDocument.

**Tableau 35 – Attributs de MarketManagement::MarketDocument**

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 36 présente toutes les extrémités d'association de MarketDocument avec d'autres classes.

**Tableau 36 – Extrémités d’association de MarketManagement:: MarketDocument avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | Period | 0..* | Period | |
| 0..* | SelfMarketDocument | 0..* | MarketDocument | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..* | AceTariffType | 0..* | AceTariffType | |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Process | 0..* | Process | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.12 MarketEvaluationPoint

Identification d'une entité dans laquelle les produits énergétiques sont mesurés ou calculés.

Le Tableau 37 présente tous les attributs de MarketEvaluationPoint.

**Tableau 37 – Attributs de MarketManagement::MarketEvaluationPoint**

| nom | mult | type | description |
|---|---|---|---|
| isSdp | 0..1 | Boolean | Hérité de: UsagePoint |
| isVirtual | 0..1 | Boolean | Hérité de: UsagePoint |
| phaseCode | 0..1 | PhaseCode | Hérité de: UsagePoint |
| grounded | 0..1 | Boolean | Hérité de: UsagePoint |
| servicePriority | 0..1 | String | Hérité de: UsagePoint |
| serviceDeliveryRemark | 0..1 | String | Hérité de: UsagePoint |
| estimatedLoad | 0..1 | CurrentFlow | Hérité de: UsagePoint |
| checkBilling | 0..1 | Boolean | Hérité de: UsagePoint |
| ratedCurrent | 0..1 | CurrentFlow | Hérité de: UsagePoint |
| nominalServiceVoltage | 0..1 | Voltage | Hérité de: UsagePoint |
| ratedPower | 0..1 | ActivePower | Hérité de: UsagePoint |
| outageRegion | 0..1 | String | Hérité de: UsagePoint |
| readCycle | 0..1 | String | Hérité de: UsagePoint |
| readRoute | 0..1 | String | Hérité de: UsagePoint |
| amiBillingReady | 0..1 | AmiBillingReadyKind | Hérité de: UsagePoint |
| connectionState | 0..1 | UsagePointConnectedKind | Hérité de: UsagePoint |
| minimalUsageExpected | 0..1 | Boolean | Hérité de: UsagePoint |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 38 présente toutes les extrémités d'association de MarketEvaluationPoint avec d'autres classes.

**Tableau 38 – Extrémités d’association de MarketManagement:: MarketEvaluationPoint avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Equipments | 0..* | Equipment | Hérité de: UsagePoint |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: UsagePoint |
| 0..* | ServiceCategory | 0..1 | ServiceCategory | Hérité de: UsagePoint |
| 0..1 | EndDevices | 0..* | EndDevice | Hérité de: UsagePoint |
| 0..1 | ServiceMultipliers | 0..* | ServiceMultiplier | Hérité de: UsagePoint |
| 0..* | Outage | 0..* | Outage | Hérité de: UsagePoint |
| 0..* | Outage | 0..* | Outage | Hérité de: UsagePoint |
| 0..* | CustomerAgreement | 0..1 | CustomerAgreement | Hérité de: UsagePoint |
| 0..* | PricingStructures | 0..* | PricingStructure | Hérité de: UsagePoint |
| 0..* | ServiceLocation | 0..1 | ServiceLocation | Hérité de: UsagePoint |
| 0..* | EndDeviceControls | 0..* | EndDeviceControl | Hérité de: UsagePoint |
| 0..1 | EndDeviceEvents | 0..* | EndDeviceEvent | Hérité de: UsagePoint |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MeterReadings | 0..* | MeterReading | Hérité de: UsagePoint |
| 0..1 | MeterServiceWorkTasks | 0..* | MeterWorkTask | Hérité de: UsagePoint |
| 0..* | MetrologyRequirements | 0..* | MetrologyRequirement | Hérité de: UsagePoint |
| 0..* | ServiceSupplier | 0..1 | ServiceSupplier | Hérité de: UsagePoint |
| 0..* | UsagePointGroups | 0..* | UsagePointGroup | Hérité de: UsagePoint |
| 0..* | UsagePointLocation | 0..1 | UsagePointLocation | Hérité de: UsagePoint |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.13 Classe racine MarketObjectStatus

Condition ou position d'un objet par rapport à son statut.

Le Tableau 39 présente tous les attributs de MarketObjectStatus.

**Tableau 39 – Attributs de MarketManagement::MarketObjectStatus**

| nom | mult | type | description |
|---|---|---|---|
| status | 0..1 | String | Condition ou position codée d'un objet par rapport à son statut. |

Le Tableau 40 présente toutes les extrémités d'association de MarketObjectStatus avec d'autres classes.

**Tableau 40 – Extrémités d'association de MarketManagement:: MarketObjectStatus avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.14 MktPSRType

Type de ressource du système électrique.

Le Tableau 41 présente tous les attributs de MktPSRType.

**Tableau 41 – Attributs de MarketManagement::MktPSRType**

| nom | mult | type | description |
|---|---|---|---|
| psrType | 0..1 | String | Type codé de ressource de système électrique. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 42 présente toutes les extrémités d'association de MktPSRType avec d'autres classes.

**Tableau 42 – Extrémités d’association de MarketManagement:: MktPSRType avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..1 | PowerSystemResources | 0..* | PowerSystemResource | Hérité de: PSRType |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.15 Classe racine Period

Identification d'un intervalle de temps qui peut avoir une résolution donnée.

Le Tableau 43 présente tous les attributs de Period.

**Tableau 43 – Attributs de MarketManagement::Period**

| nom | mult | type | description |
|---|---|---|---|
| resolution | 0..1 | Duration | Nombre d'unités de temps qui composent une étape individuelle dans une période. |
| timeInterval | 0..1 | DateTimeInterval | Date et heure de début et de fin d'un intervalle donné. |

Le Tableau 44 présente toutes les extrémités d'association de Period avec d'autres classes.

**Tableau 44 – Extrémités d’association de MarketManagement:: Period avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Reason | 0..* | Reason | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 1..1 | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.16 Classe racine Point

Identification d'un ensemble de valeurs traitées dans un intervalle de temps spécifique.

Le Tableau 45 présente tous les attributs de Point.

**Tableau 45 – Attributs de MarketManagement::Point**

| nom | mult | type | description |
|---|---|---|---|
| position | 0..1 | Integer | Valeur séquentielle représentant la position relative dans un intervalle de temps donné. |
| quality | 0..1 | String | Qualité de l’information fournie. Cette qualité peut être estimée, non disponible, telle que fournie, etc. |
| quantity | 0..1 | Decimal | Quantité principale identifiée à un point. |
| secondaryQuantity | 0..1 | Decimal | Quantité secondaire identifiée à un point. |

Le Tableau 46 présente toutes les extrémités d'association de Point avec d'autres classes.

**Tableau 46 – Extrémités d’association de MarketManagement:: Point avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AceTariffType | 0..* | AceTariffType | |
| 0..* | Period | 1..1 | Period | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | FlowDirection | 0..* | FlowDirection | |
| 0..1 | Price | 0..* | Price | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | Quantity | 0..* | Quantity | |

### 6.4.17 Classe racine Price

Coût correspondant à une mesure spécifique et exprimé dans une devise.

Le Tableau 47 présente tous les attributs de Price.

**Tableau 47 – Attributs de MarketManagement::Price**

| nom | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Decimal | Nombre d'unités monétaires spécifié dans une unité monétaire. |
| category | 0..1 | String | Catégorie de prix à utiliser pour calculer un prix. La catégorie de prix est définie par accord mutuel entre les opérateurs du système. |
| direction | 0..1 | String | Le sens indique si un opérateur du système paie les entités du marché ou inversement. |

Le Tableau 48 présente toutes les extrémités d'association de Price avec d'autres classes.

**Tableau 48 – Extrémités d'association de MarketManagement:: Price avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Domain | 0..* | Domain | |
| 0..* | Point | 0..1 | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.18 Process

Spécification formelle d'un ensemble de transactions commerciales ayant le même objectif.

Le Tableau 49 présente tous les attributs de Process.

**Tableau 49 – Attributs de MarketManagement::Process**

| nom | mult | type | description |
|---|---|---|---|
| classificationType | 0..1 | String | Mécanisme de classification utilisé pour regrouper un ensemble d'objets dans un processus métier. Le regroupement peut être détaillé ou résumé. |
| processType | 0..1 | String | Type de processus métier. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 50 présente toutes les extrémités d'association de Process avec d'autres classes.

**Tableau 50 – Extrémités d'association de MarketManagement:: Process avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.19 Classe racine Quantity

Description des quantités nécessaires à l'échange de données.

Le type de quantité est décrit soit par le rôle de l'association, soit par l'attribut Type.

L'attribut Quality fournit les informations relatives à la qualité de la quantité (mesurée, estimée, etc.).

Le Tableau 51 présente tous les attributs de Quantity.

**Tableau 51 – Attributs de MarketManagement::Quantity**

| nom | mult | type | description |
|---|---|---|---|
| quantity | 0..1 | Decimal | Valeur de quantité.<br>Le rôle de l’association fournit les informations relatives à ce qui est exprimé. |
| type | 0..1 | String | Description du type de quantité. |
| quality | 0..1 | String | Qualité des informations fournies. Cette qualité peut être estimée, non disponible, telle que fournie, etc. |

Le Tableau 52 présente toutes les extrémités d’association de Quantity avec d’autres classes.

**Tableau 52 – Extrémités d’association de MarketManagement:: Quantity avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Domain | 0..* | Domain | |
| 0..* | Point | 0..* | Point | |
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | Quantity | 0..1 | Quantity | |
| 0..1 | Detail_Quantity | 0..* | Quantity | Informations supplémentaires relatives à la quantité associée. |

### 6.4.20 Classe racine Reason

Motivation d'un acte.

Le Tableau 53 présente tous les attributs de Reason.

**Tableau 53 – Attributs de MarketManagement::Reason**

| nom | mult | type | description |
|---|---|---|---|
| code | 0..1 | String | Motivation d'un acte sous forme codée. |
| text | 0..1 | String | Explication textuelle correspondant au code de cause. |

Le Tableau 54 présente toutes les extrémités d'association de Reason avec d'autres classes.

**Tableau 54 – Extrémités d’association de MarketManagement:: Reason avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | Point | 0..* | Point | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | Period | 0..* | Period | |
| 0..* | TimeSeries | 0..* | TimeSeries | |

### 6.4.21 Series

Ensemble d'objets physiques ou conceptuels définis pour la même période ou le même instant.

Le Tableau 55 présente tous les attributs de Series.

**Tableau 55 – Attributs de MarketManagement::Series**

| nom | mult | type | description |
|---|---|---|---|
| lastUpdateDate | 0..1 | Date | Date de la dernière mise à jour relative à cet objet du marché. |
| methodType | 0..1 | String | Type de méthode employée dans le processus de métier lié à cette Series; par exemple: méthode de comptage. |
| registrationDate | 0..1 | Date | Pour un objet de marché, date d'enregistrement d'un contrat; par exemple, date de changement de fournisseur pour un client. |
| businessType | 0..1 | String | Hérité de: TimeSeries |
| cancelledTS | 0..1 | String | Hérité de: TimeSeries |
| curveType | 0..1 | String | Hérité de: TimeSeries |
| objectAggregation | 0..1 | String | Hérité de: TimeSeries |
| product | 0..1 | String | Hérité de: TimeSeries |
| version | 0..1 | String | Hérité de: TimeSeries |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 56 présente toutes les extrémités d'association de Series avec d'autres classes.

**Tableau 56 – Extrémités d'association de MarketManagement:: Series avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | SelfSeries | 0..* | Series | |
| 0..* | Series | 0..* | Series | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | Hérité de: TimeSeries |
| 0..* | EnvironmentalMonitoringStation | 0..* | EnvironmentalMonitoring Station | Hérité de: TimeSeries |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | Hérité de: TimeSeries |
| 0..* | Domain | 0..* | Domain | Hérité de: TimeSeries |
| 0..* | FlowDirection | 0..* | FlowDirection | Hérité de: TimeSeries |
| 0..* | MarketDocument | 0..* | MarketDocument | Hérité de: TimeSeries |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | Hérité de: TimeSeries |
| 0..* | Period | 0..* | Period | Hérité de: TimeSeries |
| 0..* | Reason | 0..* | Reason | Hérité de: TimeSeries |
| 0..* | RegisteredResource | 0..* | RegisteredResource | Hérité de: TimeSeries |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AttributeInstanceComponent | 0..* | AttributeInstanceComponent | Hérité de: TimeSeries |
| 0..* | Auction | 0..* | Auction | Hérité de: TimeSeries |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | Hérité de: TimeSeries |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: TimeSeries |
| 0..* | MktPSRType | 0..* | MktPSRType | Hérité de: TimeSeries |
| 0..* | Point | 0..* | Point | Hérité de: TimeSeries |
| 0..* | Price | 0..* | Price | Hérité de: TimeSeries |
| 0..* | Quantity | 0..* | Quantity | Hérité de: TimeSeries |
| 0..* | Unit | 0..* | Unit | Hérité de: TimeSeries |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.22 TimeSeries

Série Chronologique, ensemble de mesures chronologiques ordinaires ou de valeurs quantitatives d'un phénomène individuel ou collectif pris à des périodes/instants successifs, et souvent, équidistants.

Le Tableau 57 présente tous les attributs de TimeSeries.

**Tableau 57 – Attributs de MarketManagement::TimeSeries**

| nom | mult | type | description |
|---|---|---|---|
| businessType | 0..1 | String | Identification de la nature de la série chronologique. |
| cancelledTS | 0..1 | String | Indicateur selon lequel la TimeSeries, identifiée par le mRID, ainsi que toutes les valeurs envoyées dans une version précédente de la TimeSeries dans un document précédent, sont annulées. |
| curveType | 0..1 | String | Représentation codée du type de courbe décrit. |
| objectAggregation | 0..1 | String | Identification de l'objet qui est le dénominateur commun utilisé pour agréger une série chronologique. |
| product | 0..1 | String | Type de produit, tel que la puissance, l'énergie, la puissance réactive, la capacité de transport, qui fait l'objet de la série chronologique. |
| version | 0..1 | String | Version de la série chronologique. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 58 présente toutes les extrémités d'association de TimeSeries avec d'autres classes.

**Tableau 58 – Extrémités d'association de MarketManagement:: TimeSeries avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..* | EnvironmentalMonitoring Station | 0..* | EnvironmentalMonitorin gStation | |
| 0..* | DateAndOrTime | 0..* | DateAndOrTime | |
| 0..* | Domain | 0..* | Domain | |
| 0..* | FlowDirection | 0..* | FlowDirection | |
| 0..* | MarketDocument | 0..* | MarketDocument | |
| 0..* | MarketEvaluationPoint | 0..* | MarketEvaluationPoint | |
| 0..* | Period | 0..* | Period | |
| 0..* | Reason | 0..* | Reason | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | AttributeInstanceCompon ent | 0..* | AttributeInstanceCompo nent | |
| 0..* | Auction | 0..* | Auction | |
| 0..* | ConstraintDuration | 0..* | ConstraintDuration | |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | |
| 0..* | MktPSRType | 0..* | MktPSRType | |
| 0..* | Point | 0..* | Point | |
| 0..* | Price | 0..* | Price | |
| 0..* | Quantity | 0..* | Quantity | |
| 0..* | Unit | 0..* | Unit | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.4.23 Classe racine Unit

Identification du nom de l'unité pour les grandeurs de série chronologique.

Le Tableau 59 présente tous les attributs de Unit.

**Tableau 59 – Attributs de MarketManagement::Unit**

| nom | mult | type | description |
|---|---|---|---|
| name | 0..1 | String | Représentation codée de l'unité. |

Le Tableau 60 présente toutes les extrémités d'association de Unit avec d'autres classes.

**Tableau 60 – Extrémités d'association de MarketManagement::
Unit avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TimeSeries | 0..* | TimeSeries | |
| 0..* | AceTariffType | 0..* | AceTariffType | |

## 6.5 Paquetage MarketOperations

### 6.5.1 Généralités

Ce paquetage contient toutes les principales extensions de marché du CIM exigées pour les systèmes des opérations du marché.

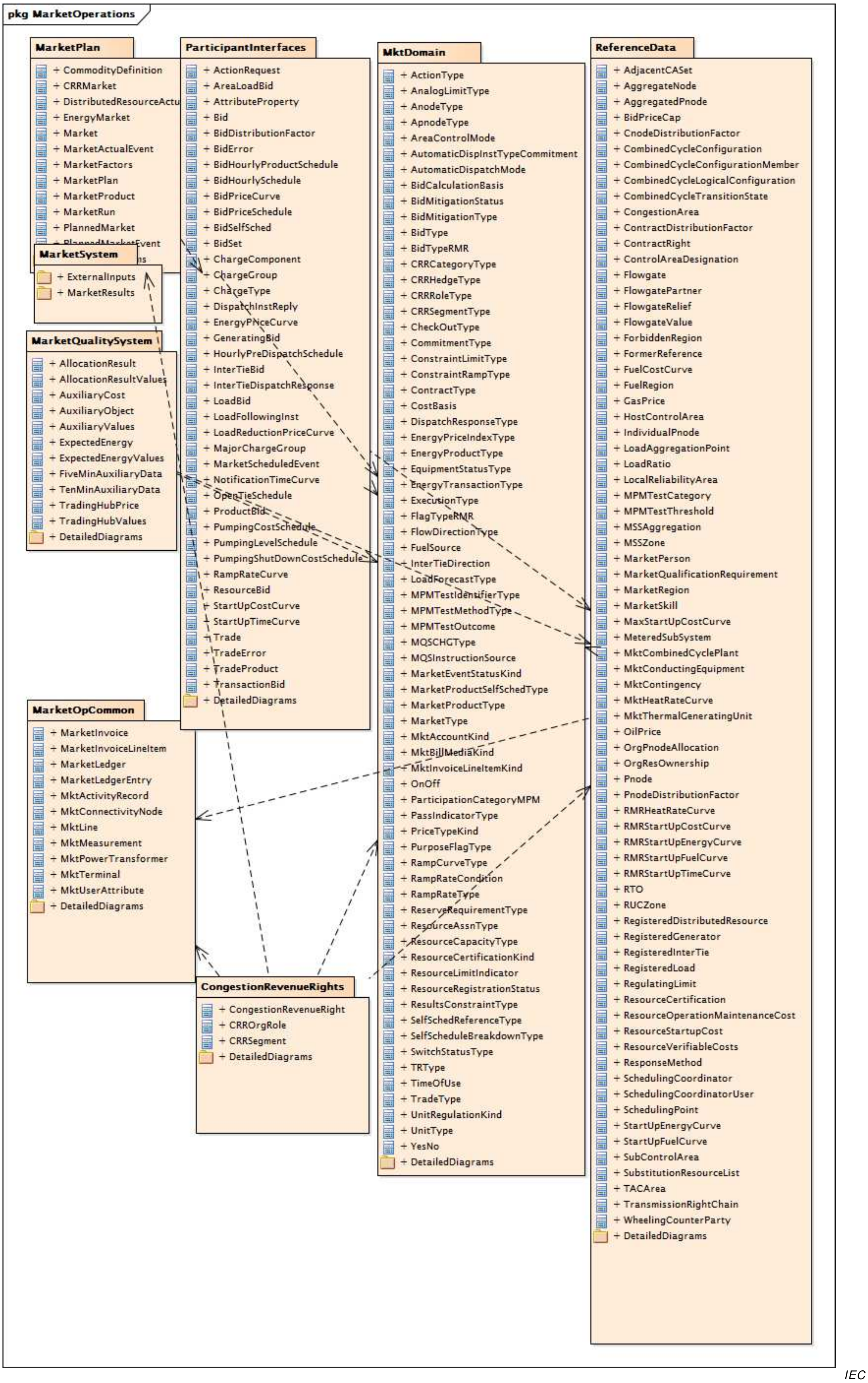
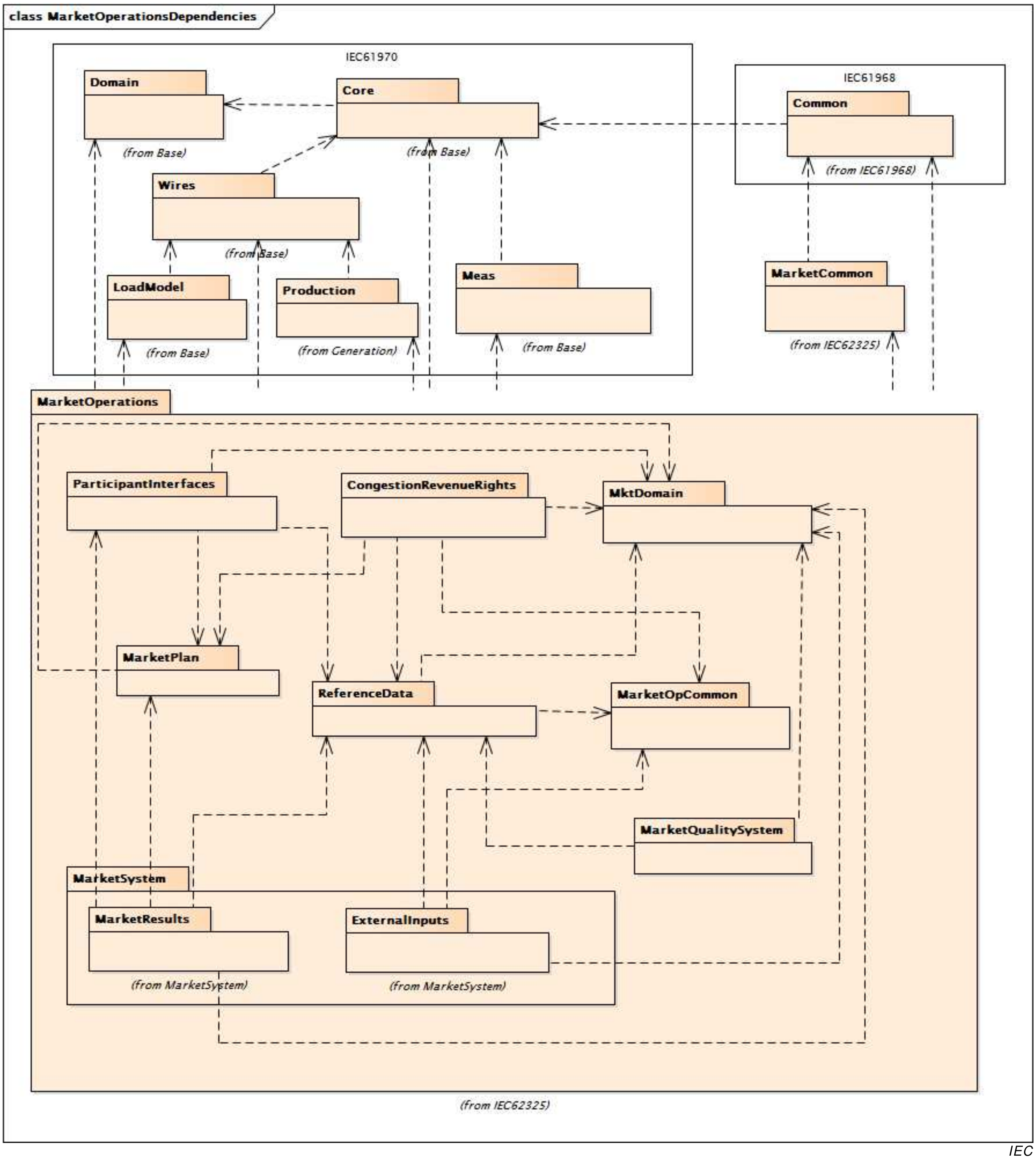



**Figure 32 – Diagramme de paquetage MarketOperations::MarketOperations**

Figure 32: représente le paquetage contenant toutes les principales extensions de marché du CIM exigées pour les systèmes des opérations du marché.




| Anglais | Français |
|---|---|
| from | provenant de |

**Figure 33 – Diagramme de classe MarketOperations::MarketOperationsDependencies**

Figure 33: représente les dépendances des opérations du marché internes au paquetage et dépendances externes.

### 6.5.2 Paquetage CongestionRevenueRights

#### 6.5.2.1 Généralités

Les frais de congestion sont une charge importante et très volatile qui concerne aujourd'hui de nombreux participants dans les marchés de l'énergie électrique basés sur les prix marginaux locaux (LMP). Pour cette raison, les ISO offrent des droits à revenu de congestion (CRR), également appelées droits de transport financiers ou contrats de congestion de transport. Ce sont des instruments financiers qui permettent aux participants du marché de se couvrir des frais de congestion lors de la programmation des transactions de production, de charge et d'énergie bilatérales.

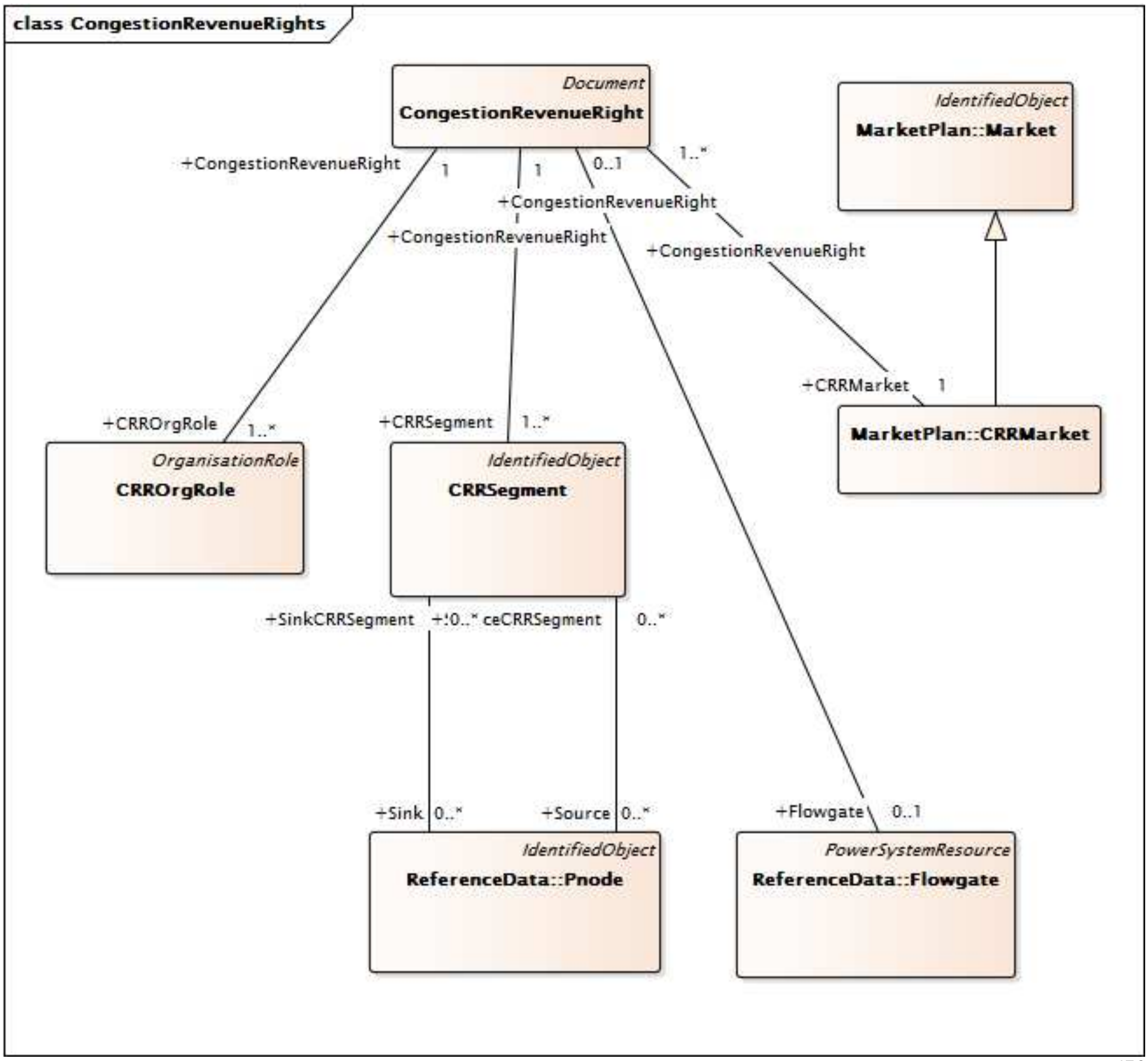



**Figure 34 – Diagramme de classe CongestionRevenueRights::CongestionRevenueRights**

Figure 34: représente un droit à revenu de congestion (CRR) qui est un instrument financier qui permet à son détenteur de recevoir un versement de CRR lorsque la congestion s'effectue de la source vers la source de CRR vers le puits de CRR. Les classes représentées sont utilisées pour modéliser les données de CRR. La classe CRRSegment contient des informations sur les segments de CRR. La classe MktOrganization contient des informations sur l'entité qui détient la CRR. Les associations de CRRSegment vers ReferenceData mettent en évidence la source et le puits Pnodes.

### 6.5.2.2 CongestionRevenueRight

Classe de droit à revenu de congestion (CRR) héritée de la classe Document.

Une CRR est un concept financier qui est utilisé pour se couvrir des frais de congestion.

La CRR est généralement réglée selon les prix marginaux locaux (LMP) qui sont calculés dans le marché à J-1. Ces LMP sont définis par les programmes/offres des ressources à J-1. Les CRR ne couvriront pas les pertes marginales. Si le composant de congestion du LMP au puits est supérieur à celui à la source, le propriétaire de la CRR est en droit de recevoir une partie des revenus de congestion. Si le composant de congestion au puits est inférieur à celui à la source, alors le propriétaire de CRR sous forme d'obligations sera facturé, tandis que le propriétaire de CRR sous forme d'options ne le sera pas.

Le Tableau 61 présente tous les attributs de CongestionRevenueRight.

**Tableau 61 – Attributs de CongestionRevenueRights:: CongestionRevenueRight**

| nom | mult | type | description |
|---|---|---|---|
| cRRcategory | 0..1 | CRRCategoryType | Catégorie de CRR définie sur "PTP" (*point-to-point*) pour une CRR point à point ou sur "NSR" (*Network Service Right*) pour un droit de service réseau. Si la catégorie de CRR est "PTP", les champs ID source et ID puits sont exigés. Si la catégorie de CRR est "NSR", un champ (ID source ou ID puits) doit être défini sur zéro et l'autre ne doit pas être défini sur zéro. Cependant, la catégorie "NSR" comprendra au moins trois enregistrements. |
| cRRtype | 0..1 | CRRSegmentType | Type de CRR, dans les définitions types possibles du système de CRR (par exemple, "LSE","ETC"). |
| hedgeType | 0..1 | CRRHedgeType | Type de couverture. Obligation ou Option. La couverture Obligation exige que le détenteur reçoive ou paie des frais de congestion. La couverture Option donne au détenteur le choix de recevoir ou de payer les frais de congestion. |
| timeOfUse | 0..1 | TimeOfUse | Indicateur des heures d'utilisation de la CRR – heures de pointe (ON), heures creuses (OFF) ou 24 h/24 (24HR). |
| tradeSliceID | 0..1 | String | Segment de la CRR décrit dans l'enregistrement actuel. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 62 présente toutes les extrémités d'association de CongestionRevenueRight avec d'autres classes.

**Tableau 62 – Extrémités d’association de CongestionRevenueRights:: CongestionRevenueRight avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | Flowgate | 0..1 | Flowgate | |
| 1..* | CRRMarket | 1..1 | CRRMarket | |
| 1..1 | CRRSegment | 1..* | CRRSegment | |
| 1..1 | CRROrgRole | 1..* | CRROrgRole | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.2.3 CRROrgRole

Identifie comment une organisation peut participer avec un droit à revenu de congestion défini (CRR).

Le Tableau 63 présente tous les attributs de CRROrgRole.

**Tableau 63 – Attributs de CongestionRevenueRights::CRROrgRole**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | CRRRoleType | Type de rôle de l'organisation quant aux droits à revenu de congestion. |
| status | 0..1 | Status | Statut du rôle de l'organisation quant aux droits à revenu de congestion. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 64 présente toutes les extrémités d'association de CRROrgRole avec d'autres classes.

**Tableau 64 – Extrémités d’association de CongestionRevenueRights:: CRROrgRole avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | CongestionRevenueRight | 1..1 | CongestionRevenueRight | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | Hérité de: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.2.4 CRRSegment

CRRSegment représente un segment de la CRR dans un cadre temporel particulier.

La classe de segment contient le montant, le prix d’équilibre, la date et l’heure de début et la date et l’heure de fin.

Le Tableau 65 présente tous les attributs de CRRSegment.

**Tableau 65 – Attributs de CongestionRevenueRights::CRRSegment**

| nom | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Money | Montant en dollars = quantité x clearingPrice |
| clearingPrice | 0..1 | Money | Prix d'équilibre d'une CRR |
| endDateTime | 0..1 | DateTime | Date et heure de fin de segment |
| quantity | 0..1 | Float | Quantité de MW associée à la CRR |
| startDateTime | 0..1 | DateTime | Date et heure de début de segment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 66 présente toutes les extrémités d'association de CRRSegment avec d'autres classes.

**Tableau 66 – Extrémités d'association de CongestionRevenueRights:: CRRSegment avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | CongestionRevenueRight | 1..1 | CongestionRevenue Right | |
| 0..* | Source | 0..* | Pnode | |
| 0..* | Sink | 0..* | Pnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3 Paquetage MarketOpCommon

#### 6.5.3.1 Généralités

Ce paquetage contient les objets communs partagés par les paquetages MarketOperations.

#### 6.5.3.2 Classe racine MarketInvoice

Ensemble des éléments de ligne d'une facture. Toutes les factures comportent une échéance, un montant à régler et des informations (client, banques, etc.) obtenues grâce à des associations. L'ensemble de la facture est fondé sur des éléments de ligne individuels, qui contiennent chacun des montants et des descriptions pour des services ou produits spécifiques.

Le Tableau 67 présente tous les attributs de MarketInvoice.

**Tableau 67 – Attributs de MarketOpCommon::MarketInvoice**

| nom | mult | type | description |
|---|---|---|---|
| amount | 0..1 | Money | Montant total à régler sur cette facture basé sur les éléments de ligne et les ajustements applicables. |
| billMediaKind | 0..1 | MktBillMediaKind | Type de support utilisé pour fournir les CustomerBillingInfo. |
| dueDate | 0..1 | Date | Date calculée à partir de laquelle le montant de la facture est dû. |
| kind | 0..1 | MktAccountKind | Type de facture ("vente" par défaut). |
| mailedDate | 0..1 | Date | Date à laquelle le relevé de facturation/la facture client a été imprimé/envoyé par courrier. |
| proForma | 0..1 | Boolean | True si le paiement doit être réglé par le client pour accepter un ErpQuote particulier (avec la conception associée) et lancer le travail, au moment où il convient de générer automatiquement un ErpInvoice associé. EprPayment.subjectStatus satisfait aux conditions spécifiées dans ErpQuote. |
| referenceNumbe r | 0..1 | String | Numéro de facture référencé par cette facture. |
| transactionDate Time | 0..1 | DateTime | Date et heure d'émission de la facture. |
| transferType | 0..1 | String | Type de transfert de facture. |

Le Tableau 68 présente toutes les extrémités d'association de MarketInvoice avec d'autres classes.

**Tableau 68 – Extrémités d’association de MarketOpCommon:: MarketInvoice avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MajorChargeGroup | 1..* | MajorChargeGroup | |
| 1..1 | MarketInvoiceLineItems | 0..* | MarketInvoiceLineItem | |

### 6.5.3.3 Classe racine MarketInvoiceLineItem

Élément de ligne individuel sur une facture.

Le Tableau 69 présente tous les attributs de MarketInvoiceLineItem.

**Tableau 69 – Attributs de MarketOpCommon::MarketInvoiceLineItem**

| nom | mult | type | description |
|---|---|---|---|
| billPeriod | 0..1 | DateTimeInterval | Période de facturation de l'élément de ligne. |
| glAccount | 0..1 | String | Le code de facturation du grand livre général doit être une combinaison valide. |
| glDateTime | 0..1 | DateTime | Les éléments de ligne de date et d'heure seront publiés dans le grand livre général. |
| kind | 0..1 | MktInvoiceLineItemKind | Type d'élément de ligne. |
| lineAmount | 0..1 | Float | Montant dû pour cet élément de ligne. |
| lineNumber | 0..1 | String | Numéro d'élément de ligne sur le relevé de facturation. |
| lineVersion | 0..1 | String | Numéro de version de la facture. |
| netAmount | 0..1 | Float | Montant net des charges de l'élément de ligne. |
| previousAmount | 0..1 | Float | Montant précédent des charges de l'élément de ligne. |

Le Tableau 70 présente toutes les extrémités d'association de MarketInvoiceLineItem avec d'autres classes.

**Tableau 70 – Extrémités d’association de MarketOpCommon:: MarketInvoiceLineItem avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketInvoice | 1..1 | MarketInvoice | |
| 0..* | Settlement | 0..* | Settlement | |
| 0..* | ContainerMarketInvoiceLineItem | 0..1 | MarketInvoiceLineItem | |
| 0..1 | ComponentMarketInvoiceLineItems | 0..* | MarketInvoiceLineItem | |

### 6.5.3.4 Classe racine MarketLedger

Dans les transactions de comptabilité, un grand livre est un registre contenant les comptes sur lesquels des débits et des crédits sont publiés depuis des journaux, dans lesquels des transactions sont enregistrées au préalable. Les entrées du journal sont régulièrement publiées sur le grand livre. Le grand livre réel contient les montants réels des comptes d'une société ou d'un secteur de marché. Les montants réels peuvent être générés dans une application source, puis chargés dans un grand livre spécifique appartenant au grand livre général ou à l'application budgétaire d'une entreprise.

Le Tableau 71 présente toutes les extrémités d'association de MarketLedger avec d'autres classes.

**Tableau 71 – Extrémités d'association de MarketOpCommon:: MarketLedger avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | MarketLedgerEntries | 0..* | MarketLedgerEntry | |

### 6.5.3.5 Classe racine MarketLedgerEntry

Détails d'une entrée individuelle dans un grand livre provenant d'un journal à la date attribuée.

Le Tableau 72 présente tous les attributs de MarketLedgerEntry.

**Tableau 72 – Attributs de MarketOpCommon::MarketLedgerEntry**

| nom | mult | type | description |
|---|---|---|---|
| accountID | 0..1 | String | Identifiant de compte de cette entrée. |
| accountKind | 0..1 | MktAccountKind | Type de compte de cette entrée. |
| amount | 0..1 | Money | Montant du débit/crédit de ce compte. |
| postedDateTime | 0..1 | DateTime | Date et heure auxquelles cette entrée a été publiée sur le grand livre. |
| status | 0..1 | Status | Statut de l'entrée du grand livre. |
| transactionDateTime | 0..1 | DateTime | Date et heure de l'enregistrement de l'entrée du journal. |

Le Tableau 73 présente toutes les extrémités d'association de MarketLedgerEntry avec d'autres classes.

**Tableau 73 – Extrémités d'association de MarketOpCommon:: MarketLedgerEntry avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Settlement | 0..* | Settlement | |
| 0..* | MarketLedger | 1..1 | MarketLedger | |

### 6.5.3.6 MktActivityRecord

Sous-classe de IEC61968:Common:ActivityRecord

Le Tableau 74 présente tous les attributs de MktActivityRecord.

**Tableau 74 – Attributs de MarketOpCommon::MktActivityRecord**

| nom | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | Hérité de: ActivityRecord |
| type | 0..1 | String | Hérité de: ActivityRecord |
| severity | 0..1 | String | Hérité de: ActivityRecord |
| reason | 0..1 | String | Hérité de: ActivityRecord |
| status | 0..1 | Status | Hérité de: ActivityRecord |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 75 présente toutes les extrémités d'association de MktActivityRecord avec d'autres classes.

**Tableau 75 – Extrémités d'association de MarketOpCommon:: MktActivityRecord avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketFactors | 0..* | MarketFactors | |
| 0..* | Assets | 0..* | Asset | Hérité de: ActivityRecord |
| 0..* | Author | 0..1 | Author | Hérité de: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.7 MktConnectivityNode

Sous-classe de IEC61970:Topology:ConnectivityNode

Le Tableau 76 présente tous les attributs de MktConnectivityNode.

**Tableau 76 – Attributs de MarketOpCommon::MktConnectivityNode**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 77 présente toutes les extrémités d'association de MktConnectivityNode avec d'autres classes.

**Tableau 77 – Extrémités d'association de MarketOpCommon:: MktConnectivityNode avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 1..1 | NodeConstraintTerm | 0..* | NodeConstraintTerm | |
| 0..* | RTO | 1..1 | RTO | |
| 1..1 | SysLoadDistribuFactor | 0..1 | SysLoadDistributionFactor | |
| 1..1 | LossPenaltyFactor | 0..* | LossSensitivity | |
| 1..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 1..1 | IndividualPnode | 0..1 | IndividualPnode | |
| 0..* | TopologicalNode | 0..1 | TopologicalNode | Hérité de: ConnectivityNode |
| 0..* | ConnectivityNodeContainer | 1..1 | ConnectivityNodeContainer | Hérité de: ConnectivityNode |
| 0..1 | Terminals | 0..* | Terminal | Hérité de: ConnectivityNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.8 MktLine

Sous-classe de IEC61970:Wires:Line

Le Tableau 78 présente tous les attributs de MktLine.

**Tableau 78 – Attributs de MarketOpCommon::MktLine**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 79 présente toutes les extrémités d'association de MktLine avec d'autres classes.

**Tableau 79 – Extrémités d'association de MarketOpCommon:: MktLine avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..* | Flowgate | |
| 0..* | Region | 0..1 | SubGeographicalRegion | Hérité de: Line |
| 0..1 | Equipments | 0..* | Equipment | Hérité de: EquipmentContainer |
| 0.. * | AdditionalGroupedEquipment | 0..* | Equipment | Hérité de: EquipmentContainer |
| 1..1 | ConnectivityNodes | 0..* | ConnectivityNode | Hérité de: ConnectivityNodeContainer |
| 0..1 | TopologicalNode | 0..* | TopologicalNode | Hérité de: ConnectivityNodeContainer |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.9 MktMeasurement

Sous-classe de IEC61970:Meas:Measurement

Le Tableau 80 présente tous les attributs de MktMeasurement.

**Tableau 80 – Attributs de MarketOpCommon::MktMeasurement**

| nom | mult | type | description |
|---|---|---|---|
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 1..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 81 présente toutes les extrémités d'association de MktMeasurement avec d'autres classes.

**Tableau 81 – Extrémités d'association de MarketOpCommon:: MktMeasurement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..1 | Pnode | Permet à Pnode de s'associer aux mesures d'interconnexions à courant continu externes des ressources agrégées ou de pseudo-interconnexions. |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.10 MktPowerTransformer

Sous-classe de IEC61970:Wires:PowerTransformer

Le Tableau 82 présente tous les attributs de MktPowerTransformer.

**Tableau 82 – Attributs de MarketOpCommon::MktPowerTransformer**

| nom | mult | type | description |
|---|---|---|---|
| beforeShCircuitHighestOperatingCurrent | 0..1 | CurrentFlow | Hérité de: PowerTransformer |
| beforeShCircuitHighestOperatingVoltage | 0..1 | Voltage | Hérité de: PowerTransformer |
| beforeShortCircuitAnglePf | 0..1 | AngleDegrees | Hérité de: PowerTransformer |
| highSideMinOperatingU | 0..1 | Voltage | Hérité de: PowerTransformer |
| isPartOfGeneratorUnit | 0..1 | Boolean | Hérité de: PowerTransformer |
| operationalValuesConsidered | 0..1 | Boolean | Hérité de: PowerTransformer |
| vectorGroup | 0..1 | String | Hérité de: PowerTransformer |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 83 présente toutes les extrémités d'association de MktPowerTransformer avec d'autres classes.

**Tableau 83 – Extrémités d'association de MarketOpCommon:: MktPowerTransformer avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..* | Flowgate | |
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..1 | PowerTransformerEnd | 0..* | PowerTransformerEnd | Hérité de: PowerTransformer |
| 0..1 | TransformerTanks | 0..* | TransformerTank | Hérité de: PowerTransformer |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContained | 0..* | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.11 MktTerminal

Sous-classe de IEC61970:Core:Terminal

Le Tableau 84 présente tous les attributs de MktTerminal.

**Tableau 84 – Attributs de MarketOpCommon::MktTerminal**

| nom | mult | type | description |
|---|---|---|---|
| phases | 0..1 | PhaseCode | Hérité de: Terminal |
| connected | 0..1 | Boolean | Hérité de: ACDCTerminal |
| sequenceNumber | 0..1 | Integer | Hérité de: ACDCTerminal |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 85 présente toutes les extrémités d'association de MktTerminal avec d'autres classes.

**Tableau 85 – Extrémités d’association de MarketOpCommon:: MktTerminal avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 0..1 | Flowgate | |
| 1..1 | TerminalConstraintTerm | 0..* | TerminalConstraintTerm | |
| 0..* | ConductingEquipment | 1..1 | ConductingEquipment | Hérité de: Terminal |
| 0..* | ConnectivityNode | 0..1 | ConnectivityNode | Hérité de: Terminal |
| 0..* | TopologicalNode | 0..1 | TopologicalNode | Hérité de: Terminal |
| 0..1 | RegulatingControl | 0..* | RegulatingControl | Hérité de: Terminal |
| 1.. * | NormaHeadFeeder | 0..1 | Feeder | Hérité de: Terminal |
| 1..1 | HasFirstMutualCoupling | 0..* | MutualCoupling | Hérité de: Terminal |
| 0..1 | TransformerEnd | 0..* | TransformerEnd | Hérité de: Terminal |
| 1..1 | BranchGroupTerminal | 0..* | BranchGroupTerminal | Hérité de: Terminal |
| 1..1 | SvPowerFlow | 0..* | SvPowerFlow | Hérité de: Terminal |
| 1..1 | AuxiliaryEquipment | 0..* | AuxiliaryEquipment | Hérité de: Terminal |
| 1..1 | HasSecondMutualCoupling | 0..* | MutualCoupling | Hérité de: Terminal |
| 1..1 | TieFlow | 0..2 | TieFlow | Hérité de: Terminal |
| 0..1 | EquipmentFaults | 0..* | EquipmentFault | Hérité de: Terminal |
| 0..1 | ConverterDCSides | 0..* | ACDCConverter | Hérité de: Terminal |
| 1..1 | RemoteInputSignal | 0..* | RemoteInputSignal | Hérité de: Terminal |
| 1..* | BusNameMarker | 0..1 | BusNameMarker | Hérité de: ACDCTerminal |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: ACDCTerminal |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: ACDCTerminal |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.3.12 MktUserAttribute

Sous-classe de IEC61968:Domain2:UserAttribute

Le Tableau 86 présente tous les attributs de MktUserAttribute.

**Tableau 86 – Attributs de MarketOpCommon::MktUserAttribute**

| nom | mult | type | description |
|---|---|---|---|
| sequenceNumber | 0..1 | Integer | Hérité de: UserAttribute |
| name | 0..1 | String | Hérité de: UserAttribute |
| value | 0..1 | StringQuantity | Hérité de: UserAttribute |

Le Tableau 87 présente toutes les extrémités d'association de MktUserAttribute avec d'autres classes.

**Tableau 87 – Extrémités d'association de MarketOpCommon:: MktUserAttribute avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ChargeType | 0..* | ChargeType | |
| 0..* | MarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..* | PassThroughBill | 0..* | PassThroughBill | |
| 0..* | ChargeGroup | 0..* | ChargeGroup | |
| 1..1 | AttributeProperty | 0..* | AttributeProperty | |
| 0..* | BillDeterminant | 0..* | BillDeterminant | |
| 0..* | Transaction | 0..1 | Transaction | Hérité de: UserAttribute |

## 6.5.4 Paquetage MarketPlan

### 6.5.4.1 Généralités

Définitions du plan du marché pour les marchés planifiés, les événements de marché planifié, les processus de marché réels et les événements de marché réels.

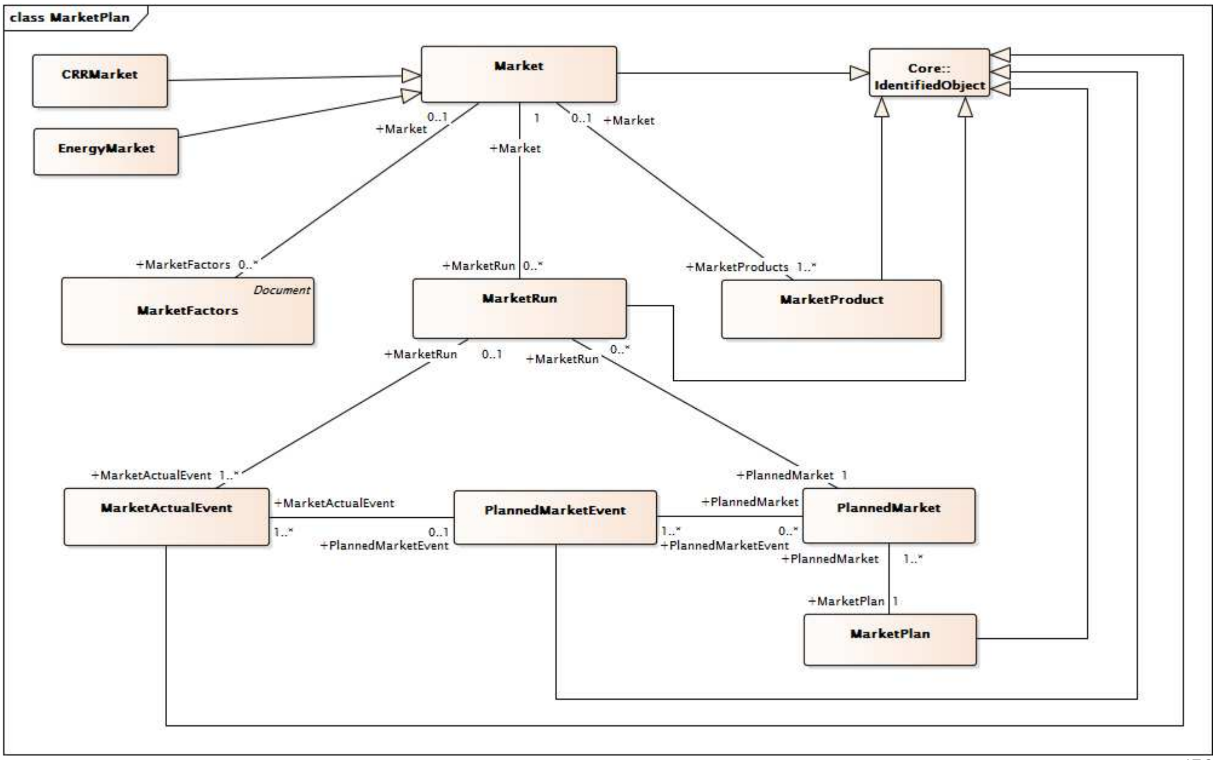



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**Figure 35 – Diagramme de classe MarketPlan::MarketPlan**

Figure 35: représente les classes principales utilisées pour identifier une instance d'une solution d'équilibre du marché. La classe Market permet d'identifier si l'instance est un marché à J-1 (DAM), heure d'avant (intrajournalier) ou temps réel (RTM). La classe MarketFactors décrit un intervalle de marché comme la date/heure de début et de fin du marché. La classe MarketProduct indique le produit du marché, tel que le service énergétique ou auxiliaire. La classe PlannedMarket décrit les heures de début et de fin du marché. Les relations entre les classes sont également représentées.

#### 6.5.4.2 CommodityDefinition

Dans le cadre de l'IEC 62325, les marchandises (Commodities) sont les produits du marché (MarketProducts) (énergie, régulation, réserve, etc.) échangés à un emplacement spécifique. Dans le cas présent, il s'agit d'un Pnode (un nœud de tarification spécifique ou une zone de tarification ou une zone définie comme étant une collection de nœuds de tarification). La CommodityDefinition contient deux paramètres, plus l'unité de mesure et la devise dans laquelle la marchandise est échangée. Il convient que chaque CommodityDefinition soit relativement statique, définie une seule fois et rarement modifiée.

Le Tableau 88 présente tous les attributs de CommodityDefinition.

**Tableau 88 – Attributs de MarketPlan::CommodityDefinition**

| nom | mult | type | description |
|---|---|---|---|
| commodityCurrency | 0..1 | Currency | Devise dans laquelle la marchandise est échangée, conformément aux conventions normatives ainsi qu'à l'énumération Currency. |
| commodityUnit | 0..1 | UnitSymbol | Unité de mesure dans laquelle la marchandise est échangée, conformément aux conventions normatives ainsi qu'à l'énumération UnitSymbol. |
| commodityUnitMultiplier | 0..1 | UnitMultiplier | Multiplicateur de l'unité, par exemple "k" pour convertir l'unité "W-h" en "kW-h", conformément aux conventions normatives ainsi qu'à l'énumération UnitMultiplier. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 89 présente toutes les extrémités d'association de CommodityDefinition avec d'autres classes.

**Tableau 89 – Extrémités d'association de MarketPlan:: CommodityDefinition avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 1..1 | Pnode | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..1 | CommodityPrice | 1..* | CommodityPrice | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.4.3 CRRMarket

Modèle décrivant le marché de vente aux enchères des droits à revenu de congestion

Le Tableau 90 présente tous les attributs de CRRMarket.

**Tableau 90 – Attributs de MarketPlan::CRRMarket**

| nom | mult | type | description |
|---|---|---|---|
| labelID | 0..1 | String | labelID – Identifiant pour un ensemble d'apnodes/pnodes utilisé dans un marché de CRR |
| actualEnd | 0..1 | DateTime | Hérité de: Market |
| actualStart | 0..1 | DateTime | Hérité de: Market |
| dst | 0..1 | Boolean | Hérité de: Market |
| end | 0..1 | DateTime | Hérité de: Market |
| localTimeZone | 0..1 | String | Hérité de: Market |
| start | 0..1 | DateTime | Hérité de: Market |
| status | 0..1 | String | Hérité de: Market |
| timeIntervalLength | 0..1 | Float | Hérité de: Market |
| tradingDay | 0..1 | DateTime | Hérité de: Market |
| tradingPeriod | 0..1 | String | Hérité de: Market |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 91 présente toutes les extrémités d'association de CRRMarket avec d'autres classes.

**Tableau 91 – Extrémités d’association de MarketPlan::CRRMarket avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | CongestionRevenueRight | 1..* | CongestionRevenueRight | |
| 0..1 | MarketFactors | 0..* | MarketFactors | Hérité de: Market |
| 0..1 | MarketProducts | 1..* | MarketProduct | Hérité de: Market |
| 1..1 | MarketRun | 0..* | MarketRun | Hérité de: Market |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.4 DistributedResourceActualEvent

Un événement de réponse à une demande est créé en cas de besoin de ressources répondant à des requêtes d’ajustement de la demande. Ces événements sont créés par les opérations du système ISO/RTO et gérés par un système de gestion des réponses à la demande (DRMS – demand response management system). Ces événements peuvent être ou ne pas être coordonnés avec les événements du marché (Market Events) et un marché d’énergie (Energy Market) défini. L’événement sollicite le déploiement d’un certain nombre de ressources enregistrées, ou le déploiement de ressources au sein d’une zone (une zone organisationnelle dans le système d’alimentation qui contient un certain nombre de ressources).

Le Tableau 92 présente tous les attributs de DistributedResourceActualEvent.

**Tableau 92 – Attributs de MarketPlan::DistributedResourceActualEvent**

| nom | mult | type | description |
|---|---|---|---|
| totalPowerAdjustment | 0..1 | ActivePower | Ajustement complet de la puissance active (par exemple, réduction de charge) demandé pour cet événement de réponse à la demande. |
| eventComments | 1..1 | String | Hérité de: MarketActualEvent |
| eventEndTime | 0..1 | DateTime | Hérité de: MarketActualEvent |
| eventStartTime | 0..1 | DateTime | Hérité de: MarketActualEvent |
| eventStatus | 1..1 | MarketEventStatusKind | Hérité de: MarketActualEvent |
| eventType | 1..1 | String | Hérité de: MarketActualEvent |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 93 présente toutes les extrémités d'association de DistributedResourceActualEvent avec d'autres classes.

**Tableau 93 – Extrémités d'association de MarketPlan:: DistributedResourceActualEvent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | InstructionClearing | 0..* | InstructionClearing | ActualDemandResponseEvents pouvant exister sans faire partie d'un MarketActualEvent coordonné associé à un marché. Ces ActualDemandResponseEvents peuvent avoir plusieurs instructions InstructionClearing pour les RegisteredResources ou AggregateNodes DemandResponse spécifiés. |
| 1..1 | ResourcePerformanceEvaluations | 0..* | ResourcePerformanceEvaluation | |
| 0..1 | InstructionClearingDOT | 0..* | InstructionClearingDOT | |
| 1..* | MarketRun | 0..1 | MarketRun | Hérité de: MarketActualEvent |
| 1..* | PlannedMarketEvent | 0..1 | PlannedMarketEvent | Hérité de: MarketActualEvent |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.5 EnergyMarket

Marché énergétique et auxiliaire (par exemple, énergie, réserve tournante, réserve non tournante) avec une description des paramètres de réglage des opérations du marché.

Le Tableau 94 présente tous les attributs de EnergyMarket.

**Tableau 94 – Attributs de MarketPlan::EnergyMarket**

| nom | mult | type | description |
|---|---|---|---|
| actualEnd | 0..1 | DateTime | Hérité de: Market |
| actualStart | 0..1 | DateTime | Hérité de: Market |
| dst | 0..1 | Boolean | Hérité de: Market |
| end | 0..1 | DateTime | Hérité de: Market |
| localTimeZone | 0..1 | String | Hérité de: Market |
| start | 0..1 | DateTime | Hérité de: Market |
| status | 0..1 | String | Hérité de: Market |
| timeIntervalLength | 0..1 | Float | Hérité de: Market |
| tradingDay | 0..1 | DateTime | Hérité de: Market |
| tradingPeriod | 0..1 | String | Hérité de: Market |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 95 présente toutes les extrémités d'association de EnergyMarket avec d'autres classes.

**Tableau 95 – Extrémités d'association de MarketPlan:: EnergyMarket avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResources | 0..* | RegisteredResource | |
| 0..* | RTO | 0..1 | RTO | |
| 0..1 | Settlements | 0..* | Settlement | |
| 1..1 | MarketResults | 0..1 | MarketResults | |
| 1..1 | Bids | 0..* | Bid | |
| 0..1 | MarketFactors | 0..* | MarketFactors | Hérité de: Market |
| 0..1 | MarketProducts | 1..* | MarketProduct | Hérité de: Market |
| 1..1 | MarketRun | 0..* | MarketRun | Hérité de: Market |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.4.6 Market

Marché (à J-1 ou à temps réel, par exemple) avec une description des paramètres de réglage des opérations du marché.

Le Tableau 96 présente tous les attributs de Market.

**Tableau 96 – Attributs de MarketPlan::Market**

| nom | mult | type | description |
|---|---|---|---|
| actualEnd | 0..1 | DateTime | Heure de fin du marché – fin réelle du marché |
| actualStart | 0..1 | DateTime | Heure de début du marché – début réel du marché |
| dst | 0..1 | Boolean | True si l'heure d'été (DST – Daylight Saving Time) est appliquée. |
| end | 0..1 | DateTime | Heure de fin du marché. |
| localTimeZone | 0..1 | String | Fuseau horaire local. |
| start | 0..1 | DateTime | Heure de début du marché. |
| status | 0..1 | String | Statut du marché<br>"OPEN" (OUVERT), "CLOSED" (FERMÉ), "CLEARED" (ÉQUILIBRÉ), "BLOCKED" (BLOQUÉ) |
| timeIntervalLength | 0..1 | Float | Longueur de l'intervalle de temps entre les transactions. |
| tradingDay | 0..1 | DateTime | Date de transaction sur le marché |
| tradingPeriod | 0..1 | String | Période d'échange décrivant le marché. Pour un marché de l'énergie:<br>Jour<br>Heure<br>Pour un marché de CRR:<br>Année<br>Mois<br>Saison |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 97 présente toutes les extrémités d'association de Market avec d'autres classes.

**Tableau 97 – Extrémités d’association de MarketPlan::Market avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MarketFactors | 0..* | MarketFactors | |
| 0..1 | MarketProducts | 1..* | MarketProduct | |
| 1..1 | MarketRun | 0..* | MarketRun | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.4.7 MarketActualEvent

Cette classe représente l'instance réelle d'un événement.

Le Tableau 98 présente tous les attributs de MarketActualEvent.

**Tableau 98 – Attributs de MarketPlan::MarketActualEvent**

| nom | mult | type | description |
|---|---|---|---|
| eventComments | 1..1 | String | Commentaires à structure non imposée concernant l'événement, pour tout besoin quelconque. |
| eventEndTime | 0..1 | DateTime | Heure de fin de l'événement. |
| eventStartTime | 0..1 | DateTime | Heure de début de l'événement. |
| eventStatus | 1..1 | MarketEventStatusKind | Statut de l'événement, par exemple, actif, annulé, expiré, etc. |
| eventType | 1..1 | String | Type réel d'événement. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 99 présente toutes les extrémités d'association de MarketActualEvent avec d'autres classes.

**Tableau 99 – Extrémités d'association de MarketPlan:: MarketActualEvent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | MarketRun | 0..1 | MarketRun | Processus de marché déclenché par cet événement réel. Par exemple, le processus de marché DA est déclenché par l'événement de l'ouverture d'un appel d'offres réel et se termine par l'exécution et la fin réelles du processus de marché DA saisi par le runState du RunMarket |
| 1..* | PlannedMarketEvent | 0..1 | PlannedMarketEvent | Événement planifié exécuté par cet événement réel. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.8 MarketFactors

Agrégation des informations du marché relatives à un intervalle de temps spécifique.

Le Tableau 100 présente tous les attributs de MarketFactors.

**Tableau 100 – Attributs de MarketPlan::MarketFactors**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Fin de l'intervalle de temps pour lequel une exigence est définie. |
| intervalStartTime | 0..1 | DateTime | Début de l'intervalle de temps pour lequel une exigence est définie. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 101 présente toutes les extrémités d'association de MarketFactors avec d'autres classes.

**Tableau 101 – Extrémités d'association de MarketPlan:: MarketFactors avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | |
| 0..* | Market | 0..1 | Market | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.9 MarketPlan

Cette classe identifie un ensemble de marchés planifiés.

Le Tableau 102 présente tous les attributs de MarketPlan.

**Tableau 102 – Attributs de MarketPlan::MarketPlan**

| nom | mult | type | description |
|---|---|---|---|
| tradingDay | 0..1 | DateTime | Jour d'échange du marché planifié. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 103 présente toutes les extrémités d'association de MarketPlan avec d'autres classes.

**Tableau 103 – Extrémités d'association de MarketPlan:: MarketPlan avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | PlannedMarket | 1..* | PlannedMarket | Un plan de marché comprend plusieurs marchés (DA, HA, RT). |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.10 MarketProduct

Produit commercialisé par un RTO (par exemple, énergie, réserve tournante de 10 min). La régulation à la hausse, la régulation à la baisse, la réserve tournante, la réserve non tournante, etc., sont des exemples de produits de services auxiliaires.

Le Tableau 104 présente tous les attributs de MarketProduct.

**Tableau 104 – Attributs de MarketPlan::MarketProduct**

| nom | mult | type | description |
|---|---|---|---|
| marketProductType | 0..1 | MarketProductType | Exemples de type de produit du marché:<br>EN (Energy, Énergie)<br>RU (Regulation Up – Régulation à la hausse)<br>RD (Regulation Dn – Régulation à la baisse)<br>SR (Spinning Reserve – Réserve tournante)<br>NR (Non-Spinning Reserve – Réserve non tournante)<br>RC (RUC, Residual Unit Commitment – Gestion de la production résiduelle) |
| rampInterval | 0..1 | Float | Intervalle de temps de rampe pour le type de produit du marché spécifié par l'attribut marketProductType. Par exemple, si marketProductType = EN (énumération MarketProductType), alors rampInterval est l'intervalle de temps de rampe de Energy. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 105 présente toutes les extrémités d'association de MarketProduct avec d'autres classes.

**Tableau 105 – Extrémités d'association de MarketPlan:: MarketProduct avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | Market | 0..1 | Market | |
| 1..1 | ProductBids | 0..* | ProductBid | |
| 1..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | |
| 0..1 | BidError | 0..* | BidError | |
| 0..1 | BidPriceCap | 0..* | BidPriceCap | |
| 0..1 | MarketRegionResults | 0..1 | MarketRegionResults | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.11 MarketRun

Cette classe représente une instance réelle d'un marché planifié. Par exemple, un marché à J-1 s'ouvre avec l'ouverture d'un appel d'offres (Bid Submission) et se termine par la clôture de cet appel d'offres. Le processus de marché représente l'ensemble du processus. MarketRuns peut être défini pour les marchés tels que Day Ahead Market (marché à J-1), Real Time Market (marché en temps réel), Hour Ahead Market (marché à H-1) et Week Ahead Market (marché de la semaine d'avant).

Le Tableau 106 présente tous les attributs de MarketRun.

**Tableau 106 – Attributs de MarketPlan::MarketRun**

| nom | mult | type | description |
|---|---|---|---|
| executionType | 0..1 | ExecutionType | Type d'exécution: J-1, intrajournalier, prérépartition en temps réel, répartition en temps réel |
| marketApprovalTime | 0..1 | DateTime | Heure approuvée de la répartition. Identifie l'heure à laquelle le répartiteur approuve un cas de répartition d'unités en temps réel spécifique. |
| marketApprovedStatus | 0..1 | Boolean | Défini sur True lorsque le plan est approuvé par une autorité. Celui-ci devient le plan officiel du marché à J-1. Identifie le cas approuvé du marché pour l'intervalle de temps spécifié. |
| marketEndTime | 0..1 | DateTime | Heure de fin définie comme la fin du marché: heure de fin du marché. |
| marketStartTime | 0..1 | DateTime | Heure de début définie comme le début du marché: heure de début du marché. |
| marketType | 0..1 | MarketType | Type de marché: marché à J-1ou marché en temps réel. |
| reportedState | 0..1 | String | État des activités de processus de marché telles que rapportées par les systèmes de marché aux services de définition du marché. |

| nom | mult | type | description |
|---|---|---|---|
| runState | 0..1 | String | État contrôlé par le service de définition du marché.<br>Exemples de valeurs possibles: Open (Ouvert), Close (Fermé). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 107 présente toutes les extrémités d'association de MarketRun avec d'autres classes.

**Tableau 107 – Extrémités d’association de MarketPlan:: MarketRun avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Market | 1..1 | Market | |
| 0..1 | MarketActualEvent | 1..* | MarketActualEvent | Tous les événements réels qui déclenchent ce processus de marché. |
| 0..* | PlannedMarket | 1..1 | PlannedMarket | Un marché planifié peut avoir plusieurs processus de marché, car il peut présenter un nouveau processus. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.4.12 Classe racine PlannedMarket

Modélise un marché planifié. Par exemple, un marché DA/HA/RT planifié.

Le Tableau 108 présente tous les attributs de PlannedMarket.

**Tableau 108 – Attributs de MarketPlan::PlannedMarket**

| nom | mult | type | description |
|---|---|---|---|
| marketEndTime | 0..1 | DateTime | Heure de fin du marché. |
| marketStartTime | 0..1 | DateTime | Heure de début du marché. |
| marketType | 0..1 | MarketType | Type de marché. |

Le Tableau 109 présente toutes les extrémités d'association de PlannedMarket avec d'autres classes.

**Tableau 109 – Extrémités d'association de MarketPlan:: PlannedMarket avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | MarketRun | 0..* | MarketRun | Un marché planifié peut avoir plusieurs processus de marché, car il peut présenter un nouveau processus. |
| 0..* | PlannedMarketEvent | 1..* | PlannedMarketEvent | Un marché planifié doit avoir un ensemble d'événements planifiés. |
| 1..* | MarketPlan | 1..1 | MarketPlan | Un plan de marché comprend plusieurs marchés (DA, HA, RT). |

### 6.5.4.13 PlannedMarketEvent

Cette classe représente des événements planifiés. Elle est utilisée pour modéliser les différents événements planifiés dans un marché (heure de fermeture, temps de mise en équilibre, etc.).

Le Tableau 110 présente tous les attributs de PlannedMarketEvent.

**Tableau 110 – Attributs de MarketPlan::PlannedMarketEvent**

| nom | mult | type | description |
|---|---|---|---|
| eventType | 0..1 | String | Type d'événement planifié. |
| plannedTime | 0..1 | Integer | Il s'agit d'un temps relatif pour que cet attribut puisse être utilisé par plusieurs marchés planifiés. Par exemple, les appels d'offres s'effectuent tous les jours à 10h. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 111 présente toutes les extrémités d'association de PlannedMarketEvent avec d'autres classes.

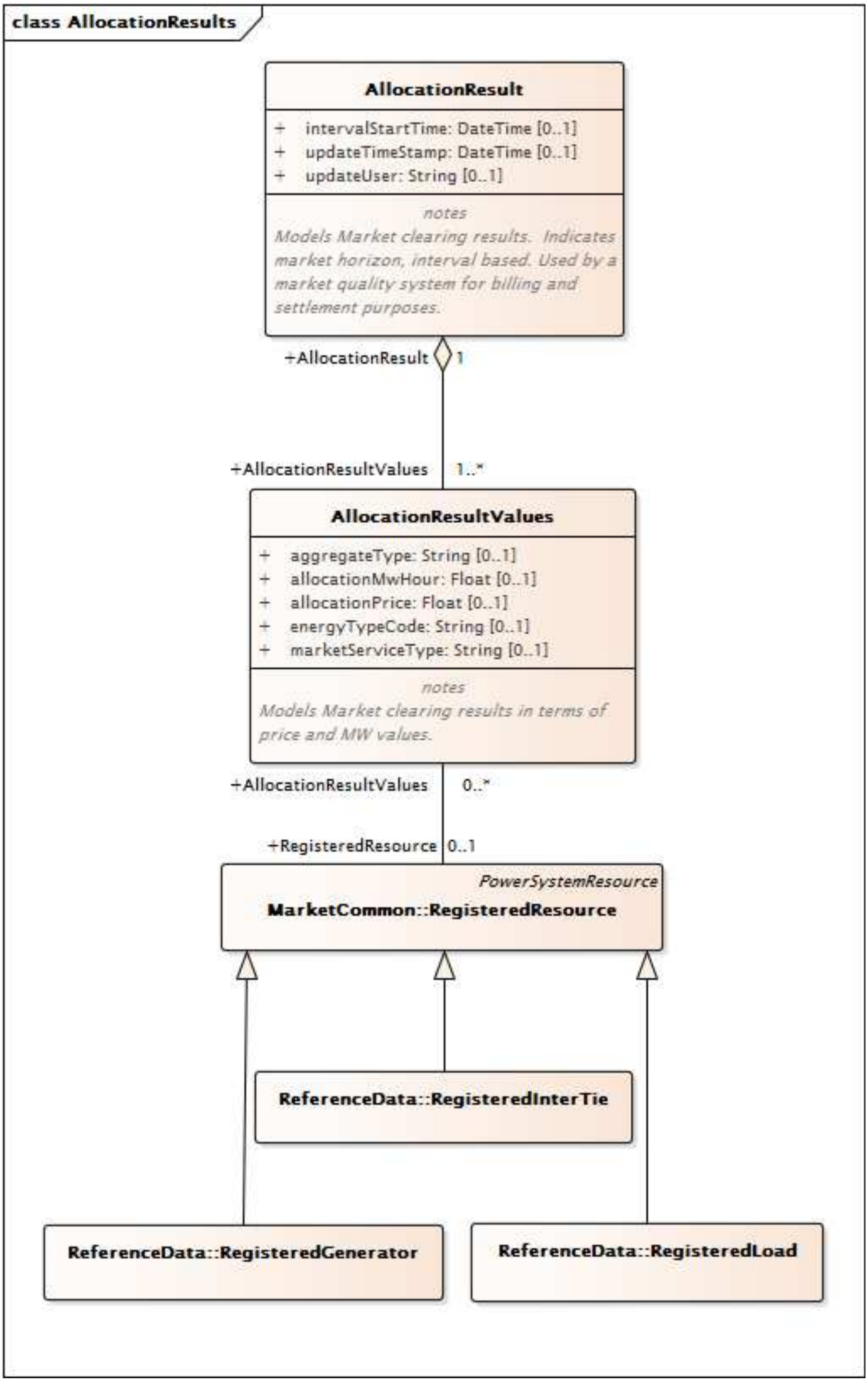
**Tableau 111 – Extrémités d'association de MarketPlan:: PlannedMarketEvent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MarketActualEvent | 1..* | MarketActualEvent | Tous les événements réels qui exécutent cet événement planifié. |
| 1..* | PlannedMarket | 0..* | PlannedMarket | Un marché planifié doit avoir un ensemble d'événements planifiés. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.5 Paquetage MarketQualitySystem

#### 6.5.5.1 Généralités

Comptabilité, calcul et corrections des données de compteur post-commercialisation pour réduire les erreurs et les litiges relatifs à la facturation. Réduit la validation, la vérification et la correction manuelles des données de transaction qui pourraient influer sur les règlements du marché. Nouvelle publication des résultats du marché après correction des données concernées.


class AllocationResults
AllocationResult
+ intervalStartTime: DateTime [0..1]
+ updateTimeStamp: DateTime [0..1]
+ updateUser: String [0..1]
notes
Models Market clearing results. Indicates market horizon, interval based. Used by a market quality system for billing and settlement purposes.
+AllocationResult
1
+AllocationResultValues
1..*
AllocationResultValues
+ aggregateType: String [0..1]
+ allocationMwHour: Float [0..1]
+ allocationPrice: Float [0..1]
+ energyTypeCode: String [0..1]
+ marketServiceType: String [0..1]
notes
Models Market clearing results in terms of price and MW values.
+AllocationResultValues
0..*
+RegisteredResource
0..1
PowerSystemResource
MarketCommon::RegisteredResource
ReferenceData::RegisteredInterTie
ReferenceData::RegisteredGenerator
ReferenceData::RegisteredLoad

IEC

| Anglais | Français |
|---|---|
| Class AllocationResults | Classe AllocationResults |
| notes<br><br>Models Market clearing results. Indicates market horizon, interval based. Used by a market quality system for billing and settlement purposes | notes<br><br>Modélise les résultats d'équilibre du marché. Indique l'horizon du marché sur la base d'intervalles. Utilisée par un système de qualité du marché à des fins de facturation et de règlement |
| notes<br><br>Models Market clearing results in terms of price and MW values | notes<br><br>Modélise les résultats d'équilibre du marché en termes de prix et de MW |

**Figure 36 – Diagramme de classe MarketQualitySystem::AllocationResults**

Figure 36: représente les principales classes associées au système de qualité du marché (MQS – Market Quality System) qui est généralement utilisé pour effectuer des analyses avant la facturation et le règlement. Elle décrit les résultats envoyés au MQS. Les principales classes incluent AllocationResult, qui définit l'horizon du marché. Les résultats attribués sont décrits dans les classes AllocationResultsValues et RegisteredResource, ainsi que dans les sous-classes pour les générateurs, les interconnexions et les charges.

#### 6.5.5.2 Classe racine AllocationResult

Modélise les résultats d'équilibre du marché. Indique l'horizon du marché sur la base d'intervalles. Utilisée par un système de qualité du marché à des fins de facturation et de règlement.

Le Tableau 112 présente tous les attributs de AllocationResult.

**Tableau 112 – Attributs de MarketQualitySystem::AllocationResult**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Le Tableau 113 présente toutes les extrémités d'association de AllocationResult avec d'autres classes.

**Tableau 113 – Extrémités d’association de MarketQualitySystem:: AllocationResult avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | AllocationResultValues | 1..* | AllocationResultValues | |

#### 6.5.5.3 Classe racine AllocationResultValues

Modélise les résultats d'équilibre du marché en termes de prix et de MW.

Le Tableau 114 présente tous les attributs de AllocationResultValues.

**Tableau 114 – Attributs de MarketQualitySystem::AllocationResultValues**

| nom | mult | type | description |
|---|---|---|---|
| aggregateType | 0..1 | String | "1" – "Detail",<br>"2" – "Aggregate by Market service type", auquel cas le champ "AllocationEnergyType" ne sera pas renseigné;<br>"3" – "Aggregate by "AllocationEnergyType", auquel cas "MarketServiceType" ne sera pas renseigné. |
| allocationMwHour | 0..1 | Float | |
| allocationPrice | 0..1 | Float | |
| energyTypeCode | 0..1 | String | |
| marketServiceType | 0..1 | String | Les différentes possibilités sont:<br>ME – Market Energy (capacité d'énergie du marché);<br>SR – Spinning Reserve (capacité de réserve tournante);<br>NR – Non-Spinning Reserve (capacité de réserve non tournante);<br>DAC – Day Ahead Capacity (capacité à J-1);<br>DEC – Derate Capacity (capacité de réduction). |

Le Tableau 115 présente toutes les extrémités d'association de AllocationResultValues avec d'autres classes.

**Tableau 115 – Extrémités d'association de MarketQualitySystem::AllocationResultValues avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | AllocationResult | 1..1 | AllocationResult | |

### 6.5.5.4 Classe racine AuxiliaryCost

Modélise les résultats d'équilibre du marché pour les coûts auxiliaires.

Le Tableau 116 présente tous les attributs de AuxiliaryCost.

**Tableau 116 – Attributs de MarketQualitySystem::AuxiliaryCost**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| marketType | 0..1 | MarketType | |
| updateTimeStamp | 0..1 | DateTime | |
| updateUser | 0..1 | String | |

Le Tableau 117 présente toutes les extrémités d'association de AuxiliaryCost avec d'autres classes.

**Tableau 117 – Extrémités d'association de MarketQualitySystem:: AuxiliaryCost avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryValues | 1..* | AuxiliaryValues | |

### 6.5.5.5 Classe racine AuxiliaryObject

Modélise les valeurs auxiliaires.

Le Tableau 118 présente toutes les extrémités d'association de AuxiliaryObject avec d'autres classes.

**Tableau 118 – Extrémités d'association de MarketQualitySystem:: AuxiliaryObject avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |

### 6.5.5.6 AuxiliaryValues

Modélise les valeurs auxiliaires.

Le Tableau 119 présente tous les attributs de AuxiliaryValues.

**Tableau 119 – Attributs de MarketQualitySystem::AuxiliaryValues**

| nom | mult | type | description |
|---|---|---|---|
| minExpostCapacity | 0..1 | Float | |
| maxExpostCapacity | 0..1 | Float | |
| availUndispatchedQ | 0..1 | Float | |
| incrementalORAvail | 0..1 | Float | |
| startUpCost | 0..1 | Float | |
| startUpCostEligibilityFlag | 0..1 | YesNo | |
| noLoadCost | 0..1 | Float | |
| noLoadCostEligibilityFlag | 0..1 | YesNo | |

Le Tableau 120 présente toutes les extrémités d'association de AuxiliaryValues avec d'autres classes.

**Tableau 120 – Extrémités d'association de MarketQualitySystem:: AuxiliaryValues avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | AuxillaryCost | 1..1 | AuxiliaryCost | |
| 1..* | TenMinAuxillaryData | 1..1 | TenMinAuxiliaryData | |
| 1..* | FiveMinAuxillaryData | 1..1 | FiveMinAuxiliaryData | |
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | Hérité de: AuxiliaryObject |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | Hérité de: AuxiliaryObject |

#### 6.5.5.7 Classe racine ExpectedEnergy

Modélise l'énergie prévue à partir de l'équilibre du marché sur la base d'intervalles.

Le Tableau 121 présente tous les attributs de ExpectedEnergy.

**Tableau 121 – Attributs de MarketQualitySystem::ExpectedEnergy**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Le Tableau 122 présente toutes les extrémités d'association de ExpectedEnergy avec d'autres classes.

**Tableau 122 – Extrémités d'association de MarketQualitySystem:: ExpectedEnergy avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ExpectedEnergyValues | 1..* | ExpectedEnergyValues | |

#### 6.5.5.8 Classe racine ExpectedEnergyValues

Modélise l'énergie prévue à partir de l'équilibre du marché.

Le Tableau 123 présente tous les attributs de ExpectedEnergyValues.

**Tableau 123 – Attributs de MarketQualitySystem::ExpectedEnergyValues**

| nom | mult | type | description |
|---|---|---|---|
| energyTypeCode | 0..1 | String | |
| expectedMwh | 0..1 | Float | |

Le Tableau 124 présente toutes les extrémités d'association de ExpectedEnergyValues avec d'autres classes.

**Tableau 124 – Extrémités d’association de MarketQualitySystem::ExpectedEnergyValues avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ExpectedEnergy | 1..1 | ExpectedEnergy | |

#### 6.5.5.9 Classe racine FiveMinAuxiliaryData

Modélise des données auxiliaires de 5 min.

Le Tableau 125 présente tous les attributs de FiveMinAuxiliaryData.

**Tableau 125 – Attributs de MarketQualitySystem::FiveMinAuxiliaryData**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Le Tableau 126 présente toutes les extrémités d'association de FiveMinAuxiliaryData avec d'autres classes.

**Tableau 126 – Extrémités d’association de MarketQualitySystem:: FiveMinAuxiliaryData avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryValues | 1..* | AuxiliaryValues | |

#### 6.5.5.10 Classe racine TenMinAuxiliaryData

Modélise des données auxiliaires de 10 min.

Le Tableau 127 présente tous les attributs de TenMinAuxiliaryData.

**Tableau 127 – Attributs de MarketQualitySystem::TenMinAuxiliaryData**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Le Tableau 128 présente toutes les extrémités d'association de TenMinAuxiliaryData avec d'autres classes.

**Tableau 128 – Extrémités d'association de MarketQualitySystem:: TenMinAuxiliaryData avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | AuxillaryData | 1..* | AuxiliaryValues | |

### 6.5.5.11 Classe racine TradingHubPrice

Modélise les prix sur les plates-formes de négociation sur la base d'intervalles.

Le Tableau 129 présente tous les attributs de TradingHubPrice.

**Tableau 129 – Attributs de MarketQualitySystem::TradingHubPrice**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| marketType | 0..1 | MarketType | |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |

Le Tableau 130 présente toutes les extrémités d'association de TradingHubPrice avec d'autres classes.

**Tableau 130 – Extrémités d'association de MarketQualitySystem:: TradingHubPrice avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TradingHubValues | 1..* | TradingHubValues | |

### 6.5.5.12 Classe racine TradingHubValues

Modélise les prix sur les plates-formes de négociation.

Le Tableau 131 présente tous les attributs de TradingHubValues.

**Tableau 131 – Attributs de MarketQualitySystem::TradingHubValues**

| nom | mult | type | description |
|---|---|---|---|
| price | 0..1 | Float | Utilise le type de marché. Pour DA, le prix est calculé sur une base horaire. Pour RTM, le prix est défini pour 5 min. |

Le Tableau 132 présente toutes les extrémités d'association de TradingHubValues avec d'autres classes.

**Tableau 132 – Extrémités d'association de MarketQualitySystem:: TradingHubValues avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 1..* | TradingHubPrice | 1..1 | TradingHubPrice | |

### 6.5.6 Paquetage MarketSystem

#### 6.5.6.1 Généralités

Résultats du marché et entrées externes consommées par les marchés.

#### 6.5.6.2 Paquetage ExternalInputs

##### 6.5.6.2.1 Généralités

Entrées du système de marché depuis les sources externes.

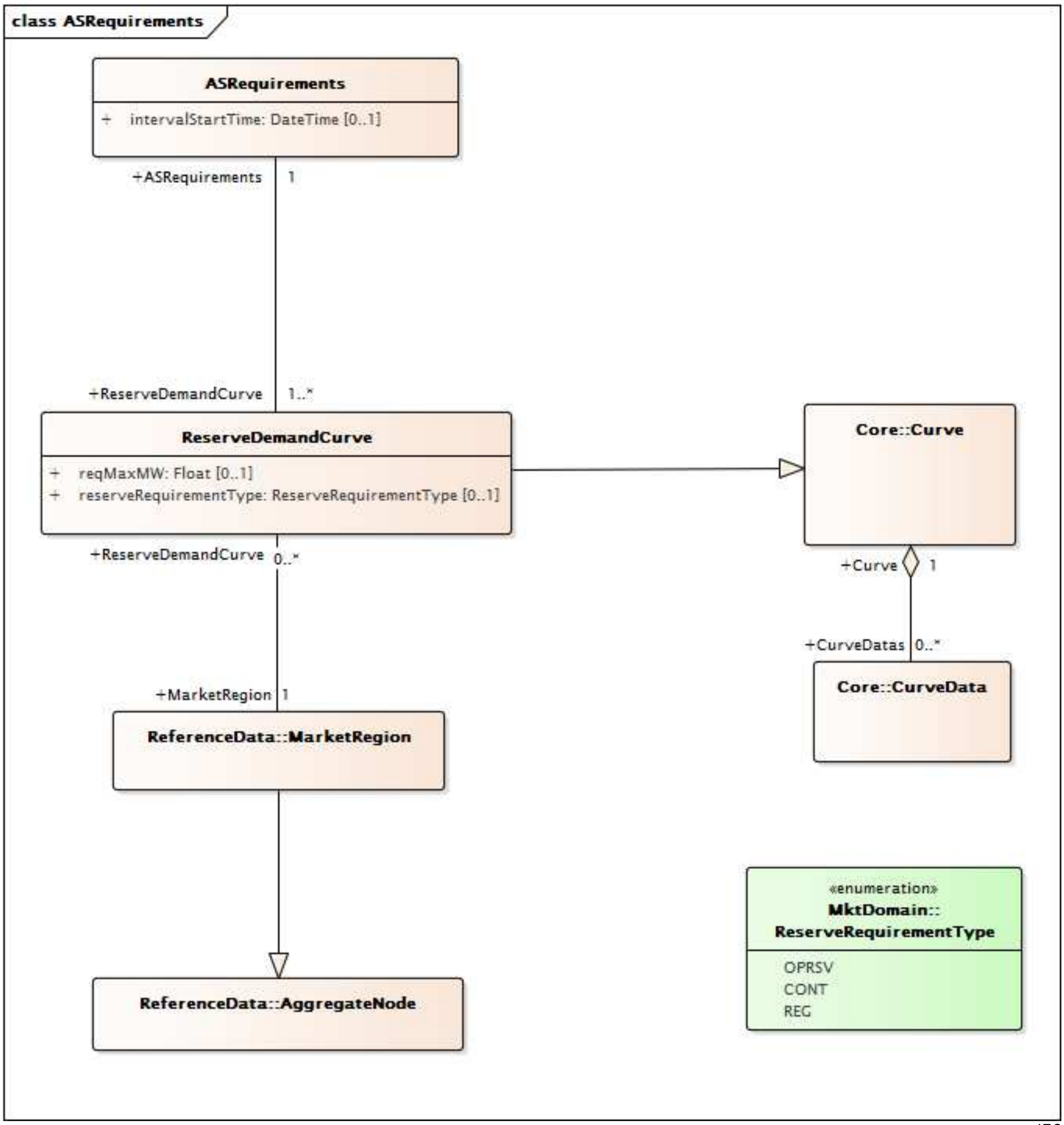



**Figure 37 – Diagramme de classe ExternalInputs::ASRequirements**

Figure 37: modélise les exigences des services auxiliaires. La classe ASRequirements est utilisée pour décrire l'intervalle pour lequel l'AncillaryService (service auxiliaire) est exigé et pour indiquer les quantités de services auxiliaires exigées. L'association à MarketRegion modélise les exigences régionales des services auxiliaires. La classe MarketProduct est utilisée pour décrire la classe particulière de services auxiliaires.

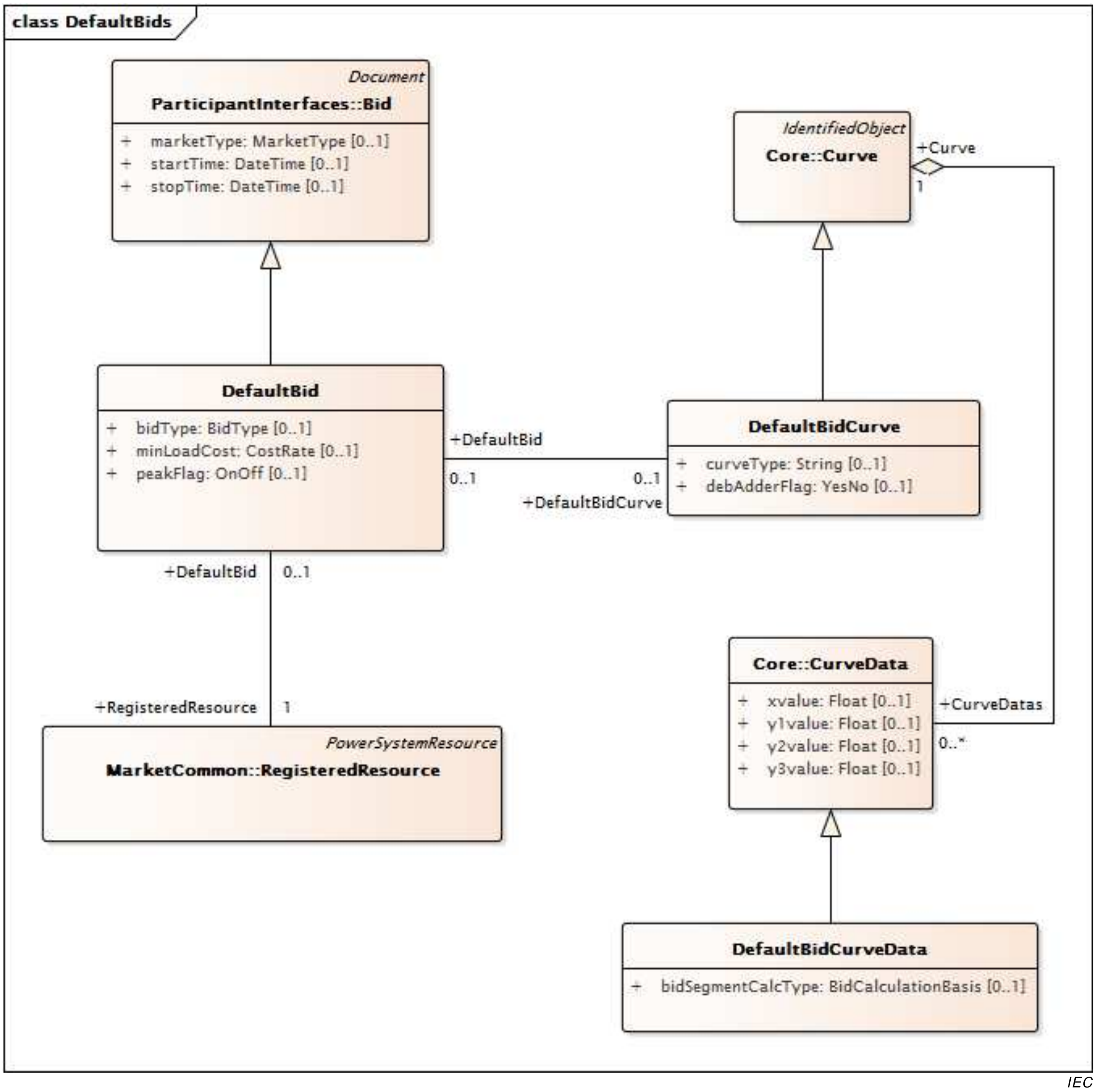



**Figure 38 – Diagramme de classe ExternalInputs::DefaultBids**

Figure 38: représente les classes utilisées pour modéliser une offre par défaut. L'atténuation de l'emprise sur le marché (Market Power Mitigation) est un processus qui permet de détecter la tentative d'un participant d'exercer un pouvoir de marché. La détection est suivie de mesures correctives. Il s'agit par exemple de remplacer l'offre d'origine par une offre par défaut. Les classes comprennent Curve et Curve Data (du paquetage principal IEC 61970), ainsi que la classe générique DefaultBid qui peut être utilisée pour décrire davantage les méthodes de mesures correctives spécifiques du marché. Trois méthodes types sont utilisées: LMP, coût basé sur l'indice du gaz ou négociation hors marché.

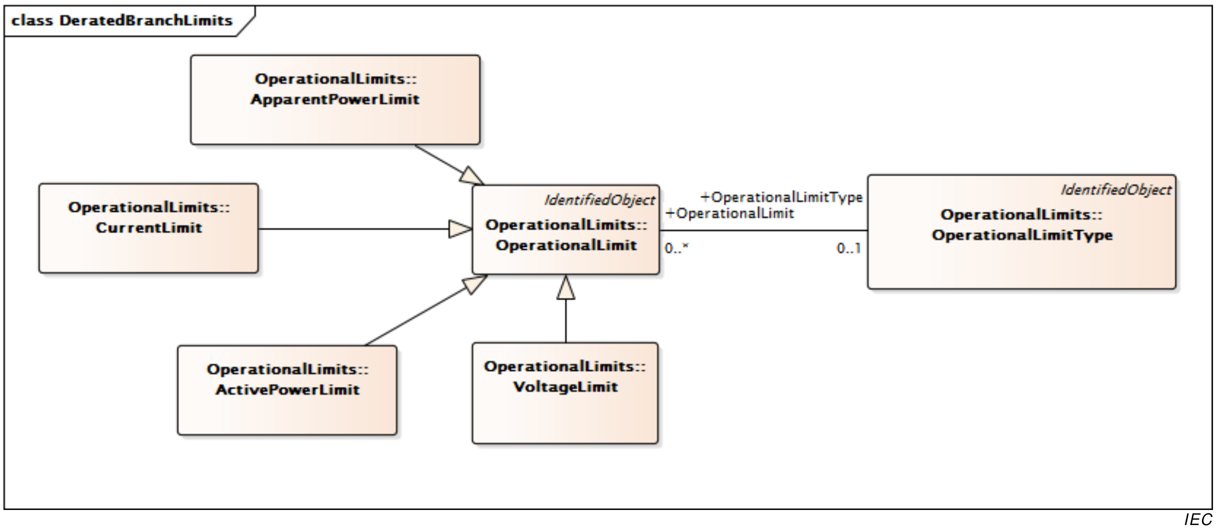



**Figure 39 – Diagramme de classe ExternalInputs::DeratedBranchLimits**

Figure 39: représente les principales classes utilisées pour modéliser les limites opérationnelles. Les classes sont réutilisées à partir du paquetage IEC 61970 OperationalLimits et comprennent la ActivePower (puissance active), la Voltage (tension) et les CurrentLimits (limites de courant).

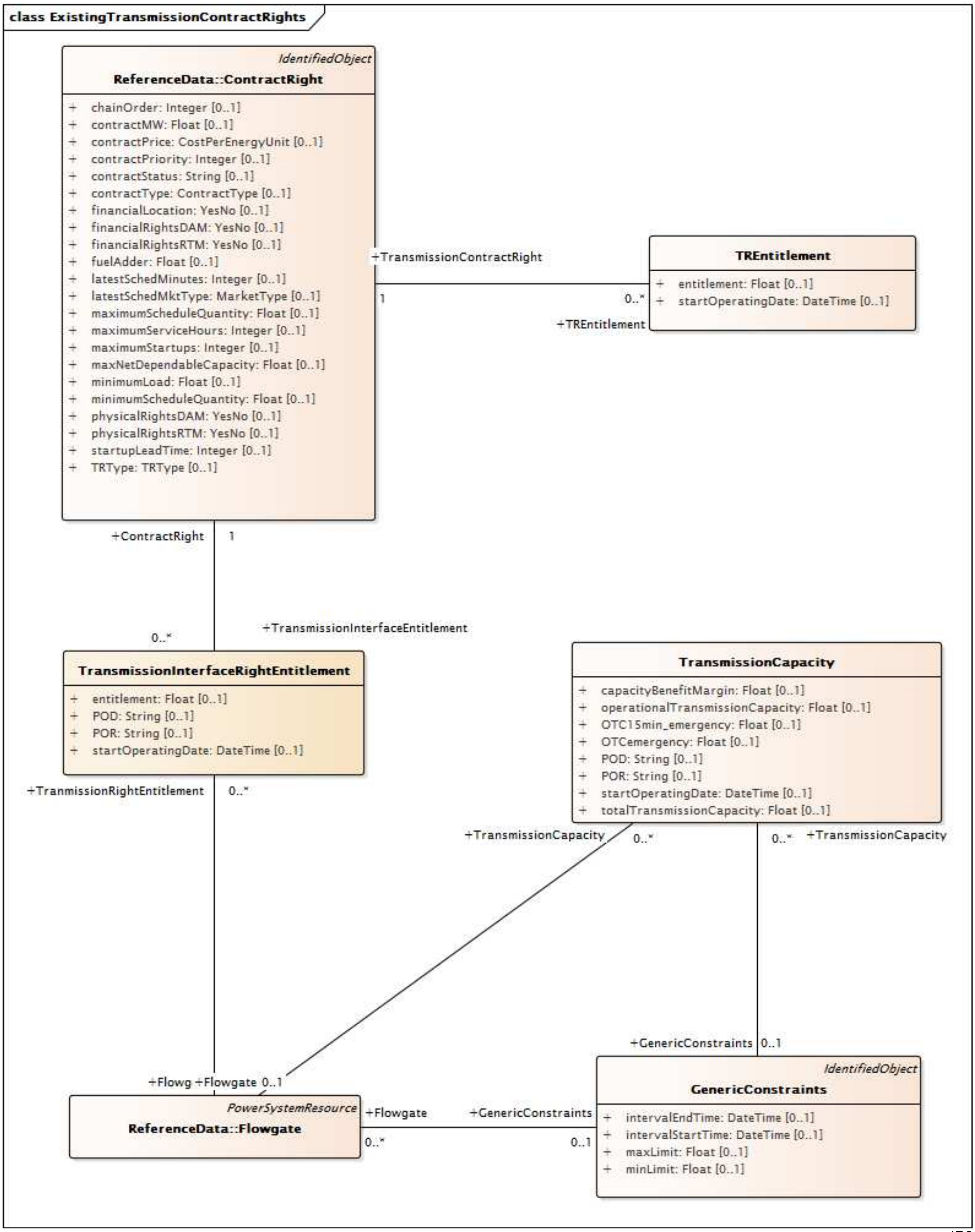



**Figure 40 – Diagramme de classe ExternalInputs::ExistingTransmissionContractRights**

Figure 40: représente les principales classes utilisées pour modéliser ces droits existants. Lors du démarrage et de l'évolution des marchés LMP, il s'est avéré nécessaire de prendre en compte les droits de transport existants dans la solution du marché. La classe ContractRight inclut des informations sur les droits spécifiques qui sont contractés en termes de puissance active et d'autres informations contractuelles. La classe TransmissionInterfaceRightEntitlement décrit les aspects physiques du droit dans les points de livraison et de réception associés de la puissance active.

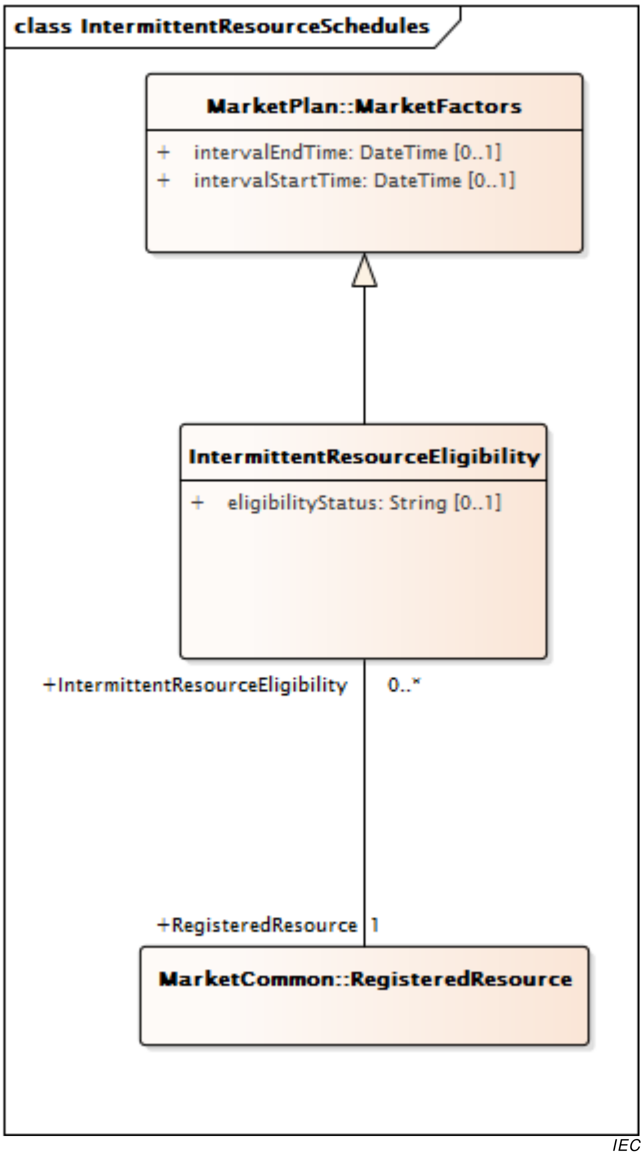



**Figure 41 – Diagramme de classe ExternalInputs::IntermittentResourceSchedules**

Figure 41: représente les classes principales utilisées pour modéliser les ressources (variables) intermittentes. Elle est habituellement utilisée pour modéliser les énergies renouvelables, telles que l'énergie éolienne. La classe IntermittentResourceEligibility permet d'indiquer si les ressources sont en mesure de participer en tant que ressources renouvelables régies par un régime générique de ressources renouvelables.

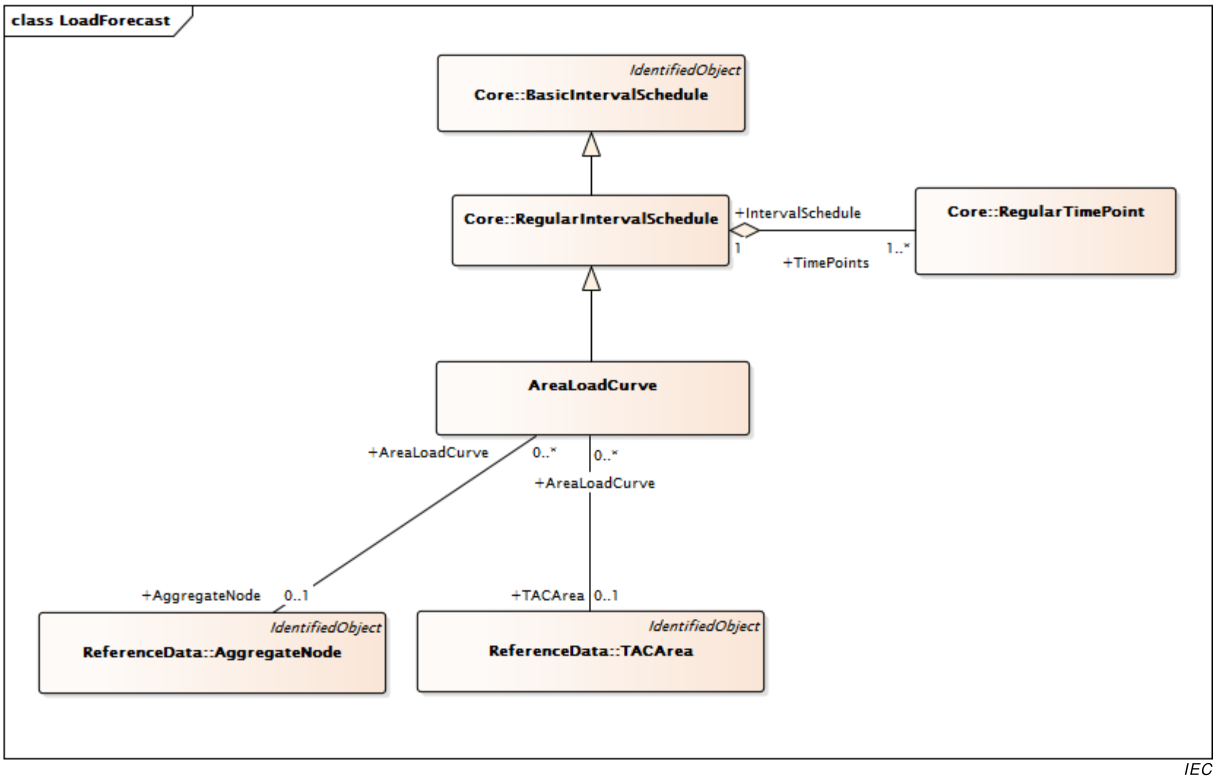



**Figure 42 – Diagramme de classe ExternalInputs::LoadForecast**

Figure 42: représente les classes utilisées pour modéliser la prévision de charge dans l'horizon temporel du marché. Le modèle réutilisé procède au classement dans le paquetage Core IEC 61970 et ajoute des modèles spécifiques aux marchés.

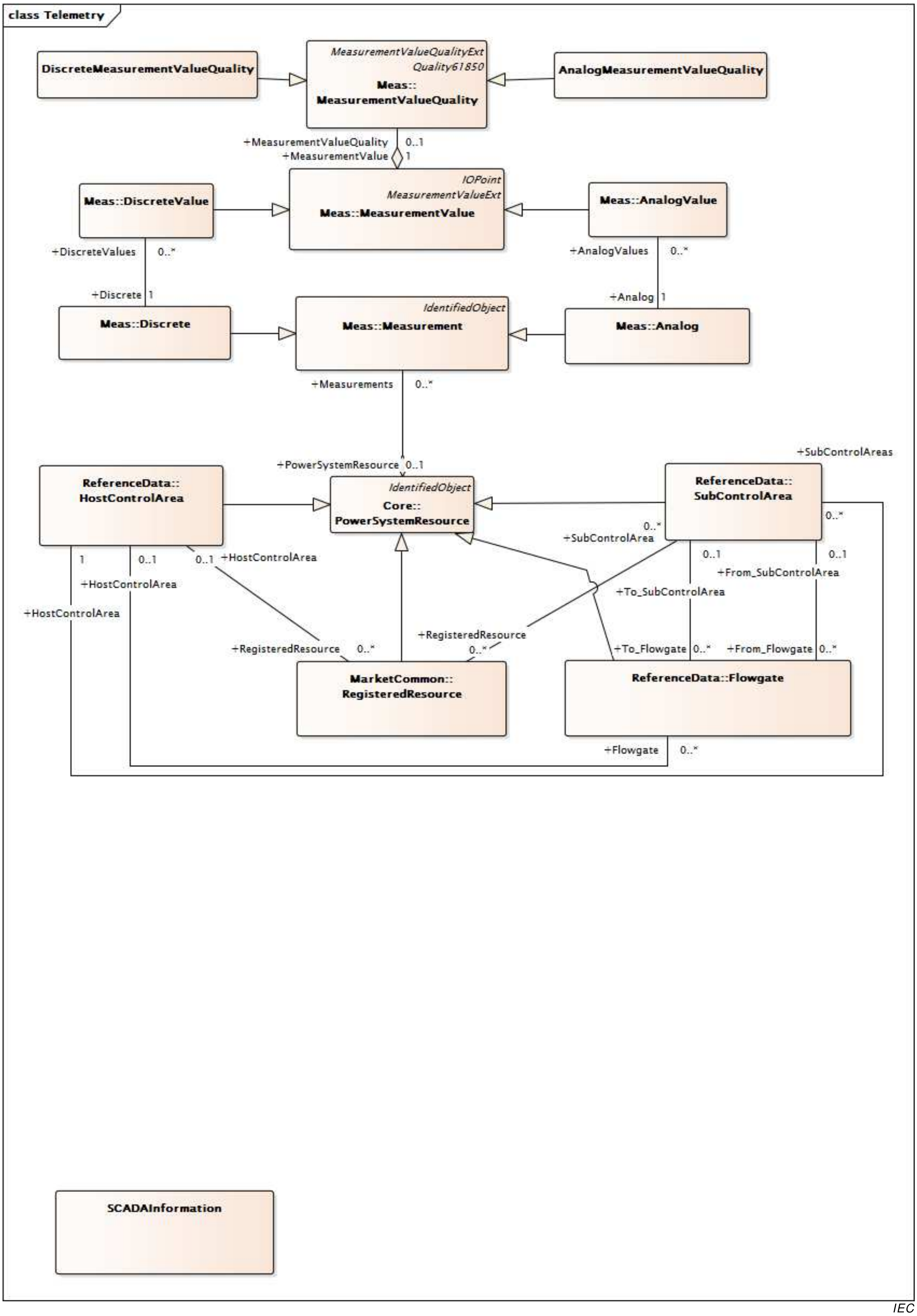



**Figure 43 – Diagramme de classe ExternalInputs::Telemetry**

Figure 43: représente les classes permettant de modéliser la télémétrie qui peut être utilisée (en cas de défaillance ou d'indisponibilité d'une solution d'estimateur d'état) pour lancer un

processus de marché en temps réel. La figure représente les classes du paquetage IEC 61970 et d'autres classes RegisteredResource et Flowgate spécifiques au marché, ainsi que leurs relations avec les classes des paquetages IEC 61970.

#### 6.5.6.2.2 Classe racine ASRequirements

Modélise les exigences relatives aux services auxiliaires (Ancillary Service Requirements). Décrit l'intervalle pour lequel l'exigence est applicable.

Le Tableau 133 présente tous les attributs de ASRequirements.

**Tableau 133 – Attributs de ExternalInputs::ASRequirements**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | Début de l'intervalle de temps pour lequel l'exigence est définie. |

Le Tableau 134 présente toutes les extrémités d'association de ASRequirements avec d'autres classes.

**Tableau 134 – Extrémités d'association de ExternalInputs:: ASRequirements avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ReserveDemandCurve | 1..* | ReserveDemandCurve | |

#### 6.5.6.2.3 AnalogMeasurementValueQuality

Indicateurs de qualité de mesure des valeurs Analog (analogiques).

Le Tableau 135 représente tous les attributs de AnalogMeasurementValueQuality.

**Tableau 135 – Attributs de ExternalInputs::AnalogMeasurementValueQuality**

| nom | mult | type | description |
|---|---|---|---|
| scadaQualityCode | 0..1 | String | Code de qualité d'une grandeur analogique donnée. |
| badReference | 0..1 | Boolean | Hérité de: Quality61850 |
| estimatorReplaced | 0..1 | Boolean | Hérité de: Quality61850 |
| failure | 0..1 | Boolean | Hérité de: Quality61850 |
| oldData | 0..1 | Boolean | Hérité de: Quality61850 |
| operatorBlocked | 0..1 | Boolean | Hérité de: Quality61850 |
| oscillatory | 0..1 | Boolean | Hérité de: Quality61850 |
| outOfRange | 0..1 | Boolean | Hérité de: Quality61850 |
| overFlow | 0..1 | Boolean | Hérité de: Quality61850 |
| source | 0..1 | Source | Hérité de: Quality61850 |
| suspect | 0..1 | Boolean | Hérité de: Quality61850 |
| test | 0..1 | Boolean | Hérité de: Quality61850 |
| validity | 0..1 | Validity | Hérité de: Quality61850 |

Le Tableau 136 présente toutes les extrémités d'association de AnalogMeasurementValueQuality avec d'autres classes.

**Tableau 136 – Extrémités d’association de ExternalInputs:: AnalogMeasurementValueQuality avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MeasurementValue | 1..1 | MeasurementValue | Hérité de: MeasurementValueQuality |

#### 6.5.6.2.4 AreaLoadCurve

Définition de courbe de charge de zone.

Le Tableau 137 présente tous les attributs de AreaLoadCurve.

**Tableau 137 – Attributs de ExternalInputs::AreaLoadCurve**

| nom | mult | type | description |
|---|---|---|---|
| forecastType | 0..1 | LoadForecastType | Type de zone de prévision de charge. |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 138 présente toutes les extrémités d'association de AreaLoadCurve avec d'autres classes.

**Tableau 138 – Extrémités d’association de ExternalInputs:: AreaLoadCurve avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AggregateNode | 0..1 | AggregateNode | |
| 0..* | TACArea | 0..1 | TACArea | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.5 BaseCaseConstraintLimit

Valeur maximale en MW ou MVA pouvant varier dans le temps et limite minimale facultative en MW ou MVA (Y1 et Y2, respectivement) attribuée à un scénario de base d'analyse de contingence. Utiliser CurveSchedule XAxisUnits pour indiquer des MW ou des MVA. À utiliser uniquement si BaseCaseConstraintLimit diffère de DefaultConstraintLimit.

Le Tableau 139 présente tous les attributs de BaseCaseConstraintLimit.

**Tableau 139 – Attributs de ExternalInputs::BaseCaseConstraintLimit**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 140 présente toutes les extrémités d'association de BaseCaseConstraintLimit avec d'autres classes.

**Tableau 140 – Extrémités d’association de ExternalInputs:: BaseCaseConstraintLimit avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.6 Classe racine BranchEndFlow

Flux et caractéristiques assignées dynamiques associés à l'extrémité d'une branche.

Le Tableau 141 présente tous les attributs de BranchEndFlow.

**Tableau 141 – Attributs de ExternalInputs::BranchEndFlow**

| nom | mult | type | description |
|---|---|---|---|
| mwFlow | 0..1 | Float | Flux de MW sur la branche<br>Utilisation de l'attribut: Flux de puissance active au niveau de l'appareil, du transformateur, du déphaseur en série ou de l'extrémité ligne |
| mVARFlow | 0..1 | Float | Flux MVAR sur la branche<br>Utilisation de l'attribut: Flux de puissance réactive au niveau de l'appareil, du transformateur, du déphaseur en série ou de l'extrémité ligne |
| normalRating | 0..1 | Float | Valeur assignée normale de la branche |
| longTermRating | 0..1 | Float | Valeur assignée à long terme de la branche |
| shortTermRating | 0..1 | Float | Valeur assignée à court terme de la branche |
| loadDumpRating | 0..1 | Float | Valeur assignée de perte de charge de la branche |

Le Tableau 142 présente toutes les extrémités d'association de BranchEndFlow avec d'autres classes.

**Tableau 142 – Extrémités d'association de ExternalInputs:: BranchEndFlow avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MktPowerTransformerEndBFlow | 0..* | MktPowerTransformer | |
| 0..1 | MktPowerTransformerEndAFlow | 0..* | MktPowerTransformer | |
| 0..1 | MktACLineSegmentEndAFlow | 0..* | MktACLineSegment | |
| 0..1 | MktACLineSegmentEndBFlow | 0..* | MktACLineSegment | |
| 0..1 | MktSeriesCompensatorEndBFlow | 0..* | MktSeriesCompensator | |
| 0..1 | MktSeriresCompensatorEndAFlow | 0..* | MktSeriesCompensator | |

### 6.5.6.2.7 ConstraintTerm

Une condition de contrainte est un élément d'une contrainte linéaire.

Le Tableau 143 présente tous les attributs de ConstraintTerm.

**Tableau 143 – Attributs de ExternalInputs::ConstraintTerm**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | |
| function | 0..1 | String | La fonction est une valeur énumérée qui peut être "active", "reactive" (réactive) ou "VA" pour indiquer le type de flux. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 144 présente toutes les extrémités d'association de ConstraintTerm avec d'autres classes.

**Tableau 144 – Extrémités d’association de ExternalInputs:: ConstraintTerm avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.8 ContingencyConstraintLimit

Valeur maximale en MW ou MVA pouvant varier dans le temps et limite minimale facultative en MW ou MVA (Y1 et Y2, respectivement) attribuées à une contrainte pour une contingence spécifique. Utiliser CurveSchedule XAxisUnits pour indiquer des MW ou des MVA.

Le Tableau 145 présente tous les attributs de ContingencyConstraintLimit.

**Tableau 145 – Attributs de ExternalInputs::ContingencyConstraintLimit**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 146 présente toutes les extrémités d'association de ContingencyConstraintLimit avec d'autres classes.

**Tableau 146 – Extrémités d’association de ExternalInputs:: ContingencyConstraintLimit avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktContingency | 1..1 | MktContingency | |
| 1..1 | MWLimitSchedules | 1..1 | MWLimitSchedule | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.9 Classe racine ControlAreaSolutionData

Échange et pertes du pool de solution d'estimateur d'état

Le Tableau 147 présente tous les attributs de ControlAreaSolutionData.

**Tableau 147 – Attributs de ExternalInputs::ControlAreaSolutionData**

| nom | mult | type | description |
|---|---|---|---|
| solvedLosses | 0..1 | Float | MW de pertes de pool<br><br>Utilisation de l'attribut: Pertes de puissance active du pool en MW |
| solvedInterchange | 0..1 | Float | Échange en MW du pool<br><br>Utilisation de l'attribut: Échange de puissance active du pool |

Le Tableau 148 présente toutes les extrémités d'association de ControlAreaSolutionData avec d'autres classes.

**Tableau 148 – Extrémités d’association de ExternalInputs:: ControlAreaSolutionData avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktControlArea | 0..1 | MktControlArea | |

#### 6.5.6.2.10 DefaultBid

DefaultBid (offre par défaut) est une classe générique regroupant Default Energy Bid, Default Startup Bid et Default Minimum Load Bid:

Default Energy Bid

Une DefaultEnergyBid (offre d'énergie par défaut) est une fonction en escalier croissante de façon monotone composée au maximum de 10 segments d'offre valuée ou de 10 paires ($/MW, MW). Il existe trois méthodes pour déterminer la DefaultEnergyBid:

– selon les coûts: dérivée de la consommation de chaleur ou du coût moyen multiplié par l'indice de prix du gaz plus 10 %.
– selon les LMP: une moyenne pondérée des LMP au cours des 90 derniers jours
– négocié: un montant négocié avec l'entité indépendante désignée.

Default Startup Bid

Une DSUB (Default Startup Bid – offre de démarrage par défaut) doit être calculée pour chaque unité RMR en fonction du StartUpCost (coût de démarrage) indiqué dans le Master File (fichier maître) et dans le GPI et l'EPI applicables.

Default Minimum Load Bid

Une DMLB (Default Minimum Load Bid – offre minimale de charge par défaut) doit être calculée pour chaque unité RMR en fonction du Minimum Load Cost (coût de charge minimal) indiqué dans le Master File (fichier maître) et dans le GPI applicable.

Le Tableau 149 présente tous les attributs de DefaultBid.

**Tableau 149 – Attributs de ExternalInputs::DefaultBid**

| nom | mult | type | description |
|---|---|---|---|
| bidType | 0..1 | BidType | Type d'offre par défaut tel que Default Energy Bid, Default Minimum Load Bid et Default Startup Bid |
| minLoadCost | 0..1 | CostRate | Coût de charge minimal en $/h |
| peakFlag | 0..1 | OnOff | heures de pointe, heures creuses ou toutes les heures |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 150 présente toutes les extrémités d'association de DefaultBid avec d'autres classes.

**Tableau 150 – Extrémités d’association de ExternalInputs:: DefaultBid avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DefaultBidCurve | 0..1 | DefaultBidCurve | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.11 DefaultBidCurve

Courbe d'offre par défaut pour la courbe d'offre d'énergie par défaut et pour les courbes de démarrage par défaut (coût et temps)

Le Tableau 151 présente tous les attributs de DefaultBidCurve.

**Tableau 151 – Attributs de ExternalInputs::DefaultBidCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveType | 0..1 | String | Pour indiquer un type utilisé pour une courbe d'offre d'énergie par défaut, tel qu'un type basé sur les LMP, les coûts ou la consultation. |
| debAdderFlag | 0..1 | YesNo | Indicateur d'additionneur d'offre d'énergie par défaut |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |

| nom | mult | type | description |
|---|---|---|---|
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 152 présente toutes les extrémités d'association de DefaultBidCurve avec d'autres classes.

**Tableau 152 – Extrémités d’association de ExternalInputs::DefaultBidCurve avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | DefaultBid | 0..1 | DefaultBid | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.12 DefaultBidCurveData

Données de courbe pour la courbe d'offre par défaut et pour la courbe de coût de démarrage.

Le Tableau 153 présente tous les attributs de DefaultBidCurveData.

**Tableau 153 – Attributs de ExternalInputs::DefaultBidCurveData**

| nom | mult | type | description |
|---|---|---|---|
| bidSegmentCalcType | 0..1 | BidCalculationBasis | Type de base de calcul utilisé pour définir la courbe de segment d'offre par défaut. |
| xvalue | 0..1 | Float | Hérité de: CurveData |
| y1value | 0..1 | Float | Hérité de: CurveData |
| y2value | 0..1 | Float | Hérité de: CurveData |
| y3value | 0..1 | Float | Hérité de: CurveData |

Le Tableau 154 présente toutes les extrémités d'association de DefaultBidCurveData avec d'autres classes.

**Tableau 154 – Extrémités d’association de ExternalInputs:: DefaultBidCurveData avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Curve | 1..1 | Curve | Hérité de: CurveData |

### 6.5.6.2.13 DefaultConstraintLimit

Valeur maximale en MW ou MVA pouvant varier dans le temps et limite minimale facultative en MW ou MVA (Y1 et Y2, respectivement) attribuée à une valeur par défaut si aucune limite de contrainte spécifique n'est indiquée pour une analyse de contingence. Utiliser CurveSchedule XAxisUnits pour indiquer des MW ou des MVA.

Le Tableau 155 présente tous les attributs de DefaultConstraintLimit.

**Tableau 155 – Attributs de ExternalInputs::DefaultConstraintLimit**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 156 présente toutes les extrémités d'association de DefaultConstraintLimit avec d'autres classes.

**Tableau 156 – Extrémités d’association de ExternalInputs:: DefaultConstraintLimit avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraintSum | 1..1 | SecurityConstraintSum | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.14 DiscreteMeasurementValueQuality

Indicateurs de qualité de mesure des valeurs Discrete (discrètes).

Le Tableau 157 présente tous les attributs de DiscreteMeasurementValueQuality.

**Tableau 157 – Attributs de ExternalInputs::DiscreteMeasurementValueQuality**

| nom | mult | type | description |
|---|---|---|---|
| manualReplaceIndicator | 0..1 | Boolean | Indicateur de remplacement manuel du commutateur. Indicateur précisant que le commutateur est à remplacement manuel. |
| removeFromOperationIndicator | 0..1 | Boolean | Indicateur de mise hors service. Indicateur précisant que le commutateur a été mis hors service. |
| badReference | 0..1 | Boolean | Hérité de: Quality61850 |
| estimatorReplaced | 0..1 | Boolean | Hérité de: Quality61850 |
| failure | 0..1 | Boolean | Hérité de: Quality61850 |
| oldData | 0..1 | Boolean | Hérité de: Quality61850 |
| operatorBlocked | 0..1 | Boolean | Hérité de: Quality61850 |
| oscillatory | 0..1 | Boolean | Hérité de: Quality61850 |
| outOfRange | 0..1 | Boolean | Hérité de: Quality61850 |
| overFlow | 0..1 | Boolean | Hérité de: Quality61850 |
| source | 0..1 | Source | Hérité de: Quality61850 |
| suspect | 0..1 | Boolean | Hérité de: Quality61850 |
| test | 0..1 | Boolean | Hérité de: Quality61850 |
| validity | 0..1 | Validity | Hérité de: Quality61850 |

Le Tableau 158 présente toutes les extrémités d'association de DiscreteMeasurementValueQuality avec d'autres classes.

**Tableau 158 – Extrémités d'association de ExternalInputs::DiscreteMeasurementValueQuality avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MeasurementValue | 1..1 | MeasurementValue | Hérité de: MeasurementValueQuality |

#### 6.5.6.2.15 Classe racine DistributionFactorSet

Classe principale regroupant tous les facteurs de distribution d'un marché.

Elle est calculée tous les jours pour les facteurs DA et toutes les heures pour les facteurs RT.

Le Tableau 159 présente tous les attributs de DistributionFactorSet.

**Tableau 159 – Attributs de ExternalInputs::DistributionFactorSet**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | Début de l'intervalle de temps pour lequel une exigence est définie. |
| intervalEndTime | 0..1 | DateTime | Fin de l'intervalle de temps pour lequel une exigence est définie. |
| marketType | 0..1 | MarketType | Type de marché. |

Le Tableau 160 présente toutes les extrémités d'association de DistributionFactorSet avec d'autres classes.

**Tableau 160 – Extrémités d'association de ExternalInputs:: DistributionFactorSet avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | GenDistributionFactor | 0..* | GenDistributionFactor | |
| 0..* | SysLoadDistribuFactor | 0..* | SysLoadDistributionFactor | |
| 0..* | LoadDistributionFactor | 0..* | LoadDistributionFactor | |

### 6.5.6.2.16 EnergyPriceIndex

Un EnergyPriceIndex (indice du prix de l'énergie) pour chaque ressource est valide pour une période (quotidiennement, par exemple) identifiée par une Valid Period Start Time (heure de démarrage de période valide) et par une Valid Period End Time (heure de fin de période valide). L'indice du prix de l'énergie est exprimé en $/MWh.

Le Tableau 161 présente tous les attributs de EnergyPriceIndex.

**Tableau 161 – Attributs de ExternalInputs::EnergyPriceIndex**

| nom | mult | type | description |
|---|---|---|---|
| energyPriceIndex | 0..1 | Float | Indice du prix de l'énergie |
| energyPriceIndexType | 0..1 | EnergyPriceIndexType | Type d'EPI (gros ou détail) |
| lastModified | 0..1 | DateTime | Mise à jour de l'heure |
| validPeriod | 0..1 | DateTimeInterval | Période de validité de l'indice du prix de l'énergie. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 162 présente toutes les extrémités d'association de EnergyPriceIndex avec d'autres classes.

**Tableau 162 – Extrémités d'association de ExternalInputs:: EnergyPriceIndex avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | RegisteredGenerator | 1..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.17 EnergyProfile

Indique l'heure de début, l'heure d'arrêt et le niveau d'EnergyTransaction.
Le Tableau 163 présente tous les attributs de EnergyProfile.

**Tableau 163 – Attributs de ExternalInputs::EnergyProfile**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 164 présente toutes les extrémités d'association de EnergyProfile avec d'autres classes.

**Tableau 164 – Extrémités d'association de ExternalInputs:: EnergyProfile avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | TransactionBid | 1..1 | TransactionBid | |
| 1..* | EnergyTransaction | 1..1 | EnergyTransaction | Une EnergyTransaction doit avoir au moins un EnergyProfile. |
| 0..* | ProfileDatas | 0..* | ProfileData | Hérité de: Profile |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.18 EnergyTransaction

Indique le programme des transferts d'énergie entre les zones d'échange qui sont nécessaires pour réaliser la transaction d'échange associée.

Le Tableau 165 présente tous les attributs de EnergyTransaction.

**Tableau 165 – Attributs de ExternalInputs::EnergyTransaction**

| nom | mult | type | description |
|---|---|---|---|
| capacityBacked | 0..1 | Boolean | Indicateur de capacité d'échange. Lorsque l'indicateur est défini sur True, cela signifie qu'une transaction est suivie par la fourniture de la capacité demandée. |
| congestChargeMax | 0..1 | Money | Frais de congestion maximum en unités monétaires. |
| deliveryPointP | 0..1 | ActivePower | Puissance active au point de livraison. |
| energyMin | 0..1 | ActivePower | Puissance active minimale d'une transaction si elle est répartissable. |
| firmInterchangeFlag | 0..1 | Boolean | L'indicateur d'échange ferme précise si cette transaction d'énergie peut être modifiée sans conséquences financières éventuelles. |
| payCongestion | 0..1 | Boolean | Indicateur de congestion Willing to Pay (Disposé à payer). |
| reason | 0..1 | String | Motif de la transaction d'énergie. |
| receiptPointP | 0..1 | ActivePower | Puissance active au point de réception. |
| state | 0..1 | EnergyTransactionType | { Approbation \| Négation \| Étude } |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 166 présente toutes les extrémités d'association de EnergyTransaction avec d'autres classes.

**Tableau 166 – Extrémités d'association de ExternalInputs:: EnergyTransaction avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnergyPriceCurves | 0..* | EnergyPriceCurve | |
| 1..1 | EnergyProfiles | 1..* | EnergyProfile | Une EnergyTransaction doit avoir au moins un EnergyProfile. |
| 0..* | Export_SubControlArea | 1..1 | SubControlArea | L'énergie est transférée entre les zones d'échange. |
| 0..* | Import_SubControlArea | 1..1 | SubControlArea | L'énergie est transférée entre les zones d'échange. |
| 0..1 | TransmissionReservation | 0..1 | TransmissionReservation | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.19 Classe racine GenDistributionFactor

Cette classe modélise les facteurs de distribution de production. Elle nécessite d'être utilisée avec AggregatedPnode et IndividualPnode pour indiquer la distribution de chaque partie individuelle.

Le Tableau 167 présente tous les attributs de GenDistributionFactor.

**Tableau 167 – Attributs de ExternalInputs::GenDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Utilisé pour calculer la "participation" de production d'un Pnode individuel dans un AggregatePnode. |

Le Tableau 168 présente toutes les extrémités d'association de GenDistributionFactor avec d'autres classes.

**Tableau 168 – Extrémités d'association de ExternalInputs:: GenDistributionFactor avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 1..* | AggregatedPnode | 0..1 | AggregatedPnode | |
| 0..1 | IndividualPnode | 0..1 | IndividualPnode | |

### 6.5.6.2.20 Classe racine GeneratingUnitDynamicValues

Optimal Power Flow ou State Estimator Unit Data pour le simulateur de formation des exploitants (OTS – Operator Training Simulator). Utilisée pour les utilisateurs RealTime, Study et Maintenance.

Le Tableau 169 présente tous les attributs de GeneratingUnitDynamicValues.

**Tableau 169 – Attributs de ExternalInputs::GeneratingUnitDynamicValues**

| nom | mult | type | description |
|---|---|---|---|
| lossFactor | 0..1 | Float | Facteur de perte |
| maximumMW | 0..1 | Float | Production de puissance active maximale de l'unité en MW |
| minimumMW | 0..1 | Float | Production de puissance active minimale de l'unité en MW |
| mVAR | 0..1 | Float | Production de puissance réactive de l'unité en MVAR |
| mw | 0..1 | Float | Production de puissance active de l'unité en MW |
| sensitivity | 0..1 | Float | Facteur de sensibilité de l'unité. Facteurs de distribution (DFAX) de l'unité |

Le Tableau 170 présente toutes les extrémités d'association de GeneratingUnitDynamicValues avec d'autres classes.

**Tableau 170 – Extrémités d'association de ExternalInputs::GeneratingUnitDynamicValues avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktGeneratingUnit | 1..1 | MktGeneratingUnit | |
| 0..* | Flowgate | 0..1 | Flowgate | |

### 6.5.6.2.21 GenericConstraints

Les contraintes génériques peuvent représenter des zones sécurisées, un profil de tension, une stabilité transitoire et des limites de chute de tension.

Les contraintes génériques peuvent avoir l'une des formes suivantes:

a) Limites thermiques en MW
b) Contrainte type de flux de ligne de groupe

Le Tableau 171 présente tous les attributs de GenericConstraints.

**Tableau 171 – Attributs de ExternalInputs::GenericConstraints**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Heure de fin de l'intervalle |
| intervalStartTime | 0..1 | DateTime | Heure de début de l'intervalle |
| maxLimit | 0..1 | Float | Limite maximale (MW) |
| minLimit | 0..1 | Float | Limite minimale (MW) |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 172 présente toutes les extrémités d'association de GenericConstraints avec d'autres classes.

**Tableau 172 – Extrémités d'association de ExternalInputs:: GenericConstraints avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | Flowgate | 0..* | Flowgate | |
| 0..1 | TransmissionCapacity | 0..* | TransmissionCapacity | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.22 Classe racine InterchangeETCData

Données de contrat de transport existant pour un programme d'échange

Le Tableau 173 présente tous les attributs de InterchangeETCData.

**Tableau 173 – Attributs de ExternalInputs::InterchangeETCData**

| nom | mult | type | description |
|---|---|---|---|
| contractNumber | 0..1 | String | Numéro de contrat de transport existant |
| usageMW | 0..1 | Float | Valeur en MW de l'utilisation du contrat de transport existant |

Le Tableau 174 présente toutes les extrémités d'association de InterchangeETCData avec d'autres classes.

**Tableau 174 – Extrémités d’association de ExternalInputs:: InterchangeETCData avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | InterchangeSchedule | 0..1 | InterchangeSchedule | |

### 6.5.6.2.23 InterchangeSchedule

Classe de programme d'échange pour stocker des informations sur les programmes d'échange de type import/export, énergie, etc.

Le Tableau 175 présente tous les attributs de InterchangeSchedule.

**Tableau 175 – Attributs de ExternalInputs::InterchangeSchedule**

| nom | mult | type | description |
|---|---|---|---|
| checkOutType | 0..1 | CheckOutType | Pour indiquer un type de vérification tel que la capacité ajustée ou la capacité de répartition. |
| directionType | 0..1 | InterTieDirection | Import ou export. |
| energyType | 0..1 | MarketProductType | Type de produit énergétique. |
| intervalLength | 0..1 | Integer | Longueur de l'intervalle. |
| marketType | 0..1 | MarketType | Type de marché. |
| operatingDate | 0..1 | DateTime | Date/heure d'exploitation. |
| outOfMarketType | 0..1 | Boolean | Pour indiquer un programme hors marché (OOM – Out-Of-Market). |
| scheduleType | 0..1 | EnergyProductType | Type de programme. |
| wcrID | 0..1 | String | Identifiant compteur-ressource de transit (exigé lorsque Type de programme = Transit). |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 176 présente toutes les extrémités d'association de InterchangeSchedule avec d'autres classes.

**Tableau 176 – Extrémités d'association de ExternalInputs:: InterchangeSchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | InterchangeETCData | 0..* | InterchangeETCData | |
| 0..* | InterTie | 0..1 | SchedulingPoint | |
| 0..* | RegisteredInterTie | 0..1 | RegisteredInterTie | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.24 IntermittentResourceEligibility

Indique si l'unité est susceptible d'être considérée comme une ressource renouvelable variable intermittente

Le Tableau 177 présente tous les attributs de IntermittentResourceEligibility.

**Tableau 177 – Attributs de ExternalInputs::IntermittentResourceEligibility**

| nom | mult | type | description |
|---|---|---|---|
| eligibilityStatus | 0..1 | String | Indique si une ressource est susceptible d'être admise dans le programme PIRP à une heure donnée |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 178 présente toutes les extrémités d'association de IntermittentResourceEligibility avec d'autres classes.

**Tableau 178 – Extrémités d'association de ExternalInputs::IntermittentResourceEligibility avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.25 Classe racine LoadDistributionFactor

Cette classe modélise les facteurs de distribution de charge. Il convient de l'utiliser de l'une des manières suivantes:

Utilisation avec AggregatedPnode et IndividualPnode pour indiquer la distribution de chaque partie individuelle

OU

Utilisation avec Mkt_EnergyConsumer pour représenter la distribution actuelle MW/Mvar dans son groupe de charge parent.

Le Tableau 179 présente tous les attributs de LoadDistributionFactor.

**Tableau 179 – Attributs de ExternalInputs::LoadDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| pDistFactor | 0..1 | Float | Facteur de distribution de charge de puissance réelle (MW) |
| qDistFactor | 0..1 | Float | Facteur de distribution de charge de puissance réactive (MVAr) |

Le Tableau 180 présente toutes les extrémités d'association de LoadDistributionFactor avec d'autres classes.

**Tableau 180 – Extrémités d’association de ExternalInputs:: LoadDistributionFactor avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 0..1 | IndividualPnode | 0..1 | IndividualPnode | |
| 1..* | AggregatedPnode | 0..1 | AggregatedPnode | |

#### 6.5.6.2.26 LossSensitivity

Sensibilité de perte appliquée à ConnectivityNode pour un intervalle de temps donné.

Le Tableau 181 présente tous les attributs de LossSensitivity.

**Tableau 181 – Attributs de ExternalInputs::LossSensitivity**

| nom | mult | type | description |
|---|---|---|---|
| lossFactor | 0..1 | Float | Facteur de pénalité de perte.<br>Défini de la manière suivante: 1 / (1 – perte de transport différentielle); la perte de transport différentielle est exprimée en valeur plus/moins. Les facteurs de pénalité types sont compris entre (0,9 et 1,1). |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 182 présente toutes les extrémités d'association de LossSensitivity avec d'autres classes.

**Tableau 182 – Extrémités d'association de ExternalInputs:: LossSensitivity avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.27 Classe racine MWLimitSchedule

Valeur maximale de MW et valeur facultative minimale de MW (Y1 et Y2, respectivement)

Le Tableau 183 présente toutes les extrémités d'association de MWLimitSchedule avec d'autres classes.

**Tableau 183 – Extrémités d'association de ExternalInputs:: MWLimitSchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | SecurityConstraintLimit | 1..1 | ContingencyConstraintLimit | |

#### 6.5.6.2.28 MktACLineSegment

Sous-classe de IEC61970:Wires:ACLineSegment

Le Tableau 184 présente tous les attributs de MktACLineSegment.

**Tableau 184 – Attributs de ExternalInputs::MktACLineSegment**

| nom | mult | type | description |
|---|---|---|---|
| b0ch | 0..1 | Susceptance | Hérité de: ACLineSegment |
| bch | 0..1 | Susceptance | Hérité de: ACLineSegment |
| g0ch | 0..1 | Conductance | Hérité de: ACLineSegment |
| gch | 0..1 | Conductance | Hérité de: ACLineSegment |
| r | 0..1 | Resistance | Hérité de: ACLineSegment |
| r0 | 0..1 | Resistance | Hérité de: ACLineSegment |
| shortCircuitEndTemperature | 0..1 | Temperature | Hérité de: ACLineSegment |
| x | 0..1 | Reactance | Hérité de: ACLineSegment |

| nom | mult | type | description |
|---|---|---|---|
| x0 | 0..1 | Reactance | Hérité de: ACLineSegment |
| length | 0..1 | Length | Hérité de: Conductor |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 185 présente toutes les extrémités d'association de MktACLineSegment avec d'autres classes.

**Tableau 185 – Extrémités d’association de ExternalInputs:: MktACLineSegment avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 1..1 | Clamp | 0..* | Clamp | Hérité de: ACLineSegment |
| 0..* | PerLengthImpedance | 0..1 | PerLengthImpedance | Hérité de: ACLineSegment |
| 1..1 | Cut | 0..* | Cut | Hérité de: ACLineSegment |
| 0..1 | LineFaults | 0..* | LineFault | Hérité de: ACLineSegment |
| 1..1 | ACLineSegmentPhases | 0..* | ACLineSegmentPhases | Hérité de: ACLineSegment |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContained | 0..* | EquipmentContained | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.29 MktAnalogLimit

Sous-classe de IEC 61970:Meas:AnalogLimit

Le Tableau 186 présente tous les attributs de MktAnalogLimit.

**Tableau 186 – Attributs de ExternalInputs::MktAnalogLimit**

| nom | mult | type | description |
|---|---|---|---|
| exceededLimit | 0..1 | Boolean | True si les limites sont dépassées |
| limitType | 0..1 | AnalogLimitType | Le type de limite que la valeur représente<br>Types de limites de branche:<br>Court terme<br>Moyen terme<br>Long terme<br>Limites de tension:<br>Haute<br>Basse |
| value | 0..1 | Float | Hérité de: AnalogLimit |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 187 présente toutes les extrémités d'association de MktAnalogLimit avec d'autres classes.

**Tableau 187 – Extrémités d’association de ExternalInputs:: MktAnalogLimit avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | LimitSet | 1..1 | AnalogLimitSet | Hérité de: AnalogLimit |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.30 MktAnalogLimitSet

Sous-classe de IEC 61970:Meas:AnalogLimitSet

Le Tableau 188 présente tous les attributs de MktAnalogLimitSet.

**Tableau 188 – Attributs de ExternalInputs::MktAnalogLimitSet**

| nom | mult | type | description |
|---|---|---|---|
| ratingSet | 0..1 | Integer | Nombre de l’ensemble de valeurs assignées |
| isPercentageLimits | 0..1 | Boolean | Hérité de: LimitSet |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 189 présente toutes les extrémités d'association de MktAnalogLimitSet avec d'autres classes.

**Tableau 189 – Extrémités d’association de ExternalInputs:: MktAnalogLimitSet avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Limits | 0..* | AnalogLimit | Hérité de: AnalogLimitSet |
| 0..* | Measurements | 0..* | Analog | Hérité de: AnalogLimitSet |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.31 MktControlArea

Sous-classe de marché de IEC 61970:ControlArea

Le Tableau 190 présente tous les attributs de MktControlArea.

**Tableau 190 – Attributs de ExternalInputs::MktControlArea**

| nom | mult | type | description |
|---|---|---|---|
| netInterchange | 0..1 | ActivePower | Hérité de: ControlArea |
| pTolerance | 0..1 | ActivePower | Hérité de: ControlArea |
| type | 0..1 | ControlAreaTypeKind | Hérité de: ControlArea |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 191 présente toutes les extrémités d'association de MktControlArea avec d'autres classes.

**Tableau 191 – Extrémités d’association de ExternalInputs:: MktControlArea avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ControlAreaSolutionData | 0..* | ControlAreaSolutionData | |
| 0..1 | EnergyArea | 0..1 | EnergyArea | Hérité de: ControlArea |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | Hérité de: ControlArea |
| 1..1 | TieFlow | 0..* | TieFlow | Hérité de: ControlArea |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.32 MktSeriesCompensator

Sous-classe de IEC 61970:Wires:SeriesCompensator

Le Tableau 192 présente tous les attributs de MktSeriesCompensator.

**Tableau 192 – Attributs de ExternalInputs::MktSeriesCompensator**

| nom | mult | type | description |
|---|---|---|---|
| r | 0..1 | Resistance | Hérité de: SeriesCompensator |
| r0 | 0..1 | Resistance | Hérité de: SeriesCompensator |
| x | 0..1 | Reactance | Hérité de: SeriesCompensator |
| x0 | 0..1 | Reactance | Hérité de: SeriesCompensator |
| varistorPresent | 0..1 | Boolean | Hérité de: SeriesCompensator |
| varistorRatedCurrent | 0..1 | CurrentFlow | Hérité de: SeriesCompensator |
| varistorVoltageThreshold | 0..1 | Voltage | Hérité de: SeriesCompensator |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 193 présente toutes les extrémités d'association de MktSeriesCompensator avec d'autres classes.

**Tableau 193 – Extrémités d'association de ExternalInputs:: MktSeriesCompensator avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EndBFlow | 0..1 | BranchEndFlow | |
| 0..* | EndAFlow | 0..1 | BranchEndFlow | |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContainer | 0.. * | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.33 MktShuntCompensator

Sous-classe de IEC 61970:Wires:ShuntCompensator

Le Tableau 194 présente tous les attributs de MktShuntCompensator.

**Tableau 194 – Attributs de ExternalInputs::MktShuntCompensator**

| nom | mult | type | description |
|---|---|---|---|
| aVRDelay | 0..1 | Seconds | Hérité de: ShuntCompensator |
| grounded | 0..1 | Boolean | Hérité de: ShuntCompensator |
| maximumSections | 0..1 | Integer | Hérité de: ShuntCompensator |
| nomU | 0..1 | Voltage | Hérité de: ShuntCompensator |
| normalSections | 0..1 | Integer | Hérité de: ShuntCompensator |
| phaseConnection | 0..1 | PhaseShuntConnectionKind | Hérité de: ShuntCompensator |
| switchOnCount | 0..1 | Integer | Hérité de: ShuntCompensator |
| switchOnDate | 0..1 | DateTime | Hérité de: ShuntCompensator |
| voltageSensitivity | 0..1 | VoltagePerReactivePower | Hérité de: ShuntCompensator |
| sections | 0..1 | Float | Hérité de: ShuntCompensator |
| controlEnabled | 0..1 | Boolean | Hérité de: RegulatingCondEq |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 195 présente toutes les extrémités d'association de MktShuntCompensator avec d'autres classes.

**Tableau 195 – Extrémités d’association de ExternalInputs:: MktShuntCompensator avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ShuntCompensatorDynamicData | 0..* | ShuntCompensatorDynamicData | |
| 1..1 | ShuntCompensatorPhase | 0..* | ShuntCompensatorPhase | Hérité de: ShuntCompensator |
| 1..1 | SvShuntCompensatorSections | 0..1 | SvShuntCompensatorSections | Hérité de: ShuntCompensator |
| 0..* | RegulatingControl | 0..1 | RegulatingControl | Hérité de: RegulatingCondEq |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AdditionalEquipmentContainer | 0.. * | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.34 MktSwitch

Sous-classe de IEC 61970:Wires:Switch

Le Tableau 196 présente tous les attributs de MktSwitch.

**Tableau 196 – Attributs de ExternalInputs::MktSwitch**

| nom | mult | type | description |
|---|---|---|---|
| normalOpen | 0..1 | Boolean | Hérité de: Switch |
| ratedCurrent | 0..1 | CurrentFlow | Hérité de: Switch |
| switchOnCount | 0..1 | Integer | Hérité de: Switch |
| switchOnDate | 0..1 | DateTime | Hérité de: Switch |
| retained | 0..1 | Boolean | Hérité de: Switch |
| open | 0..1 | Boolean | Hérité de: Switch |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 197 présente toutes les extrémités d'association de MktSwitch avec d'autres classes.

**Tableau 197 – Extrémités d'association de ExternalInputs:: MktSwitch avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | SwitchStatus | 0..* | SwitchStatus | |
| 1..1 | SwitchSchedules | 0..* | SwitchSchedule | Hérité de: Switch |
| 1..1 | SwitchPhase | 0..* | SwitchPhase | Hérité de: Switch |
| 0..* | CompositeSwitch | 0..1 | CompositeSwitch | Hérité de: Switch |
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContainer | 0.. * | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.35 MktTapChanger

Sous-classe de IEC 61970:Wires:TapChanger

Le Tableau 198 présente tous les attributs de MktTapChanger.

**Tableau 198 – Attributs de ExternalInputs::MktTapChanger**

| nom | mult | type | description |
|---|---|---|---|
| controlEnabled | 0..1 | Boolean | Hérité de: TapChanger |
| highStep | 0..1 | Integer | Hérité de: TapChanger |
| initialDelay | 0..1 | Seconds | Hérité de: TapChanger |
| lowStep | 0..1 | Integer | Hérité de: TapChanger |
| ltcFlag | 0..1 | Boolean | Hérité de: TapChanger |
| neutralStep | 0..1 | Integer | Hérité de: TapChanger |
| neutralU | 0..1 | Voltage | Hérité de: TapChanger |
| normalStep | 0..1 | Integer | Hérité de: TapChanger |
| subsequentDelay | 0..1 | Seconds | Hérité de: TapChanger |

| nom | mult | type | description |
|---|---|---|---|
| step | 0..1 | Float | Hérité de: TapChanger |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 199 présente toutes les extrémités d'association de MktTapChanger avec d'autres classes.

**Tableau 199 – Extrémités d'association de ExternalInputs:: MktTapChanger avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TapChangerDynamicData | 0..* | TapChangerDynamicData | |
| 0..* | TapChangerControl | 0..1 | TapChangerControl | Hérité de: TapChanger |
| 1..1 | TapSchedules | 0..* | TapSchedule | Hérité de: TapChanger |
| 1..1 | SvTapStep | 0..1 | SvTapStep | Hérité de: TapChanger |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.36 NodeConstraintTerm

À utiliser uniquement pour limiter une quantité qui ne peut pas être associée à une borne. Par exemple, une unité de production enregistrée qui n'est pas connectée électriquement au réseau.

Le Tableau 200 présente tous les attributs de NodeConstraintTerm.

**Tableau 200 – Attributs de ExternalInputs::NodeConstraintTerm**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | Hérité de: ConstraintTerm |
| function | 0..1 | String | Hérité de: ConstraintTerm |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 201 présente toutes les extrémités d'association de NodeConstraintTerm avec d'autres classes.

**Tableau 201 – Extrémités d'association de ExternalInputs:: NodeConstraintTerm avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | Hérité de: ConstraintTerm |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.37 Profile

Un profil est un type de courbe plus simple.

Le Tableau 202 présente tous les attributs de Profile.

**Tableau 202 – Attributs de ExternalInputs::Profile**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 203 présente toutes les extrémités d'association de Profile avec d'autres classes.

**Tableau 203 – Extrémités d'association de ExternalInputs:: Profile avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ProfileDatas | 0..* | ProfileData | Un profil a des données de profil qui lui sont associées. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.38 Classe racine ProfileData

Données de profil.

Le Tableau 204 présente tous les attributs de ProfileData.

**Tableau 204 – Attributs de ExternalInputs::ProfileData**

| nom | mult | type | description |
|---|---|---|---|
| bidPrice | 0..1 | Float | Prix d'offre associé au contrat |
| capacityLevel | 0..1 | ActivePower | Niveau de capacité du profil, en MW. |
| energyLevel | 0..1 | RealEnergy | Niveau d'énergie du profil, en MWH. |
| minimumLevel | 0..1 | Float | Valeur minimale du contrat en MW |
| sequenceNumber | 0..1 | Integer | Séquence pour numéroter les éléments du profil. { supérieur ou égal à 1 } |
| startDateTime | 0..1 | DateTime | Date/heure de début pour ce profil. |
| stopDateTime | 0..1 | DateTime | Date/heure d'arrêt pour ce profil. |

Le Tableau 205 présente toutes les extrémités d'association de ProfileData avec d'autres classes.

**Tableau 205 – Extrémités d'association de ExternalInputs:: ProfileData avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Profile | 0..* | Profile | Un profil a des données de profil qui lui sont associées. |

### 6.5.6.2.39 ReserveDemandCurve

Courbe de demande de réserve. Modélise les quantités maximales de réserve exigées par Market Region (région de marché) et modélise une courbe de demande de réserve pour les quantités minimales de réserve. ReserveDemandCurve est une relation entre le prix de réserve d'exploitation de l'unité en $/MWh (axe Y) et les réserves de l'unité en MW (axe X).

Le Tableau 206 présente tous les attributs de ReserveDemandCurve.

**Tableau 206 – Attributs de ExternalInputs::ReserveDemandCurve**

| nom | mult | type | description |
|---|---|---|---|
| reqMaxMW | 0..1 | Float | Limite maximale de l'exigence régionale |
| reserveRequirementType | 0..1 | ReserveRequirementType | Type d'exigences de réserve auquel s'appliquent la limite max et la courbe. Par exemple, réserve d'exploitation, régulation et contingence. |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 207 présente toutes les extrémités d'association de ReserveDemandCurve avec d'autres classes.

**Tableau 207 – Extrémités d’association de ExternalInputs:: ReserveDemandCurve avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | ASRequirements | 1..1 | ASRequirements | |
| 0..* | MarketRegion | 1..1 | MarketRegion | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.40 Classe racine SCADAInformation

Contient des informations de mise à jour indiquées par SCADA

Le Tableau 208 présente tous les attributs de SCADAInformation.

**Tableau 208 – Attributs de ExternalInputs::SCADAInformation**

| nom | mult | type | description |
|---|---|---|---|
| timeStamp | 0..1 | DateTime | Heure de mise à jour indiquée par SCADA |

#### 6.5.6.2.41 SecurityConstraintSum

Généralement fournies par les systèmes des RTO, les contraintes identifiées dans le scénario de base et dans les scénarios de contingence critique doivent être transférées.

Une contrainte a N (>=1) conditions de contraintes. Une condition est représentée par une instance de TerminalConstraintTerm.

L'expression de la contrainte est la suivante:

minValue <= c1*x1 + c2*x2 + .... cn*xn + k <= maxValue

où:

– cn est ConstraintTerm.factor
– xn est le flux de la borne

Le flux de l'équipement associé est positif pour NodeConstraintTerm de ConnectivityNode

k est SecurityConstraintsLinear.resourceMW

Les unités de k sont censées être les mêmes que les unités des flux, xn. Les constantes cn n'ont pas de dimension.

Avec ces conventions, cn et k sont positives pour une contrainte type telle que: "la somme pondérée de production doit être inférieure à la limite". En outre, les cn sont toutes égales à 1.0 pour un scénario tel que "le flux de l'interface doit être inférieur à la limite", en partant du principe que les bornes sont choisies sur le côté importation de l'interface.

Le Tableau 209 présente tous les attributs de SecurityConstraintSum.

**Tableau 209 – Attributs de ExternalInputs::SecurityConstraintSum**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 210 présente toutes les extrémités d'association de SecurityConstraintSum avec d'autres classes.

**Tableau 210 – Extrémités d'association de ExternalInputs:: SecurityConstraintSum avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ConstraintTerms | 0..* | ConstraintTerm | |
| 0..* | RTO | 0..1 | RTO | |
| 1..1 | BaseCaseConstraintLimit | 0..1 | BaseCaseConstraintLimit | |
| 1..1 | ContingencyConstraintLimits | 0..* | ContingencyConstraintLimit | |
| 1..1 | DefaultConstraintLimit | 0..1 | DefaultConstraintLimit | |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de:: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.42 SecurityConstraints

Ces contraintes sont typiques pour les gestionnaires des réseaux de transport régionaux (RTO). Elles incluent des contraintes de transport et de groupe de production identifiées dans le scénario de base et dans les scénarios de contingence critique.

Le Tableau 211 présente tous les attributs de SecurityConstraints.

**Tableau 211 – Attributs de ExternalInputs::SecurityConstraints**

| nom | mult | type | description |
|---|---|---|---|
| minMW | 0..1 | ActivePower | Limite minimale en MW (uniquement pour les contraintes de transport). |
| maxMW | 0..1 | ActivePower | Limite maximale en MW |
| actualMW | 0..1 | ActivePower | Flux d'une branche réelle ou d'un groupe de branches en MW (uniquement pour les contraintes de transport) |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 212 présente toutes les extrémités d’association de SecurityConstraints avec d’autres classes.

**Tableau 212 – Extrémités d'association de ExternalInputs:: SecurityConstraints avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RTO | 0..1 | RTO | |
| 0..* | GeneratingBid | 0..1 | GeneratingBid | |
| 0..1 | Flowgate | 0..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.43 ServicePoint

Points terminaux définis d'un trajet de transmission. Les points de service sont définis du point de vue du service de transport. Chaque point de service est contenu dans (ou à la limite de) la zone d'échange. Un point de service est la source ou la destination d'une transaction.

Le Tableau 213 présente tous les attributs de ServicePoint.

**Tableau 213 – Attributs de ExternalInputs::ServicePoint**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 214 présente toutes les extrémités d'association de ServicePoint avec d'autres classes.

**Tableau 214 – Extrémités d'association de ExternalInputs:: ServicePoint avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | PORTransmissionPath | 0..* | TransmissionPath | Un trajet de transmission a un point de service appelé "point de réception" |
| 1..1 | PODTransmissionPath | 0..* | TransmissionPath | Un trajet de transmission a un point de service "point de livraison" |
| 0..1 | SourceReservation | 0..* | TransmissionReservation | |
| 0..1 | SinkReservation | 0..* | TransmissionReservation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.2.44 Classe racine ShuntCompensatorDynamicData

Optimal Power Flow (calcul de répartition optimal) ou State Estimator Filter Bank Data (données relatives au banc de filtres de l’estimateur d’état) pour le simulateur de formation des exploitants (OTS – Operator Training Simulator). Utilisée pour les utilisateurs RealTime, Study et Maintenance.

Le Tableau 215 présente tous les attributs de ShuntCompensatorDynamicData.

**Tableau 215 – Attributs de ExternalInputs::ShuntCompensatorDynamicData**

| nom | mult | type | description |
|---|---|---|---|
| mVARInjection | 0..1 | Float | Injection de puissance réactive du banc de filtres dans la solution NA ou production de puissance réactive VCS |
| connectionStatus | 0..1 | Integer | Statut courant du condensateur de régulation de tension 1 = Connecté 0 = Déconnecté |
| desiredVoltage | 0..1 | Float | Tension souhaitée du condensateur de régulation de tension |
| voltageRegulationStatus | 0..1 | Boolean | Indicateur spécifiant si la régulation de tension agit en tant que régulateur; True = Oui, False = Non |
| stepPosition | 0..1 | Integer | Position d'un pas de condensateur de régulation de tension |

Le Tableau 216 présente toutes les extrémités d'association de ShuntCompensatorDynamicData avec d'autres classes.

**Tableau 216 – Extrémités d’association de ExternalInputs:: ShuntCompensatorDynamicData avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktShuntCompensator | 1..1 | MktShuntCompensator | |

#### 6.5.6.2.45 Classe racine SwitchStatus

Optimal Power Flow (calcul de répartition optimal) ou State Estimator Circuit Breaker Status (statut du disjoncteur de l’estimateur d’état).

Le Tableau 217 présente tous les attributs de SwitchStatus.

**Tableau 217 – Attributs de ExternalInputs::SwitchStatus**

| nom | mult | type | description |
|---|---|---|---|
| switchStatus | 0..1 | SwitchStatusType | Statut du disjoncteur (fermé ou ouvert) du flux de puissance (*power flow*). |

Le Tableau 218 présente toutes les extrémités d'association de SwitchStatus avec d'autres classes.

**Tableau 218 – Extrémités d'association de ExternalInputs::SwitchStatus avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktSwitch | 1..1 | MktSwitch | |

#### 6.5.6.2.46 Classe racine SysLoadDistributionFactor

Cette classe modélise les facteurs de distribution du système. Elle nécessite d'être utilisée avec HostControlArea et ConnectivityNode pour indiquer la distribution de chaque partie individuelle.

Le Tableau 219 présente tous les attributs de SysLoadDistributionFactor.

**Tableau 219 – Attributs de ExternalInputs::SysLoadDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Utilisé pour calculer la "participation" de charge d'un nœud de connectivité dans une zone de régulation hôte |

Le Tableau 220 présente toutes les extrémités d'association de SysLoadDistributionFactor avec d'autres classes.

**Tableau 220 – Extrémités d'association de ExternalInputs:: SysLoadDistributionFactor avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | DistributionFactorSet | 0..* | DistributionFactorSet | |
| 0..* | HostControlArea | 1..1 | HostControlArea | |

#### 6.5.6.2.47 Classe racine TREntitlement

Un Transmission Right (TR) (droit de transport) peut être une chaîne de TR ou un TR individuel.

Lorsqu'un droit de transport n'est pas une chaîne, il est officiellement le droit ETC/TOR de chaque contrat ETC/TOR avec CVR (Converted Rights – droits convertis) comme ETC. Il s'agit de la somme de tous les droits sur toutes les interfaces de transport liées pour un même TR.

Lorsque TR est une chaîne, son droit est le minimum de tous les droits pour les TR individuels de la chaîne.

Le Tableau 221 présente tous les attributs de TREntitlement.

**Tableau 221 – Attributs de ExternalInputs::TREntitlement**

| nom | mult | type | description |
|---|---|---|---|
| entitlement | 0..1 | Float | Le droit |
| startOperatingDate | 0..1 | DateTime | Date et heure d'exploitation lorsque le droit s'applique |

Le Tableau 222 présente toutes les extrémités d’association de TREntitlement avec d’autres classes.

**Tableau 222 – Extrémités d’association de ExternalInputs:: TREntitlement avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TransmissionContractRight | 1..1 | ContractRight | |

### 6.5.6.2.48 Classe racine TapChangerDynamicData

Optimal Power Flow (calcul de répartition optimal) ou State Estimator Phase Shifter Data (données relatives au déphaseur de l’estimateur d’état). Utilisée pour les utilisateurs RealTime, Study et Maintenance. Mesures du déphaseur de solution SE du dernier processus de SE

Le Tableau 223 présente tous les attributs de TapChangerDynamicData.

**Tableau 223 – Attributs de ExternalInputs::TapChangerDynamicData**

| nom | mult | type | description |
|---|---|---|---|
| tapPosition | 0..1 | Float | Position de la prise du déphaseur, position supérieure de la prise du transformateur ou position inférieure de la prise du transformateur |
| desiredVoltage | 0..1 | Float | Tension souhaitée du LTC (changeur de prise en charge) |
| voltageRegulationStatus | 0..1 | Boolean | Indicateur spécifiant si le transformateur LTC agit en tant que régulateur; True = Oui, False = Non |
| angleRegulationStatus | 0..1 | Boolean | True signifie que le déphaseur régule. |
| desiredMW | 0..1 | Float | Quantité de MW souhaitée pour le déphaseur. Point de consigne de régulation de la puissance active du déphaseur |
| solvedAngle | 0..1 | Float | Angle du déphaseur. Déphasage angulaire résolu du déphaseur |
| minimumAngle | 0..1 | Float | Déphasage angulaire minimal du déphaseur |
| maximumAngle | 0..1 | Float | Déphasage angulaire maximal du déphaseur |

Le Tableau 224 présente toutes les extrémités d'association de TapChangerDynamicData avec d'autres classes.

**Tableau 224 – Extrémités d'association de ExternalInputs:: TapChangerDynamicData avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktTapChanger | 1..1 | MktTapChanger | |

### 6.5.6.2.49 TerminalConstraintTerm

Une condition de contrainte associée à une borne spécifique sur un équipement physique.

Le Tableau 225 présente tous les attributs de TerminalConstraintTerm.

**Tableau 225 – Attributs de ExternalInputs::TerminalConstraintTerm**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | String | Hérité de: ConstraintTerm |
| function | 0..1 | String | Hérité de: ConstraintTerm |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 226 présente toutes les extrémités d'association de TerminalConstraintTerm avec d'autres classes.

**Tableau 226 – Extrémités d'association de ExternalInputs:: TerminalConstraintTerm avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktTerminal | 1..1 | MktTerminal | |
| 0..* | SecurityConstraintSum | 1..1 | SecurityConstraintSum | Hérité de: ConstraintTerm |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.50 TransferInterface

Une interface de transfert est composée de branches telles que les lignes de transmission et les transformateurs.

Le Tableau 227 présente tous les attributs de TransferInterface.

**Tableau 227 – Attributs de ExternalInputs::TransferInterface**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 228 présente toutes les extrémités d'association de TransferInterface avec d'autres classes.

**Tableau 228 – Extrémités d'association de ExternalInputs:: TransferInterface avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TransferInterfaceSolution | 1..1 | TransferInterfaceSolution | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.51 Classe racine TransferInterfaceSolution

Définitions d'interface TNA de OPF pour VSA

Le Tableau 229 présente tous les attributs de TransferInterfaceSolution.

**Tableau 229 – Attributs de ExternalInputs::TransferInterfaceSolution**

| nom | mult | type | description |
|---|---|---|---|
| interfaceMargin | 0..1 | Float | Marge de l'interface |
| transferLimit | 0..1 | Float | Interface de transfert + Limite<br>Utilisation de l'attribut: Valeur absolue du flux maximum sur l'interface de transfert. Il s'agit d'une valeur positive en MW. |
| postTransferMW | 0..1 | Float | MW post-transfert pour une étape |

Le Tableau 230 présente toutes les extrémités d'association de TransferInterfaceSolution avec d'autres classes.

**Tableau 230 – Extrémités d'association de ExternalInputs:: TransferInterfaceSolution avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TransferInterface | 1..1 | TransferInterface | |
| 0..1 | MktContingencyA | 0..1 | MktContingency | |
| 0..1 | MktContingencyB | 0..1 | MktContingency | |

#### 6.5.6.2.52 Classe racine TransmissionCapacity

Cette classe modélise la capacité de transport (une interface de transport ou une paire POR/POD) comprenant la capacité de transfert totale (TTC – Total Transfer Capacity), la capacité de transfert opérationnelle (OTC – Operating Transfer Capacity) et la marge bénéficiaire de capacité (CBM – Capacity Benefit Margin).

Le Tableau 231 présente tous les attributs de TransmissionCapacity.

**Tableau 231 – Attributs de ExternalInputs::TransmissionCapacity**

| nom | mult | type | description |
|---|---|---|---|
| capacityBenefitMargin | 0..1 | Float | La marge bénéficiaire de capacité (CBM) est utilisée par les marchés (*Markets*) pour calculer les limites de l'interface de transport. Ce nombre peut être déterminé manuellement ou selon une procédure. La CBM est définie par interface de transport (groupe de branches). |
| operationalTransmissionCapacity | 0..1 | Float | La capacité de transport opérationnelle (OTC) est la capacité de transport dans les conditions opérationnelles pendant une période de temps déterminée, comprenant les effets de réduction et les paramètres actuels de contrôle des opérations. Les OTC des interfaces de transport (groupe de branches) sont toujours fournies quelles que soient les conditions d'indisponibilité ou de commutation. |
| OTC15min_emergency | 0..1 | Float | Le seuil d'urgence de 15 min de la capacité de transport opérationnelle (OTC) |
| OTCemergency | 0..1 | Float | Le seuil d'urgence de la capacité de transport opérationnelle (OTC). |
| POD | 0..1 | String | Point de livraison |
| POR | 0..1 | String | Point de réception |
| startOperatingDate | 0..1 | DateTime | Date et heure d'exploitation lorsque le droit s'applique |
| totalTransmissionCapacity | 0..1 | Float | Capacité de transport totale |

Le Tableau 232 présente toutes les extrémités d'association de TransmissionCapacity avec d'autres classes.

**Tableau 232 – Extrémités d’association de ExternalInputs:: TransmissionCapacity avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | GenericConstraints | 0..1 | GenericConstraints | |
| 0..* | Flowgate | 0..1 | Flowgate | |

#### 6.5.6.2.53 Classe racine TransmissionInterfaceRightEntitlement

Elle est officiellement appelée le droit ETC/TOR du groupe de branches avec CVR en tant que ETC. Elle peut également être utilisée pour représenter le droit TR d'un POR/POD.

Le Tableau 233 présente tous les attributs de TransmissionInterfaceRightEntitlement.

**Tableau 233 – Attributs de ExternalInputs::TransmissionInterfaceRightEntitlement**

| nom | mult | type | description |
|---|---|---|---|
| entitlement | 0..1 | Float | Droit |
| POD | 0..1 | String | Point de livraison |
| POR | 0..1 | String | Point de réception |
| startOperatingDate | 0..1 | DateTime | Date et heure d'exploitation lorsque le droit s'applique |

Le Tableau 234 présente toutes les extrémités d'association de TransmissionInterfaceRightEntitlement avec d'autres classes.

**Tableau 234 – Extrémités d’association de ExternalInputs:: TransmissionInterfaceRightEntitlement avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ContractRight | 1..1 | ContractRight | |
| 0..* | Flowgate | 0..1 | Flowgate | |

#### 6.5.6.2.54 TransmissionPath

Une connexion, un lien ou une ligne électrique composé(e) d'un ou de plusieurs éléments de transport parallèles entre deux zones des systèmes électriques interconnectés ou d'une partie d'entre eux. TransmissionCorridor et TransmissionRightOfWay font partie des aspects juridiques. TransmissionPath fait référence aux segments entre les ServicePoints d'un TransmissionProvider.

Le Tableau 235 présente tous les attributs de TransmissionPath.

**Tableau 235 – Attributs de ExternalInputs::TransmissionPath**

| nom | mult | type | description |
|---|---|---|---|
| availTransferCapability | 0..1 | ActivePower | Capacité de transport disponible d'un trajet de transmission pour la direction de référence. |
| parallelPathFlag | 0..1 | Boolean | Indicateur précisant si le trajet de transmission est également une "voie parallèle" d'interconnexion désignée. |
| totalTransferCapability | 0..1 | ActivePower | La capacité de transport totale d'un trajet de transmission dans la direction de référence. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 236 présente toutes les extrémités d'association de TransmissionPath avec d'autres classes.

**Tableau 236 – Extrémités d'association de ExternalInputs:: TransmissionPath avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TransmissionReservation | 0..* | TransmissionReservation | |
| 0..* | PointOfReceipt | 1..1 | ServicePoint | Un trajet de transmission a un point de service appelé "point de réception" |
| 0..* | DeliveryPoint | 1..1 | ServicePoint | Un trajet de transmission a un point de service "point de livraison" |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.2.55 Classe racine TransmissionReservation

Une réservation de transport est obtenue du système OASIS pour réserver le transport pour une période de temps, un trajet de transmission et un produit de transport donnés.

Le Tableau 237 présente toutes les extrémités d'association de TransmissionReservation avec d'autres classes.

**Tableau 237 – Extrémités d'association de ExternalInputs:: TransmissionReservation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | EnergyTransaction | 0..1 | EnergyTransaction | |
| 0..1 | TransactionBid | 0..1 | TransactionBid | |
| 0..* | Source | 0..1 | ServicePoint | |
| 0..* | Sink | 0..1 | ServicePoint | |
| 0..* | TransmissionPath | 1..1 | TransmissionPath | |

### 6.5.6.2.56 UnitInitialConditions

Statut de ressource à la fin d'une période d'équilibre donnée.

Le Tableau 238 présente tous les attributs de UnitInitialConditions.

**Tableau 238 – Attributs de ExternalInputs::UnitInitialConditions**

| nom | mult | type | description |
|---|---|---|---|
| cumEnergy | 0..1 | RealEnergy | Production d'énergie cumulée sur une période d'échange. |
| cumStatusChanges | 0..1 | Integer | Nombre cumulé de changements de statut de la ressource. |
| numberOfStartups | 0..1 | Integer | Nombre de démarrages par jour d'exploitation (Operating Day) jusqu'à la fin de l'heure précédente. |
| onlineStatus | 0..1 | Boolean | "True" si le GeneratingUnit est actuellement en ligne. |
| resourceMW | 0..1 | ActivePower | Sortie en MW de ressource à la fin de la période d'équilibre précédente. |
| resourceStatus | 0..1 | Integer | Statut de ressource à la fin de la période d'équilibre précédente:<br>0 – hors ligne<br>1 – production en ligne<br>2 – en cours d'arrêt<br>3 – en cours de démarrage |
| statusDate | 0..1 | DateTime | Heure et date de resourceStatus |
| timeInStatus | 0..1 | Float | Heure dans les intervalles d'échange de marché à laquelle la ressource revêt un statut à la fin de la période d'équilibre précédente. |
| timeInterval | 0..1 | DateTime | Intervalle de temps |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 239 présente toutes les extrémités d'association de UnitInitialConditions avec d'autres classes.

**Tableau 239 – Extrémités d'association de ExternalInputs:: UnitInitialConditions avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | GeneratingUnit | 0..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3 Paquetage MarketResults

#### 6.5.6.3.1 Généralités

Résultats de l'exécution d'un marché.

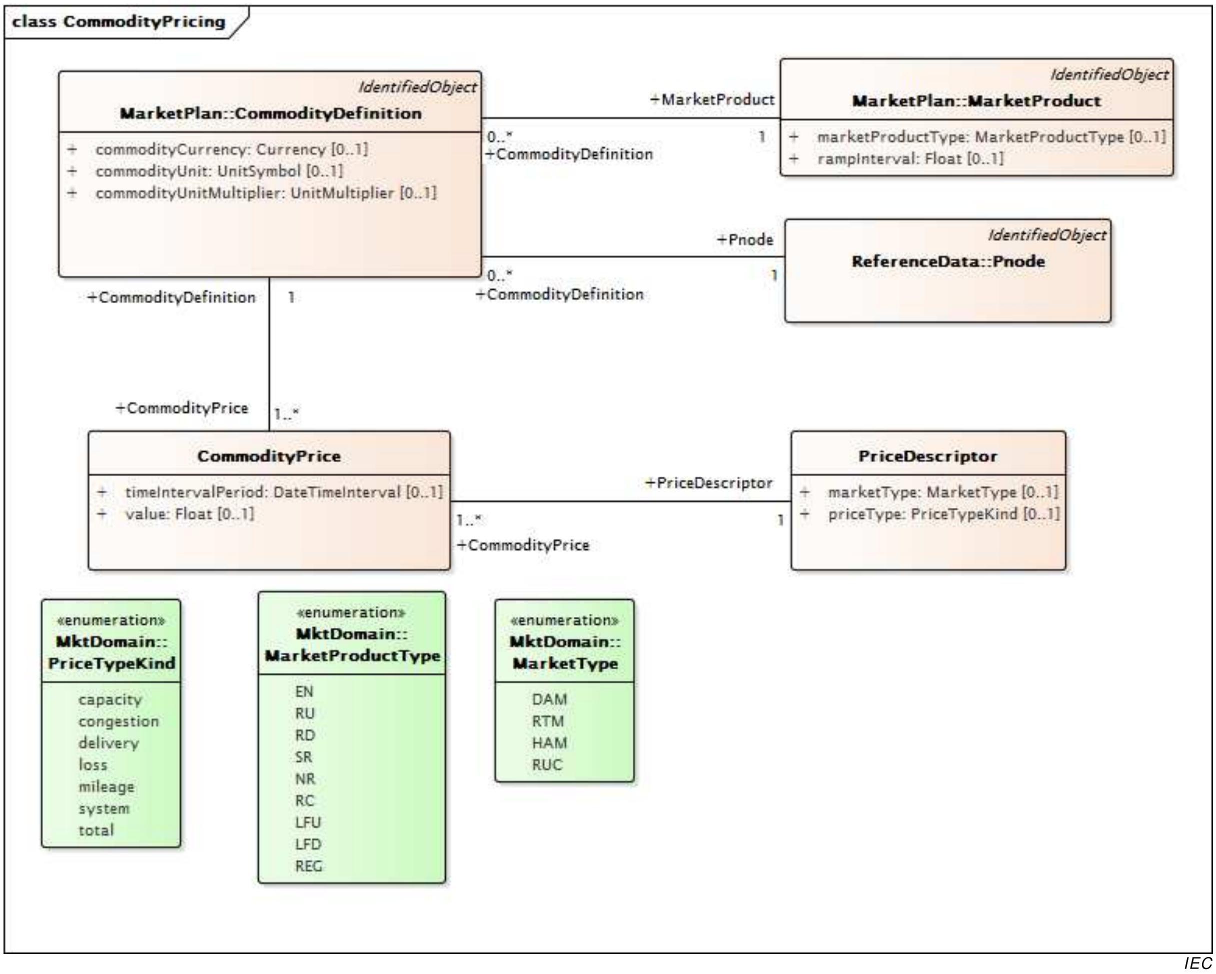



**Figure 44 – Diagramme de classe MarketResults::CommodityPricing**

Figure 44: représente la manière dont les CommodityDefinitions sont définies pour un produit du marché à un nœud de tarification. Les CommodityPrices pour des intervalles de temps spécifiques peuvent être définis pour une CommodityDefinition. Chaque CommodityPrice a un PriceDescriptor qui spécifie le type de marché et le type de tarification.

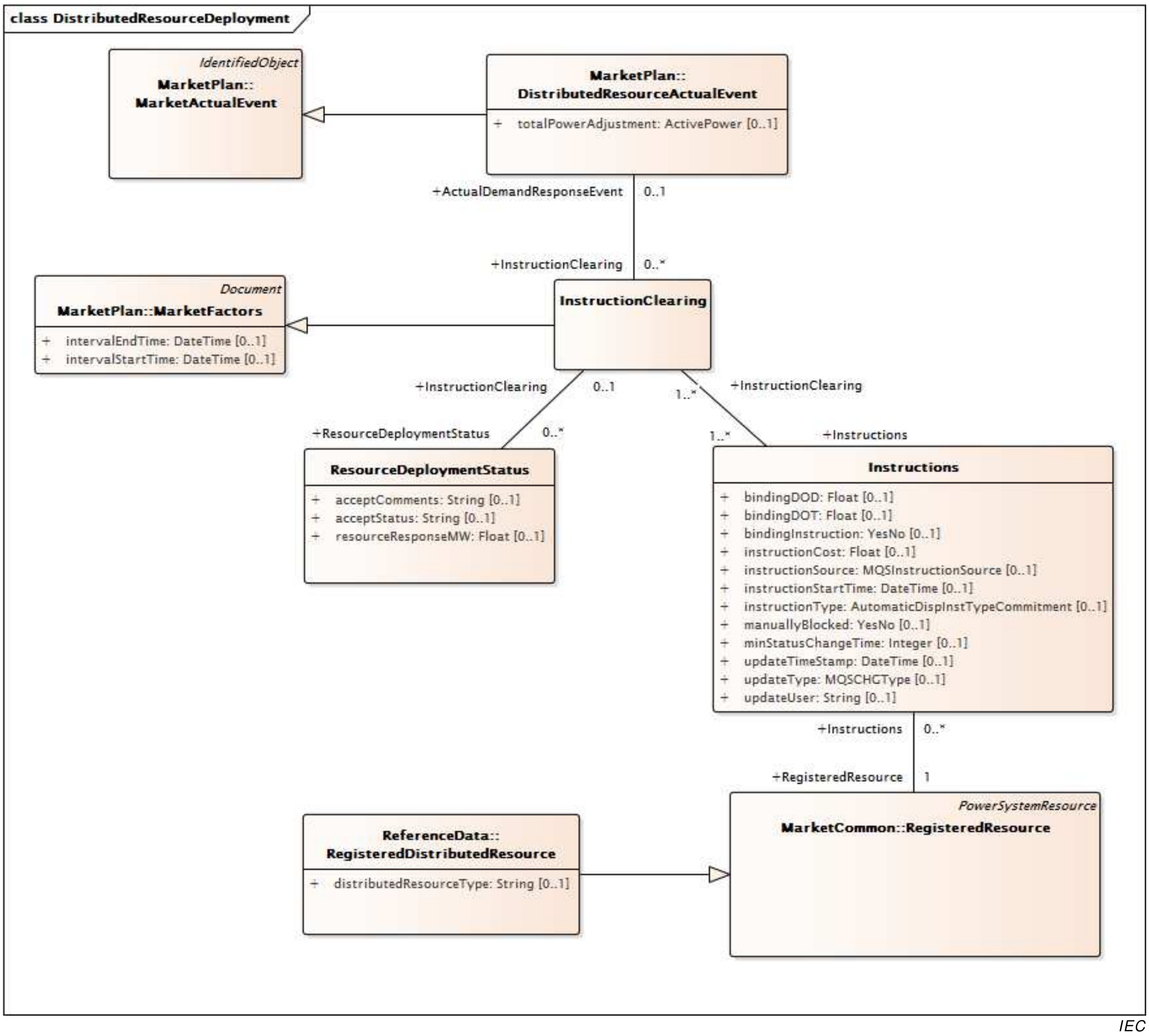



**Figure 45 – Diagramme de classe MarketResults::DistributedResourceDeployment**

Figure 45: représente les classes impliquées dans le déploiement d'une ressource répartie enregistrée. Les déploiements pilotés par des événements sont définis par la classe DistributedResourceActualEvent qui hérite de la classe MarketActualEvent. Des instructions comprenant la cible d'exploitation de répartition par obligation ou le delta de répartition d'exploitation par obligation avec l'heure de démarrage et le type d'instruction sont envoyées aux RegisteredDistributedResources. Les déploiements qui sont spécifiques aux ressources, pilotés par des événements ou en vrac sont tous pris en charge par ces classes.

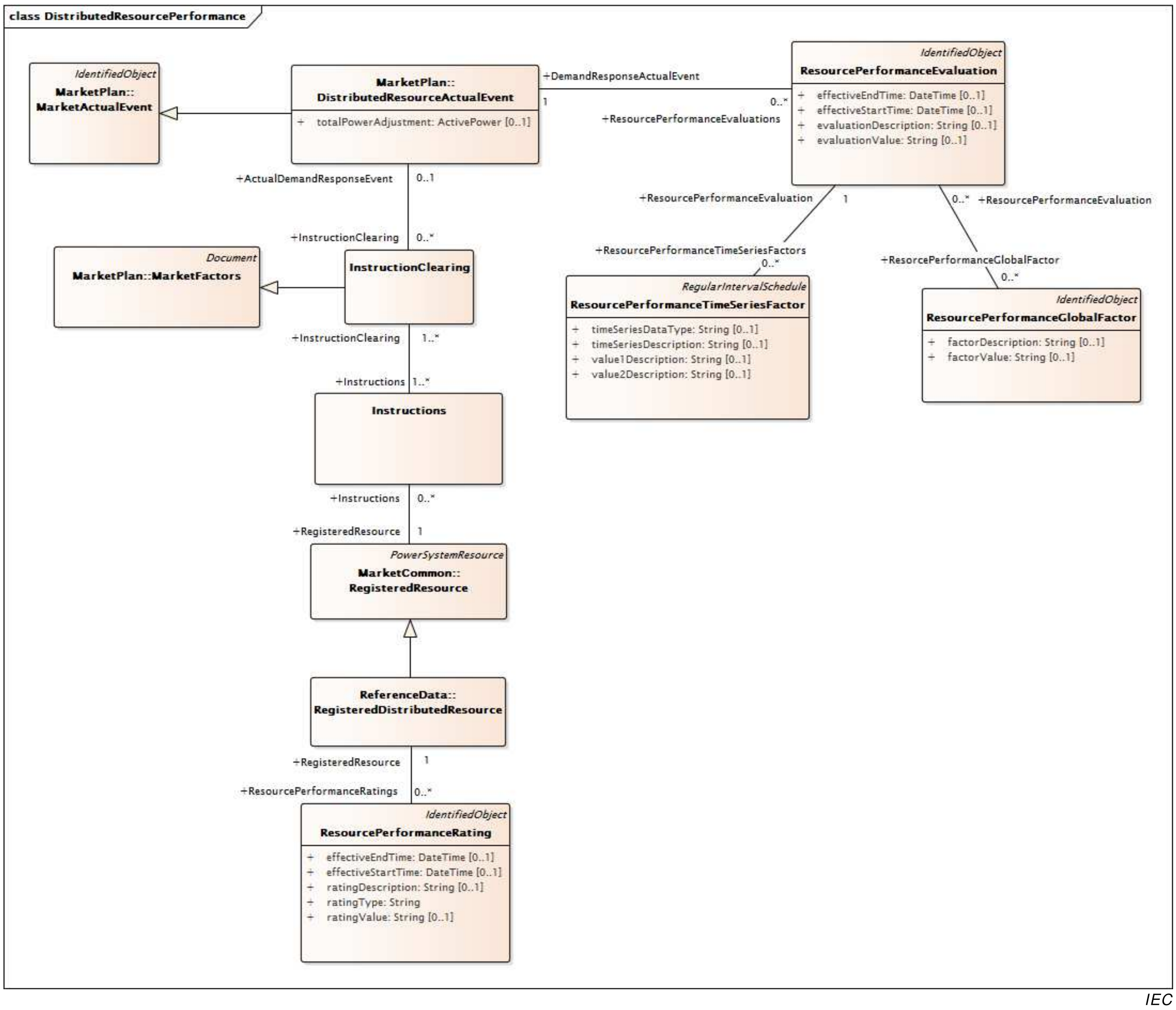



**Figure 46 – Diagramme de classe MarketResults::DistributedResourcePerformance**

Figure 46: représente la prise en charge de l'évaluation des performances pour les RegisteredDistributedResources. Après reception de l’évaluation de la qualité des revenus, une instance ResourcePerformanceEvaluation peut être créée pour chaque ressource par événement DR.

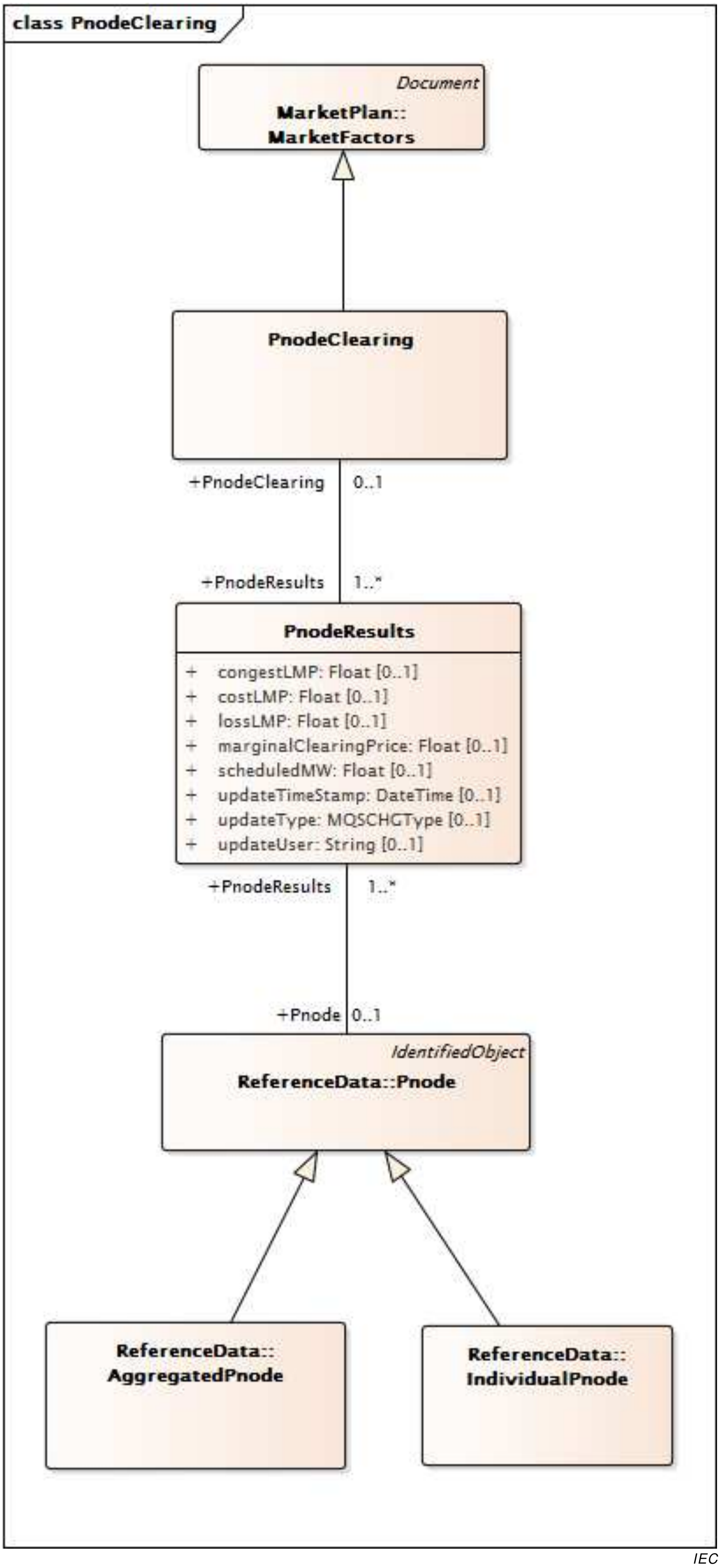



**Figure 47 – Diagramme de classe MarketResults::PnodeClearing**

Figure 47: représente les classes et les relations utilisées pour modéliser les résultats du marché nodal pour chaque Pnode par intervalle de temps. Les intervalles sont définis par PnodeClearing qui hérite de MarketFactors. Les PnodeResults comprennent les résultats pour chaque Pnode.

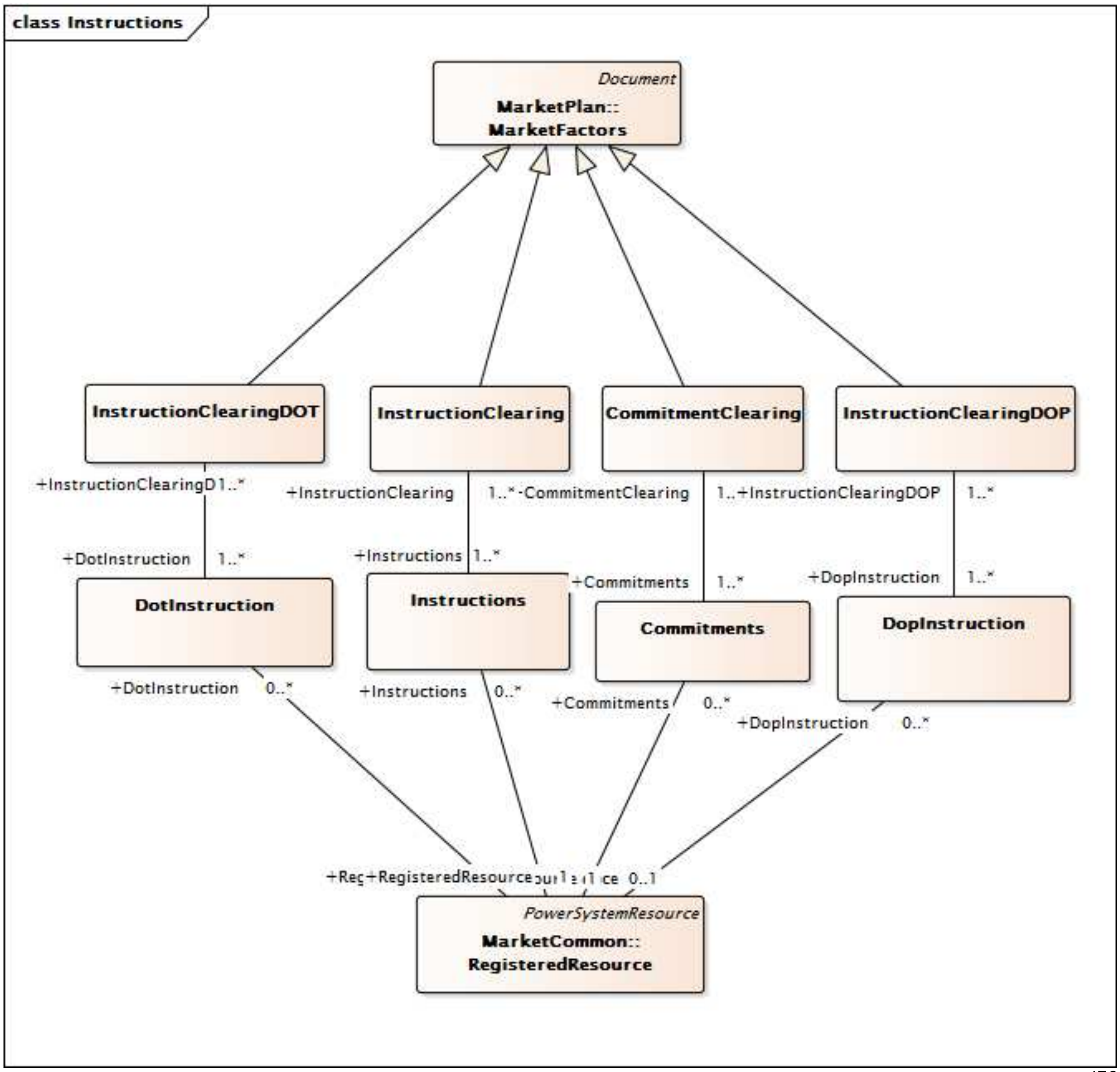



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**Figure 48 – Diagramme de classe MarketResults::Instructions**

Figure 48: représente les classes utilisées pour modéliser les instructions envoyées aux participants du marché après la mise en équilibre du marché. Les principales classes et leurs relations incluent la classe RegisteredResource qui modélise la ressource participante, ainsi que les classes Commitment et CommitmentClearing qui modélisent les résultats qui demandent l'engagement de la ressource. DotInstruction et DopInstruction autorisent la modélisation des résultats d'une répartition dynamique (ou anticipée) qui peut être utilisée dans le marché en temps réel.

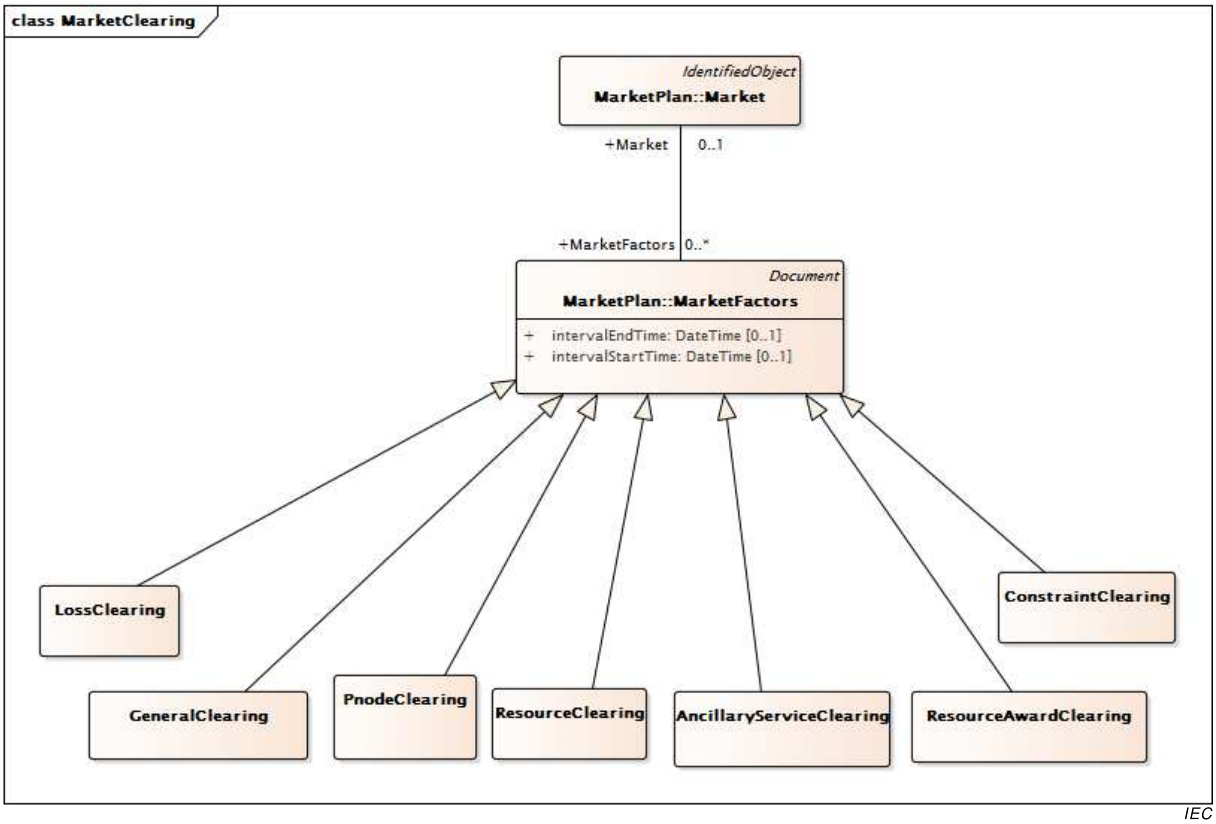


**Figure 49 – Diagramme de classe MarketResults::MarketClearing**

Figure 49: Chaque classe d'équilibre du marché définie hérite de la classe MarketFactors pour définir les intervalles de temps et le type d'exécution du marché.

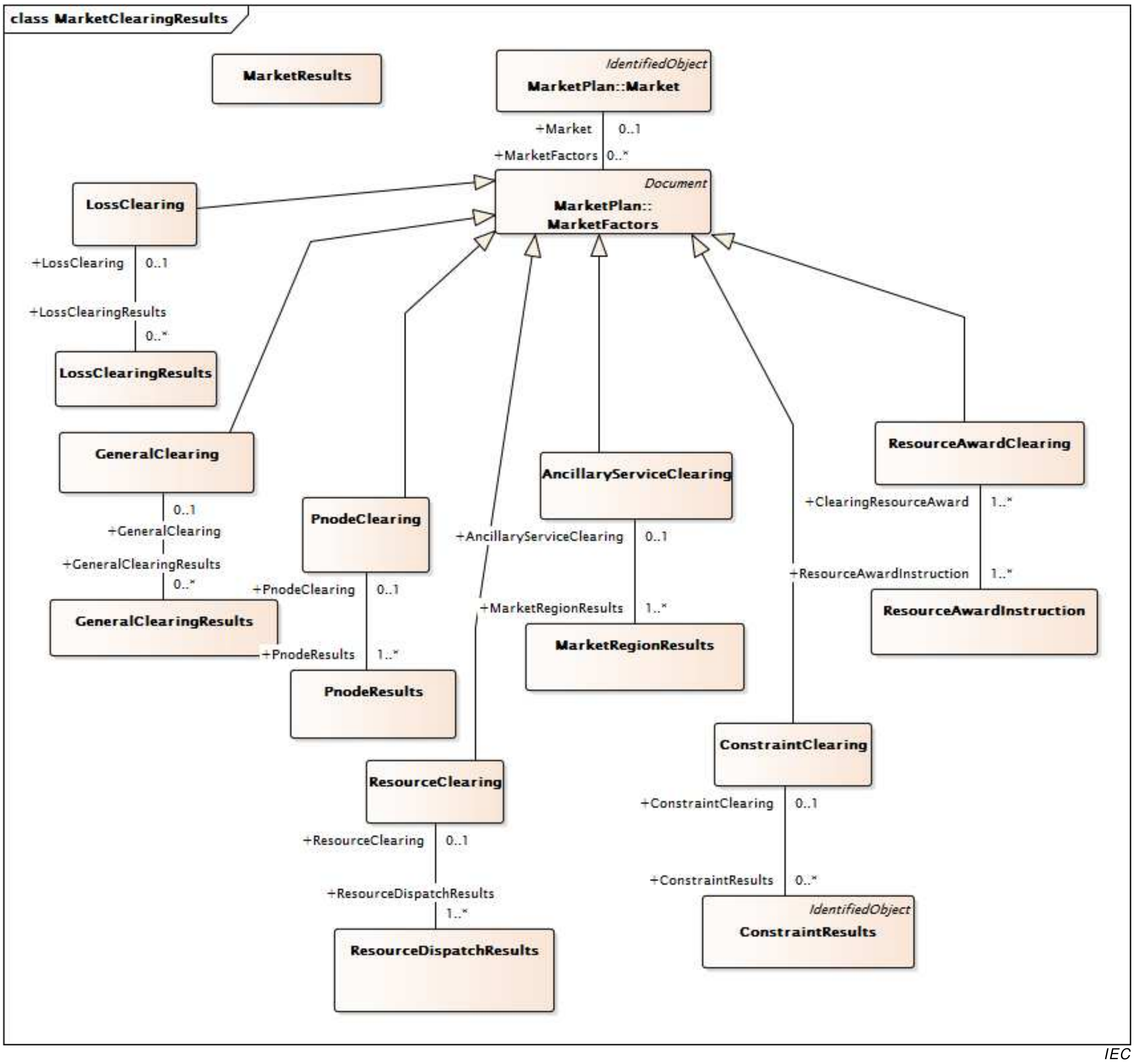


**Figure 50 – Diagramme de classe MarketResults::MarketClearingResults**

Figure 50: représente les classes et les relations utilisées pour modéliser les résultats de l'équilibre du marché de manière générale. La classe MarketResults résume le processus de marché. La classe LossClearingResults récapitule les pertes. GeneralClearingResults est utilisé pour modéliser la charge d'équilibre sur une base de zone de charge. Les classes PnodeClearing et PnodeClearingResults sont utilisées pour modéliser les prix marginaux locaux (LMP) au niveau des nœuds de tarification individuels. Les classes ResourceClearing et ResourceDispatchInstruction comprennent les instructions de répartition et l'indicateur d'une répartition dans un cas normal ou dans un cas de contingence. La classe AncillaryServiceClearing comprend des informations sur les services auxiliaires. La classe ConstraintResults comprend des contraintes de liaison et leurs coûts fictifs associés. ResourceAwardInstruction inclut des attributions de puissance active et des informations LMP.

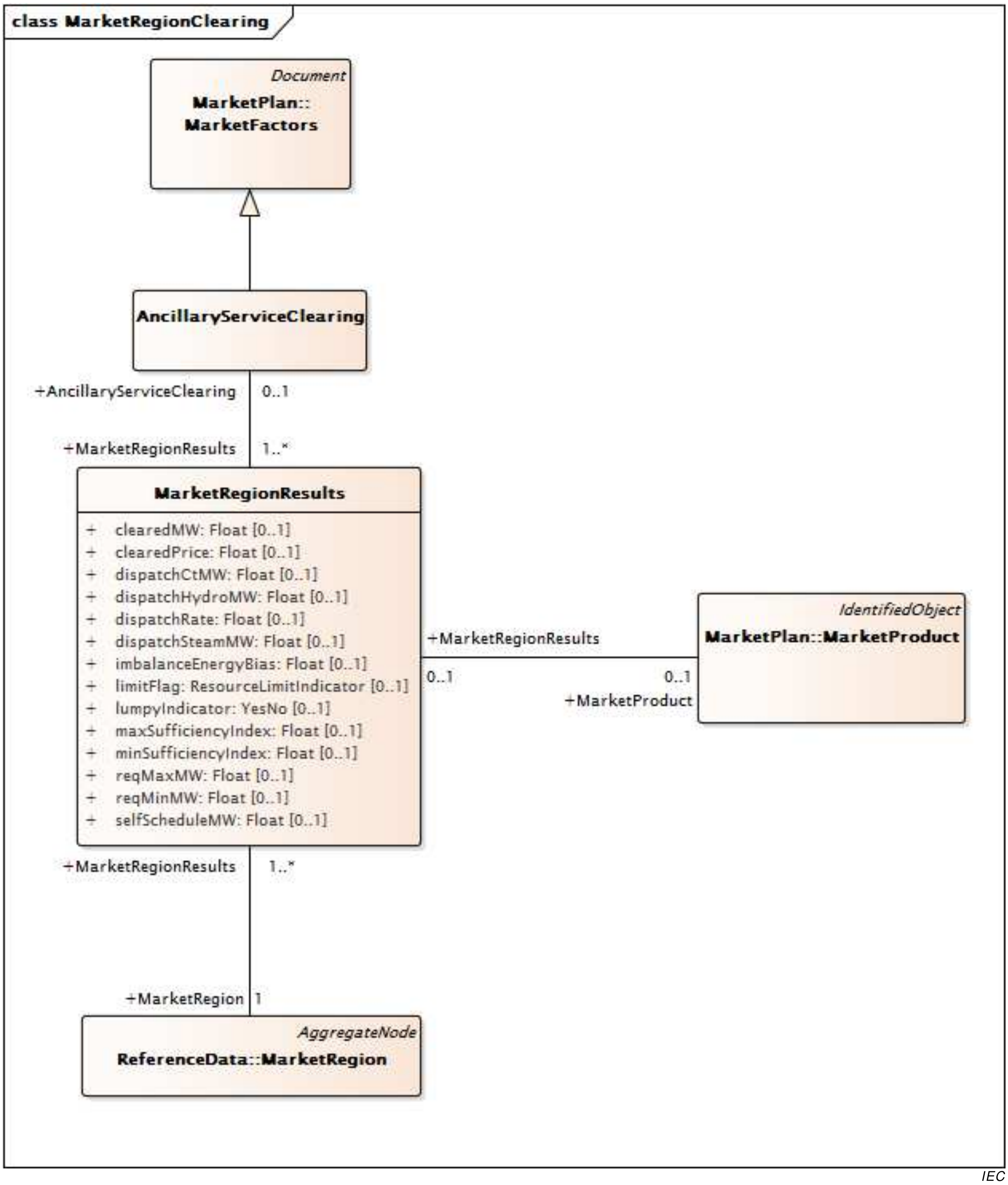


**Figure 51 – Diagramme de classe MarketResults::MarketRegionClearing**

Figure 51: représente les classes et les relations utilisées pour modéliser les résultats de l'équilibre du marché des services auxiliaires au niveau d'une région du marché. Le modèle inclut les résultats au niveau des produits de services auxiliaires spécifiques grâce à la classe MarketProduct et à l'association à MarketRegionResults.

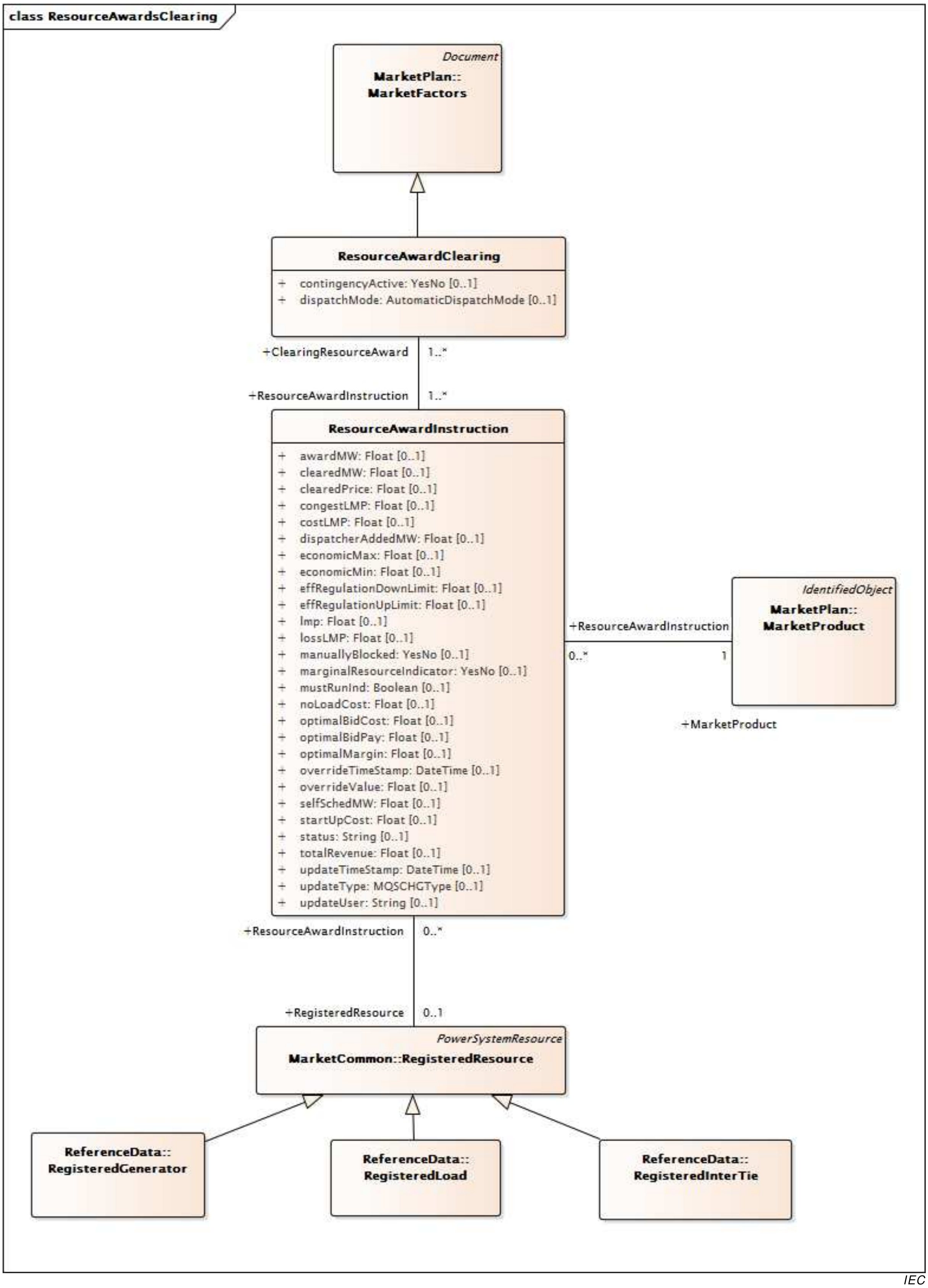


**Figure 52 – Diagramme de classe MarketResults::ResourceAwardsClearing**

Figure 52: représente les classes et les relations utilisées pour modéliser les résultats de l'équilibre du marché en mettant l'accent sur ResourceAwardInstruction. La classe principale de ce diagramme est ResourceAwardInstruction, qui modélise les détails de l'attribution à la ressource comprenant la puissance active, le prix marginal local (LMP) et ses composants

d'énergie, de congestion et de coût. Le modèle de ressource inclut les sous-classes RegisteredGenerator, RegisteredLoad et RegisteredInterTie.

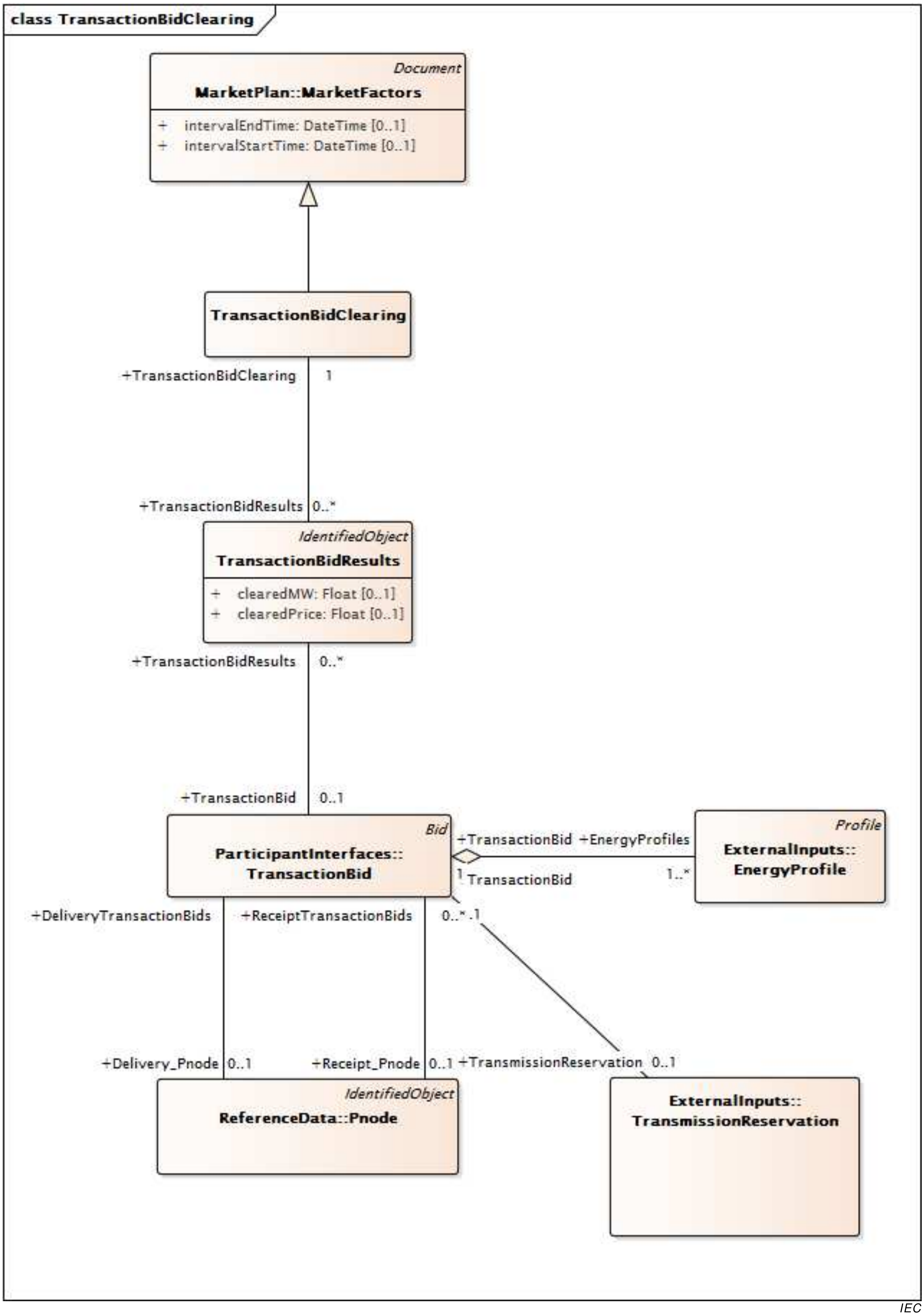


**Figure 53 – Diagramme de classe MarketResults::TransactionBidClearing**

Figure 53: représente les classes et les relations utilisées pour modéliser les résultats de l'équilibre du marché en mettant l'accent sur les résultats de transaction. Ce modèle inclut la classe TransactionBid et ses relations avec les Pnodes Point of Delivery (point de livraison) et Point of Receipt (point de réception).

#### 6.5.6.3.2 AncillaryServiceClearing

Modélise les résultats de l'équilibre du marché par rapport aux produits des services auxiliaires

Le Tableau 240 présente tous les attributs de AncillaryServiceClearing.

**Tableau 240 – Attributs de MarketResults::AncillaryServiceClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 241 présente toutes les extrémités d'association de AncillaryServiceClearing avec d'autres classes.

**Tableau 241 – Extrémités d'association de MarketResults:: AncillaryServiceClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MarketRegionResults | 1..* | MarketRegionResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.3 BillDeterminant

Modélise différents frais pour prendre en charge la facturation et le règlement

Le Tableau 242 présente tous les attributs de BillDeterminant.

**Tableau 242 – Attributs de MarketResults::BillDeterminant**

| nom | mult | type | description |
|---|---|---|---|
| calculationLevel | 0..1 | String | Niveau de l'ordre de calcul des frais. |
| configVersion | 0..1 | String | Version de configuration de la logique de calcul pour le règlement. |
| deleteStatus | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| exception | 0..1 | String | |
| factor | 0..1 | String | |
| frequency | 0..1 | String | |
| numberInterval | 0..1 | Integer | Nombre d'intervalles de l'élément déterminant de facturation le jour de l'opération, par exemple: 300 pour des intervalles de cinq minutes. |
| offset | 0..1 | String | |
| precisionLevel | 0..1 | String | Niveau de précision de la valeur actuelle. |
| primaryYN | 0..1 | String | |
| referenceFlag | 0..1 | String | |
| reportable | 0..1 | String | |
| roundOff | 0..1 | String | |
| source | 0..1 | String | |
| terminationDate | 0..1 | DateTime | |
| unitOfMeasure | 0..1 | String | Unité de mesure de la valeur actuelle de l'élément déterminant de facturation. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 243 présente toutes les extrémités d'association de BillDeterminant avec d'autres classes.

**Tableau 243 – Extrémités d'association de MarketResults:: BillDeterminant avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..1 | ChargeProfileData | 0..* | ChargeProfileData | |
| 0..* | ChargeComponents | 0..* | ChargeComponent | Un BillDeterminant peut avoir 0-n ChargeComponent et un ChargeComponent peut être associé à 0-n BillDeterminant. |
| 0..1 | ChargeProfile | 0..1 | ChargeProfile | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.4 ChargeProfile

Type de profil pour les charges financières

Le Tableau 244 présente tous les attributs de ChargeProfile.

**Tableau 244 – Attributs de MarketResults::ChargeProfile**

| nom | mult | type | description |
|---|---|---|---|
| type | 0..1 | String | Type de profil. Peut être le montant, le prix ou la quantité. |
| frequency | 0..1 | String | Fréquence de calcul: quotidienne ou mensuelle. |
| numberInterval | 0..1 | Integer | Nombre d'intervalles des données de profil. |
| unitOfMeasure | 0..1 | String | Unité de mesure appliquée à l'attribut de valeur des données de profil. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 245 présente toutes les extrémités d'association de ChargeProfile avec d'autres classes.

**Tableau 245 – Extrémités d'association de MarketResults:: ChargeProfile avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | BillDeterminant | 0..1 | BillDeterminant | |
| 0..1 | ChargeProfileData | 0..* | ChargeProfileData | |
| 0..* | PassTroughBill | 0..1 | PassThroughBill | |
| 0..* | Bid | 0..1 | Bid | |
| 0..* | ProfileDatas | 0..* | ProfileData | Hérité de: Profile |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.5 Classe racine ChargeProfileData

Modèle de différents frais associés à un profil d'énergie pour prendre en charge la facturation et le règlement

Le Tableau 246 présente tous les attributs de ChargeProfileData.

**Tableau 246 – Attributs de MarketResults::ChargeProfileData**

| nom | mult | type | description |
|---|---|---|---|
| sequence | 0..1 | Integer | Numéro de séquence du profil. |
| timeStamp | 0..1 | DateTime | Date et heure d'un intervalle. |
| value | 0..1 | Float | Valeur d'un intervalle selon un type de profil donné (montant, prix ou quantité), en fonction de l'unité de mesure. |

Le Tableau 247 présente toutes les extrémités d'association de ChargeProfileData avec d'autres classes.

**Tableau 247 – extrémités d'association de MarketResults:: ChargeProfileData avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BillDeterminant | 0..1 | BillDeterminant | |
| 0..* | ChargeProfile | 0..1 | ChargeProfile | |

### 6.5.6.3.6 CommitmentClearing

Modélise les résultats de l'équilibre du marché qui demandent l'engagement des unités.

Le Tableau 248 présente tous les attributs de CommitmentClearing.

**Tableau 248 – Attributs de MarketResults::CommitmentClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de:: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 249 présente toutes les extrémités d'association de CommitmentClearing avec d'autres classes.

**Tableau 249 – Extrémités d'association de MarketResults:: CommitmentClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | Commitments | 1..* | Commitments | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.7 Classe racine Commitments

Apporte les informations nécessaires (sur la base de ressources) pour la collecte des résultats d'engagement de Démarrage/Arrêt. Ces informations sont utiles pour tous les marchés.

Le Tableau 250 présente tous les attributs de Commitments.

**Tableau 250 – Attributs de MarketResults::Commitments**

| nom | mult | type | description |
|---|---|---|---|
| commitmentType | 0..1 | CommitmentType | Type de statut UC (engagement volontaire, engagement ISO ou engagement SCUC) |
| instructionCost | 0..1 | Float | Coût total associé au changement du statut de la ressource. |
| instructionType | 0..1 | AutomaticDispInstTypeCommitment | Indique s'il s'agit d'un démarrage ou d'un arrêt. |
| intervalEndTime | 0..1 | DateTime | Heure de fin de la période d'engagement. Ce sera sur une limite d'intervalle. |
| intervalStartTime | 0..1 | DateTime | Heure de début de la période d'engagement. Ce sera sur une limite d'intervalle. |
| minStatusChangeTime | 0..1 | Integer | Temps de démarrage de la période d'engagement SCUC. Temps de démarrage calculé en fonction de StartUpTimeCurve, fourni avec la Bid.<br>Il s'agit d'une combinaison de l'offre du temps de démarrage et du temps d'indisponibilité de l'unité.<br>Unités en minutes |
| noLoadCost | 0..1 | Float | Coût à charge nulle de l'unité en cas de marchandise énergétique |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Le Tableau 251 présente toutes les extrémités d'association de Commitments avec d'autres classes.

**Tableau 251 – Extrémités d’association de MarketResults:: Commitments avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 1..* | CommitmentClearing | 1..* | CommitmentClearing | |

#### 6.5.6.3.8 Classe racine CommodityPrice

La classe CommodityPrice est utilisée afin de définir le prix d’une marchandise pendant un intervalle de temps donné. L’intervalle peut être long (par exemple, un an), ou très court (par exemple, 5 min). Il existe de nombreuses instances de la classe CommodityPrice pour chaque instance de la CommodityDefinition à laquelle elles sont associées. À noter qu’il peut y avoir plus d’un prix associé à un intervalle donné et que ces variances sont décrites par l’association (ou les associations) avec la classe PriceDescriptor.

Le Tableau 252 présente tous les attributs de CommodityPrice.

**Tableau 252 – Attributs de MarketResults::CommodityPrice**

| nom | mult | type | description |
|---|---|---|---|
| timeIntervalPeriod | 0..1 | DateTimeInterval | Intervalle de temps pendant lequel CommodityPrice est valide, conformément aux conventions normalisées associées à la classe DateTimeInterval. |
| value | 0..1 | Float | Prix de la Commodity (marchancise), exprimé en valeur à virgule flottante. La devise et l'unité de mesure sont définies dans la classe CommodityDefinition associée. |

Le Tableau 253 présente toutes les extrémités d'association de CommodityPrice avec d'autres classes.

**Tableau 253 – Extrémités d'association de MarketResults:: CommodityPrice avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | CommodityDefinition | 1..1 | CommodityDefinition | |
| 1..* | PriceDescriptor | 1..1 | PriceDescriptor | |
| 0..* | PnodeClearing | 0..1 | PnodeClearing | |

#### 6.5.6.3.9 ConstraintClearing

Regroupe tous les éléments associés aux contraintes avec obligation et aux violations de contraintes par intervalle et par marché.

Le Tableau 254 présente tous les attributs de ConstraintClearing.

**Tableau 254 – Attributs de MarketResults::ConstraintClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |

| nom | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 255 présente toutes les extrémités d'association de ConstraintClearing avec d'autres classes.

**Tableau 255 – Extrémités d'association de MarketResults:: ConstraintClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ConstraintResults | 0..* | ConstraintResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.10 ConstraintResults

Fournit les résultats du marché sur le traitement des contraintes pour le DAM ou le RTM. Les données incluent le type de contrainte (avec obligation ou violée), la valeur résolue de la contrainte et le coût implicite associé.

Le Tableau 256 présente tous les attributs de ConstraintResults.

**Tableau 256 – Attributs de MarketResults::ConstraintResults**

| nom | mult | type | description |
|---|---|---|---|
| baseFlow | 0..1 | Float | Flux de puissance de la base de la branche. |
| BGLimit | 0..1 | Float | Cette valeur est déterminée dans le DA et le RTM. L'optimisation SCUC garantit que le flux de MW du groupe de branches ne dépassera pas cette limite dans la bonne direction. |
| BGTRResCap | 0..1 | Float | Capacité de réservation TR du groupe de branches – Cette valeur est déterminée dans le DA et le RTM. Cela correspond à la quantité de capacité de transport restante utilisable par le détenteur des TR. |
| bindingLimit | 0..1 | Float | Limite en MW. |
| clearedValue | 0..1 | Float | MW octroyés. |
| competitivePathConstraint | 0..1 | YesNo | Indicateur de contrainte de voie non concurrentielle (Y/N) précisant si le coût implicite d'une voie non concurrentielle est différent de zéro. |

| nom | mult | type | description |
|---|---|---|---|
| constraintType | 0..1 | ResultsConstraint Type | Type de contrainte. |
| limitFlag | 0..1 | ConstraintLimitTyp e | Indicateur de limite ("Maximum", "Minimum"). |
| optimizationFlag | 0..1 | YesNo | Inclus dans l'optimisation Y/N. |
| overloadMW | 0..1 | Float | MW de surcharge de transport. |
| percentMW | 0..1 | Float | Flux de MW réel en pourcentage de limite. |
| shadowPrice | 0..1 | Float | Coût implicite ($/MW) du produit. Coût implicite de la contrainte correspondante. |
| updateTimeStamp | 0..1 | DateTime | Horodatage de mise à jour. |
| updateType | 0..1 | MQSCHGType | Type de changement du MQS. |
| updateUser | 0..1 | String | Utilisateur mis à jour. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 257 présente toutes les extrémités d'association de ConstraintResults avec d'autres classes.

**Tableau 257 – Extrémités d’association de MarketResults:: ConstraintResults avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktContingency | 1..1 | MktContingency | |
| 0..* | ConstraintClearing | 0..1 | ConstraintClearing | |
| 1..* | Flowgate | 1..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObj ect | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObj ect | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.11 Classe racine DopInstruction

Apporte les informations nécessaires (sur la base de ressources) pour la collecte des résultats du point d'exploitation de répartition (DOP – Dispatch Operating Point) sur un intervalle de répartition. Ces informations ne sont utiles que pour le marché d'intervalle RT.

Le Tableau 258 présente tous les attributs de DopInstruction.

**Tableau 258 – Attributs de MarketResults::DopInstruction**

| nom | mult | type | description |
|---|---|---|---|
| mwDOP | 0..1 | ActivePower | Point d'exploitation réparti (MW) |
| timestampDOP | 0..1 | DateTime | Horodatage du DOP |
| plotPriority | 0..1 | Integer | Valeur utilisée pour établir la priorité du DOP lors du traçage. Ceci s'applique uniquement lorsque deux DOP existent pour une même heure, mais avec des valeurs en MW différentes. Par exemple, lors de l'indication d'une étape sur la courbe. Elle est utilisée pour déterminer si la courbe monte ou descend. |
| runIndicatorDOP | 0..1 | YesNo | Indication de la validité du DOP.<br>Indique que le DOP est calculé à partir du dernier processus (YES). L'indicateur NO indique que le DOP est copié depuis une exécution précédente.<br>2 intervalles maximum peuvent être manqués. |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Le Tableau 259 présente toutes les extrémités d'association de DopInstruction avec d'autres classes.

**Tableau 259 – Extrémités d’association de MarketResults:: DopInstruction avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResouce | 0..1 | RegisteredResource | |
| 1..* | InstructionClearingDOP | 1..* | InstructionClearingDOP | |

### 6.5.6.3.12 Classe racine DotInstruction

Apporte les informations nécessaires (sur la base de ressources) pour la collecte des résultats de la cible d'exploitation de répartition (DOT – Dispatch Operating Target) sur un intervalle de répartition. Ces informations ne sont utiles que pour le marché d'intervalle RT.

Le Tableau 260 présente tous les attributs de DotInstruction.

**Tableau 260 – Attributs de MarketResults::DotInstruction**

| nom | mult | type | description |
|---|---|---|---|
| actualRampRate | 0..1 | Float | Taux de rampe réel. |
| compliantIndicator | 0..1 | YesNo | Indicateur précisant si la ressource est conforme à l'instruction (plus/moins 10 %).<br>Indique s’il est admis que le prix soit fixé par une unité (tarification ex-post). |
| DOT | 0..1 | Float | Valeur de la cible d'exploitation de répartition. |
| economicMaxOverride | 0..1 | Float | Remplacement de la limite maximale économique d'une unité: cette valeur est nulle. Sinon, cette valeur remplace la valeur de la colonne Énergie (*Energy*).<br>Autorise le répartiteur à remplacer la valeur énergétique de l'unité. |
| expectedEnergy | 0..1 | Float | Énergie prévue. |
| generatorPerformanceDegree | 0..1 | Float | Degré de performances du générateur (DGP – *Degree of Generator Performance*) utilisé pour l'unité. Mesure de la façon dont un générateur répond pour augmenter/réduire les signaux. Calculé toutes les cinq minutes. |
| hourAheadSchedEnergy | 0..1 | Float | Résultats du HASP. |
| hourlySchedule | 0..1 | Float | Programme horaire (programme énergétique du DA). |
| instructionTime | 0..1 | DateTime | Date/Heure pour l'instruction. |
| maximumEmergencyInd | 0..1 | Boolean | True si le seuil d'urgence maximal est activé; sinon False. S'il est demandé à une unité d'atteindre son seuil d'urgence maximal, cet indicateur est défini sur True. |
| meterLoadFollowing | 0..1 | Float | Suivi de charge du sous-système du compteur. |
| nonRampRestrictedMW | 0..1 | Float | Quantité de MW souhaitée qui n'est pas restreinte par une rampe. Si aucune limite de taux de rampe n'existe pour l'unité, il s'agit de la valeur en MW sur laquelle l'unité a dû se positionner. |
| nonSpinReserve | 0..1 | Float | Réserve non tournante utilisée pour fournir de l'énergie. |
| previousDOTTimeStamp | 0..1 | DateTime | Horodatage de l’émission de la valeur DOT précédente. |
| rampRateLimit | 0..1 | Float | Limite du taux de rampe pour l'unité en MW par minute. Données d'offre du participant. |
| regulationStatus | 0..1 | YesNo | Statut de régulation (Yes/No). |
| spinReserve | 0..1 | Float | Réserve tournante utilisée pour fournir de l'énergie. |
| standardRampEnergy | 0..1 | Float | Énergie de rampe normalisée (MWh). |
| supplementalEnergy | 0..1 | Float | Énergie supplémentaire fournie par la répartition en temps réel. |
| unitStatus | 0..1 | Integer | Résultats de sortie du cas identifiant la raison pour laquelle l'unité a été engagée par le logiciel. |

Le Tableau 261 présente toutes les extrémités d'association de DotInstruction avec d'autres classes.

**Tableau 261 – Extrémités d'association de MarketResults::DotInstruction avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | InstructionClearingDOT | 1..* | InstructionClearingDOT | |

### 6.5.6.3.13 ExPostLoss

Modèle de calcul ex-post des pertes de MW.

Le Tableau 262 présente tous les attributs de ExPostLoss.

**Tableau 262 – Attributs de MarketResults::ExPostLoss**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 263 présente toutes les extrémités d'association de ExPostLoss avec d'autres classes.

**Tableau 263 – Extrémités d'association de MarketResults:: ExPostLoss avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ExPostLossResults | 0..* | ExPostLossResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.14 Classe racine ExPostLossResults

Modélise les résultats de calcul ex-post des pertes de MW. Résume les pertes en deux catégories: pertes liées au transport très haute tension et pertes totales. Calcul effectué pour chaque zone de sous-contrôle.

Le Tableau 264 présente tous les attributs de ExPostLossResults.

**Tableau 264 – Attributs de MarketResults::ExPostLossResults**

| nom | mult | type | description |
|---|---|---|---|
| totalLossMW | 0..1 | Float | Pertes totales de MW dans l'entreprise<br>Utilisation de l'attribut: À titre informatif – Sortie du moteur LPA. |
| ehvLossMW | 0..1 | Float | Pertes de MW THT dans l'entreprise<br>Utilisation de l'attribut: À titre informatif – Sortie du moteur LPA. |

Le Tableau 265 présente toutes les extrémités d'association de ExPostLossResults avec d'autres classes.

**Tableau 265 – Extrémités d'association de MarketResults:: ExPostLossResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ExPostLoss | 1..1 | ExPostLoss | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |

#### 6.5.6.3.15 ExPostMarketRegion

Modèle de calcul ex-post des MW octroyés sur une base régionale

Le Tableau 266 présente tous les attributs de ExPostMarketRegion.

**Tableau 266 – Attributs de MarketResults::ExPostMarketRegion**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 267 présente toutes les extrémités d'association de ExPostMarketRegion avec d'autres classes.

**Tableau 267 – Extrémités d'association de MarketResults:: ExPostMarketRegion avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ExPostMarketRegionResults | 0..1 | ExPostMarketRegionResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.16 Classe racine ExPostMarketRegionResults

Modèle de calcul ex-post des MW octroyés sur une base régionale. Comprend le prix d'équilibre

Le Tableau 268 présente tous les attributs de ExPostMarketRegionResults.

**Tableau 268 – Attributs de MarketResults::ExPostMarketRegionResults**

| nom | mult | type | description |
|---|---|---|---|
| exPostClearedPrice | 0..1 | Float | |

Le Tableau 269 présente toutes les extrémités d'association de ExPostMarketRegionResults avec d'autres classes.

**Tableau 269 – Extrémités d'association de MarketResults:: ExPostMarketRegionResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ExPostMarketRegion | 1..1 | ExPostMarketRegion | |
| 0..* | MarketRegion | 1..1 | MarketRegion | |

#### 6.5.6.3.17 ExPostPricing

Modèle de tarification ex-post des nœuds

Le Tableau 270 présente tous les attributs de ExPostPricing.

**Tableau 270 – Attributs de MarketResults::ExPostPricing**

| nom | mult | type | description |
|---|---|---|---|
| energyPrice | 0..1 | Float | Prix de l'énergie sur le marché |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 271 présente toutes les extrémités d'association de ExPostPricing avec d'autres classes.

**Tableau 271 – Extrémités d'association de MarketResults:: ExPostPricing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.18 Classe racine ExPostPricingResults

Modèle de tarification ex-post des nœuds. Comprend les informations LMP, fondées sur Pnode.

Le Tableau 272 présente tous les attributs de ExPostPricingResults.

**Tableau 272 – Attributs de MarketResults::ExPostPricingResults**

| nom | mult | type | description |
|---|---|---|---|
| congestLMP | 0..1 | Float | Composant de congestion d'un prix marginal local (LMP) en unités monétaires par MW; composant de congestion du LMP horaire à un nœud de tarification spécifique<br>Utilisation de l'attribut: Résultat du logiciel de sécurité, tarification et répartition (SPD)/essai de faisabilité simultané (SFT) et correspond au composant de congestion horaire du LMP pour chaque nœud de tarification. |
| lmp | 0..1 | Float | LMP moyen pondéré de 5 min; prix marginal local du Pnode pour lequel le calcul de tarification est en cours d'exécution.<br>Utilisation de l'attribut: LMP moyen pondéré de 5 min à afficher sur l'interface utilisateur |
| lossLMP | 0..1 | Float | Composant de perte d'un prix marginal local (LMP) en unités monétaires par MW; composant de perte du LMP horaire à un nœud de tarification spécifique<br>Utilisation de l'attribut: Résultat du logiciel de sécurité, tarification et répartition (SPD – Security, Pricing, and Dispatch)/essai de faisabilité simultané (SFT – Simultaneous Feasibility Test) et correspond au composant de perte horaire du LMP pour chaque nœud de tarification. |

Le Tableau 273 présente toutes les extrémités d'association de ExPostPricingResults avec d'autres classes.

**Tableau 273 – Extrémités d’association de MarketResults:: ExPostPricingResults avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ExPostPricing | 1..1 | ExPostPricing | |
| 0..* | Pnode | 1..1 | Pnode | |

### 6.5.6.3.19 ExPostResource

Modèle de tarification ex-post des ressources.

Le Tableau 274 présente tous les attributs de ExPostResource.

**Tableau 274 – Attributs de MarketResults::ExPostResource**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 275 présente toutes les extrémités d'association de ExPostResource avec d'autres classes.

**Tableau 275 – Extrémités d’association de MarketResults:: ExPostResource avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ExPostResourceResults | 0..* | ExPostResourceResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.20 Classe racine ExPostResourceResults

Modèle de tarification ex-post des ressources contenant des composants de LMP: énergie, congestion, perte. Fondé sur les ressources.

Le Tableau 276 présente tous les attributs de ExPostResourceResults.

**Tableau 276 – Attributs de MarketResults::ExPostResourceResults**

| nom | mult | type | description |
|---|---|---|---|
| congestionLMP | 0..1 | Float | Composant LMP en dollars américains (déconseillé) |
| desiredMW | 0..1 | Float | Sortie souhaitée d'unités |
| dispatchRate | 0..1 | Float | Taux de répartition d'unité issu de la répartition d'unités en temps réel. |
| lmp | 0..1 | Float | LMP (*Local Marginal Price*) en dollars américains au niveau de l'équipement (déconseillé) |
| lossLMP | 0..1 | Float | lmp des pertes (déconseillé) |
| maxEconomicMW | 0..1 | Float | MW économique maximum |
| minEconomicMW | 0..1 | Float | MW économique minimum |
| resourceMW | 0..1 | Float | Sortie MW actuelle de l'équipement<br>Utilisation de l'attribut: À titre informatif – À titre informatif – Sortie du moteur LPA. |
| status | 0..1 | EquipmentStatusType | Statut de l'équipement |

Le Tableau 277 présente toutes les extrémités d'association de ExPostResourceResults avec d'autres classes.

**Tableau 277 – Extrémités d'association de MarketResults:: ExPostResourceResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | ExPostResource | 1..1 | ExPostResource | |

#### 6.5.6.3.21 GeneralClearing

Modèle de résultat d'équilibre du processus de marché au niveau du marché. Identifie l'intervalle

Le Tableau 278 présente tous les attributs de GeneralClearing.

**Tableau 278 – Attributs de MarketResults::GeneralClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 279 présente toutes les extrémités d'association de GeneralClearing avec d'autres classes.

**Tableau 279 – Extrémités d'association de MarketResults:: GeneralClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneralClearingResults | 0..* | GeneralClearingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.22 Classe racine GeneralClearingResults

Fournit la valeur prévisionnelle de la charge ajustée sur une base de zone de prévision de charge.

Le Tableau 280 présente tous les attributs de GeneralClearingResults.

**Tableau 280 – Attributs de MarketResults::GeneralClearingResults**

| nom | mult | type | description |
|---|---|---|---|
| loadForecast | 0..1 | ActivePower | Prévision de charge (MW), par période de temps (5', 10', 15') |
| totalLoad | 0..1 | Float | Quantité de charge dans la Zone de réglage<br>Utilisation de l'attribut: valeur de charge horaire pour le domaine spécifique |
| totalNetInterchange | 0..1 | Float | Quantité d'échange pour la Zone de réglage<br>Utilisation de l'attribut: valeur d'échange horaire pour le domaine spécifique |

Le Tableau 281 présente toutes les extrémités d'association de GeneralClearingResults avec d'autres classes.

**Tableau 281 – Extrémités d'association de MarketResults:: GeneralClearingResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | GeneralClearing | 0..1 | GeneralClearing | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |

#### 6.5.6.3.23 InstructionClearing

Modèle de mise en équilibre du marché relatif aux instructions d'engagement. Identifie l'intervalle

Le Tableau 282 présente tous les attributs de InstructionClearing.

**Tableau 282 – Attributs de MarketResults::InstructionClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 283 présente toutes les extrémités d'association de InstructionClearing avec d'autres classes.

**Tableau 283 – Extrémités d'association de MarketResults:: InstructionClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ResourceDeployment Status | 0..* | ResourceDeploymentSta tus | |
| 0..* | ActualDemandRespon seEvent | 0..1 | DistributedResourceActu alEvent | ActualDemandResponseEvents pouvant exister sans faire partie d'un MarketActualEvent coordonné associé à un marché. Ces ActualDemandResponseEvents peuvent avoir plusieurs instructions InstructionClearing pour le type spécifié de RegisteredResources ou de ressource d'énergie distribuée de AggregateNodes. |
| 1..* | Instructions | 1..* | Instructions | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.24 InstructionClearingDOP

Modèle de mise en équilibre du marché relatif à DispatchOperatingPoint (point d'exploitation de répartition). Identifie l'intervalle

Le Tableau 284 présente tous les attributs de InstructionClearingDOP.

**Tableau 284 – Attributs de MarketResults::InstructionClearingDOP**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 285 présente toutes les extrémités d'association de InstructionClearingDOP avec d'autres classes.

**Tableau 285 – Extrémités d'association de MarketResults:: InstructionClearingDOP avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | DopInstruction | 1..* | DopInstruction | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.25 InstructionClearingDOT

Modèle de mise en équilibre du marché, relatif à Dispatch Operating Target (cible de répartition d'exploitation) (modèle de répartition anticipée). Identifie l'intervalle

Le Tableau 286 présente tous les attributs de InstructionClearingDOT.

**Tableau 286 – Attributs de MarketResults::InstructionClearingDOT**

| nom | mult | type | description |
|---|---|---|---|
| contingencyActive | 0..1 | YesNo | Indique que le système fonctionne actuellement dans un mode avec contingence. |
| dispatchMode | 0..1 | AutomaticDispatchMode | |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 287 présente toutes les extrémités d'association de InstructionClearingDOT avec d'autres classes.

**Tableau 287 – Extrémités d’association de MarketResults:: InstructionClearingDOT avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | DotInstruction | 1..* | DotInstruction | |
| 0..* | DemandResponseActualEvent | 0..1 | DistributedResourceActualEvent | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.26 Classe racine Instructions

Apporte les informations nécessaires (sur la base de ressources) pour la collecte des résultats de l'instruction Démarrage/Arrêt. Ces informations sont utiles pour le DAM (RUC exclusivement) et le RTM (HASP, prérépartition et intervalle).

Le Tableau 288 présente tous les attributs de Instructions.

**Tableau 288 – Attributs de MarketResults::Instructions**

| nom | mult | type | description |
|---|---|---|---|
| bindingDOD | 0..1 | Float | Delta de répartition d’exploitation par obligation qui fournit un delta relatif à appliquer. Généralement utilisé dans les instructions de réponse à une demande. Il est possible de cumuler les instructions bindingDOD, c’est-à-dire qu’une deuxième instruction DOD ne remplace pas la DOD précédente. Au lieu de cela, la deuxième DOD s’ajoute aux DOD précédentes. |
| bindingDOT | 0..1 | Float | |
| bindingInstruction | 0..1 | YesNo | |
| instructionCost | 0..1 | Float | Coût total associé au changement du statut de la ressource. |
| instructionSource | 0..1 | MQSInstructionSource | source de l'instruction pour les résultats qualité du marché (INS, ACT) |
| instructionStartTime | 0..1 | DateTime | Moment auquel il convient que la ressource soit sur Pmin (pour les démarrages).<br>Moment auquel la ressource est hors ligne. |

| nom | mult | type | description |
|---|---|---|---|
| instructionType | 0..1 | AutomaticDispInstTypeCommitment | Indique s'il s'agit d'un démarrage ou d'un arrêt. |
| manuallyBlocked | 0..1 | YesNo | Indicateur de blocage manuel (Yes/No). L'instruction a été bloquée par un opérateur. |
| minStatusChangeTime | 0..1 | Integer | Temps de démarrage minimal exigé pour mettre l'unité en ligne (minutes).<br>Temps de démarrage de la période d'engagement SCUC. Temps de démarrage calculé en fonction de StartUpTimeCurve, fourni avec la Bid.<br>Il s'agit d'une combinaison de l'offre du temps de démarrage et du temps d'arrêt de l'unité.<br>Unités en minutes |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Le Tableau 289 présente toutes les extrémités d'association de Instructions avec d'autres classes.

**Tableau 289 – Extrémités d'association de MarketResults:: Instructions avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 1..* | InstructionClearing | 1..* | InstructionClearing | |
| 0..* | AggregateNode | 0..1 | AggregateNode | |

### 6.5.6.3.27 Classe racine LoadFollowingOperatorInput

Modèle de capacités de suivi charge entrées par les opérateurs sur une base temporaire. Relatif aux ressources enregistrées dans les sous-systèmes mesurés

Le Tableau 290 présente tous les attributs de LoadFollowingOperatorInput.

**Tableau 290 – Attributs de MarketResults::LoadFollowingOperatorInput**

| nom | mult | type | description |
|---|---|---|---|
| dataEntryTimeStamp | 0..1 | DateTime | Moment auquel la saisie de données a été réalisée. |
| tempLoadFollowingUpManualCap | 0..1 | Float | Capacité LFU entrée manuellement sur une base temporaire. |
| tempLoadFollowingDownManualCap | 0..1 | Float | Capacité LFD entrée manuellement sur une base temporaire. |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Le Tableau 291 présente toutes les extrémités d'association de LoadFollowingOperatorInput avec d'autres classes.

**Tableau 291 – Extrémités d'association de MarketResults:: LoadFollowingOperatorInput avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |

#### 6.5.6.3.28 LossClearing

RT uniquement; n'est publié que toutes les 5 min pour les résultats des précédents intervalles de temps RT.

Le Tableau 292 présente tous les attributs de LossClearing.

**Tableau 292 – Attributs de MarketResults::LossClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 293 présente toutes les extrémités d'association de LossClearing avec d'autres classes.

**Tableau 293 – Extrémités d'association de MarketResults:: LossClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.29 Classe racine LossClearingResults

Indique la quantité perdue de MW pour les zones RUC, les zones de sous-contrôle et la perte totale.

Le Tableau 294 présente tous les attributs de LossClearingResults.

**Tableau 294 – Attributs de MarketResults::LossClearingResults**

| nom | mult | type | description |
|---|---|---|---|
| lossMW | 0..1 | Float | |

Le Tableau 295 présente toutes les extrémités d'association de LossClearingResults avec d'autres classes.

**Tableau 295 – Extrémités d'association de MarketResults:: LossClearingResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | LossClearing | 0..1 | LossClearing | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | RUCZone | 0..1 | RUCZone | |

#### 6.5.6.3.30 MPMClearing

Modèle des résultats des essais de l'emprise sur le marché et de l'atténuation éventuelle. Sur la base d'intervalles

Le Tableau 296 présente tous les attributs de MPMClearing.

**Tableau 296 – Attributs de MarketResults::MPMClearing**

| nom | mult | type | description |
|---|---|---|---|
| LMPMFinalFlag | 0..1 | YesNo | |
| SMPMFinalFlag | 0..1 | YesNo | |
| mitigationOccuredFlag | 0..1 | YesNo | |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 297 présente toutes les extrémités d'association de MPMClearing avec d'autres classes.

**Tableau 297 – Extrémités d'association de MarketResults:: MPMClearing avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MPMTestResults | 0..* | MPMTestResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.31 Classe racine MPMResourceStatus

Modèle des résultats des essais de la puissance du marché. Indique le statut de la ressource pour l'intervalle associé

Le Tableau 298 présente tous les attributs de MPMResourceStatus.

**Tableau 298 – Attributs de MarketResults::MPMResourceStatus**

| nom | mult | type | description |
|---|---|---|---|
| resourceStatus | 0..1 | String | Interval Test Status (Statut de l'essai de l'intervalle)<br>'N' – non applicable |

Le Tableau 299 présente toutes les extrémités d'association de MPMResourceStatus avec d'autres classes.

**Tableau 299 – Extrémités d'association de MarketResults:: MPMResourceStatus avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |

#### 6.5.6.3.32 Classe racine MPMTestResults

Fournit les résultats des essais MPM sous forme de résultat et de pourcentage de marge (le cas échéant). A des relations avec la Zone des Designated Congestion Area Tests (essais de la zone de congestion désignée), CurveSchedData pour les essais du segment d'offres, avec la SubControlArea pour les essais à l'échelle du système, et avec les Pnodes pour les essais d'impact LMPM.

Le Tableau 300 présente tous les attributs de MPMTestResults.

**Tableau 300 – Attributs de MarketResults::MPMTestResults**

| nom | mult | type | description |
|---|---|---|---|
| outcome | 0..1 | MPMTestOutcome | Résultats de l'essai. Pour les essais du prix, d'impact et conduits, les valeurs types sont NA, Pass (succès), Fail (échec), Disable (désactiver), ou Skip (ignorer). |
| marginPercent | 0..1 | PerCent | Utilisé pour représenter le résultat en % de marge à l'essai d'impact |

Le Tableau 301 présente toutes les extrémités d'association de MPMTestResults avec d'autres classes.

**Tableau 301 – Extrémités d'association de MarketResults:: MPMTestResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MPMClearing | 0..1 | MPMClearing | |
| 1..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |

### 6.5.6.3.33 Classe racine MarketRegionResults

Fournit tous les résultats des services auxiliaires régionaux pour le DAM et le RTM. Les données spécifiques sont de type marchandises (régulation à la hausse, régulation à la baisse, réserve tournante, réserve non tournante ou total majoré des réserves) en fonction des MW octroyés, du prix d'équilibre et de la capacité totale exigée pour la région.

Le Tableau 302 présente tous les attributs de MarketRegionResults.

**Tableau 302 – Attributs de MarketResults::MarketRegionResults**

| nom | mult | type | description |
|---|---|---|---|
| clearedMW | 0..1 | Float | Valeur de production octroyée en MW. Pour AS, cette valeur est clearedMW = AS Total. Pour AS, clearedMW – selfScheduleMW = AS Procured |
| clearedPrice | 0..1 | Float | Prix marginal ($/MW) pour la marchandise (énergie, régulation à la hausse, régulation à la baisse, réserve tournante ou réserve non tournante) en fonction du calcul du prix. |
| dispatchCtMW | 0..1 | Float | MW répartissables pour les unités de combustion. |
| dispatchHydroMW | 0..1 | Float | MW répartissables pour les unités hydrauliques. |
| dispatchRate | 0..1 | Float | Taux de répartition en MW/minute. |
| dispatchSteamMW | 0..1 | Float | MW répartissables pour les unités à vapeur. |
| imbalanceEnergyBias | 0..1 | Float | Influence énergétique du déséquilibre (MW) par période de temps (5 s uniquement) |
| limitFlag | 0..1 | ResourceLimitIndicator | Indicateurs AS locaux révélant si la limite supérieure ou inférieure de l'approvisionnement régional AS est contraignante |
| lumpyIndicator | 0..1 | YesNo | L'indicateur "arrondi" (Y/N) indique si la ressource qui définit le prix est un générateur irrégulier par heure pendant l'horizon temporel.<br>Applicable uniquement au DAM |
| maxSufficiencyIndex | 0..1 | Float | Limite maximale de l'exigence régionale |
| minSufficiencyIndex | 0..1 | Float | Limite minimale de l'exigence régionale |
| reqMaxMW | 0..1 | Float | Limite maximale de l'exigence régionale |
| reqMinMW | 0..1 | Float | Limite minimale de l'exigence régionale |
| selfScheduleMW | 0..1 | Float | Aof AS, selfScheduleMW = AS Self-Provided |

Le Tableau 303 présente toutes les extrémités d'association de MarketRegionResults avec d'autres classes.

**Tableau 303 – Extrémités d'association de MarketResults:: MarketRegionResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MarketProduct | 0..1 | MarketProduct | |
| 1..* | AncillaryServiceClearing | 0..1 | AncillaryServiceClearing | |
| 1..* | MarketRegion | 1..1 | MarketRegion | |

### 6.5.6.3.34 Classe racine MarketResults

Cette classe contient des éléments constitués de valeurs uniques pour l'ensemble de l'horizon temporel du marché. C'est-à-dire que pour le DAM, il y a une valeur pour chaque élément, non élaborée sur une base horaire. Récapitulatif du processus de marché

Le Tableau 304 présente tous les attributs de MarketResults.

**Tableau 304 – Attributs de MarketResults::MarketResults**

| nom | mult | type | description |
|---|---|---|---|
| startUpCost | 0..1 | Float | Coût total de démarrage ($) sur l'horizon temporel |
| minimumLoadCost | 0..1 | Float | Coût de charge minimal ($) sur l'horizon temporel |
| contingentOperatingResAvail | 0..1 | YesNo | Indicateur global de disponibilité de réserve d'exploitation contingente (Yes/No) |
| energyCost | 0..1 | Float | Coût minimal en énergie ($) sur l'horizon temporel |
| totalCost | 0..1 | Float | Coût total (énergie + AS) ($) par heure pendant l'horizon temporel |
| ancillarySvcCost | 0..1 | Float | Coût AS total (c'est-à-dire le règlement) pendant l'horizon temporel |
| totalRucCost | 0..1 | Float | Capacité RUC totale pour cet intervalle |

Le Tableau 305 présente toutes les extrémités d'association de MarketResults avec d'autres classes.

**Tableau 305 – Extrémités d'association de MarketResults:: MarketResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | EnergyMarket | 1..1 | EnergyMarket | |

### 6.5.6.3.35 MarketStatement

Une déclaration (statement) est un ensemble d'éléments de lignes de déclaration. Chaque déclaration et ses éléments de lignes donnent le détail des charges spécifiques à tout moment. Utilisée par la facturation et le règlement

Le Tableau 306 présente tous les attributs de MarketStatement.

**Tableau 306 – Attributs de MarketResults::MarketStatement**

| nom | mult | type | description |
|---|---|---|---|
| tradeDate | 0..1 | DateTime | Date à laquelle le règlement est effectué. |
| referenceNumber | 0..1 | String | Numéro de version de la précédente déclaration (en cas d'ajustement). |
| start | 0..1 | DateTime | Début d'une période de facturation. |
| end | 0..1 | DateTime | Fin d'une période de facturation. |
| transactionDate | 0..1 | DateTime | Date à laquelle la déclaration est émise. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 307 présente toutes les extrémités d'association de MarketStatement avec d'autres classes.

**Tableau 307 – Extrémités d'association de MarketResults:: MarketStatement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | MarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.36 MarketStatementLineItem

Élément individuel de ligne sur une déclaration de règlement ISO.

Le Tableau 308 présente tous les attributs de MarketStatementLineItem.

**Tableau 308 – Attributs de MarketResults::MarketStatementLineItem**

| nom | mult | type | description |
|---|---|---|---|
| currentAmount | 0..1 | Float | Montant du règlement actuel. |
| currentISOAmount | 0..1 | Float | Montant du règlement ISO actuel. |
| currentISOQuantity | 0..1 | Float | Quantité du règlement ISO actuel. |
| currentPrice | 0..1 | Float | Prix du règlement actuel. |
| currentQuantity | 0..1 | Float | Quantité du règlement actuel, en fonction de l'unité de mesure. |
| intervalDate | 0..1 | DateTime | Date à laquelle le règlement est effectué. |
| intervalNumber | 0..1 | String | Nombre d'intervalles. |
| netAmount | 0..1 | Float | Montant net du règlement. |
| netISOAmount | 0..1 | Float | Montant net du règlement ISO. |
| netISOQuantity | 0..1 | Float | Quantité nette du règlement ISO. |
| netPrice | 0..1 | Float | Prix net du règlement. |
| netQuantity | 0..1 | Float | Quantité nette du règlement, en fonction de l'unité de mesure. |
| previousAmount | 0..1 | Float | Montant du précédent règlement. |
| previousISOAmount | 0..1 | Float | Montant du précédent règlement ISO. |
| previousISOQuantity | 0..1 | Float | Quantité du précédent règlement ISO. |
| previousPrice | 0..1 | Float | Prix du précédent règlement. |
| previousQuantity | 0..1 | Float | Quantité du règlement précédent, en fonction de l'unité de mesure. |
| quantityUOM | 0..1 | String | Unité de mesure de la quantité de l'élément de ligne. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 309 présente toutes les extrémités d'association de MarketStatementLineItem avec d'autres classes.

**Tableau 309 – Extrémités d'association de MarketResults:: MarketStatementLineItem avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | MarketStatement | 1..1 | MarketStatement | |
| 0..* | ContainerMarketStatementLineItem | 0..1 | MarketStatementLineItem | |
| 0..1 | PassThroughBill | 0..1 | PassThroughBill | |
| 0..1 | ComponentMarketStatementLineItem | 0..* | MarketStatementLineItem | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.37 MitigatedBid

Résultats d'une offre atténuée publiés pour une période de règlement précise.

Le Tableau 310 présente tous les attributs de MitigatedBid.

**Tableau 310 – Attributs de MarketResults::MitigatedBid**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 311 présente toutes les extrémités d'association de MitigatedBid avec d'autres classes.

**Tableau 311 – Extrémités d'association de MarketResults:: MitigatedBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Bid | 0..1 | Bid | |
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.38 MitigatedBidClearing

Modèle d'atténuation de l'emprise sur le marché fondé sur des offres de références ou des offres atténuées. Sur la base d'intervalles.

Le Tableau 312 présente tous les attributs de MitigatedBidClearing.

**Tableau 312 – Attributs de MarketResults::MitigatedBidClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 313 présente toutes les extrémités d'association de MitigatedBidClearing avec d'autres classes.

**Tableau 313 – Extrémités d’association de MarketResults:: MitigatedBidClearing avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 1..* | MitigatedBid | 0..* | MitigatedBid | |
| 1..* | RMRDetermination | 0..* | RMRDetermination | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.39 Classe racine MitigatedBidSegment

Modèle d'offre atténuée. Indique un segment d'offre linéaire par morceaux qui a été atténué

Le Tableau 314 présente tous les attributs de MitigatedBidSegment.

**Tableau 314 – Attributs de MarketResults::MitigatedBidSegment**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | |
| thresholdType | 0..1 | String | |
| segmentNumber | 0..1 | Integer | Numéro de segment de l'offre atténuée |
| segmentMW | 0..1 | Float | Valeur MW du segment de l'offre atténuée |

Le Tableau 315 présente toutes les extrémités d'association de MitigatedBidSegment avec d'autres classes.

**Tableau 315 – Extrémités d'association de MarketResults:: MitigatedBidSegment avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | |

#### 6.5.6.3.40 PassThroughBill

PassThroughBill est utilisé pour:

1) Des transactions de charge dans les deux sens avec ou sans implication ISO
2) Des charges ou paiements directs spécifiques calculés de façon externe ou fournis directement aux règlements
3) Des éléments déterminants spécifiques à la facturation des charges fournis de façon externe et utilisés pour le calcul des frais

Le Tableau 316 présente tous les attributs de PassThroughBill.

**Tableau 316 – Attributs de MarketResults::PassThroughBill**

| nom | mult | type | description |
|---|---|---|---|
| adjustedAmount | 0..1 | Money | |
| amount | 0..1 | Money | Montant des frais du produit/service. |
| billedTo | 0..1 | String | Entreprise à laquelle la transaction PTB est facturée. |
| billEnd | 0..1 | DateTime | Date de fin de la période de facturation |
| billRunType | 0..1 | String | Type de processus de règlement; par exemple, prélim., final et nouveau processus. |
| billStart | 0..1 | DateTime | Date de début de la période de facturation |
| effectiveDate | 0..1 | DateTime | Date effective de la transaction |
| isDisputed | 0..1 | Boolean | Indicateur de transaction litigieuse |
| isProfiled | 0..1 | Boolean | Indicateur selon lequel des données de profil sont associées ou non au PTB. |
| paidTo | 0..1 | String | Entreprise à laquelle la transaction PTB est réglée. |
| previousEnd | 0..1 | DateTime | Date de fin de la précédente période de facturation |
| previousStart | 0..1 | DateTime | Date de début de la précédente période de facturation |
| price | 0..1 | Money | Prix du produit/service. |
| productCode | 0..1 | String | Identifiant du produit pour la détermination du type de charge de la transaction. |
| providedBy | 0..1 | String | Entreprise auprès de laquelle la transaction PTB est effectuée. |
| quantity | 0..1 | FloatQuantity | Quantité du produit. |
| serviceEnd | 0..1 | DateTime | Date de fin de la prestation de service, si périodique. |
| serviceStart | 0..1 | DateTime | Date de début de la prestation de service, si périodique. |
| soldTo | 0..1 | String | Entreprise à laquelle la transaction PTB est vendue. |
| taxAmount | 0..1 | Money | Part des taxes sur les services sélectionnés. |
| timeZone | 0..1 | String | Code du fuseau horaire |
| tradeDate | 0..1 | DateTime | Date d'opération |
| transactionDate | 0..1 | DateTime | Date à laquelle a lieu la transaction. |
| transactionType | 0..1 | String | Type de la transaction. Par exemple, taxe client, facturation client, correspondance créances clients/comptes créditeurs ou éléments déterminants de facturation |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 317 présente toutes les extrémités d'association de PassThroughBill avec d'autres classes.

**Tableau 317 – Extrémités d'association de MarketResults:: PassThroughBill avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..1 | MarketStatementLineItem | 0..1 | MarketStatementLineItem | |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.41 PnodeClearing

Résultats d'équilibre d'un nœud de tarification publiés pour une période de règlement précise.

Le Tableau 318 présente tous les attributs de PnodeClearing.

**Tableau 318 – Attributs de MarketResults::PnodeClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 319 présente toutes les extrémités d'association de PnodeClearing avec d'autres classes.

**Tableau 319 – Extrémités d’association de MarketResults:: PnodeClearing avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | CommodityPrice | 0..* | CommodityPrice | |
| 0..1 | PnodeResults | 1..* | PnodeResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.42 Classe racine PnodeResults

Fournit le prix total, le composant de coût, le composant de perte et le composant de congestion pour les Pnodes des DAM / RTM. Plusieurs prix sont produits en fonction du type de processus (MPM, RUC, tarification ou programmation/répartition).

Le Tableau 320 présente tous les attributs de PnodeResults.

**Tableau 320 – Attributs de MarketResults::PnodeResults**

| nom | mult | type | description |
|---|---|---|---|
| marginalClearingPrice | 0..1 | Float | Prix marginal local (LMP) ($/MWh) |
| costLMP | 0..1 | Float | Composant coût du prix marginal local (LMP) en unités monétaires par MW. |
| lossLMP | 0..1 | Float | Composant perte du prix marginal local (LMP) en unités monétaires par MW. |
| congestLMP | 0..1 | Float | Composant congestion du prix marginal local (LMP) en unités monétaires par MW. |
| scheduledMW | 0..1 | Float | Programme MW total au niveau de Pnode |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |

Le Tableau 321 présente toutes les extrémités d'association de PnodeResults avec d'autres classes.

**Tableau 321 – Extrémités d’association de MarketResults:: PnodeResults avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | PnodeClearing | 0..1 | PnodeClearing | |
| 1..* | Pnode | 0..1 | Pnode | |

### 6.5.6.3.43 Classe racine PriceDescriptor

Le prix d’une marchandise peut varier dans le temps au cours d’un intervalle donné. Par exemple, un prix peut être prévu un an à l’avance, un mois à l’avance, un jour à l’avance, une heure à l’avance et en temps réel; ceci est défini à l’aide du MarketType. En outre, un prix peut avoir un ou plusieurs composants. Par exemple, un prix marginal local d’énergie peut être calculé comme la somme arithmétique du prix du système, du prix de congestion et du prix-réclame. L’énumération priceType est utilisée pour déterminer si le prix est le prix total (priceType="total") ou l’un des composants constituants potentiellement nombreux.

Le Tableau 322 présente tous les attributs de PriceDescriptor.

**Tableau 322 – Attributs de MarketResults::PriceDescriptor**

| nom | mult | type | description |
|---|---|---|---|
| marketType | 0..1 | MarketType | Cadre temporel correspondant au prix, conformément aux conventions normalisées associées à l’énumération MarketType. |
| priceType | 0..1 | PriceTypeKind | "kind" (type) de prix décrit. En général, le priceType est soit "total", c’est-à-dire que le prix est le prix réglé pour acheter ou vendre la marchandise (parfois appelé prix "all-in" (prix global)), soit l’un des composants potentiellement nombreux. |

Le Tableau 323 présente toutes les extrémités d’association de PriceDescriptor avec d’autres classes.

**Tableau 323 – Extrémités d’association de MarketResults:: PriceDescriptor avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | CommodityPrice | 1..* | CommodityPrice | |

#### 6.5.6.3.44 Classe racine RMRDetermination

Indique s'il s'agit d'une unité RMR (Reliablity Must Run – marche obligatoire pour fiabilité): son activation est exigée pour satisfaire aux critères de fiabilité du code de réseau, à la demande de charge ou à la prise en charge de la tension.

Le Tableau 324 présente toutes les extrémités d’association de RMRDetermination avec d’autres classes.

**Tableau 324 – Extrémités d’association de MarketResults:: RMRDetermination avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MitigatedBidClearing | 1..* | MitigatedBidClearing | |
| 0..* | Bid | 0..1 | Bid | |

#### 6.5.6.3.45 RMROperatorInput

Saisie par l'opérateur RMR des exigences RMR par intervalle de marché.

Le Tableau 325 présente tous les attributs de RMROperatorInput.

**Tableau 325 – Attributs de MarketResults::RMROperatorInput**

| nom | mult | type | description |
|---|---|---|---|
| manuallySchedRMRMw | 0..1 | Float | La valeur inférieure de la prérépartition originelle ou du processus à courant alternatif (ou exigence RMR) devient la valeur de prérépartition. |
| updateUser | 0..1 | String | |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 326 présente toutes les extrémités d'association de RMROperatorInput avec d'autres classes.

**Tableau 326 – Extrémités d'association de MarketResults:: RMROperatorInput avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.46 Classe racine RUCAwardInstruction

Cette classe modélise les informations relatives aux attributions RUC

Le Tableau 327 présente tous les attributs de RUCAwardInstruction.

**Tableau 327 – Attributs de MarketResults::RUCAwardInstruction**

| nom | mult | type | description |
|---|---|---|---|
| clearedPrice | 0..1 | Float | Prix marginal ($/MW) pour la marchandise (régulation à la hausse, régulation à la baisse, réserve tournante ou réserve non tournante) pour le calcul du prix. |
| marketProductType | 0..1 | MarketProductType | Le type de produit majeur peut comprendre, entre autres, les éléments suivants:<br>Énergie<br>Régulation à la hausse<br>Régulation à la baisse<br>Réserve tournante<br>Réserve non tournante<br>Réserve d'exploitation |
| RUCAward | 0..1 | Float | L'attribution RUC d'une ressource est la part de capacité RUC qui n'est pas soumise à un contrat RA ou RMR. L'attribution RUC d'une ressource est la part de capacité RUC qui est éligible pour un règlement de disponibilité RUC. |
| RUCCapacity | 0..1 | Float | La capacité RUC d'une ressource est la différence entre (i) le RUC Schedule et (ii) la valeur supérieure du programme DA et la charge minimum. |
| RUCSchedule | 0..1 | Float | RUCSchedule d'une ressource constitue son niveau de sortie qui équilibre la prévision de charge utilisée dans RUC. La RUCSchedule dans RUC est similaire au DA Schedule dans DAM. |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Le Tableau 328 présente toutes les extrémités d'association de RUCAwardInstruction avec d'autres classes.

**Tableau 328 – Extrémités d'association de MarketResults:: RUCAwardInstruction avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ClearingResourceAward | 1..* | ResourceAwardClearing | |

#### 6.5.6.3.47 ResourceAwardClearing

Modélise les détails de l'offre et de la mise en équilibre du marché. La classe indique si une contingence est active et si le système de répartition automatique est actif pour cet intervalle de la solution de marché.

Le Tableau 329 présente tous les attributs de ResourceAwardClearing.

**Tableau 329 – Attributs de MarketResults::ResourceAwardClearing**

| nom | mult | type | description |
|---|---|---|---|
| dispatchMode | 0..1 | AutomaticDispatchMode | |
| contingencyActive | 0..1 | YesNo | Indique que le système fonctionne actuellement dans un mode avec contingence. |
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 330 présente toutes les extrémités d'association de ResourceAwardClearing avec d'autres classes.

**Tableau 330 – Extrémités d’association de MarketResults:: ResourceAwardClearing avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | RUCAwardInstruction | 1..* | RUCAwardInstruction | |
| 1..* | ResourceAwardInstruction | 1..* | ResourceAwardInstruction | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.48 Classe racine ResourceAwardInstruction

Modèle des résultats du marché, instruction pour la ressource. Contient les détails d'attribution comme attributs

Le Tableau 331 présente tous les attributs de ResourceAwardInstruction.

**Tableau 331 – Attributs de MarketResults::ResourceAwardInstruction**

| nom | mult | type | description |
|---|---|---|---|
| awardMW | 0..1 | Float | Pour énergie DA: Non applicable;<br>Pour AS DA: Attribution marché AS DA;<br>Pour énergie RT: Non applicable;<br>Pour AS RT: Attribution marché AS RT (à l'exception des attributions relatives au marché AS DA et à l'alimentation automatique) |
| clearedMW | 0..1 | Float | Pour énergie DA: Programme total = programme du marché à J-1 (DAM) + attribution à programmation automatique à J-1 (DA);<br>Pour AS DA: Attributions services auxiliaires DA = attribution marché AS DA + attribution à programmation automatique AS DA;<br>Pour énergie RT: Programme total = programme RTM + attribution planification automatique RT;<br>Pour AS RT: Attributions services auxiliaires RT = attribution à programmation automatique RT + attribution marché AS RT + attribution marché AS DA + attribution à programmation automatique AS DA; |
| clearedPrice | 0..1 | Float | Prix marginal ($/MW) pour la marchandise (régulation à la hausse, régulation à la baisse, réserve tournante ou réserve non tournante) pour le calcul du prix. |
| congestLMP | 0..1 | Float | Composant congestion du prix marginal local (LMP) en unités monétaires par MW. |
| costLMP | 0..1 | Float | Composant coût du prix marginal local (LMP) en unités monétaires par MW. |
| dispatcherAddedMW | 0..1 | Float | Quantité MW tier2 ajoutée par le répartiteur<br>Résultats du marché du marché des réserves synchronisées |
| economicMax | 0..1 | Float | Sortie max. de l'unité pour la répartition; offre comprise dans le maximum économique |
| economicMin | 0..1 | Float | Sortie min. de l'unité pour la répartition; offre comprise dans le minimum économique |
| effRegulationDownLimit | 0..1 | Float | Limite effective de régulation à la baisse (MW) |
| effRegulationUpLimit | 0..1 | Float | Limite effective de régulation à la hausse |
| lmp | 0..1 | Float | Valeur du prix marginal local |
| lossLMP | 0..1 | Float | Composant perte du prix marginal local (LMP) en unités monétaires par MW. |
| manuallyBlocked | 0..1 | YesNo | Indique si une attribution a été bloquée manuellement (Y/N). Valide pour la réserve tournante et non tournante. |
| marginalResourceIndicator | 0..1 | YesNo | Indicateur (Yes/No) selon lequel cette ressource définit le prix pour la répartition/programme |
| mustRunInd | 0..1 | Boolean | Identifie si l'unité a été configurée pour devoir être exécutée par le participant du marché en charge des offres dans l'unité |

| nom | mult | type | description |
|---|---|---|---|
| noLoadCost | 0..1 | Float | Coût à charge nulle de l'unité en cas de marchandise énergétique |
| optimalBidCost | 0..1 | Float | Coût optimal de l'offre |
| optimalBidPay | 0..1 | Float | Règlement optimal de réalisation de l'offre en fonction du prix marginal local |
| optimalMargin | 0..1 | Float | Marge optimale de réalisation de l'offre |
| overrideTimeStamp | 0..1 | DateTime | Heure de la saisie manuelle de données. |
| overrideValue | 0..1 | Float | Fournit la capacité pour l'opérateur de réseau de remplacer des éléments, tels que les exigences en rotation, avant d'exécuter l'algorithme. Cette valeur est fondée sur les produits du marché (rotation, non-rotation, régulation à la hausse, régulation à la baisse ou RUC). |
| selfSchedMW | 0..1 | Float | Pour énergie DA: Attribution à programmation automatique totale DA;<br>Pour AS DA: Attribution alimentation automatique AS DA;<br>Pour énergie RT: Attribution à programmation automatique totale RT;<br>Pour AS RT: Attribution alimentation automatique AS RT (à l'exception des attributions relatives au marché AS DA et à alimentation automatique) |
| startUpCost | 0..1 | Float | Coût de démarrage de l'unité en cas de marchandise énergétique |
| status | 0..1 | String | Dans ou en dehors du statut de la ressource |
| totalRevenue | 0..1 | Float | Revenu total de l'offre (startup_cost + no_load_cost + bid_pay) |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |

Le Tableau 332 présente toutes les extrémités d'association de ResourceAwardInstruction avec d'autres classes.

**Tableau 332 – Extrémités d’association de MarketResults:: ResourceAwardInstruction avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..1 | SelfScheduleBreakdown | 0..* | SelfScheduleBreakdown | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..* | ClearingResourceAward | 1..* | ResourceAwardClearing | |

#### 6.5.6.3.49 ResourceClearing

Modèle des résultats du marché, y compris les résultats équilibrés des ressources. Associé à ResourceDispatchResults.

Le Tableau 333 présente tous les attributs de ResourceClearing.

**Tableau 333 – Attributs de MarketResults::ResourceClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 334 présente toutes les extrémités d'association de ResourceClearing avec d'autres classes.

**Tableau 334 – Extrémités d’association de MarketResults:: ResourceClearing avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ResourceDispatchResults | 1..* | ResourceDispatchResults | |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.50 Classe racine ResourceDeploymentStatus

Le Tableau 335 présente tous les attributs de ResourceDeploymentStatus.

**Tableau 335 – Attributs de MarketResults::ResourceDeploymentStatus**

| nom | mult | type | description |
|---|---|---|---|
| acceptComments | 0..1 | String | Commentaires expliquant le motif du statut d'acceptation, par exemple, expliquant pourquoi une demande n'est acceptée que partiellement et pas complètement. |
| acceptStatus | 0..1 | String | Statut de la ressource pour ce déploiement. Les valeurs comprennent (full compliance (conformité totale), partial compliance, (conformité partielle), opt-out (désistement), etc.) |
| resourceResponseMW | 0..1 | Float | Le montant en MW de la ressource peut contribuer à ce déploiement. Il provient du fournisseur DR – comme réponse? Ou comme valeur réelle? Cela intervient-il dans le cadre de la liquidation? À voir. |

Le Tableau 336 présente toutes les extrémités d'association de ResourceDeploymentStatus avec d'autres classes.

**Tableau 336 – Extrémités d'association de MarketResults:: ResourceDeploymentStatus avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | InstructionClearing | 0..1 | InstructionClearing | |

### 6.5.6.3.51 Classe racine ResourceDispatchResults

La classe ResourceDispatchResults comprend les résultats de marché qui peuvent être transmis à un coordinateur de programmation (SC). Les données spécifiques fournies se composent de plusieurs indicateurs tels que les indicateurs de contingence, le démarrage bloqué et la répartition RMR. Elles fournissent par ailleurs le statut prévisionnel global et le statut de régulation de la ressource.

Le Tableau 337 présente tous les attributs de ResourceDispatchResults.

**Tableau 337 – Attributs de MarketResults::ResourceDispatchResults**

| nom | mult | type | description |
|---|---|---|---|
| blockedDispatch | 0..1 | String | Indicateur de répartition bloquée (Yes/No). |
| blockedPublishDOP | 0..1 | String | Blocage envoyant DOP vers ADS (Y/N) |
| contingencyFlag | 0..1 | YesNo | Indicateur de réserve d'exploitation contingente (Yes/No). Ressource participant à la capacité AS dans la répartition de la contingence. |
| limitIndicator | 0..1 | String | Indique quelle limite constitue la contrainte |
| lowerLimit | 0..1 | Float | Limite inférieure de rampe d'énergie de la ressource |
| maxRampRate | 0..1 | Float | Taux de rampe maximum |
| operatingLimitHigh | 0..1 | Float | Limite opérationnelle supérieure intégrant toute réduction utilisée par la répartition en temps réel (RTD) pour l'intervalle de liaison. |

| nom | mult | type | description |
|---|---|---|---|
| operatingLimitLow | 0..1 | Float | Limite opérationnelle inférieure intégrant toute réduction utilisée par la répartition en temps réel pour l'intervalle de liaison. |
| penaltyDispatchIndicator | 0..1 | YesNo | Indicateur de répartition à pénalité (Yes / No) révélant une régulation non économique. |
| regulatingLimitHigh | 0..1 | Float | Limite de régulation supérieure intégrant toute réduction utilisée par la répartition en temps réel (RTD) pour l'intervalle de liaison. |
| regulatingLimitLow | 0..1 | Float | Limite de régulation inférieure intégrant toute réduction utilisée par la répartition en temps réel pour l'intervalle de liaison. |
| resourceStatus | 0..1 | String | Statut d'engagement de l'unité (Marche/Arrêt/Démarrage) |
| totalSchedule | 0..1 | Float | Programme ascendant total des ressources. Programme total = En + toutes les AS par ressource et par intervalle |
| updateTimeStamp | 0..1 | DateTime | |
| updateType | 0..1 | MQSCHGType | |
| updateUser | 0..1 | String | |
| upperLimit | 0..1 | Float | Limite supérieure de rampe d'énergie de la ressource |

Le Tableau 338 présente toutes les extrémités d'association de ResourceDispatchResults avec d'autres classes.

**Tableau 338 – Extrémités d'association de MarketResults:: ResourceDispatchResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 1..* | ResourceClearing | 0..1 | ResourceClearing | |

#### 6.5.6.3.52 Classe racine ResourceLoadFollowingInst

Modèle des résultats d'équilibre du marché pour les ressources réalisant des offres pour être conformes à la charge

Le Tableau 339 présente tous les attributs de ResourceLoadFollowingInst.

**Tableau 339 – Attributs de MarketResults::ResourceLoadFollowingInst**

| nom | mult | type | description |
|---|---|---|---|
| instructionID | 0..1 | String | Identifiant d'instruction unique pour chaque instruction; affecté par le coordinateur de programmation et fourni à l'ADS. L'ADS passe à travers. |
| intervalStartTime | 0..1 | DateTime | Début de l'intervalle de temps pour lequel une exigence est définie. |
| calcLoadFollowingMW | 0..1 | Float | Moyenne pondérée pour RTPD et RTCD et identique pour RTID |
| dispWindowLowLimt | 0..1 | Float | |
| dispWindowHighLimt | 0..1 | Float | |

Le Tableau 340 présente toutes les extrémités d'association de ResourceLoadFollowingInst avec d'autres classes.

**Tableau 340 – Extrémités d’association de MarketResults:: ResourceLoadFollowingInst avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | ResourceClearing | 0..1 | ResourceClearing | |

#### 6.5.6.3.53 ResourcePerformanceEvaluation

Représente une évaluation de performances du déploiement d’une ressource. Chaque déploiement de ressource peut avoir plusieurs évaluations de performances effectuées par le biais de différentes mesures d’évaluation ou de différents algorithmes, ou par différentes autorités d’évaluation.

Le Tableau 341 présente tous les attributs de ResourcePerformanceEvaluation.

**Tableau 341 – Attributs de MarketResults::ResourcePerformanceEvaluation**

| nom | mult | type | description |
|---|---|---|---|
| effectiveEndTime | 0..1 | DateTime | |
| effectiveStartTime | 0..1 | DateTime | |
| evaluationDescription | 0..1 | String | Description de l’évaluation de performances, par exemple, la classification des valeurs assignées utilisée (elles sont toutes admises), la raison pour laquelle l’évaluation a été effectuée, tout élément décrivant l’évaluation de performances de réponse à la demande. |
| evaluationValue | 0..1 | String | Valeurs des performances. Étant donné qu’il s’agit d’une valeur String (chaîne de caractères), tous les systèmes de valeurs assignées sont pris en charge (par exemple, "1","2","3" ou "low" (faible), "medium" (moyen), "high" (élevé)). Le système de valeurs assignées est décrit dans l’attribut performanceValueDescription. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 342 présente toutes les extrémités d’association de ResourcePerformanceEvaluation avec d’autres classes.

**Tableau 342 – Extrémités d'association de MarketResults:: ResourcePerformanceEvaluation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ResourcePerformance TimeSeriesFactors | 0..* | ResourcePerformanceTi meSeriesFactor | |
| 0..* | DemandResponseAct ualEvent | 1..1 | DistributedResourceActu alEvent | |
| 0..* | ResorcePerformance GlobalFactor | 0..* | ResourcePerformanceGl obalFactor | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObj ect | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObj ect | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.54 ResourcePerformanceGlobalFactor

Les facteurs globaux sont des paires propriété/valeur utilisées pour ajuster les valeurs des performances de la ressource; par exemple, les facteurs d'échelle (par exemple, passage de la base de référence à l'échelle supérieure ou inférieure), les additionneurs (positifs ou négatifs), etc.

Le Tableau 343 présente tous les attributs de ResourcePerformanceGlobalFactor.

**Tableau 343 – Attributs de MarketResults::ResourcePerformanceGlobalFactor**

| nom | mult | type | description |
|---|---|---|---|
| factorDescription | 0..1 | String | Description (nom) de la priorité (facteur). |
| factorValue | 0..1 | String | Valeur de la priorité (facteur). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 344 présente toutes les extrémités d'association de ResourcePerformanceGlobalFactor avec d'autres classes.

**Tableau 344 – Extrémités d’association de MarketResults::ResourcePerformanceGlobalFactor avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ResourcePerformanceEvaluation | 0..* | ResourcePerformanceEvaluation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.55 ResourcePerformanceRating

Caractéristiques assignées d’une ressource pour des performances de réponse à la demande, par exemple, pour un ensemble d'échantillonnement soumis à des évaluations mensuelles de performances de réponse à la demande, la ressource peut avoir la valeur assignée “excellent”, “average” (moyen) ou “poor performance” (performances médiocres) pour l’ensemble échantillon

Le Tableau 345 présente tous les attributs de ResourcePerformanceRating.

**Tableau 345 – Attributs de MarketResults::ResourcePerformanceRating**

| nom | mult | type | description |
|---|---|---|---|
| effectiveEndTime | 0..1 | DateTime | Date et heure de début de validité des caractéristiques assignées |
| effectiveStartTime | 0..1 | DateTime | Date et heure de fin de validité des caractéristiques assignées |
| ratingDescription | 0..1 | String | Description des caractéristiques assignées de réponse à la demande de la ressource |
| ratingType | 1..1 | String | Type de performances assignées, par exemple, à quel marché ou à quel produit s’applique l’assignation |
| ratingValue | 0..1 | String | Caractéristiques assignées de réponse à la demande de la ressource |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 346 présente toutes les extrémités d’association de ResourcePerformanceRating avec d’autres classes.

**Tableau 346 – Extrémités d'association de MarketResults:: ResourcePerformanceRating avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredDistributedResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.56 ResourcePerformanceTimeSeriesFactor

Représente les performances d'une ressource sous forme de données de série chronologique pour une période de temps, un intervalle de temps et des critères d'évaluation spécifiques.

Le Tableau 347 présente tous les attributs de ResourcePerformanceTimeSeriesFactor.

**Tableau 347 – Attributs de MarketResults::ResourcePerformanceTimeSeriesFactor**

| nom | mult | type | description |
|---|---|---|---|
| timeSeriesDataType | 0..1 | String | Type de données de la série chronologique, par exemple, données de base, données relevées sur le compteur, données de performances calculées. |
| timeSeriesDescription | 0..1 | String | Description facultative des données de la série chronologique, par exemple, données de base, données relevées sur le compteur, données de performances calculées. |
| value1Description | 0..1 | String | Description de la value1 contenue dans le TimeSeriesFactor. |
| value2Description | 0..1 | String | Description de la value2 contenue dans le TimeSeriesFactor. |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 348 présente toutes les extrémités d'association de ResourcePerformanceTimeSeriesFactor avec d'autres classes.

**Tableau 348 – Extrémités d'association de MarketResults:: ResourcePerformanceTimeSeriesFactor avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ResourcePerformance Evaluation | 1..1 | ResourcePerformanceEv aluation | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObj ect | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObj ect | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.6.3.57 Classe racine SelfScheduleBreakdown

Modèle des résultats à programmations automatiques. Comprend les quantités de MW à programmation automatique et le type de programmation automatique pour chaque type à programmation automatique inclus dans la valeur totale des MW à programmation automatique trouvée dans ResourceAwardInstruction.

Le Tableau 349 présente tous les attributs de SelfScheduleBreakdown.

**Tableau 349 – Attributs de MarketResults::SelfScheduleBreakdown**

| nom | mult | type | description |
|---|---|---|---|
| selfSchedMW | 0..1 | Float | Valeur d'équilibre pour le type à programmation automatique spécifique répertorié. |
| selfSchedType | 0..1 | SelfScheduleBreakdow nType | Type de panne de programmation automatique. |

Le Tableau 350 présente toutes les extrémités d'association de SelfScheduleBreakdown avec d'autres classes.

**Tableau 350 – Extrémités d'association de MarketResults:: SelfScheduleBreakdown avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ResourceAwardInstruc tion | 1..1 | ResourceAwardInstructi on | |

### 6.5.6.3.58 Settlement

Spécifie un processus de règlement.

Le Tableau 351 présente tous les attributs de Settlement.

**Tableau 351 – Attributs de MarketResults::Settlement**

| nom | mult | type | description |
|---|---|---|---|
| tradeDate | 0..1 | DateTime | Date d'opération à laquelle le règlement est effectué. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 352 présente toutes les extrémités d'association de Settlement avec d'autres classes.

**Tableau 352 – Extrémités d'association de MarketResults:: Settlement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketInvoiceLineItem | 0..* | MarketInvoiceLineItem | |
| 0..* | MarketLedgerEntry | 0..* | MarketLedgerEntry | |
| 0..* | EnergyMarket | 0..1 | EnergyMarket | |
| 0..* | MajorChargeGroup | 1..* | MajorChargeGroup | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.59 TransactionBidClearing

Contient les intervalles utiles pour le TransactionBidResults associé. Par exemple, les résultats d'équilibre à J-1 pour les offres des transactions de chaque intervalle du jour de marché.

Le Tableau 353 présente tous les attributs de TransactionBidClearing.

**Tableau 353 – Attributs de MarketResults::TransactionBidClearing**

| nom | mult | type | description |
|---|---|---|---|
| intervalEndTime | 0..1 | DateTime | Hérité de: MarketFactors |
| intervalStartTime | 0..1 | DateTime | Hérité de: MarketFactors |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 354 présente toutes les extrémités d'association de TransactionBidClearing avec d'autres classes.

**Tableau 354 – Extrémités d’association de MarketResults:: TransactionBidClearing avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TransactionBidResults | 0..* | TransactionBidResults | |
| 0..* | MktActivityRecord | 0..* | MktActivityRecord | Hérité de: MarketFactors |
| 0..* | Market | 0..1 | Market | Hérité de: MarketFactors |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.6.3.60 TransactionBidResults

Contient les résultats d'équilibre pour chaque TransactionBid soumise au et acceptée par le marché.

Le Tableau 355 présente tous les attributs de TransactionBidResults.

**Tableau 355 – Attributs de MarketResults::TransactionBidResults**

| nom | mult | type | description |
|---|---|---|---|
| clearedMW | 0..1 | Float | Quantité de mégawatts de la transaction du marché |
| clearedPrice | 0..1 | Float | Prix de la transaction de marché |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 356 présente toutes les extrémités d'association de TransactionBidResults avec d'autres classes.

**Tableau 356 – Extrémités d'association de MarketResults:: TransactionBidResults avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TransactionBidClearing | 1..1 | TransactionBidClearing | |
| 0..* | TransactionBid | 0..1 | TransactionBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.7 Paquetage MktDomain

#### 6.5.7.1 Généralités

Le paquetage MktDomain est une bibliothèque de données regroupant les grandeurs et les unités qui définissent les types de données des attributs (propriétés) pouvant être repris par n'importe quelle classe dans n'importe quel autre paquetage au sein de MarketOperations.

#### 6.5.7.2 Énumération ActionType

Type d'action associé avec un ActionRequest par rapport à un ParticipantInterfaces::Trade.

Le Tableau 357 présente tous les libellés de ActionType.

**Tableau 357 – Libellés de MktDomain::ActionType**

| libellé | valeur | description |
|---|---|---|
| CANCEL | | Annule une opération. |

### 6.5.7.3 Énumération AnalogLimitType

Type de limite spécifié pour AnalogLimits.

Le Tableau 358 présente tous les libellés de AnalogLimitType.

**Tableau 358 – Libellés de MktDomain::AnalogLimitType**

| libellé | valeur | description |
|---|---|---|
| BranchMediumTerm | | Limite à moyen terme de l'agence |
| BranchLongTerm | | Limite à long terme de l'agence |
| BranchShortTerm | | Limite à court terme de l'agence |
| VoltageHigh | | Limite supérieure de tension |
| VoltageLow | | Limite inférieure de tension |

### 6.5.7.4 Énumération AnodeType

Types de nœuds agrégés, par exemple:

SYS – System Zone/Region (Zone/région système);

RUC – Zone RUC;

LFZ – Load Forecast Zone (Zone de prévision de charge);

REG – Région énergie/services auxiliaires du marché;

AGR – Aggregate Generation Resource (Ressource de production agrégée);

POD – Point of Delivery (Point de livraison);

ALR – Aggregate Load Resource (Ressource de charge agrégée);

LTAC – Load TransmissionAccessCharge (TAC) Group (Groupe de charge TransmissionAccessCharge);

ACA – Adjacent Control Area (Zone de réglage adjacente);

ASR – Aggregated System Resource (Ressource système agrégée);

ECA – Embedded Control Area (Zone de réglage embarquée).

Le Tableau 359 présente tous les libellés de AnodeType.

**Tableau 359 – Libellés de MktDomain::AnodeType**

| libellé | valeur | description |
|---|---|---|
| SYS | | Zone/région système |
| RUC | | Zone RU |
| LFZ | | Zone de prévision de charge |
| REG | | Région énergie/services auxiliaires du marché |
| AGR | | Ressource de production agrégée |
| POD | | Point de livraison |
| ALR | | Ressource de charge agrégée |

| libellé | valeur | description |
|---|---|---|
| LTAC | | Groupe de charge TransmissionAccessCharge (TAC) |
| ACA | | Zone de réglage adjacente |
| ASR | | Ressource système agrégée |
| ECA | | Zone de réglage embarquée |
| DER | | Ressource d’énergie distribuée |

### 6.5.7.5 Énumération ApnodeType

Types de nœuds agrégés, par exemple:

AG – Aggregated Generation (Production agrégée)
CPZ – Custom Price Zone (Zone de tarification personnalisée)
DPZ – Default Price Zone (Zone de tarification par défaut)
LAP – Load Aggregation Point (Point d'agrégation de charge)
TH – Trading Hub (Plate-forme de négociation)
SYS – System Zone (Zone système)
CA – Control Area (Zone de réglage)
GA – Generic Aggregation (Agrégation générique)
EHV – 500 kV
GH – Generic Hub (Plate-forme générique)
ZN – Zone
INT – Interface
BUS – Bus

Le Tableau 360 présente tous les libellés de ApnodeType.

**Tableau 360 – Libellés de MktDomain::ApnodeType**

| libellé | valeur | description |
|---|---|---|
| AG | | Production agrégée |
| CPZ | | Zone de tarification personnalisée |
| DPZ | | Zone de tarification par défaut |
| TH | | Plate-forme de négociation |
| SYS | | Zone système |
| CA | | Zone de réglage |
| DCA | | Zone de congestion désignée |
| GA | | Agrégation générique |
| GH | | Plate-forme générique |
| EHV | | 500 kV – Nœuds de tarification agrégés très haute tension |
| ZN | | Zone |
| INT | | Interface |
| BUS | | Bus |

### 6.5.7.6 Énumération AreaControlMode

Mode de contrôle actuel de la zone

Le Tableau 361 présente tous les libellés de AreaControlMode.

**Tableau 361 – Libellés de MktDomain::AreaControlMode**

| libellé | valeur | description |
|---|---|---|
| CF | | CF = Constant Frequency (Fréquence constante) |
| CTL | | Ligne d'interconnexion constante |
| TLB | | Conditionnement par ligne d'interconnexion |
| OFF | | Contrôle d'arrêt |

### 6.5.7.7 Énumération AutomaticDispInstTypeCommitment

Types d'instructions des engagements.

Le Tableau 362 présente tous les libellés de AutomaticDispInstTypeCommitment.

**Tableau 362 – Libellés de MktDomain::AutomaticDispInstTypeCommitment**

| libellé | valeur | description |
|---|---|---|
| START_UP | | Type d’instruction de démarrage |
| SHUT_DOWN | | Type d’instruction d’arrêt |
| DR_DEPLOY | | Déploiement de la ressource distribuée |
| DR_RELEASE | | Libération de la ressource distribuée |
| DR_ADJUSTMENT | | Ajustement de la ressource distribuée pour une ressource distribuée déjà déployée (engagée). |

### 6.5.7.8 Énumération AutomaticDispatchMode

Mode de répartition automatique

Le Tableau 363 présente tous les libellés de AutomaticDispatchMode.

**Tableau 363 – Libellés de MktDomain::AutomaticDispatchMode**

| libellé | valeur | description |
|---|---|---|
| INTERVAL | | |
| CONTINGENCY | | Apparition d’une contingence, nouvelle répartition des réserves de contingence |
| MANUAL | | Forçage par l’opérateur |

### 6.5.7.9 Énumération BidCalculationBasis

La base est utilisée pour calculer la courbe du prix de l'offre pour une offre d'énergie par défaut.

Le Tableau 364 présente tous les libellés de BidCalculationBasis.

**Tableau 364 – Libellés de MktDomain::BidCalculationBasis**

| libellé | valeur | description |
|---|---|---|
| LMP_BASED | | Fondé sur les prix payés à un emplacement de tarification en particulier. |
| COST_BASED | | Fondé sur les caractéristiques de production de l'unité et sur le coût du combustible. |
| NEGOTIATED | | Montant négocié avec l'entité indépendante désignée. |

### 6.5.7.10 Énumération BidMitigationStatus

Par exemple,

'S' – Atténuation par SMPM en cas de "faute"

'L' – Atténuation par LMPM en cas de "faute"

'R' – Modification par LMPM en raison des règles RMR

'M' – Atténuation en cas de "faute" par SMPM et LMPM

'B' – Atténuation en cas de "faute" par SMPM et modification par LMLM en raison des règles RMR

'O' – Original

Le Tableau 365 présente tous les libellés de BidMitigationStatus.

**Tableau 365 – Libellés de MktDomain::BidMitigationStatus**

| libellé | valeur | description |
|---|---|---|
| S | | |
| L | | |
| R | | |
| M | | |
| B | | |
| O | | |

### 6.5.7.11 Énumération BidMitigationType

Par exemple,

Initial

Final

Le Tableau 366 présente tous les libellés de BidMitigationType.

**Tableau 366 – Libellés de MktDomain::BidMitigationType**

| libellé | valeur | description |
|---|---|---|
| I | | |
| F | | |

### 6.5.7.12 Énumération BidType

Par exemple:

DEFAULT_ENERGY_BID
DEFAULT_STARTUP_BID
DEFAULT_MINIMUM_LOAD_BID

Le Tableau 367 présente tous les libellés de BidType.

**Tableau 367 – Libellés de MktDomain::BidType**

| libellé | valeur | description |
|---|---|---|
| DEFAULT_ENERGY_BID | | |
| DEFAULT_STARTUP_BID | | |
| DEFAULT_MINIMUM_LOAD_BID | | |

### 6.5.7.13 Énumération BidTypeRMR

Le type de programmes automatiques d'offre présente deux types correspondant à la sortie exigée pour les exigences et la prérépartition qualifiée.

Le Tableau 368 présente tous les libellés de BidTypeRMR.

**Tableau 368 – Libellés de MktDomain::BidTypeRMR**

| libellé | valeur | description |
|---|---|---|
| REQUIREMENTS | | Type de programmes automatiques d'offre pour la prérépartition qualifiée. |
| QUALIFIED_PREDISPATCH | | Type de programmes automatiques d'offre pour les exigences en sortie. |

### 6.5.7.14 Énumération CRRCategoryType

Types de catégories de droits à revenu de congestion

Le Tableau 369 présente tous les libellés de CRRCategoryType.

**Tableau 369 – Libellés de MktDomain::CRRCategoryType**

| libellé | valeur | description |
|---|---|---|
| NSR | | Service réseau |
| PTP | | Point à Point |

### 6.5.7.15 Énumération CRRHedgeType

Types de couvertures des droits à revenu de congestion

Le Tableau 370 présente tous les libellés de CRRHedgeType.

**Tableau 370 – Libellés de MktDomain::CRRHedgeType**

| libellé | valeur | description |
|---|---|---|
| OBLIGATION | | |
| OPTION | | |

### 6.5.7.16 Énumération CRRRoleType

Types de rôles qu'une organisation peut jouer vis-à-vis d'un droit à revenu de congestion.

Le Tableau 371 présente tous les libellés de CRRRoleType.

**Tableau 371 – Libellés de MktDomain::CRRRoleType**

| libellé | valeur | description |
|---|---|---|
| SELLER | | |
| BUYER | | |
| OWNER | | |

### 6.5.7.17 Énumération CRRSegmentType

Type de CRR, dans les définitions types possibles du système de CRR (par exemple, 'LSE', 'ETC').

Le Tableau 372 présente tous les libellés de CRRSegmentType.

**Tableau 372 – Libellés de MktDomain::CRRSegmentType**

| libellé | valeur | description |
|---|---|---|
| AUC | | |
| CAP | | |
| CF | | |
| CVR | | Droits convertis. |
| ETC | | Contrat de transport existant. |
| LSE | | Société d'approvisionnement. |
| MT | | Transport commercial. |
| TOR | | Droits de propriété de transport. |

### 6.5.7.18 Énumération CheckOutType

Pour indiquer un type de vérification tel que la capacité ajustée ou la capacité de répartition

Le Tableau 373 présente tous les libellés de CheckOutType.

**Tableau 373 – Libellés de MktDomain::CheckOutType**

| libellé | valeur | description |
|---|---|---|
| PRE_HOUR | | |
| PRE_SCHEDULE | | |
| AFTER_THE_FACT | | |

### 6.5.7.19 Énumération CommitmentType

Par exemple:

SELF – Engagement volontaire

ISO – Nouvel engagement pour cette période de marché

UC – Engagement existant qui était un vestige d'un précédent marché.

Le Tableau 374 présente tous les libellés de CommitmentType.

**Tableau 374 – Libellés de MktDomain::CommitmentType**

| libellé | valeur | description |
|---|---|---|
| SELF | | |
| ISO | | |
| UC | | |

### 6.5.7.20 Énumération ConstraintLimitType

Type de limite des résultats des contraintes avec obligation, par exemple:

MAXIMUM
MINIMUM

Le Tableau 375 présente tous les libellés de ConstraintLimitType.

**Tableau 375 – Libellés de MktDomain::ConstraintLimitType**

| libellé | valeur | description |
|---|---|---|
| MAXIMUM | | |
| MINIMUM | | |

### 6.5.7.21 Énumération ConstraintRampType

Type de rampe de contrainte

Le Tableau 376 présente tous les libellés de ConstraintRampType.

**Tableau 376 – Libellés de MktDomain::ConstraintRampType**

| libellé | valeur | description |
|---|---|---|
| FAST | | |
| SLOW | | |

### 6.5.7.22 Énumération ContractType

Type de contrat de transport, par exemple:

O – Autre
TE- Transmission Export (Export de transport)
TI – Transmission Import (Import de transport)
ETC – Existing Transmission Contract (Contrat de transport existant)
RMT – Contrat RMT
TOR – Transmission Ownership Right (Droit de propriété de transport)
RMR – Contrat RMR (Reliability Must Run)
CVR – Contrat converti.

Le Tableau 377 présente tous les libellés de ContractType.

**Tableau 377 – Libellés de MktDomain::ContractType**

| libellé | valeur | description |
|---|---|---|
| ETC | | ETC – Contrat de transport existant |
| TOR | | TOR – Droit de propriété de transport |
| RMR | | RMR – Contrat RMR |
| RMT | | RMT – Contrat RMT |
| O | | O – Autre |
| TE | | TE – Export de transport |
| TI | | TI – Import de transport |
| CVR | | CVR – Contrat converti |

### 6.5.7.23 Énumération CostBasis

Par exemple:

Bid Cost (Coût de l'offre)
Proxy Cost (Coût du proxy)
Registered Cost (Coût enregistré)

Le Tableau 378 présente tous les libellés de CostBasis.

**Tableau 378 – Libellés de MktDomain::CostBasis**

| libellé | valeur | description |
|---|---|---|
| BIDC | | |
| PRXC | | |
| REGC | | |

### 6.5.7.24 Énumération DispatchResponseType

Par exemple:

NON_RESPONSE
ACCEPT
DECLINE
PARTIAL.

Le Tableau 379 présente tous les libellés de DispatchResponseType.

**Tableau 379 – Libellés de MktDomain::DispatchResponseType**

| libellé | valeur | description |
|---|---|---|
| NON_RESPONSE | | |
| ACCEPT | | |
| DECLINE | | |
| PARTIAL | | |

#### 6.5.7.25 Énumération EnergyPriceIndexType

Par exemple:

WHOLESALE
RETAIL
BOTH

Le Tableau 380 présente tous les libellés de EnergyPriceIndexType.

**Tableau 380 – Libellés de MktDomain::EnergyPriceIndexType**

| libellé | valeur | description |
|---|---|---|
| WHOLESALE | | |
| RETAIL | | |
| BOTH | | |

#### 6.5.7.26 Énumération EnergyProductType

Type de produit énergétique

Le Tableau 381 présente tous les libellés de EnergyProductType.

**Tableau 381 – Libellés de MktDomain::EnergyProductType**

| libellé | valeur | description |
|---|---|---|
| FIRM | | Ferme |
| NFRM | | Non ferme |
| DYN | | Dynamique |
| WHL | | Transit |

#### 6.5.7.27 Énumération EquipmentStatusType

Statut de l'équipement

Le Tableau 382 présente tous les libellés de EquipmentStatusType.

**Tableau 382 – Libellés de MktDomain::EquipmentStatusType**

| libellé | valeur | description |
|---|---|---|
| In | | L'équipement est en marche. |
| Out | | L'équipement n'est pas en marche. |

#### 6.5.7.28 Énumération EnergyTransactionType

Définit l’état d'une transaction.

Le Tableau 383 présente tous les libellés de EnergyTransactionType.

**Tableau 383 – Libellés de MktDomain::EnergyTransactionType**

| libellé | valeur | description |
|---|---|---|
| approve | | Approbation |
| deny | | Négation |
| study | | Étude |

### 6.5.7.29 Énumération ExecutionType

Types d'exécutions des processus de marché

Le Tableau 384 présente tous les libellés de ExecutionType.

**Tableau 384 – Libellés de MktDomain::ExecutionType**

| libellé | valeur | description |
|---|---|---|
| DA | | J-1 |
| HASP | | Exécution à H-1 en temps réel |
| RTPD | | Prérépartition en temps réel |
| RTD | | Répartition en temps réel |

### 6.5.7.30 Énumération FlagTypeRMR

Indique si l'unité est de type RMR et son type de condition, par exemple:

'N' – l'unité n'est pas de type RMR
'1' – Unité RMR de condition 1
'2' – Unité RMR de condition 2

Le Tableau 385 présente tous les libellés de FlagTypeRMR.

**Tableau 385 – Libellés de MktDomain::FlagTypeRMR**

| libellé | valeur | description |
|---|---|---|
| N | | 'N' – L'unité n'est pas de type RMR |
| 1 | | '1' – Unité RMR de condition 1 |
| 2 | | '2' – Unité RMR de condition 2 |

### 6.5.7.31 Énumération FlowDirectionType

Indique le sens du flux d'énergie dans la Flowgate.

Le Tableau 386 présente tous les libellés de FlowDirectionType.

**Tableau 386 – Libellés de MktDomain::FlowDirectionType**

| libellé | valeur | description |
|---|---|---|
| Forward | | Le flux va dans le sens aval. |
| Reverse | | Le flux va dans le sens amont. |

#### 6.5.7.32 Énumération FuelSource

Par exemple:

Bio Gas (décharge, eaux usées, digesteur, etc.)
Biomass (Biomasse)
Coal (Charbon)
DIST
Natural Gas (Gaz naturel)
Geothermal (Énergie géothermique)
HRCV
None (Aucun)
Nuclear (Énergie nucléaire)
Oil (Pétrole)
Other (Autre)
Solar (Énergie solaire)
Waste to Energy (Transformation des déchets en énergie)
Water (Énergie hydraulique)
Wind (Énergie éolienne)

Le Tableau 387 présente tous les libellés de FuelSource.

**Tableau 387 – Libellés de MktDomain::FuelSource**

| libellé | valeur | description |
|---|---|---|
| NG | | Gaz naturel |
| NNG | | Gaz non naturel |
| BGAS | | Biogaz (décharge, eaux usées, digesteur, etc.) |
| BIOM | | Biomasse |
| COAL | | Charbon |
| DIST | | |
| GAS | | |
| GEOT | | Énergie géothermique |
| HRCV | | |
| NONE | | aucun |
| NUCL | | Énergie nucléaire |
| OIL | | Pétrole |
| OTHR | | Autre |
| SOLR | | Énergie solaire |
| WAST | | Transformation des déchets en énergie |
| WATR | | Énergie hydraulique |
| WIND | | Énergie éolienne |

#### 6.5.7.33 Énumération InterTieDirection

Sens d'une ligne d'interconnexion.

Le Tableau 388 présente tous les libellés de InterTieDirection.

**Tableau 388 – Libellés de MktDomain::InterTieDirection**

| libellé | valeur | description |
|---|---|---|
| E | | Exportation. |
| .I | | Importation. |

### 6.5.7.34 Énumération LoadForecastType

Types de zones de prévision de charge.

Le Tableau 389 présente tous les libellés de LoadForecastType.

**Tableau 389 – Libellés de MktDomain::LoadForecastType**

| libellé | valeur | description |
|---|---|---|
| LFZ | | Zone de prévision de charge. |
| LZMS | | Zone du sous-système mesuré. |

### 6.5.7.35 Énumération MPMTestIdentifierType

Type d'identifiant d'essai d'atténuation de l'emprise sur le marché, par exemple:

1 – Essai de tarification globale

2 – Essai de conductivité globale

3 – Essai d'impact global

4 – Essai de tarification locale

5 – Essai de conductivité locale

6 – Essai d'impact local

Le Tableau 390 présente tous les libellés de MPMTestIdentifierType.

**Tableau 390 – Libellés de MktDomain::MPMTestIdentifierType**

| libellé | valeur | description |
|---|---|---|
| 1 | | 1 – Essai de tarification globale. |
| 2 | | 2 – Essai de conductivité globale. |
| 3 | | 3 – Essai d'impact global. |
| 4 | | 4 – Essai de tarification locale. |
| 5 | | 5 – Essai de conductivité locale. |
| 6 | | 6 – Essai d'impact local. |

### 6.5.7.36 Énumération MPMTestMethodType

Type de méthode d'essai d'atténuation d'emprise sur le marché.

Essais avec les seuils normalisés (par défaut) ou essais avec les seuils alternatifs.

Le Tableau 391 présente tous les libellés de MPMTestMethodType.

**Tableau 391 – Libellés de MktDomain::MPMTestMethodType**

| libellé | valeur | description |
|---|---|---|
| NORMAL | | Normal. |
| ALTERNATE | | Alternatif. |

### 6.5.7.37 Énumération MPMTestOutcome

Par exemple:

Passed (Succès)
Failed (Échec)
Disabled (Désactivé)
Skipped (Ignoré)

Le Tableau 392 présente tous les libellés de MPMTestOutcome.

**Tableau 392 – Libellés de MktDomain::MPMTestOutcome**

| libellé | valeur | description |
|---|---|---|
| P | | Succès |
| F | | Échec |
| D | | Désactivé |
| S | | Ignoré |

### 6.5.7.38 Énumération MQSCHGType

Par exemple:

ADD – ajouter
CHG – modifier

Le Tableau 393 présente tous les libellés de MQSCHGType.

**Tableau 393 – Libellés de MktDomain::MQSCHGType**

| libellé | valeur | description |
|---|---|---|
| ADD | | |
| CHG | | |

### 6.5.7.39 Énumération MQSInstructionSource

Valeurs valides, par exemple:

INS – Instruction de RTM
ACT – Instruction réelle après les faits

Le Tableau 394 présente tous les libellés de MQSInstructionSource.

**Tableau 394 – Libellés de MktDomain::MQSInstructionSource**

| libellé | valeur | description |
|---|---|---|
| INS | | |
| ACT | | |

### 6.5.7.40 Énumération MarketEventStatusKind

Types de statuts des événements du marché.

Le Tableau 395 présente tous les libellés de MarketEventStatusKind.

**Tableau 395 – Libellés de MktDomain::MarketEventStatusKind**

| libellé | valeur | description |
|---|---|---|
| active | | Le statut de l'événement est actuellement dans un état actif.<br>Actif (lorsque sysdate est supérieure ou égale à l'heure de début planifiée) |
| cancelled | | Le statut de l'événement est actuellement dans un état annulé.<br>Annulé (arrêté avant l'heure de début planifiée ou l'heure de fin planifiée) |
| completed | | Le statut de l'événement est actuellement dans un état terminé.<br>Terminé (lorsque sysdate est égale à l'heure de libération) |
| planned | | Le statut de l'événement est actuellement dans un état planifié.<br>Planifié (sysdate est inférieure à l'heure de début planifiée) |

### 6.5.7.41 Énumération MarketProductSelfSchedType

Types d'offres à programmation automatique de produits du marché

Le Tableau 396 présente tous les libellés de MarketProductSelfSchedType.

**Tableau 396 – Libellés de MktDomain::MarketProductSelfSchedType**

| libellé | valeur | description |
|---|---|---|
| ETC | | Contrat de transport existant. |
| TOR | | Droit de propriété de transport. |
| RMR | | Marche obligatoire (must run) pour fiabilité |
| RGMR | | Marche obligatoire (must run) pour régulation |
| RMT | | Prise obligatoire (must take) pour fiabilité |
| PT | | Preneur de prix. |
| LPT | | Preneur de prix inférieur. |
| SP | | Alimentation automatique. |
| RA | | Adéquation de la ressource. |

#### 6.5.7.42 Énumération MarketProductType

Par exemple:

Énergie, régulation à la hausse, régulation à la baisse, réserve tournante, réserve non tournante, RUC, suivi de charge (ascendant et descendant).

Le Tableau 397 présente tous les libellés de MarketProductType.

**Tableau 397 – Libellés de MktDomain::MarketProductType**

| libellé | valeur | description |
|---|---|---|
| EN | | Type d'énergie |
| RU | | Régulation à la hausse |
| RD | | Régulation à la baisse |
| SR | | Réserve tournante |
| NR | | Réserve non tournante |
| RC | | Engagement de l'unité résiduelle |
| LFU | | Suivi de charge ascendant |
| LFD | | Suivi de charge descendant |
| REG | | Régulation |

#### 6.5.7.43 Énumération MarketType

Type de marché.

Le Tableau 398 présente tous les libellés de MarketType.

**Tableau 398 – Libellés de MktDomain::MarketType**

| libellé | valeur | description |
|---|---|---|
| DAM | | Marché à J-1 |
| RTM | | Marché en temps réel |
| HAM | | Marché à H-1 |
| RUC | | Engagement de l'unité résiduelle. |

#### 6.5.7.44 Énumération MktAccountKind

Type de compte de marché.

Le Tableau 399 présente tous les libellés de MktAccountKind.

**Tableau 399 – Libellés de MktDomain::MktAccountKind**

| libellé | valeur | description |
|---|---|---|
| normal | | |
| reversal | | |
| statistical | | |
| estimate | | |

#### 6.5.7.45 Énumération MktBillMediaKind

Type de support de facturation.

Le Tableau 400 présente tous les libellés de MktBillMediaKind.

**Tableau 400 – Libellés de MktDomain::MktBillMediaKind**

| libellé | valeur | description |
|---|---|---|
| paper | | |
| electronic | | |
| other | | |

#### 6.5.7.46 Énumération MktInvoiceLineItemKind

Type d'élément de ligne de facturation.

Le Tableau 401 présente tous les libellés de MktInvoiceLineItemKind.

**Tableau 401 – Libellés de MktDomain::MktInvoiceLineItemKind**

| libellé | valeur | description |
|---|---|---|
| initial | | |
| recalculation | | |
| other | | |

#### 6.5.7.47 Énumération OnOff

ON

OFF

Le Tableau 402 présente tous les libellés de OnOff.

**Tableau 402 – Libellés de MktDomain::OnOff**

| libellé | valeur | description |
|---|---|---|
| ON | | |
| OFF | | |

#### 6.5.7.48 Énumération ParticipationCategoryMPM

Par exemple:

'Y' – Participe à LMPM et à SMPM

'N' – N'est pas inclus dans les mesures de tarification LMP

'S' – Participe aux mesures de tarification SMPM

'L' – Participe aux mesures de tarification LMPM

Le Tableau 403 présente tous les libellés de ParticipationCategoryMPM.

**Tableau 403 – Libellés de MktDomain::ParticipationCategoryMPM**

| libellé | valeur | description |
|---|---|---|
| Y | | |
| N | | |
| S | | |
| L | | |

#### 6.5.7.49 Énumération PassIndicatorType

Définit les acceptations individuelles qui produisent des résultats par type d'exécution/type de marché.

Le Tableau 404 présente tous les libellés de PassIndicatorType.

**Tableau 404 – Libellés de MktDomain::PassIndicatorType**

| libellé | valeur | description |
|---|---|---|
| MPM-1 | | Acceptation de l'atténuation de l'emprise sur le marché 1 |
| MPM-2 | | Acceptation de l'atténuation de l'emprise sur le marché 2 |
| MPM-3 | | Acceptation de l'atténuation de l'emprise sur le marché 3 |
| MPM-4 | | Acceptation de l'atténuation de l'emprise sur le marché 4 |
| RUC | | Engagement de l'unité résiduelle |
| RTPD | | Prérépartition en temps réel |
| RTED | | Répartition économique en temps réel |
| HA-SCUC | | Engagement d'unité à contrainte de sécurité à H-1 |
| DA | | J-1 |

#### 6.5.7.50 Énumération PriceTypeKind

La valeur de cette énumération pour différents prix est "total" pour le prix complet/fort/global price, "congestion" pour le coût de congestion associé au prix total, "loss" pour le prix-réclame associé au prix total, "capacity" pour les prix relatifs à la capacité installée ou réservée, "mileage" pour la comptabilité en fonction de l'utilisation, "system" pour les prix relatifs au système global/plaque de cuivre, et "delivery" pour les prix relatifs à la distribution.

Le Tableau 405 présente tous les libellés de PriceTypeKind.

**Tableau 405 – Libellés de MktDomain::PriceTypeKind**

| libellé | valeur | description |
|---|---|---|
| capacity | | |
| congestion | | |
| delivery | | |
| loss | | |
| mileage | | |
| system | | |
| total | | |

#### 6.5.7.51 Énumération PurposeFlagType

Indicateur d'objectif MPM, par exemple:

Nature des données de seuil:

'M' – Seuil d'atténuation
'R' – Seuil de rapport

Le Tableau 406 présente tous les libellés de PurposeFlagType.

**Tableau 406 – Libellés de MktDomain::PurposeFlagType**

| libellé | valeur | description |
|---|---|---|
| M | | |
| R | | |

#### 6.5.7.52 Énumération RampCurveType

Par exemple:

0 – Taux de rampe fixe indépendant de la fonction de taux de la sortie MW de l'unité
1 – Taux de rampe statiques comme fonction de la sortie MW de l'unité uniquement
2 – Taux de rampe dynamiques comme fonction de la sortie MW de l'unité et du temps de rampe

Le Tableau 407 présente tous les libellés de RampCurveType.

**Tableau 407 – Libellés de MktDomain::RampCurveType**

| libellé | valeur | description |
|---|---|---|
| 0 | | Taux de rampe fixe indépendant de la fonction de taux de la sortie MW de l'unité |
| 1 | | Taux de rampe statiques comme fonction de la sortie MW de l'unité uniquement |
| 2 | | Taux de rampe dynamiques comme fonction de la sortie MW de l'unité et du temps de rampe |

#### 6.5.7.53 Énumération RampRateCondition

Condition du taux de rampe

Le Tableau 407 présente tous les libellés de RampRateCondition.

**Tableau 408 – Libellés de MktDomain::RampRateCondition**

| libellé | valeur | description |
|---|---|---|
| WORST | | |
| BEST | | |
| NORMAL | | |
| NA | | Non applicable |

#### 6.5.7.54 Énumération RampRateType

Type de courbe du taux de rampe.

Le Tableau 409 présente tous les libellés de RampRateType.

**Tableau 409 – Libellés de MktDomain::RampRateType**

| libellé | valeur | description |
|---|---|---|
| OP | | Taux de rampe opérationnel. |
| REG | | Taux de rampe régulateur. |
| OP_RES | | Taux de rampe opérationnel de réserve. |
| LD_DROP | | Taux de rampe de chute de charge. |
| LD_PICKUP | | Taux de reprise de charge. |
| INTERTIE | | Taux de rampe de la ligne d'interconnexion. |

#### 6.5.7.55 Énumération ReserveRequirementType

Par exemple:

Réserve d'exploitation, régulation, contingence

Le Tableau 410 présente tous les libellés de ReserveRequirementType.

**Tableau 410 – Libellés de MktDomain::ReserveRequirementType**

| libellé | valeur | description |
|---|---|---|
| OPRSV | | Reserve d’exploitation |
| CONT | | Contingence |
| REG | | Régulation |

#### 6.5.7.56 Énumération ResourceAssnType

Par exemple:

Indicateur de puits de propriétaire d'actifs pour utilisation par CRR
Indicateur de source de propriétaire d'actifs pour utilisation par CRR
Marche obligatoire pour fiabilité
Coordinateur de programmation
Société d'approvisionnement

Le Tableau 411 présente tous les libellés de ResourceAssnType.

**Tableau 411 – Libellés de MktDomain::ResourceAssnType**

| libellé | valeur | description |
|---|---|---|
| CSNK | | |
| CSRC | | |
| RMR | | |
| SC | | |
| LSE | | |

### 6.5.7.57 Énumération ResourceCapacityType

Type de capacité de la ressource.

Le Tableau 412 présente tous les libellés de ResourceCapacityType.

**Tableau 412 – Libellés de MktDomain::ResourceCapacityType**

| libellé | valeur | description |
|---|---|---|
| RU | | Régulation à la hausse. |
| RD | | Régulation à la baisse. |
| SR | | Réserve tournante. |
| NR | | Réserve non tournante. |
| MO | | Offre obligatoire (must offer). |
| FO | | Offre flexible. |
| RA | | Adéquation de la ressource. |
| RMR | | Marche obligatoire pour fiabilité. |

### 6.5.7.58 Énumération ResourceCertificationKind

Types utilisés pour la certification des ressources.

Le Tableau 413 présente tous les libellés de ResourceCertificationKind.

**Tableau 413 – Libellés de MktDomain::ResourceCertificationKind**

| libellé | valeur | description |
|---|---|---|
| RegulationUp | | Régulation à la hausse (RU, REGUP) |
| RegulationDown | | Régulation à la baisse (RD, REGDN) |
| SpinningReserve | | Réserve tournante (SR, RRSPIN) |
| NonSpinningReserve | | Réserve non tournante (NR, NONSPIN) |
| ReliabilityMustRun | | Marche obligatoire pour fiabilité (RMR) |
| BLACKSTART | | Démarrage en autonome |
| DemandSideResponse | | Réponse côté demande (DSR) |
| SynchronousCondenser | | Compensateur synchrone (SYNCCOND) |
| IntermittentResource | | Ressource intermittente |
| ReliabilityUnitCommitment | | Engagement d'unité pour fiabilité (RUC) |
| Energy | | |
| Capacity | | |

### 6.5.7.59 Énumération ResourceLimitIndicator

Indicateurs AS locaux révélant si la limite supérieure ou inférieure de l'approvisionnement régional AS est contraignante.

Le Tableau 414 présente tous les libellés de ResourceLimitIndicator.

**Tableau 414 – Libellés de MktDomain::ResourceLimitIndicator**

| libellé | valeur | description |
|---|---|---|
| UPPER | | |
| LOWER | | |

### 6.5.7.60 Énumération ResourceRegistrationStatus

Type de statut d'enregistrement de la ressource, par exemple:

Active (actif)
Mothballed (suspendu)
Planned (planifié)
Decommissioned (mis hors service)

Le Tableau 415 présente tous les libellés de ResourceRegistrationStatus.

**Tableau 415 – Libellés de MktDomain::ResourceRegistrationStatus**

| libellé | valeur | description |
|---|---|---|
| Active | | L'enregistrement de la ressource est actif |
| Planned | | Le statut d'enregistrement est en phase de planification |
| Mothballed | | L'enregistrement de la ressource a été suspendu |
| Decommissioned | | Le statut d'enregistrement de la ressource est mis hors service |

### 6.5.7.61 Énumération ResultsConstraintType

Types de contraintes avec obligation des résultats du marché.

Le Tableau 416 présente tous les libellés de ResultsConstraintType.

**Tableau 416 – Libellés de MktDomain::ResultsConstraintType**

| libellé | valeur | description |
|---|---|---|
| FG_act | | Scénario de base réel de l'interface de transit. |
| Contingency | | Contingence. |
| Interface | | Interface. |
| Actual | | Réel. |

### 6.5.7.62 Énumération SelfSchedReferenceType

Indication sur le type de programme automatique en cours de référencement.

Le Tableau 417 présente tous les libellés de SelfSchedReferenceType.

**Tableau 417 – Libellés de MktDomain::SelfSchedReferenceType**

| libellé | valeur | description |
|---|---|---|
| ETC | | Contrat de transport existant. |
| TOR | | Droit de propriété de transport. |

### 6.5.7.63 Énumération SelfScheduleBreakdownType

Type de panne à programmation automatique.

Le Tableau 418 présente tous les libellés de SelfScheduleBreakdownType.

**Tableau 418 – Libellés de MktDomain::SelfScheduleBreakdownType**

| libellé | valeur | description |
|---|---|---|
| ETC | | Contrat de transport existant. |
| TOR | | Droit de propriété de transport. |
| LPT | | Preneur de prix inférieur. |

### 6.5.7.64 Énumération SwitchStatusType

Statut du disjoncteur (fermé ou ouvert).

Le Tableau 419 présente tous les libellés de SwitchStatusType.

**Tableau 419 – Libellés de MktDomain::SwitchStatusType**

| libellé | valeur | description |
|---|---|---|
| Closed | | Statut fermé. |
| Open | | Statut ouvert. |

### 6.5.7.65 Énumération TRType

Type de droit de contrat de transport, par exemple:

droits de contrats individuels ou en chaîne

Le Tableau 420 présente tous les libellés de TRType.

**Tableau 420 – Libellés de MktDomain::TRType**

| libellé | valeur | description |
|---|---|---|
| CHAIN | | Chaîne TR |
| INDIVIDUAL | | TR individuel |

### 6.5.7.66 Énumération TimeOfUse

Temps d'utilisation utilisé par une définition CRR pour la spécification de la durée d'étendue CRR. ON – La CRR s'étend sur les heures de pointe de la journée, OFF – La CRR s'étend sur les heures creuses de la journée, 24HR – La CRR s'étend sur toute la journée.

Le Tableau 421 présente tous les libellés de TimeOfUse.

**Tableau 421 – Libellés de MktDomain::TimeOfUse**

| libellé | valeur | description |
|---|---|---|
| ON | | Le temps d'utilisation s'étend uniquement sur les heures de pointe de la journée. |
| OFF | | Le temps d'utilisation s'étend uniquement sur les heures creuses de la journée. |
| 24HR | | Le temps d'utilisation s'étend sur toute la journée, soit 24 h. |

### 6.5.7.67 Énumération TradeType

Type d'opération.

Le Tableau 422 présente tous les libellés de TradeType.

**Tableau 422 – Libellés de MktDomain::TradeType**

| libellé | valeur | description |
|---|---|---|
| IST | | Opération InterSC. |
| AST | | Opération commerciale relative aux services auxiliaires. |
| UCT | | Opération commerciale d'engagement d'unité. |

### 6.5.7.68 Énumération UnitRegulationKind

Type de régulation d'unité.

Le Tableau 423 présente tous les libellés de UnitRegulationKind.

**Tableau 423 – Libellés de MktDomain::UnitRegulationKind**

| libellé | valeur | description |
|---|---|---|
| 0 | | L'unité n'est pas en régulation. |
| 1 | | L'unité est en AGC et en régulation. |
| 2 | | L'unité est censée être en régulation, mais ne l'est pas actuellement. |

### 6.5.7.69 Énumération UnitType

Combined Cycle (cycle combiné)

Gas Turbine (turbine à gaz)

Hydro Turbine (turbine hydraulique)

Other (autre)

Photovoltaic (photovoltaïque)

Hydro Pump-Turbine (turbine à pompe hydraulique)

Reciprocating Engine (moteur à explosion)

Steam Turbine (turbine à vapeur)

Synchronous Condenser (compensateur synchrone)

Wind Turbine (éolienne)

Le Tableau 424 présente tous les libellés de UnitType.

**Tableau 424 – Libellés de MktDomain::UnitType**

| libellé | valeur | description |
|---|---|---|
| CCYC | | Cycle combiné |
| GTUR | | Turbine à gaz |
| HYDR | | Turbine hydraulique |
| OTHR | | Autre |
| PHOT | | Photovoltaïque |
| PTUR | | Turbine à pompe hydraulique |
| RECP | | Moteur à explosion |
| STUR | | Turbine à vapeur |
| SYNC | | Compensateur synchrone |
| WIND | | Éolienne |

#### 6.5.7.70 Énumération YesNo

Utilisée comme indicateur défini sur Yes (Oui) ou No (Non).

Le Tableau 425 présente tous les libellés de YesNo.

**Tableau 425 – Libellés de MktDomain::YesNo**

| libellé | valeur | description |
|---|---|---|
| YES | | |
| NO | | |

### 6.5.8 Paquetage ParticipantInterfaces

#### 6.5.8.1 Généralités

Interfaces des participants du marché pour les offres et les échanges commerciaux.

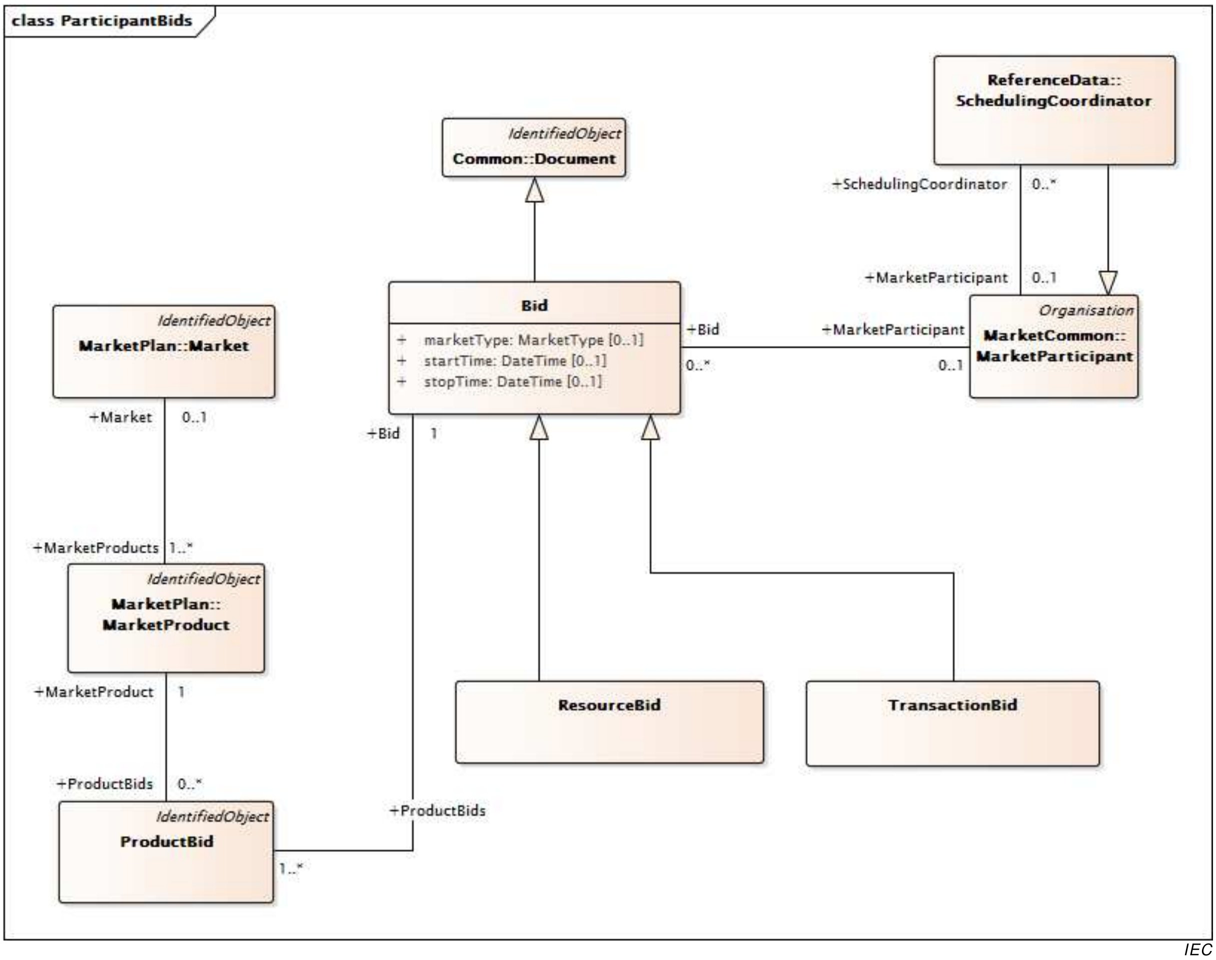


**Figure 54 – Diagramme de classe ParticipantInterfaces::ParticipantBids**

Figure 54: représente une vue générale de la classe et des associations qui sont utilisées pour modéliser les offres et les enchères. Le terme "Offre" (*Bid*) inclut les offres de vente et d'achat d'un ou plusieurs produits d'électricité. Bid est une sous-classe de la classe Document du paquetage IEC 61968. Les Bid sont elles-mêmes classées en ResourceBid et TransactionBid. Les enchères sont associées aux coordinateurs de programmation qui les soumet au nom des participants du marché, comme l'indique l'association entre la classe Bid et la classe SchedulingCoordinator.

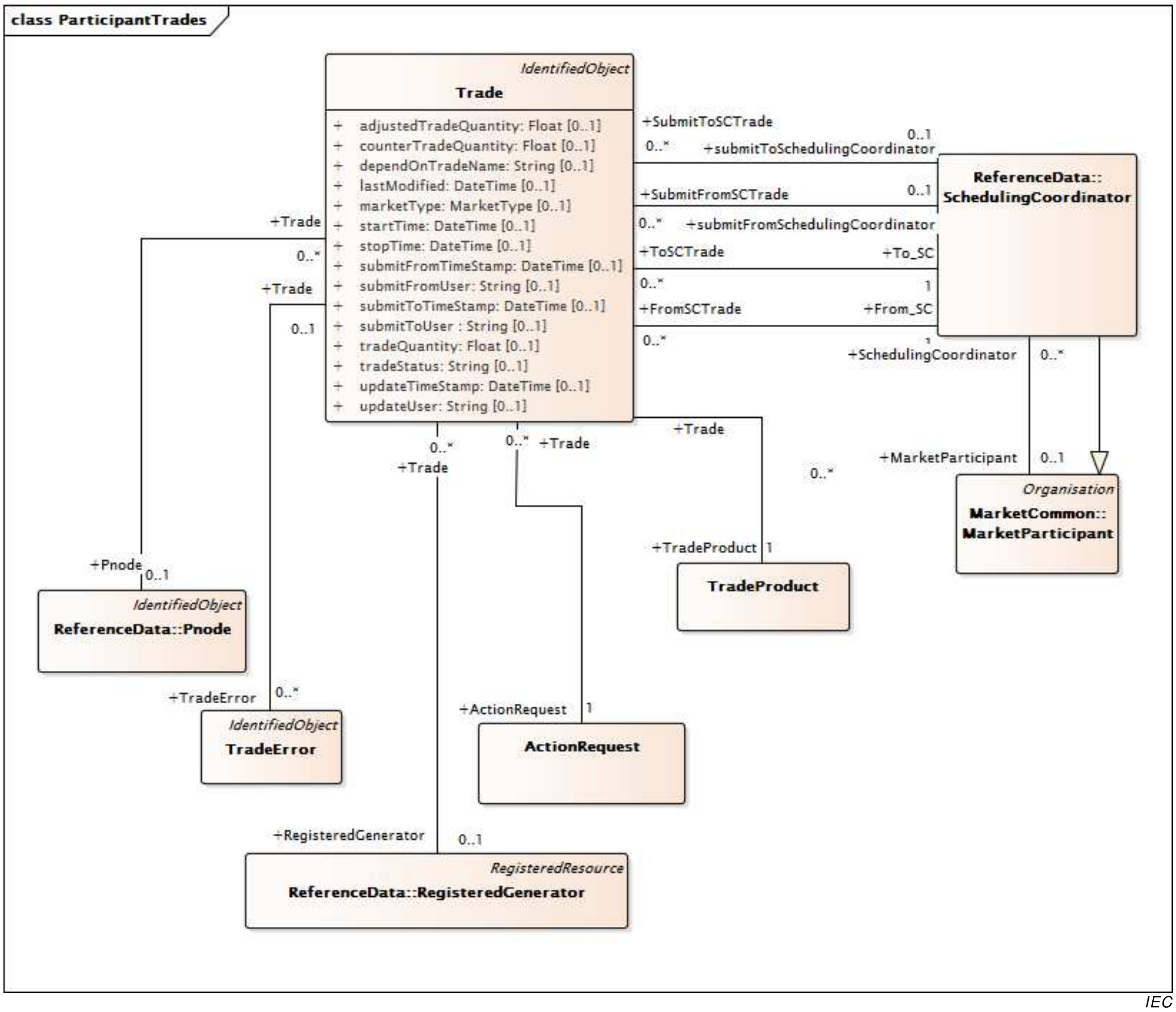


**Figure 55 – Diagramme de classe ParticipantInterfaces::ParticipantTrades**

Figure 55: représente la classe et les associations utilisées pour modéliser les opérations financières entre coordinateurs de programmation. La classe TradeProduct est utilisée pour indiquer le produit électrique inclus dans l'opération. La classe SchedulingCoordinator est utilisée pour indiquer les coordinateurs de programmation impliqués dans l'opération.

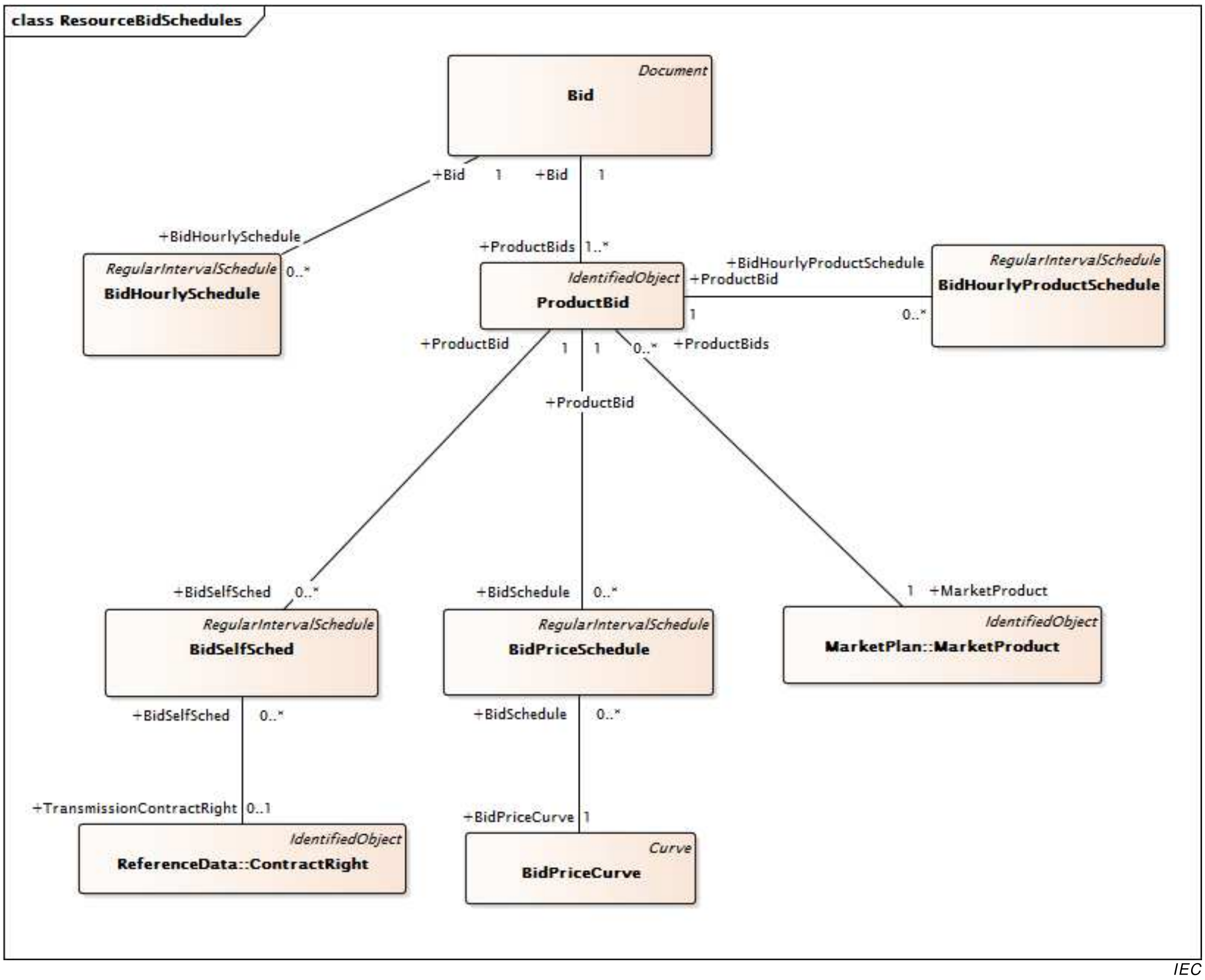


**Figure 56 – Diagramme de classe ParticipantInterfaces::ResourceBidSchedules**

Figure 56: représente les classes et les associations utilisées pour modéliser les programmations d’offres. Un coordinateur de programmation peut décider de soumettre une offre valide pour plusieurs intervalles de l’horizon temporel du marché. Les classes de ce modèle prennent en charge ce concept.

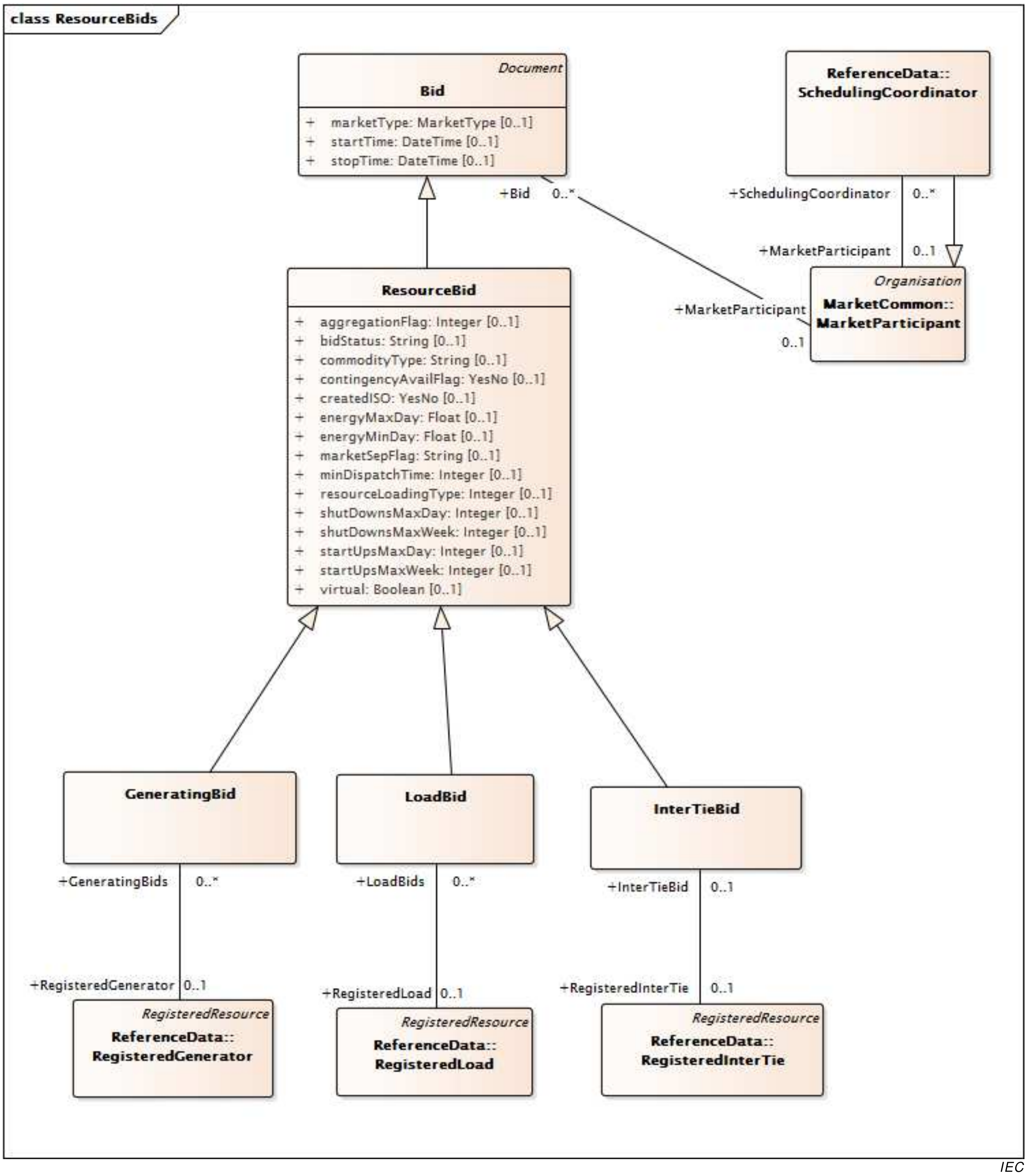


**Figure 57 – Diagramme de classe ParticipantInterfaces::ResourceBids**

Figure 57: représente les classes et les associations utilisées pour modéliser les offres de ressources. Le modèle d'offre de ressource comprend les sous-classes GeneratingBid, LoadBid et IntertieBid pour modéliser les relations avec les classes ResourceBid et Bid.

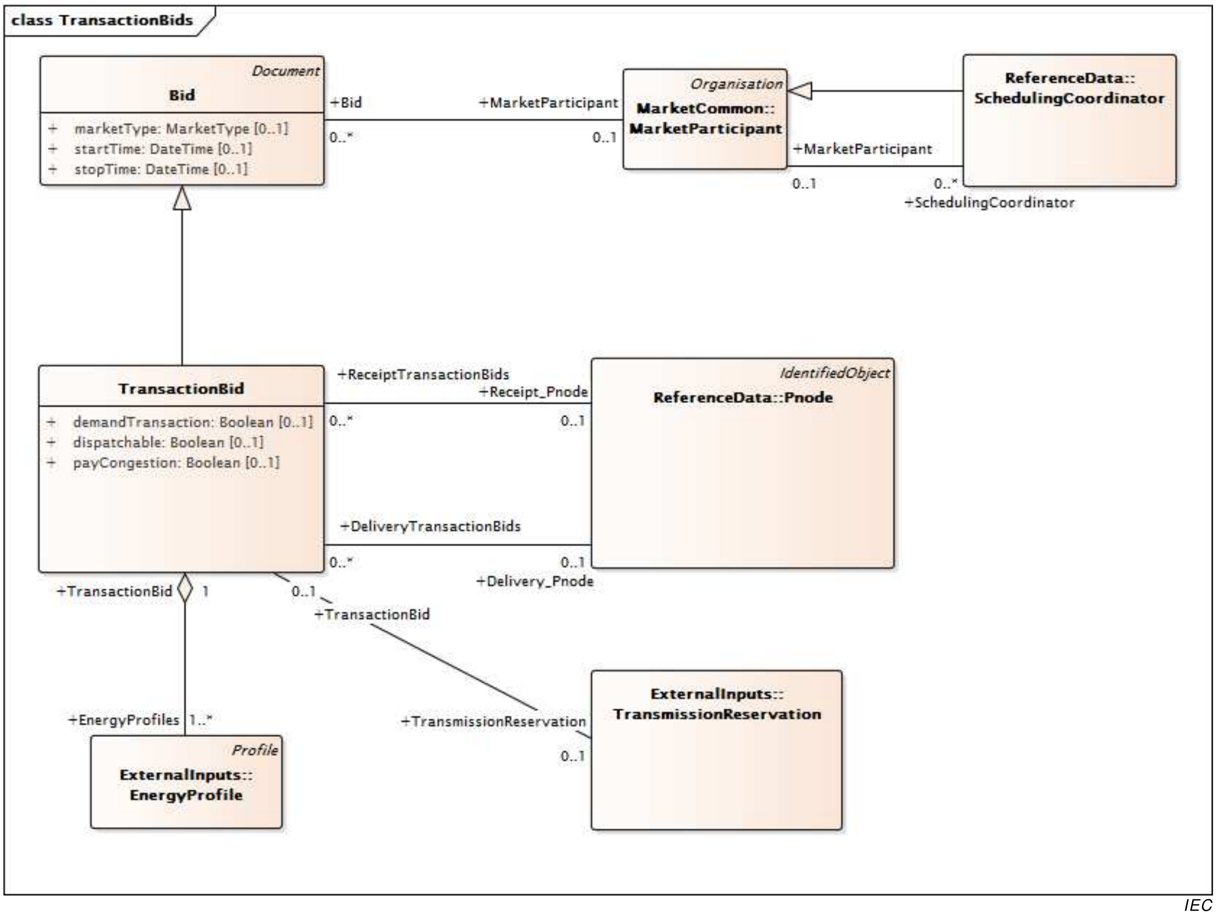


**Figure 58 – Diagramme de classe ParticipantInterfaces::TransactionBids**

Figure 58: représente les classes et les associations utilisées pour modéliser les offres de transactions. Le modèle TransactionBid intègre les sous-classes GeneratingBid, LoadBid et IntertieBid pour modéliser les relations avec les classes ResourceBid et Bid.

### 6.5.8.2 Classe racine ActionRequest

Requête d'action par rapport à une opération existante.

Le Tableau 426 présente tous les attributs de ActionRequest.

**Tableau 426 – Attributs de ParticipantInterfaces::ActionRequest**

| nom | mult | type | description |
|---|---|---|---|
| actionName | 0..1 | ActionType | Type de nom d'action pour la requête d'action. |

Le Tableau 427 présente toutes les extrémités d'association de ActionRequest avec d'autres classes.

**Tableau 427 – Extrémités d’association de ParticipantInterfaces:: ActionRequest avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Bid | 0..* | Bid | |
| 1..1 | Trade | 0..* | Trade | |

### 6.5.8.3 AreaLoadBid

AreaLoadBid n'est pas soumise aux marchés par un participant de marché. À la place, elle consiste simplement en une agrégation de toutes les LoadBids contenues dans une SubControlArea spécifique. Il convient de faire en sorte que cette entité hérite de Bid pour la représentation du cadre temporel (startTime, stopTime) et du type de marché.

Le Tableau 428 présente tous les attributs de AreaLoadBid.

**Tableau 428 – Attributs de ParticipantInterfaces::AreaLoadBid**

| nom | mult | type | description |
|---|---|---|---|
| demandBidMW | 0..1 | Float | Quantité de Mégawatts de l'offre de la demande pour la zone.<br>Utilisation de l'attribut: Il s'agit de la quantité MW de la demande planifiée à J-1 |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 429 présente toutes les extrémités d'association de AreaLoadBid avec d'autres classes.

**Tableau 429 – Extrémités d'association de ParticipantInterfaces:: AreaLoadBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | LoadBid | 0..* | LoadBid | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.4 Classe racine AttributeProperty

Propriété d'un attribut en particulier contenant le nom et la valeur

Le Tableau 430 présente tous les attributs de AttributeProperty.

**Tableau 430 – Attributs de ParticipantInterfaces::AttributeProperty**

| nom | mult | type | description |
|---|---|---|---|
| sequence | 0..1 | String | |
| propertyName | 0..1 | String | |
| propertyValue | 0..1 | String | |

Le Tableau 431 présente toutes les extrémités d'association de AttributeProperty avec d'autres classes.

**Tableau 431 – Extrémités d'association de ParticipantInterfaces:: AttributeProperty avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 1..1 | MktUserAttribute | |

### 6.5.8.5 Bid

Représente les offres d'achat et les offres de vente des services auxiliaires ou d'énergie dans un marché sponsorisé par un RTO.

Le Tableau 432 présente tous les attributs de Bid.

**Tableau 432 – Attributs de ParticipantInterfaces::Bid**

| nom | mult | type | description |
|---|---|---|---|
| startTime | 0..1 | DateTime | Heure et date de début auxquelles s'applique l'offre. |
| stopTime | 0..1 | DateTime | Heure et date de fin auxquelles s'applique l'offre. |
| marketType | 0..1 | MarketType | Type de marché: DAM ou RTM. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 433 présente toutes les extrémités d'association de Bid avec d'autres classes.

**Tableau 433 – Extrémités d'association de ParticipantInterfaces:: Bid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | |
| 0..* | ActionRequest | 1..1 | ActionRequest | |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | |
| 1..1 | ProductBids | 1..* | ProductBid | Une offre comprend au moins une offre de produit pour les produits de marché. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.6 Classe racine BidDistributionFactor

Cette classe permet au coordinateur de programmation de saisir différents intervalles de temps pour les facteurs de répartition

Le Tableau 434 présente tous les attributs de BidDistributionFactor.

**Tableau 434 – Attributs de ParticipantInterfaces::BidDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| timeIntervalStart | 0..1 | DateTime | Début de l'intervalle de temps dans lequel l'offre est valide (aaa-mm-jj hh24: mi: ss). |
| timeIntervalEnd | 0..1 | DateTime | Fin de l'intervalle de temps dans lequel l'offre est valide (aaa-mm-jj hh24: mi: ss). |

Le Tableau 435 présente toutes les extrémités d'association de BidDistributionFactor avec d'autres classes.

**Tableau 435 – Extrémités d'association de ParticipantInterfaces:: BidDistributionFactor avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | PnodeDistributionFactor | 0..* | PnodeDistributionFactor | |
| 0..* | ProductBid | 1..1 | ProductBid | |

### 6.5.8.7 BidError

Cette classe représente les informations d'erreur pour une offre détectée lors de la validation de la soumission

Le Tableau 436 présente tous les attributs de BidError.

**Tableau 436 – Attributs de ParticipantInterfaces::BidError**

| nom | mult | type | description |
|---|---|---|---|
| errPriority | 0..1 | Integer | Numéro de priorité pour le message d'erreur |
| errMessage | 0..1 | String | message d'erreur |
| ruleID | 0..1 | Integer | |
| startTime | 0..1 | DateTime | heure au sein de l'offre à laquelle s'applique l'erreur |
| endTime | 0..1 | DateTime | heure au sein de l'offre à laquelle s'applique l'erreur |
| logTimeStamp | 0..1 | DateTime | |
| componentType | 0..1 | String | |
| msgLevel | 0..1 | Integer | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 437 présente toutes les extrémités d'association de BidError avec d'autres classes.

**Tableau 437 – Extrémités d'association de ParticipantInterfaces:: BidError avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketProduct | 0..1 | MarketProduct | |
| 0..* | ResourceBid | 0..* | ResourceBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.8 BidHourlyProductSchedule

Emboîtement pour les paramètres d'offre qui dépendent d'un type de produit de marché.

Le Tableau 438 présente tous les attributs de BidHourlyProductSchedule.

**Tableau 438 – Attributs de ParticipantInterfaces::BidHourlyProductSchedule**

| nom | mult | type | description |
|---|---|---|---|
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 439 présente toutes les extrémités d'association de BidHourlyProductSchedule avec d'autres classes.

**Tableau 439 – Extrémités d'association de ParticipantInterfaces:: BidHourlyProductSchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.9 BidHourlySchedule

Emboîtement pour les paramètres horaires des offres qui ne dépendent pas des produits.

Le Tableau 440 présente tous les attributs de BidHourlySchedule.

**Tableau 440 – Attributs de ParticipantInterfaces::BidHourlySchedule**

| nom | mult | type | description |
|---|---|---|---|
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 441 présente toutes les extrémités d'association de BidHourlySchedule avec d'autres classes.

**Tableau 441 – Extrémités d'association de ParticipantInterfaces:: BidHourlySchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.10 BidPriceCurve

Relation entre le coût d'exploitation de l'unité en $/h (axe Y) et la sortie de l'unité en MW (axe X).

Le Tableau 442 présente tous les attributs de BidPriceCurve.

**Tableau 442 – Attributs de ParticipantInterfaces::BidPriceCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 443 présente toutes les extrémités d'association de BidPriceCurve avec d'autres classes.

**Tableau 443 – Extrémités d'association de ParticipantInterfaces:: BidPriceCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | BidSchedule | 0..* | BidPriceSchedule | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.11 BidPriceSchedule

Définit des programmes d'offre pour autoriser une offre de produit à utiliser des courbes de prix d'offre spécifiques sur plusieurs intervalles de temps.

Le Tableau 444 présente tous les attributs de BidPriceSchedule.

**Tableau 444 – Attributs de ParticipantInterfaces::BidPriceSchedule**

| nom | mult | type | description |
|---|---|---|---|
| bidType | 0..1 | BidMitigationType | Type d'offre:<br>I – Offre initiale;<br>F – Offre finale |
| mitigationStatus | 0..1 | BidMitigationStatus | Statut d'atténuation:<br>'S' – Atténuation par SMPM en cas de "faute"<br>'L' – Atténuation par LMPM en cas de "faute"<br>'R' – Modification par LMPM en raison des règles RMR<br>'M' – Atténuation en cas de "faute" par SMPM et LMPM<br>'B' – Atténuation en cas de "faute" par SMPM et modification par LMLM en raison des règles RMR<br>'O' – Original |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 445 présente toutes les extrémités d'association de BidPriceSchedule avec d'autres classes.

**Tableau 445 – Extrémités d'association de ParticipantInterfaces:: BidPriceSchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BidPriceCurve | 1..1 | BidPriceCurve | |
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.12 BidSelfSched

Définit les valeurs de programmation automatique à utiliser pour des intervalles de temps en spécifiés.

Le Tableau 446 présente tous les attributs de BidSelfSched.

**Tableau 446 – Attributs de ParticipantInterfaces::BidSelfSched**

| nom | mult | type | description |
|---|---|---|---|
| balancingFlag | 0..1 | YesNo | Il s'agit d'un indicateur Y/N pour la programmation automatique d'une ressource par marché, par date et par heure, utilisant un identifiant TR spécifique. Il indique si une programmation automatique utilisant un TR est en équilibre avec une autre programmation automatique utilisant le même identifiant TR. |
| bidType | 0..1 | BidTypeRMR | bidType a deux types conformément aux exigences de sortie exigées et à la prérépartition qualifiée. |
| priorityFlag | 0..1 | YesNo | Il s'agit d'un indicateur Y/N pour une programmation automatique d'une ressource par marché, par date et par heure, utilisant un identifiant TR spécifique. Il indique si une programmation automatique utilisant un TR sera programmée en priorité dans un DAM/RTM. |
| pumpSelfSchedMw | 0..1 | Float | Contient la quantité de programmation automatique de pompage pour PriceTaker, ExistingTransmissionContract et TransmissionOwnershipRights. Si cette valeur est nulle, alors l'unité est en mode pompage. |
| referenceType | 0..1 | SelfSchedReferenceType | Indication sur le type de programmation automatique en cours de référencement. |
| selfSchedMw | 0..1 | Float | Valeur de programmation automatique |
| selfSchedSptResource | 0..1 | String | Ressource à programmation automatique pour la prise en charge de l'export du preneur de prix |
| selfSchedType | 0..1 | MarketProductSelfSchedType | Cet attribut est utilisé pour spécifier si une offre intègre une offre à programmation automatique. Si cela est le cas, le type de programmation automatique est indiqué. Les valeurs potentielles sont, entre autres:<br>'ETC' – Existing transmission contract (Contrat de transport existant)<br>'TOR' – Transmission ownership right (Droit de propriété de transport)<br>'RMR' – Reliability must run (Marche obligatoire pour fiabilité)<br>'RGMR' – Regulatory must run (Marche obligatoire pour régulation)<br>"RMT" – Reliability must take (Prise obligatoire pour Fiabilité)<br>"PT" – Price taker (Preneur de prix)<br>"LPT" – Low price taker (Preneur de prix inférieur)<br>"SP" – Self provision (Alimentation automatique)<br>"RA" – Resource adequacy (Adéquation de la ressource)<br>Cet attribut est initialement défini dans la classe BidSelfSched |
| updateType | 0..1 | MQSCHGType | |

| nom | mult | type | description |
|---|---|---|---|
| wheelingTransactionReference | 0..1 | String | Identifiant unique pour une transaction de transit. Une transaction de transit consiste en un échange d'énergie en équilibre parmi les ressources d'offre et de demande. |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 447 présente toutes les extrémités d'association de BidSelfSched avec d'autres classes.

**Tableau 447 – Extrémités d’association de ParticipantInterfaces:: BidSelfSched avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | ProductBid | 1..1 | ProductBid | |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.13 BidSet

Ensemble d'offres exclusives mutuellement pour lequel une offre maximum peut être planifiée.

Parmi ces offres de production, une seule peut être planifiée à la fois.

Le Tableau 448 présente tous les attributs de BidSet.

**Tableau 448 – Attributs de ParticipantInterfaces::BidSet**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 449 présente toutes les extrémités d'association de BidSet avec d'autres classes.

**Tableau 449 – Extrémités d’association de ParticipantInterfaces:: BidSet avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 1..* | GeneratingBid | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.14 ChargeComponent

Un élément d’imposition (ChargeComponent) est une liste d'éléments de qualité de charge configurables à intégrer au calcul du règlement et/ou aux éléments déterminants de facturation (BillDeterminant).

Le Tableau 450 présente tous les attributs de ChargeComponent.

**Tableau 450 – Attributs de ParticipantInterfaces::ChargeComponent**

| nom | mult | type | description |
|---|---|---|---|
| deleteStatus | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| message | 0..1 | String | |
| type | 0..1 | String | |
| sum | 0..1 | String | |
| roundOff | 0..1 | String | |
| equation | 0..1 | String | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 451 présente toutes les extrémités d'association de ChargeComponent avec d'autres classes.

**Tableau 451 – Extrémités d'association de ParticipantInterfaces:: ChargeComponent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BillDeterminants | 0..* | BillDeterminant | Un BillDeterminant peut avoir 0-n ChargeComponent et un ChargeComponent peut être associé à 0-n BillDeterminant. |
| 0..* | ChargeTypes | 0..* | ChargeType | Un ChargeType peut avoir 0-n ChargeComponent et un ChargeComponent peut être associé à 0-n ChargeType. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.15 ChargeGroup

ChargeGroup constitue le regroupement des types de charges pour la facturation des règlements. Exemples: services auxiliaires, intérêts, etc.

Le Tableau 452 présente tous les attributs de ChargeGroup.

**Tableau 452 – Attributs de ParticipantInterfaces::ChargeGroup**

| nom | mult | type | description |
|---|---|---|---|
| marketCode | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 453 présente toutes les extrémités d'association de ChargeGroup avec d'autres classes.

**Tableau 453 – Extrémités d'association de ParticipantInterfaces:: ChargeGroup avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | ChargeGroupParent | 0..1 | ChargeGroup | Une instance ChargeGroup peut avoir plusieurs relations avec d'autres instances ChargeGroup. |
| 0..* | ChargeType | 0..* | ChargeType | Une ChargeGroup peut avoir 0-n ChargeType. Une ChargeType peut avoir 0-n ChargeGroup. |
| 0..1 | ChargeGroupChild | 0..* | ChargeGroup | Une instance ChargeGroup peut avoir plusieurs relations avec d'autres instances ChargeGroup. |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.8.16 ChargeType

ChargeType représente la configuration de niveau de base permettant au règlement de traiter des frais spécifiques dans le cadre de la facturation. Exemples: Day Ahead Spinning Reserve Default Invoice Interest Charge (frais d'intérêts de facturation par défaut de la réserve tournante à J-1), etc.

Le Tableau 454 présente tous les attributs de ChargeType.

**Tableau 454 – Attributs de ParticipantInterfaces::ChargeType**

| nom | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| factor | 0..1 | String | |
| chargeOrder | 0..1 | String | |
| frequencyType | 0..1 | String | |
| chargeVersion | 0..1 | String | |
| totalInterval | 0..1 | String | |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 455 présente toutes les extrémités d'association de ChargeType avec d'autres classes.

**Tableau 455 – Extrémités d'association de ParticipantInterfaces:: ChargeType avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktUserAttribute | 0..* | MktUserAttribute | |
| 0..* | ChargeComponents | 0..* | ChargeComponent | Un ChargeType peut avoir 0-n ChargeComponent et un ChargeComponent peut être associé à 0-n ChargeType. |
| 0..* | ChargeGroup | 0..* | ChargeGroup | Un ChargeGroup peut avoir 0-n ChargeType. Un ChargeType peut avoir 0-n ChargeGroup. |
| 0..* | MajorChargeGroup | 0..* | MajorChargeGroup | Un MajorChargeGroup peut avoir 0-n ChargeType. Un ChargeType peut avoir 0-n MajorChargeGroup. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.17 DispatchInstReply

Réponse de la ressource enregistrée accusant réception des instructions de répartition

Le Tableau 456 présente tous les attributs de DispatchInstReply.

**Tableau 456 – Attributs de ParticipantInterfaces::DispatchInstReply**

| nom | mult | type | description |
|---|---|---|---|
| acceptMW | 0..1 | ActivePower | Quantité de MW acceptée par le répondeur. MW de la réponse aka. |
| acceptStatus | 0..1 | DispatchResponseType | Statut d'acceptation soumis par le répondeur. Il est nécessaire de définir le type d'énumération. |
| certificationName | 0..1 | String | Le Subject DN est le X509 Certificate Subject DN. Il s'agit essentiellement du nom de certificat présenté par le client. Dans le cas des certificats ADS, il s'agira du nom de l'utilisateur. Il peut venir d'un client API ou du client MP (interface graphique utilisateur).<br>Le Subject ID comprend normalement plus d'un nom d'utilisateur (Nom Commun); il peut également contenir des informations comme la ville, l'identifiant de l'entreprise, etc. |

| nom | mult | type | description |
|---|---|---|---|
| clearedMW | 0..1 | ActivePower | Quantité de MW associée à l'instruction. Pour les répartitions de liaisons de cinq minutes, il s'agit du Goto MW ou DOT. |
| instructionTime | 0..1 | DateTime | Date/Heure cible pour l'instruction reçue. |
| instructionType | 0..1 | String | type d'instruction:<br>engagement<br>hors de la séquence<br>répartition |
| passIndicator | 0..1 | PassIndicatorType | Type de processus pour la mise en équilibre du marché. |
| receivedTime | 0..1 | DateTime | Marqueur temporel indiquant l'heure à laquelle l'instruction a été reçue. |
| startTime | 0..1 | DateTime | heure de début |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 457 présente toutes les extrémités d'association de DispatchInstReply avec d'autres classes.

**Tableau 457 – Extrémités d'association de ParticipantInterfaces:: DispatchInstReply avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.18 Classe racine EnergyPriceCurve

Relation entre un prix en $ (ou autre unité monétaire)/h (axe Y) et une valeur en MW (axe X).

Le Tableau 458 présente toutes les extrémités d'association de EnergyPriceCurve avec d'autres classes.

**Tableau 458 – Extrémités d'association de ParticipantInterfaces:: EnergyPriceCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnergyTransactions | 0..* | EnergyTransaction | |

#### 6.5.8.19 GeneratingBid

Offre pour fournir de l'énergie/des services auxiliaires à partir d'une unité ou d'une ressource de production

Le Tableau 459 présente tous les attributs de GeneratingBid.

**Tableau 459 – Attributs de ParticipantInterfaces::GeneratingBid**

| nom | mult | type | description |
|---|---|---|---|
| combinedCycleUnitOffer | 0..1 | String | Indiquera si l'unité fait partie ou non d'une offre CC |
| downTimeMax | 0..1 | Float | Temps d'indisponibilité maximum. |
| installedCapacity | 0..1 | Float | Valeur de la capacité installée |
| lowerRampRate | 0..1 | ActivePowerChangeRate | Taux maximum de rampe d'arrêt en MW/min |
| maxEmergencyMW | 0..1 | ActivePower | Puissance assignée disponible pour l'unité en cas d'urgence; supérieure ou égale à la limite économique maximale. |
| maximumEconomicMW | 0..1 | Float | Limite économique supérieure maximale en MW; il convient qu'elle ne dépasse pas la limite MW opérationnelle maximale. |
| minEmergencyMW | 0..1 | ActivePower | Puissance assignée minimale pour l'unité en cas d'urgence; est inférieure ou égale au minimum économique. |
| minimumEconomicMW | 0..1 | Float | Limite économique inférieure en MW devant être supérieure ou égale à la limite MW opérationnelle minimale |
| noLoadCost | 0..1 | Float | Coût à charge nulle fixe de la ressource. |
| notificationTime | 0..1 | Float | Temps exigé pour la notification de l'équipe avant le démarrage de l'unité. |
| operatingMode | 0..1 | String | Mode d'exploitation de l'offre ('C' – cyclique, 'F' – fixe, 'M' – marche obligatoire, 'U' – indisponible) |
| raiseRampRate | 0..1 | ActivePowerChangeRate | Taux maximum de rampe de démarrage en MW/min |
| rampCurveType | 0..1 | Integer | Type de courbe de rampe:<br>0 – Taux de rampe fixe indépendant de la fonction de taux de la sortie MW de l'unité<br>1 – Taux de rampe statiques comme fonction de la sortie MW de l'unité uniquement<br>2 – Taux de rampe dynamiques comme fonction de la sortie MW de l'unité et du temps de rampe |
| startupCost | 0..1 | Float | Coût/prix de démarrage |
| startUpRampRate | 0..1 | ActivePowerChangeRate | Taux de rampe de démarrage de la ressource (MW/min) |
| startUpType | 0..1 | Integer | Type de démarrage de la ressource:<br>1 – Temps de démarrage fixe et coût de démarrage fixe<br>2 – Temps de démarrage en fonction du temps d'indisponibilité et du coût de démarrage fixe<br>3 – Coût de démarrage en fonction du temps d'indisponibilité |
| upTimeMax | 0..1 | Float | Temps de fonctionnement maximum. |

| nom | mult | type | description |
|---|---|---|---|
| aggregationFlag | 0..1 | Integer | Hérité de: ResourceBid |
| bidStatus | 0..1 | String | Hérité de: ResourceBid |
| commodityType | 0..1 | String | Hérité de: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: ResourceBid |
| createdISO | 0..1 | YesNo | Hérité de: ResourceBid |
| energyMaxDay | 0..1 | Float | Hérité de: ResourceBid |
| energyMinDay | 0..1 | Float | Hérité de: ResourceBid |
| marketSepFlag | 0..1 | String | Hérité de: ResourceBid |
| minDispatchTime | 0..1 | Integer | Hérité de: ResourceBid |
| resourceLoadingType | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| virtual | 0..1 | Boolean | Hérité de: ResourceBid |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 460 présente toutes les extrémités d'association de GeneratingBid avec d'autres classes.

**Tableau 460 – Extrémités d'association de ParticipantInterfaces:: GeneratingBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | SecurityConstraints | 0..* | SecurityConstraints | |
| 1..* | BidSet | 0..1 | BidSet | |
| 0..* | NotificationTimeCurve | 0..1 | NotificationTimeCurve | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..* | StartUpCostCurve | 0..1 | StartUpCostCurve | |
| 0..* | StartUpTimeCurve | 0..1 | StartUpTimeCurve | |
| 0..* | BidError | 0..* | BidError | Hérité de: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.20 HourlyPreDispatchSchedule

Indicateur spécifiant qu'une ressource doit avoir une prérépartition horaire. La ressource peut être RegisteredGenerator ou RegisteredInterTie.

Cette programmation est associée avec les paramètres horaires dans une offre de ressource.

Le Tableau 461 présente tous les attributs de HourlyPreDispatchSchedule.

**Tableau 461 – Attributs de ParticipantInterfaces::HourlyPreDispatchSchedule**

| nom | mult | type | description |
|---|---|---|---|
| value | 0..1 | Boolean | Indicateur définissant que pour cette heure dans l'offre de ressource la ressource doit avoir une prérépartition horaire. |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 462 présente toutes les extrémités d'association de HourlyPreDispatchSchedule avec d'autres classes.

**Tableau 462 – Extrémités d’association de ParticipantInterfaces::HourlyPreDispatchSchedule avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | Hérité de: BidHourlySchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.21 InterTieBid

Cette classe représente l'offre de ligne d'interconnexion.

Le Tableau 463 présente tous les attributs de InterTieBid.

**Tableau 463 – Attributs de ParticipantInterfaces::InterTieBid**

| nom | mult | type | description |
|---|---|---|---|
| minHourlyBlock | 0..1 | Integer | Bloc horaire minimal pour une ressource de ligne d'interconnexion fournie au sein de l'offre. |
| aggregationFlag | 0..1 | Integer | Hérité de: ResourceBid |
| bidStatus | 0..1 | String | Hérité de: ResourceBid |
| commodityType | 0..1 | String | Hérité de: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: ResourceBid |
| createdISO | 0..1 | YesNo | Hérité de: ResourceBid |
| energyMaxDay | 0..1 | Float | Hérité de: ResourceBid |
| energyMinDay | 0..1 | Float | Hérité de: ResourceBid |
| marketSepFlag | 0..1 | String | Hérité de: ResourceBid |
| minDispatchTime | 0..1 | Integer | Hérité de: ResourceBid |
| resourceLoadingType | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| virtual | 0..1 | Boolean | Hérité de: ResourceBid |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 464 présente toutes les extrémités d'association de InterTieBid avec d'autres classes.

**Tableau 464 – Extrémités d'association de ParticipantInterfaces:: InterTieBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredInterTie | 0..1 | RegisteredInterTie | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | BidError | 0..* | BidError | Hérité de: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.22 Classe racine InterTieDispatchResponse

Réponse de la ressource de ligne d'interconnexion accusant réception des instructions de répartition.

Le Tableau 465 présente tous les attributs de InterTieDispatchResponse.

**Tableau 465 – Attributs de ParticipantInterfaces::InterTieDispatchResponse**

| nom | mult | type | description |
|---|---|---|---|
| acceptStatus | 0..1 | DispatchResponseType | Statut d'acceptation soumis par le répondeur. Les valeurs valides sont NON-RESPONSE (pas de réponse), ACCEPT (accepter), DECLINE (refuser), PARTIAL (partiel). |
| acceptMW | 0..1 | Float | Quantité de MW acceptée par le répondeur. MW de la réponse aka. |
| clearedMW | 0..1 | Float | Quantité de MW associée à l'instruction. Pour les répartitions de liaisons de 5 min, il s'agit du Goto MW ou DOT |
| startTime | 0..1 | DateTime | Partie de la clé composite que l'application en aval utilise pour se conformer à l'instruction |
| passIndicator | 0..1 | PassIndicatorType | Partie de la clé composite que l'application en aval utilise pour se conformer à l'instruction |

Le Tableau 466 présente toutes les extrémités d'association de InterTieDispatchResponse avec d'autres classes.

**Tableau 466 – Extrémités d’association de ParticipantInterfaces:: InterTieDispatchResponse avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredInterTie | 1..1 | RegisteredInterTie | |

### 6.5.8.23 LoadBid

Offre pour fournir de l'énergie/des services auxiliaires à partir d'une ressource de charge (la charge participante réduit la consommation).

Le Tableau 467 présente tous les attributs de LoadBid.

**Tableau 467 – Attributs de ParticipantInterfaces::LoadBid**

| nom | mult | type | description |
|---|---|---|---|
| dropRampRate | 0..1 | ActivePowerChangeRate | Taux maximal selon lequel la charge peut être réduite (MW/min) |
| loadRedInitiationCost | 0..1 | Money | coût d'initiation de la réduction de charge |
| loadRedInitiationTime | 0..1 | Float | temps d'initiation de la réduction de charge |
| marketDate | 0..1 | DateTime | La date représente le NextMarketDate pour lequel les offres de réponse de charge s'appliquent. |
| meteredValue | 0..1 | Boolean | Indicateur selon lequel la réduction de charge est mesurée. (Voir ci-dessus.)<br><br>Si priceSetting et meteredValue sont tous deux égaux à 1, le système est en mesure de définir le LMP dans le marché en temps réel. |
| minLoad | 0..1 | ActivePower | Seuil minimal de la charge MW en dessous duquel elle ne peut être réduite. |
| minLoadReduction | 0..1 | ActivePower | Quantité MW minimale pour une réduction de charge (par exemple, puissance assignée en MW d'une pompe discrète). |
| minLoadReductionCost | 0..1 | Money | Coût en dollars au niveau de la charge réduite minimale |
| minLoadReductionInterval | 0..1 | Float | La plus petite réduction de charge de la période doit être conservée avant que la charge ne puisse être restaurée à des niveaux normalisés. |
| minTimeBetLoadRed | 0..1 | Float | Temps le plus court pendant lequel la charge doit être maintenue à des niveaux normalisés avant une nouvelle réduction de charge. |
| pickUpRampRate | 0..1 | ActivePowerChangeRate | Taux maximal auquel la charge peut être restaurée (MW/min) |
| priceSetting | 0..1 | Boolean | Indicateur selon lequel le système peut définir les tâches LMP en accord avec la valeur mesurée. Il y a plus de chances que ceci soit dynamique en comparaison avec la valeur mesurée; toutefois, il est obligatoire que la définition des prix et la valeur mesurée restent à la même source. Actuellement, aucun client n'a fait appel à la capacité de mesure, mais si cette option est mise en œuvre, la définition des prix (*Price Setting*) pourrait devenir dynamique. Cependant, la valeur mesurée restera statique. |

| nom | mult | type | description |
|---|---|---|---|
| reqNoticeTime | 0..1 | Float | Période de temps exigée depuis une commande pour réduire une charge à la durée nécessaire pour l'obtention de la réduction de charge minimale. |
| shutdownCost | 0..1 | Money | Coût fixe associé à l'engagement d'une réduction de charge. |
| aggregationFlag | 0..1 | Integer | Hérité de: ResourceBid |
| bidStatus | 0..1 | String | Hérité de: ResourceBid |
| commodityType | 0..1 | String | Hérité de: ResourceBid |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: ResourceBid |
| createdISO | 0..1 | YesNo | Hérité de: ResourceBid |
| energyMaxDay | 0..1 | Float | Hérité de: ResourceBid |
| energyMinDay | 0..1 | Float | Hérité de: ResourceBid |
| marketSepFlag | 0..1 | String | Hérité de: ResourceBid |
| minDispatchTime | 0..1 | Integer | Hérité de: ResourceBid |
| resourceLoadingType | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| shutDownsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxDay | 0..1 | Integer | Hérité de: ResourceBid |
| startUpsMaxWeek | 0..1 | Integer | Hérité de: ResourceBid |
| virtual | 0..1 | Boolean | Hérité de: ResourceBid |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 468 présente toutes les extrémités d'association de LoadBid avec d'autres classes.

**Tableau 468 – Extrémités d'association de ParticipantInterfaces:: LoadBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | LoadReductionPriceCurve | 0..* | LoadReductionPriceCurve | |
| 0..1 | RampRateCurve | 0..* | RampRateCurve | |
| 0..* | AreaLoadBid | 0..1 | AreaLoadBid | |
| 0..* | RegisteredLoad | 0..1 | RegisteredLoad | |
| 0..* | BidError | 0..* | BidError | Hérité de: ResourceBid |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.24 Classe racine LoadFollowingInst

Charge de sous-système mesurée suivant l'instruction

Le Tableau 469 présente tous les attributs de LoadFollowingInst.

**Tableau 469 – Attributs de ParticipantInterfaces::LoadFollowingInst**

| nom | mult | type | description |
|---|---|---|---|
| mssInstructionID | 0..1 | String | Identifiant d'instruction unique pour chaque instruction; affecté par le coordinateur de programmation et fourni à l'ADS. L'ADS passe à travers. |
| startTime | 0..1 | DateTime | Heure de début de l'instruction |
| endTime | 0..1 | DateTime | Heure de fin de l'instruction |
| loadFollowingMW | 0..1 | Float | Élément MW positif pour le suivi de charge ascendant et négatif pour le suivi de charge descendant |

Le Tableau 470 présente toutes les extrémités d'association de LoadFollowingInst avec d'autres classes.

**Tableau 470 – Extrémités d'association de ParticipantInterfaces:: LoadFollowingInst avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |

### 6.5.8.25 LoadReductionPriceCurve

Il s'agit de la sensibilité du prix que le soumissionnaire exprime pour permettre l'interruption de charge du marché. Relation entre le prix (axe Y1) et les MW (axe X).

Le Tableau 471 présente tous les attributs de LoadReductionPriceCurve.

**Tableau 471 – Attributs de ParticipantInterfaces::LoadReductionPriceCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 472 présente toutes les extrémités d'association de LoadReductionPriceCurve avec d'autres classes.

**Tableau 472 – Extrémités d'association de ParticipantInterfaces:: LoadReductionPriceCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | LoadBid | 1..1 | LoadBid | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.26 MajorChargeGroup

Un groupe de charge majeur (Major Charge Group) est identique au type de facturation qui fournit le plus haut niveau de regroupement pour la configuration des types de frais. Exemples: marché, FERC, RMR.

Le Tableau 473 présente tous les attributs de MajorChargeGroup.

**Tableau 473 – Attributs de ParticipantInterfaces::MajorChargeGroup**

| nom | mult | type | description |
|---|---|---|---|
| runType | 0..1 | String | |
| runVersion | 0..1 | String | |
| frequencyType | 0..1 | String | |
| invoiceType | 0..1 | String | |
| effectiveDate | 0..1 | DateTime | |
| terminationDate | 0..1 | DateTime | |
| requireAutorun | 0..1 | String | |
| revisionNumber | 0..1 | String | Numéro de révision pour le groupe de charge majeur |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 474 présente toutes les extrémités d'association de MajorChargeGroup avec d'autres classes.

**Tableau 474 – Extrémités d’association de ParticipantInterfaces:: MajorChargeGroup avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | Settlement | 0..* | Settlement | |
| 0..* | ChargeType | 0..* | ChargeType | Un MajorChargeGroup peut avoir 0-n ChargeType. Un ChargeType peut être associé à 0-n MajorChargeGroup. |
| 0..1 | MktScheduledEvent | 0..* | MarketScheduledEvent | |
| 1..* | MarketInvoice | 0..* | MarketInvoice | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.27 MarketScheduledEvent

Indique à un événement de déclencher une ou plusieurs activité(s), telles que la lecture d'un compteur, le recalcul d'une facture, la requête d'une tâche, lorsque les unités de production doivent être planifiées pour la maintenance, lorsqu'un transformateur est prévu pour une remise à neuf, etc.

Le Tableau 475 présente tous les attributs de MarketScheduledEvent.

**Tableau 475 – Attributs de ParticipantInterfaces::MarketScheduledEvent**

| nom | mult | type | description |
|---|---|---|---|
| category | 0..1 | String | Catégorie de l'événement planifié. |
| duration | 0..1 | Seconds | Durée de l'événement planifié, par exemple, temps de rampe entre les valeurs. |
| status | 0..1 | Status | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 476 présente toutes les extrémités d'association de MarketScheduledEvent avec d'autres classes.

**Tableau 476 – Extrémités d’association de ParticipantInterfaces:: MarketScheduledEvent avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MajorChargeGroup | 0..1 | MajorChargeGroup | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.28 NotificationTimeCurve

Courbe de temps de notification en fonction du temps d'indisponibilité. Relation entre le temps de notification de l'équipe (axe Y1) et le temps de démarrage de l'unité (axe Y2) par rapport au temps d'indisponibilité écoulé de l'unité (axe X).

Le Tableau 477 présente tous les attributs de NotificationTimeCurve.

**Tableau 477 – Attributs de ParticipantInterfaces::NotificationTimeCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 478 présente toutes les extrémités d'association de NotificationTimeCurve avec d'autres classes.

**Tableau 478 – Extrémités d'association de ParticipantInterfaces:: NotificationTimeCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 0..* | GeneratingBid | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.29 OpenTieSchedule

Résultat de la validation de l'offre par rapport aux conditions qui peuvent exister sur un échange qui devient déconnecté ou considérablement réduit comparé au flux de MW.

Cette programmation est associée avec les paramètres horaires dans une offre de ressource.

Le Tableau 479 présente tous les attributs de OpenTieSchedule.

**Tableau 479 – Attributs de ParticipantInterfaces::OpenTieSchedule**

| nom | mult | type | description |
|---|---|---|---|
| value | 0..1 | Boolean | |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 480 présente toutes les extrémités d'association de OpenTieSchedule avec d'autres classes.

**Tableau 480 – Extrémités d'association de ParticipantInterfaces:: OpenTieSchedule avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Bid | 1..1 | Bid | Hérité de: BidHourlySchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.30 ProductBid

Composant d'une offre qui se rapporte à un produit de marché.

Le Tableau 481 présente tous les attributs de ProductBid.

**Tableau 481 – Attributs de ParticipantInterfaces::ProductBid**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 482 présente toutes les extrémités d'association de ProductBid avec d'autres classes.

**Tableau 482 – Extrémités d'association de ParticipantInterfaces:: ProductBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..* | Bid | 1..1 | Bid | Une offre comprend au moins une offre de produit pour les produits de marché |
| 1..1 | BidDistributionFactor | 0..* | BidDistributionFactor | |
| 1..1 | BidSchedule | 0..* | BidPriceSchedule | |
| 1..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..* | MarketProduct | 1..1 | MarketProduct | |
| 1..1 | BidHourlyProductSchedule | 0..* | BidHourlyProductSchedule | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.8.31 PumpingCostSchedule

Coût d'exploitation d'une unité hydraulique de stockage à pompe fonctionnant en tant que pompe hydraulique.

Ce programme est associé aux paramètres horaires dans une offre de ressource associée à un produit spécifique compris dans l'offre.

Le Tableau 483 présente tous les attributs de PumpingCostSchedule.

**Tableau 483 – Attributs de ParticipantInterfaces::PumpingCostSchedule**

| nom | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 484 présente toutes les extrémités d'association de PumpingCostSchedule avec d'autres classes.

**Tableau 484 – Extrémités d’association de ParticipantInterfaces:: PumpingCostSchedule avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | Hérité de: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.8.32 PumpingLevelSchedule

Niveau d'exploitation fixe d'une unité hydraulique de stockage à pompe fonctionnant en tant que pompe hydraulique. Associé au type de produit du marché de l'énergie.

Ce programme est associé aux paramètres horaires dans une offre de ressource associée à un produit spécifique compris dans l'offre.

Le Tableau 485 présente tous les attributs de PumpingLevelSchedule.

**Tableau 485 – Attributs de ParticipantInterfaces::PumpingLevelSchedule**

| nom | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 486 présente toutes les extrémités d'association de PumpingLevelSchedule avec d'autres classes.

**Tableau 486 – Extrémités d’association de ParticipantInterfaces:: PumpingLevelSchedule avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | Hérité de: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.33 PumpingShutDownCostSchedule

Coût d'arrêt d'une unité hydraulique de stockage à pompe (en mode pompage) ou d'une pompe.

Ce programme est associé aux paramètres horaires dans une offre de ressource associée à un produit spécifique compris dans la soumission.

Le Tableau 487 présente tous les attributs de PumpingShutDownCostSchedule.

**Tableau 487 – Attributs de ParticipantInterfaces::PumpingShutDownCostSchedule**

| nom | mult | type | description |
|---|---|---|---|
| value | 0..1 | Float | |
| endTime | 0..1 | DateTime | Hérité de: RegularIntervalSchedule |
| timeStep | 0..1 | Seconds | Hérité de: RegularIntervalSchedule |
| startTime | 0..1 | DateTime | Hérité de: BasicIntervalSchedule |
| value1Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |

| nom | mult | type | description |
|---|---|---|---|
| Value1Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| value2Multiplier | 0..1 | UnitMultiplier | Hérité de: BasicIntervalSchedule |
| Value2Unit | 0..1 | UnitSymbol | Hérité de: BasicIntervalSchedule |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 488 présente toutes les extrémités d'association de PumpingShutDownCostSchedule avec d'autres classes.

**Tableau 488 – Extrémités d’association de ParticipantInterfaces::PumpingShutDownCostSchedule avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ProductBid | 1..1 | ProductBid | Hérité de: BidHourlyProductSchedule |
| 1..1 | TimePoints | 1..* | RegularTimePoint | Hérité de: RegularIntervalSchedule |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.34 RampRateCurve

Taux de rampe en fonction de la sortie en MW de la ressource

Le Tableau 489 présente tous les attributs de RampRateCurve.

**Tableau 489 – Attributs de ParticipantInterfaces::RampRateCurve**

| nom | mult | type | description |
|---|---|---|---|
| condition | 0..1 | RampRateCondition | condition du taux de rampe |
| constraintRampType | 0..1 | ConstraintRampType | Condition qui identifie s'il convient de contraindre une ressource de production de l'approvisionnement en services auxiliaires si son changement de programmation ou de répartition au cours des heures ou des intervalles commerciaux exige plus qu'une fraction spécifiée de la durée des heures ou intervalles commerciaux.<br>Les valeurs valides sont Fast/Slow (Rapide/Lent). |
| rampRateType | 0..1 | RampRateType | Manière dont le taux de rampe est appliqué (par exemple, augmentation ou diminution, comme lors de l'application à une ressource de production) |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |

| nom | mult | type | description |
|---|---|---|---|
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 490 présente toutes les extrémités d'association de RampRateCurve avec d'autres classes.

**Tableau 490 – Extrémités d'association de ParticipantInterfaces:: RampRateCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | GeneratingBid | 0..1 | GeneratingBid | |
| 0..* | InterTieBid | 0..1 | InterTieBid | |
| 0..* | LoadBid | 0..1 | LoadBid | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.35 ResourceBid

Offre d'énergie pour production, charge ou type virtuel de l'ensemble de la période de transaction sur le marché (c'est-à-dire un jour pour le marché à J-1 et une heure pour le marché en temps réel)

Le Tableau 491 présente tous les attributs de ResourceBid.

**Tableau 491 – Attributs de ParticipantInterfaces::ResourceBid**

| nom | mult | type | description |
|---|---|---|---|
| aggregationFlag | 0..1 | Integer | Indicateur d'agrégation<br>0: niveau de ressource individuel<br>1: emplacement du nœud agrégé<br>2: emplacement du prix agrégé |
| bidStatus | 0..1 | String | |
| commodityType | 0..1 | String | Type de produit énergétique:<br>'En' – Énergie<br>'RU' – Regulation Up (Régulation à la hausse)<br>'Rd' – Regulation Dn (Régulation à la baisse)<br>'Sr' – Spinning Reserve (Réserve tournante)<br>'Nr' – Non-Spinning Reserve (Réserve non tournante)<br>'Or' – Operating Reserve (Réserve d'exploitation) |
| contingencyAvailFlag | 0..1 | YesNo | Disponibilité de la réserve d'exploitation contingente (Yes/No). La ressource est disponible pour participer uniquement avec la capacité dans la répartition de la contingence. |
| createdISO | 0..1 | YesNo | Un Yes indique que cette offre a été créée par l'ISO. |
| energyMaxDay | 0..1 | Float | Quantité maximale d'énergie par jour pouvant être produite au cours de la période d'échange en MWh |
| energyMinDay | 0..1 | Float | Quantité minimale d'énergie par jour qui doit être produite au cours de la période d'échange en MWh |
| marketSepFlag | 0..1 | String | Indicateur de séparation de marché<br>'Y' – Appliquer les contraintes de séparation de marché pour cette offre.<br>'N' – Ne pas appliquer les contraintes de séparation de marché pour cette offre. |
| minDispatchTime | 0..1 | Integer | nombre minimal d'heures consécutives pendant lesquelles une ressource doit être répartie en cas d'acceptation de l'offre |
| resourceLoadingType | 0..1 | Integer | Type de courbe de chargement de ressource<br>1 – chargement continu progressif<br>2 – chargement continu linéaire par morceaux<br>3 – chargement en masse |
| shutDownsMaxDay | 0..1 | Integer | Nombre maximal d'arrêts par jour. |
| shutDownsMaxWeek | 0..1 | Integer | Nombre maximal d'arrêts par semaine. |
| startUpsMaxDay | 0..1 | Integer | Nombre maximal de démarrages par jour. |
| startUpsMaxWeek | 0..1 | Integer | Nombre maximal de démarrages par semaine. |
| virtual | 0..1 | Boolean | True si l'offre est virtuelle. L'offre est considérée comme non virtuelle si cet attribut est manquant. |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | Status | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 492 présente toutes les extrémités d'association de ResourceBid avec d'autres classes.

**Tableau 492 – Extrémités d'association de ParticipantInterfaces:: ResourceBid avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BidError | 0..* | BidError | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.8.36 StartUpCostCurve

Coûts et temps de démarrage en fonction du temps d'indisponibilité. Relation entre le coût de démarrage de l'unité (axe Y1) et le temps d'indisponibilité écoulé de l'unité (axe X).

Le Tableau 493 présente tous les attributs de StartUpCostCurve.

**Tableau 493 – Attributs de ParticipantInterfaces::StartUpCostCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 494 présente toutes les extrémités d'association de StartUpCostCurve avec d'autres classes.

**Tableau 494 – Extrémités d’association de ParticipantInterfaces:: StartUpCostCurve avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBid | 0..* | GeneratingBid | |
| 0..* | RegisteredGenerators | 0..* | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.37 StartUpTimeCurve

Courbe du temps de démarrage en fonction du temps d'indisponibilité, dans laquelle le temps est spécifié en minutes. Relation entre le temps de démarrage de l'unité (axe Y1) et le temps d'indisponibilité écoulé de l'unité (axe X).

Le Tableau 495 présente tous les attributs de StartUpTimeCurve.

**Tableau 495 – Attributs de ParticipantInterfaces::StartUpTimeCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 496 présente toutes les extrémités d'association de StartUpTimeCurve avec d'autres classes.

**Tableau 496 – Extrémités d'association de ParticipantInterfaces:: StartUpTimeCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBid | 0..* | GeneratingBid | |
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerato r | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.38 Trade

Opérations entre coordinateurs de programmation pour la modélisation des opérations financières qui peuvent avoir des répercussions sur le règlement

Le Tableau 497 présente tous les attributs de Trade.

**Tableau 497 – Attributs de ParticipantInterfaces::Trade**

| nom | mult | type | description |
|---|---|---|---|
| adjustedTradeQuantity | 0..1 | Float | Quantité d'opérations validées et acceptées sur le marché actuel d'un échange physique d'énergie. |
| counterTradeQuantity | 0..1 | Float | Quantité de MW soumise par le coordinateur de programmation du compteur pour un même échange |
| dependOnTradeName | 0..1 | String | DependOnISTName renvoie directement au nom IST unique dans la chaîne d'échanges physiques d'énergie. |
| lastModified | 0..1 | DateTime | Date et heure de la dernière modification de l'opération. |
| marketType | 0..1 | MarketType | |
| startTime | 0..1 | DateTime | Heure et date de début auxquelles s'applique l'opération. |
| stopTime | 0..1 | DateTime | Heure et date de fin auxquelles s'applique l'opération. |
| submitFromTimeStamp | 0..1 | DateTime | Horodatage de l'envoi de l'offre de l'opération du coordinateur de programmation au participant de marché |
| submitFromUser | 0..1 | String | Identifiant utilisateur de l'offre de l'opération du coordinateur de programmation |
| submitToTimeStamp | 0..1 | DateTime | Horodatage de l'envoi de l'offre pour l'opération du coordinateur de programmation au participant de marché |
| submitToUser | 0..1 | String | Identifiant utilisateur de l'offre pour l'opération du coordinateur de programmation |
| tradeQuantity | 0..1 | Float | tradeQuantity:<br>Si tradeType = IST, quantité d'un échange d'énergie.<br>Si tradeType = AST, quantité d'un échange obligatoire de services auxiliaires.<br>Si tradeType = UCT, quantité d'un échange obligatoire d'engagement d'unité. |
| tradeStatus | 0..1 | String | Statut résultant de l'opération à la suite du traitement du moteur de règles. |
| updateTimeStamp | 0..1 | DateTime | |
| updateUser | 0..1 | String | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 498 présente toutes les extrémités d'association de Trade avec d'autres classes.

**Tableau 498 – Extrémités d'association de ParticipantInterfaces::
Trade avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | TradeError | 0..* | TradeError | |
| 0..* | ActionRequest | 1..1 | ActionRequest | |
| 0..* | Pnode | 0..1 | Pnode | |
| 0..* | submitToSchedulingCoordinator | 0..1 | SchedulingCoordinator | |
| 0..* | To_SC | 1..1 | SchedulingCoordinator | |
| 0..* | From_SC | 1..1 | SchedulingCoordinator | |
| 0..* | submitFromSchedulingCoordinator | 0..1 | SchedulingCoordinator | |
| 0..* | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..* | TradeProduct | 1..1 | TradeProduct | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.39 TradeError

Erreur d'opération et messages d'avertissement associés au traitement du moteur de règles de l'opération soumise.

Le Tableau 499 présente tous les attributs de TradeError.

**Tableau 499 – Attributs de ParticipantInterfaces::TradeError**

| nom | mult | type | description |
|---|---|---|---|
| errPriority | 0..1 | Integer | Numéro de priorité pour le message d'erreur |
| errMessage | 0..1 | String | message d'erreur |
| ruleID | 0..1 | Integer | Identifiant de règle qui déclenche le message d'erreur/d'avertissement |
| startTime | 0..1 | DateTime | heure au sein de l'opération à laquelle s'applique l'erreur |
| endTime | 0..1 | DateTime | heure au sein de l'opération à laquelle s'applique l'erreur |
| logTimeStamp | 0..1 | DateTime | Horodatage du message d'erreur/d'avertissement enregistré |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 500 présente toutes les extrémités d'association de TradeError avec d'autres classes.

**Tableau 500 – Extrémités d'association de ParticipantInterfaces:: TradeError avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Trade | 0..1 | Trade | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.8.40 Classe racine TradeProduct

| | |
|---|---|
| TradeType | TradeProduct |
| IST (Commerce entre coordinateurs de programmation) | PHY (Physical Energy Trade – Échange physique d'énergie) |
| IST | APN (Échanges d'énergie au niveau des nœuds de tarification agrégés) |
| IST | CPT (Échange d'énergie physique convertie) |
| AST (Commerce de services auxiliaires) | RUT (Regulation Up Trade – Opération de régulation à la hausse) |
| AST | RDT (Regulation Down Trade – Opération de régulation à la baisse) |
| AST | SRT (Spinning Reserve Trade – Opération de réserve tournante) |
| AST | NRT (Non-Spinning Reserve Trade – Opération de réserve non tournante) |
| UCT (Unit Commitment Trade – Opération d'engagement d'unité) | nulle |

Le Tableau 501 présente tous les attributs de TradeProduct.

**Tableau 501 – Attributs de ParticipantInterfaces::TradeProduct**

| nom | mult | type | description |
|---|---|---|---|
| tradeType | 0..1 | TradeType | IST (InterSc Trade) – Commerce entre coordinateurs de programmation;<br>AST *(*Ancilliary Services Trade) – Commerce de services auxiliaires;<br>UCT (Unit Commitment Trade) – Commerce d'engagement d'unité |
| tradeProductType | 0..1 | String | PHY (Échange d'énergie physique);<br>APN (Échanges d'énergie au niveau des nœuds de tarification agrégés)<br>CPT (Échange d'énergie physique convertie)<br>RUT (Regulation Up Trade – Opération de régulation à la hausse);<br>RDT (Regulation Down Trade – Opération de régulation à la baisse);<br>SRT (Spinning Reserve Trade – Opération de réserve tournante)<br>NRT (Non-Spinning Reserve Trade – Opération de réserve non tournante) |

Le Tableau 502 présente toutes les extrémités d'association de TradeProduct avec d'autres classes.

**Tableau 502 – Extrémités d'association de ParticipantInterfaces:: TradeProduct avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Trade | 0..* | Trade | |

### 6.5.8.41 TransactionBid

Transactions bilatérales ou programmées pour l'énergie et les services auxiliaires pris en considération par le processus de mise en équilibre du marché

Le Tableau 503 présente tous les attributs de TransactionBid.

**Tableau 503 – Attributs de ParticipantInterfaces::TransactionBid**

| nom | mult | type | description |
|---|---|---|---|
| demandTransaction | 0..1 | Boolean | Indiquer True s'il s'agit d'une transaction de demande. |
| dispatchable | 0..1 | Boolean | Indiquer True s'il s'agit d'une transaction répartissable. |
| payCongestion | 0..1 | Boolean | Indiquer True s'il s'agit d'une transaction de règlement volontaire (*willing to pay*). Cet indicateur est utilisé pour déterminer si un programme est prêt à payer une congestion. |
| startTime | 0..1 | DateTime | Hérité de: Bid |
| stopTime | 0..1 | DateTime | Hérité de: Bid |
| marketType | 0..1 | MarketType | Hérité de: Bid |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |

| nom | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 504 présente toutes les extrémités d'association de TransactionBid avec d'autres classes.

**Tableau 504 – Extrémités d’association de ParticipantInterfaces:: TransactionBid avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | EnergyProfiles | 1..* | EnergyProfile | |
| 0..1 | TransmissionReservation | 0..1 | TransmissionReservation | |
| 0..1 | TransactionBidResults | 0..* | TransactionBidResults | |
| 0..* | Receipt_Pnode | 0..1 | Pnode | |
| 0..* | Delivery_Pnode | 0..1 | Pnode | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: Bid |
| 0..* | EnergyMarket | 1..1 | EnergyMarket | Hérité de: Bid |
| 0..1 | ChargeProfiles | 0..* | ChargeProfile | Hérité de: Bid |
| 1..1 | MitigatedBidSegment | 0..* | MitigatedBidSegment | Hérité de: Bid |
| 0..1 | MitigatedBid | 0..* | MitigatedBid | Hérité de: Bid |
| 0..1 | RMRDetermination | 0..* | RMRDetermination | Hérité de: Bid |
| 0..* | ActionRequest | 1..1 | ActionRequest | Hérité de: Bid |
| 1..1 | BidHourlySchedule | 0..* | BidHourlySchedule | Hérité de: Bid |
| 1..1 | ProductBids | 1..* | ProductBid | Hérité de: Bid |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

## 6.5.9 Paquetage ReferenceData

### 6.5.9.1 Généralités

Données de référence statiques du marché.

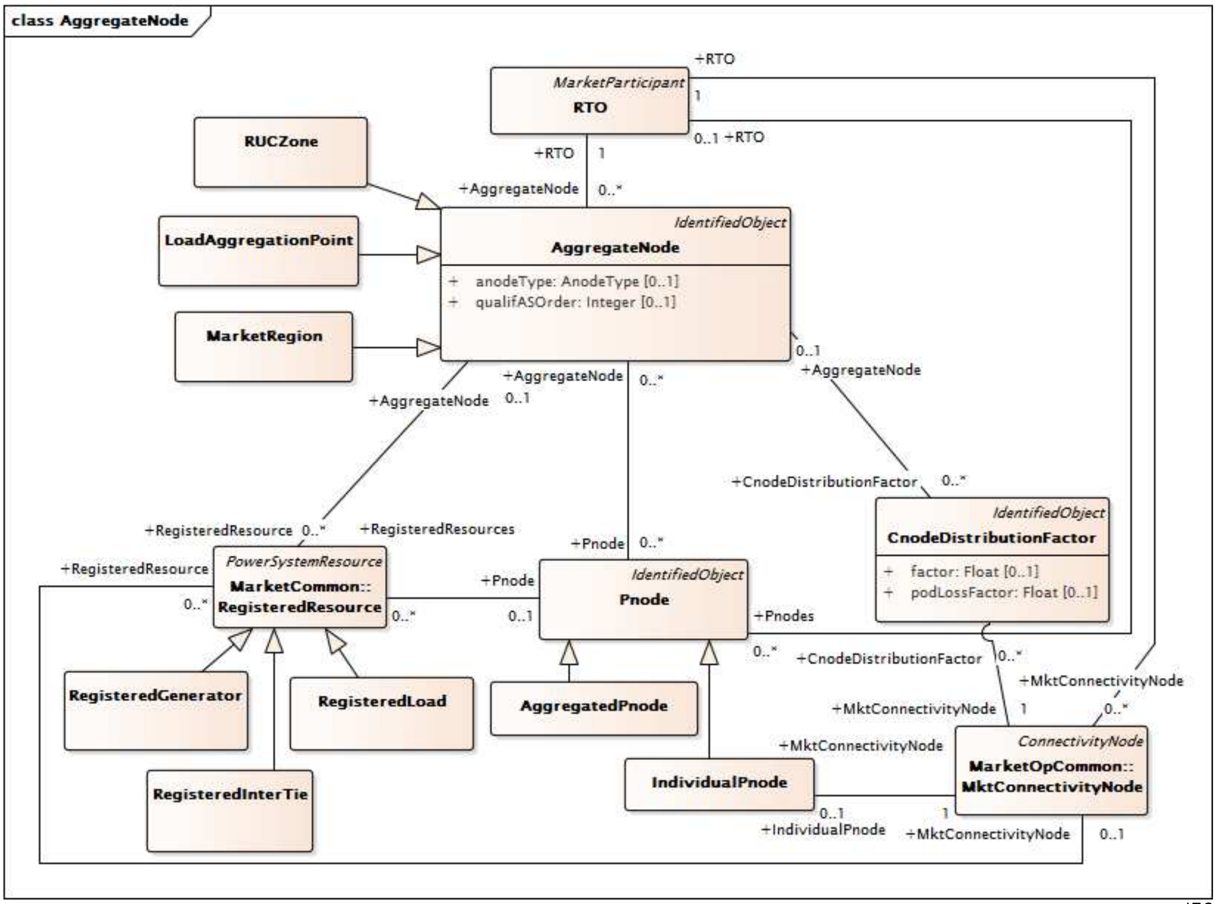


Figure 59 – Diagramme de classe ReferenceData::AggregateNode

Figure 59: représente les classes et les associations utilisées pour modéliser les nœuds agrégés. Les nœuds agrégés permettent le regroupement de Pnode et des Cnodes qui est défini davantage par AnodeType. Les ReliabilityUnitCommitment Zone (zones d'engagement d'unité pour fiabilité) sont modélisées par la sous-classe RUCZone. Les points d'agrégation de charge sont modélisés par la sous-classe LoadAggregationPoint; les régions de marché sont modélisées par la sous-classe MarketRegion. Les ressources enregistrées sont associées aux nœuds agrégés.

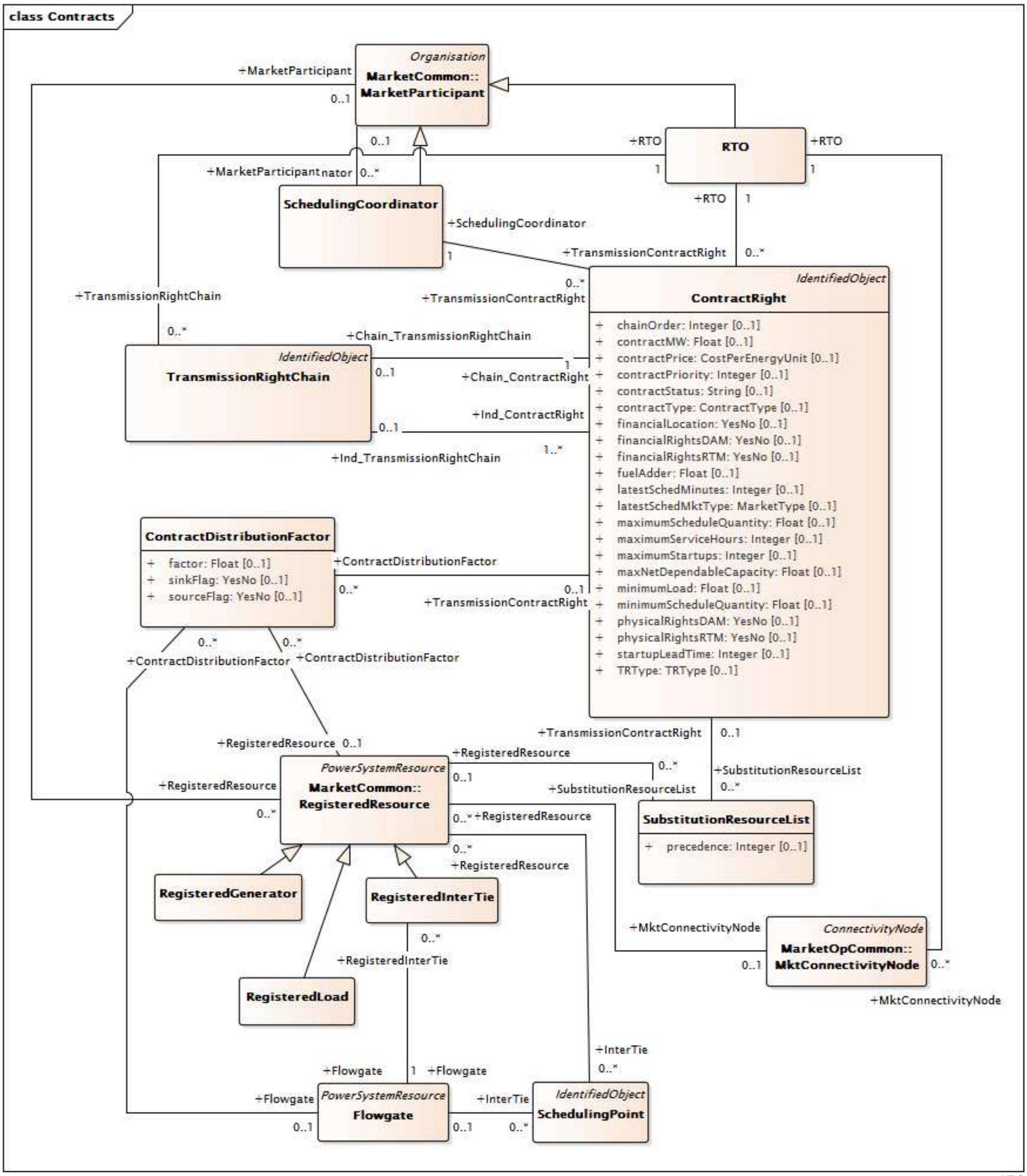


**Figure 60 – Diagramme de classe ReferenceData::Contracts**

Figure 60: représente les classes et les associations utilisées pour modéliser les droits de contrats. La classe ContractRight est utilisée pour récapituler les informations contractuelles. Les types de contrats comprennent les contrats de transport existants (ETC), les droits de propriété de transport (TOR), les contrats RMR et les contrats RMT. Les ETC et les TOR représentent les contrats d'utilisation de transport maintenus en vertu d'une clause d'antériorité qui existent lors du démarrage d'un ISO. Les RMR et RMT ont un impact sur la production. La figure représente les associations avec TransmissionRightChain qui peuvent être utilisées pour modéliser les maillons des ETC et avec les coordinateurs de programmation. Les associations sont par ailleurs prises en considération pour la modélisation de la répartition vers les sources et les puits (classe DistributionFactor de contrat) et vers les ressources enregistrées (classe RegisteredResource).

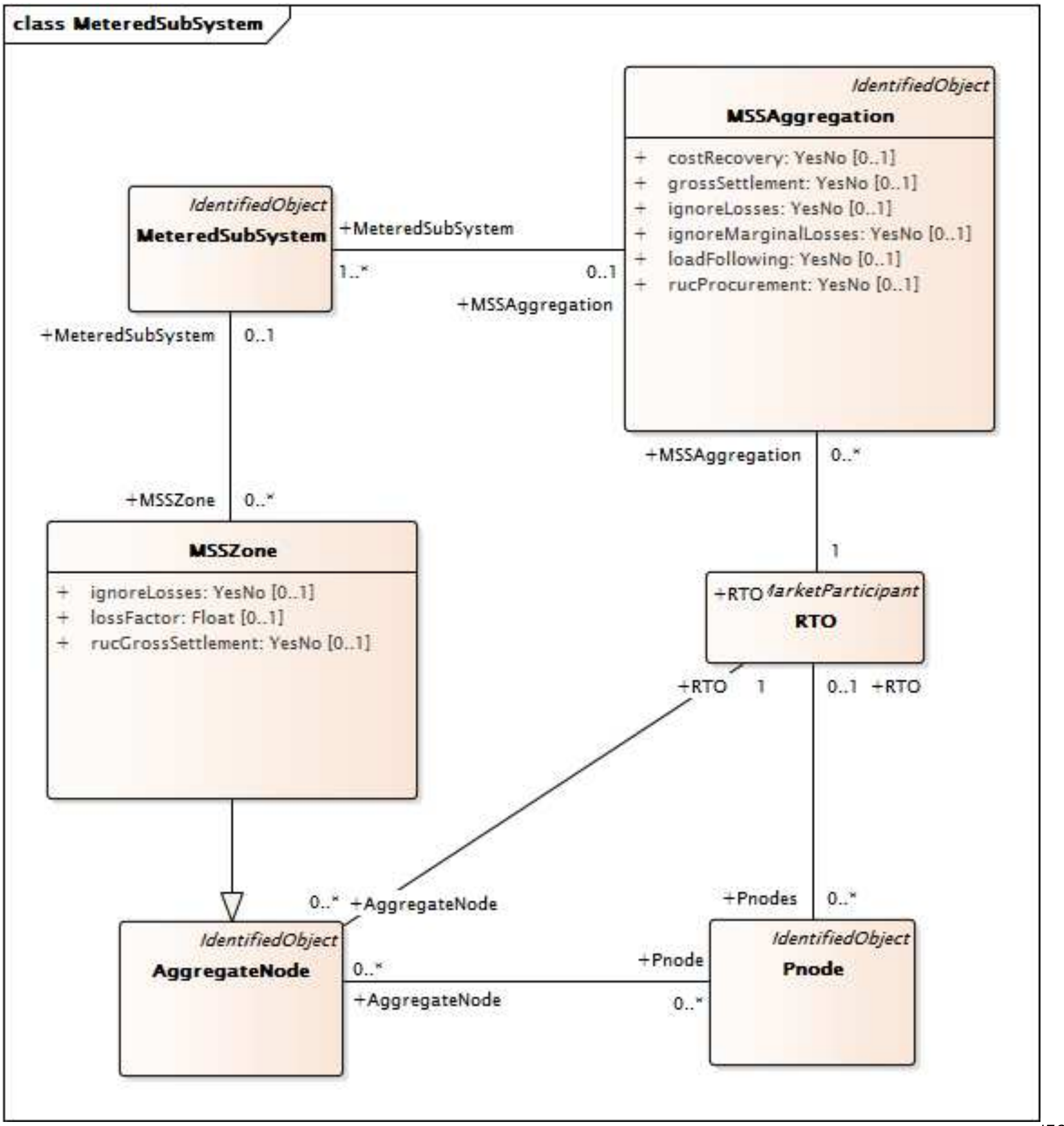


**Figure 61 – Diagramme de classe ReferenceData::MeteredSubSystem**

Figure 61: représente les classes et les associations utilisées pour modéliser les sous-systèmes avec comptage. Un sous-système mesuré (MSS – Metered SubSystem) est un système à localisation géographique contiguë qui était auparavant exploité en tant que zone de réglage séparée et qui est à présent utilisé conformément à un accord relatif aux MSS. Les associations sont représentées à l'aide de la classe MSSZone, qui est utilisée pour modéliser les zones avec MSS. MSSZone est un sous-type du AggregatedNode.

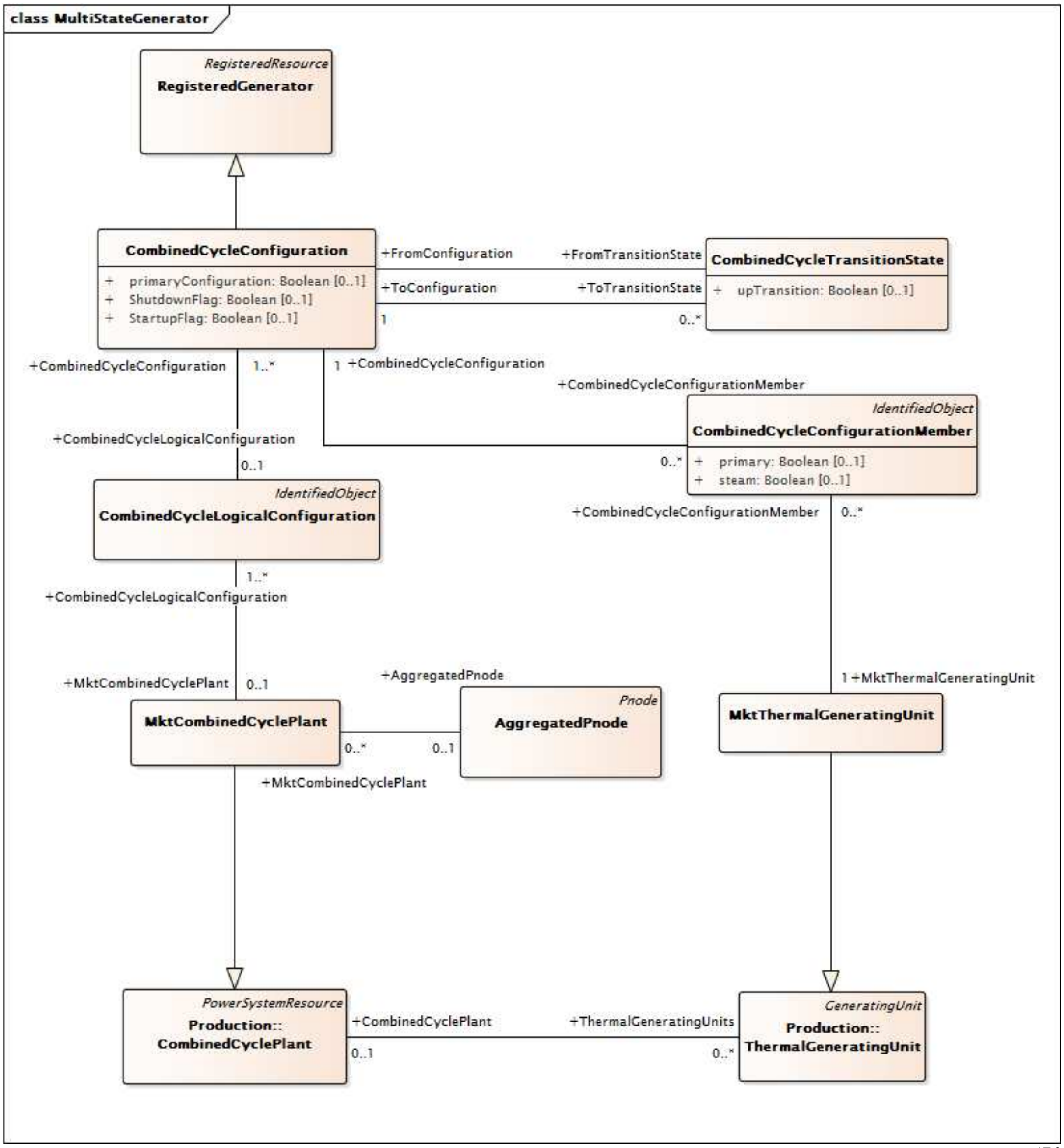


**Figure 62 – Diagramme de classe ReferenceData::MultiStateGenerator**

Figure 62: représente les classes et les associations utilisées pour modéliser les unités de production dans plusieurs états (MultiState Generating Units), tels que les centrales à cycle combiné. La centrale à cycle combiné est associée aux classes LogicalConfiguration, Configuration et ConfigurationMember, qui sont utilisées pour modéliser les différents modes d'exploitation de la centrale. Les classes MktCombinedCyclePlant et MktThermalGeneratingUnit sont des sous-classes des classes parents du paquetage Production 61970.

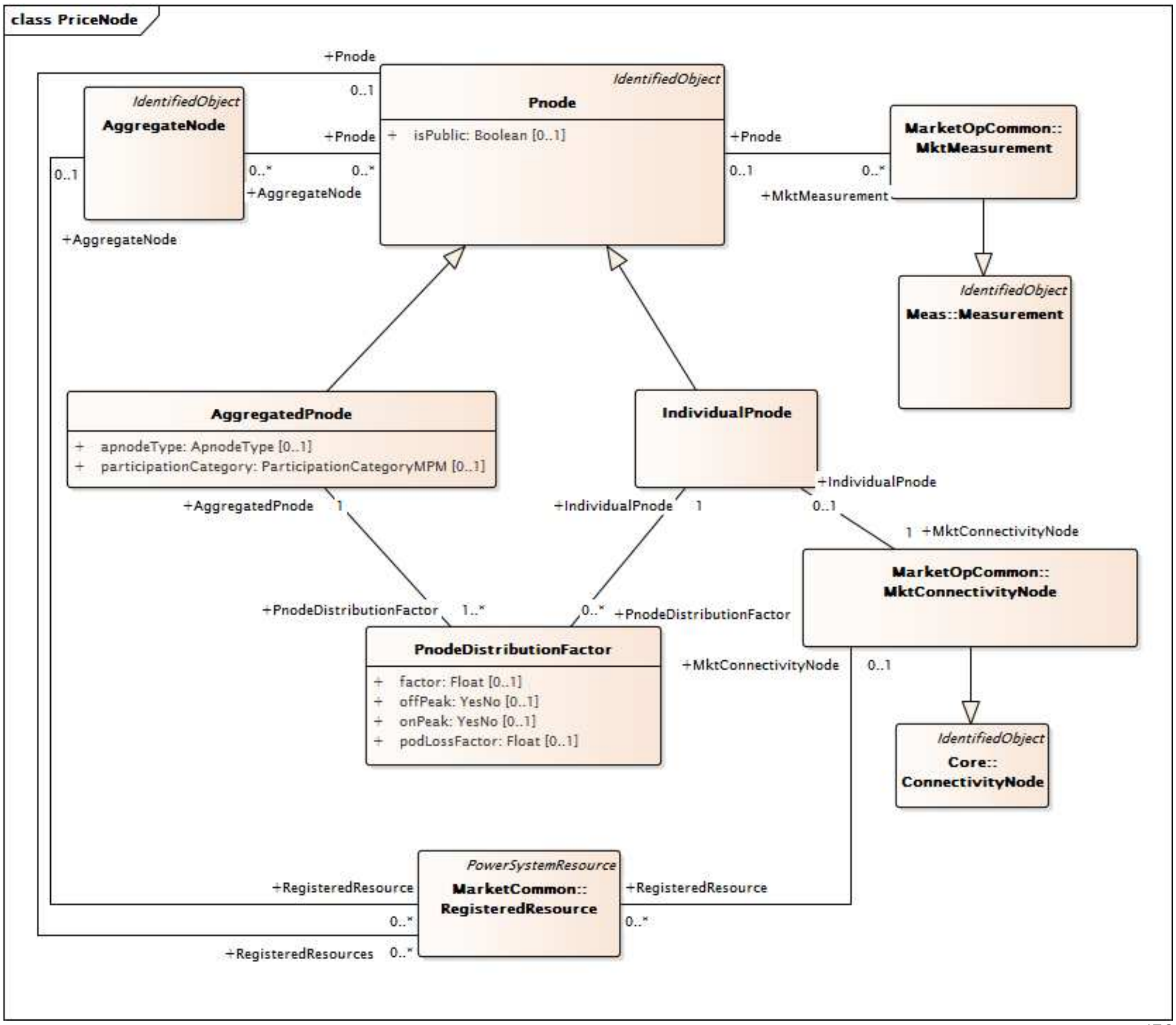


**Figure 63 – Diagramme de classe ReferenceData::PriceNode**

Figure 63: représente les classes et les associations utilisées pour modéliser les nœuds de tarification. Le nœud de tarification est référencé par les participants de marché, sous forme d'enchères/d'offres pour l'énergie, les services auxiliaires et les CRR. AggregatePnodes et IndividualPnodes constituent des sous-classes de la classe Pnode. Les Pnodes sont associés aux ressources enregistrées et utilisés dans les enchères/offres. Les Pnodes sont par ailleurs associés à la classe MktMeasurement utilisée dans la facturation et le règlement.

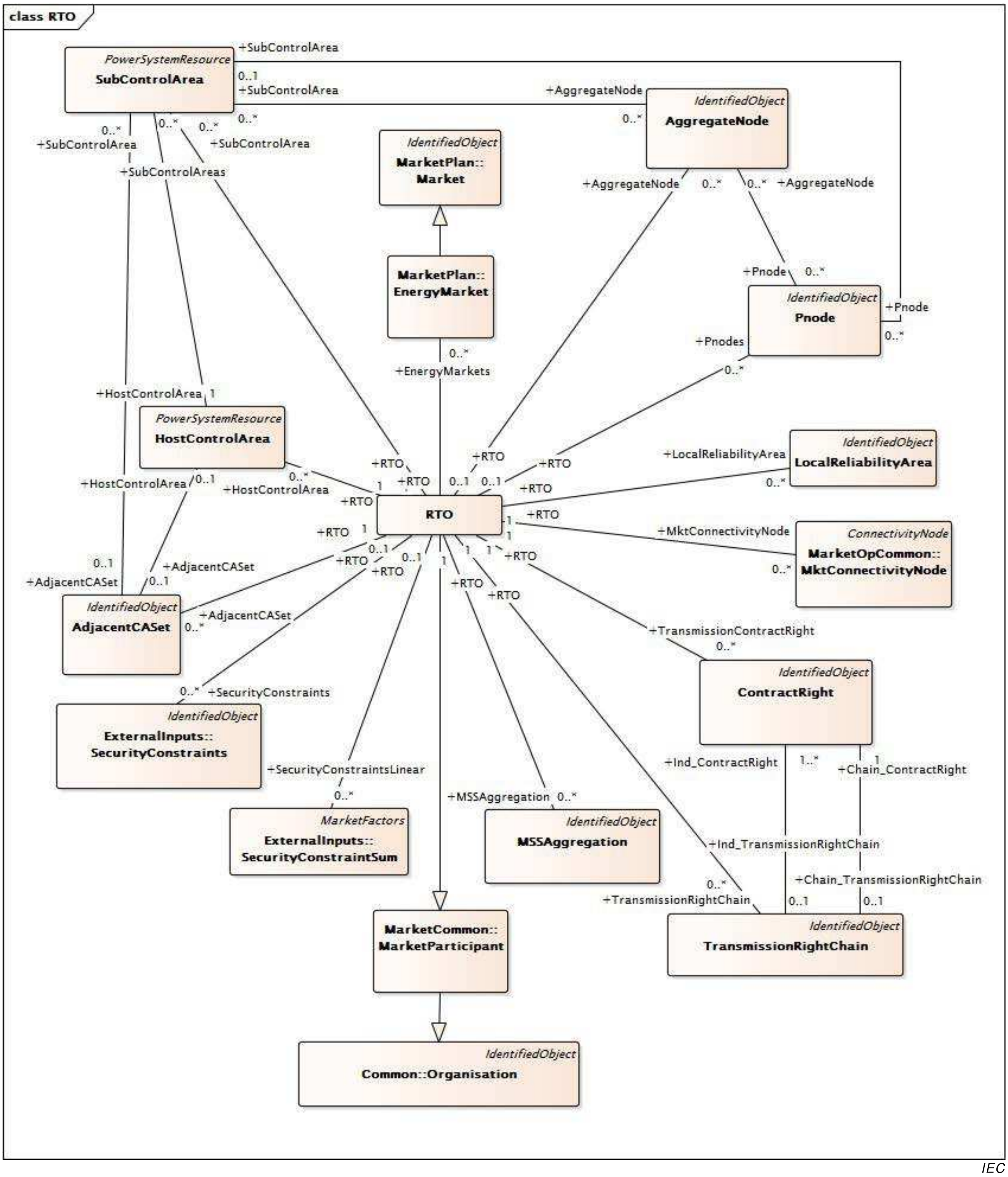


**Figure 64 – Diagramme de classe ReferenceData::RTO**

Figure 64: représente les classes et les associations utilisées pour modéliser une organisation de transport régional (Regional Transmission Organization) exploitant un marché d’électricité. Les principales classes et relations sont représentées d’un point de vue général. Les figures à l’appui apparaissent tout au long de la norme.

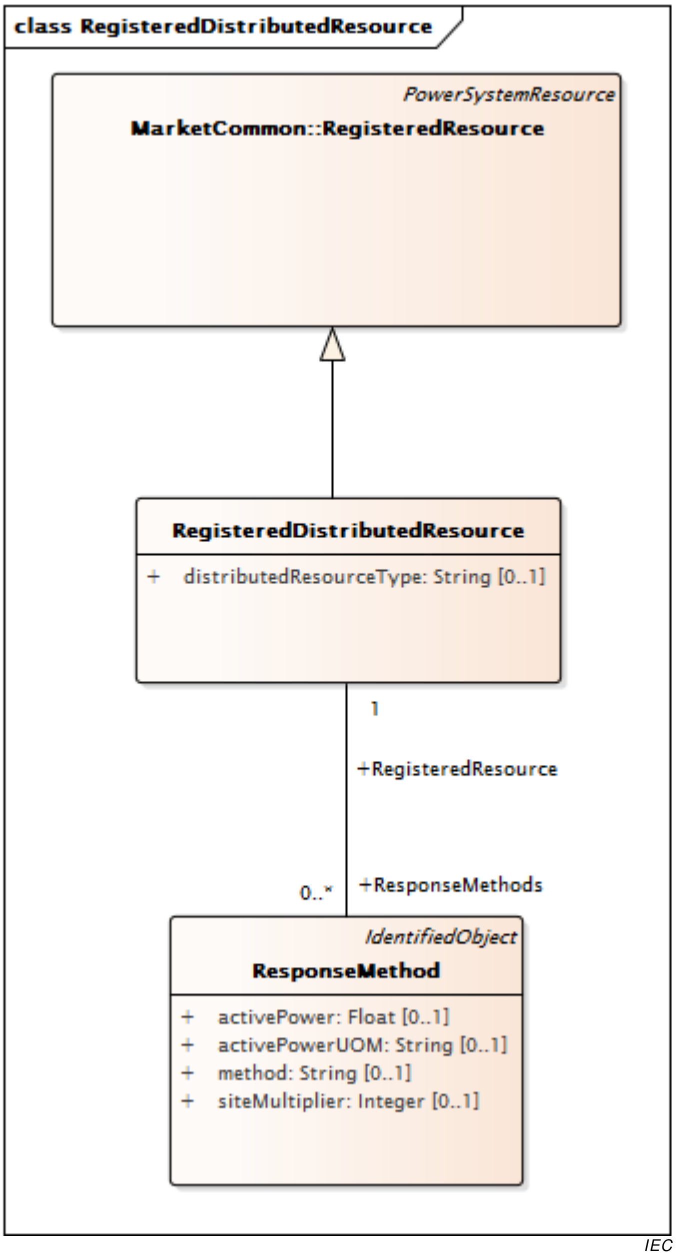


**Figure 65 – Diagramme de classe ReferenceData::RegisteredDistributedResource**

Figure 65: représente les classes et les associations utilisées pour modéliser les ressources réparties enregistrées. Les RegisteredDistributedResources peuvent représenter les ressources de réponse à la demande, les ressources de stockage, les ressources de production de distribution, etc. La RegisteredDistributedResource hérite de RegisteredResource et peut être associée aux ResponseMethods.

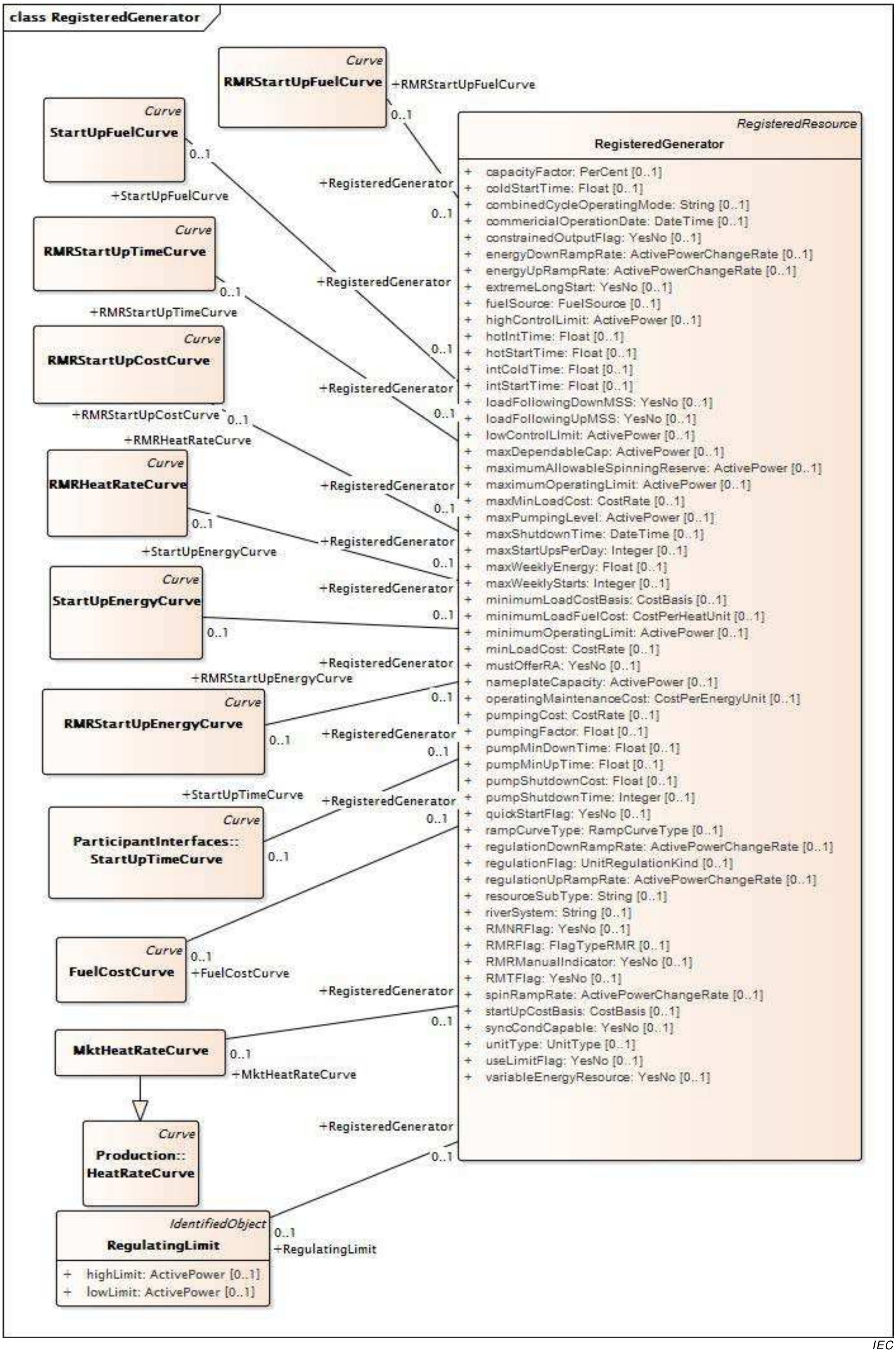


**Figure 66 – Diagramme de classe ReferenceData::RegisteredGenerator**

Figure 66: représente les classes et les associations utilisées pour modéliser les générateurs enregistrés. RMRStartUpFuelCurve est utilisée pour modéliser le coût de démarrage des unités de production classées comme unités RMR. Les autres classes sont utilisées pour modéliser le coût de démarrage, la courbe d'énergie de démarrage, les taux de chaleur et le temps de démarrage pour les unités RMR. La classe FuelCostCurve est utilisée pour modéliser la relation entre le coût de combustible au démarrage et la puissance de sortie active de l'unité.

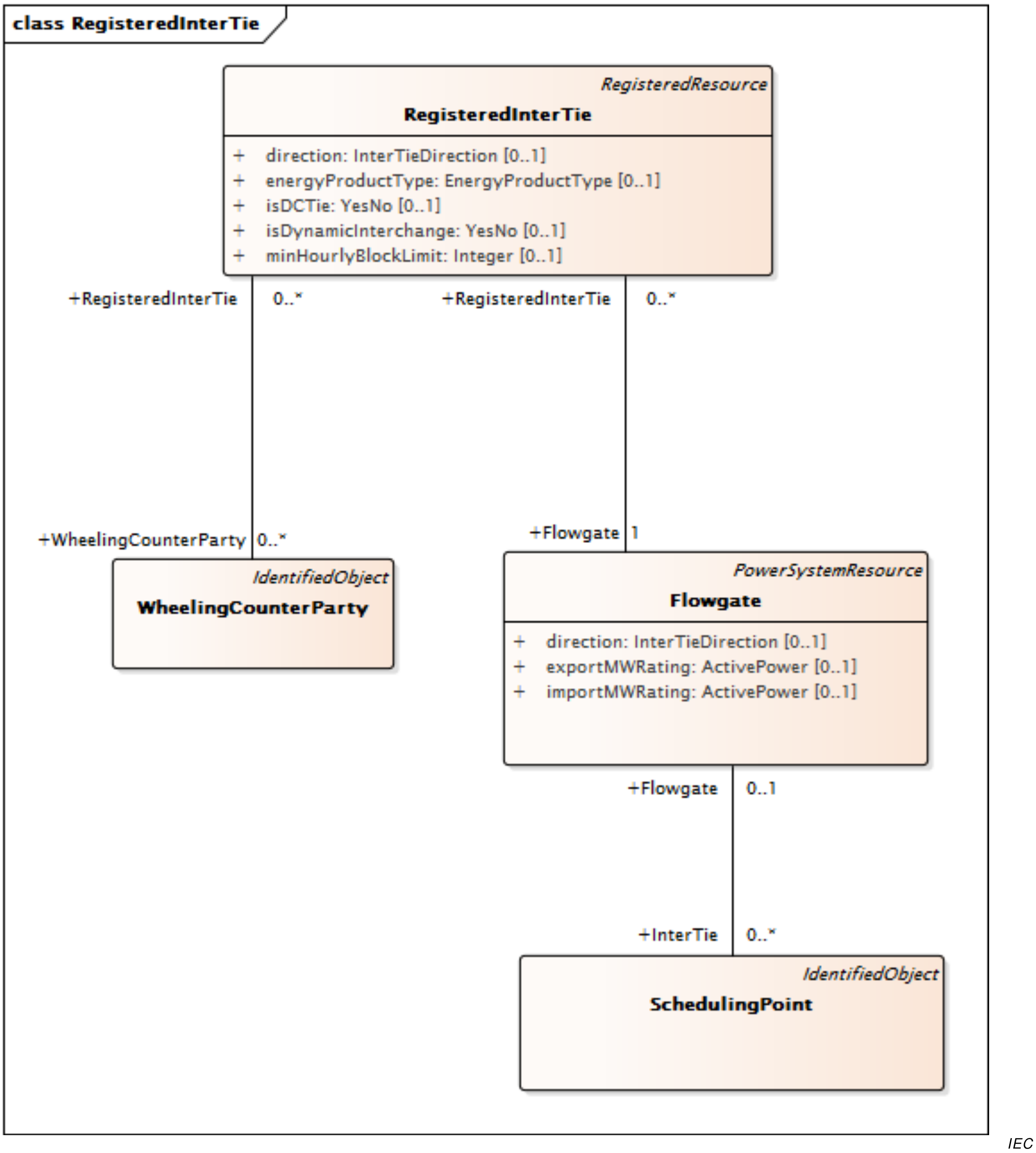


**Figure 67 – Diagramme de classe ReferenceData::RegisteredInterTie**

Figure 67: représente les classes et les associations utilisées pour modéliser les inerties enregistrées. La classe SchedulingPoint est associée à une interface de transit (Flowgate) qui est à son tour associée à RegisteredInterTie. RegisteredInterTie peut être une ligne à courant alternatif ou à courant continu vers une entité voisine des services publics.

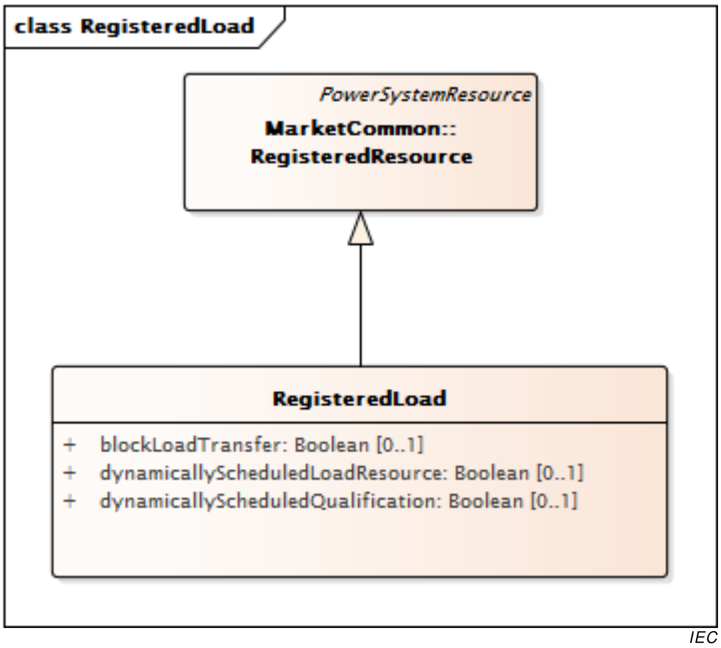


**Figure 68 – Diagramme de classe ReferenceData::RegisteredLoad**

Figure 68: représente les classes et les associations utilisées pour modéliser les charges enregistrées. Les RegisterLoads peuvent participer au marché en offrant de réduire la consommation énergétique de base et un ou plusieurs produit(s) électrique(s) (énergie, services auxiliaires) en retour d'un paiement. La classe MktEnergyConsumer modélise l'emplacement de la charge sur le réseau.

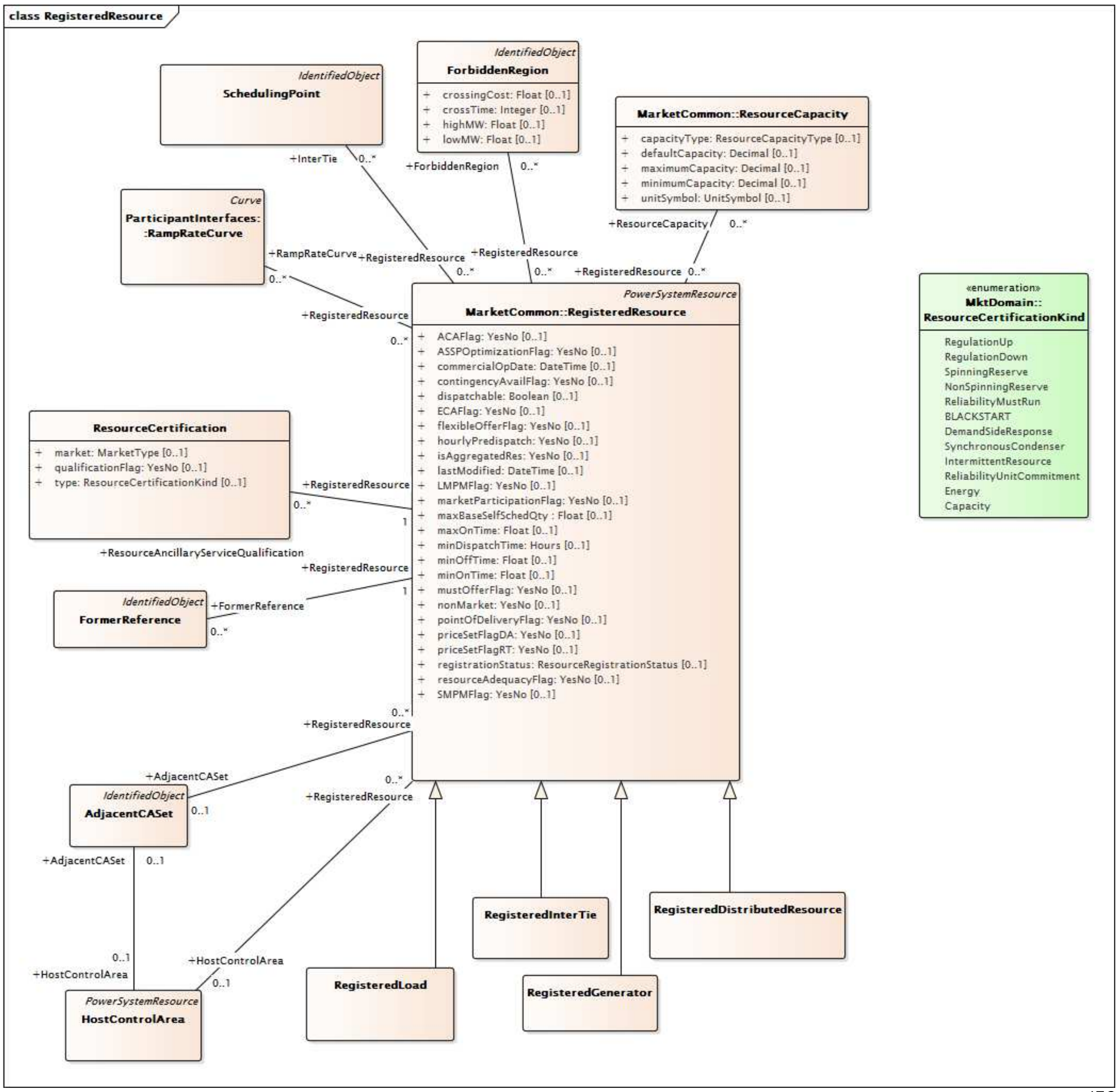


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**Figure 69 – Diagramme de classe ReferenceData::RegisteredResource**

Figure 69: représente les classes et les associations utilisées pour modéliser les ressources enregistrées. La classe RegisteredResource est la classe parent des sous-classes RegisteredGenerator, RegisteredInterTie et RegisteredLoad. Les classes principales intègrent la classe ResourceAncillaryServiceQualification qui est utilisée pour décrire les services auxiliaires que la ressource est en mesure d'offrir au titre de sa qualification. Les classes ResourceCapability, ForbiddenRegion et RampRateCurve sont utilisées pour modéliser les capacités techniques de la ressource. La classe SchedulingPoint modélise les points à la frontière avec d'autres entités de service public au niveau desquels la ressource peut fournir des produits de marché au RTO.

### 6.5.9.2 AdjacentCASet

Zones de réglage adjacentes aux groupes

Le Tableau 505 présente tous les attributs de AdjacentCASet.

**Tableau 505 – Attributs de ReferenceData::AdjacentCASet**

| nom | mult | type | description |
|---|---|---|---|
| lossPercentage | 0..1 | Float | Pourcentage de perte |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 506 présente toutes les extrémités d'association de AdjacentCASet avec d'autres classes.

**Tableau 506 – Extrémités d'association de ReferenceData:: AdjacentCASet avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | SubControlArea | 0..* | SubControlArea | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | HostControlArea | 0..1 | HostControlArea | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.3 AggregateNode

Un nœud agrégé peut définir un regroupement par type davantage défini par l'énumération AnodeType. Les types vont de System Zone/Regions (zone/régions système) aux Market Energy Regions (régions d'énergie du marché), en passant par les Aggregated Loads (charges agrégées) et les Aggregated Generators (générateurs agrégés

Le Tableau 507 présente tous les attributs de AggregateNode.

**Tableau 507 – Attributs de ReferenceData::AggregateNode**

| nom | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | Type de nœud agrégé |
| qualifASOrder | 0..1 | Integer | Ordre de traitement pour les alimentations automatiques AS de cette région. La priorité de cet attribut détermine les attributions de toute ressource située dans des régions se chevauchant. Ces régions sont traitées par ordre de priorité. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 508 présente toutes les extrémités d'association de AggregateNode avec d'autres classes.

**Tableau 508 – Extrémités d'association de ReferenceData:: AggregateNode avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..* | Pnode | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..* | SubControlArea | 0..* | SubControlArea | |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Une RegisteredResource ne peut être associée qu'à un AggregateNode si elle n'est pas reliée à un Pnode ou à un MktConnectivityNode. |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | |
| 0..1 | Instruction | 0..* | Instructions | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.4 AggregatedPnode

Un nœud de tarification agrégé est un type spécialisé de nœud de tarification utilisé pour modéliser des éléments comme la System Zone (zone système), la Default Price Zone (zone de tarification par défaut), la Custom Price Zone (zone de tarification personnalisée), la Control Area (zone de réglage), la Aggregated Generation (production agrégée), la Aggregated Participating Load (charge participante agrégée), la Aggregated Non-Participating Load (charge non participante agrégée), la Trading Hub (plate-forme de négociation) et la Designated Control Area Zone (DCA – zone de réglage désignée).

Le Tableau 509 présente tous les attributs de AggregatedPnode.

**Tableau 509 – Attributs de ReferenceData::AggregatedPnode**

| nom | mult | type | description |
|---|---|---|---|
| apnodeType | 0..1 | ApnodeType | Types de nœuds de tarification agrégés |
| participationCategory | 0..1 | ParticipationCategoryMPM | Participation de la zone de réglage désignée dans la mesure de la tarification LMP<br>'Y' – Participe à l'atténuation de l'emprise sur le marché local (LMPM – Local Market Power Mitigation) et à l'atténuation de l'emprise sur le marché du système (SMPM – System Market Power Mitigation)<br>'N' – N'est pas inclus dans les mesures de tarification LMP<br>'S' – Participe aux mesures de tarification SMPM<br>'L' – Participe aux mesures de tarification LMPM |
| isPublic | 0..1 | Boolean | Hérité de: Pnode |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 510 présente toutes les extrémités d'association de AggregatedPnode avec d'autres classes.

**Tableau 510 – Extrémités d'association de ReferenceData:: AggregatedPnode avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GenDistributionFactor | 1..* | GenDistributionFactor | |
| 0..1 | LoadDistributionFactor | 1..* | LoadDistributionFactor | |
| 1..1 | MPMTestResults | 1..* | MPMTestResults | |
| 0..* | TACArea | 0..* | TACArea | |
| 1..1 | PnodeDistributionFactor | 1..* | PnodeDistributionFactor | |
| 1..1 | TradingHubValues | 0..* | TradingHubValues | |
| 0..1 | MktCombinedCyclePlant | 0..* | MktCombinedCyclePlant | |
| 0..* | MPMTestThreshold | 1..* | MPMTestThreshold | |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | Hérité de: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | Hérité de: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | Hérité de: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | Hérité de: Pnode |
| 0..1 | Trade | 0..* | Trade | Hérité de: Pnode |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | Hérité de: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | Hérité de: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | Hérité de: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | Hérité de: Pnode |
| 0..* | RTO | 0..1 | RTO | Hérité de: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.5 Classe racine BidPriceCap

Cette classe représente le plafonnement du prix de l'offre.

Le Tableau 511 présente tous les attributs de BidPriceCap.

**Tableau 511 – Attributs de ReferenceData::BidPriceCap**

| nom | mult | type | description |
|---|---|---|---|
| marketType | 0..1 | MarketType | Type de marché du plafonnement (DAM ou RTM) |
| bidFloor | 0..1 | CostPerEnergyUnit | Plancher de l'offre ($/MWh) |
| bidCeiling | 0..1 | CostPerEnergyUnit | Plafond de l'offre ($/MWh) |
| defaultPrice | 0..1 | CostPerEnergyUnit | Prix par défaut de l'offre ($/MWh) |
| bidFloorAS | 0..1 | CostPerEnergyUnit | Plancher de l'offre ($/MWh) pour les AS génériques par rapport à un produit de marché spécifique |
| bidCeilingAS | 0..1 | CostPerEnergyUnit | Plafond de l'offre ($/MWh) pour les AS génériques par rapport à un produit de marché spécifique |

Le Tableau 512 présente toutes les extrémités d'association de BidPriceCap avec d'autres classes.

**Tableau 512 – Extrémités d'association de ReferenceData:: BidPriceCap avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketProduct | 0..1 | MarketProduct | |

#### 6.5.9.6 CnodeDistributionFactor

Facteurs de participation par Cnode. Utilisé pour calculer la "participation" d'un Cnode dans un AggregateNode. Chaque Cnode associé à un AggregateNode est affecté à un facteur de participation pour sa participation au sein de l'AggregateNode.

Le Tableau 513 présente tous les attributs de CnodeDistributionFactor.

**Tableau 513 – Attributs de ReferenceData::CnodeDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Utilisé pour calculer la "participation" d'un Cnode dans un AggregateNode. |
| podLossFactor | 0..1 | Float | Point du facteur de perte de livraison. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 514 présente toutes les extrémités d'association de CnodeDistributionFactor avec d'autres classes.

**Tableau 514 – Extrémités d’association de ReferenceData:: CnodeDistributionFactor avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 0..* | AggregateNode | 0..1 | AggregateNode | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.7 CombinedCycleConfiguration

Options de configuration pour les unités à cycle combiné.

Par exemple, un cycle combiné avec (CT1, CT2, ST1) aura des configurations (CT1, ST1) et (CT2, ST1) faisant partie de (1CT + 1STlogicalconfiguration).

Le Tableau 515 présente tous les attributs de CombinedCycleConfiguration.

**Tableau 515 – Attributs de ReferenceData::CombinedCycleConfiguration**

| nom | mult | type | description |
|---|---|---|---|
| primaryConfiguration | 0..1 | Boolean | Cette CombinedCycleConfiguration est-elle la configuration primaire dans la configuration logique associée? |
| ShutdownFlag | 0..1 | Boolean | Une centrale à cycle combiné peut-elle être arrêtée dans cette configuration? |
| StartupFlag | 0..1 | Boolean | Une centrale à cycle combiné peut-elle être démarrée dans cette configuration? |
| capacityFactor | 0..1 | PerCent | Hérité de: RegisteredGenerator |
| coldStartTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| combinedCycleOperatingMode | 0..1 | String | Hérité de: RegisteredGenerator |
| commericialOperationDate | 0..1 | DateTime | Hérité de: RegisteredGenerator |
| constrainedOutputFlag | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| energyDownRampRate | 0..1 | ActivePowerChangeRate | Hérité de: RegisteredGenerator |
| energyUpRampRate | 0..1 | ActivePowerChangeRate | Hérité de: RegisteredGenerator |
| extremeLongStart | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| fuelSource | 0..1 | FuelSource | Hérité de: RegisteredGenerator |
| highControlLimit | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| hotIntTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| hotStartTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| intColdTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| intStartTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| loadFollowingDownMSS | 0..1 | YesNo | Hérité de: RegisteredGenerator |

| nom | mult | type | description |
|---|---|---|---|
| loadFollowingUpMSS | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| lowControlLImit | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| maxDependableCap | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| maximumOperatingLimit | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| maxMinLoadCost | 0..1 | CostRate | Hérité de: RegisteredGenerator |
| maxPumpingLevel | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| maxShutdownTime | 0..1 | DateTime | Hérité de: RegisteredGenerator |
| maxStartUpsPerDay | 0..1 | Integer | Hérité de: RegisteredGenerator |
| maxWeeklyEnergy | 0..1 | Float | Hérité de: RegisteredGenerator |
| maxWeeklyStarts | 0..1 | Integer | Hérité de: RegisteredGenerator |
| minimumLoadCostBasis | 0..1 | CostBasis | Hérité de: RegisteredGenerator |
| minimumLoadFuelCost | 0..1 | CostPerHeatUnit | Hérité de: RegisteredGenerator |
| minimumOperatingLimit | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| minLoadCost | 0..1 | CostRate | Hérité de: RegisteredGenerator |
| mustOfferRA | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| nameplateCapacity | 0..1 | ActivePower | Hérité de: RegisteredGenerator |
| operatingMaintenanceCost | 0..1 | CostPerEnergyUnit | Hérité de: RegisteredGenerator |
| pumpingCost | 0..1 | CostRate | Hérité de: RegisteredGenerator |
| pumpingFactor | 0..1 | Float | Hérité de: RegisteredGenerator |
| pumpMinDownTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| pumpMinUpTime | 0..1 | Float | Hérité de: RegisteredGenerator |
| pumpShutdownCost | 0..1 | Float | Hérité de: RegisteredGenerator |
| pumpShutdownTime | 0..1 | Integer | Hérité de: RegisteredGenerator |
| quickStartFlag | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| rampCurveType | 0..1 | RampCurveType | Hérité de: RegisteredGenerator |
| regulationDownRampRate | 0..1 | ActivePowerChangeRate | Hérité de: RegisteredGenerator |
| regulationFlag | 0..1 | UnitRegulationKind | Hérité de: RegisteredGenerator |
| regulationUpRampRate | 0..1 | ActivePowerChangeRate | Hérité de: RegisteredGenerator |
| resourceSubType | 0..1 | String | Hérité de: RegisteredGenerator |
| riverSystem | 0..1 | String | Hérité de: RegisteredGenerator |
| RMNRFlag | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| RMRFlag | 0..1 | FlagTypeRMR | Hérité de: RegisteredGenerator |
| RMRManualIndicator | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| RMTFlag | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| spinRampRate | 0..1 | ActivePowerChangeRate | Hérité de: RegisteredGenerator |
| startUpCostBasis | 0..1 | CostBasis | Hérité de: RegisteredGenerator |
| syncCondCapable | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| unitType | 0..1 | UnitType | Hérité de: RegisteredGenerator |
| useLimitFlag | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| variableEnergyResource | 0..1 | YesNo | Hérité de: RegisteredGenerator |
| ACAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |

| nom | mult | type | description |
|---|---|---|---|
| ASSPOptimizationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | Hérité de: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| dispatchable | 0..1 | Boolean | Hérité de: RegisteredResource |
| ECAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | Hérité de: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | Hérité de: RegisteredResource |
| lastModified | 0..1 | DateTime | Hérité de: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | Hérité de: RegisteredResource |
| maxOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| minDispatchTime | 0..1 | Hours | Hérité de: RegisteredResource |
| minOffTime | 0..1 | Float | Hérité de: RegisteredResource |
| minOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| nonMarket | 0..1 | YesNo | Hérité de: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | Hérité de: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationSt atus | Hérité de: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 516 présente toutes les extrémités d'association de CombinedCycleConfiguration avec d'autres classes.

**Tableau 516 – Extrémités d’association de ReferenceData:: CombinedCycleConfiguration avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ToTransitionState | 0..* | CombinedCycleTransitio nState | |
| 1..1 | FromTransitionState | 0..* | CombinedCycleTransitio nState | |
| 1..1 | CombinedCycleConfig urationMember | 0..* | CombinedCycleConfigur ationMember | |
| 1..* | CombinedCycleLogica lConfiguration | 0..1 | CombinedCycleLogicalC onfiguration | |
| 0..1 | GeneratingBids | 0..* | GeneratingBid | Hérité de: RegisteredGenerator |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | StartUpTimeCurve | 0..1 | StartUpTimeCurve | Hérité de: RegisteredGenerator |
| 0..1 | Trade | 0..* | Trade | Hérité de: RegisteredGenerator |
| 0..1 | MktHeatRateCurve | 0..1 | MktHeatRateCurve | Hérité de: RegisteredGenerator |
| 0..1 | RMRHeatRateCurve | 0..1 | RMRHeatRateCurve | Hérité de: RegisteredGenerator |
| 0..1 | RMRStartUpCostCurve | 0..1 | RMRStartUpCostCurve | Hérité de: RegisteredGenerator |
| 0..1 | RMRStartUpEnergyCurve | 0..1 | RMRStartUpEnergyCurve | Hérité de: RegisteredGenerator |
| 0..1 | RMRStartUpFuelCurve | 0..1 | RMRStartUpFuelCurve | Hérité de: RegisteredGenerator |
| 0..1 | RMRStartUpTimeCurve | 0..1 | RMRStartUpTimeCurve | Hérité de: RegisteredGenerator |
| 0..1 | RegulatingLimit | 0..1 | RegulatingLimit | Hérité de: RegisteredGenerator |
| 0..1 | StartUpFuelCurve | 0..1 | StartUpFuelCurve | Hérité de: RegisteredGenerator |
| 0..1 | StartUpEnergyCurve | 0..1 | StartUpEnergyCurve | Hérité de: RegisteredGenerator |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | Hérité de: RegisteredGenerator |
| 1..1 | EnergyPriceIndex | 1..1 | EnergyPriceIndex | Hérité de: RegisteredGenerator |
| 0..1 | UnitInitialConditions | 0..* | UnitInitialConditions | Hérité de: RegisteredGenerator |
| 0..* | StartUpCostCurves | 0..* | StartUpCostCurve | Hérité de: RegisteredGenerator |
| 0..1 | FuelCostCurve | 0..1 | FuelCostCurve | Hérité de: RegisteredGenerator |
| 0..* | FuelRegion | 0..1 | FuelRegion | Hérité de: RegisteredGenerator |
| 0..* | LocalReliabilityArea | 0..1 | LocalReliabilityArea | Hérité de: RegisteredGenerator |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | Hérité de: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | Hérité de: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | Hérité de: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | Hérité de: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | Hérité de: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | Hérité de: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | Hérité de: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | Hérité de: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | Hérité de: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | Hérité de: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | Hérité de: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | Hérité de: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | Hérité de: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | Hérité de: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | Hérité de: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | Hérité de: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Hérité de: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | Hérité de: RegisteredResource |
| 0..* | Reason | 0..* | Reason | Hérité de: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | Hérité de: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | Hérité de: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | Hérité de: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | Hérité de: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | Hérité de: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | Hérité de: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | Hérité de: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | Hérité de: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | Hérité de: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | Hérité de: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | Hérité de: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | Hérité de: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | Hérité de: RegisteredResource |
| 0..* | Domain | 0..* | Domain | Hérité de: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | Hérité de: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | Hérité de: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.8 CombinedCycleConfigurationMember

Membre de configuration de configuration CCP.

Le Tableau 517 présente tous les attributs de CombinedCycleConfigurationMember.

**Tableau 517 – Attributs de ReferenceData::CombinedCycleConfigurationMember**

| nom | mult | type | description |
|---|---|---|---|
| primary | 0..1 | Boolean | Configuration primaire. |
| steam | 0..1 | Boolean | Centrale à vapeur. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 518 présente toutes les extrémités d'association de CombinedCycleConfigurationMember avec d'autres classes.

**Tableau 518 – Extrémités d'association de ReferenceData:: CombinedCycleConfigurationMember avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktThermalGeneratingUnit | 1..1 | MktThermalGeneratingUnit | |
| 0..* | CombinedCycleConfiguration | 1..1 | CombinedCycleConfiguration | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.9 CombinedCycleLogicalConfiguration

Configuration logique d'une centrale à cycle combiné.

Les configurations d'exploitation des centrales à cycle combiné (CCP – Combined Cycle Plant) sont représentées sous forme de ressources CCP logiques. La représentation logique doit être utilisée pour les applications de marché afin d'optimiser et de contrôler les opérations de marché. La représentation logique est par ailleurs nécessaire pour contrôler le nombre de configurations CCP et pour tempérer les problèmes de performances qui peuvent se produire dans d'autres circonstances.

Par exemple, (2CT configuration),(1CT + 1ST configuration) sont des exemples de configurations logiques, sans pour autant spécifier les CT et ST spécifiques qui entrent dans cette configuration.

Le Tableau 519 présente tous les attributs de CombinedCycleLogicalConfiguration.

**Tableau 519 – Attributs de ReferenceData::CombinedCycleLogicalConfiguration**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 520 présente toutes les extrémités d'association de CombinedCycleLogicalConfiguration avec d'autres classes.

**Tableau 520 – Extrémités d'association de ReferenceData:: CombinedCycleLogicalConfiguration avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | CombinedCycleConfiguration | 1..* | CombinedCycleConfiguration | |
| 1..* | MktCombinedCyclePlant | 0..1 | MktCombinedCyclePlant | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.10 Classe racine CombinedCycleTransitionState

Définit les statuts de transition disponible de et pour relatifs aux configurations à cycle combiné.

Le Tableau 521 présente tous les attributs de CombinedCycleTransitionState.

**Tableau 521 – Attributs de ReferenceData::CombinedCycleTransitionState**

| nom | mult | type | description |
|---|---|---|---|
| upTransition | 0..1 | Boolean | Indicateur montrant s'il s'agit d'une transition ascendante (UP).<br>Si ce n'est pas le cas, il s'agit d'une transition descendante (DOWN). |

Le Tableau 522 présente toutes les extrémités d'association de CombinedCycleTransitionState avec d'autres classes.

**Tableau 522 – Extrémités d’association de ReferenceData::CombinedCycleTransitionState avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ToConfiguration | 1..1 | CombinedCycleConfiguration | |
| 0..* | FromConfiguration | 1..1 | CombinedCycleConfiguration | |

### 6.5.9.11 CongestionArea

Définition de la zone de congestion désignée (DCA – *Designated Congestion Area*)

Le Tableau 523 présente tous les attributs de CongestionArea.

**Tableau 523 – Attributs de ReferenceData::CongestionArea**

| nom | mult | type | description |
|---|---|---|---|
| apnodeType | 0..1 | ApnodeType | Hérité de: AggregatedPnode |
| participationCategory | 0..1 | ParticipationCategoryMPM | Hérité de: AggregatedPnode |
| isPublic | 0..1 | Boolean | Hérité de: Pnode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 524 présente toutes les extrémités d'association de CongestionArea avec d'autres classes.

**Tableau 524 – Extrémités d’association de ReferenceData:: CongestionArea avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | IndividualPnode | 0..* | IndividualPnode | |
| 0..1 | GenDistributionFactor | 1..* | GenDistributionFactor | Hérité de: AggregatedPnode |
| 0..1 | LoadDistributionFactor | 1..* | LoadDistributionFactor | Hérité de: AggregatedPnode |
| 1..1 | MPMTestResults | 1..* | MPMTestResults | Hérité de: AggregatedPnode |
| 0..* | TACArea | 0..* | TACArea | Hérité de: AggregatedPnode |
| 1..1 | PnodeDistributionFactor | 1..* | PnodeDistributionFactor | Hérité de: AggregatedPnode |
| 1..1 | TradingHubValues | 0..* | TradingHubValues | Hérité de: AggregatedPnode |
| 0..1 | MktCombinedCyclePlant | 0..* | MktCombinedCyclePlant | Hérité de: AggregatedPnode |
| 0..* | MPMTestThreshold | 1..* | MPMTestThreshold | Hérité de: AggregatedPnode |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | Hérité de: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | SinkCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | Hérité de: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | Hérité de: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | Hérité de: Pnode |
| 0..1 | Trade | 0..* | Trade | Hérité de: Pnode |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | Hérité de: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | Hérité de: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | Hérité de: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | Hérité de: Pnode |
| 0..* | RTO | 0..1 | RTO | Hérité de: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.12 Classe racine ContractDistributionFactor

Répartition parmi les ressources au point puits ou au point source

Le Tableau 525 présente tous les attributs de ContractDistributionFactor.

**Tableau 525 – Attributs de ReferenceData::ContractDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Valeur MW que fournit cette ressource à l'ensemble du contrat. |
| sourceFlag | 0..1 | YesNo | Cette valeur sera définie sur YES (Oui) si le Cnode référencé est défini en tant que point source dans le contrat. |
| sinkFlag | 0..1 | YesNo | Cette valeur sera définie sur YES (Oui) si le Cnode référencé est défini en tant que point puits dans le contrat. |

Le Tableau 526 présente toutes les extrémités d'association de ContractDistributionFactor avec d'autres classes.

**Tableau 526 – Extrémités d'association de ReferenceData:: ContractDistributionFactor avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |
| 0..* | Flowgate | 0..1 | Flowgate | |

### 6.5.9.13 ContractRight

Fournit une définition du droit de propriété de transport et les identifiants des contrats de transport existants pour une utilisation par SCUC. Hébergement de contrat RMR: Délai de démarrage, limites des services de contrat, heures max. travaillées, quantité max. de MWh, nombre max. de démarrages, taux de rampe, capacité fiable nette max., capacité minimale et substitution d'unité pour le DAM/RTM à retrouver;

Le Tableau 527 présente tous les attributs de ContractRight.

**Tableau 527 – Attributs de ReferenceData::ContractRight**

| nom | mult | type | description |
|---|---|---|---|
| chainOrder | 0..1 | Integer | En cas d'utilisation combinée à une chaîne de contrat de droit de transport, il s'agit de la priorité pour les contrats. |
| contractMW | 0..1 | Float | Valeur en MW du contrat |
| contractPrice | 0..1 | CostPerEnergyUnit | Valeur financière du contrat |
| contractPriority | 0..1 | Integer | Priorité pour le contrat. Il convient qu'elle soit unique parmi tous les contrats pour une ressource spécifique. Cette valeur constitue la directive pour l'algorithme SCUC en cas d'ordre de réalisation/suppression de contrats. |
| contractStatus | 0..1 | String | Statut du contrat |
| contractType | 0..1 | ContractType | Type du contrat. Exemples de valeurs possibles:<br>ETC, TOR ou RMR et RMT (programmations automatiques) |
| financialLocation | 0..1 | YesNo | Indicateur spécifiant si l'emplacement associé à ce contrat est financier (par exemple, nœuds de tarification) ou physique (par exemple, nœuds de connectivité). |
| financialRightsDAM | 0..1 | YesNo | Indicateur selon lequel ce contrat attribue des droits financiers sur le DAM |
| financialRightsRTM | 0..1 | YesNo | Indicateur selon lequel ce contrat attribue des droits financiers sur le RTM |
| fuelAdder | 0..1 | Float | Mélangeur de combustible estimé |
| latestSchedMinutes | 0..1 | Integer | Indique les dernières minutes de programmation (par exemple, t – xx) pour lesquelles il peut être demandé à la ressource de répondre. Cet attribut est uniquement utilisé si le type de marché n'est pas fourni. |
| latestSchedMktType | 0..1 | MarketType | Indique le dernier type de marché de programmation auquel un contrat peut être appliqué. Utilisé en combinaison avec l'attribut latestSchedMinutes afin de déterminer le moment le plus tardif auquel ce contrat peut être appelé. Les valeurs possibles pour cet attribut sont les suivantes: DAM ou RTM; cet attribut peut également être omis. Dans ce cas, l'attribut latestSchedMinutes définit la valeur. |
| maximumScheduleQuantity | 0..1 | Float | Quantité MW maximale planifiée |

| nom | mult | type | description |
|---|---|---|---|
| maximumServiceHours | 0..1 | Integer | Nombre maximal d'heures travaillées |
| maximumStartups | 0..1 | Integer | Nombre maximal de démarrages |
| maxNetDependableCapacity | 0..1 | Float | Capacité fiable nette maximale |
| minimumLoad | 0..1 | Float | Charge minimale |
| minimumScheduleQuantity | 0..1 | Float | Quantité minimale planifiée |
| physicalRightsDAM | 0..1 | YesNo | Indicateur selon lequel ce contrat attribue des droits physiques sur le DAM |
| physicalRightsRTM | 0..1 | YesNo | Indicateur selon lequel ce contrat attribue des droits physiques sur le RTM |
| startupLeadTime | 0..1 | Integer | Délai de démarrage |
| TRType | 0..1 | TRType | Type de droit de transport – s'agit-il d'un droit contractuel individuel ou en chaîne?<br>Types = CHAIN (en chaîne) ou INDIVIDUAL (individuel) |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 528 présente toutes les extrémités d'association de ContractRight avec d'autres classes.

**Tableau 528 – Extrémités d’association de ReferenceData:: ContractRight avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | TREntitlement | 0..* | TREntitlement | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | |
| 1..1 | TransmissionInterfaceEntitlement | 0..* | TransmissionInterfaceRightEntitlement | |
| 1..* | Ind_TransmissionRightChain | 0..1 | TransmissionRightChain | |
| 1..1 | Chain_TransmissionRightChain | 0..1 | TransmissionRightChain | |
| 0..* | SchedulingCoordinator | 1..1 | SchedulingCoordinator | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.14 Classe racine ControlAreaDesignation

Indique la Zone de réglage associée à la programmation automatique.

Le Tableau 529 présente tous les attributs de ControlAreaDesignation.

**Tableau 529 – Attributs de ReferenceData::ControlAreaDesignation**

| nom | mult | type | description |
|---|---|---|---|
| attained | 0..1 | YesNo | Atteint. |
| native | 0..1 | YesNo | Natif. |

Le Tableau 530 présente toutes les extrémités d'association de ControlAreaDesignation avec d'autres classes.

**Tableau 530 – Extrémités d’association de ReferenceData:: ControlAreaDesignation avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..* | SubControlArea | 0..* | SubControlArea | |

#### 6.5.9.15 Flowgate

Une interface de transit (Flowgate) se compose d'un ou de plusieurs éléments de transport destinés à modéliser l'impact du flux de MW relatif aux limites de transport et à l'utilisation du service de transport.

Le Tableau 531 présente tous les attributs de Flowgate.

**Tableau 531 – Attributs de ReferenceData::Flowgate**

| nom | mult | type | description |
|---|---|---|---|
| direction | 0..1 | InterTieDirection | Sens de l'interface de transit: exportation ou importation |
| exportMWRating | 0..1 | ActivePower | Puissance assignée en MW d'exportation |
| importMWRating | 0..1 | ActivePower | Puissance assignée en MW d'importation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | | |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 532 présente toutes les extrémités d'association de Flowgate avec d'autres classes.

**Tableau 532 – Extrémités d'association de ReferenceData:: Flowgate avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MktLine | 0..* | MktLine | |
| 0..* | MktPowerTransformer | 0..* | MktPowerTransformer | |
| 0..1 | SecurityConstraints | 0..1 | SecurityConstraints | |
| 0..1 | TransmissionCapacity | 0..* | TransmissionCapacity | |
| 0..1 | TranmissionRightEntitlement | 0..* | TransmissionInterfaceRightEntitlement | |
| 1..1 | ConstraintResults | 1..* | ConstraintResults | |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | |
| 1..1 | FlowgateRelief | 0..* | FlowgateRelief | |
| 0..1 | InterTie | 0..* | SchedulingPoint | |
| 1..1 | RegisteredInterTie | 0..* | RegisteredInterTie | |
| 0..* | To_SubControlArea | 0..1 | SubControlArea | |
| 0..* | From_SubControlArea | 0..1 | SubControlArea | |
| 1..1 | FlowgateValue | 0..* | FlowgateValue | |
| 0..1 | CongestionRevenueRight | 0..1 | CongestionRevenueRight | |
| 0..1 | MktTerminal | 0..* | MktTerminal | |
| 0..1 | GeneratingUnitDynamicValues | 0..* | GeneratingUnitDynamicValues | |
| 0..* | GenericConstraints | 0..1 | GenericConstraints | |
| 0..* | HostControlArea | 0..1 | HostControlArea | |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.16 FlowgatePartner

Partenaire défini de l'interface de transit

Le Tableau 533 présente tous les attributs de FlowgatePartner.

**Tableau 533 – Attributs de ReferenceData::FlowgatePartner**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 534 présente toutes les extrémités d'association de FlowgatePartner avec d'autres classes.

**Tableau 534 – Extrémités d'association de ReferenceData:: FlowgatePartner avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | FlowgateValue | 0..1 | FlowgateValue | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.17 Classe racine FlowgateRelief

L'IDC (Interchange Distribution Calculator – calculateur de la répartition des échanges) envoie les données pour un TLR (Transmission Loading Relief – allègement de la charge des installations de transport).

Le Tableau 535 présente tous les attributs de FlowgateRelief.

**Tableau 535 – Attributs de ReferenceData::FlowgateRelief**

| nom | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | Date/heure de prise d'effet de l'enregistrement<br>Utilisé pour déterminer quand un enregistrement prend effet. |
| terminateDate | 0..1 | DateTime | Date/heure de fin de la prise d'effet de l'enregistrement<br>Utilisé pour déterminer quand la prise d'effet d'un enregistrement se termine. |
| idcTargetMktFlow | 0..1 | Integer | Niveau de flux d'énergie qu'il convient de maintenir conformément aux règles TLR tel que spécifié par l'IDC.<br>Pour les marchés en temps réel, à utiliser lors de la répartition afin de contrôler les contraintes sous TLR et de calculer les flux de marché sans contraintes |

Le Tableau 536 présente toutes les extrémités d'association de FlowgateRelief avec d'autres classes.

**Tableau 536 – Extrémités d'association de ReferenceData:: FlowgateRelief avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 1..1 | Flowgate | |

### 6.5.9.18 Classe racine FlowgateValue

J-1, charge native réseau, répartition économique, valeurs utilisées pour le calcul du processus de détermination de charge native réseau (NNL – Network Native Load).

Le Tableau 537 présente tous les attributs de FlowgateValue.

**Tableau 537 – Attributs de ReferenceData::FlowgateValue**

| nom | mult | type | description |
|---|---|---|---|
| economicDispatchLimit | 0..1 | Integer | Limite pour le flux d'énergie de la priorité 6 de répartition économique sur l'interface de transit indiquée pour la période de temps spécifiée. |
| effectiveDate | 0..1 | DateTime | Date/heure de prise d'effet de l'enregistrement<br>Utilisé pour déterminer quand un enregistrement prend effet |
| firmNetworkLimit | 0..1 | Integer | Limite pour le flux ferme sur l'interface de transit indiquée pour la période de temps spécifiée. Quantité de flux d'énergie sur une interface de transit indiquée résultant d'une production dans le marché qui peut être considérée comme priorité de réseau ferme. |
| flowDirectionFlag | 0..1 | FlowDirectionType | Indique le sens du flux d'énergie dans l'interface de transit |
| mktFlow | 0..1 | Integer | Quantité de flux d'énergie sur une interface de transit indiquée résultant d'une production dans le marché. |
| netFirmNetworkLimit | 0..1 | Integer | Flux d'énergie net dans l'interface de transit pour le FlowgatePartner associé |

Le Tableau 538 présente toutes les extrémités d'association de FlowgateValue avec d'autres classes.

**Tableau 538 – Extrémités d'association de ReferenceData:: FlowgateValue avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Flowgate | 1..1 | Flowgate | |
| 0..1 | FlowgatePartner | 0..1 | FlowgatePartner | |

### 6.5.9.19 ForbiddenRegion

La région interdite (Forbbiden region) exploite des plages dans lesquelles les unités ne sont pas en mesure de maintenir une exploitation constante sans endommager l'équipement. Les quatre attributs qui définissent une région interdite sont la quantité MW inférieure, la quantité MW supérieure, le temps de traversée et le coût de traversée.

Le Tableau 539 présente tous les attributs de ForbiddenRegion.

**Tableau 539 – Attributs de ReferenceData::ForbiddenRegion**

| nom | mult | type | description |
|---|---|---|---|
| crossingCost | 0..1 | Float | Coût associé à la traversée de la région interdite |
| crossTime | 0..1 | Integer | Temps (minutes) nécessaire pour traverser la région interdite. |
| highMW | 0..1 | Float | Extrémité supérieure de la définition de la région |
| lowMW | 0..1 | Float | Extrémité inférieure de la définition de la région |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 540 présente toutes les extrémités d'association de ForbiddenRegion avec d'autres classes.

**Tableau 540 – Extrémités d’association de ReferenceData:: ForbiddenRegion avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.20 FormerReference

Utilisé pour indiquer d'anciennes références pour le même équipement. L'identifiant, le nom et les dates effectives sont utilisés.

Le Tableau 541 présente tous les attributs de FormerReference.

**Tableau 541 – Attributs de ReferenceData::FormerReference**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 542 présente toutes les extrémités d'association de FormerReference avec d'autres classes.

**Tableau 542 – Extrémités d'association de ReferenceData:: FormerReference avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.21 FuelCostCurve

Relation entre le coût de combustible de l'unité en $/kWh (axe Y) et la puissance de sortie de l'unité en MW (axe X).

Le Tableau 543 présente tous les attributs de FuelCostCurve.

**Tableau 543 – Attributs de ReferenceData::FuelCostCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 544 présente toutes les extrémités d'association de FuelCostCurve avec d'autres classes.

**Tableau 544 – Extrémités d'association de ReferenceData:: FuelCostCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.22 FuelRegion

Indication d'une région à des fins d'inventaire de combustible

Le Tableau 545 présente tous les attributs de FuelRegion.

**Tableau 545 – Attributs de ReferenceData::FuelRegion**

| nom | mult | type | description |
|---|---|---|---|
| fuelRegionType | 0..1 | String | Type de région combustible |
| lastModified | 0..1 | DateTime | Heure de la dernière mise à jour |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 546 présente toutes les extrémités d'association de FuelRegion avec d'autres classes.

**Tableau 546 – Extrémités d'association de ReferenceData:: FuelRegion avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..* | RegisteredGenerator | |
| 1..1 | GasPrice | 1..1 | GasPrice | |
| 1..1 | OilPrice | 1..1 | OilPrice | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.23 Classe racine GasPrice

Prix du gaz en unités monétaires

Le Tableau 547 présente tous les attributs de GasPrice.

**Tableau 547 – Attributs de ReferenceData::GasPrice**

| nom | mult | type | description |
|---|---|---|---|
| gasPriceIndex | 0..1 | Float | Prix moyen du gaz naturel dans une région combustible définie. |

Le Tableau 548 présente toutes les extrémités d'association de GasPrice avec d'autres classes.

**Tableau 548 – Extrémités d'association de ReferenceData:: GasPrice avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | FuelRegion | 1..1 | FuelRegion | |

### 6.5.9.24 HostControlArea

Une HostControlArea (zone de réglage hôte) comporte un ensemble de points d'attache et d'un ensemble de commandes de générateur (c'est-à-dire l'AGC). Elle possède en outre une charge totale comprenant les pertes de transport et de distribution.

Le Tableau 549 présente tous les attributs de HostControlArea.

**Tableau 549 – Attributs de ReferenceData::HostControlArea**

| nom | mult | type | description |
|---|---|---|---|
| areaControlMode | 0..1 | AreaControlMode | Mode de réglage actuel de la zone:<br>(CF = fréquence constante) ou (CTL = ligne d'interconnexion constante) ou<br>(TLB = conditionnement par ligne d'interconnexion) ou (OFF = contrôle d'arrêt) |
| freqSetPoint | 0..1 | Frequency | Point de consigne actuel de fréquence du système électrique pour un contrôle de production automatique |
| frequencyBiasFactor | 0..1 | Float | Facteur de polarisation de la fréquence de la zone de réglage, en MW/0,1 Hz, pour le contrôle de la production automatique (AGC – *automatic generation control*) |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 550 présente toutes les extrémités d'association de HostControlArea avec d'autres classes.

**Tableau 550 – Extrémités d’association de ReferenceData:: HostControlArea avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 0..* | RegisteredResource | |
| 1..1 | SysLoadDistribuFactor | 0..* | SysLoadDistributionFactor | |
| 0..1 | TransferInterface | 0..* | TransferInterface | |
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..1 | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..1 | Flowgate | 0..* | Flowgate | |
| 1..1 | SubControlAreas | 0..* | SubControlArea | La zone d'échange peut fonctionner en tant que zone de réglage. |
| 0..* | RTO | 1..1 | RTO | |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.25 IndividualPnode

Nœud de tarification individuel fondé sur Pnode

Le Tableau 551 présente tous les attributs de IndividualPnode.

**Tableau 551 – Attributs de ReferenceData::IndividualPnode**

| nom | mult | type | description |
|---|---|---|---|
| isPublic | 0..1 | Boolean | Hérité de: Pnode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 552 présente toutes les extrémités d'association de IndividualPnode avec d'autres classes.

**Tableau 552 – Extrémités d'association de ReferenceData:: IndividualPnode avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MktConnectivityNode | 1..1 | MktConnectivityNode | |
| 0..1 | GenDistributionFactor | 0..1 | GenDistributionFactor | |
| 0..1 | LoadDistributionFactor | 0..1 | LoadDistributionFactor | |
| 1..1 | PnodeDistributionFactor | 0..* | PnodeDistributionFactor | |
| 0..* | CongestionArea | 0..* | CongestionArea | |
| 0..1 | RegisteredResources | 0..* | RegisteredResource | Hérité de: Pnode |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | Hérité de: Pnode |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | Hérité de: Pnode |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | Hérité de: Pnode |
| 0..1 | PnodeResults | 1..* | PnodeResults | Hérité de: Pnode |
| 0..1 | Trade | 0..* | Trade | Hérité de: Pnode |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | Hérité de: Pnode |
| 0..* | SubControlArea | 0..1 | SubControlArea | Hérité de: Pnode |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | Hérité de: Pnode |
| 0..* | AggregateNode | 0..* | AggregateNode | Hérité de: Pnode |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | Hérité de: Pnode |
| 0..* | RTO | 0..1 | RTO | Hérité de: Pnode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.26 LoadAggregationPoint

Classe spécialisée de type AggregatedNode. Définit les points d'agrégation de charge (Load Aggregation Points).

Le Tableau 553 présente tous les attributs de LoadAggregationPoint.

**Tableau 553 – Attributs de ReferenceData::LoadAggregationPoint**

| nom | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | Hérité de: AggregateNode |
| qualifASOrder | 0..1 | Integer | Hérité de: AggregateNode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 554 présente toutes les extrémités d'association de LoadAggregationPoint avec d'autres classes.

**Tableau 554 – Extrémités d’association de ReferenceData:: LoadAggregationPoint avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 0..* | Pnode | Hérité de: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | Hérité de: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | Hérité de: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | Hérité de: AggregateNode |
| 0..* | RTO | 1..1 | RTO | Hérité de: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.27 Classe racine LoadRatio

Représente le rapport de la part de charge pour le coordinateur de programmation associé.

Le Tableau 555 présente tous les attributs de LoadRatio.

**Tableau 555 – Attributs de ReferenceData::LoadRatio**

| nom | mult | type | description |
|---|---|---|---|
| intervalStartTime | 0..1 | DateTime | Heure de début de l'intervalle |
| intervalEndTime | 0..1 | DateTime | Heure de fin de l'intervalle |
| share | 0..1 | PerCent | Part en pourcentage de la charge totale de marché pour l'intervalle de temps sélectionné. |

Le Tableau 556 présente toutes les extrémités d'association de LoadRatio avec d'autres classes.

**Tableau 556 – Extrémités d’association de ReferenceData:: LoadRatio avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | SchedulingCoordinator | 0..1 | SchedulingCoordinator | |

### 6.5.9.28 LocalReliabilityArea

Permet la définition de zones de fiabilité (par exemple, les paquets de charge) dans ISO/RTO

Le Tableau 557 présente tous les attributs de LocalReliabilityArea.

**Tableau 557 – Attributs de ReferenceData::LocalReliabilityArea**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 558 présente toutes les extrémités d'association de LocalReliabilityArea avec d'autres classes.

**Tableau 558 – Extrémités d'association de ReferenceData:: LocalReliabilityArea avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..* | RegisteredGenerator | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.29 MPMTestCategory

Fournit une référence aux méthodes et identificateurs d'essai d'atténuation de l'emprise sur le marché *(Market Power Mitigation*) pour les résultats du DAM et du RTM. Les données spécifiques sont l'identificateur d'essai (prix, impact ou conductivité) et la méthode d'essai (système MPM, MPM local, système alternatif MPM ou MPM local alternatif).

Le Tableau 559 présente tous les attributs de MPMTestCategory.

**Tableau 559 – Attributs de ReferenceData::MPMTestCategory**

| nom | mult | type | description |
|---|---|---|---|
| testIdentifier | 0..1 | MPMTestIdentifierType | 1 – Essai de tarification globale<br>2 – Essai de conductivité globale<br>3 – Essai d'impact global<br>4 – Essai de tarification locale<br>5 – Essai de conductivité locale<br>6 – Essai d'impact local |
| testMethod | 0..1 | MPMTestMethodType | Méthode d'exécution de la surveillance de la puissance sur le marché. Par exemple, seuils Normal (par défaut) ou Alternatif. |
| purposeFlag | 0..1 | PurposeFlagType | Nature des données de seuil:<br>'M' – Seuil d'atténuation<br>'R' – Seuil de rapport |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 560 présente toutes les extrémités d'association de MPMTestCategory avec d'autres classes.

**Tableau 560 – Extrémités d'association de ReferenceData:: MPMTestCategory avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | MPMResourceStatus | 0..* | MPMResourceStatus | |
| 1..1 | MPMTestResults | 0..* | MPMTestResults | |
| 1..1 | MPMTestThreshold | 0..* | MPMTestThreshold | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.30 Classe racine MPMTestThreshold

Seuils d'essai d'atténuation de l'emprise sur le marché (MPM) pour les ressources ainsi que les zones de congestion désignées (DCA)

Le Tableau 561 présente tous les attributs de MPMTestThreshold.

**Tableau 561 – Attributs de ReferenceData::MPMTestThreshold**

| nom | mult | type | description |
|---|---|---|---|
| price | 0..1 | CostPerEnergyUnit | Seuil de prix en $/MW |
| percent | 0..1 | PerCent | Seuil de prix en % |
| marketType | 0..1 | MarketType | Type de marché (DAM, RTM) |

Le Tableau 562 présente toutes les extrémités d'association de MPMTestThreshold avec d'autres classes.

**Tableau 562 – Extrémités d'association de ReferenceData:: MPMTestThreshold avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 1..* | AggregatedPnode | 0..* | AggregatedPnode | |
| 0..* | MPMTestCategory | 1..1 | MPMTestCategory | |

### 6.5.9.31 MSSAggregation

Agrégation MSS (Metered Sub-System – sous-système mesuré) de zones MSS.

Le Tableau 563 présente tous les attributs de MSSAggregation.

**Tableau 563 – Attributs de ReferenceData::MSSAggregation**

| nom | mult | type | description |
|---|---|---|---|
| costRecovery | 0..1 | YesNo | Charge pour les coûts d'émission, les coûts de démarrage ou les coûts de charge minimale. |
| grossSettlement | 0..1 | YesNo | Le suivi de la charge MSS peut sélectionner le règlement net ou brut. Le règlement net exige que la demande nette soit établie au point d'agrégation de charge MSS et l'offre nette nécessite d'être établie au même montant que le prix moyen pondéré de la production MSS. La charge brute est établie au point d'agrégation de charge du système et l'offre brute est établie au LMP. L'agrégation MSS qui choisit le règlement brut doit identifier si ses ressources sont du suivi de charge ou non. |
| ignoreLosses | 0..1 | YesNo | Indique si les pertes doivent être ignorées pour cette zone. Également appelé Exclude Marginal Losses (Exclure les pertes marginales). |
| ignoreMarginalLosses | 0..1 | YesNo | Indique si les pertes marginales doivent être ignorées pour cette zone. |
| loadFollowing | 0..1 | YesNo | Indique que cette agrégation MSS spécifique participe à la fonction de suivi de la charge. |
| rucProcurement | 0..1 | YesNo | Indique que RUC sera fourni par l'ISO ou bien autoprocuré. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 564 présente toutes les extrémités d'association de MSSAggregation avec d'autres classes.

**Tableau 564 – Extrémités d'association de ReferenceData:: MSSAggregation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MeteredSubSystem | 1..* | MeteredSubSystem | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.32 MSSZone

Modèle permettant de définir une zone au sein d'un système MSS

Le Tableau 565 présente tous les attributs de MSSZone.

**Tableau 565 – Attributs de ReferenceData::MSSZone**

| nom | mult | type | description |
|---|---|---|---|
| ignoreLosses | 0..1 | YesNo | Indique si les pertes doivent être ignorées pour cette zone MSS. |
| lossFactor | 0..1 | Float | Il s'agit du facteur de perte par défaut pour la zone MSS. Les pertes réelles sont calculées au cours du RTM. |
| rucGrossSettlement | 0..1 | YesNo | Le suivi de la charge MSS peut sélectionner le règlement net ou brut. Le règlement net exige que la demande nette soit établie au point d'agrégation de charge (LAP) MSS et l'offre nette nécessite d'être établie au même montant que le prix moyen pondéré de la production MSS. La charge brute est établie au point d'agrégation de charge du système et l'offre brute est établie au LMP. L'agrégation MSS qui choisit le règlement brut doit identifier si ses ressources sont du suivi de charge ou non. |
| anodeType | 0..1 | AnodeType | Hérité de: AggregateNode |
| qualifASOrder | 0..1 | Integer | Hérité de: AggregateNode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 566 présente toutes les extrémités d'association de MSSZone avec d'autres classes.

**Tableau 566 – Extrémités d'association de ReferenceData:: MSSZone avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MeteredSubSystem | 0..1 | MeteredSubSystem | |
| 0..* | Pnode | 0..* | Pnode | Hérité de: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | Hérité de: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | Hérité de: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | Hérité de: AggregateNode |
| 0..* | RTO | 1..1 | RTO | Hérité de: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.33 MarketPerson

Informations générales relatives au nom et autres informations pour contacter des personnes.

Le Tableau 567 présente tous les attributs de MarketPerson.

**Tableau 567 – Attributs de ReferenceData::MarketPerson**

| nom | mult | type | description |
|---|---|---|---|
| category | 0..1 | String | Catégorie de cette personne relative aux opérations des entreprises d'électricité, classées conformément aux normes et aux pratiques de l'entreprise. Exemples: employé, prestataire, agent, non affilié, etc.<br>Ce champ n'est pas utilisé pour indiquer si cette personne est un client pour l'entreprise d'électricité. Souvent, employé et prestataire sont également clients. Les informations clients sont obtenues en lien avec Organisation et CustomerData. De la même manière, ce champ n'indique pas les différents rôles que peut jouer cette personne dans le cadre des opérations relatives aux entreprises d'électricité. |
| electronicAddressAlternate | 0..1 | ElectronicAddress | Adresse électronique alternative. |
| electronicAddressPrimary | 0..1 | ElectronicAddress | Adresse électronique primaire. |
| firstName | 0..1 | String | Prénom de la personne. |
| governmentID | 0..1 | String | Identifiant unique pour la personne relatif à l'autorité gouvernementale qui la gouverne, par exemple, le "federal tax identifier" (équivalent américain du numéro de sécurité sociale). |
| landlinePhone | 0..1 | TelephoneNumber | Numéro de téléphone fixe. |
| lastName | 0..1 | String | Nom de famille de la personne. |
| mName | 0..1 | String | Deuxième prénom ou initiale(s). |
| mobilePhone | 0..1 | TelephoneNumber | Numéro de téléphone mobile. |
| prefix | 0..1 | String | Titre ou civilité (Mademoiselle, Monsieur, Docteur, etc.) |
| specialNeed | 0..1 | String | Les besoins de services spéciaux pour la personne (contact) sont décrits; par exemple, soutien vital, etc. |
| status | 0..1 | Status | |
| suffix | 0..1 | String | Suffixe pour le nom de la personne, comme II, III, etc. |
| userID | 0..1 | String | Nom d'utilisateur pour la personne; exigé pour la connexion. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 568 présente toutes les extrémités d'association de MarketPerson avec d'autres classes.

**Tableau 568 – Extrémités d'association de ReferenceData:: MarketPerson avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MarketSkills | 0..* | MarketSkill | |
| 0..* | MarketParticipant | 0..* | MarketParticipant | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.34 MarketQualificationRequirement

Certaines compétences sont exigées et doivent faire l'objet d'une certification pour qu'une personne (généralement un membre de l'équipe) soit qualifiée et apte à effectuer des tâches sur certains types d'équipements.

Le Tableau 569 présente tous les attributs de MarketQualificationRequirement.

**Tableau 569 – Attributs de ReferenceData::MarketQualificationRequirement**

| nom | mult | type | description |
|---|---|---|---|
| effectiveDate | 0..1 | DateTime | Date effective du privilège, date de fin du privilège ou date effective de l'application pour l'organisation |
| expirationDate | 0..1 | DateTime | Il s'agit de la date de fin de l'application pour l'organisation<br>L'organisation spécifique ne peut plus accéder à l'application à partir de la date de fin |
| qualificationID | 0..1 | String | Identifiant de qualification. |
| status | 0..1 | Integer | Statut du privilège. Indique le statut de qualification de l'utilisateur.<br>Les statuts actuels sont: 1=Nouveau, 2=Actif, 3=Refusé, 4=Terminé, 5=Supprimé<br>Ces statuts sont soumis à des mises à jour. |
| statusType | 0..1 | String | Nom du statut de la qualification; utilisé pour afficher le statut de l'utilisateur ou de l'organisation. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 570 présente toutes les extrémités d'association de MarketQualificationRequirement avec d'autres classes.

**Tableau 570 – Extrémités d'association de ReferenceData:: MarketQualificationRequirement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketSkills | 0..* | MarketSkill | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.35 MarketRegion

Classe spécialisée de type AggregatedNode. Définit les MarketRegions. Les régions peuvent être les régions de marché, les régions d'énergie ou les régions de services auxiliaires du système.

Le Tableau 571 présente tous les attributs de MarketRegion.

**Tableau 571 – Attributs de ReferenceData::MarketRegion**

| nom | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | Hérité de: AggregateNode |
| qualifASOrder | 0..1 | Integer | Hérité de: AggregateNode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 572 présente toutes les extrémités d'association de MarketRegion avec d'autres classes.

**Tableau 572 – Extrémités d'association de ReferenceData:: MarketRegion avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ReserveDemandCurve | 0..* | ReserveDemandCurve | |
| 1..1 | MarketRegionResults | 1..* | MarketRegionResults | |
| 1..1 | ExPostMarketRegionResults | 0..* | ExPostMarketRegionResults | |
| 0..* | Pnode | 0..* | Pnode | Hérité de: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | Hérité de: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | Hérité de: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | Hérité de: AggregateNode |
| 0..* | RTO | 1..1 | RTO | Hérité de: AggregateNode |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.36 MarketSkill

Niveau de compétences métier exigé pour utiliser ou entretenir un certain type d'actif et/ou réaliser des tâches précises.

Le Tableau 573 présente tous les attributs de MarketSkill.

**Tableau 573 – Attributs de ReferenceData::MarketSkill**

| nom | mult | type | description |
|---|---|---|---|
| certificationPeriod | 0..1 | DateTimeInterval | Intervalle entre la certification et sa date d'expiration. |
| effectiveDateTime | 0..1 | DateTime | Date et heure de prise d'effet de la compétence. |
| level | 0..1 | String | Niveau de compétence pour un métier. |
| type | 0..1 | String | Hérité de: Document |
| authorName | 0..1 | String | Hérité de: Document |
| createdDateTime | 0..1 | DateTime | Hérité de: Document |
| lastModifiedDateTime | 0..1 | DateTime | Hérité de: Document |
| revisionNumber | 0..1 | String | Hérité de: Document |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Document |
| subject | 0..1 | String | Hérité de: Document |
| title | 0..1 | String | Hérité de: Document |
| docStatus | 0..1 | Status | Hérité de: Document |
| status | 0..1 | Status | Hérité de: Document |
| comment | 0..1 | String | Hérité de: Document |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 574 présente toutes les extrémités d'association de MarketSkill avec d'autres classes.

**Tableau 574 – Extrémités d'association de ReferenceData:: MarketSkill avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketQualificationRequirements | 0..* | MarketQualificationRequirement | |
| 0..* | MarketPerson | 0..1 | MarketPerson | |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: Document |
| 0..* | Editor | 0..1 | Editor | Hérité de: Document |
| 0..* | Approver | 0..1 | Approver | Hérité de: Document |
| 0..* | Author | 0..1 | Author | Hérité de: Document |
| 0..* | Issuer | 0..1 | Issuer | Hérité de: Document |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.37 MaxStartUpCostCurve

Coûts et temps maximum de démarrage en fonction du temps d'indisponibilité. Relation entre le coût de démarrage de l'unité (axe Y1) et le temps d'indisponibilité écoulé de l'unité (axe X). Ceci est utilisé pour valider les informations fournies dans l'offre.

Le Tableau 575 présente tous les attributs de MaxStartUpCostCurve.

**Tableau 575 – Attributs de ReferenceData::MaxStartUpCostCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 576 présente toutes les extrémités d'association de MaxStartUpCostCurve avec d'autres classes.

**Tableau 576 – Extrémités d'association de ReferenceData:: MaxStartUpCostCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.38 MeteredSubSystem

Sous-système mesuré

Le Tableau 577 présente tous les attributs de MeteredSubSystem.

**Tableau 577 – Attributs de ReferenceData::MeteredSubSystem**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 578 présente toutes les extrémités d'association de MeteredSubSystem avec d'autres classes.

**Tableau 578 – Extrémités d'association de ReferenceData:: MeteredSubSystem avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MSSZone | 0..* | MSSZone | |
| 1..* | MSSAggregation | 0..1 | MSSAggregation | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.39 MktCombinedCyclePlant

Sous-classe de Production:CombinedCyclePlant issue du paquetage IEC 61970.

Ensemble de turbines à combustion et de turbines à vapeur dans lequel la chaleur évacuée des turbines à combustion est récupérée pour générer de la vapeur destinée aux turbines à vapeur, entraînant ainsi un accroissement global du rendement de la centrale

Le Tableau 579 présente tous les attributs de MktCombinedCyclePlant.

**Tableau 579 – Attributs de ReferenceData::MktCombinedCyclePlant**

| nom | mult | type | description |
|---|---|---|---|
| combCyclePlantRating | 0..1 | ActivePower | Hérité de: CombinedCyclePlant |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 580 présente toutes les extrémités d'association de MktCombinedCyclePlant avec d'autres classes.

**Tableau 580 – Extrémités d’association de ReferenceData:: MktCombinedCyclePlant avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AggregatedPnode | 0..1 | AggregatedPnode | |
| 0..1 | CombinedCycleLogicalConfiguration | 1..* | CombinedCycleLogicalConfiguration | |
| 0..1 | ThermalGeneratingUnits | 0..* | ThermalGeneratingUnit | Hérité de: CombinedCyclePlant |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.40 MktConductingEquipment

Sous-classe de IEC 61970:Core:ConductingEquipment

Le Tableau 581 présente tous les attributs de MktConductingEquipment.

**Tableau 581 – Attributs de ReferenceData::MktConductingEquipment**

| nom | mult | type | description |
|---|---|---|---|
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |

| nom | mult | type | description |
|---|---|---|---|
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 582 présente toutes les extrémités d'association de MktConductingEquipment avec d'autres classes.

**Tableau 582 – Extrémités d'association de ReferenceData:: MktConductingEquipment avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BaseVoltage | 0..1 | BaseVoltage | Hérité de: ConductingEquipment |
| 1..1 | SvStatus | 0..* | SvStatus | Hérité de: ConductingEquipment |
| 0..* | ProtectionEquipments | 0..* | ProtectionEquipment | Hérité de: ConductingEquipment |
| 1..1 | Terminals | 0..* | Terminal | Hérité de: ConductingEquipment |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipmen t | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentC ontainer | 0.. * | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObj ect | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObj ect | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.41 MktContingency

Sous-classe de IEC 61970:Contingency

Le Tableau 583 présente tous les attributs de MktContingency.

**Tableau 583 – Attributs de ReferenceData::MktContingency**

| nom | mult | type | description |
|---|---|---|---|
| loadRolloverFlag | 0..1 | Boolean | indicateur de variation de charge<br>Indicateur qui spécifie s'il convient d'effectuer le retournement de charge et la lecture de charge pour cette contingence. |
| ltcControlFlag | 0..1 | Boolean | indicateur d'activation ltc<br>Indicateur qui spécifie si les changeurs de prise en charge régulent la tension pendant la solution de la contingence. |
| participationFactorSet | 0..1 | String | Indicateur du facteur de participation<br>Indique quel ensemble de facteurs de participation du générateur il convient d'utiliser pour réattribuer de la production dans la contingence. |
| screeningFlag | 0..1 | Boolean | indicateur de contrôle des indisponibilités<br>Indicateur qui spécifie si le contrôle est ignoré pour la contingence. |
| mustStudy | 0..1 | Boolean | Hérité de: Contingency |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 584 présente toutes les extrémités d'association de MktContingency avec d'autres classes.

**Tableau 584 – Extrémités d'association de ReferenceData:: MktContingency avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ContingencyConstraintLimit | 0..* | ContingencyConstraintLimit | |
| 0..1 | TransferInterfaceSolutionA | 0..1 | TransferInterfaceSolution | |
| 0..1 | TransferInterfaceSolutionB | 0..1 | TransferInterfaceSolution | |
| 1..1 | ConstraintResults | 0..* | ConstraintResults | |
| 1..1 | ContingencyElement | 0..* | ContingencyElement | Hérité de: Contingency |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.42 MktHeatRateCurve

Sous-classe de IEC 61970:Generation:Production:HeatRateCurve

Le Tableau 585 présente tous les attributs de MktHeatRateCurve.

**Tableau 585 – Attributs de ReferenceData::MktHeatRateCurve**

| nom | mult | type | description |
|---|---|---|---|
| isNetGrossP | 0..1 | Boolean | Hérité de: HeatRateCurve |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 586 présente toutes les extrémités d'association de MktHeatRateCurve avec d'autres classes.

**Tableau 586 – Extrémités d'association de ReferenceData:: MktHeatRateCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 0..1 | ThermalGeneratingUnit | 1..1 | ThermalGeneratingUnit | Hérité de: HeatRateCurve |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.43 MktThermalGeneratingUnit

Sous-classe de ThermalGeneratingUnit issue du paquetage Production dans IEC 61970.

Le Tableau 587 présente tous les attributs de MktThermalGeneratingUnit.

**Tableau 587 – Attributs de ReferenceData::MktThermalGeneratingUnit**

| nom | mult | type | description |
|---|---|---|---|
| oMCost | 0..1 | CostPerHeatUnit | Hérité de: ThermalGeneratingUnit |
| allocSpinResP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| autoCntrlMarginP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| baseP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| controlDeadband | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| controlPulseHigh | 0..1 | Seconds | Hérité de: GeneratingUnit |
| controlPulseLow | 0..1 | Seconds | Hérité de: GeneratingUnit |
| controlResponseRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| efficiency | 0..1 | PerCent | Hérité de: GeneratingUnit |
| genControlMode | 0..1 | GeneratorControlMode | Hérité de: GeneratingUnit |
| genControlSource | 0..1 | GeneratorControlSource | Hérité de: GeneratingUnit |
| governorMPL | 0..1 | PU | Hérité de: GeneratingUnit |
| governorSCD | 0..1 | PerCent | Hérité de: GeneratingUnit |
| highControlLimit | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| initialP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| longPF | 0..1 | Float | Hérité de: GeneratingUnit |
| lowControlLimit | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| lowerRampRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| maxEconomicP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| maxOperatingP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| minEconomicP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| minimumOffTime | 0..1 | Seconds | Hérité de: GeneratingUnit |
| minOperatingP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| modelDetail | 0..1 | Classification | Hérité de: GeneratingUnit |
| nominalP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| normalPF | 0..1 | Float | Hérité de: GeneratingUnit |
| penaltyFactor | 0..1 | Float | Hérité de: GeneratingUnit |
| raiseRampRate | 0..1 | ActivePowerChangeRate | Hérité de: GeneratingUnit |
| ratedGrossMaxP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| ratedGrossMinP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| ratedNetMaxP | 0..1 | ActivePower | Hérité de: GeneratingUnit |
| shortPF | 0..1 | Float | Hérité de: GeneratingUnit |
| startupCost | 0..1 | Money | Hérité de: GeneratingUnit |
| startupTime | 0..1 | Seconds | Hérité de: GeneratingUnit |
| tieLinePF | 0..1 | Float | Hérité de: GeneratingUnit |
| totalEfficiency | 0..1 | PerCent | Hérité de: GeneratingUnit |
| variableCost | 0..1 | Money | Hérité de: GeneratingUnit |
| aggregate | 0..1 | Boolean | Hérité de: Equipment |
| inService | 0..1 | Boolean | Hérité de: Equipment |
| networkAnalysisEnabled | 0..1 | Boolean | Hérité de: Equipment |
| normallyInService | 0..1 | Boolean | Hérité de: Equipment |

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 588 présente toutes les extrémités d'association de MktThermalGeneratingUnit avec d'autres classes.

**Tableau 588 – Extrémités d'association de ReferenceData:: MktThermalGeneratingUnit avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | CombinedCycleConfigurationMember | 0..* | CombinedCycleConfigurationMember | |
| 0..1 | CAESPlant | 0..1 | CAESPlant | Hérité de: ThermalGeneratingUnit |
| 0..* | CogenerationPlant | 0..1 | CogenerationPlant | Hérité de: ThermalGeneratingUnit |
| 0..* | CombinedCyclePlant | 0..1 | CombinedCyclePlant | Hérité de: ThermalGeneratingUnit |
| 1..1 | EmmissionAccounts | 0..* | EmissionAccount | Hérité de: ThermalGeneratingUnit |
| 1..1 | EmissionCurves | 0..* | EmissionCurve | Hérité de: ThermalGeneratingUnit |
| 1..1 | FossilFuels | 0..* | FossilFuel | Hérité de: ThermalGeneratingUnit |
| 1..1 | FuelAllocationSchedules | 0..* | FuelAllocationSchedule | Hérité de: ThermalGeneratingUnit |
| 1..1 | HeatInputCurve | 0..1 | HeatInputCurve | Hérité de: ThermalGeneratingUnit |
| 1..1 | HeatRateCurve | 0..1 | HeatRateCurve | Hérité de: ThermalGeneratingUnit |
| 1..1 | IncrementalHeatRateCurve | 0..1 | IncrementalHeatRateCurve | Hérité de: ThermalGeneratingUnit |
| 1..1 | ShutdownCurve | 0..1 | ShutdownCurve | Hérité de: ThermalGeneratingUnit |
| 1..1 | StartupModel | 0..1 | StartupModel | Hérité de: ThermalGeneratingUnit |
| 0..1 | RotatingMachine | 0..* | RotatingMachine | Hérité de: GeneratingUnit |
| 1..1 | GenUnitOpCostCurves | 0..* | GenUnitOpCostCurve | Hérité de: GeneratingUnit |
| 1..1 | GenUnitOpSchedule | 0..1 | GenUnitOpSchedule | Hérité de: GeneratingUnit |
| 1..1 | ControlAreaGeneratingUnit | 0..* | ControlAreaGeneratingUnit | Hérité de: GeneratingUnit |
| 1..1 | GrossToNetActivePowerCurves | 0..* | GrossToNetActivePowerCurve | Hérité de: GeneratingUnit |
| 0..1 | OperationalLimitSet | 0..* | OperationalLimitSet | Hérité de: Equipment |
| 1..1 | ContingencyEquipment | 0..* | ContingencyEquipment | Hérité de: Equipment |
| 0..* | EquipmentContainer | 0..1 | EquipmentContainer | Hérité de: Equipment |
| 0..1 | Faults | 0..* | Fault | Hérité de: Equipment |
| 0..* | AdditionalEquipmentContainer | 0.. * | EquipmentContainer | Hérité de: Equipment |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.44 Classe racine OilPrice

Prix du pétrole en unités monétaires

Le Tableau 589 présente tous les attributs de OilPrice.

**Tableau 589 – Attributs de ReferenceData::OilPrice**

| nom | mult | type | description |
|---|---|---|---|
| oilPriceIndex | 0..1 | Float | Prix moyen du pétrole dans une région combustible définie. |

Le Tableau 590 présente toutes les extrémités d'association de OilPrice avec d'autres classes.

**Tableau 590 – Extrémités d’association de ReferenceData:: OilPrice avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | FuelRegion | 1..1 | FuelRegion | |

#### 6.5.9.45 OrgPnodeAllocation

Cette classe modélise l'attribution entre les propriétaires d'actifs et les nœuds de tarification.

Le Tableau 591 présente tous les attributs de OrgPnodeAllocation.

**Tableau 591 – Attributs de ReferenceData::OrgPnodeAllocation**

| nom | mult | type | description |
|---|---|---|---|
| maxMWAllocation | 0..1 | ActivePower | MW maximum pour la source/le puits de l'attribution |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 592 présente toutes les extrémités d'association de OrgPnodeAllocation avec d'autres classes.

**Tableau 592 – Extrémités d'association de ReferenceData:: OrgPnodeAllocation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Pnode | 1..1 | Pnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.46 OrgResOwnership

Cette classe modélise le pourcentage et le type de propriété entre la ressource et l'organisation.

Le Tableau 593 présente tous les attributs de OrgResOwnership.

**Tableau 593 – Attributs de ReferenceData::OrgResOwnership**

| nom | mult | type | description |
|---|---|---|---|
| asscType | 0..1 | ResourceAssnType | Type d'association pour l'association entre l'organisation et la ressource. |
| masterSchedulingCoordinatorFlag | 0..1 | YesNo | Indicateur spécifiant que le coordinateur de programmation représentant la ressource est le coordinateur de programmation principal. |
| ownershipPercent | 0..1 | PerCent | Pourcentage de propriété pour chaque ressource |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 594 présente toutes les extrémités d'association de OrgResOwnership avec d'autres classes.

**Tableau 594 – Extrémités d'association de ReferenceData:: OrgResOwnership avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.47 Pnode

Un nœud de tarification est directement associé à un nœud de connectivité. Il s'agit d'un emplacement de tarification auquel les participants du marché soumettent leurs offres d'achat/de vente, leurs droits à revenu de congestion (CRR) achat/vente et leur règlement.

Le Tableau 595 présente tous les attributs de Pnode.

**Tableau 595 – Attributs de ReferenceData::Pnode**

| nom | mult | type | description |
|---|---|---|---|
| isPublic | 0..1 | Boolean | Si True, alors ce Pnode est public (les prix sont publiés pour les marchés DA/RT et FTR); sinon, il est privé (l'emplacement n'est pas utilisable par le marché pour les appels d'offres/FTR/transactions). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 596 présente toutes les extrémités d'association de Pnode avec d'autres classes.

**Tableau 596 – Extrémités d'association de ReferenceData::Pnode avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResources | 0..* | RegisteredResource | Une ressource enregistrée injecte de la puissance dans un ou plusieurs nœuds de connectivité relatifs à un nœud de tarification. |
| 0..* | SourceCRRSegment | 0..* | CRRSegment | |
| 0..* | SinkCRRSegment | 0..* | CRRSegment | |
| 0..1 | MktMeasurement | 0..* | MktMeasurement | Permet d'associer Measurements à des Pnodes. |
| 1..1 | ExPostResults | 0..* | ExPostPricingResults | |
| 0..1 | PnodeResults | 1..* | PnodeResults | |
| 0..1 | Trade | 0..* | Trade | |
| 0..1 | ReceiptTransactionBids | 0..* | TransactionBid | |
| 0..1 | DeliveryTransactionBids | 0..* | TransactionBid | |
| 0..* | SubControlArea | 0..1 | SubControlArea | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 0..* | AggregateNode | 0..* | AggregateNode | |
| 1..1 | OrgPnodeAllocation | 0..* | OrgPnodeAllocation | |
| 0..* | RTO | 0..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.48 Classe racine PnodeDistributionFactor

Cette classe permet au coordinateur de programmation d'introduire plusieurs facteurs de distribution pour un nœud de tarification.

Le Tableau 597 présente tous les attributs de PnodeDistributionFactor.

**Tableau 597 – Attributs de ReferenceData::PnodeDistributionFactor**

| nom | mult | type | description |
|---|---|---|---|
| factor | 0..1 | Float | Utilisé pour calculer la "participation" d'un Pnode dans un AggregatePnode. Par exemple, pour une région de régulation, ce facteur est défini sur 1 et le total de tous les facteurs d'une région de régulation spécifique ne doit pas être défini sur 1. Pour une zone de tarification, le total de tous les facteurs doit être défini sur 1. |
| offPeak | 0..1 | YesNo | Indique que ce facteur de distribution doit être appliqué pendant les périodes creuses. |
| onPeak | 0..1 | YesNo | Indique que ce facteur doit être appliqué pendant les périodes de pointe. |
| podLossFactor | 0..1 | Float | Facteur de perte du point de livraison. |

Le Tableau 598 présente toutes les extrémités d'association de PnodeDistributionFactor avec d'autres classes.

**Tableau 598 – Extrémités d’association de ReferenceData:: PnodeDistributionFactor avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | BidDistributionFactor | 0..1 | BidDistributionFactor | |
| 1..* | AggregatedPnode | 1..1 | AggregatedPnode | |
| 0..* | IndividualPnode | 1..1 | IndividualPnode | |

### 6.5.9.49 RMRHeatRateCurve

Modèle prenant en charge le traitement des unités RMR.

Le Tableau 599 présente tous les attributs de RMRHeatRateCurve.

**Tableau 599 – Attributs de ReferenceData::RMRHeatRateCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |

| nom | mult | type | description |
|---|---|---|---|
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 600 présente toutes les extrémités d'association de RMRHeatRateCurve avec d'autres classes.

**Tableau 600 – Extrémités d'association de ReferenceData:: RMRHeatRateCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.50 RMRStartUpCostCurve

Modèle prenant en charge le traitement des unités RMR.

Le Tableau 601 présente tous les attributs de RMRStartUpCostCurve.

**Tableau 601 – Attributs de ReferenceData::RMRStartUpCostCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 602 présente toutes les extrémités d'association de RMRStartUpCostCurve avec d'autres classes.

**Tableau 602 – Extrémités d'association de ReferenceData:: RMRStartUpCostCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.51 RMRStartUpEnergyCurve

Modèle prenant en charge le traitement des unités RMR.

Le Tableau 603 présente tous les attributs de RMRStartUpEnergyCurve.

**Tableau 603 – Attributs de ReferenceData::RMRStartUpEnergyCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 604 présente toutes les extrémités d'association de RMRStartUpEnergyCurve avec d'autres classes.

**Tableau 604 – Extrémités d'association de ReferenceData:: RMRStartUpEnergyCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.52 RMRStartUpFuelCurve

Modèle prenant en charge le traitement des unités RMR.

Le Tableau 605 présente tous les attributs de RMRStartUpFuelCurve.

**Tableau 605 – Attributs de ReferenceData::RMRStartUpFuelCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 606 présente toutes les extrémités d'association de RMRStartUpFuelCurve avec d'autres classes.

**Tableau 606 – Extrémités d'association de ReferenceData:: RMRStartUpFuelCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.53 RMRStartUpTimeCurve

Modèle prenant en charge le traitement des unités RMR.

Le Tableau 607 présente tous les attributs de RMRStartUpTimeCurve.

**Tableau 607 – Attributs de ReferenceData::RMRStartUpTimeCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 608 présente toutes les extrémités d'association de RMRStartUpTimeCurve avec d'autres classes.

**Tableau 608 – Extrémités d'association de ReferenceData:: RMRStartUpTimeCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.54 RTO

Gestionnaire des réseaux de transport régionaux.

Le Tableau 609 présente tous les attributs de RTO.

**Tableau 609 – Attributs de ReferenceData::RTO**

| nom | mult | type | description |
|---|---|---|---|
| streetAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| postalAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| phone1 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| phone2 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Organisation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 610 présente toutes les extrémités d'association de RTO avec d'autres classes.

**Tableau 610 – Extrémités d'association de ReferenceData::RTO avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | MktConnectivityNode | 0..* | MktConnectivityNode | |
| 1..1 | CommodityDefinition | 0..* | CommodityDefinition | |
| 1..1 | AdjacentCASet | 0..* | AdjacentCASet | |
| 1..1 | AggregateNode | 0..* | AggregateNode | |
| 1..1 | TransmissionContractRight | 0..* | ContractRight | |
| 1..1 | FuelRegion | 0..* | FuelRegion | |
| 1..1 | HostControlArea | 0..* | HostControlArea | |
| 1..1 | LocalReliabilityArea | 0..* | LocalReliabilityArea | |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | Pnodes | 0..* | Pnode | |
| 1..1 | TransmissionRightChain | 0..* | TransmissionRightChain | |
| 1..1 | SubControlArea | 0..* | SubControlArea | |
| 0..1 | EnergyMarkets | 0..* | EnergyMarket | |
| 0..1 | SecurityConstraintsLinear | 0..* | SecurityConstraintSum | |
| 0..1 | SecurityConstraints | 0..* | SecurityConstraints | |
| 1..1 | MSSAggregation | 0..* | MSSAggregation | |
| 0..* | MarketPerson | 0..* | MarketPerson | Hérité de: MarketParticipant |
| 0..* | MarketDocument | 0..* | MarketDocument | Hérité de: MarketParticipant |
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | Hérité de: MarketParticipant |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: MarketParticipant |
| 0..1 | Bid | 0..* | Bid | Hérité de: MarketParticipant |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: MarketParticipant |
| 0..* | MarketRole | 0..* | MarketRole | Hérité de: MarketParticipant |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | Hérité de: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | Hérité de: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.55 RUCZone

Classe spécialisée de type AggregatedNode. Définit les zones RUC. Une région de prévisions représente une collection de nœuds pour lesquels l'opérateur du marché a développé suffisamment de demandes historiques et de données climatiques pertinentes pour effectuer une prévision de la demande pour cette région. L'opérateur du marché peut ajuster cette prévision pour garantir que l'engagement d'unité pour fiabilité (RUC – *Reliability Unit Commitment*) fournit l'approvisionnement en capacité locale adéquat.

Le Tableau 611 présente tous les attributs de RUCZone.

**Tableau 611 – Attributs de ReferenceData::RUCZone**

| nom | mult | type | description |
|---|---|---|---|
| anodeType | 0..1 | AnodeType | Hérité de: AggregateNode |
| qualifASOrder | 0..1 | Integer | Hérité de: AggregateNode |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 612 présente toutes les extrémités d'association de RUCZone avec d'autres classes.

**Tableau 612 – Extrémités d'association de ReferenceData:: RUCZone avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | LossClearingResults | 0..* | LossClearingResults | |
| 0..* | Pnode | 0..* | Pnode | Hérité de: AggregateNode |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | Hérité de: AggregateNode |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: AggregateNode |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: AggregateNode |
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | Hérité de: AggregateNode |
| 0..1 | Instruction | 0..* | Instructions | Hérité de: AggregateNode |
| 0..* | RTO | 1..1 | RTO | Hérité de: AggregateNode |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.56 RegisteredDistributedResource

Ressource enregistrée qui représente une source d'énergie distribuée, telle qu'un microgénérateur, une pile à combustible, une source d'énergie photovoltaïque, etc.

Le Tableau 613 présente tous les attributs de RegisteredDistributedResource.

**Tableau 613 – Attributs de ReferenceData::RegisteredDistributedResource**

| nom | mult | type | description |
|---|---|---|---|
| distributedResourceType | 0..1 | String | Type de ressource. Par exemple: pile à combustible, volant d'inertie, énergie photovoltaïque, microturbine, production combinée de chaleur et d'électricité, *Vehicle to grid* (déchargement d'un véhicule vers le réseau) et le stockage décentralisé d'énergie, entre autres. |
| ACAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | Hérité de: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| dispatchable | 0..1 | Boolean | Hérité de: RegisteredResource |
| ECAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | Hérité de: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | Hérité de: RegisteredResource |
| lastModified | 0..1 | DateTime | Hérité de: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |

| nom | mult | type | description |
|---|---|---|---|
| marketParticipationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | Hérité de: RegisteredResource |
| maxOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| minDispatchTime | 0..1 | Hours | Hérité de: RegisteredResource |
| minOffTime | 0..1 | Float | Hérité de: RegisteredResource |
| minOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| nonMarket | 0..1 | YesNo | Hérité de: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | Hérité de: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationSt atus | Hérité de: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 614 présente toutes les extrémités d'association de RegisteredDistributedResource avec d'autres classes.

**Tableau 614 – Extrémités d'association de ReferenceData:: RegisteredDistributedResource avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ResourcePerformanceR atings | 0..* | ResourcePerformanceR ating | |
| 1..1 | ResponseMethods | 0..* | ResponseMethod | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | Hérité de: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | Hérité de: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | Hérité de: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | Hérité de: RegisteredResource |
| 0..1 | ContractDistributionFact or | 0..* | ContractDistributionFact or | Hérité de: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | Hérité de: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | Hérité de: RegisteredResource |
| 0..1 | ResourceDispatchResul ts | 0..* | ResourceDispatchResult s | Hérité de: RegisteredResource |
| 0..1 | SubstitutionResourceLis t | 0..* | SubstitutionResourceLis t | Hérité de: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | Hérité de: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: RegisteredResource |
| 1..1 | IntermittentResourceEli gibility | 0..* | IntermittentResourceElig ibility | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | FormerReference | 0..* | FormerReference | Hérité de: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | Hérité de: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | Hérité de: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | Hérité de: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | Hérité de: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | Hérité de: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Hérité de: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | Hérité de: RegisteredResource |
| 0..* | Reason | 0..* | Reason | Hérité de: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | Hérité de: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | Hérité de: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | Hérité de: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | Hérité de: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | Hérité de: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | Hérité de: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | Hérité de: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | Hérité de: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | Hérité de: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | Hérité de: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | Hérité de: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | Hérité de: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | Hérité de: RegisteredResource |
| 0..* | Domain | 0..* | Domain | Hérité de: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | Hérité de: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | Hérité de: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.57 RegisteredGenerator

Modèle d'un générateur enregistré pour participer au marché

Le Tableau 615 présente tous les attributs de RegisteredGenerator.

**Tableau 615 – Attributs de ReferenceData::RegisteredGenerator**

| nom | mult | type | description |
|---|---|---|---|
| capacityFactor | 0..1 | PerCent | Rapport de l'énergie réelle produite par la ressource divisé par l'énergie potentielle maximale si la ressource est complètement utilisée. Par exemple: centrales éoliennes. |
| coldStartTime | 0..1 | Float | Temps de démarrage à froid. |
| combinedCycleOperatingMode | 0..1 | String | Mode d'exploitation à cycle combiné. |
| commericialOperationDate | 0..1 | DateTime | |
| constrainedOutputFlag | 0..1 | YesNo | Indicateur (Yes/No) du générateur à sortie restreinte (COG – Constrained Output Generator), par ressource de production |
| energyDownRampRate | 0..1 | ActivePowerChangeRate | Taux de réponse en MW par minute pour une rampe décroissante de l'énergie. |
| energyUpRampRate | 0..1 | ActivePowerChangeRate | Taux de réponse en MW par minute pour une rampe croissante de l'énergie. |
| extremeLongStart | 0..1 | YesNo | Certaines unités qui mettent beaucoup de temps à démarrer peuvent nécessiter de recevoir des instructions de démarrage avant que les résultats du marché DA ne soient disponibles. Les ressources dont le démarrage est long peuvent être des ressources physiques au sein de la commande dont le temps de démarrage est supérieur à 18 heures ou l'engagement de ligne d'interconnexion contractuelle à démarrage long qui doit être terminé à 6 h le jour précédent. Par conséquent, il est nécessaire qu'un processus détermine l'engagement de telles ressources avant le marché DA. |
| fuelSource | 0..1 | FuelSource | Valeurs: ressources à base de gaz naturel, ressource à base de gaz non naturel<br>"NG": ressource à base de gaz naturel; ressource alimentée au gaz naturel<br>"NNG": ressource à base de gaz non naturel; ressource alimentée par un autre combustible que le gaz naturel |
| highControlLimit | 0..1 | ActivePower | Limite supérieure du contrôle secondaire (AGC) |
| hotIntTime | 0..1 | Float | Temps de chaud à intermédiaire (saisonnier) |
| hotStartTime | 0..1 | Float | Temps de démarrage à chaud |
| intColdTime | 0..1 | Float | Temps d'intermédiaire à froid (saisonnier) |
| intStartTime | 0..1 | Float | Temps de démarrage à température intermédiaire. |
| loadFollowingDownMSS | 0..1 | YesNo | Certifie les ressources à utiliser dans le suivi de charge descendant MSS. |
| loadFollowingUpMSS | 0..1 | YesNo | Certifie les ressources à utiliser dans le suivi de charge ascendant MSS. |
| lowControlLImit | 0..1 | ActivePower | Limite inférieure du contrôle secondaire (AGC) |
| maxDependableCap | 0..1 | ActivePower | Capacité fiable maximale (MNDC). La capacité fiable maximale nette est utilisée conjointement avec un contrat RMR. |
| maximumAllowableSpinningReserve | 0..1 | ActivePower | Réserve tournante maximale autorisée. La réserve tournante ne sera jamais considérée comme supérieure à cette valeur sans tenir compte du point d'exploitation en cours. |
| maximumOperatingLimit | 0..1 | ActivePower | Limite opérationnelle maximale en MW pouvant être entrée dans l'unité par le répartiteur |

| nom | mult | type | description |
|---|---|---|---|
| maxMinLoadCost | 0..1 | CostRate | Valeur maximale enregistrée du coût de charge minimal d'une ressource de production enregistré à l'aide d'une base de coût "Bid Cost" (coût d'offre). |
| maxPumpingLevel | 0..1 | ActivePower | Niveau de pompage maximal d'une unité à pompe hydraulique |
| maxShutdownTime | 0..1 | DateTime | Heure maximale à laquelle cet appareil peut être arrêté. |
| maxStartUpsPerDay | 0..1 | Integer | Nombre maximal de démarrages par jour |
| maxWeeklyEnergy | 0..1 | Float | Énergie hebdomadaire maximale (saisonnière) |
| maxWeeklyStarts | 0..1 | Integer | Nombre maximal de démarrages hebdomadaires (paramètre saisonnier) |
| minimumLoadCostBasis | 0..1 | CostBasis | Coût de base pour la charge minimale. |
| minimumLoadFuelCost | 0..1 | CostPerHeatUnit | Coût du combustible exigé pour l'exploitation d'une ressource de production au niveau de charge minimal. |
| minimumOperatingLimit | 0..1 | ActivePower | Limite opérationnelle minimale en MW pouvant être entrée dans l'unité par le répartiteur |
| minLoadCost | 0..1 | CostRate | Coût de charge minimal. Valeur: (devise/h) |
| mustOfferRA | 0..1 | YesNo | Indicateur qui spécifie que cette unité est une ressource d'adéquation de ressource et d'offre obligatoire. |
| nameplateCapacity | 0..1 | ActivePower | Valeur en MW indiquée sur la plaque signalétique du générateur; valeur que ce dernier peut fournir. |
| operatingMaintenanceCost | 0..1 | CostPerEnergyUnit | Partie du coût d'exploitation d'une ressource de production qui n'est pas relative au coût de combustible. |
| pumpingCost | 0..1 | CostRate | |
| pumpingFactor | 0..1 | Float | Facteur de pompage pour les unités de stockage à pompe, facteur de conversion entre la production et le pompage. |
| pumpMinDownTime | 0..1 | Float | Temps d'indisponibilité minimal de la pompe dans une unité de stockage à pompe. |
| pumpMinUpTime | 0..1 | Float | Aspect de temps de disponibilité minimal de la pompe dans une unité de stockage à pompe. |
| pumpShutdownCost | 0..1 | Float | Coût d'arrêt d'une pompe lors de l'aspect de la pompe d'une unité de stockage à pompe. |
| pumpShutdownTime | 0..1 | Integer | Temps d'arrêt (en minutes) de l'aspect de la pompe d'une unité de stockage à pompe. |
| quickStartFlag | 0..1 | YesNo | Indicateur de démarrage rapide (Yes/No). Identifie le générateur enregistré comme unité de démarrage rapide. Une unité de démarrage rapide est une unité ayant la capacité d'être disponible pour une charge sous une période de 30 min. |
| rampCurveType | 0..1 | RampCurveType | Type de courbe de rampe. Identifie le type de courbe pouvant être fixe, statique ou dynamique. |
| regulationDownRampRate | 0..1 | ActivePowerChangeRate | Taux de réponse de régulation à la baisse en MW par minute. |
| regulationFlag | 0..1 | UnitRegulationKind | Indique si l'unité est en cours de régulation ou non ou si elle est censée être en cours de régulation mais ne l'est pas. |
| regulationUpRampRate | 0..1 | ActivePowerChangeRate | Taux de réponse de régulation à la hausse en MW par minute. |

| nom | mult | type | description |
|---|---|---|---|
| resourceSubType | 0..1 | String | Sous-type d'unité utilisé par Settlements ou par une application de programmation. L'utilisation d'applications par le sous-type d'unité peut définir les types nécessaires, le cas échéant. |
| riverSystem | 0..1 | String | Réseau fluvial auquel la ressource est reliée. |
| RMNRFlag | 0..1 | YesNo | Indicateur RMNR (*Reliability Must Not Run*): indique si l'unité RMR est définie comme RMNR dans le marché actuel. |
| RMRFlag | 0..1 | FlagTypeRMR | Indicateur RMR (marche obligatoire pour fiabilité): indique si l'unité est du type RMR:<br>N' – l'unité n'est pas de type RMR<br>'1' – Unité RMR de condition 1<br>'2' – Unité RMR de condition 2 |
| RMRManualIndicator | 0..1 | YesNo | Indique le statut de prédétermination manuelle RMR [Y/N] |
| RMTFlag | 0..1 | YesNo | Indicateur RMT (Prise obligatoire pour Fiabilité) (Yes/No): indique si l'unité est du type RMR |
| spinRampRate | 0..1 | ActivePowerChangeRate | Taux de réponse de la réserve tournante en MW par minute. |
| startUpCostBasis | 0..1 | CostBasis | Coût de base pour le démarrage. |
| syncCondCapable | 0..1 | YesNo | Le compensateur synchrone de ressource est-il une ressource capable? |
| unitType | 0..1 | UnitType | Type d'unité de production: à cycle combiné, turbine à gaz, turbine hydraulique, autre, photovoltaïque, turbine à pompe hydraulique, moteur à explosion, turbine à vapeur, compensateur synchrone, éolienne. |
| useLimitFlag | 0..1 | YesNo | Indicateur de limite d'utilisation: indique si la ressource à utilisation limitée est entièrement planifiée (ou si elle contient des excédents pour la répartition en temps réel) (Y/N) |
| variableEnergyResource | 0..1 | YesNo | Indique que cette ressource a l'intention de participer au programme de ressource intermittente. |
| ACAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | Hérité de: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| dispatchable | 0..1 | Boolean | Hérité de: RegisteredResource |
| ECAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | Hérité de: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | Hérité de: RegisteredResource |
| lastModified | 0..1 | DateTime | Hérité de: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | Hérité de: RegisteredResource |
| maxOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| minDispatchTime | 0..1 | Hours | Hérité de: RegisteredResource |
| minOffTime | 0..1 | Float | Hérité de: RegisteredResource |
| minOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |

| nom | mult | type | description |
|---|---|---|---|
| nonMarket | 0..1 | YesNo | Hérité de: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | Hérité de: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | Hérité de: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 616 présente toutes les extrémités d'association de RegisteredGenerator avec d'autres classes.

**Tableau 616 – Extrémités d'association de ReferenceData:: RegisteredGenerator avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | GeneratingBids | 0..* | GeneratingBid | |
| 0..1 | StartUpTimeCurve | 0..1 | StartUpTimeCurve | |
| 0..1 | Trade | 0..* | Trade | |
| 0..1 | MktHeatRateCurve | 0..1 | MktHeatRateCurve | |
| 0..1 | RMRHeatRateCurve | 0..1 | RMRHeatRateCurve | |
| 0..1 | RMRStartUpCostCurve | 0..1 | RMRStartUpCostCurve | |
| 0..1 | RMRStartUpEnergyCurve | 0..1 | RMRStartUpEnergyCurve | |
| 0..1 | RMRStartUpFuelCurve | 0..1 | RMRStartUpFuelCurve | |
| 0..1 | RMRStartUpTimeCurve | 0..1 | RMRStartUpTimeCurve | |
| 0..1 | RegulatingLimit | 0..1 | RegulatingLimit | |
| 0..1 | StartUpFuelCurve | 0..1 | StartUpFuelCurve | |
| 0..1 | StartUpEnergyCurve | 0..1 | StartUpEnergyCurve | |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | |
| 1..1 | EnergyPriceIndex | 1..1 | EnergyPriceIndex | |
| 0..1 | UnitInitialConditions | 0..* | UnitInitialConditions | |
| 0..* | StartUpCostCurves | 0..* | StartUpCostCurve | |
| 0..1 | FuelCostCurve | 0..1 | FuelCostCurve | |
| 0..* | FuelRegion | 0..1 | FuelRegion | |
| 0..* | LocalReliabilityArea | 0..1 | LocalReliabilityArea | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | Hérité de: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | Hérité de: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | Hérité de: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | Hérité de: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | Hérité de: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | Hérité de: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | Hérité de: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | Hérité de: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | Hérité de: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | Hérité de: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | Hérité de: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | Hérité de: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | Hérité de: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | Hérité de: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | Hérité de: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | Hérité de: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Hérité de: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | Hérité de: RegisteredResource |
| 0..* | Reason | 0..* | Reason | Hérité de: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | Hérité de: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | Hérité de: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | Hérité de: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | Hérité de: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | Hérité de: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | Hérité de: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | Hérité de: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | Hérité de: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | Hérité de: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | Hérité de: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | Hérité de: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | HostControlArea | 0..1 | HostControlArea | Hérité de: RegisteredResource |
| 0..* | Domain | 0..* | Domain | Hérité de: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | Hérité de: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | Hérité de: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.58 RegisteredInterTie

Cette classe représente la ressource de ligne d'interconnexion.

Le Tableau 617 présente tous les attributs de RegisteredInterTie.

**Tableau 617 – Attributs de ReferenceData::RegisteredInterTie**

| nom | mult | type | description |
|---|---|---|---|
| direction | 0..1 | InterTieDirection | Indique la direction (exportation/importation) d'une ressource d'interconnexion. |
| energyProductType | 0..1 | EnergyProductType | Dans chaque principal type de produit, le type de marchandise peut être appliqué pour spécifier davantage le type. |
| isDCTie | 0..1 | YesNo | Indicateur spécifiant si cette interconnexion est une connexion à courant continu. |
| isDynamicInterchange | 0..1 | YesNo | Spécifie si la ressource d'interconnexion est enregistrée pour l'échange dynamique. |
| minHourlyBlockLimit | 0..1 | Integer | Limite supérieure enregistrée du bloc horaire minimal pour une ressource d'interconnexion. |
| ACAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | Hérité de: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| dispatchable | 0..1 | Boolean | Hérité de: RegisteredResource |
| ECAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | Hérité de: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | Hérité de: RegisteredResource |
| lastModified | 0..1 | DateTime | Hérité de: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |

| nom | mult | type | description |
|---|---|---|---|
| marketParticipationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | Hérité de: RegisteredResource |
| maxOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| minDispatchTime | 0..1 | Hours | Hérité de: RegisteredResource |
| minOffTime | 0..1 | Float | Hérité de: RegisteredResource |
| minOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| nonMarket | 0..1 | YesNo | Hérité de: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | Hérité de: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | Hérité de: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| aliasName | 0..1 | String | Hérité de: RegisteredResource |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 618 présente toutes les extrémités d'association de RegisteredInterTie avec d'autres classes.

**Tableau 618 – Extrémités d'association de ReferenceData:: RegisteredInterTie avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | InterTieDispatchResponse | 0..* | InterTieDispatchResponse | |
| 0..1 | InterchangeSchedule | 0..* | InterchangeSchedule | |
| 0..1 | InterTieBid | 0..1 | InterTieBid | |
| 0..* | Flowgate | 1..1 | Flowgate | |
| 0..* | WheelingCounterParty | 0..* | WheelingCounterParty | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | Hérité de: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | Hérité de: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | Hérité de: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | Hérité de: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | Hérité de: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | Hérité de: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | Hérité de: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | Hérité de: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | Hérité de: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | Hérité de: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | Hérité de: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | Hérité de: RegisteredResource |
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | Hérité de: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | Hérité de: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | Hérité de: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | Hérité de: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Hérité de: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | Hérité de: RegisteredResource |
| 0..* | Reason | 0..* | Reason | Hérité de: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | Hérité de: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | Hérité de: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | Hérité de: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | Hérité de: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | Hérité de: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | Hérité de: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | Hérité de: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | Hérité de: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | Hérité de: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | Hérité de: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | Hérité de: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | Hérité de: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | Hérité de: RegisteredResource |
| 0..* | Domain | 0..* | Domain | Hérité de: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | Hérité de: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | Hérité de: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.59 RegisteredLoad

Modèle d'une charge enregistrée pour participer au marché.

RegisteredLoad est utilisée pour modéliser toute charge desservie directement par le marché de vente en gros. Les RegisteredLoads peuvent être répartissables ou non-répartissables et peuvent avoir ou ne pas avoir de courbes d'offre. Par exemple: charge d'une société de distribution, charge d'un détaillant d'énergie, charge d'une installation raccordée à un système d'alimentation en vrac important.

Il convient de modéliser les charges desservies de manière indirecte, par exemple, par le biais d'un détaillant d'énergie ou d'une entreprise intégrée d'électricité, comme des RegisteredDistributedResources. Par exemple: stockage au niveau distribution, production au niveau distribution et réponse à la demande au niveau distribution.

Le Tableau 619 présente tous les attributs de RegisteredLoad

**Tableau 619 – Attributs de ReferenceData::RegisteredLoad**

| nom | mult | type | description |
|---|---|---|---|
| blockLoadTransfer | 0..1 | Boolean | Procédure d'exploitation d'urgence – Indicateur spécifiant que la ressource est une pseudo-ressource de charge en masse. |
| dynamicallyScheduledLoadResource | 0..1 | Boolean | Indicateur spécifiant qu'une ressource de charge fait partie d'une charge DSR. |
| dynamicallyScheduledQualification | 0..1 | Boolean | Statut de qualification (utilisé pour la qualification DSR) |
| ACAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| ASSPOptimizationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| commercialOpDate | 0..1 | DateTime | Hérité de: RegisteredResource |
| contingencyAvailFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| dispatchable | 0..1 | Boolean | Hérité de: RegisteredResource |
| ECAFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| flexibleOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| hourlyPredispatch | 0..1 | YesNo | Hérité de: RegisteredResource |
| isAggregatedRes | 0..1 | YesNo | Hérité de: RegisteredResource |
| lastModified | 0..1 | DateTime | Hérité de: RegisteredResource |
| LMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| marketParticipationFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| maxBaseSelfSchedQty | 0..1 | Float | Hérité de: RegisteredResource |
| maxOnTime | 0..1 | Float | Hérité de: RegisteredResource |

| nom | mult | type | description |
|---|---|---|---|
| minDispatchTime | 0..1 | Hours | Hérité de: RegisteredResource |
| minOffTime | 0..1 | Float | Hérité de: RegisteredResource |
| minOnTime | 0..1 | Float | Hérité de: RegisteredResource |
| mustOfferFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| nonMarket | 0..1 | YesNo | Hérité de: RegisteredResource |
| pointOfDeliveryFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagDA | 0..1 | YesNo | Hérité de: RegisteredResource |
| priceSetFlagRT | 0..1 | YesNo | Hérité de: RegisteredResource |
| registrationStatus | 0..1 | ResourceRegistrationStatus | Hérité de: RegisteredResource |
| resourceAdequacyFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| SMPMFlag | 0..1 | YesNo | Hérité de: RegisteredResource |
| aliasName | 0..1 | String | Hérité de: RegisteredResource |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 620 présente toutes les extrémités d'association de RegisteredLoad avec d'autres classes.

**Tableau 620 – Extrémités d’association de ReferenceData:: RegisteredLoad avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | LoadBids | 0..* | LoadBid | |
| 0..1 | AuxillaryObject | 0..* | AuxiliaryObject | |
| 0..* | MarketParticipant | 0..1 | MarketParticipant | Hérité de: RegisteredResource |
| 0..* | ForbiddenRegion | 0..* | ForbiddenRegion | Hérité de: RegisteredResource |
| 0..* | AggregateNode | 0..1 | AggregateNode | Hérité de: RegisteredResource |
| 1..1 | DispatchInstReply | 0..* | DispatchInstReply | Hérité de: RegisteredResource |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: RegisteredResource |
| 0..* | RampRateCurve | 0..* | RampRateCurve | Hérité de: RegisteredResource |
| 0..1 | ContractDistributionFactor | 0..* | ContractDistributionFactor | Hérité de: RegisteredResource |
| 1..1 | OrgResOwnership | 0..* | OrgResOwnership | Hérité de: RegisteredResource |
| 0..1 | RMROperatorInput | 0..* | RMROperatorInput | Hérité de: RegisteredResource |
| 0..1 | ResourceDispatchResults | 0..* | ResourceDispatchResults | Hérité de: RegisteredResource |
| 0..1 | SubstitutionResourceList | 0..* | SubstitutionResourceList | Hérité de: RegisteredResource |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | Hérité de: RegisteredResource |
| 0..* | SubControlArea | 0..* | SubControlArea | Hérité de: RegisteredResource |
| 1..1 | IntermittentResourceEligibility | 0..* | IntermittentResourceEligibility | Hérité de: RegisteredResource |
| 1..1 | FormerReference | 0..* | FormerReference | Hérité de: RegisteredResource |
| 1..1 | Commitments | 0..* | Commitments | Hérité de: RegisteredResource |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | MPMResourceStatus | 0..* | MPMResourceStatus | Hérité de: RegisteredResource |
| 0..1 | RUCAwardInstruction | 0..* | RUCAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | AllocationResultValues | 0..* | AllocationResultValues | Hérité de: RegisteredResource |
| 1..1 | LoadFollowingInst | 0..* | LoadFollowingInst | Hérité de: RegisteredResource |
| 0..1 | ResourceLoadFollowingInst | 0..* | ResourceLoadFollowingInst | Hérité de: RegisteredResource |
| 1..1 | ResourceAncillaryServiceQualification | 0..* | ResourceCertification | Hérité de: RegisteredResource |
| 0..1 | ExpectedEnergyValues | 0..* | ExpectedEnergyValues | Hérité de: RegisteredResource |
| 0..* | Reason | 0..* | Reason | Hérité de: RegisteredResource |
| 0..* | InterTie | 0..* | SchedulingPoint | Hérité de: RegisteredResource |
| 0..* | MktConnectivityNode | 0..1 | MktConnectivityNode | Hérité de: RegisteredResource |
| 0..1 | DopInstruction | 0..* | DopInstruction | Hérité de: RegisteredResource |
| 0..* | ResourceCapacity | 0..* | ResourceCapacity | Hérité de: RegisteredResource |
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | Hérité de: RegisteredResource |
| 1..1 | DefaultBid | 0..1 | DefaultBid | Hérité de: RegisteredResource |
| 0..* | MarketObjectStatus | 0..* | MarketObjectStatus | Hérité de: RegisteredResource |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | Hérité de: RegisteredResource |
| 0..1 | LoadFollowingOperatorInput | 0..* | LoadFollowingOperatorInput | Hérité de: RegisteredResource |
| 0..* | MPMTestThreshold | 0..* | MPMTestThreshold | Hérité de: RegisteredResource |
| 0..1 | ResourceAwardInstruction | 0..* | ResourceAwardInstruction | Hérité de: RegisteredResource |
| 0..1 | ExPostResourceResults | 0..* | ExPostResourceResults | Hérité de: RegisteredResource |
| 1..1 | Instructions | 0..* | Instructions | Hérité de: RegisteredResource |
| 0..* | Pnode | 0..1 | Pnode | Hérité de: RegisteredResource |
| 0..* | HostControlArea | 0..1 | HostControlArea | Hérité de: RegisteredResource |
| 0..* | Domain | 0.. * | Domain | Hérité de: RegisteredResource |
| 0..* | EnergyMarkets | 0..* | EnergyMarket | Hérité de: RegisteredResource |
| 0..1 | DotInstruction | 0..* | DotInstruction | Hérité de: RegisteredResource |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.60 RegulatingLimit

Cette classe représente les caractéristiques physiques d'un générateur concernant la limite de régulation.

Le Tableau 621 présente tous les attributs de RegulatingLimit.

**Tableau 621 – Attributs de ReferenceData::RegulatingLimit**

| nom | mult | type | description |
|---|---|---|---|
| highLimit | 0..1 | ActivePower | |
| lowLimit | 0..1 | ActivePower | |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 622 présente toutes les extrémités d'association de RegulatingLimit avec d'autres classes.

**Tableau 622 – Extrémités d’association de ReferenceData:: RegulatingLimit avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.61 Classe racine ResourceCertification

Spécifie la certification permettant à une ressource de participer à des marchés spécifiques.

Le Tableau 623 présente tous les attributs de ResourceCertification.

**Tableau 623 – Attributs de ReferenceData::ResourceCertification**

| nom | mult | type | description |
|---|---|---|---|
| market | 0..1 | MarketType | Type de marché |
| qualificationFlag | 0..1 | YesNo | Statut de la qualification ("Y" = actif, "N" = inactif) |
| type | 0..1 | ResourceCertificationKind | Type de service basé sur l'énumération ResourceAncillaryServiceType |

Le Tableau 624 présente toutes les extrémités d'association de ResourceCertification avec d'autres classes.

**Tableau 624 – Extrémités d'association de ReferenceData:: ResourceCertification avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredResource | Les RegisteredResources sont qualifiées pour les types de services auxiliaires de ressource (qui incluent les types de produits du marché ainsi que d'autres types, tels que BlackStart) en étant associées à la classe ResourceAncillaryServiceQualification. |

### 6.5.9.62 ResourceOperationMaintenanceCost

Modélise les coûts d'exploitation (O) et de maintenance (M) d'une ressource de production.

Le Tableau 625 présente tous les attributs de ResourceOperationMaintenanceCost.

**Tableau 625 – Attributs de ReferenceData::ResourceOperationMaintenanceCost**

| nom | mult | type | description |
|---|---|---|---|
| gasPercentAboveLowSustainedLimit | 0..1 | PerCent | Pourcentage de prix d'indice du combustible (gaz) pour une exploitation supérieure à la limite inférieure maintenue (LSL – Low Sustained Limit) |
| oilPercentAboveLowSustainedLimit | 0..1 | PerCent | Pourcentage de prix du combustible (FOP – Fuel Oil Price) pour une exploitation supérieure à la limite inférieure maintenue. |
| omCostColdStartup | 0..1 | Float | Coût d'exploitation et de maintenance vérifiable ($), démarrage à froid |
| omCostHotStartup | 0..1 | Float | Coût d'exploitation et de maintenance vérifiable ($), démarrage à chaud |
| omCostIntermediateStartup | 0..1 | Float | Coût d'exploitation et de maintenance vérifiable ($), démarrage à température intermédiaire |
| omCostLowSustainedLimit | 0..1 | Float | Coût d'exploitation et de maintenance vérifiable ($/MWh), limite inférieure maintenue |
| solidfuelPercentAboveLowSustainedLimit | 0..1 | PerCent | Pourcentage de combustible solide pour une exploitation supérieure à la limite inférieure maintenue. |
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 626 présente toutes les extrémités d'association de ResourceOperationMaintenanceCost avec d'autres classes.

**Tableau 626 – Extrémités d’association de ReferenceData::ResourceOperationMaintenanceCost avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | ResourceVerifiableCosts | 0..1 | ResourceVerifiableCosts | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

#### 6.5.9.63 Classe racine ResourceStartupCost

Modélise les coûts de démarrage d'une ressource de production.

Le Tableau 627 présente tous les attributs de ResourceStartupCost.

**Tableau 627 – Attributs de ReferenceData::ResourceStartupCost**

| nom | mult | type | description |
|---|---|---|---|
| fuelColdStartup | 0..1 | Float | Combustible vérifiable pour démarrage à froid (MMBtu par démarrage) |
| fuelHotStartup | 0..1 | Float | Combustible vérifiable pour démarrage à chaud (MMBtu par démarrage) |
| fuelIntermediateStartup | 0..1 | Float | Combustible vérifiable pour démarrage à température intermédiaire (MMBtu par démarrage) |
| fuelLowSustainedLimit | 0..1 | Float | Quantité minimale de combustible, MMBtu/MWh |
| gasPercentColdStartup | 0..1 | PerCent | Pourcentage du prix d'indice du combustible (gaz) pour démarrage à froid |
| gasPercentHotStartup | 0..1 | PerCent | Pourcentage du prix d'indice du combustible (gaz) pour démarrage à chaud |
| gasPercentIntermediateStartup | 0..1 | PerCent | Pourcentage du prix d'indice du combustible (gaz) pour démarrage à température intermédiaire |
| gasPercentLowSustainedLimit | 0..1 | PerCent | Pourcentage du prix d'indice du combustible (gaz) pour une exploitation à la limite inférieure maintenue. |
| oilPercentColdStartup | 0..1 | PerCent | Pourcentage du prix du combustible (FOP – Fuel Oil Price) pour démarrage à froid |
| oilPercentHotStartup | 0..1 | PerCent | Pourcentage du prix du combustible (FOP) pour démarrage à chaud |
| oilPercentIntermediateStartup | 0..1 | PerCent | Pourcentage du prix du combustible (FOP) pour démarrage à température intermédiaire |
| oilPercentLowSustainedLimit | 0..1 | PerCent | Pourcentage du prix du combustible pour une exploitation à la limite inférieure maintenue. |
| solidfuelPercentColdStartup | 0..1 | PerCent | Pourcentage de combustible solide pour démarrage à froid |
| solidfuelPercentHotStartup | 0..1 | PerCent | Pourcentage de combustible solide pour démarrage à chaud |

| nom | mult | type | description |
|---|---|---|---|
| solidfuelPercentIntermediateStartup | 0..1 | PerCent | Pourcentage de combustible solide pour démarrage à température intermédiaire |
| solidfuelPercentLowSustainedLimit | 0..1 | PerCent | Pourcentage de combustible solide pour une exploitation à la limite inférieure maintenue. |

Le Tableau 628 présente toutes les extrémités d'association de ResourceStartupCost avec d'autres classes.

**Tableau 628 – Extrémités d'association de ReferenceData:: ResourceStartupCost avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ResourceVerifiableCosts | 1..1 | ResourceVerifiableCosts | |

### 6.5.9.64 Classe racine ResourceVerifiableCosts

Cette classe est définie pour décrire les coûts vérifiables associés à une ressource de production.

Le Tableau 629 présente toutes les extrémités d'association de ResourceVerifiableCosts avec d'autres classes.

**Tableau 629 – Extrémités d'association de ReferenceData:: ResourceVerifiableCosts avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredResource | 1..1 | RegisteredResource | |
| 0..1 | MktHeatRateCurve | 1..1 | MktHeatRateCurve | |
| 0..1 | ResourceOperationMaintenanceCost | 1..1 | ResourceOperationMaintenanceCost | |
| 1..1 | ResourceStartupCost | 0..* | ResourceStartupCost | |

### 6.5.9.65 ResponseMethod

Spécifie une catégorie d'utilisation d'énergie à laquelle une réponse à une demande s'applique; par exemple, l'énergie provenant de l'éclairage, les systèmes de chauffage, de ventilation et de climatisation, entre autres.

Le Tableau 630 présente tous les attributs de ResponseMethod.

**Tableau 630 – Attributs de ReferenceData::ResponseMethod**

| nom | mult | type | description |
|---|---|---|---|
| activePower | 0..1 | Float | Valeur de la puissance active pour le type d'ajustement de la demande. Elle prend en charge les requêtes d'une certaine quantité de puissance active fournie par une méthode de réponse particulière à effectuer auprès d'une ressource, tel que spécifié par l'attribut de la méthode (par exemple, éclairage, systèmes de chauffage, de ventilation et de climatisation, conditionneurs d'air muraux, etc.). |
| activePowerUOM | 0..1 | String | Unité de mesure de la puissance active, par exemple, kilowatts (kW), mégawatts (mW), etc. |
| method | 0..1 | String | Méthode de réponse (par exemple, éclairage, systèmes de chauffage, de ventilation et de climatisation, conditionneurs d'air muraux, etc.). |
| siteMultiplier | 0..1 | Integer | Cette valeur prévoit la mise à l'échelle de la puissance active d'une méthode de réponse. Par exemple, une méthode de réponse de climatisation peut utiliser une petite quantité de puissance active provenant de chaque conditionneur d'air (par exemple, 0,1 kilowatt), mais le multiplicateur du site peut être utilisé pour produire un ajustement total de la puissance active en multipliant la puissance active de la méthode de réponse par cette valeur (par exemple, pour un bâtiment comportant 100 conditionneurs d'air type fenêtre: 100 * 0,1 kW = 10 kW). |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 631 présente toutes les extrémités d'association de ResponseMethod avec d'autres classes.

**Tableau 631 – Extrémités d'association de ReferenceData:: ResponseMethod avec d'autres classes**

| mult à partir | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 1..1 | RegisteredDistributedResource | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.66 SchedulingCoordinator

Les participants du marché peuvent être représentés par des coordinateurs de programmation (SC) enregistrés à l'aide du RTO/ISO. Un participant peut enregistrer plusieurs coordinateurs de programmation avec le RTO/ISO. Plusieurs participants du marché peuvent opérer avec le RTO/ISO à l'aide d'un seul coordinateur de programmation. Un coordinateur de programmation peut planifier plusieurs générateurs. Un point de programmation de la charge peut être utilisé par plusieurs coordinateurs de programmation. Chaque coordinateur de programmation peut planifier de la charge à plusieurs points de programmation. Un point de programmation d'interconnexion peut être utilisé par plusieurs coordinateurs de programmation. Chaque coordinateur de programmation peut planifier un échange à plusieurs points de programmation d'interconnexion.

Le Tableau 632 présente tous les attributs de SchedulingCoordinator.

**Tableau 632 – Attributs de ReferenceData::SchedulingCoordinator**

| nom | mult | type | description |
|---|---|---|---|
| creditFlag | 0..1 | YesNo | Indicateur de solvabilité (Y, N) |
| creditStartEffectiveDate | 0..1 | DateTime | Date à laquelle le coordinateur de programmation devient solvable. |
| lastModified | 0..1 | DateTime | Indication de l'heure de la dernière modification des informations sur ce coordinateur de programmation. |
| qualificationStatus | 0..1 | String | Statut de qualification du coordinateur de programmation: Qualifié, Non Qualifié ou Disqualifié. |
| scid | 0..1 | String | Il s'agit du champ Nom abrégé ou ID du coordinateur de programmation. |
| streetAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| postalAddress | 0..1 | StreetAddress | Hérité de: Organisation |
| phone1 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| phone2 | 0..1 | TelephoneNumber | Hérité de: Organisation |
| electronicAddress | 0..1 | ElectronicAddress | Hérité de: Organisation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 633 présente toutes les extrémités d'association de SchedulingCoordinator avec d'autres classes.

**Tableau 633 – Extrémités d’association de ReferenceData:: SchedulingCoordinator avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | MarketParticipant | 0..1 | MarketParticipant | |
| 0..1 | SubmitToSCTrade | 0..* | Trade | |
| 1..1 | ToSCTrade | 0..* | Trade | |
| 1..1 | FromSCTrade | 0..* | Trade | |
| 0..1 | SubmitFromSCTrade | 0..* | Trade | |
| 1..1 | TransmissionContract Right | 0..* | ContractRight | |
| 0..1 | LoadRatio | 1..1 | LoadRatio | |
| 0..* | MarketPerson | 0..* | MarketPerson | Hérité de: MarketParticipant |
| 0..* | MarketDocument | 0..* | MarketDocument | Hérité de: MarketParticipant |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | SchedulingCoordinator | 0..* | SchedulingCoordinator | Hérité de: MarketParticipant |
| 0..1 | RegisteredResource | 0..* | RegisteredResource | Hérité de: MarketParticipant |
| 0..1 | Bid | 0..* | Bid | Hérité de: MarketParticipant |
| 0..* | TimeSeries | 0..* | TimeSeries | Hérité de: MarketParticipant |
| 0..* | MarketRole | 0..* | MarketRole | Hérité de: MarketParticipant |
| 0..* | ParentOrganisation | 0..1 | ParentOrganization | Hérité de: Organisation |
| 0..1 | Roles | 0..* | OrganisationRole | Hérité de: Organisation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.67 Classe racine SchedulingCoordinatorUser

Décrit les utilisateurs d'un coordinateur de programmation.

Le Tableau 634 présente tous les attributs de SchedulingCoordinatorUser.

**Tableau 634 – Attributs de ReferenceData::SchedulingCoordinatorUser**

| nom | mult | type | description |
|---|---|---|---|
| loginID | 0..1 | String | ID de connexion |
| loginRole | 0..1 | String | Rôles affectés (il s'agit de rôles avec des privilèges de lecture ou de lecture/écriture sur différents systèmes de marché) |

### 6.5.9.68 SchedulingPoint

Connexion à d'autres organisations dans les limites de l'ISO/RTO.

Le Tableau 635 présente tous les attributs de SchedulingPoint.

**Tableau 635 – Attributs de ReferenceData::SchedulingPoint**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 636 présente toutes les extrémités d'association de SchedulingPoint avec d'autres classes.

**Tableau 636 – Extrémités d'association de ReferenceData:: SchedulingPoint avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | InterchangeSchedule | 0..* | InterchangeSchedule | |
| 0..* | Flowgate | 0..1 | Flowgate | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.69 StartUpEnergyCurve

Consommation d'énergie d'une ressource de production nécessaire pour effectuer un démarrage depuis StartUpEnergyCurve. La définition de StartUpEnergyCurve inclut: xvalue comme temps de refroidissement et y1value comme valeur MW.

Le Tableau 637 présente tous les attributs de StartUpEnergyCurve.

**Tableau 637 – Attributs de ReferenceData::StartUpEnergyCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 638 présente toutes les extrémités d'association de StartUpEnergyCurve avec d'autres classes.

**Tableau 638 – Extrémités d'association de ReferenceData:: StartUpEnergyCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.70 StartUpFuelCurve

Consommation de combustible d'une ressource de production nécessaire pour effectuer un démarrage (x = temps de refroidissement) depuis StartUpFuelCurve. xAxisData -> temps de refroidissement et y1AxisData -> MBtu.

Le Tableau 639 présente tous les attributs de StartUpFuelCurve.

**Tableau 639 – Attributs de ReferenceData::StartUpFuelCurve**

| nom | mult | type | description |
|---|---|---|---|
| curveStyle | 0..1 | CurveStyle | Hérité de: Curve |
| xMultiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| xUnit | 0..1 | UnitSymbol | Hérité de: Curve |
| y1Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y1Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y2Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y2Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| y3Multiplier | 0..1 | UnitMultiplier | Hérité de: Curve |
| y3Unit | 0..1 | UnitSymbol | Hérité de: Curve |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 640 présente toutes les extrémités d'association de StartUpFuelCurve avec d'autres classes.

**Tableau 640 – Extrémités d'association de ReferenceData:: StartUpFuelCurve avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | RegisteredGenerator | 0..1 | RegisteredGenerator | |
| 1..1 | CurveDatas | 0..* | CurveData | Hérité de: Curve |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.71 SubControlArea

Zone définie afin d'assurer le suivi de l'échange avec les zones voisines via des points d'interconnexion; peut ou non être utilisée comme Zone de réglage.

Le Tableau 641 présente tous les attributs de SubControlArea.

**Tableau 641 – Attributs de ReferenceData::SubControlArea**

| nom | mult | type | description |
|---|---|---|---|
| areaShortName | 0..1 | String | Nom abrégé de la zone de marché qui est la zone de régulation. Il fait référence au nom de la zone de régulation AGC. |
| constantCoefficient | 0..1 | Float | Coefficient constant d'estimation de la perte |
| embeddedControlArea | 0..1 | YesNo | Utilisé conjointement avec l'indicateur InternalCA. Si l'indicateur InternalCA est défini sur YES, cet indicateur ne s'applique pas. Si l'indicateur InternalCA est défini sur NO, cet indicateur spécifie AdjacentCA (NO) ou EmbeddedCA (YES). |
| internalCA | 0..1 | YesNo | Indique par Yes/No si cette zone de réglage est contenue à l'intérieur du système. |
| linearCoefficient | 0..1 | Float | Coefficient linéaire d'estimation de la perte |
| localCA | 0..1 | YesNo | Indique que cette zone de réglage correspond à la zone de réglage local. |
| maxSelfSchedMW | 0..1 | Float | Quantité maximale de MW de programmation automatique autorisée pour une zone de réglage embarquée. |
| minSelfSchedMW | 0..1 | Float | Quantité minimale de MW de programmation automatique autorisée pour une zone de réglage embarquée. |
| quadraticCoefficient | 0..1 | Float | Coefficient quadratique d'estimation de la perte |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 642 présente toutes les extrémités d'association de SubControlArea avec d'autres classes.

**Tableau 642 – Extrémités d'association de ReferenceData:: SubControlArea avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 1..1 | Export_EnergyTransactions | 0..* | EnergyTransaction | L'énergie est transférée entre les zones d'échange |
| 1..1 | Import_EnergyTransactions | 0..* | EnergyTransaction | L'énergie est transférée entre les zones d'échange |
| 0..1 | GeneralClearingResults | 0..* | GeneralClearingResults | |
| 0..1 | BidSelfSched | 0..* | BidSelfSched | |
| 0..* | RegisteredResource | 0..* | RegisteredResource | |
| 0..1 | ExPostLossResults | 0..* | ExPostLossResults | |
| 0..1 | LossClearingResults | 1..* | LossClearingResults | |
| 0..* | AdjacentCASet | 0..1 | AdjacentCASet | |
| 0..* | AggregateNode | 0..* | AggregateNode | |
| 0..1 | CnodeDistributionFactor | 0..* | CnodeDistributionFactor | |
| 0..* | ControlAreaDesignation | 0..* | ControlAreaDesignation | |
| 0..1 | To_Flowgate | 0..* | Flowgate | |
| 0..1 | From_Flowgate | 0..* | Flowgate | |
| 0..* | HostControlArea | 1..1 | HostControlArea | La zone d'échange peut fonctionner en tant que zone de réglage |
| 0..1 | Pnode | 0..* | Pnode | |
| 0..* | RTO | 1..1 | RTO | |
| 0..* | PSRType | 0..1 | PSRType | Hérité de: PowerSystemResource |
| 0..1 | Controls | 0..* | Control | Hérité de: PowerSystemResource |
| 0..1 | Measurements | 0..* | Measurement | Hérité de: PowerSystemResource |
| 1..1 | OperatingShare | 0..* | OperatingShare | Hérité de: PowerSystemResource |
| 0..* | ReportingGroup | 0..* | ReportingGroup | Hérité de: PowerSystemResource |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.72 Classe racine SubstitutionResourceList

Liste des ressources qui peuvent être substituées dans les limites d'une définition de contrat. Cette classe comporte une priorité et une ressource.

Le Tableau 643 présente tous les attributs de SubstitutionResourceList.

**Tableau 643 – Attributs de ReferenceData::SubstitutionResourceList**

| nom | mult | type | description |
|---|---|---|---|
| precedence | 0..1 | Integer | Indicateur de l'ordre dans lequel il convient de substituer une ressource. Plus le nombre est faible, plus la priorité est élevée. |

Le Tableau 644 présente toutes les extrémités d'association de SubstitutionResourceList avec d'autres classes.

**Tableau 644 – Extrémités d'association de ReferenceData:: SubstitutionResourceList avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredResource | 0..1 | RegisteredResource | |
| 0..* | TransmissionContractRight | 0..1 | ContractRight | |

### 6.5.9.73 TACArea

Zone de taxe d'accès pour transport. Taxe fixée, au nom du propriétaire de transport participant, aux parties qui exigent l'accès au boîtier de commande.

Le Tableau 645 présente tous les attributs de TACArea.

**Tableau 645 – Attributs de ReferenceData::TACArea**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 646 présente toutes les extrémités d'association de TACArea avec d'autres classes.

**Tableau 646 – Extrémités d'association de ReferenceData:: TACArea avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | AreaLoadCurve | 0..* | AreaLoadCurve | |
| 0..* | AggregatedPnode | 0..* | AggregatedPnode | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.74 TransmissionRightChain

Permet la mise en chaîne de TransmissionContractRights. Plusieurs droits de contrats individuels peuvent être inclus dans la définition d'une chaîne TransmissionRightChain. Une chaîne TransmissionRightChain est également définie comme TransmissionContractRight en tant que tel.

Le Tableau 647 présente tous les attributs de TransmissionRightChain.

**Tableau 647 – Attributs de ReferenceData::TransmissionRightChain**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 648 présente toutes les extrémités d'association de TransmissionRightChain avec d'autres classes.

**Tableau 648 – Extrémités d’association de ReferenceData:: TransmissionRightChain avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | Ind_ContractRight | 1..* | ContractRight | |
| 0..1 | Chain_ContractRight | 1..1 | ContractRight | |
| 0..* | RTO | 1..1 | RTO | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.5.9.75 WheelingCounterParty

Entité contractante d'une transaction de transit.

Le Tableau 649 présente tous les attributs de WheelingCounterParty.

**Tableau 649 – Attributs de ReferenceData::WheelingCounterParty**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 650 présente toutes les extrémités d'association de WheelingCounterParty avec d'autres classes.

**Tableau 650 – Extrémités d'association de ReferenceData:: WheelingCounterParty avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | RegisteredInterTie | 0..* | RegisteredInterTie | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

## 6.6 Paquetage Environmental

### 6.6.1 Généralités

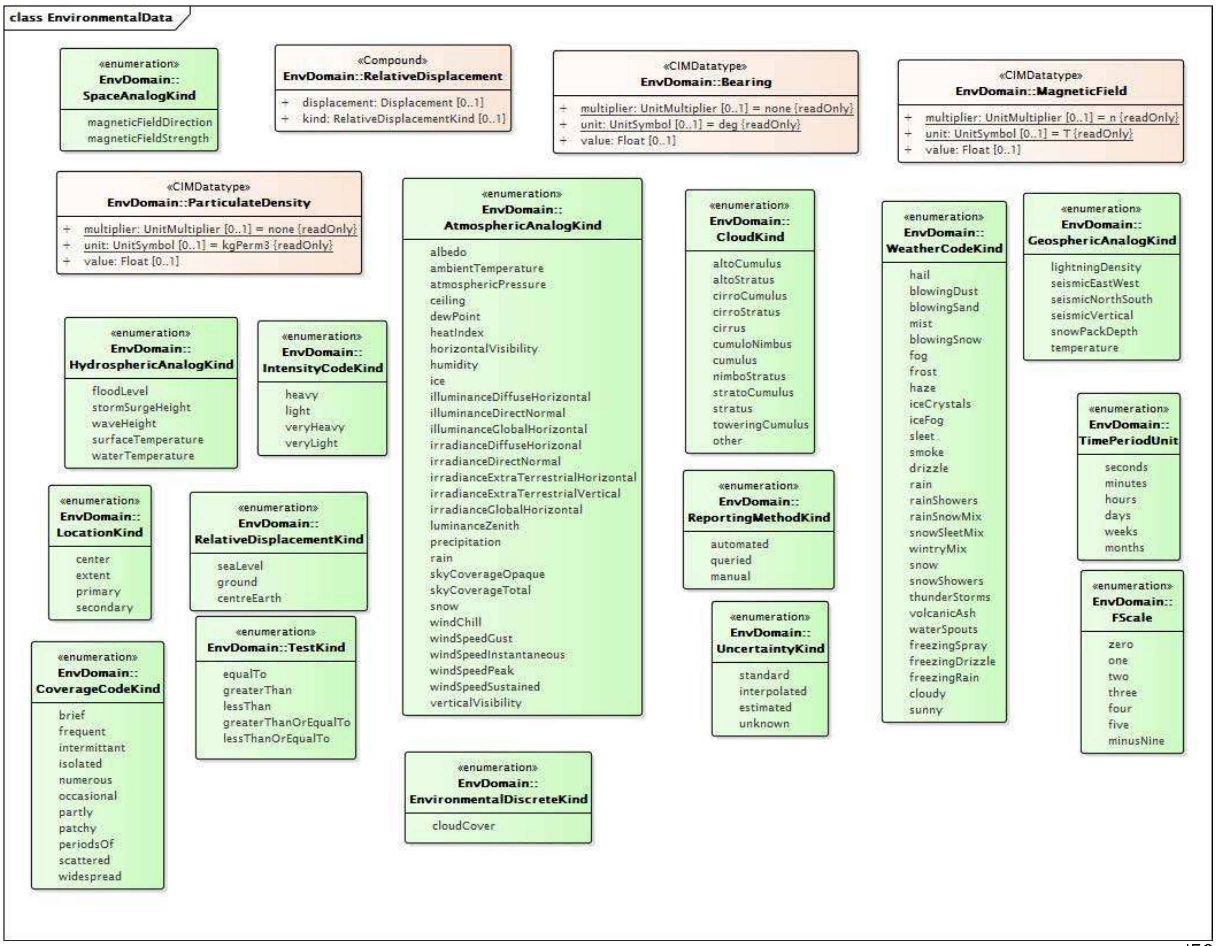



**Figure 70 – Diagramme de classe Environmental::EnvironmentalData**

Figure 70: représente les types de données et les énumérations qui prennent en charge les classes du paquetage Environmental, en plus des types de données et des énumérations contenus dans les paquetages IEC 61970, IEC 61968 et IEC 62325.

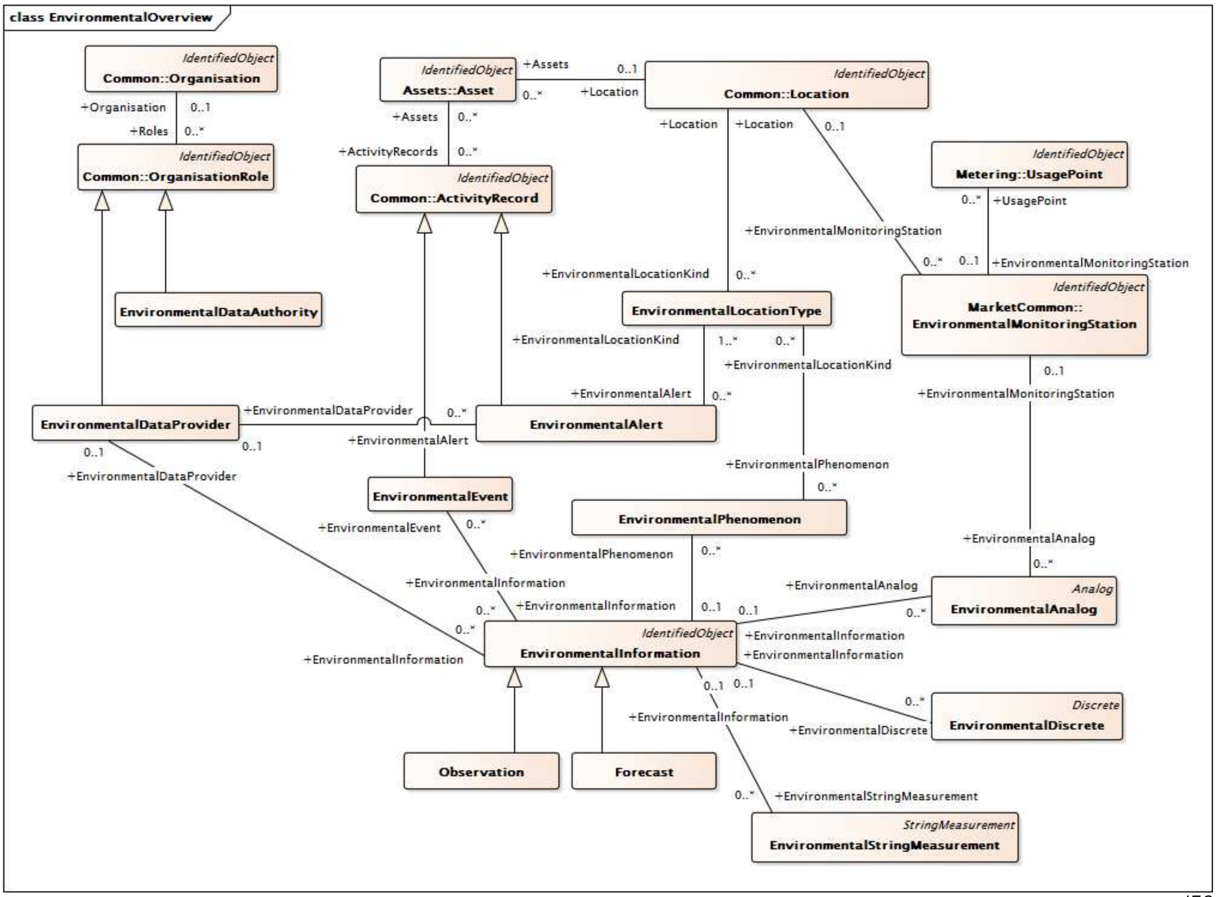
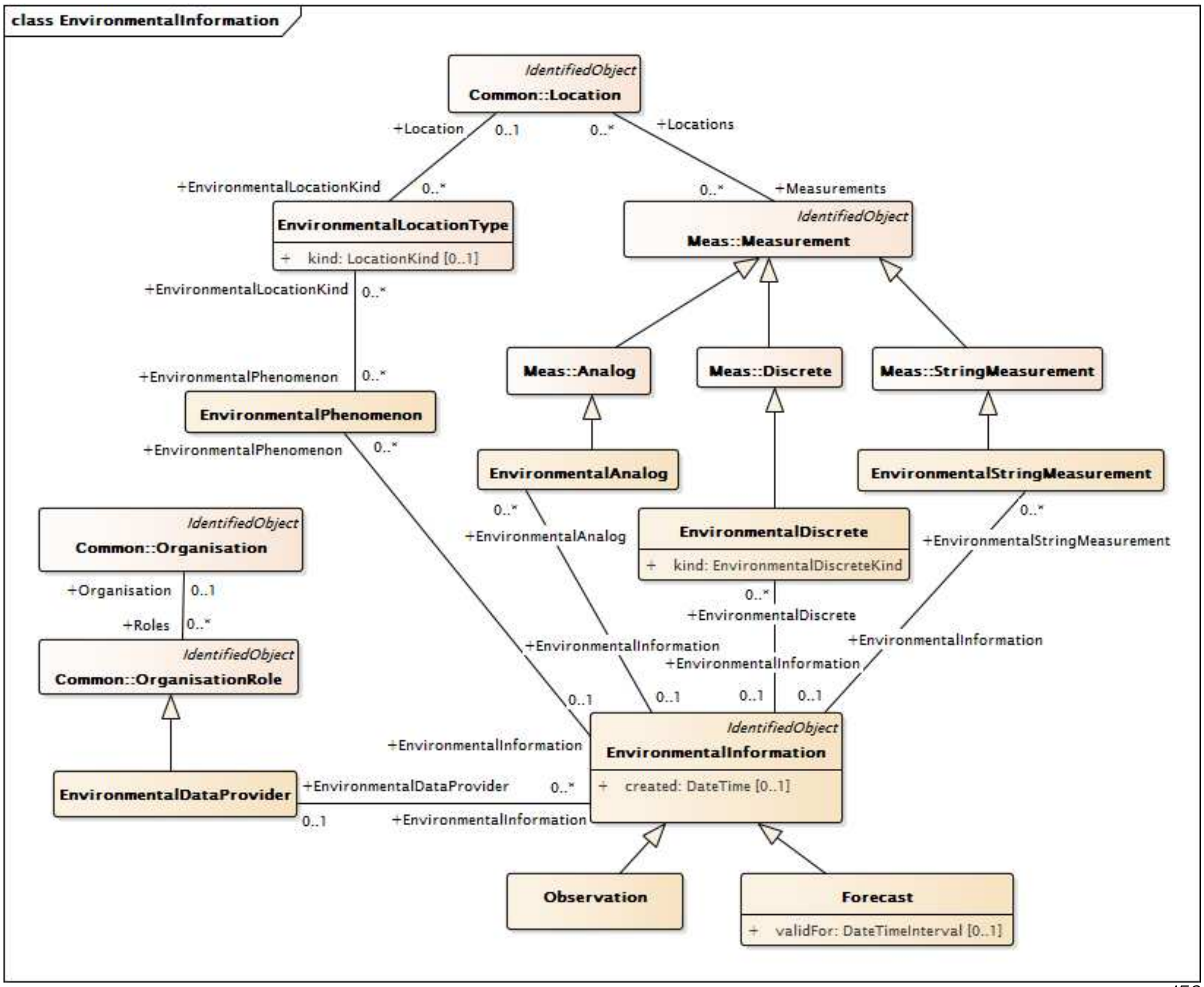



**Figure 71 – Diagramme de classe Environmental::EnvironmentalOverview**

Figure 71: représente le paquetage Environmental qui contient un modèle d'éléments généralement utilisés pour décrire les conditions environnementales (météorologiques), y compris les prévisions, les observations, les mesures, les phénomènes et les alertes.

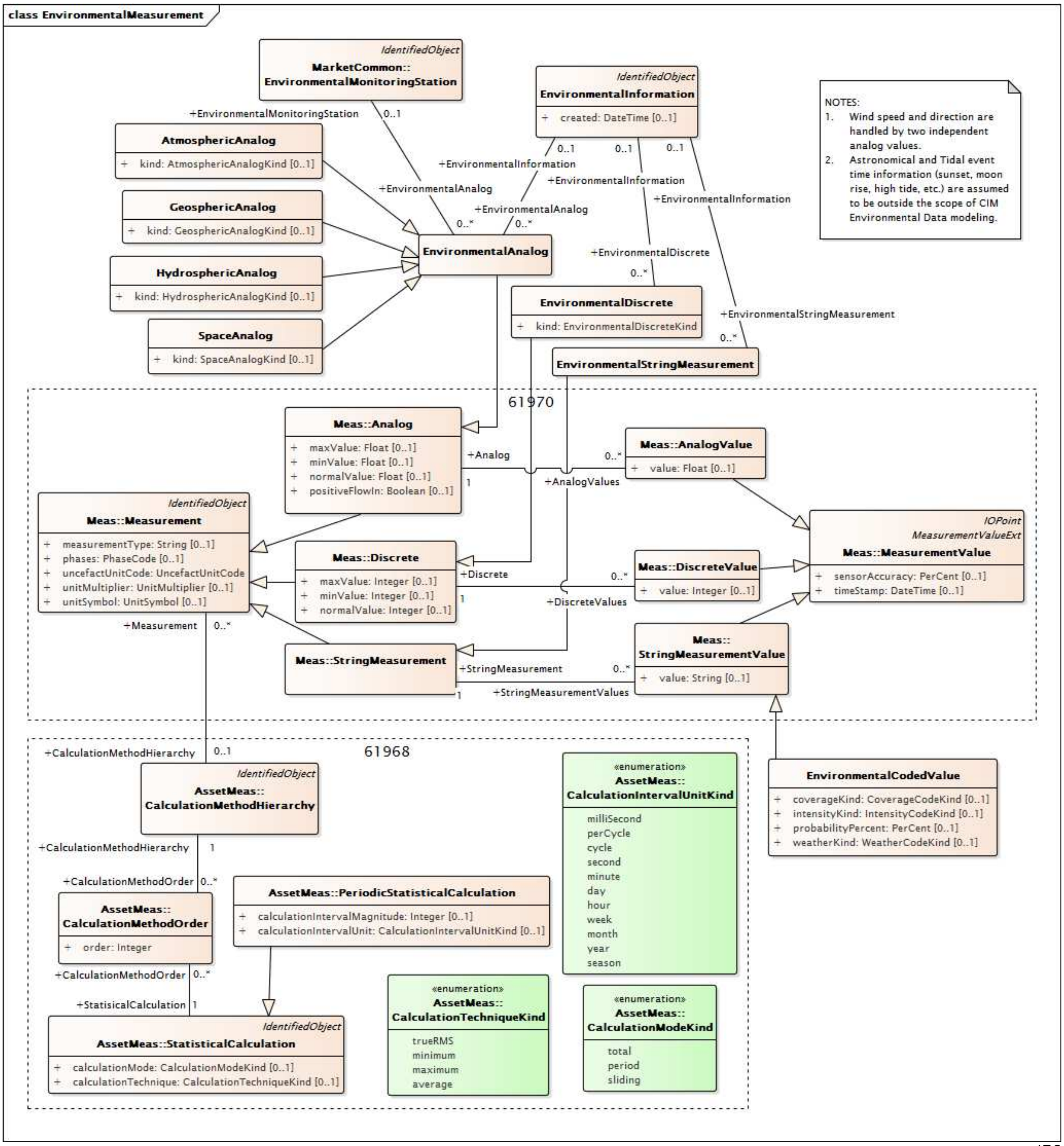
Les classes fondamentales sont Observation, Forecast, EnvironmentalAlert et EnvironmentalEvent. Les autres classes décrivent le contenu des informations relatives aux classes fondamentales.




IEC

**Figure 72 – Diagramme de classe Environmental::EnvironmentalInformation**

Figure 72: représente Observation et Forecast qui sont les deux classes enfants de EnvironmentalInformation. Elles ont chacune des données environnementales qui sont associées à EnvironmentalInformation et qui sont décrites en termes de mesures et de phénomènes. Les informations de localisation relatives aux observations et aux prévisions sont fournies au niveau de la mesure ou du phénomène, pas au niveau de l’observation ou de la prévision. Les observations et les prévisions peuvent avoir un fournisseur de données défini qui leur est propre.




IEC

| Anglais | Français |
|---|---|
| NOTES:<br><br>1. Wind speed and direction are handled by two independent analog values. | NOTES:<br><br>1. La vitesse et le sens du vent sont gérés par deux valeurs analogiques indépendantes. |
| 2. Astronomical and Tidal event time information (sunset, moon rise, high tide, etc.) are assumed to be outside the scope of CIM Environmental Data modelling. | 2. Les informations temporelles relatives aux événements Astronomical et Tidal (coucher de soleil, lever de lune, marée haute, etc.) sont considérées comme ne relevant pas du domaine d'application de la modélisation Environmental Data du CIM. |

**Figure 73 – Diagramme de classe Environmental::EnvironmentalMeasurement**

Figure 73: représente les observations et les prévisions qui peuvent comprendre un nombre quelconque de valeurs analogiques (float – à virgule flottante), discrètes (integer – entières) ou de chaîne (string) valides pendant différents instants ou différentes périodes de temps.

La nature des valeurs est décrite par les classes enfants de Measurement. Il est possible de décrire les calculs statistiques périodiques grâce aux classes de méthodes de calculs. Les valeurs réelles faisant partie des observations ou des prévisions sont représentées par les classes enfants de MeasurementValue.

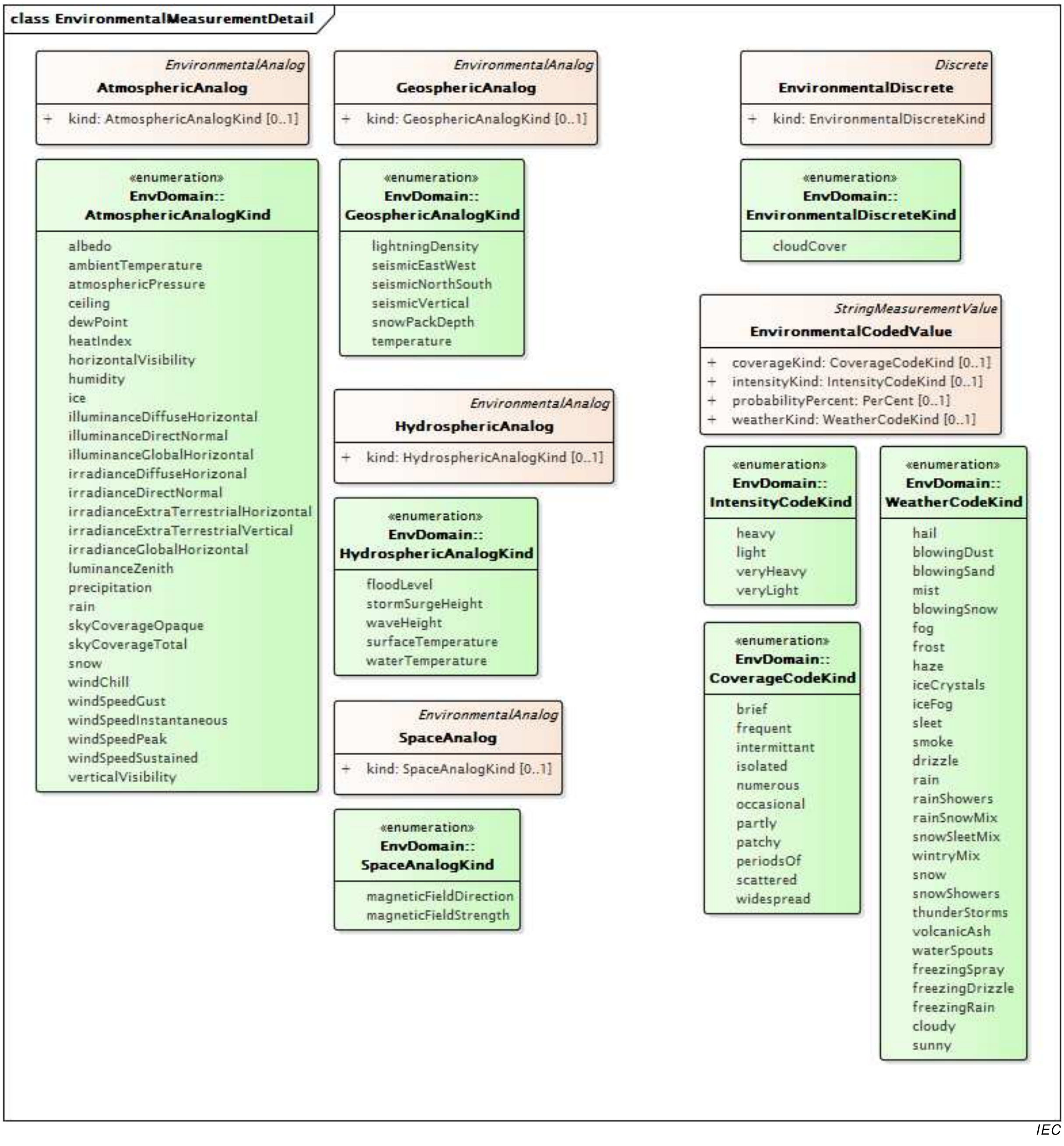



**Figure 74 – Diagramme de classe Environmental::EnvironmentalMeasurementDetail**

Figure 74: représente toutes les classes enfants de EnvironmentalAnalog dont chacune a un attribut .kind énuméré dont les valeurs reflètent ce que la valeur à virgule flottante mesure dans ce domaine (atmosphère, géosphère, hydrosphère ou espace).

La classe EnvironmentalDiscrete a un attribut .kind énuméré dont la valeur reflète ce que la valeur entière mesure.

La classe EnvironmentalCodedValue (classe enfant de StringMeasurementValue) a des valeurs de chaîne supplémentaires permettant une description claire des conditions météorologiques, telles que 'heavy isolated thunderstorms' (orages violents isolés).

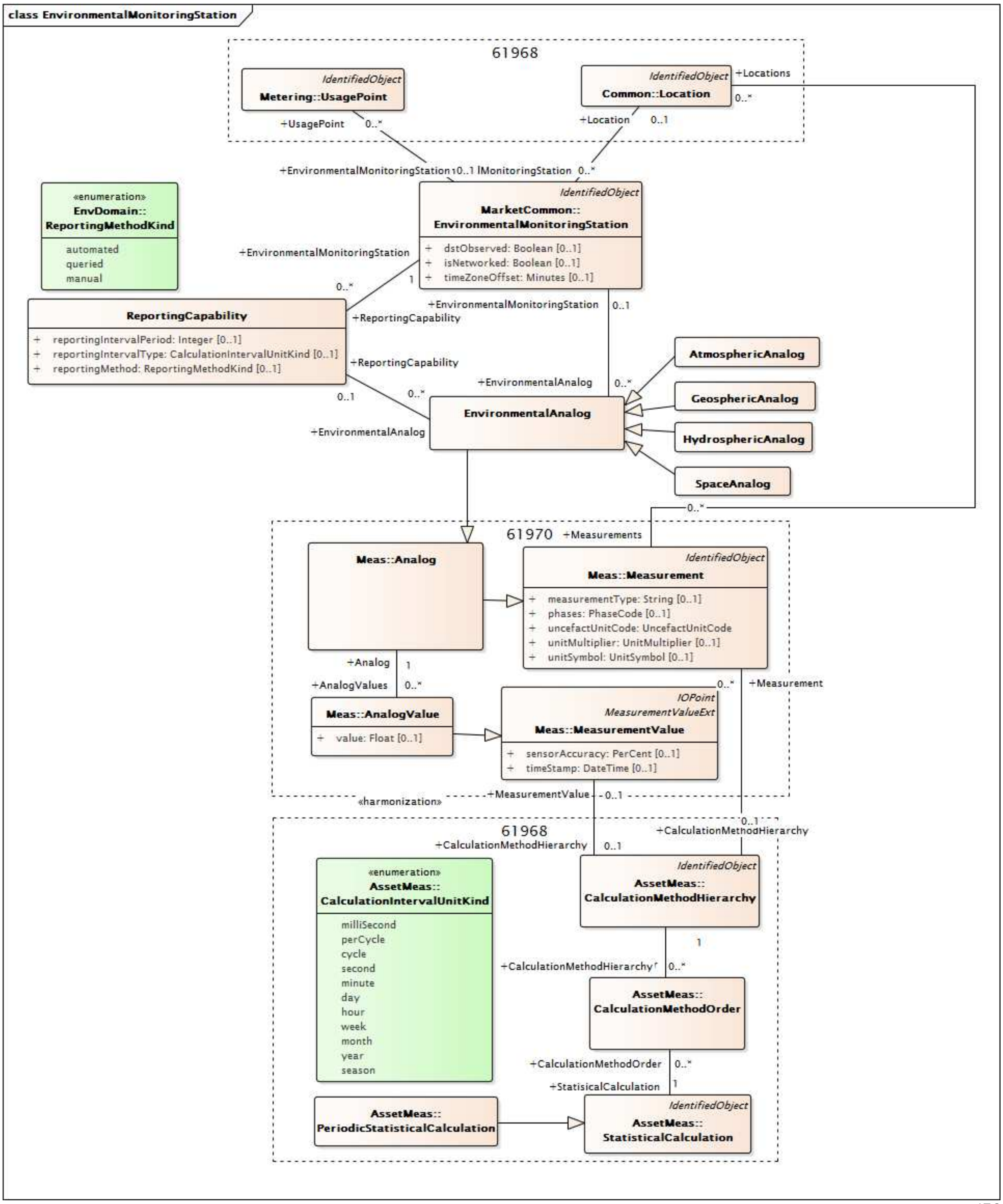



**Figure 75 – Diagramme de classe Environmental::EnvironmentalMonitoringStation**

Figure 75: représente les stations de surveillance environnementale, qui fournissent des valeurs analogiques pour les observations, qui peuvent appartenir aux services publics ou à une entité externe. Les capacités de ces stations peuvent être définies par la classe ReportingCapability. La classe ReportingCapability utilise les classes enfants EnvironmentalAnalog afin de décrire de manière spécifique les valeurs analogiques que la station de surveillance peut renvoyer. Une association entre les classes EnvironmentalMonitoringStation et UsagePoint empêche l'éventualité qu'une station de surveillance puisse être associée à un compteur ou à un client.

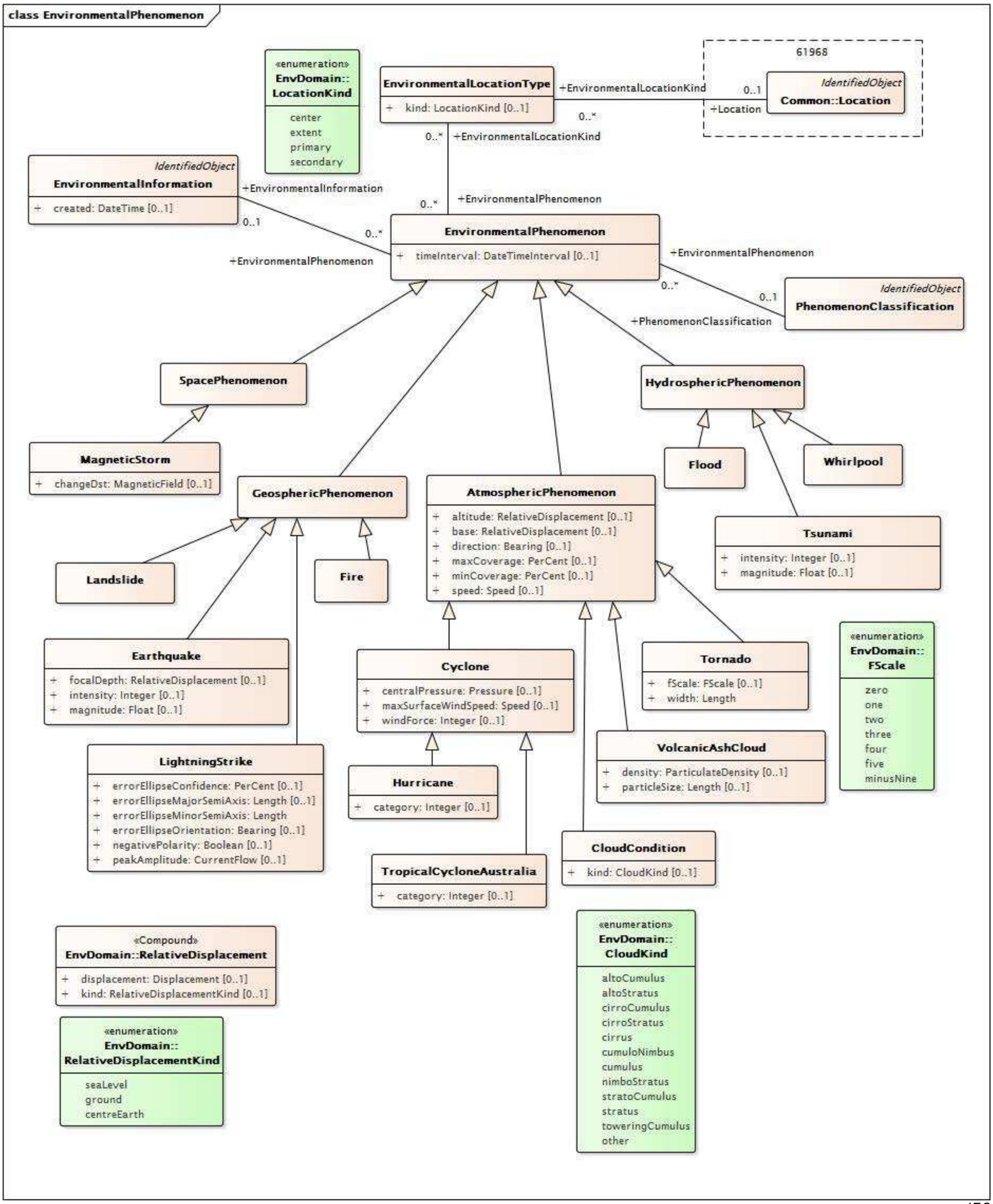



**Figure 76 – Diagramme de classe Environmental::EnvironmentalPhenomenon**

Figure 76: représente la classe EnvironmentalPhenomenon et les types spécialisés des classes Phenomenon. Les phénomènes environnementaux, comme les ouragans, les tornades, les feux de prairie, les inondations ou les tempêtes dans le champ géomagnétique, sont décrits par la classe Phenomenon et ses enfants, qui sont sous-classées à l'aide des groupements régionaux utilisés pour les valeurs analogiques (atmosphère, géosphère, hydrosphère et espace). Chaque classe enfant EnvironmentalPhenomenon a des attributs qui reflètent les informations communes utilisées pour décrire la nature ou la sévérité de ce type particulier de phénomène.

Les emplacements des phénomènes sont souvent décrits par un centre et un encombrement. Ils sont pris en charge par la classe EnvironmentalLocationKind.

Les principaux événements, comme l'ouragan Katrina ou le séisme de 2008 au Sichuan, peuvent être décrits dans le temps au moyen de diverses instances d'une classe enfant Phenomenon associées à une observation relative à un événement 'named' (nommé).

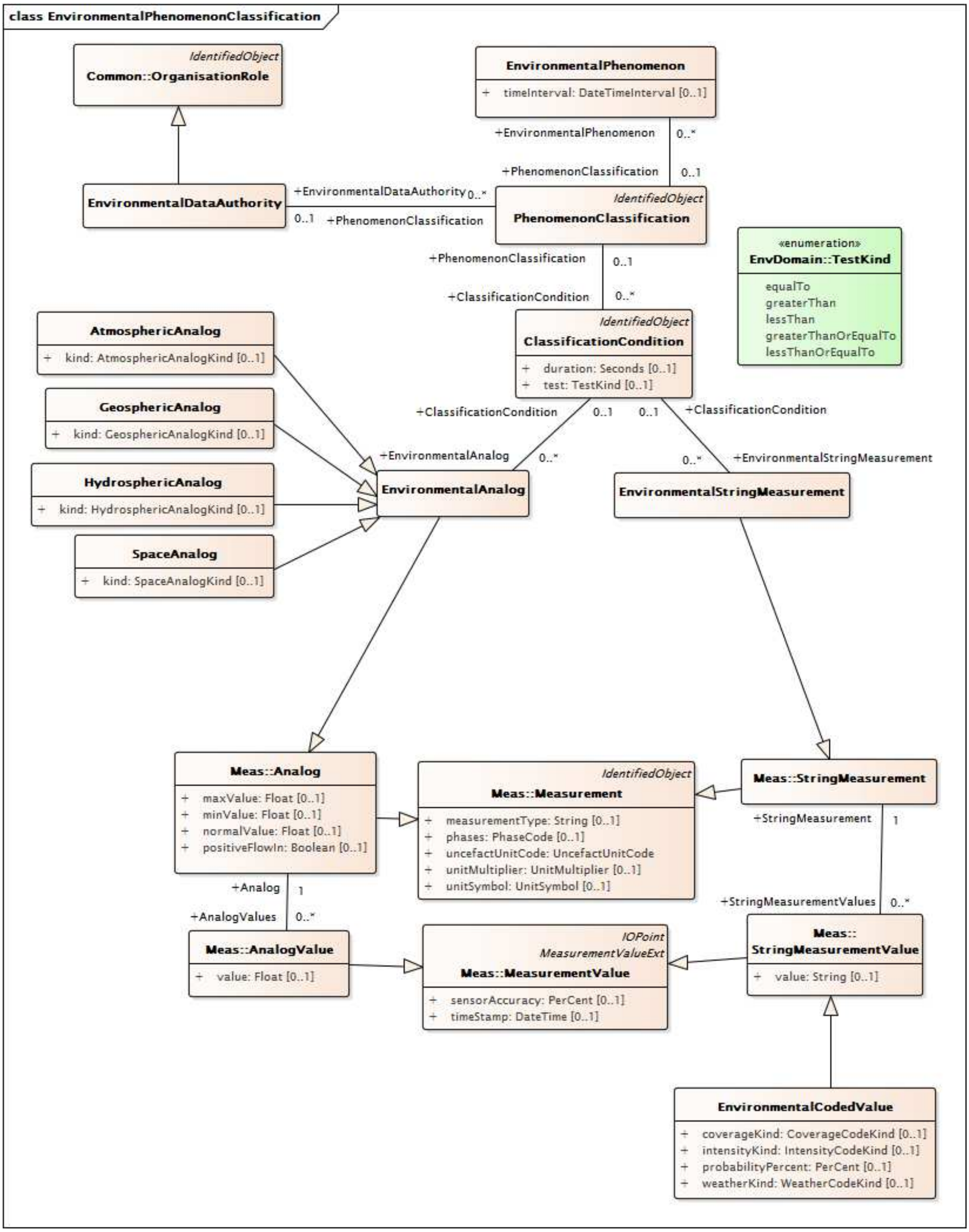



**Figure 77 – Diagramme de classe Environmental::EnvironmentalPhenomenonClassification**

Figure 77: représente la classe PhenomenonClassification et les classes associées qui prennent en charge la classification de la sévérité des phénomènes. Ceci permet à une autorité responsable des données environnementales (par exemple, un service météorologique ou une entreprise d'électricité) de définir les niveaux seuils des conditions de classification du phénomène. Ces niveaux seuils sont décrits au moyen de Measurement et de classes enfants de MeasurementValue (qui fournissent une valeur) en association avec un essai (égal à, supérieur, inférieur, etc.) et une durée.

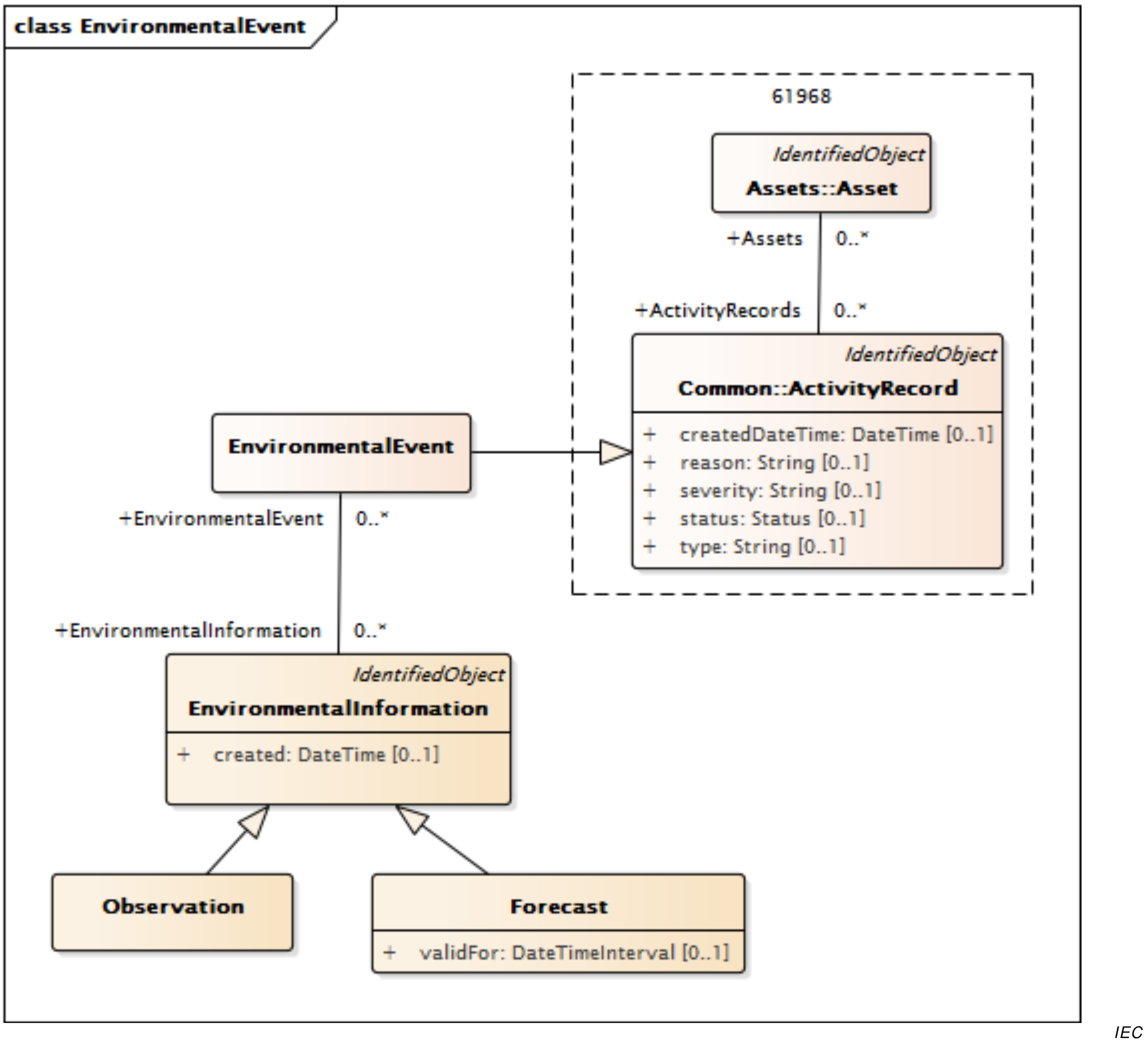



**Figure 78 – Diagramme de classe Environmental::EnvironmentalEvent**

Figure 78: représente la classe EnvironmentalEvent en tant que classes associées. Les événements environnementaux sont des occurrences auxquelles une ou plusieurs prévisions ou observations peuvent être liées et qui peuvent être associées à ou affecter un ou plusieurs actifs. Un événement peut être une occurrence 'named', comme le séisme de Tohoku de 2011 qui aurait pu avoir divers ensembles d'observations ou de prévisions associées. Il peut également s'agir d'un ensemble d'observations ou de prévisions s'appliquant simplement à un actif ou à un patrimoine.

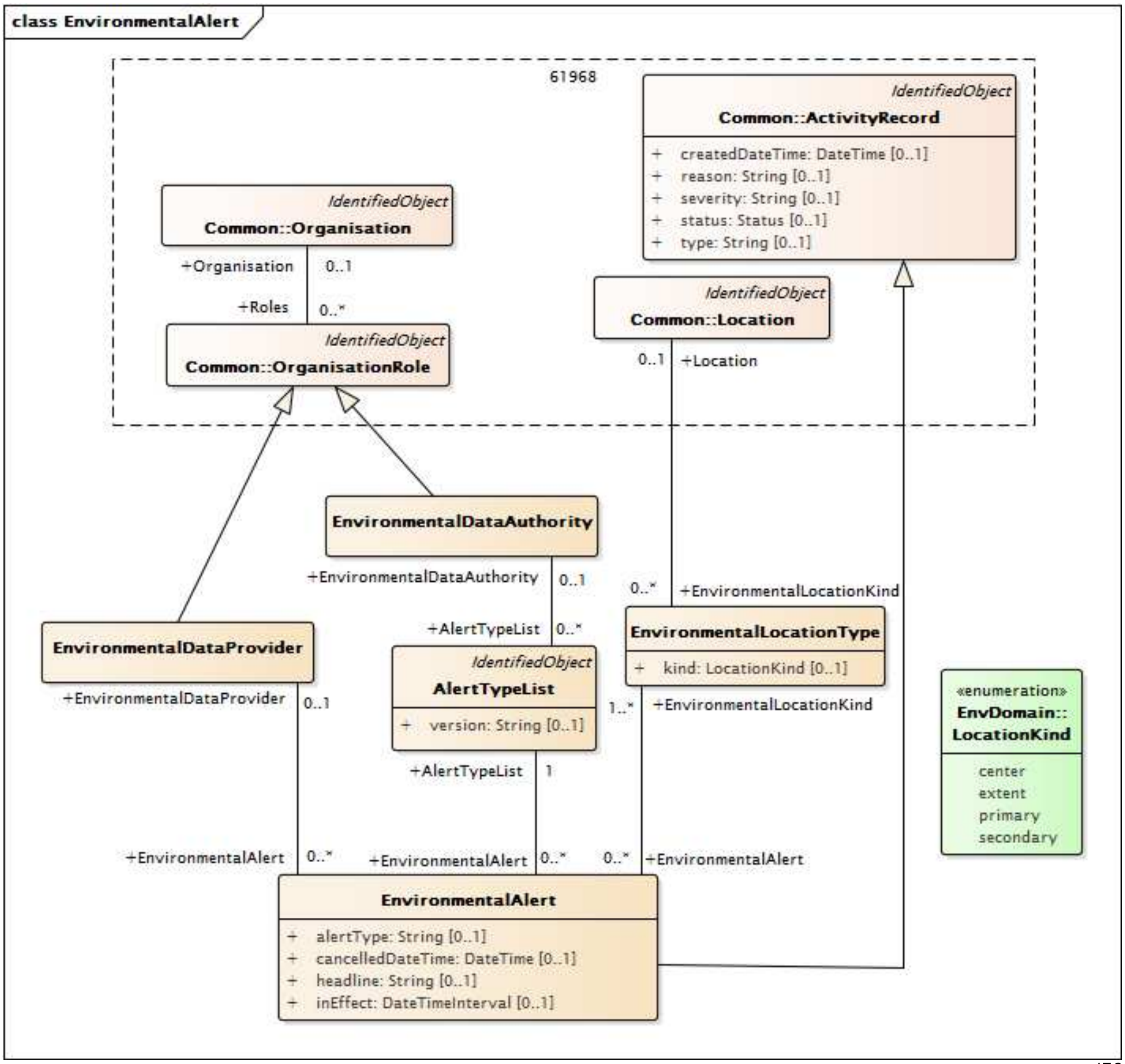



**Figure 79 – Diagramme de classe Environmental::EnvironmentalAlert**

Figure 79: représente EnvironmentalAlerts et les classes associées. Les alertes environnementales, telles que les veilles et les avertissements de tornade, sont décrites par la classe EnvironmentalAlert, qui prend en charge la définition du fournisseur d'alerte, de l'autorité compétente en matière d'alerte, et de l'emplacement de l'alerte. Les autorités compétentes en matière d'alerte (comme le service météorologique national des États-Unis – National Weather Service) mettent régulièrement à jour les types et les définitions des alertes qu'elles peuvent émettre. La classe AlertTypeList permet de spécifier la version utilisée des définitions d'alerte. Certaines alertes s'appliquent à des emplacements primaires et secondaires. La classe EnvironmentalLocationKind permet d'effectuer la distinction correspondante.

### 6.6.2 AlertTypeList

Liste nommée de types d'alertes.

NOTE Le nom de la liste est reflété dans l'attribut .name (hérité de IdentifiedObject).

Le Tableau 651 présente tous les attributs de AlertTypeList.

**Tableau 651 – Attributs de Environmental::AlertTypeList**

| nom | mult | type | description |
|---|---|---|---|
| version | 0..1 | String | Version de la liste nommée de types d’alertes. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 652 présente toutes les extrémités d’association de AlertTypeList avec d’autres classes.

**Tableau 652 – Extrémités d’association de Environmental:: AlertTypeList avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataAuthority | 0..1 | EnvironmentalDataAuthority | Autorité responsable des données environnementales chargée de publier cette liste de types d’alertes. |
| 1..1 | EnvironmentalAlert | 0..* | EnvironmentalAlert | Alerte dont le type est tiré de cette liste de types d’alertes |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.3 AtmosphericAnalog

Grandeur analogique (à virgule flottante) qui mesure une condition atmosphérique.

Le Tableau 653 présente tous les attributs de AtmosphericAnalog.

**Tableau 653 – Attributs de Environmental::AtmosphericAnalog**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | AtmosphericAnalogKind | Type de grandeur analogique atmosphérique. |
| maxValue | 0..1 | Float | Hérité de: Analog |
| minValue | 0..1 | Float | Hérité de: Analog |
| normalValue | 0..1 | Float | Hérité de: Analog |
| positiveFlowIn | 0..1 | Boolean | Hérité de: Analog |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 654 présente toutes les extrémités d'association de AtmosphericAnalog avec d'autres classes.

**Tableau 654 – Extrémités d'association de Environmental:: AtmosphericAnalog avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoringStation | Hérité de: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Hérité de: EnvironmentalAnalog |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | Hérité de: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | Hérité de: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | Hérité de: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.4 AtmosphericPhenomenon

Phénomène atmosphérique comprenant une base et une altitude lui fournissant une zone de couverture verticale (associées à Location (l'emplacement) pour couvrir un espace tridimensionnel).

Le Tableau 655 présente tous les attributs de AtmosphericPhenomenon.

**Tableau 655 – Attributs de Environmental::AtmosphericPhenomenon**

| nom | mult | type | description |
|---|---|---|---|
| altitude | 0..1 | RelativeDisplacement | Altitude maximale du phénomène. |
| base | 0..1 | RelativeDisplacement | Altitude de la base du phénomène. |
| direction | 0..1 | Bearing | Direction du movement du phénomène. |
| maxCoverage | 0..1 | PerCent | Pourcentage maximal de couverture. |
| minCoverage | 0..1 | PerCent | Pourcentage minimal de couverture. |
| speed | 0..1 | Speed | Vitesse du phénomène. |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 656 présente toutes les extrémités d'association de AtmosphericPhenomenon avec d'autres classes.

**Tableau 656 – Extrémités d'association de Environmental:: AtmosphericPhenomenon avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.5 ClassificationCondition

Condition de classification utilisée pour définir les préconditions qui doivent être satisfaites par une classification de phénomènes.

Le Tableau 657 présente tous les attributs de ClassificationCondition.

**Tableau 657 – Attributs de Environmental::ClassificationCondition**

| nom | mult | type | description |
|---|---|---|---|
| duration | 0..1 | Seconds | Durée de la condition en seconds. |
| test | 0..1 | TestKind | Essai appliqué à la valeur. |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 658 présente toutes les extrémités d'association de ClassificationCondition avec d'autres classes.

**Tableau 658 – Extrémités d'association de Environmental:: ClassificationCondition avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Classification de phénomènes à laquelle cette condition est liée. |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | StringMeasurement qui contribue à la définition de cette condition de classification. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Analog qui contribue à la définition de cette condition de classification. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.6 CloudCondition

Phénomène nuageux classé avec un type.

Le Tableau 659 présente tous les attributs de CloudCondition.

**Tableau 659 – Attributs de Environmental::CloudCondition**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | CloudKind | Type de nuage tel que défini par l'énumération CloudKind. |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 660 présente toutes les extrémités d'association de CloudCondition avec d'autres classes.

**Tableau 660 – Extrémités d'association de Environmental:: CloudCondition avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.7 Cyclone

Cyclone (ou cyclone tropical), système de tempête à rotation rapide caractérisé par un centre à basse pression, des vents violents, et une disposition d'orages en spirale qui produisent une pluie intense.

Le Tableau 661 présente tous les attributs de Cyclone.

**Tableau 661 – Attributs de Environmental::Cyclone**

| nom | mult | type | description |
|---|---|---|---|
| centralPressure | 0..1 | Pressure | Pression centrale du cyclone pendant l'intervalle de temps. |
| maxSurfaceWindSpeed | 0..1 | Speed | Vitesse du vent superficielle maximale du cyclone pendant l'intervalle de temps. |
| windForce | 0..1 | Integer | Force du vent telle que classée sur l'échelle de Beaufort (0-12) pendant l'intervalle de temps. |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 662 présente toutes les extrémités d'association de Cyclone avec d'autres classes.

**Tableau 662 – Extrémités d'association de Environmental:: Cyclone avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.8 Earthquake

Séisme.

Le Tableau 663 présente tous les attributs de Earthquake.

**Tableau 663 – Attributs de Environmental::Earthquake**

| nom | mult | type | description |
|---|---|---|---|
| focalDepth | 0..1 | RelativeDisplacement | Profondeur, en dessous de la surface de la terre, du point focal du séisme. |
| intensity | 0..1 | Integer | Intensité du séisme telle que définie par l'échelle de Mercalli. Les valeurs possibles sont 1-12, correspondant à I-XII. |
| magnitude | 0..1 | Float | Magnitude du séisme telle que définie sur l'échelle de magnitude de moment (Mw), qui mesure les dimensions des séismes en termes d'énergie libérée. Doit être supérieure à zéro. |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 664 présente toutes les extrémités d'association de Earthquake avec d'autres classes.

**Tableau 664 – Extrémités d'association de Environmental:: Earthquake avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.9 EnvironmentalAnalog

Mesure analogique (valeur à virgule flottante) pertinente dans le domaine environnemental.

Le Tableau 665 présente tous les attributs de EnvironmentalAnalog.

**Tableau 665 – Attributs de Environmental::EnvironmentalAnalog**

| nom | mult | type | description |
|---|---|---|---|
| maxValue | 0..1 | Float | Hérité de: Analog |
| minValue | 0..1 | Float | Hérité de: Analog |
| normalValue | 0..1 | Float | Hérité de: Analog |
| positiveFlowIn | 0..1 | Boolean | Hérité de: Analog |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 666 présente toutes les extrémités d'association de EnvironmentalAnalog avec d'autres classes.

**Tableau 666 – Extrémités d'association de Environmental:: EnvironmentalAnalog avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoringStation | Station de surveillance qui effectue cette mesure environnementale analogique. |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Condition de classification que cette grandeur analogique aide à définir. |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Observation ou prévision à laquelle cette mesure environnementale analogique est associée. |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | Capacité de production de rapports que cette valeur environnementale aide à définir. |
| 1..1 | AnalogValues | 0..* | AnalogValue | Hérité de: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | Hérité de: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.10 EnvironmentalAlert

Alerte environnementale émise par un fournisseur ou un système.

Le Tableau 667 présente tous les attributs de EnvironmentalAlert.

**Tableau 667 – Attributs de Environmental::EnvironmentalAlert**

| nom | mult | type | description |
|---|---|---|---|
| alertType | 0..1 | String | Type d'alerte émise tiré de la liste spécifiée de types d'alertes. |
| cancelledDateTime | 0..1 | DateTime | Heure et date auxquelles l'alerte a été annulée. Utilisé uniquement si l'alerte est annulée avant son expiration. |
| headline | 0..1 | String | Description textuelle abrégée de l'alerte émise.<br>Note: le texte complet de l'alerte apparaît dans l'attribut .description (hérité de IdentifiedObject). |
| inEffect | 0..1 | DateTimeInterval | Intervalle pendant lequel cette alerte météorologique est effective. |
| createdDateTime | 0..1 | DateTime | Hérité de: ActivityRecord |
| type | 0..1 | String | Hérité de: ActivityRecord |
| severity | 0..1 | String | Hérité de: ActivityRecord |
| reason | 0..1 | String | Hérité de: ActivityRecord |
| status | 0..1 | Status | Hérité de: ActivityRecord |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 668 présente toutes les extrémités d'association de EnvironmentalAlert avec d'autres classes.

**Tableau 668 – Extrémités d'association de Environmental:: EnvironmentalAlert avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | AlertTypeList | 1..1 | AlertTypeList | Liste de types d'alertes à partir de laquelle le type de cette alerte a été tiré. |
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Fournisseur de données environnementales pour cette alerte. |
| 0..* | EnvironmentalLocationKind | 1..* | EnvironmentalLocationType | Type d'emplacement auquel cette alerte environnementale s'applique. |
| 0..* | Assets | 0..* | Asset | Hérité de: ActivityRecord |
| 0..* | Author | 0..1 | Author | Hérité de: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.11 EnvironmentalCodedValue

Valeur environnementale décrite à l'aide d'une valeur codée. Un triplicat de valeurs énumérées représentant l'intensité, la zone de couverture et les conditions météorologiques est utilisé. Ces valeurs peuvent être concaténées dans une valeur de chaîne.

Le Tableau 669 présente tous les attributs de EnvironmentalCodedValue.

**Tableau 669 – Attributs de Environmental::EnvironmentalCodedValue**

| nom | mult | type | description |
|---|---|---|---|
| coverageKind | 0..1 | CoverageCodeKind | Code représentant la zone de couverture de la condition météorologique. |
| intensityKind | 0..1 | IntensityCodeKind | Code représentant l'intensité de la condition météorologique. |
| probabilityPercent | 0..1 | PerCent | Probabilité qu'une condition météorologique survienne pendant l'intervalle de temps, exprimée en pourcentage. Applicable uniquement lorsque la condition météorologique est liée à une prévision (pas à une observation). |
| weatherKind | 0..1 | WeatherCodeKind | Code représentant le type de condition météorologique. |
| value | 0..1 | String | Hérité de: StringMeasurementValue |
| timeStamp | 0..1 | DateTime | Hérité de: MeasurementValue |
| sensorAccuracy | 0..1 | PerCent | Hérité de: MeasurementValue |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 670 présente toutes les extrémités d'association de EnvironmentalCodedValue avec d'autres classes.

**Tableau 670 – Extrémités d'association de Environmental:: EnvironmentalCodedValue avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | StringMeasurement | 1..1 | StringMeasurement | Hérité de: StringMeasurementValue |
| 1..1 | RemoteSource | 0..1 | RemoteSource | Hérité de: MeasurementValue |
| 1..1 | MeasurementValueQuality | 0..1 | MeasurementValueQuality | Hérité de: MeasurementValue |
| 0..* | MeasurementValueSource | 1..1 | MeasurementValueSource | Hérité de: MeasurementValue |
| 0..1 | BilateralToIOPoint | 0..* | ProvidedBilateralPoint | Hérité de: IOPoint |
| 0..* | IOPointSource | 0..1 | IOPointSource | Hérité de: IOPoint |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.12 EnvironmentalDataAuthority

Entité définissant des classifications ou des catégories d'informations environnementales, comme des phénomènes ou des alertes.

Le Tableau 671 présente tous les attributs de EnvironmentalDataAuthority.

**Tableau 671 – Attributs de Environmental::EnvironmentalDataAuthority**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 672 présente toutes les extrémités d'association de EnvironmentalDataAuthority avec d'autres classes.

**Tableau 672 – Extrémités d'association de Environmental:: EnvironmentalDataAuthority avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | AlertTypeList | 0..* | AlertTypeList | Version spécifique d'une liste d'alertes publiée par cette autorité responsable des données environnementales. |
| 0..1 | PhenomenonClassification | 0..* | PhenomenonClassification | Classification de phénomènes définie par cette autorité responsable des données environnementales. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | Hérité de: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.13 EnvironmentalDataProvider

Entité fournissant des données environnementales. Il peut s'agir d'un fournisseur de données météorologiques observées, d'une entité fournissant des prévisions, d'une autorité émettant des alertes, etc.

Le Tableau 673 présente tous les attributs de EnvironmentalDataProvider.

**Tableau 673 – Attributs de Environmental::EnvironmentalDataProvider**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 674 présente toutes les extrémités d'association de EnvironmentalDataProvider avec d'autres classes.

**Tableau 674 – Extrémités d'association de Environmental:: EnvironmentalDataProvider avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | EnvironmentalAlert | 0..* | EnvironmentalAlert | Alerte émise par ce fournisseur de données environnementales. |
| 0..1 | EnvironmentalInformation | 0..* | EnvironmentalInformation | Informations environnementales fournies par ce fournisseur de données environnementales. |
| 0..1 | ConfigurationEvents | 0..* | ConfigurationEvent | Hérité de: OrganisationRole |
| 0..* | Organisation | 0..1 | Organisation | Hérité de: OrganisationRole |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.14 EnvironmentalDiscrete

Mesure discrète (valeur entière) pertinente dans le domaine environnemental.

Le Tableau 675 présente tous les attributs de EnvironmentalDiscrete.

**Tableau 675 – Attributs de Environmental::EnvironmentalDiscrete**

| nom | mult | type | description |
|---|---|---|---|
| kind | 1..1 | EnvironmentalDiscreteKind | Type de valeur environnementale discrète (entière). |
| maxValue | 0..1 | Integer | Hérité de: Discrete |
| minValue | 0..1 | Integer | Hérité de: Discrete |
| normalValue | 0..1 | Integer | Hérité de: Discrete |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 676 présente toutes les extrémités d'association de EnvironmentalDiscrete avec d'autres classes.

**Tableau 676 – Extrémités d'association de Environmental:: EnvironmentalDiscrete avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Observation ou prévision à laquelle cette valeur environnementale discrète (entière) est associée. |
| 1..1 | DiscreteValues | 0..* | DiscreteValue | Hérité de: Discrete |
| 0..* | ValueAliasSet | 0..1 | ValueAliasSet | Hérité de: Discrete |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.15 EnvironmentalEvent

Événement environnemental auquel une ou plusieurs prévisions ou observations peuvent être liées et qui peut être associé à ou affecter un ou plusieurs actifs.

Cette classe est destinée à être utilisée comme moyen de regroupement de prévisions et/ou d'observations et peut être utilisée à plusieurs fins, y compris:

– définir un événement 'named' (nommé) comme l'ouragan Katrina et permettre de l'associer à des valeurs historiques (ou prévisionnelles) correspondant aux phénomènes et aux mesures (précipitations, température) survenus/effectuées au fil du temps
– identifier les actifs qui ont été affectés (ou qui sont susceptibles d'être affectés selon une prévision) par un phénomène ou un ensemble de mesures

Le Tableau 677 présente tous les attributs de EnvironmentalEvent.

**Tableau 677 – Attributs de Environmental::EnvironmentalEvent**

| nom | mult | type | description |
|---|---|---|---|
| createdDateTime | 0..1 | DateTime | Hérité de: ActivityRecord |
| type | 0..1 | String | Hérité de: ActivityRecord |
| severity | 0..1 | String | Hérité de: ActivityRecord |
| reason | 0..1 | String | Hérité de: ActivityRecord |
| status | 0..1 | Status | Hérité de: ActivityRecord |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 678 présente toutes les extrémités d'association de EnvironmentalEvent avec d'autres classes.

**Tableau 678 – Extrémités d'association de Environmental:: EnvironmentalEvent avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..* | EnvironmentalInformation | Prévision ou observation liée à cet événement environnemental. |
| 0..* | Assets | 0..* | Asset | Hérité de: ActivityRecord |
| 0..* | Author | 0..1 | Author | Hérité de: ActivityRecord |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.16 EnvironmentalInformation

Classe abstraite (avec les classes enfants concrètes Observation et Forecast) qui regroupe des phénomènes et/ou des ensembles de valeurs environnementales.

Le Tableau 679 présente tous les attributs de EnvironmentalInformation.

**Tableau 679 – Attributs de Environmental::EnvironmentalInformation**

| nom | mult | type | description |
|---|---|---|---|
| created | 0..1 | DateTime | Horodatage de la création de la prévision |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 680 présente toutes les extrémités d'association de EnvironmentalInformation avec d'autres classes.

**Tableau 680 – Extrémités d'association de Environmental:: EnvironmentalInformation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Fournisseur de données environnementales fournissant ces informations environnementales. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Valeur environnementale analogique associée à cette observation/prévision. |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | Valeur environnementale discrète (entière) associée à cette observation/prévision. |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | Événement environnemental lié à cette prévision/observation. |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | Mesure environnementale en chaîne associée à cette prévision/observation. |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.17 Classe racine EnvironmentalLocationType

Type d'emplacement environnemental. Utilisé lorsqu'une alerte ou un phénomène environnemental(e) est associé à plusieurs emplacements

Le Tableau 681 présente tous les attributs de EnvironmentalLocationType.

**Tableau 681 – Attributs de Environmental::EnvironmentalLocationType**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | LocationKind | Type d'emplacement. Les valeurs types peuvent être center (centre), extent (encombrement), primary (primaire), secondary (secondaire), etc. |

Le Tableau 682 présente toutes les extrémités d'association de EnvironmentalLocationType avec d'autres classes.

**Tableau 682 – Extrémités d'association de Environmental:: EnvironmentalLocationType avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | Location | 0..1 | Location | Emplacement de cette instance de ce type d'emplacement environnemental. |
| 1..* | EnvironmentalAlert | 0..* | EnvironmentalAlert | Alerte environnementale s'appliquant aux emplacements de ce type. |
| 0..* | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | Phénomène environnemental pour lequel cet emplacement est pertinent. |

### 6.6.18 Classe racine EnvironmentalPhenomenon

Caractéristiques réelles ou prévues d'un phénomène environnemental à un instant spécifique (ou pendant un intervalle de temps spécifique) pouvant avoir à la fois un centre et un emplacement zone/ligne.

Le Tableau 683 présente tous les attributs de EnvironmentalPhenomenon.

**Tableau 683 – Attributs de Environmental::EnvironmentalPhenomenon**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Horodatage du phénomène comme instant unique ou intervalle de temps. |

Le Tableau 684 présente toutes les extrémités d'association de EnvironmentalPhenomenon avec d'autres classes.

**Tableau 684 – Extrémités d'association de Environmental:: EnvironmentalPhenomenon avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Prévision ou observation dont cette description de phénomène fait partie. |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Emplacement pertinent pour ce phénomène environnemental. |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Classification de ce phénomène. |

### 6.6.19 EnvironmentalStringMeasurement

Mesure en chaîne pertinente dans le domaine environnemental.

Le Tableau 685 présente tous les attributs de EnvironmentalStringMeasurement.

**Tableau 685 – Attributs de Environmental::EnvironmentalStringMeasurement**

| nom | mult | type | description |
|---|---|---|---|
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 686 présente toutes les extrémités d'association de EnvironmentalStringMeasurement avec d'autres classes.

**Tableau 686 – Extrémités d'association de Environmental::EnvironmentalStringMeasurement avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Condition de classification que cette mesure en chaîne aide à définir. |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Observation ou prévision associée à cette valeur environnementale de chaîne. |
| 1..1 | StringMeasurementValues | 0..* | StringMeasurementValue | Hérité de: StringMeasurement |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObj ect | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObj ect | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.20 Fire

Incendie, souvent non maîtrisé, couvrant une surface de terre qui contient généralement de la végétation combustible. Les informations associées à l'emplacement sont censées décrire la surface totale brûlée à partir d'un instant spécifié.

Le Tableau 687 présente tous les attributs de Fire.

**Tableau 687 – Attributs de Environmental::Fire**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 688 présente toutes les extrémités d'association de Fire avec d'autres classes.

**Tableau 688 – Extrémités d'association de Environmental::Fire avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformat ion | 0..1 | EnvironmentalInformatio n | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocatio nKind | 0..* | EnvironmentalLocationT ype | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassific ation | 0..1 | PhenomenonClassificati on | Hérité de: EnvironmentalPhenomenon |

### 6.6.21 Flood

Inondation, débordement d'une grande quantité d'eau au-delà de son espace de confinement habituel, notamment sur une terre normalement sèche.

Le Tableau 689 présente tous les attributs de Flood.

**Tableau 689 – Attributs de Environmental::Flood**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 690 présente toutes les extrémités d'association de Flood avec d'autres classes.

**Tableau 690 – Extrémités d'association de Environmental::Flood avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.22 Forecast

Groupe prévisionnel d'ensembles de valeurs et/ou de caractéristiques de phénomènes.

Le Tableau 691 présente tous les attributs de Forecast.

**Tableau 691 – Attributs de Environmental::Forecast**

| nom | mult | type | description |
|---|---|---|---|
| validFor | 0..1 | DateTimeInterval | Intervalle pendant lequel la prévision est valide. Par exemple, une prévision émise à un instant pour le lendemain peut être valide les 2 heures qui suivent son émission. |
| created | 0..1 | DateTime | Hérité de: EnvironmentalInformation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 692 présente toutes les extrémités d'association de Forecast avec d'autres classes.

**Tableau 692 – Extrémités d'association de Environmental:: Forecast avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | Hérité de: EnvironmentalInformation |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | Hérité de: EnvironmentalInformation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.23 GeosphericAnalog

Grandeur analogique (à virgule flottante) mesurant une condition géosphérique.

Le Tableau 693 présente tous les attributs de GeosphericAnalog.

**Tableau 693 – Attributs de Environmental::GeosphericAnalog**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | GeosphericAnalogKind | Type de valeur géosphérique analogique. |
| maxValue | 0..1 | Float | Hérité de: Analog |
| minValue | 0..1 | Float | Hérité de: Analog |
| normalValue | 0..1 | Float | Hérité de: Analog |
| positiveFlowIn | 0..1 | Boolean | Hérité de: Analog |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 694 présente toutes les extrémités d'association de GeosphericAnalog avec d'autres classes.

**Tableau 694 – Extrémités d'association de Environmental:: GeosphericAnalog avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoring Station | Hérité de: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Hérité de: EnvironmentalAnalog |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | Hérité de: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | Hérité de: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | Hérité de: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.24 GeosphericPhenomenon

Phénomène géosphérique.

Le Tableau 695 présente tous les attributs de GeosphericPhenomenon.

**Tableau 695 – Attributs de Environmental::GeosphericPhenomenon**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 696 présente toutes les extrémités d'association de GeosphericPhenomenon avec d'autres classes.

**Tableau 696 – Extrémités d'association de Environmental:: GeosphericPhenomenon avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.25 Hurricane

Ouragan, sous-type de cyclone survenant dans l'Atlantique Nord ou dans le nord-est de l'océan Pacifique dont l'intensité est mesurée à l'aide de l'échelle de Saffir-Simpson.

Le Tableau 697 présente tous les attributs de Hurricane.

**Tableau 697 – Attributs de Environmental::Hurricane**

| nom | mult | type | description |
|---|---|---|---|
| category | 0..1 | Integer | Catégorie d'ouragan pendant l'intervalle de temps, déterminée à l'aide de l'échelle de Saffir-Simpson, de valeur assignée comprise entre 1 et 5 selon une vitesse maintenue de vent soutenue de l'ouragan. |
| centralPressure | 0..1 | Pressure | Hérité de: Cyclone |
| maxSurfaceWindSpeed | 0..1 | Speed | Hérité de: Cyclone |
| windForce | 0..1 | Integer | Hérité de: Cyclone |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 698 présente toutes les extrémités d'association de Hurricane avec d'autres classes.

**Tableau 698 – Extrémités d'association de Environmental:: Hurricane avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.26 HydrosphericAnalog

Grandeur analogique (à virgule flottante) mesurant une condition hydrosphérique.

Le Tableau 699 présente tous les attributs de HydrosphericAnalog.

**Tableau 699 – Attributs de Environmental::HydrosphericAnalog**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | HydrosphericAnalogKind | Type de grandeur analogique hydrosphérique. |
| maxValue | 0..1 | Float | Hérité de: Analog |
| minValue | 0..1 | Float | Hérité de: Analog |
| normalValue | 0..1 | Float | Hérité de: Analog |
| positiveFlowIn | 0..1 | Boolean | Hérité de: Analog |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 700 présente toutes les extrémités d'association de HydrosphericAnalog avec d'autres classes.

**Tableau 700 – Extrémités d'association de Environmental:: HydrosphericAnalog avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoringStation | Hérité de: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Hérité de: EnvironmentalAnalog |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | Hérité de: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | Hérité de: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | Hérité de: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.27 HydrosphericPhenomenon

Phénomène hydrosphérique.

Le Tableau 701 présente tous les attributs de HydrosphericPhenomenon.

**Tableau 701 – Attributs de Environmental::HydrosphericPhenomenon**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 702 présente toutes les extrémités d'association de HydrosphericPhenomenon avec d'autres classes.

**Tableau 702 – Extrémités d'association de Environmental:: HydrosphericPhenomenon avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.28 Landslide

Glissement de terrain, gros bloc de pierres et de terre qui, de manière subite et rapide, descend d'une montagne ou d'un colline.

Le Tableau 703 présente tous les attributs de Landslide.

**Tableau 703 – Attributs de Environmental::Landslide**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 704 présente toutes les extrémités d'association de Landslide avec d'autres classes.

**Tableau 704 – Extrémités d'association de Environmental:: Landslide avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.29 LightningStrike

Éclair nuage-sol survenant à un emplacement particulier.

Le Tableau 705 présente tous les attributs de LightningStrike.

**Tableau 705 – Attributs de Environmental::LightningStrike**

| nom | mult | type | description |
|---|---|---|---|
| errorEllipseConfidence | 0..1 | PerCent | Probabilité que l'éclair soit tombé dans errorEllipse. |
| errorEllipseMajorSemiAxis | 0..1 | Length | Longueur du demi-axe majeur (rayon le plus long) de l'ellipse d'erreur. |
| errorEllipseMinorSemiAxis | 1..1 | Length | Longueur du demi-axe mineur (rayon le plus court) de l'ellipse d'erreur. |
| errorEllipseOrientation | 0..1 | Bearing | Orientation du demi-axe majeur en degrés à partir du nord réel. |
| negativePolarity | 0..1 | Boolean | Polarité de l'éclair, T indiquant une polarité négative. Environ 90 % de tous les foudroiements sont négatifs, ce qui signifie qu'ils ont été initiés par une grande concentration de charge négative dans la base des nuages; ceci tend à induire une zone de charge positive sur la terre. |
| peakAmplitude | 0..1 | CurrentFlow | Courant de crête du foudroiement. |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 706 présente toutes les extrémités d'association de LightningStrike avec d'autres classes.

**Tableau 706 – Extrémités d'association de Environmental:: LightningStrike avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.30 MagneticStorm

Orage magnétique, perturbation temporaire du champ magnétique de la terre, induite par un rayonnement et des flux de particules chargées provenant du soleil.

Le Tableau 707 présente tous les attributs de MagneticStorm.

**Tableau 707 – Attributs de Environmental::MagneticStorm**

| nom | mult | type | description |
|---|---|---|---|
| changeDst | 0..1 | MagneticField | Changement de la perturbation – indice Dst (disturbance storm time – temps de perturbation orageuse). Les dimensions d'un orage géomagnétique sont classées comme suit:<br>- modérées ( -50 nT >valeur minimale de l'indice Dst > -100 nT)<br>- intenses (-100 nT > valeur minimale de l'indice Dst > -250 nT) or<br>- forte tempête (valeur minimale de l'indice Dst < -250 nT). |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 708 présente toutes les extrémités d'association de MagneticStorm avec d'autres classes.

**Tableau 708 – Extrémités d'association de Environmental:: MagneticStorm avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.31 Observation

Ensembles de valeurs et/ou de caractéristiques de phénomènes observées (réelles non prévues).

Le Tableau 709 présente tous les attributs de Observation.

**Tableau 709 – Attributs de Environmental::Observation**

| nom | mult | type | description |
|---|---|---|---|
| created | 0..1 | DateTime | Hérité de: EnvironmentalInformation |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 710 présente toutes les extrémités d'association de Observation avec d'autres classes.

**Tableau 710 – Extrémités d'association de Environmental:: Observation avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataProvider | 0..1 | EnvironmentalDataProvider | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalDiscrete | 0..* | EnvironmentalDiscrete | Hérité de: EnvironmentalInformation |
| 0..* | EnvironmentalEvent | 0..* | EnvironmentalEvent | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | Hérité de: EnvironmentalInformation |
| 0..1 | EnvironmentalStringMeasurement | 0..* | EnvironmentalStringMeasurement | Hérité de: EnvironmentalInformation |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.32 PhenomenonClassification

Classification prédéfinie de phénomènes telle que définie par une autorité compétente particulière.

Le Tableau 711 présente tous les attributs de PhenomenonClassification.

**Tableau 711 – Attributs de Environmental::PhenomenonClassification**

| nom | mult | type | description |
|---|---|---|---|
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 712 présente toutes les extrémités d'association de PhenomenonClassification avec d'autres classes.

**Tableau 712 – Extrémités d'association de Environmental:: PhenomenonClassification avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalDataAuthority | 0..1 | EnvironmentalDataAuthority | Autorité définissant ce phénomène environnemental. |
| 0..1 | ClassificationCondition | 0..* | ClassificationCondition | Condition contribuant à la classification de ce phénomène. |
| 0..1 | EnvironmentalPhenomenon | 0..* | EnvironmentalPhenomenon | |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.33 Classe racine ReportingCapability

Définition d'un ensemble de capacités de production de rapports pour cette station de surveillance. Les EnvironmentalValueSets associées décrivent la plage maximale de valeurs environnementales possibles que la station est capable de renvoyer. Cet attribut est principalement destiné à assister une entreprise d'électricité dans la gestion de ses stations.

Le Tableau 713 présente tous les attributs de ReportingCapability.

**Tableau 713 – Attributs de Environmental::ReportingCapability**

| nom | mult | type | description |
|---|---|---|---|
| reportingIntervalPeriod | 0..1 | Integer | Nombre d'unités de temps constituant la période de production de rapports. |
| reportingIntervalType | 0..1 | CalculationIntervalUnitKind | Unité de temps servant à exprimer la période de production de rapports. |
| reportingMethod | 0..1 | ReportingMethodKind | Indique comment la station météorologique rapporte des observations. |

Le Tableau 714 présente toutes les extrémités d'association de ReportingCapability avec d'autres classes.

**Tableau 714 – Extrémités d'association de Environmental:: ReportingCapability avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 1..1 | EnvironmentalMonitoring Station | Station de surveillance environnementale à laquelle appartient cet ensemble de capacités de production de rapports. |
| 0..1 | EnvironmentalAnalog | 0..* | EnvironmentalAnalog | Un des ensembles de valeurs environnementales exprimant l'une des capacités de production de rapports. |

### 6.6.34 SpaceAnalog

Grandeur analogique (à virgule flottante) mesurant une condition spatiale (extra-terrestre).

Le Tableau 715 présente tous les attributs de SpaceAnalog.

**Tableau 715 – Attributs de Environmental::SpaceAnalog**

| nom | mult | type | description |
|---|---|---|---|
| kind | 0..1 | SpaceAnalogKind | Type de valeur spatiale analogique. |
| maxValue | 0..1 | Float | Hérité de: Analog |
| minValue | 0..1 | Float | Hérité de: Analog |
| normalValue | 0..1 | Float | Hérité de: Analog |
| positiveFlowIn | 0..1 | Boolean | Hérité de: Analog |
| measurementType | 0..1 | String | Hérité de: Measurement |
| phases | 0..1 | PhaseCode | Hérité de: Measurement |
| uncefactUnitCode | 0..1 | UncefactUnitCode | Hérité de: Measurement |
| unitMultiplier | 0..1 | UnitMultiplier | Hérité de: Measurement |
| unitSymbol | 0..1 | UnitSymbol | Hérité de: Measurement |
| aliasName | 0..1 | String | Hérité de: IdentifiedObject |
| description | 0..1 | String | Hérité de: IdentifiedObject |
| mRID | 0..1 | String | Hérité de: IdentifiedObject |
| name | 0..1 | String | Hérité de: IdentifiedObject |

Le Tableau 716 présente toutes les extrémités d'association de SpaceAnalog avec d'autres classes.

**Tableau 716 – Extrémités d’association de Environmental:: SpaceAnalog avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalMonitoringStation | 0..1 | EnvironmentalMonitoringStation | Hérité de: EnvironmentalAnalog |
| 0..* | ClassificationCondition | 0..1 | ClassificationCondition | Hérité de: EnvironmentalAnalog |
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalAnalog |
| 0..* | ReportingCapability | 0..1 | ReportingCapability | Hérité de: EnvironmentalAnalog |
| 1..1 | AnalogValues | 0..* | AnalogValue | Hérité de: Analog |
| 0..* | LimitSets | 0..* | AnalogLimitSet | Hérité de: Analog |
| 0..* | Terminal | 0..1 | ACDCTerminal | Hérité de: Measurement |
| 0..* | PowerSystemResource | 0..1 | PowerSystemResource | Hérité de: Measurement |
| 0..1 | DiagramObjects | 0..* | DiagramObject | Hérité de: IdentifiedObject |
| 1..1 | Names | 0..* | Name | Hérité de: IdentifiedObject |
| 0..1 | PropertiesCIMDataObject | 0..1 | DataSetMember | Hérité de: IdentifiedObject |
| 1..1 | TargetingCIMDataObject | 0..* | DataSetMember | Hérité de: IdentifiedObject |

### 6.6.35 SpacePhenomenon

Phénomène extra-terrestre.

Le Tableau 717 présente tous les attributs de SpacePhenomenon.

**Tableau 717 – Attributs de Environmental::SpacePhenomenon**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 718 présente toutes les extrémités d’association de SpacePhenomenon avec d’autres classes.

**Tableau 718 – Extrémités d’association de Environmental:: SpacePhenomenon avec d’autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.36 Tornado

Tornade, vent tournoyant destructeur et violent accompagné d'un nuage en forme d'entonnoir qui se déplace au-dessus de la terre en suivant une trajectoire étroite.

Le Tableau 719 présente tous les attributs de Tornado.

**Tableau 719 – Attributs de Environmental::Tornado**

| nom | mult | type | description |
|---|---|---|---|
| fScale | 0..1 | FScale | Échelle de Fujita (appelée échelle EF depuis 2007) pour la tournade. |
| width | 1..1 | Length | Largeur de la tornade pendant l'intervalle de temps. |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 720 présente toutes les extrémités d'association de Tornado avec d'autres classes.

**Tableau 720 – Extrémités d'association de Environmental:: Tornado avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.37 TropicalCycloneAustralia

Cyclone tropical, sous-type de cyclone qui se forme à l'est de 90°E dans l'hémisphère Sud, dont l'intensité est mesurée par l'*Australian tropical cyclone intensity scale* (échelle australienne d'intensité des cyclones tropicaux).

Le Tableau 721 présente tous les attributs de TropicalCycloneAustralia.

**Tableau 721 – Attributs de Environmental::TropicalCycloneAustralia**

| nom | mult | type | description |
|---|---|---|---|
| category | 0..1 | Integer | Force du cyclone tropical pendant l'intervalle de temps, d'après le système de catégorisation de l'*Australian Bureau of Meteorology* (service météorologique australien), où:<br>1 – cyclone tropical, avec des rafales types au-dessus d'un terrain plat comprises entre 90 km/h et 125 km/h<br>2 – cyclone tropical, avec des rafales types au-dessus d'un terrain plat comprises entre 125 km/h et 164 km/h<br>3 – cyclone tropical violent, avec des rafales types au-dessus d'un terrain plat comprises entre 165 km/h et 224 km/h<br>4 – cyclone tropical violent, avec des rafales types au-dessus d'un terrain plat comprises entre 225 km/h et 279 km/h<br>5 – cyclone tropical violent, avec des rafales types au-dessus d'un terrain plat supérieures à 280 km/h. |
| centralPressure | 0..1 | Pressure | Hérité de: Cyclone |
| maxSurfaceWindSpeed | 0..1 | Speed | Hérité de: Cyclone |
| windForce | 0..1 | Integer | Hérité de: Cyclone |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 722 présente toutes les extrémités d'association de TropicalCycloneAustralia avec d'autres classes.

**Tableau 722 – Extrémités d'association de Environmental:: TropicalCycloneAustralia avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.38 Tsunami

Tsunami (raz de marée), longue vague en haute mer provoquée par un séisme, un glissement de terrine sous-marin, ou d'autres perturbations

Le Tableau 723 présente tous les attributs de Tsunami.

**Tableau 723 – Attributs de Environmental::Tsunami**

| nom | mult | type | description |
|---|---|---|---|
| intensity | 0..1 | Integer | Intensité de tsunami sur l'échelle d'intensité de tsunami de Papadopoulos-Imamura. Les valeurs possibles sont 1-12, correspondant à I-XII. |
| magnitude | 0..1 | Float | Magnitude de tsunami sur l'échelle de magnitude de tsunami (Mt). Supérieure à zéro. |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 724 présente toutes les extrémités d'association de Tsunami avec d'autres classes.

**Tableau 724 – Extrémités d'association de Environmental:: Tsunami avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.39 VolcanicAshCloud

Nuages de cendres formés par suite d'une éruption volcanique.

Le Tableau 725 présente tous les attributs de VolcanicAshCloud.

**Tableau 725 – Attributs de Environmental::VolcanicAshCloud**

| nom | mult | type | description |
|---|---|---|---|
| density | 0..1 | ParticulateDensity | Densité particulaire du nuage de cendres pendant l'intervalle de temps. |
| particleSize | 0..1 | Length | Diamètres des particules pendant l'intervalle de temps. |
| altitude | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| base | 0..1 | RelativeDisplacement | Hérité de: AtmosphericPhenomenon |
| direction | 0..1 | Bearing | Hérité de: AtmosphericPhenomenon |
| maxCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| minCoverage | 0..1 | PerCent | Hérité de: AtmosphericPhenomenon |
| speed | 0..1 | Speed | Hérité de: AtmosphericPhenomenon |
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

Le Tableau 726 présente toutes les extrémités d'association de VolcanicAshCloud avec d'autres classes.

**Tableau 726 – Extrémités d'association de Environmental:: VolcanicAshCloud avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.40 Whirlpool

Tourbillon, masse d'eau en rotation rapide dans une rivière ou une mer vers laquelle des objets peuvent être entraînés, généralement provoqué par la rencontre de courants contraires.

Le Tableau 727 présente tous les attributs de Whirlpool.

**Tableau 727 – Attributs de Environmental::Whirlpool**

| nom | mult | type | description |
|---|---|---|---|
| timeInterval | 0..1 | DateTimeInterval | Hérité de: EnvironmentalPhenomenon |

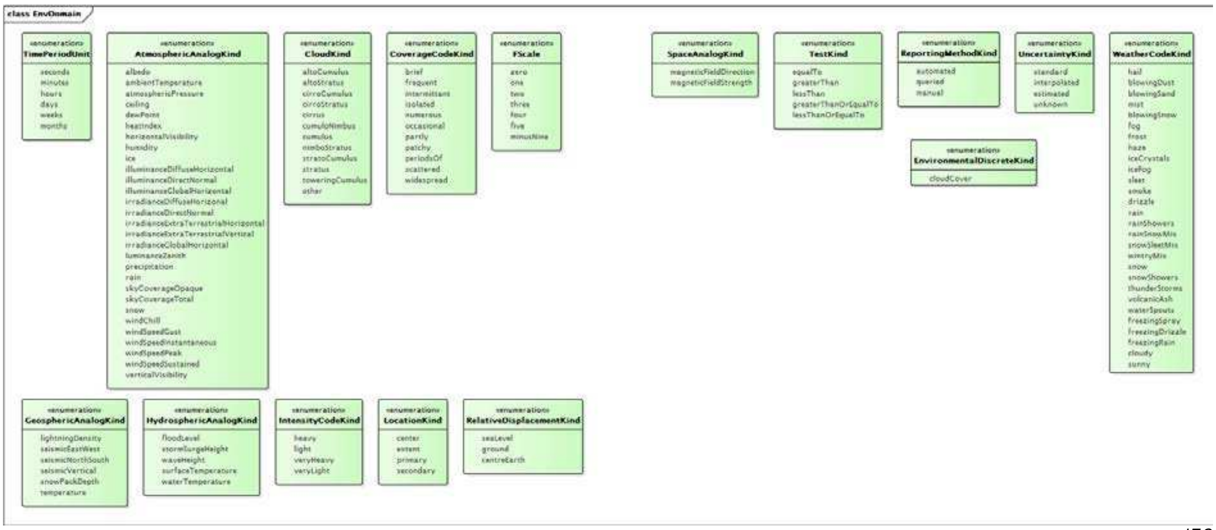
Le Tableau 728 présente toutes les extrémités d'association de Whirlpool avec d'autres classes.

**Tableau 728 – Extrémités d'association de Environmental:: Whirlpool avec d'autres classes**

| mult à partir de | nom | mult vers | type | description |
|---|---|---|---|---|
| 0..* | EnvironmentalInformation | 0..1 | EnvironmentalInformation | Hérité de: EnvironmentalPhenomenon |
| 0..* | EnvironmentalLocationKind | 0..* | EnvironmentalLocationType | Hérité de: EnvironmentalPhenomenon |
| 0..* | PhenomenonClassification | 0..1 | PhenomenonClassification | Hérité de: EnvironmentalPhenomenon |

### 6.6.41 Paquetage EnvDomain

#### 6.6.41.1 Généralités




IEC

**Figure 80 – Diagramme de classe EnvDomain::EnvDomain**

Figure 80: représente les types de données et les énumérations qui prennent en charge les classes du paquetage Environmental, en plus des types de données et des énumérations contenus dans les paquetages IEC 61970, IEC 61968 et IEC 62325.

#### 6.6.41.2 Composé RelativeDisplacement

Déplacement vertical relatif au niveau de la mer, au niveau de la terre ou au centre de la Terre.

Le Tableau 729 présente tous les attributs de RelativeDisplacement.

**Tableau 729 – Attributs de EnvDomain::RelativeDisplacement**

| nom | mult | type | description |
|---|---|---|---|
| displacement | 0..1 | Displacement | |
| kind | 0..1 | RelativeDisplacementKind | |

#### 6.6.41.3 Énumération AtmosphericAnalogKind

Types de valeurs analogiques (à virgule flottante) mesurant une condition atmosphérique.

Le Tableau 730 présente tous les libellés de AtmosphericAnalogKind.

**Tableau 730 – Libellés de EnvDomain::AtmosphericAnalogKind**

| libellé | value | description |
|---|---|---|
| albedo | | |
| ambientTemperature | | Température mesurée avec un thermomètre exposé à l'air et placé dans un lieu abrité du rayonnement direct du soleil. Également appelée température de "réservoir sec" car la température de l'air est indiquée par un thermomètre insensible à l'humidité de l'air. |
| atmosphericPressure | | |
| ceiling | | |
| dewPoint | | Température à laquelle l'air doit être refroidi à pression constante et teneur en vapeur d'eau constante pour parvenir au point de saturation. En d'autres termes, il s'agit de la température à laquelle la vapeur d'eau commence à se condenser. |
| heatIndex | | Température de la chaleur ressentie pour une combinaison donnée de température d'air chaud et d'humidité relative. |
| horizontalVisibility | | |
| humidity | | |
| ice | | |
| illuminanceDiffuseHorizontal | | |
| illuminanceDirectNormal | | |
| illuminanceGlobalHorizontal | | |
| irradianceDiffuseHorizonal | | |
| irradianceDirectNormal | | |
| irradianceExtraTerrestrialHorizontal | | |
| irradianceExtraTerrestrialVertical | | |
| irradianceGlobalHorizontal | | |
| luminanceZenith | | |
| precipitation | | |
| rain | | |
| skyCoverageOpaque | | |
| skyCoverageTotal | | |
| snow | | Quantité de neige pendant une durée spécifiée. |
| windChill | | Température du froid ressenti en fonction du taux de perte de chaleur par la peau exposée due au vent et aux températures froides. |
| windSpeedGust | | Vitesse instantanée maximale du vent au cours des 10 min précédant une durée pendant laquelle une différence de plus de 10 nœuds a été constatée entre les pointes et les accalmies durant cette période de 10 min.<br>la valeur de 0 indique l'absence de rafales pendant les 10 min précédentes. |
| windSpeedInstantaneous | | Vitesse du vent à un moment donné. |
| windSpeedPeak | | Vitesse instantanée de pointe du vent au cours des 60 min précédant une durée pendant laquelle une vitesse de pointe de plus de 25 nœuds a été constatée.<br>la valeur de 0 indique que la vitesse n'a pas dépassé 25 nœuds pendant les 60 min précédentes. |
| windSpeedSustained | | Vitesse instantanée moyenne du vent au cours des 2 min précédant un moment donné. |
| verticalVisibility | | |

### 6.6.41.4 Énumération CloudKind

Type de nuage.

Le Tableau 731 présente tous les libellés de CloudKind.

**Tableau 731 – Libellés de EnvDomain::CloudKind**

| libellé | valeur | description |
|---|---|---|
| altoCumulus | | |
| altoStratus | | |
| cirroCumulus | | |
| cirroStratus | | |
| cirrus | | |
| cumuloNimbus | | |
| cumulus | | |
| nimboStratus | | |
| stratoCumulus | | |
| stratus | | |
| toweringCumulus | | |
| other | | |

### 6.6.41.5 Énumération CoverageCodeKind

Types de couvertures de conditions météorologiques.

Le Tableau 732 présente tous les libellés de CoverageCodeKind.

**Tableau 732 – Libellés de EnvDomain::CoverageCodeKind**

| libellé | valeur | description |
|---|---|---|
| brief | | |
| frequent | | |
| intermittant | | |
| isolated | | |
| numerous | | |
| occasional | | |
| partly | | |
| patchy | | |
| periodsOf | | |
| scattered | | |
| widespread | | |

### 6.6.41.6 Énumération EnvironmentalDiscreteKind

Grandeur discrète (entière) mesurant une condition environnementale.

Le Tableau 733 présente tous les libellés de EnvironmentalDiscreteKind.

**Tableau 733 – Libellés de EnvDomain::EnvironmentalDiscreteKind**

| libellé | valeur | description |
|---|---|---|
| cloudCover | | Couverture nuageuse en octa. |

### 6.6.41.7 Énumération FScale

Échelle de Fujita (appelée échelle EF depuis 2007) relative aux dommages provoqués par les tornades. Ensemble d'estimations (pas de mesures) des vents en fonction des dommages. Elle utilise des rafales de trois secondes estimées à l'emplacement des dommages par le biais d'une appréciation à 8 niveaux de dommages selon 28 indicateurs. Ces estimations varient en fonction de la hauteur et de l'exposition.

La rafale de 3 s ne provient pas du même vent que celui spécifié dans les observations en surface normalisées.

Les énumérations sont conformes aux conventions NOAA.

Le Tableau 734 présente tous les libellés de FScale.

**Tableau 734 – Libellés de EnvDomain::FScale**

| libellé | valeur | description |
|---|---|---|
| zero | | 65 mph – 85 mph rafale de 3 s. |
| one | | 86 mph – 110 mph rafale de 3 s. |
| two | | 111 mph – 135 mph rafale de 3 s. |
| three | | 136 mph – 165 mph rafale de 3 s. |
| four | | 166 mph – 200 mph rafale de 3 s. |
| five | | Plus de 200 mph rafale de 3 s.. |
| minusNine | | Non connu. |

### 6.6.41.8 Énumération GeosphericAnalogKind

Types de valeurs analogiques (à virgule flottante) mesurant une condition géosphérique.

Le Tableau 735 présente tous les libellés de GeosphericAnalogKind.

**Tableau 735 – Libellés de EnvDomain::GeosphericAnalogKind**

| libellé | valeur | description |
|---|---|---|
| lightningDensity | | Taux d'éclair en coups de foudre/h/km2. |
| seismicEastWest | | |
| seismicNorthSouth | | |
| seismicVertical | | |
| snowPackDepth | | |
| temperature | | |

### 6.6.41.9 Énumération HydrosphericAnalogKind

Types de valeurs analogiques (à virgule flottante) mesurant une condition hydrosphérique.

Le Tableau 736 présente tous les libellés de HydrosphericAnalogKind.

**Tableau 736 – Libellés de EnvDomain::HydrosphericAnalogKind**

| libellé | valeur | description |
|---|---|---|
| floodLevel | | |
| stormSurgeHeight | | |
| waveHeight | | |
| surfaceTemperature | | |
| waterTemperature | | |

### 6.6.41.10 Énumération IntensityCodeKind

Types d'intensités de conditions météorologiques.

Le Tableau 737 présente tous les libellés de IntensityCodeKind.

**Tableau 737 – Libellés de EnvDomain::IntensityCodeKind**

| libellé | valeur | description |
|---|---|---|
| heavy | | |
| light | | |
| veryHeavy | | |
| veryLight | | |

### 6.6.41.11 Énumération LocationKind

Nature de l'emplacement défini pour une entité environnementale. Les valeurs possibles sont center (centre), perimeter (périmètre), primary (primaire), secondary (secondaire).

Le Tableau 738 présente tous les libellés de LocationKind.

**Tableau 738 – Libellés de EnvDomain::LocationKind**

| libellé | valeur | description |
|---|---|---|
| center | | Centre d'un phénomène. À utiliser généralement avec un Location avec une seule PositionPoint instance. |
| extent | | Zone ou étendue linéaire d'un phénomène, pas le centre. À utiliser généralement avec un Location avec plusieurs PositionPoint instances. |
| primary | | Zone primaire à laquelle s'applique une alerte environnementale. |
| secondary | | Zone secondaire à laquelle s'applique une alerte environnementale. |

### 6.6.41.12 Énumération RelativeDisplacementKind

Types de déplacements relatifs.

Le Tableau 739 présente tous les libellés de RelativeDisplacementKind.

**Tableau 739 – Libellés de EnvDomain::RelativeDisplacementKind**

| libellé | valeur | description |
|---|---|---|
| seaLevel | | |
| ground | | |
| centreEarth | | |

#### 6.6.41.13 Énumération SpaceAnalogKind

Types de valeurs analogiques (à virgule flottante) mesurant une condition spatiale.

Le Tableau 740 présente tous les libellés de SpaceAnalogKind.

**Tableau 740 – Libellés de EnvDomain::SpaceAnalogKind**

| libellé | valeur | description |
|---|---|---|
| magneticFieldDirection | | |
| magneticFieldStrength | | |

#### 6.6.41.14 Énumération TestKind

Essai appliqué afin de déterminer si la condition est satisfaite.

Le Tableau 741 présente tous les libellés de TestKind.

**Tableau 741 – Libellés de EnvDomain::TestKind**

| libellé | valeur | description |
|---|---|---|
| equalTo | | |
| greaterThan | | |
| lessThan | | |
| greaterThanOrEqualTo | | |
| lessThanOrEqualTo | | |

#### 6.6.41.15 Énumération TimePeriodUnit

Unités dans lesquelles la fréquence de production de rapports est spécifiée.

Le Tableau 742 présente tous les libellés de TimePeriodUnit.

**Tableau 742 – Libellés de EnvDomain::TimePeriodUnit**

| libellé | valeur | description |
|---|---|---|
| seconds | | |
| minutes | | |
| hours | | |
| days | | |
| weeks | | |
| months | | |

### 6.6.41.16 Énumération ReportingMethodKind

Méthode de rassemblement des informations à partir de la station.

Le Tableau 743 présente tous les libellés de ReportingMethodKind.

**Tableau 743 – Libellés de EnvDomain::ReportingMethodKind**

| libellé | valeur | description |
|---|---|---|
| automated | | La station transmet automatiquement des rapports. |
| queried | | La station doit être sollicitée pour obtenir des observations. |
| manual | | La station doit être physiquement visitée pour rassembler des observations. |

### 6.6.41.17 Énumération UncertaintyKind

Type d’incertitude d’un relevé.

Le Tableau 744 présente tous les libellés de UncertaintyKind.

**Tableau 744 – Libellés de EnvDomain::UncertaintyKind**

| libellé | valeur | description |
|---|---|---|
| standard | | La valeur a une incertitude type cohérente avec les pratiques du Service Météorologique National compte tenu de l’instrumentation et du mode d’observation appliqué. |
| interpolated | | La valeur est interpolée |
| estimated | | La valeur est estimée. |
| unknown | | Le processus de calcul ou de mesure de la valeur n’est pas connu. |

### 6.6.41.18 Énumération WeatherCodeKind

Types de conditions météorologiques.

Le Tableau 745 présente tous les libellés de WeatherCodeKind.

**Tableau 745 – Libellés de EnvDomain::WeatherCodeKind**

| libellé | valeur | description |
|---|---|---|
| hail | | "A" code météorologique ("grêle"). |
| blowingDust | | "BD" code météorologique ("poussière poudreuse"). |
| blowingSand | | "BN" code météorologique ("sable poudreux"). |
| mist | | "BR" code météorologique (“brume"), |
| blowingSnow | | "BS" code météorologique ("neige poudreuse"). |
| fog | | "F code météorologique ("brouillard |
| frost | | "FR" code météorologique ("givre"). |
| haze | | "H" code météorologique ("brume"). |
| iceCrystals | | "IC" code météorologique ("cristaux de glace"). |
| iceFog | | "IF" code météorologique ("brouillard givrant"). |
| sleet | | "IP" code météorologique ("grésil / neige à moitié fondue"). |
| smoke | | "K" code météorologique ("fumée"). |

| libellé | valeur | description |
|---|---|---|
| drizzle | | "L" code météorologique ("crachin"). |
| rain | | "R" code météorologique ("pluie"). |
| rainShowers | | "RW" code météorologique ("averses de pluie"). |
| rainSnowMix | | "RS" code météorologique ("pluie/neige mêlées"). |
| snowSleetMix | | "SI" code météorologique ("pluie et neige mêlées"). |
| wintryMix | | "WS" code météorologique ("mélange hivernal"). |
| snow | | "S" code météorologique ("neige"). |
| snowShowers | | "SW" code météorologique ("rafales de neige"). |
| thunderStorms | | "T" code météorologique ("orages"). |
| volcanicAsh | | "VA" code météorologique ("cendre volcanique"). |
| waterSpouts | | "WP" code météorologique ("trombes marines") |
| freezingSpray | | "ZF" code météorologique ("embrun verglaçant"). |
| freezingDrizzle | | "ZL" code météorologique ("bruine verglaçante"). |
| freezingRain | | "ZR" code météorologique ("pluie verglaçante"). |
| cloudy | | |
| sunny | | |

### 6.6.41.19 Type de données Bearing

Relèvement en degrés (360 degrés constituant le nord réel). Mesuré en degrés, dans le sens des aiguilles d'une montre, à partir du nord réel. 0 degré indique qu'aucune direction n'est fournie.

Le Tableau 746 présente tous les attributs de Bearing.

**Tableau 746 – Attributs de EnvDomain::Bearing**

| nom | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=none) |
| unit | 0..1 | UnitSymbol | (const=deg) |
| value | 0..1 | Float | |

### 6.6.41.20 Type de données MagneticField

Champ magnétique en nanotesla.

Le Tableau 747 présente tous les attributs de MagneticField.

**Tableau 747 – Attributs de EnvDomain::MagneticField**

| nom | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=n) |
| unit | 0..1 | UnitSymbol | (const=T) |
| value | 0..1 | Float | |

#### 6.6.41.21 Types de données ParticulateDensity

Densité particulaire en kg/m$^3$.

Le Tableau 748 présente tous les attributs de ParticulateDensity.

**Tableau 748 – Attributs de EnvDomain::ParticulateDensity**

| nom | mult | type | description |
|---|---|---|---|
| multiplier | 0..1 | UnitMultiplier | (const=none) |
| unit | 0..1 | UnitSymbol | (const=kgPerm3) |
| value | 0..1 | Float | |

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CIM Users Group – The community of CIM users. http://cimug.ucaiug.org (disponible en anglais seulement)

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